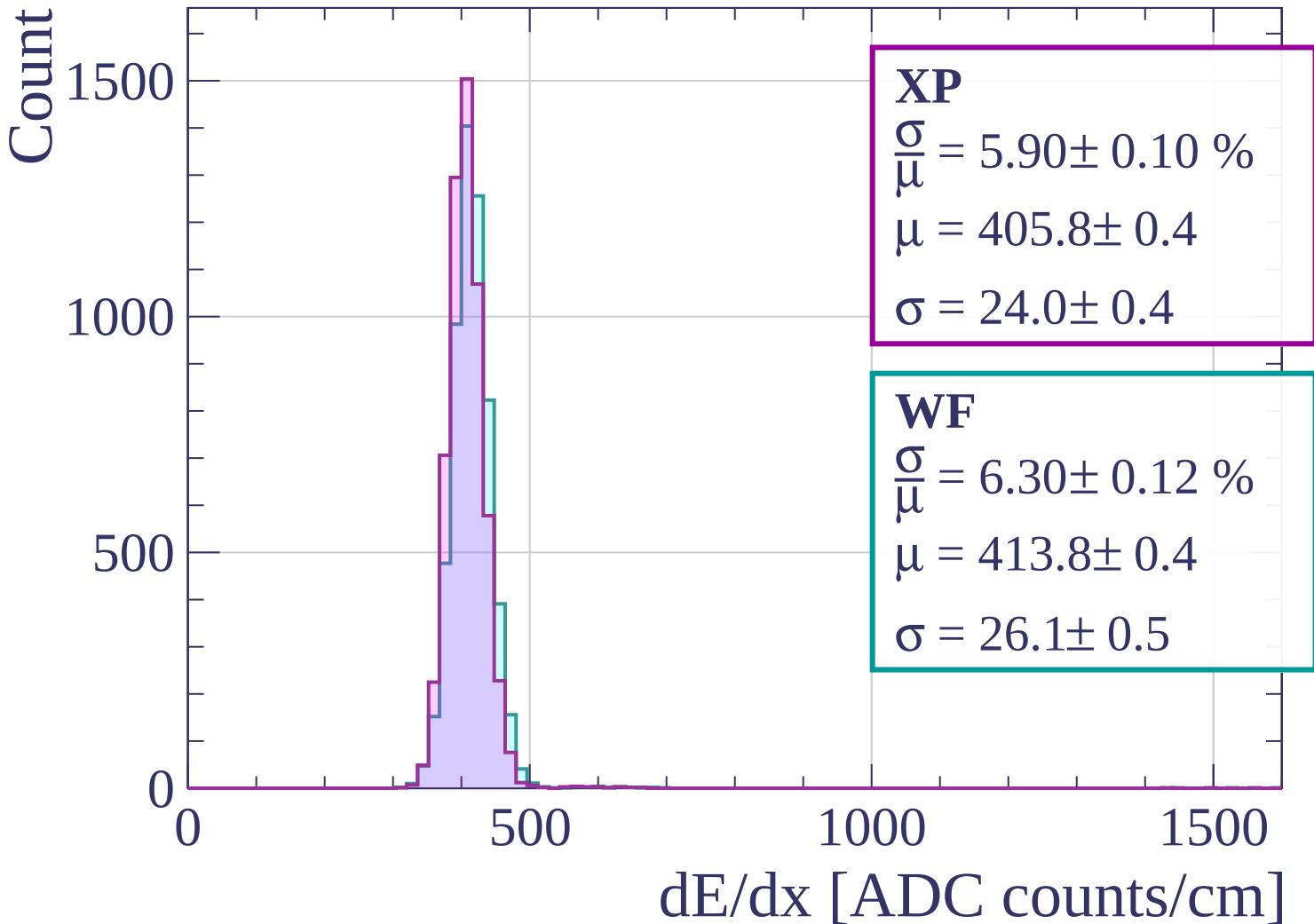
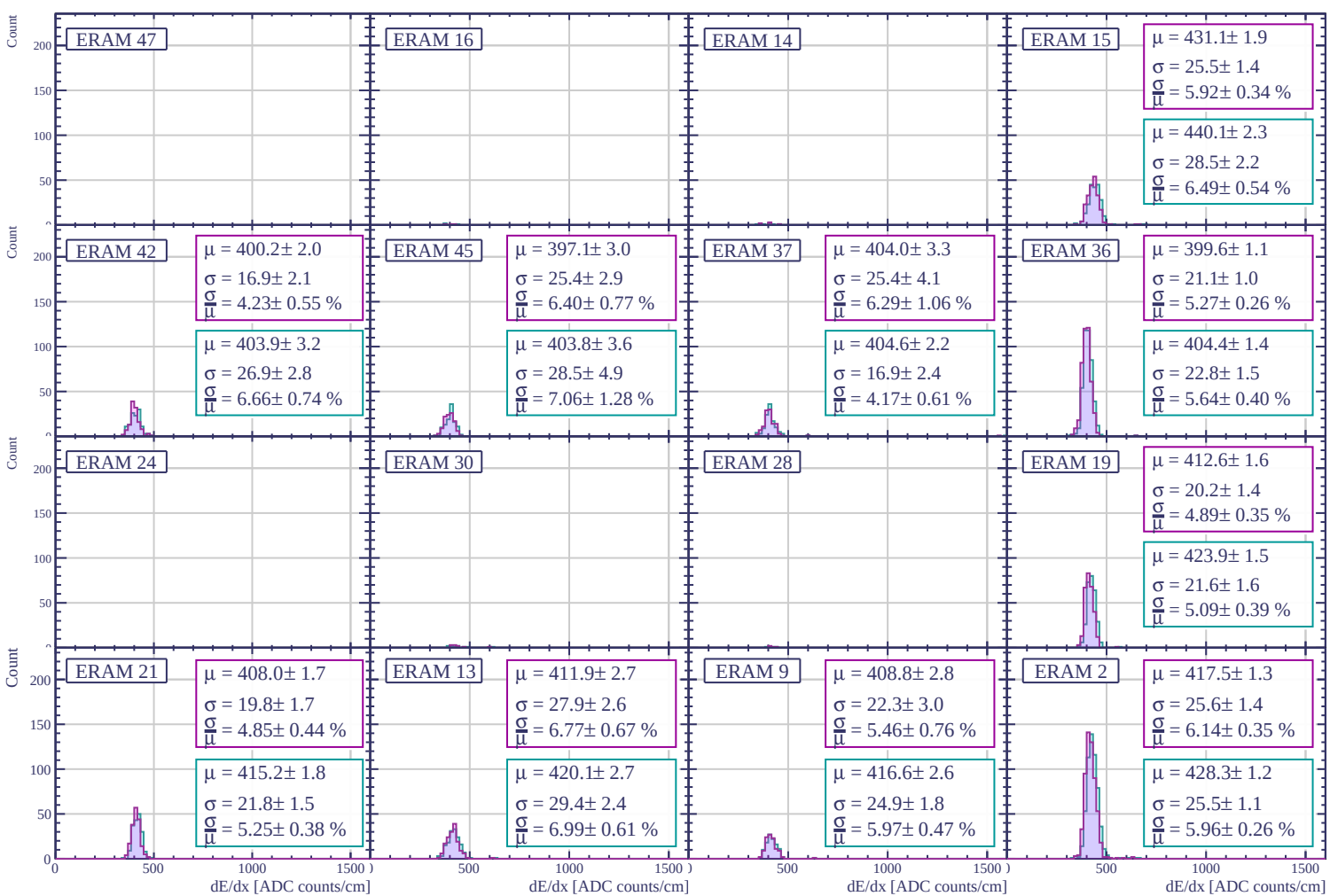
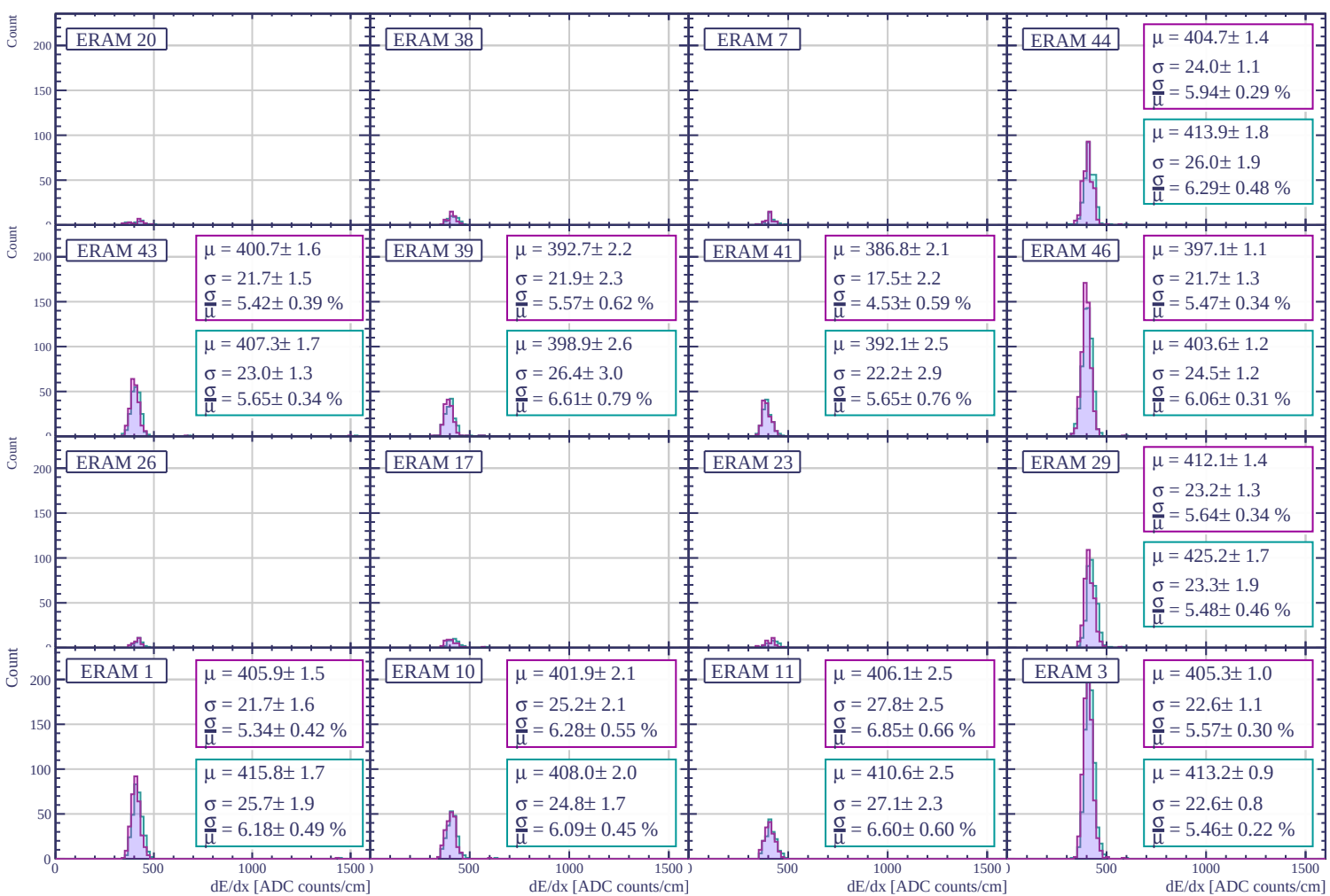
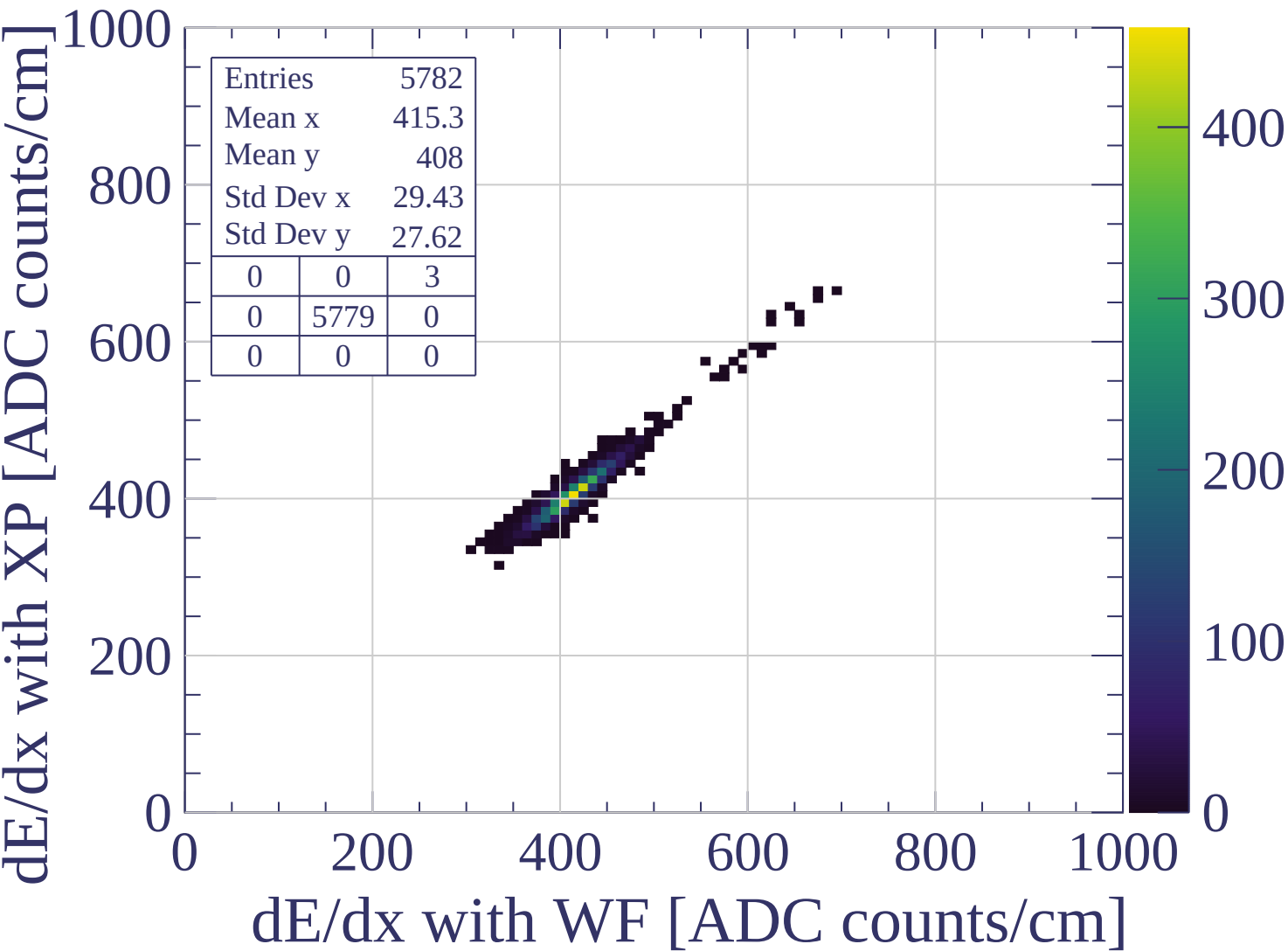
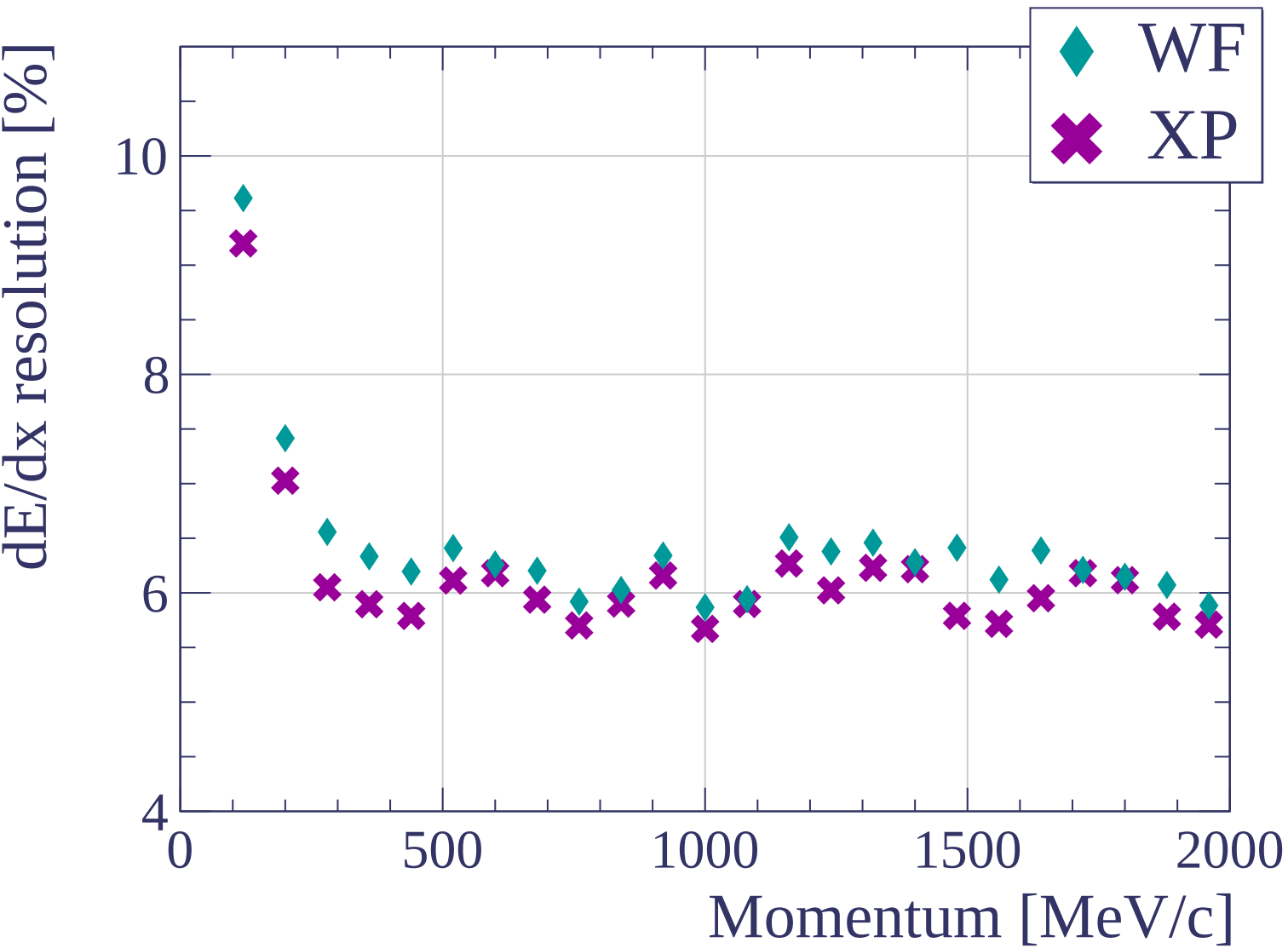


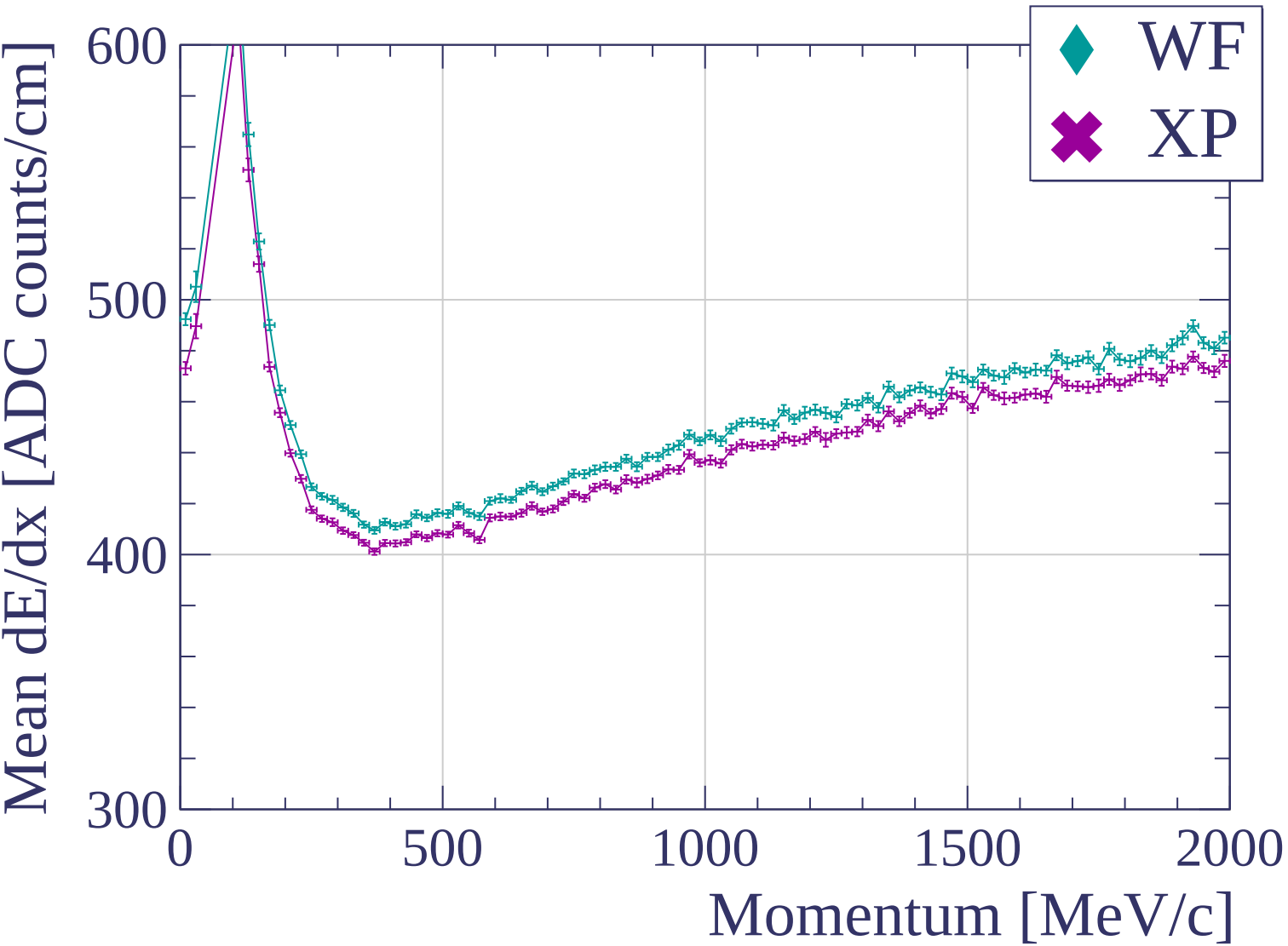


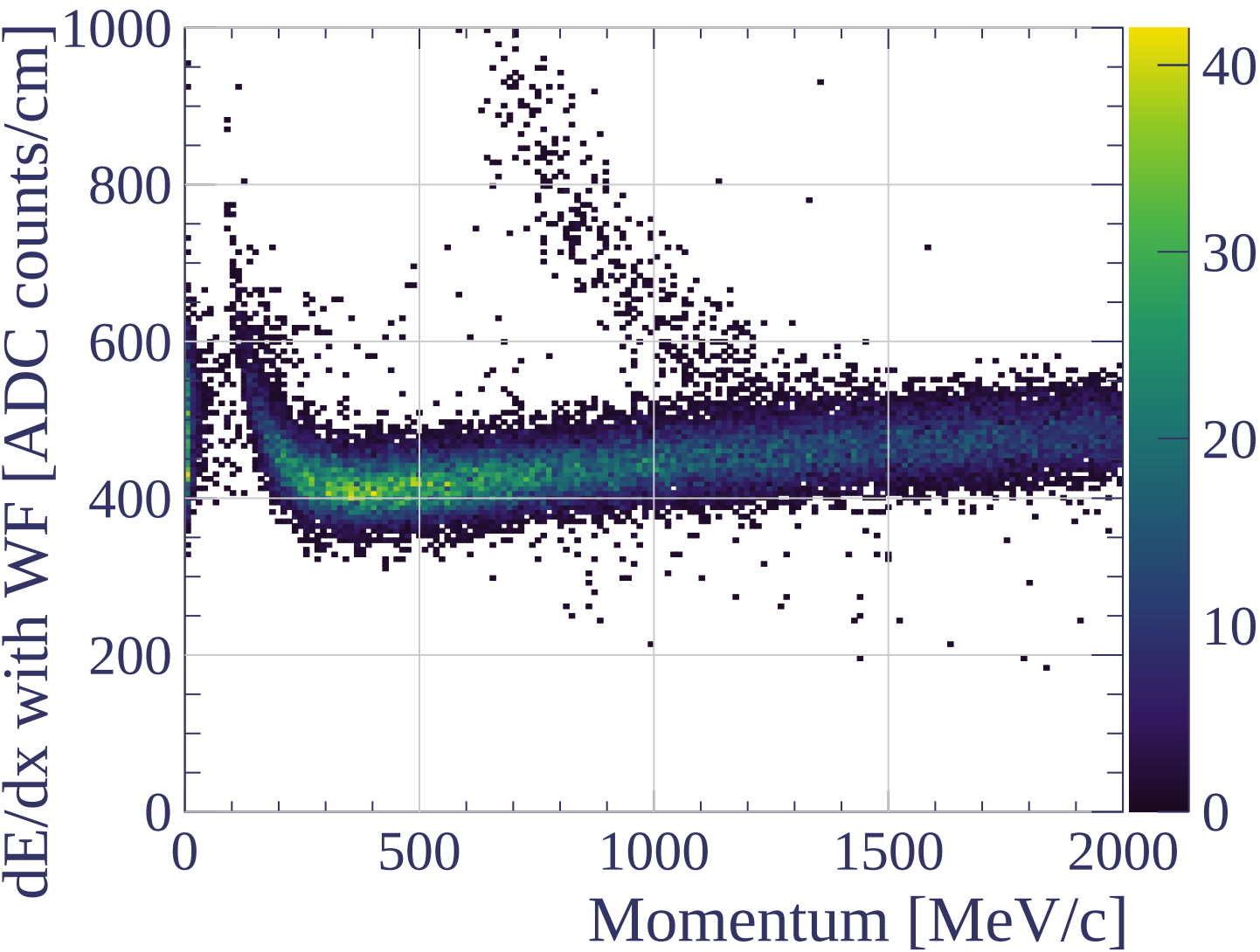


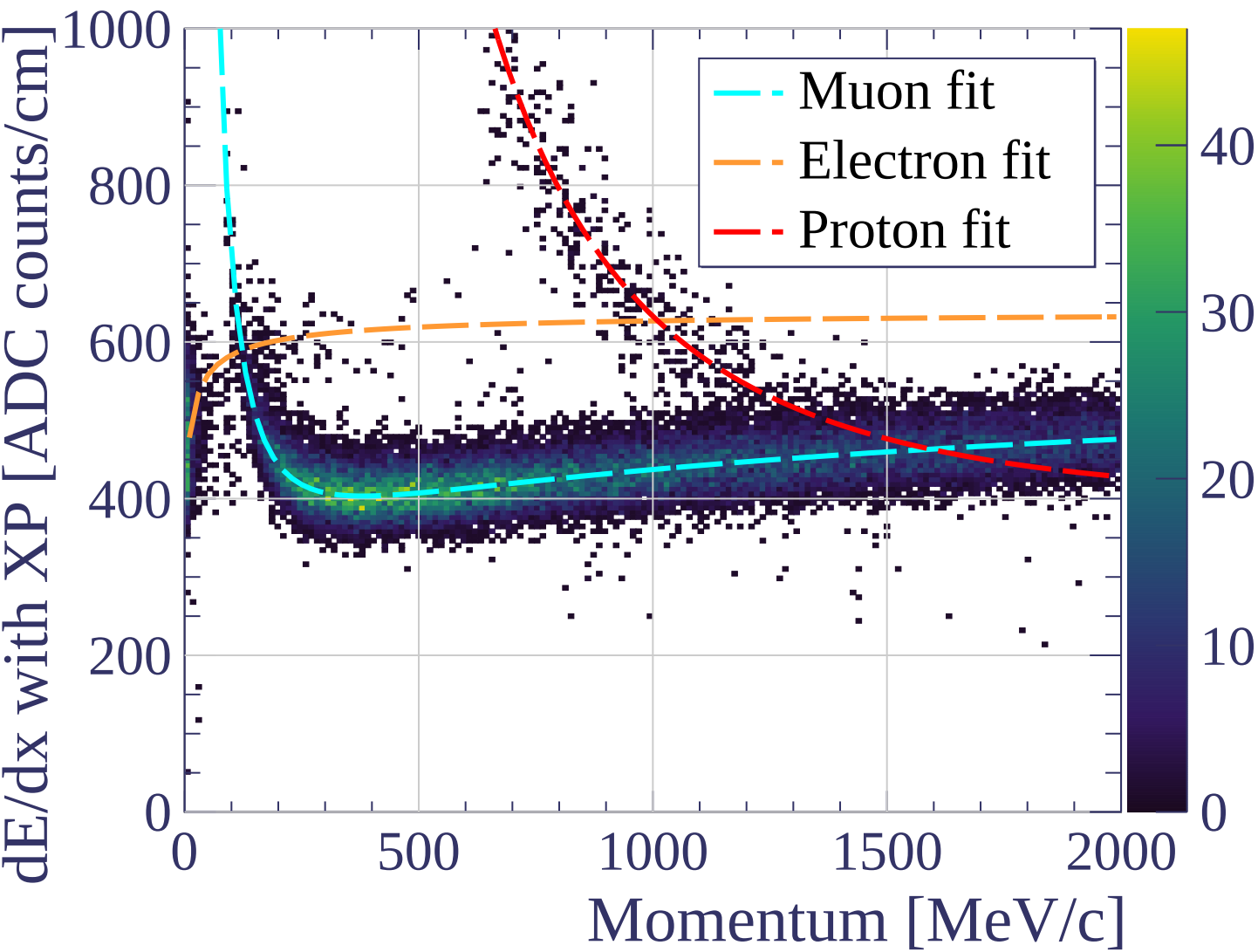




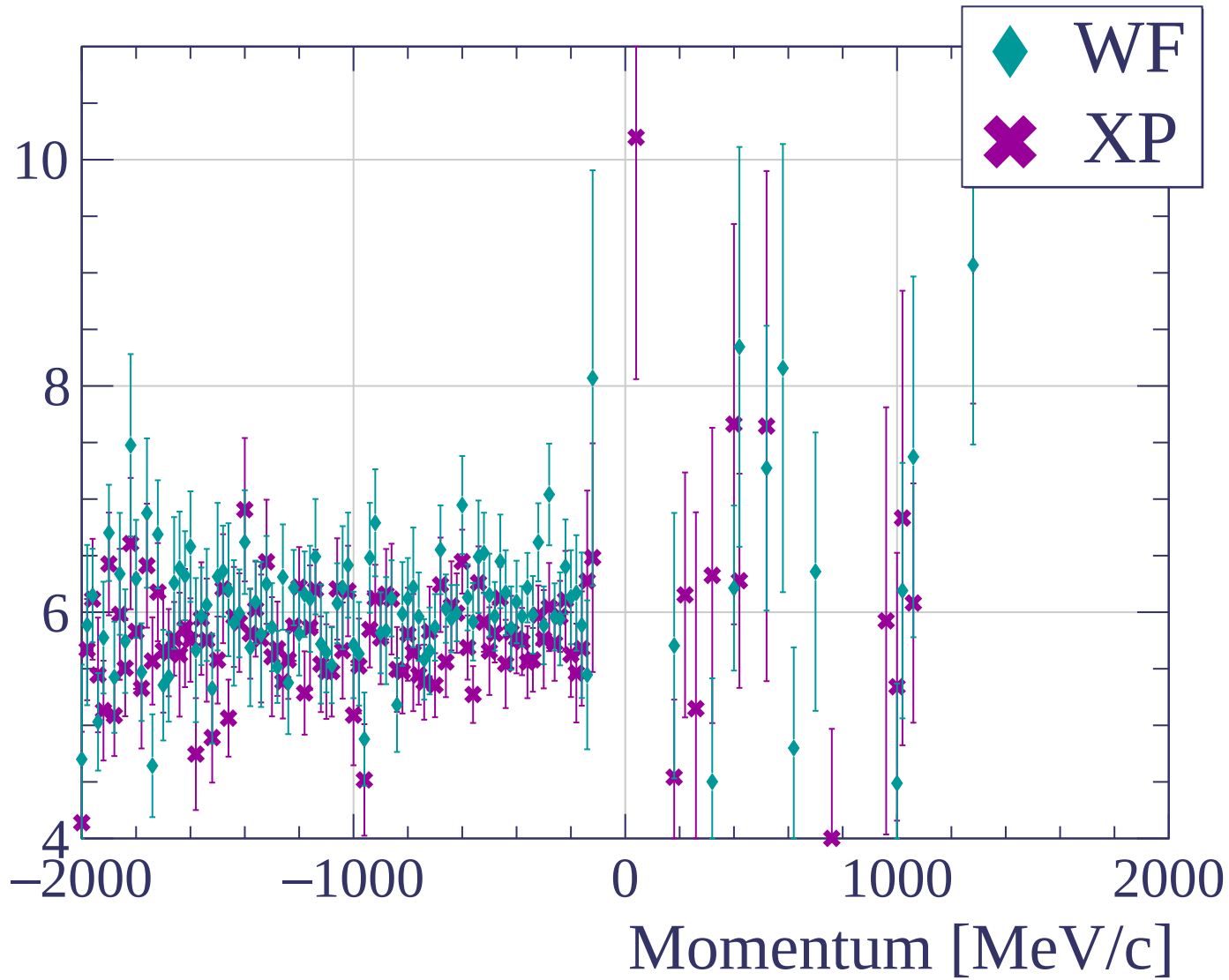


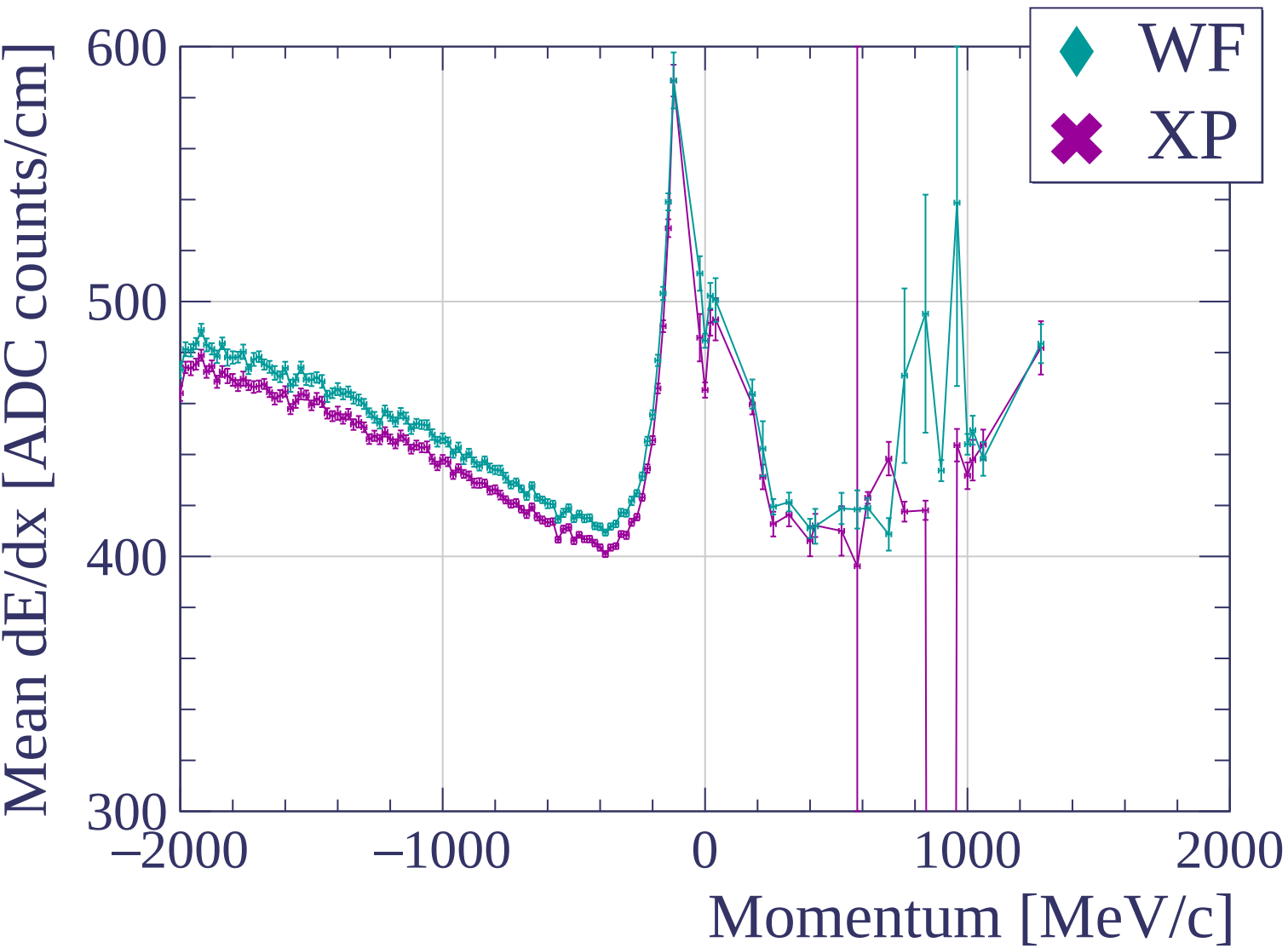


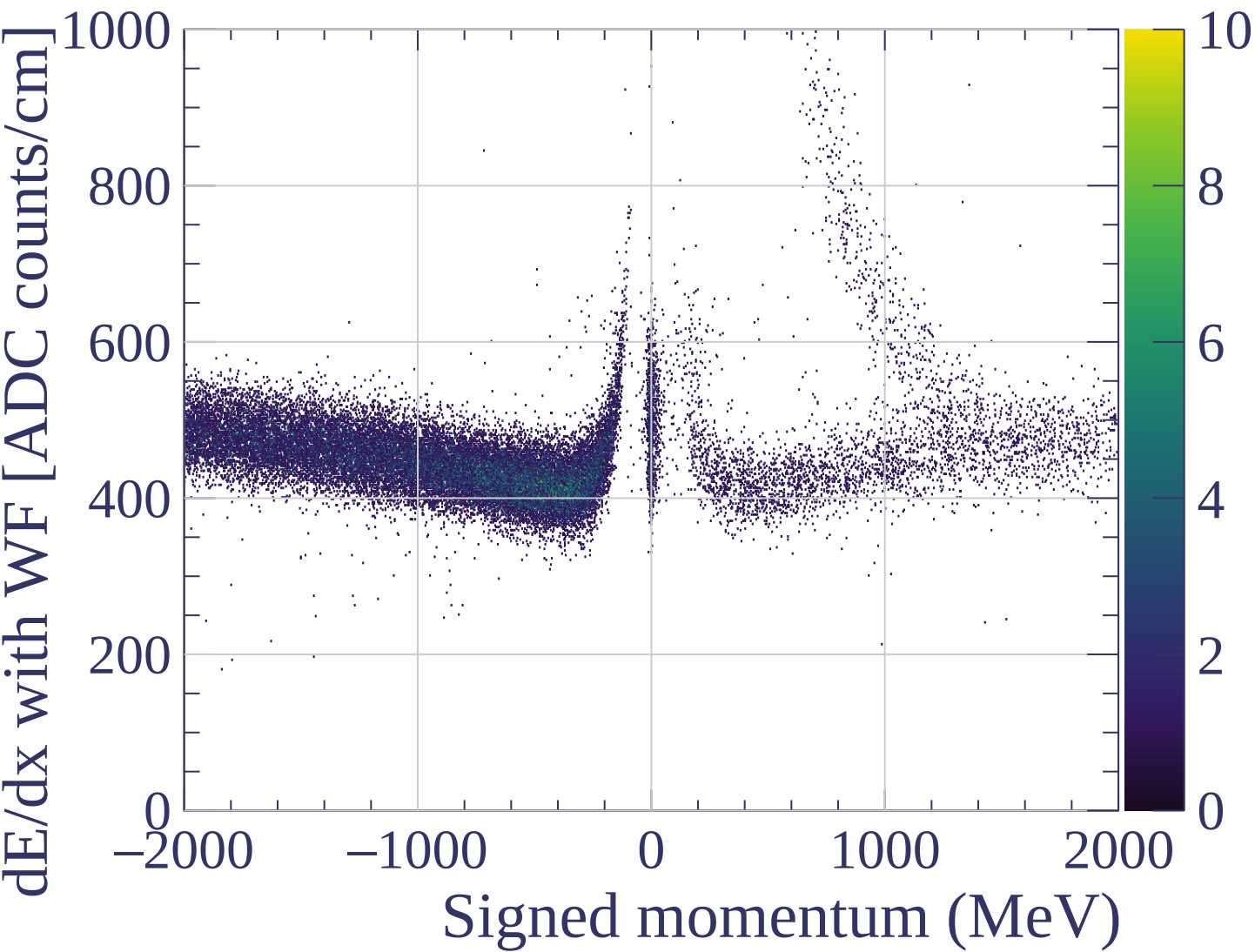


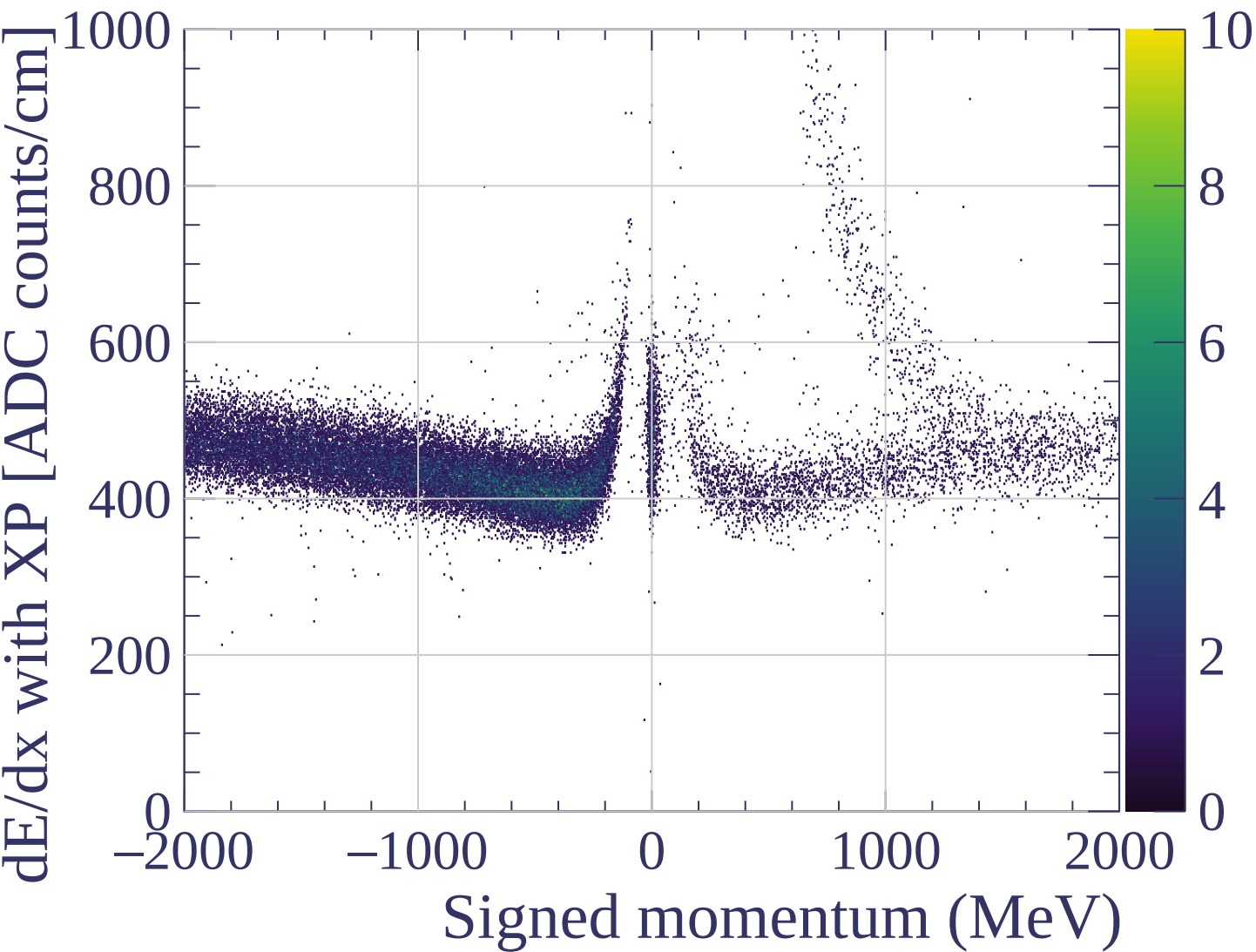


dE/dx resolution [%]









Count

Both HATs
tHAT
bHAT

200

100

0
-2000

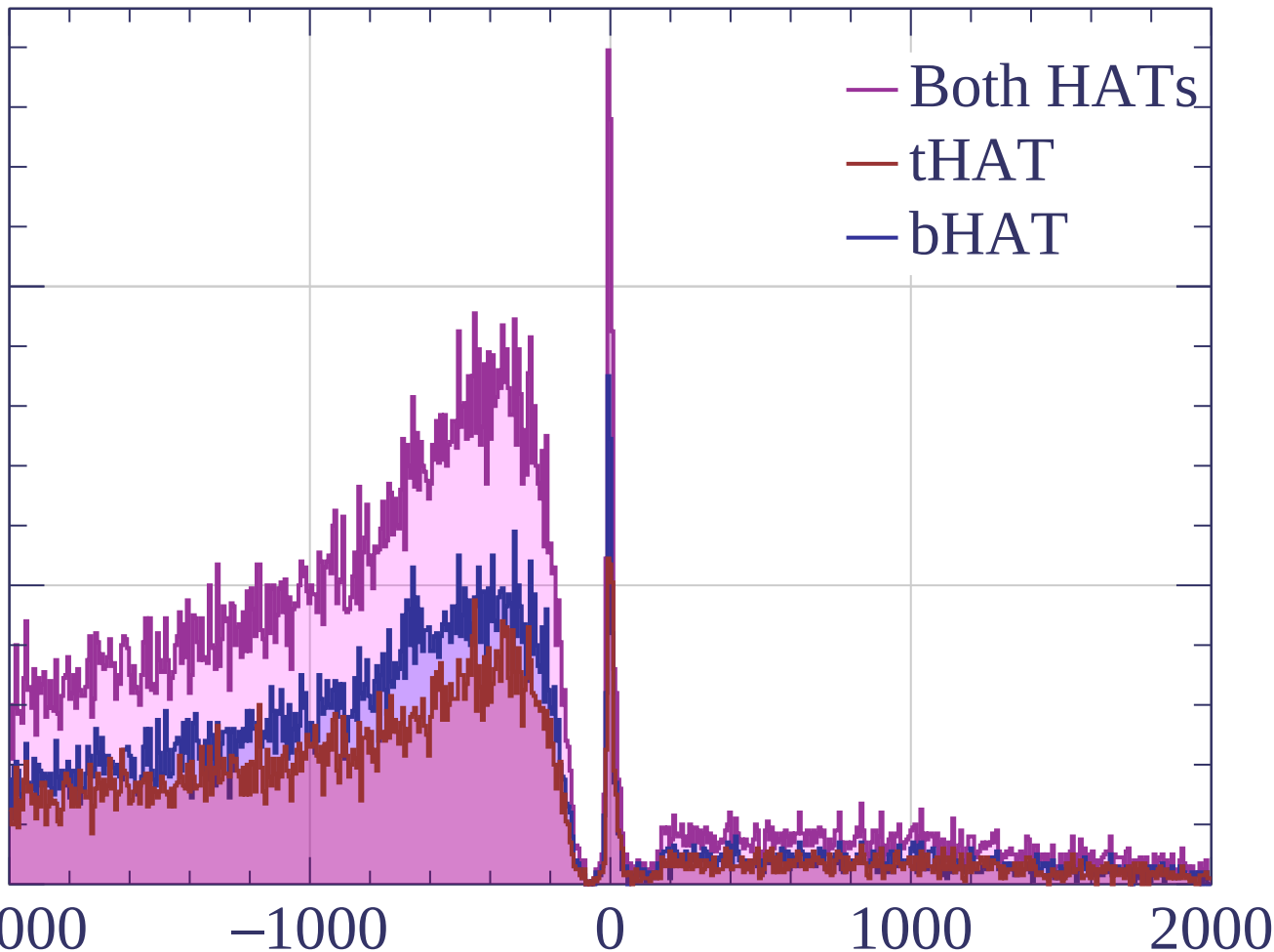
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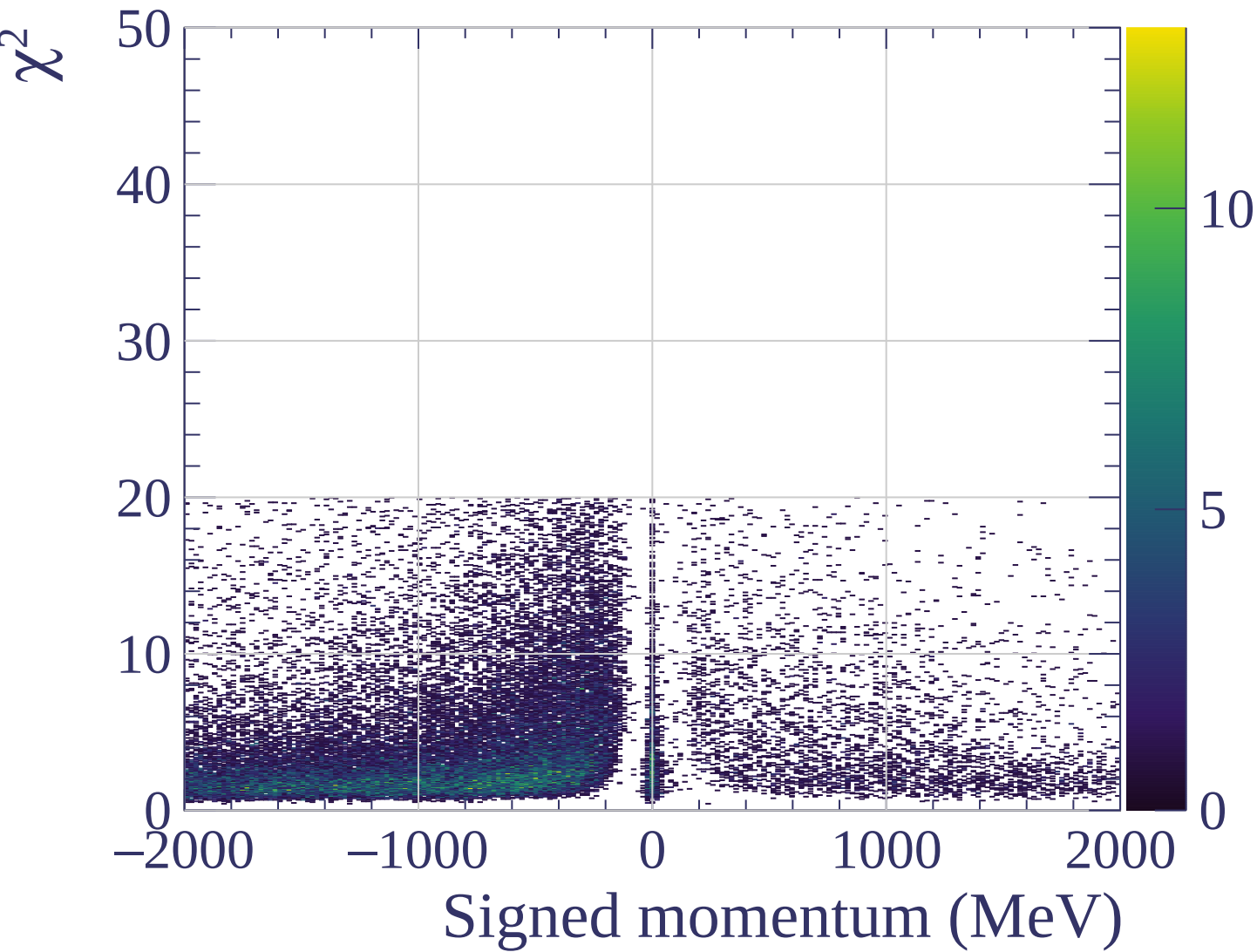
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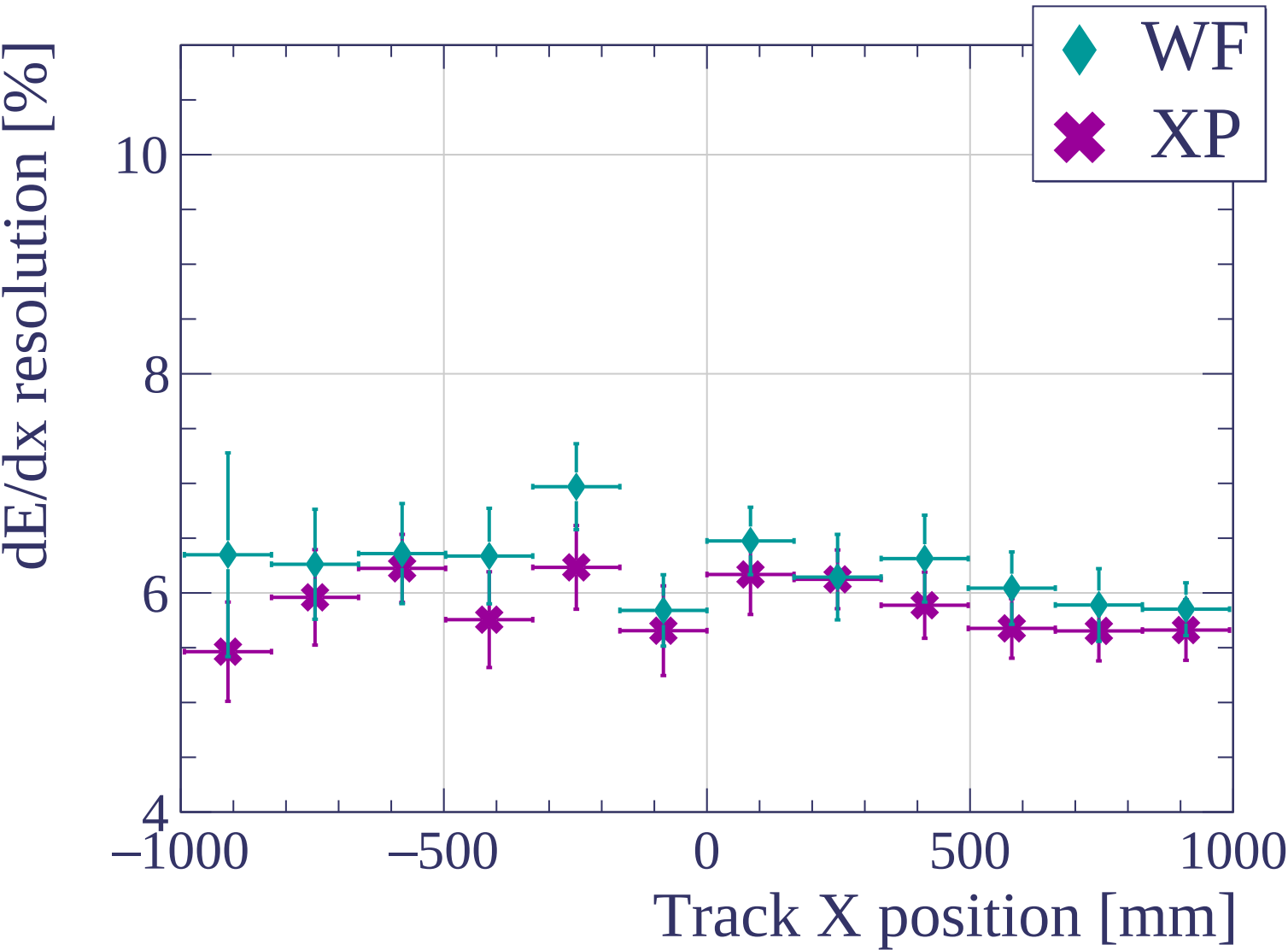
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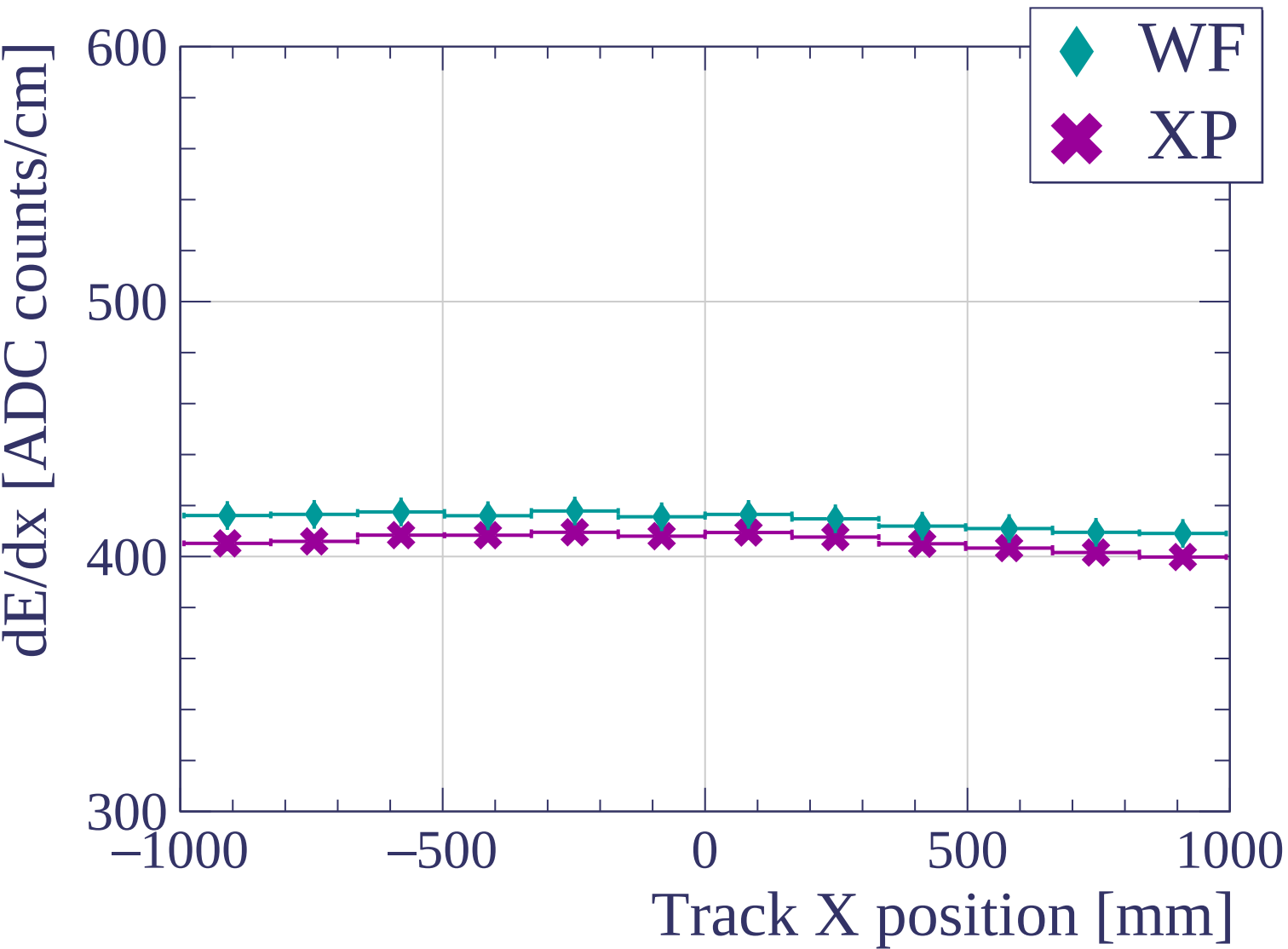
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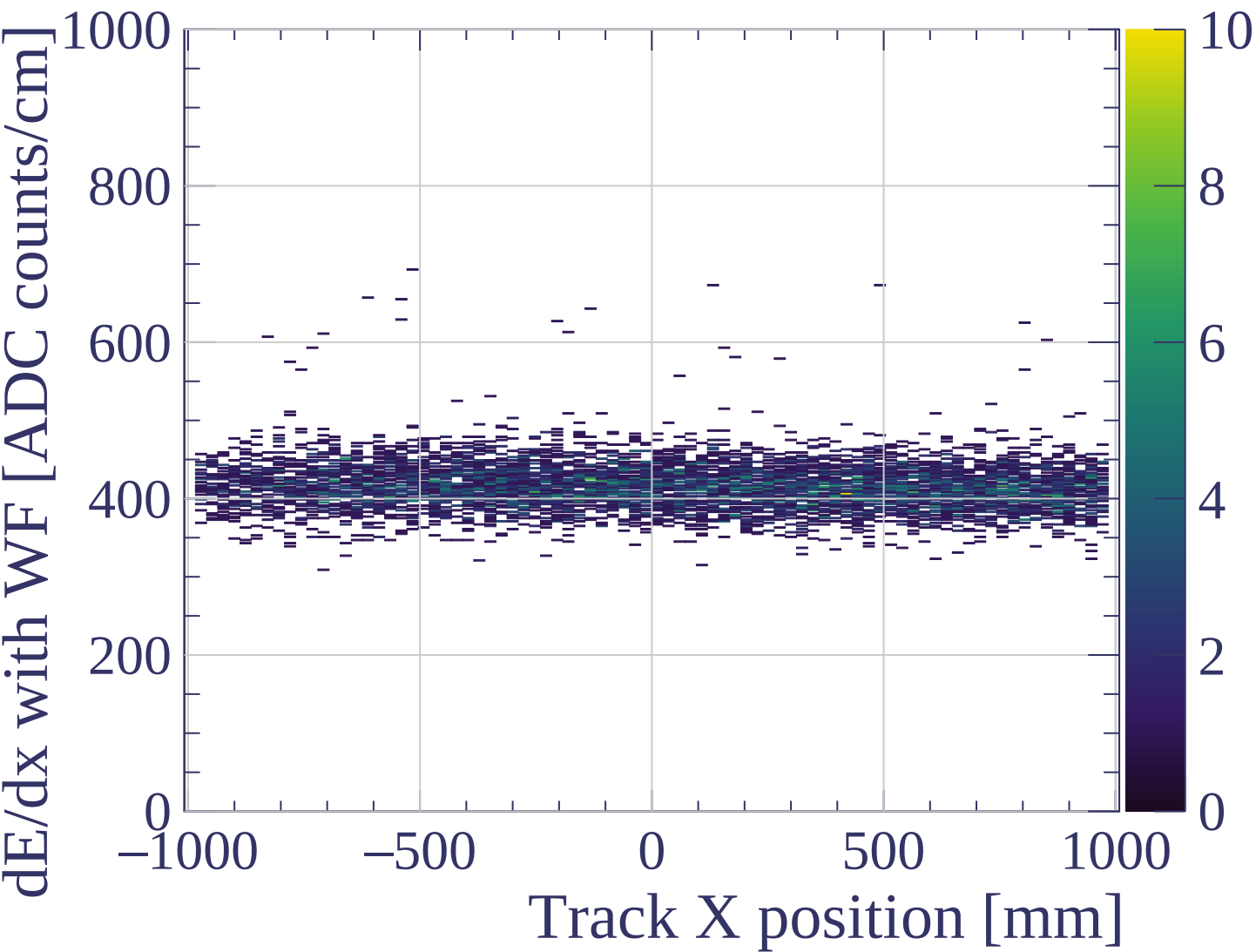
Signed momentum (MeV)

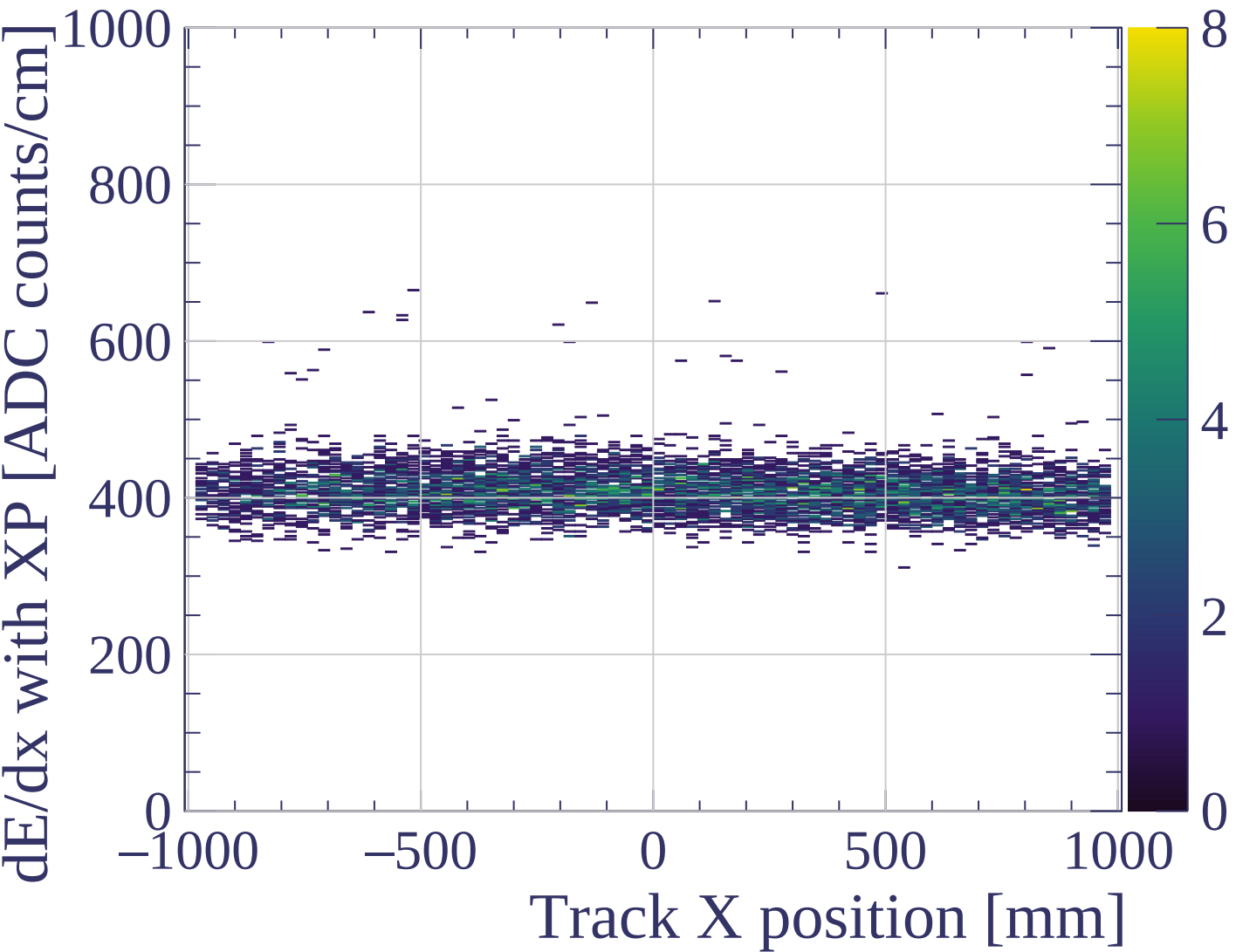




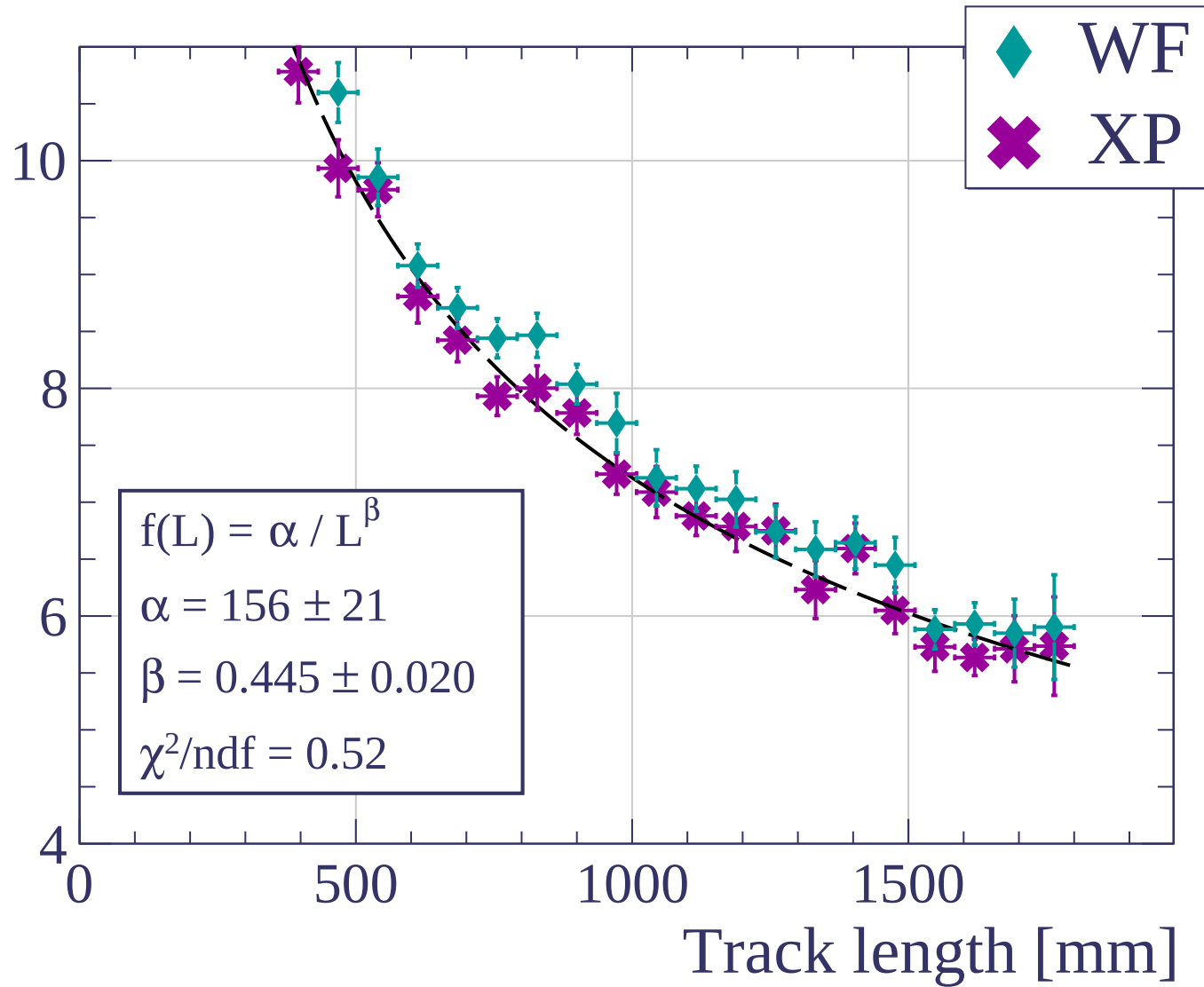


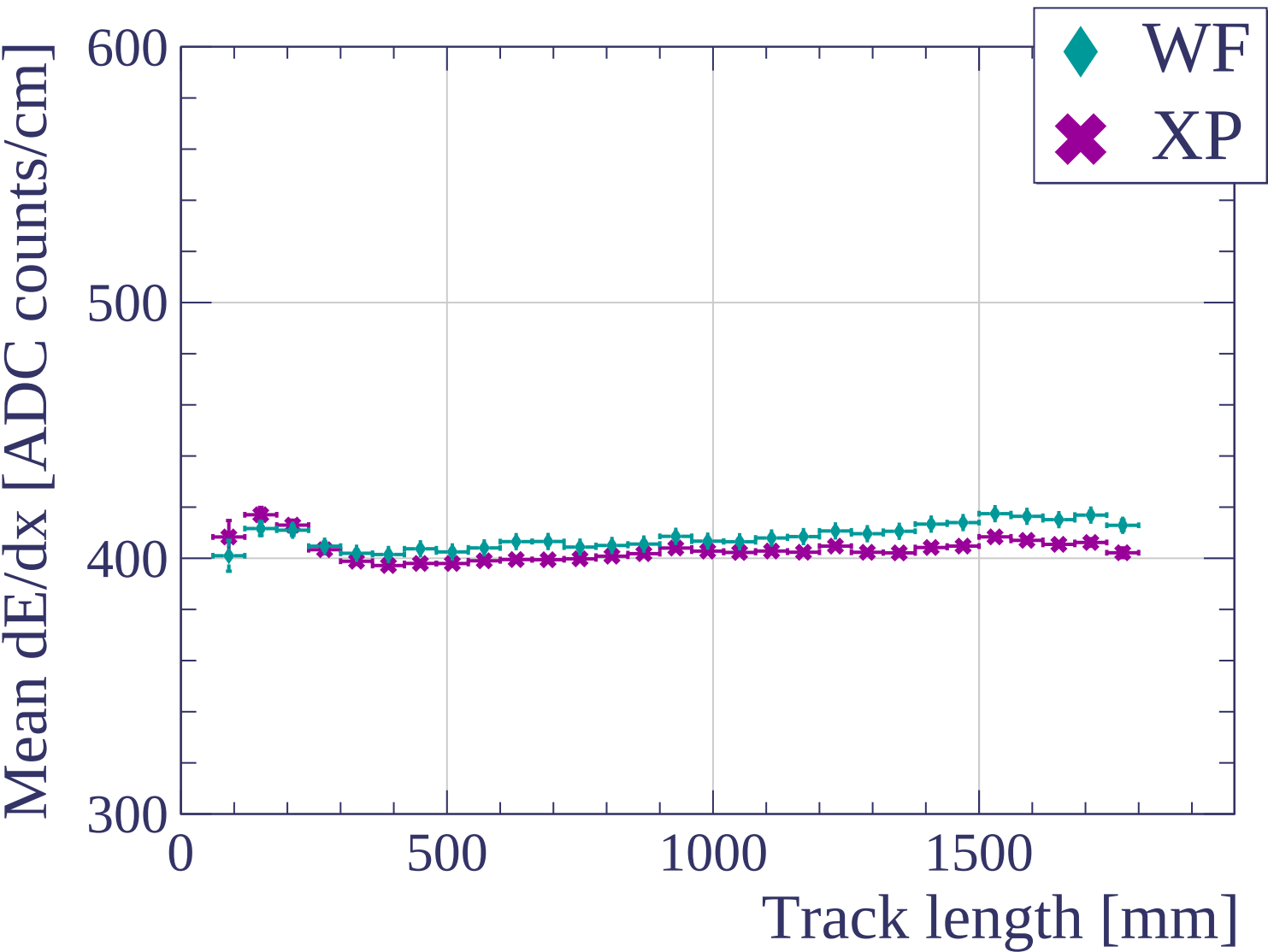


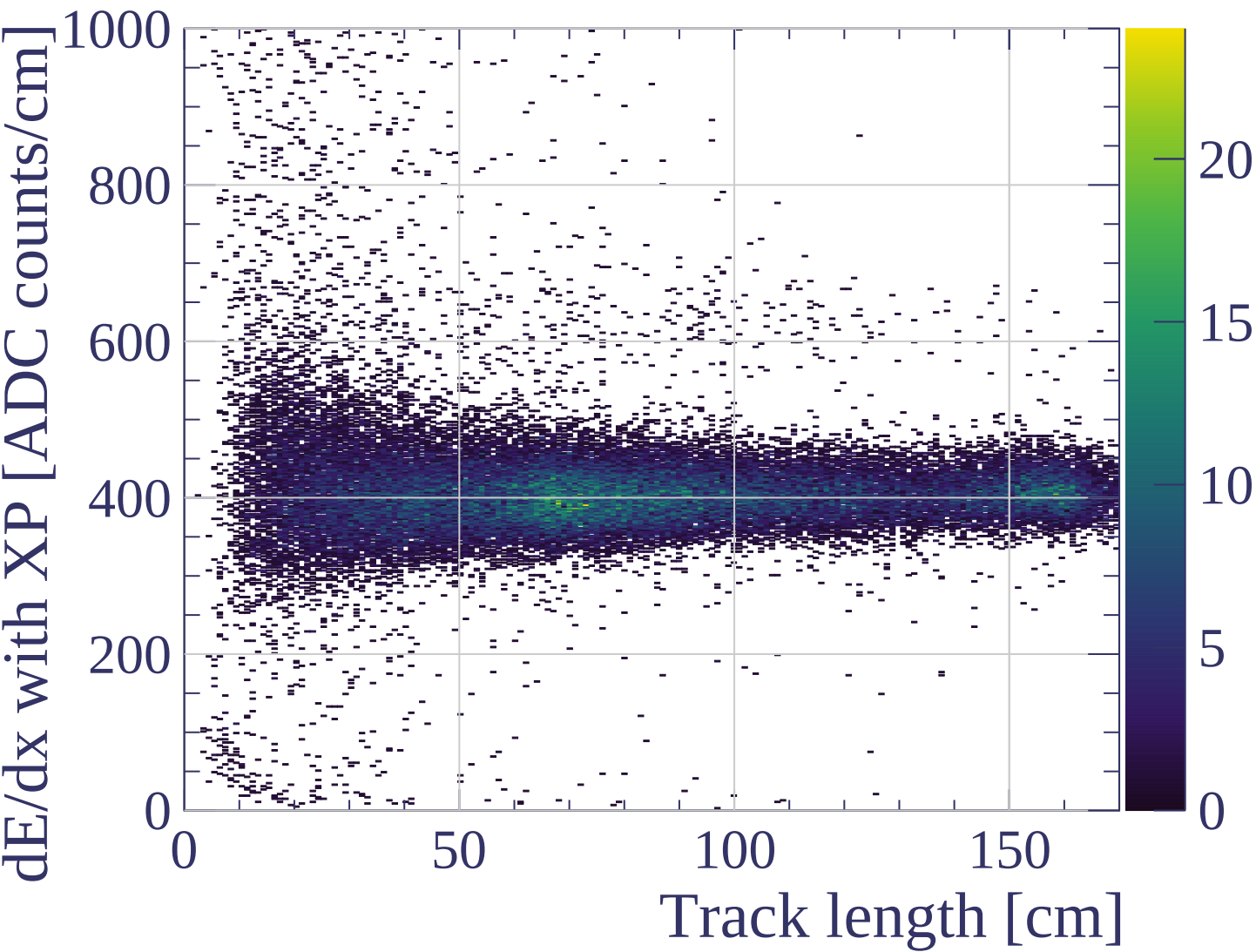


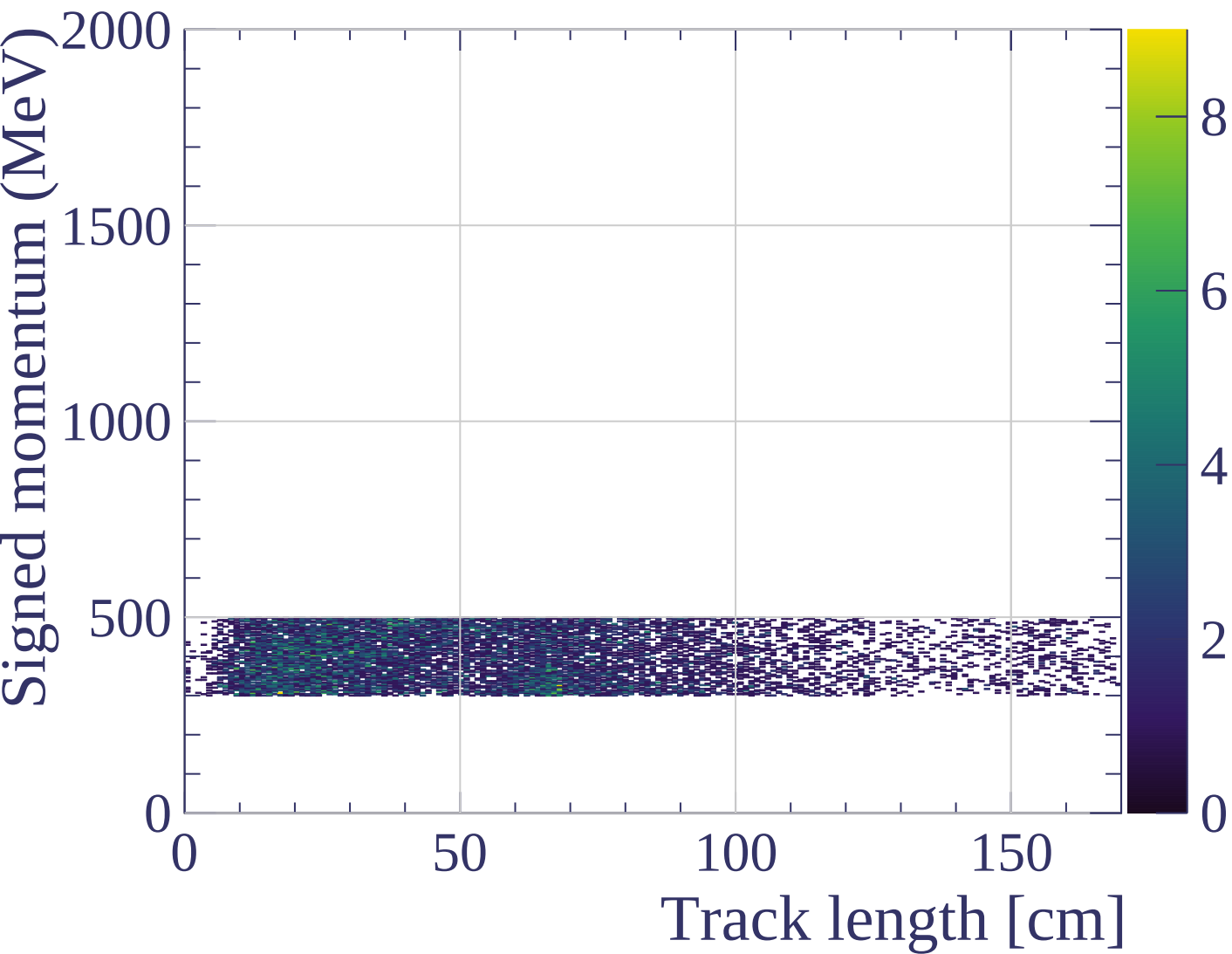


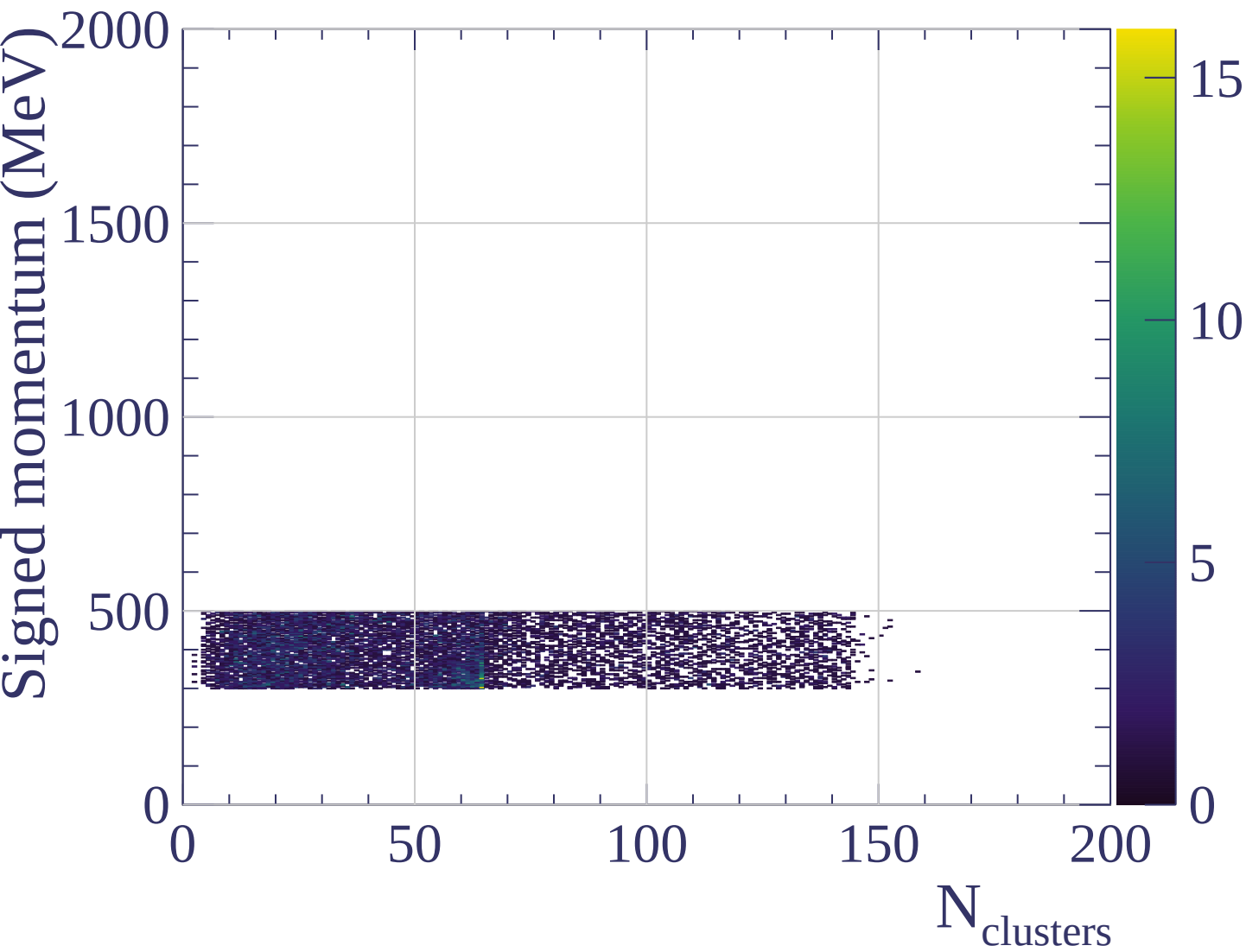
dE/dx resolution [%]



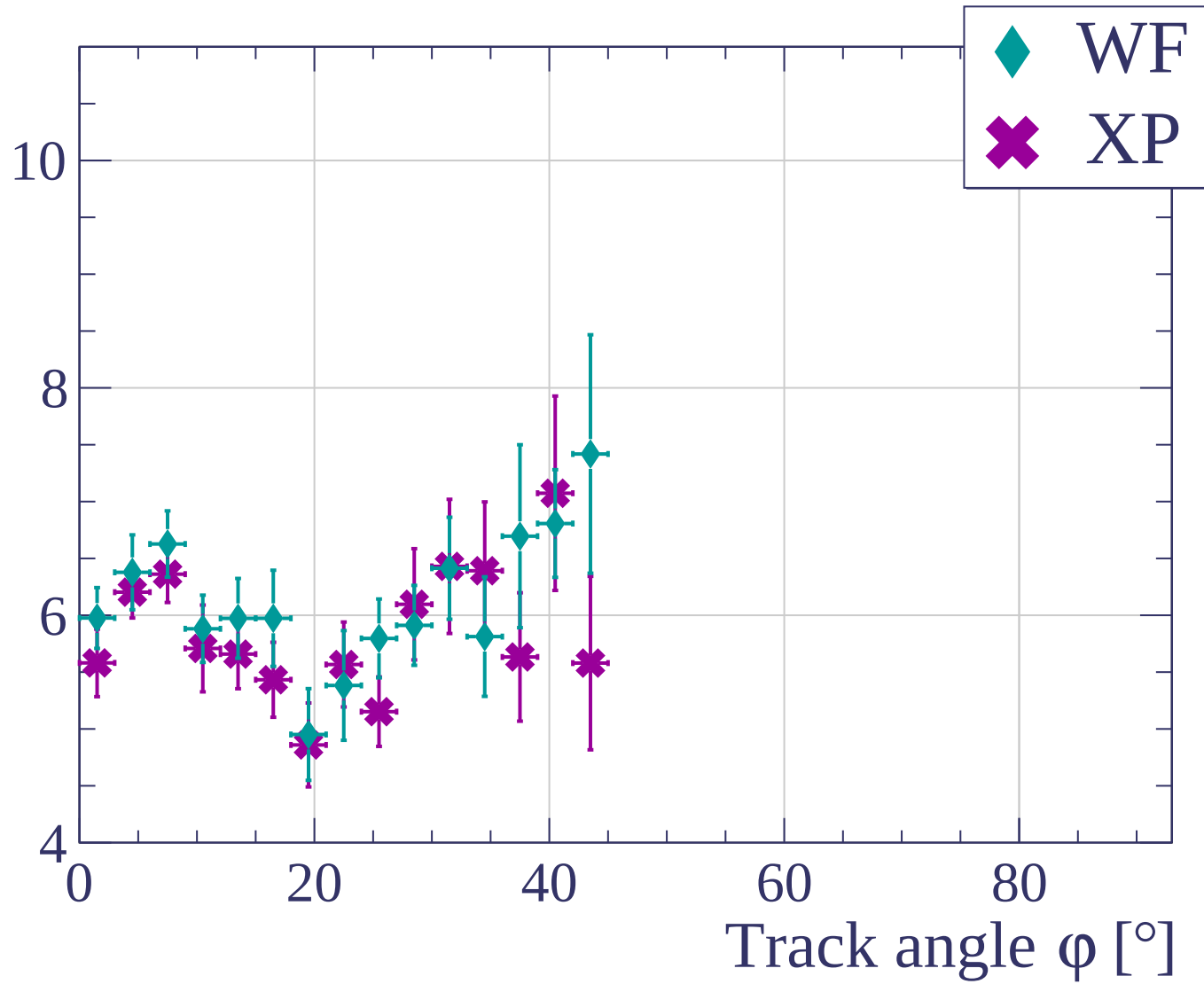


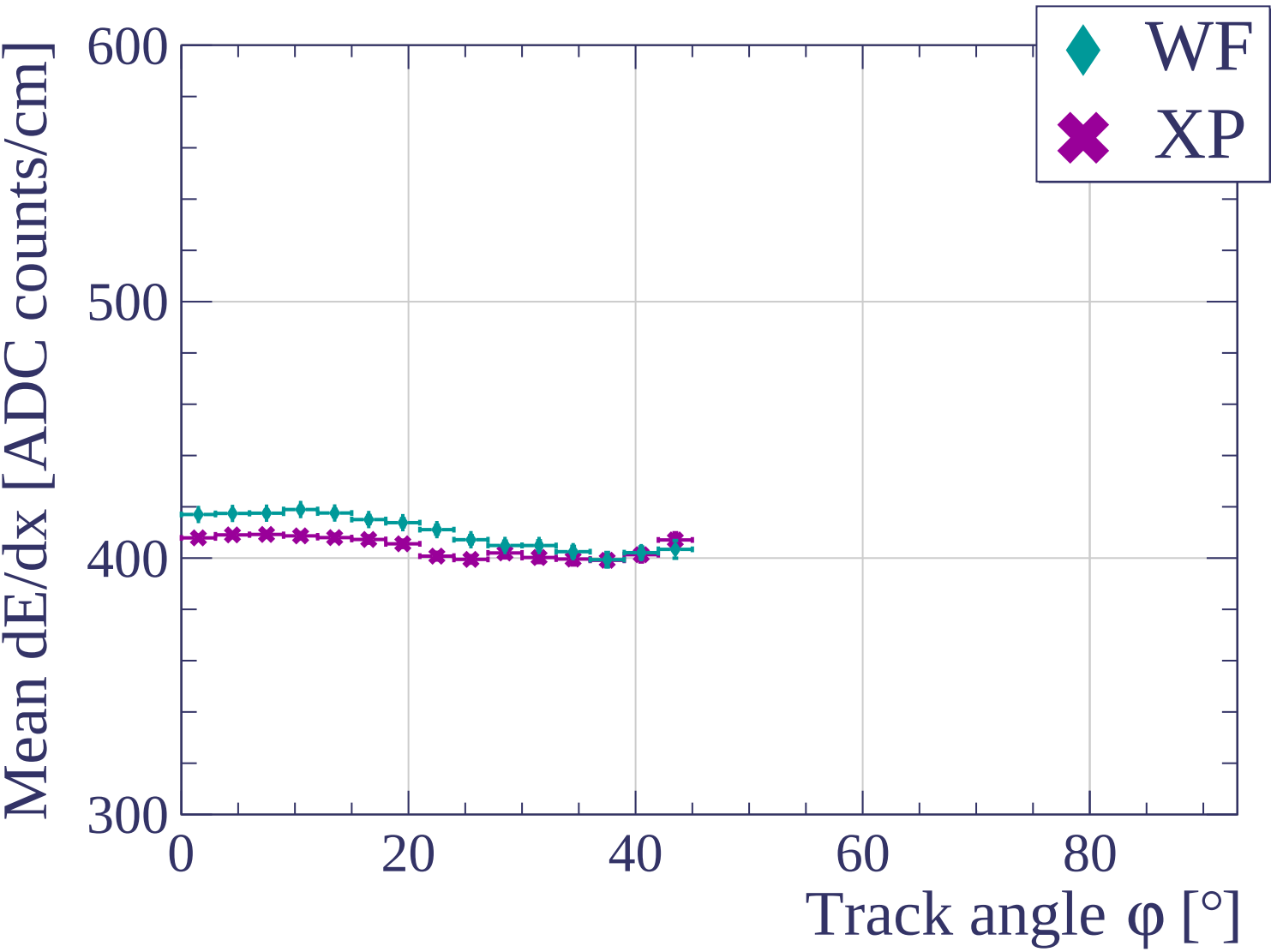






dE/dx resolution [%]





dE/dx resolution [%]

10

8

6

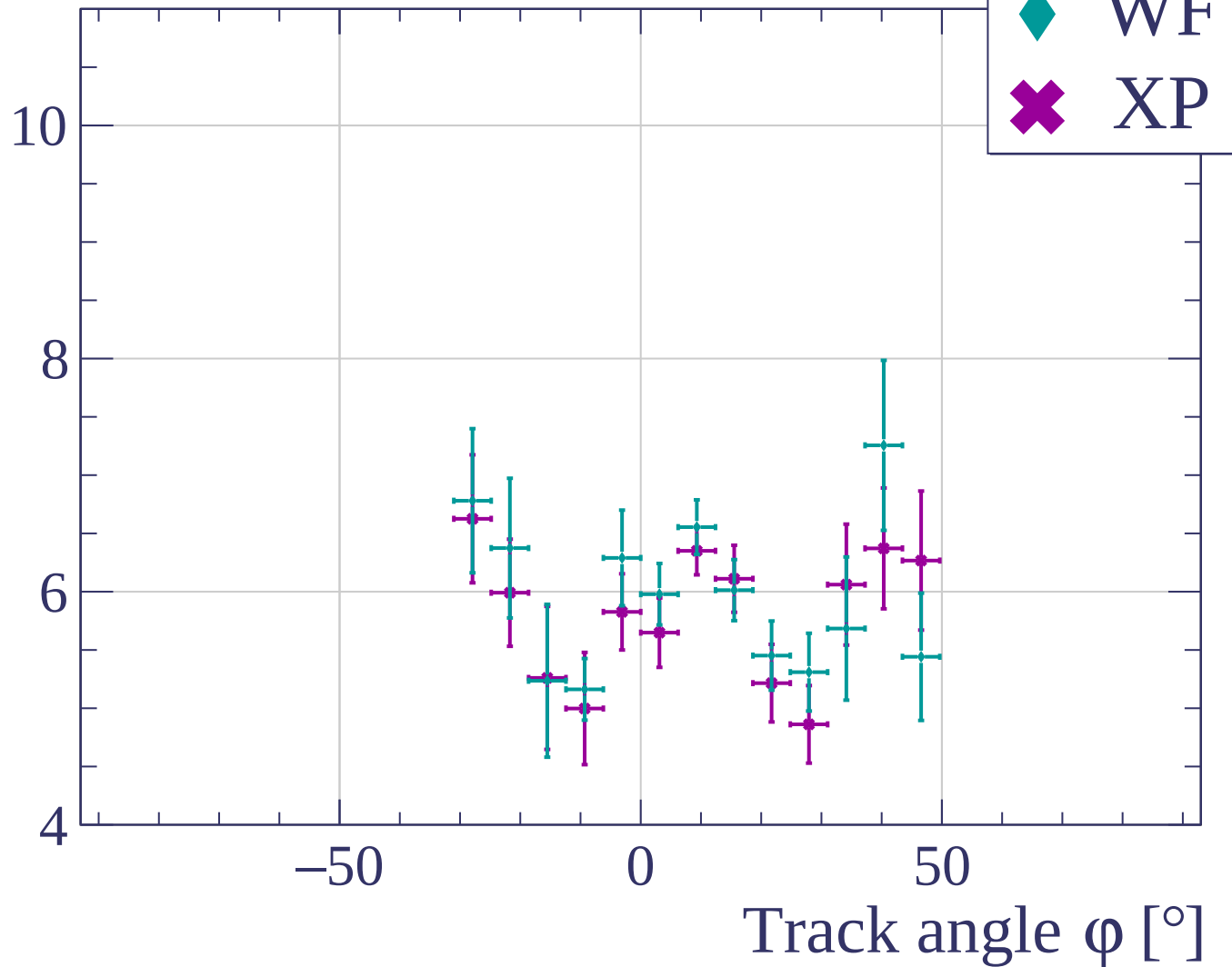
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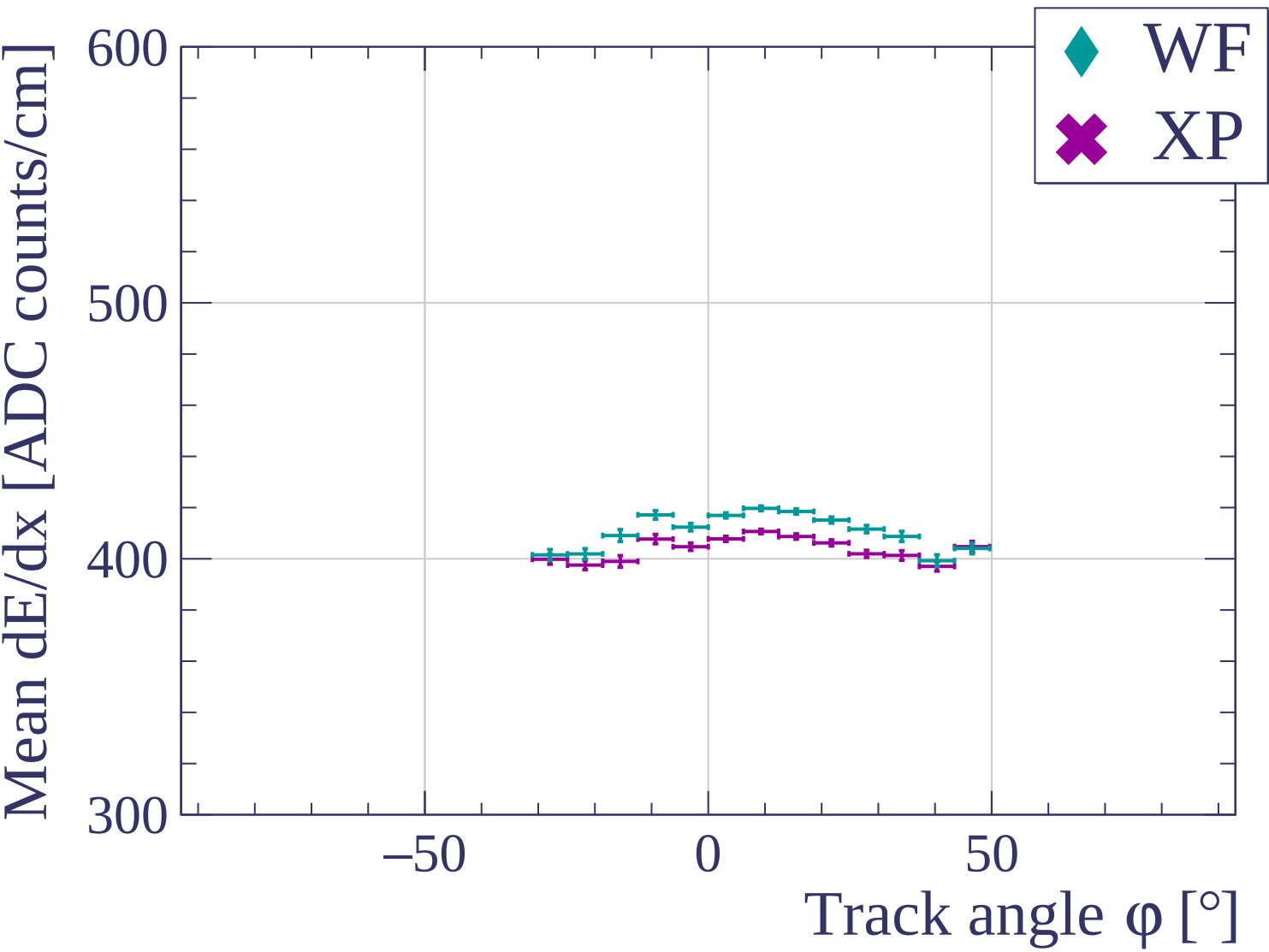
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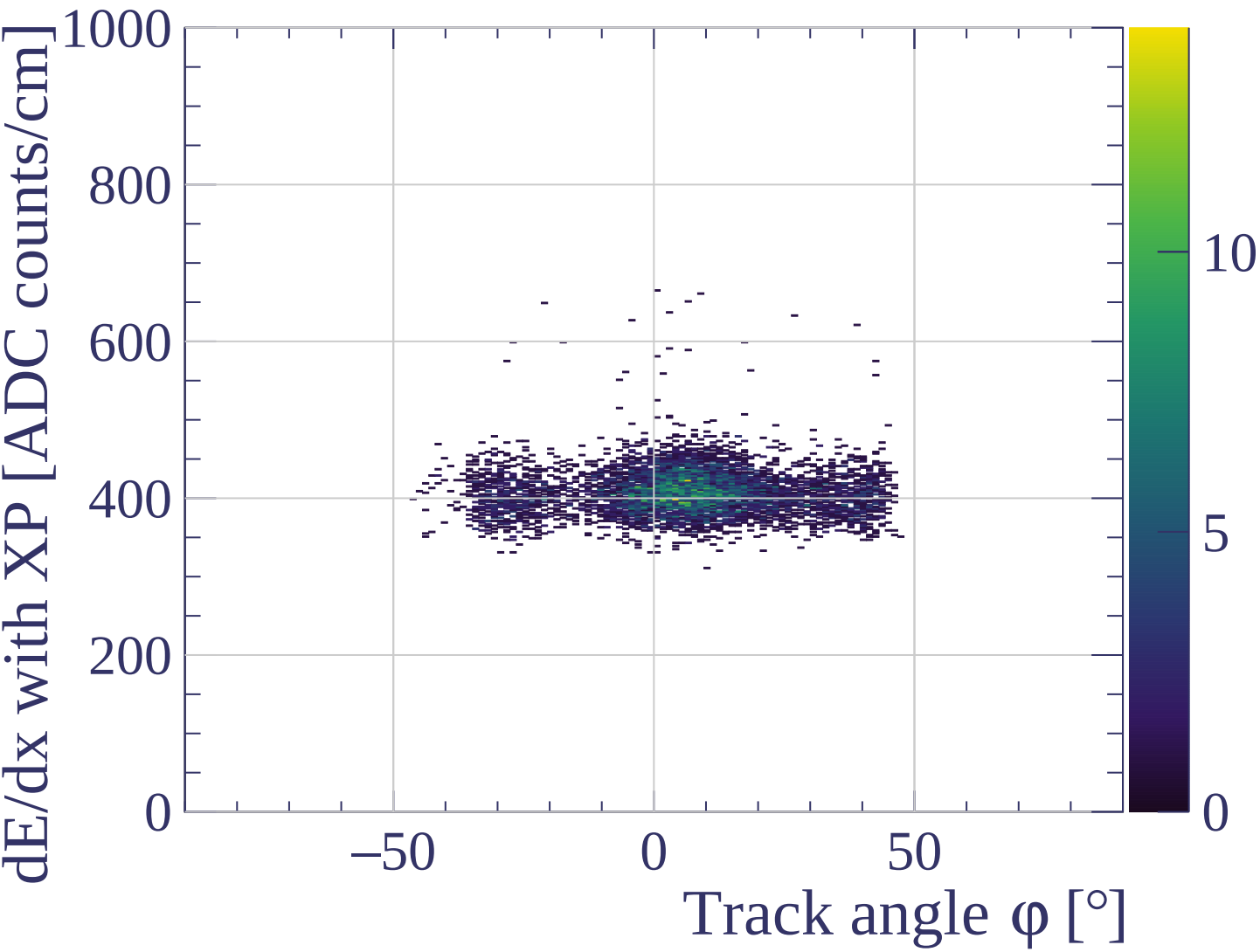
0

50

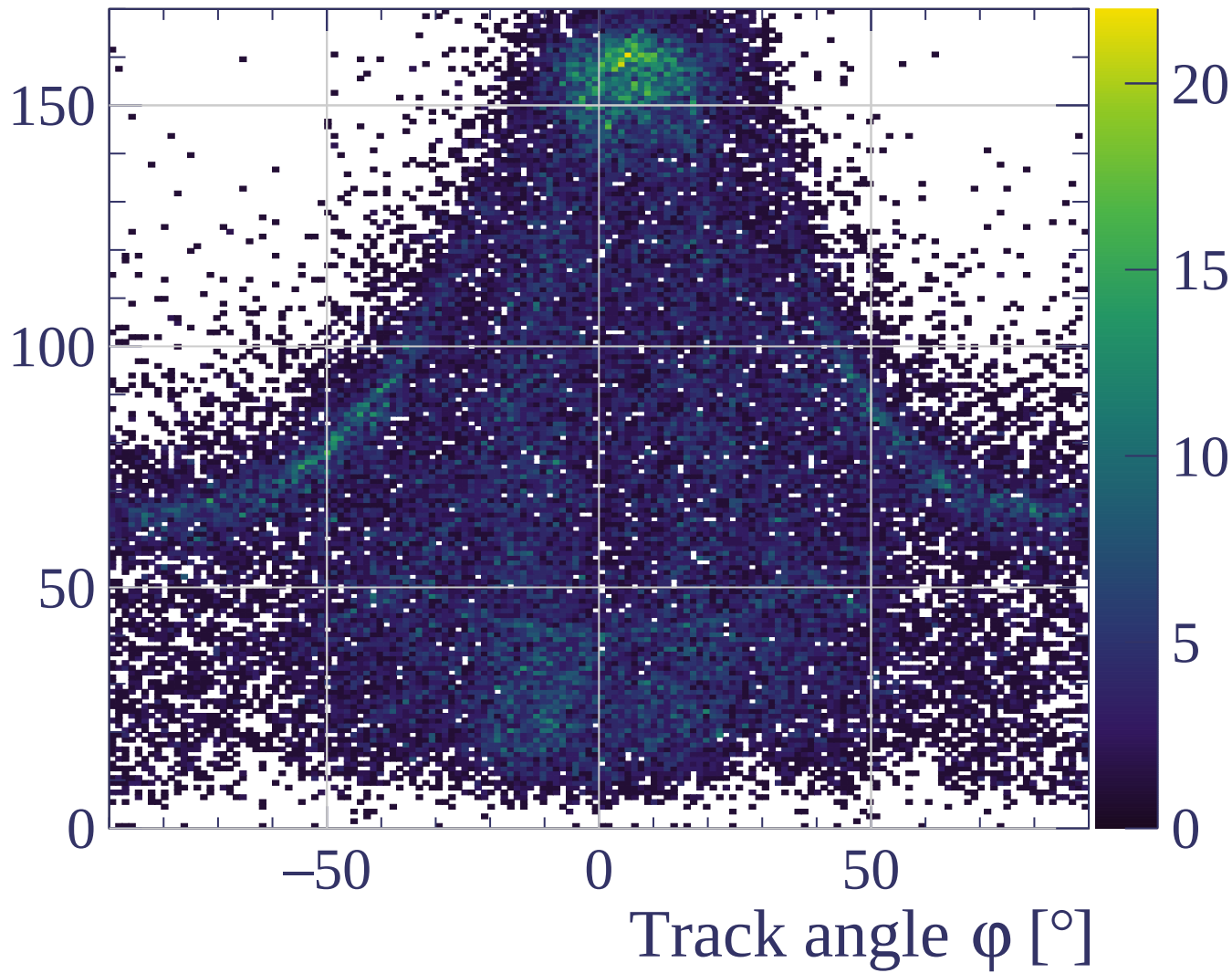
Track angle φ [°]

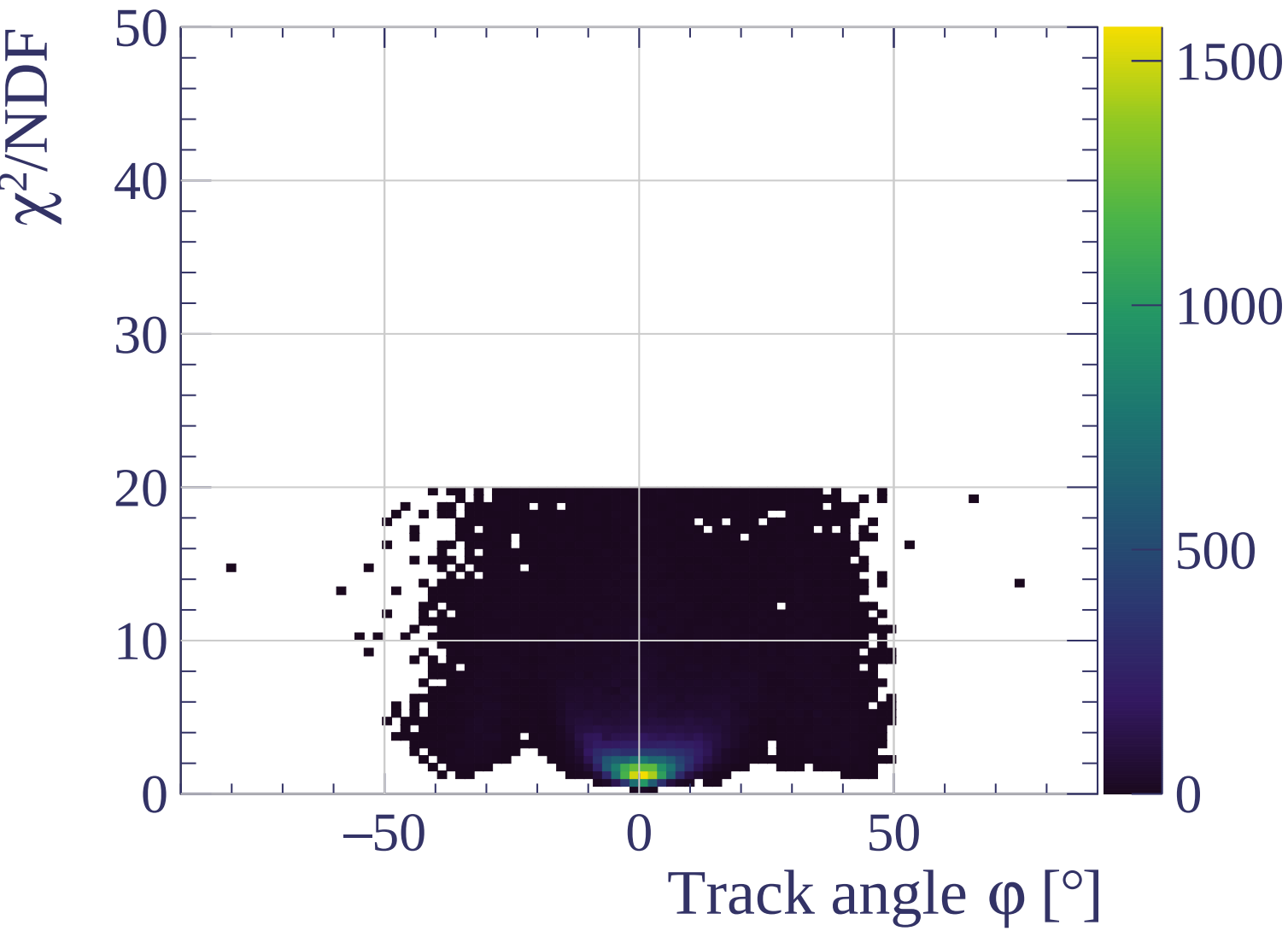


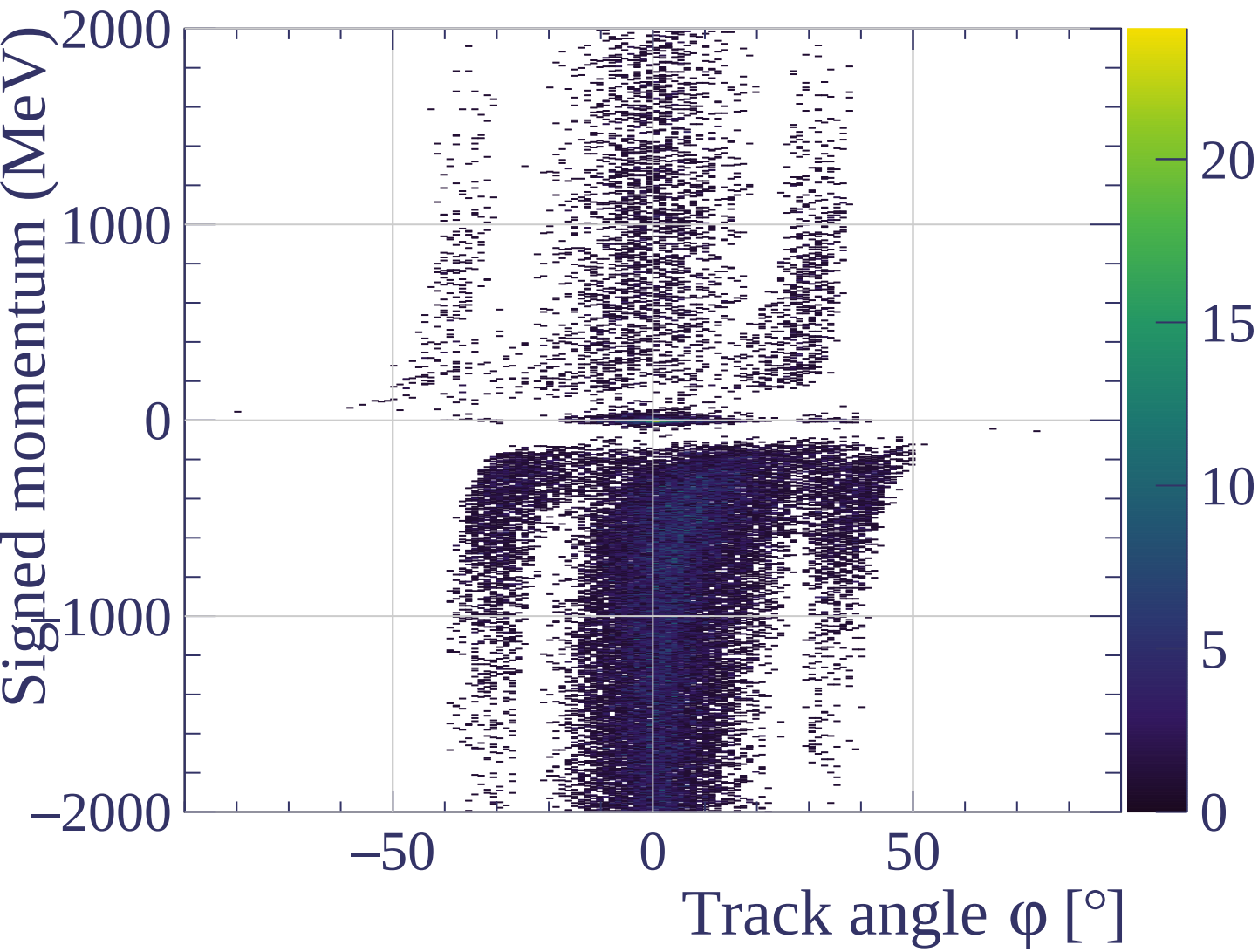




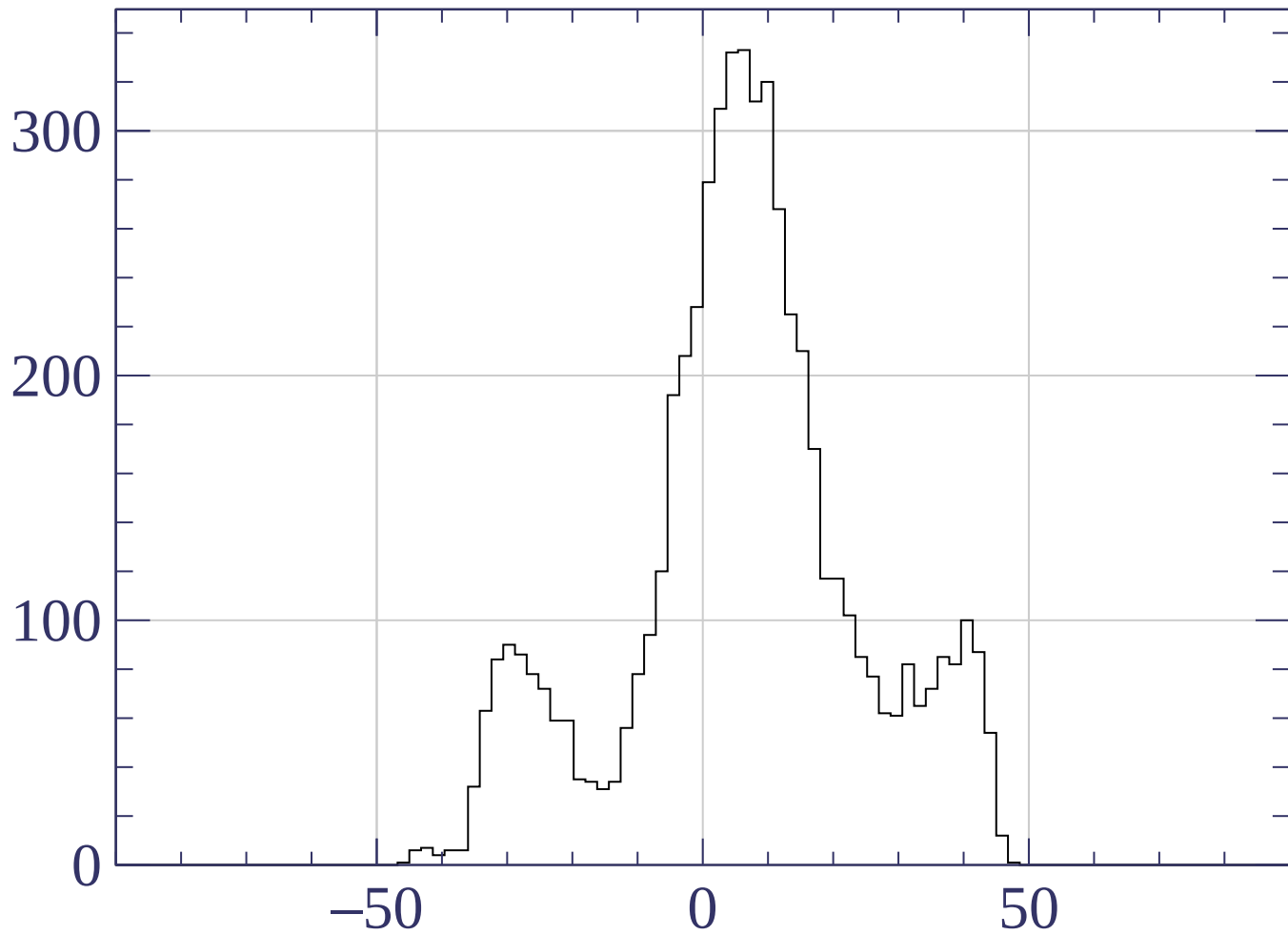
Track length [cm]



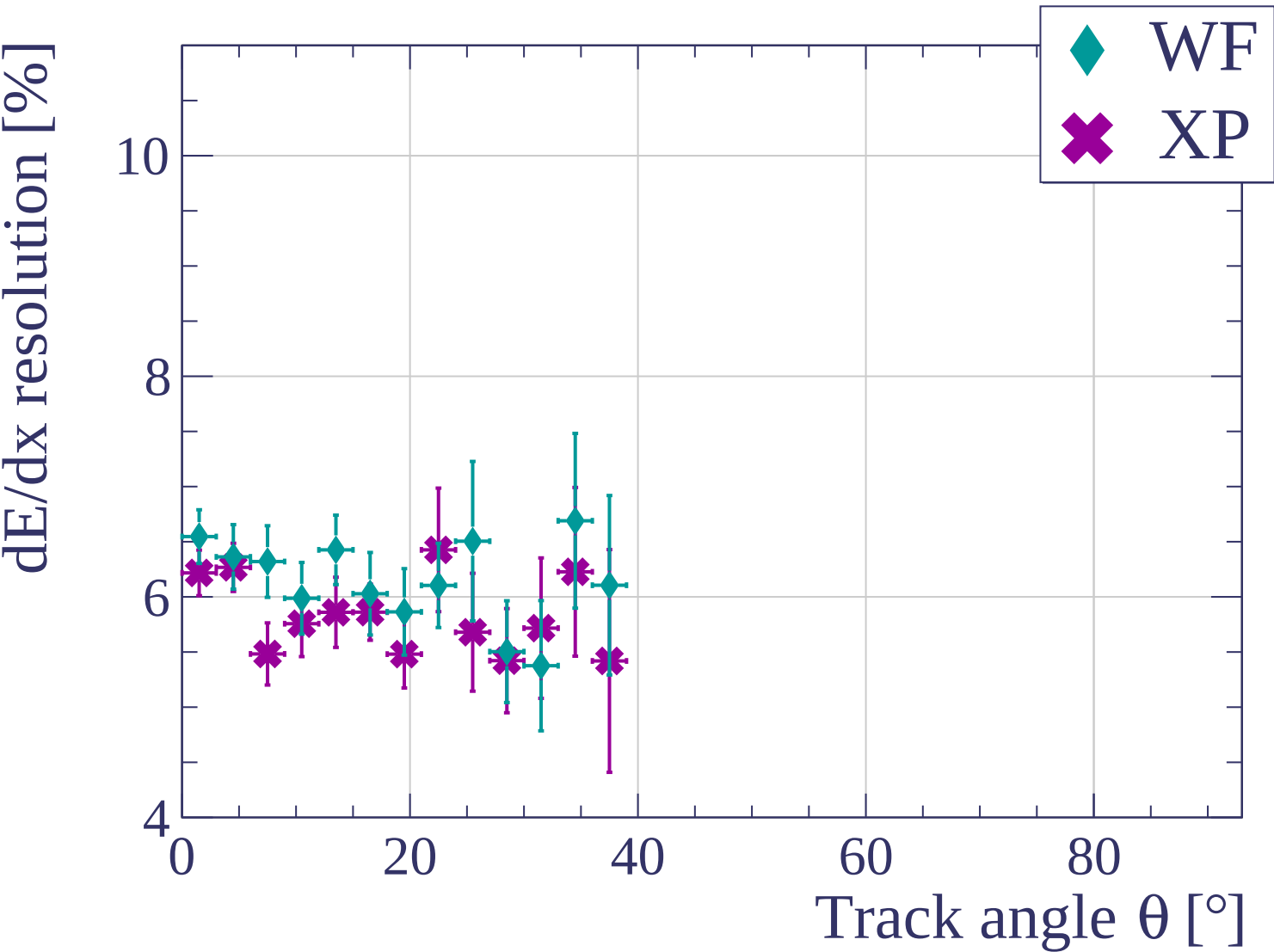


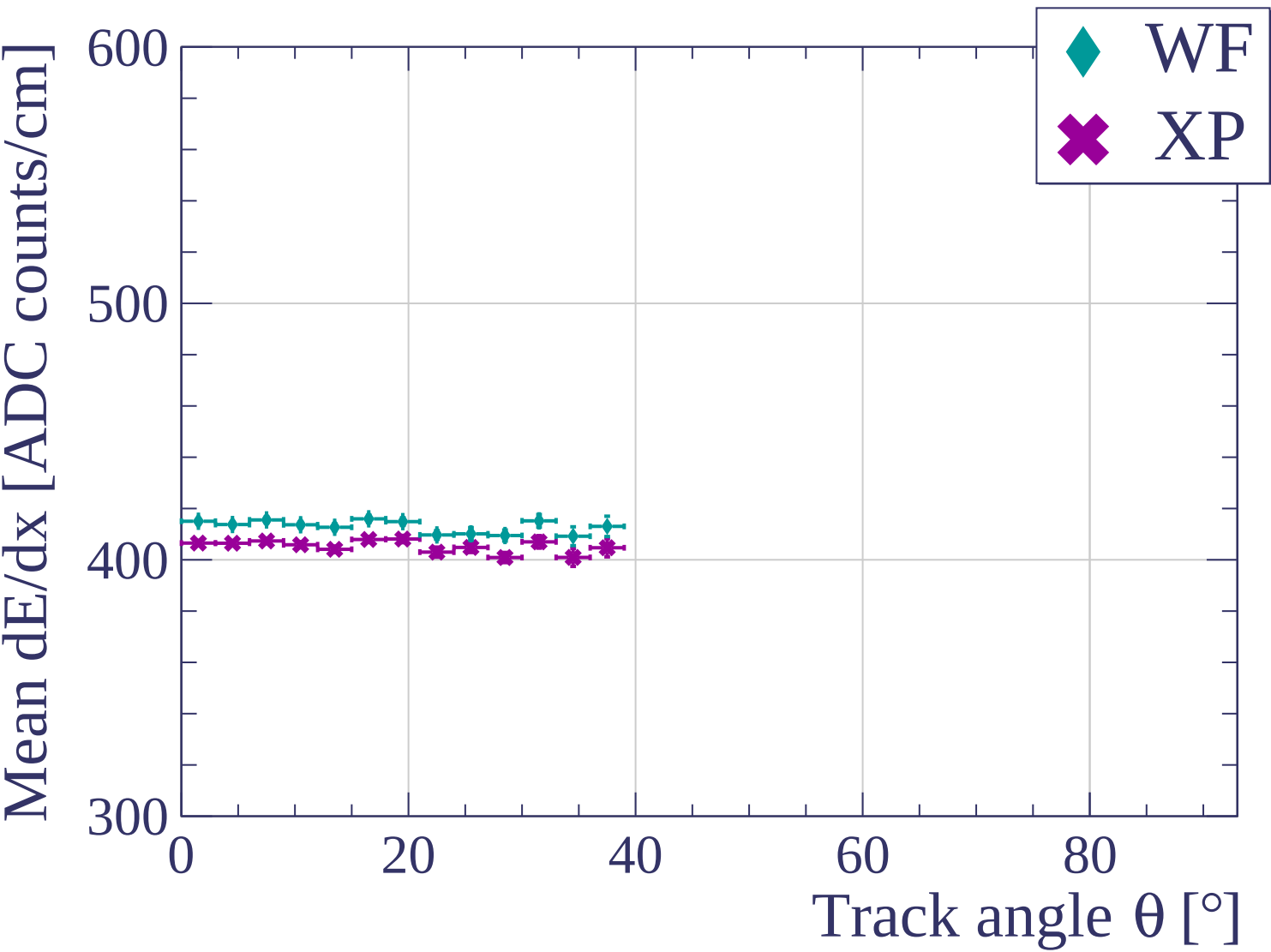


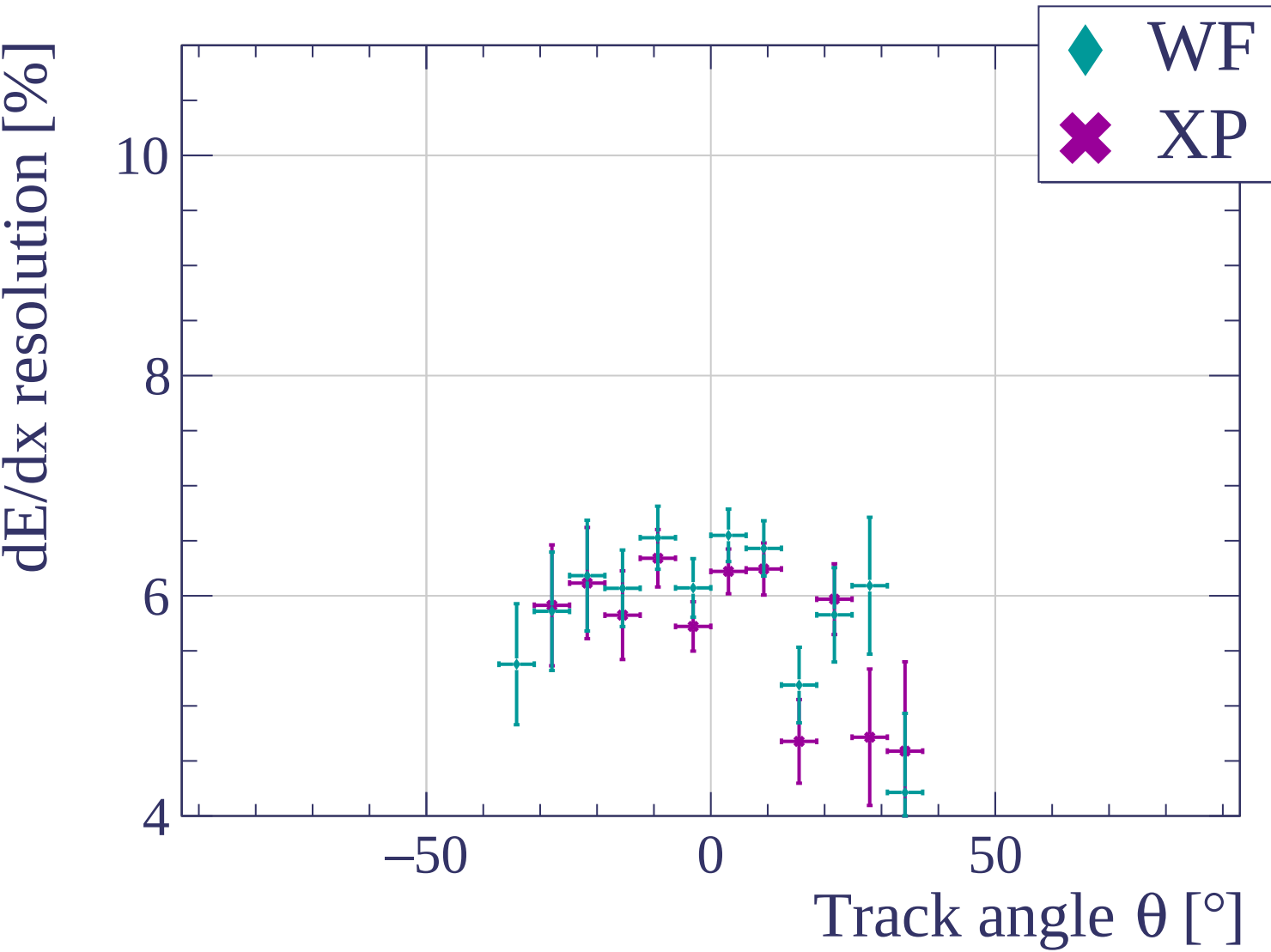
Count

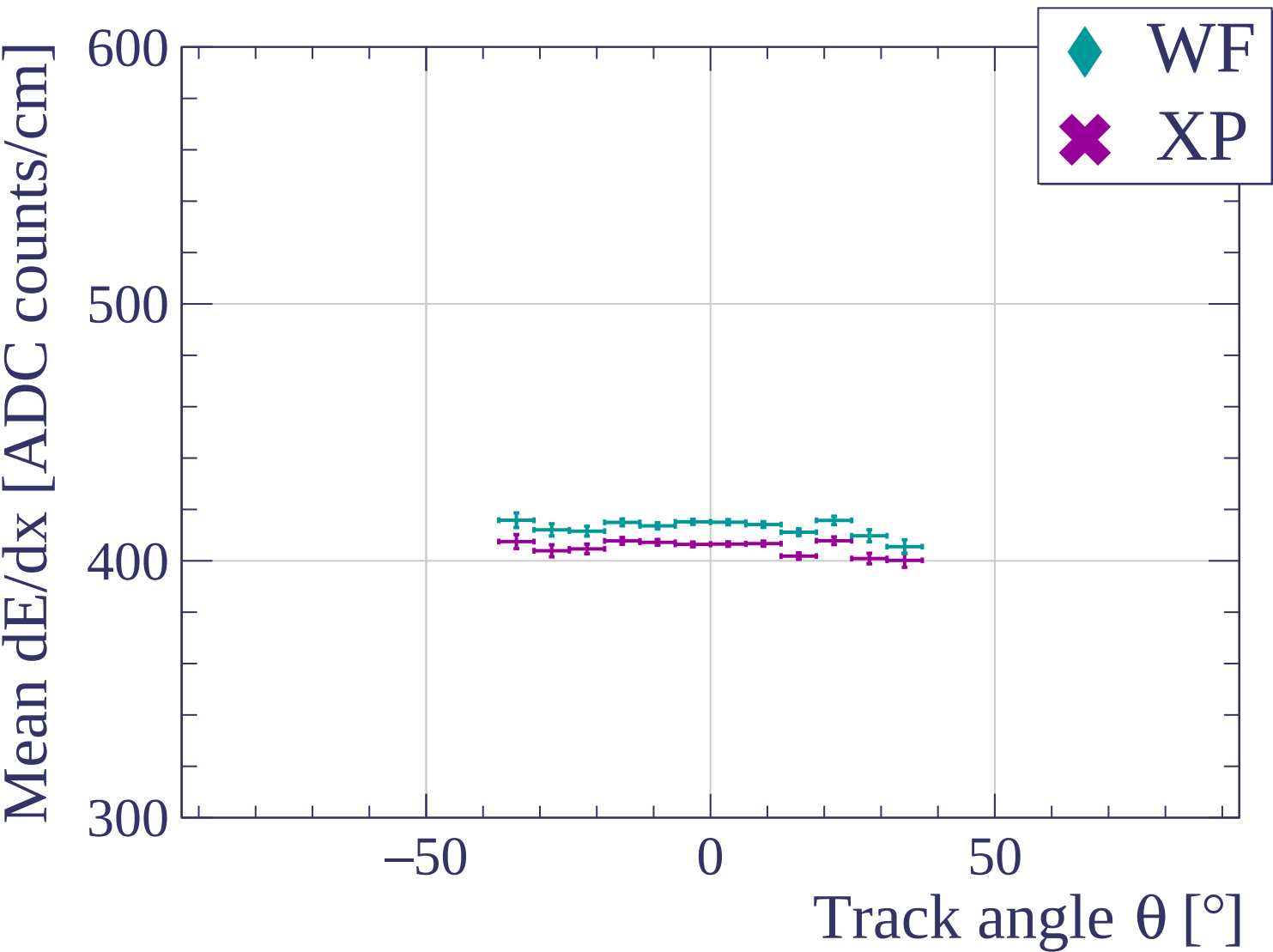


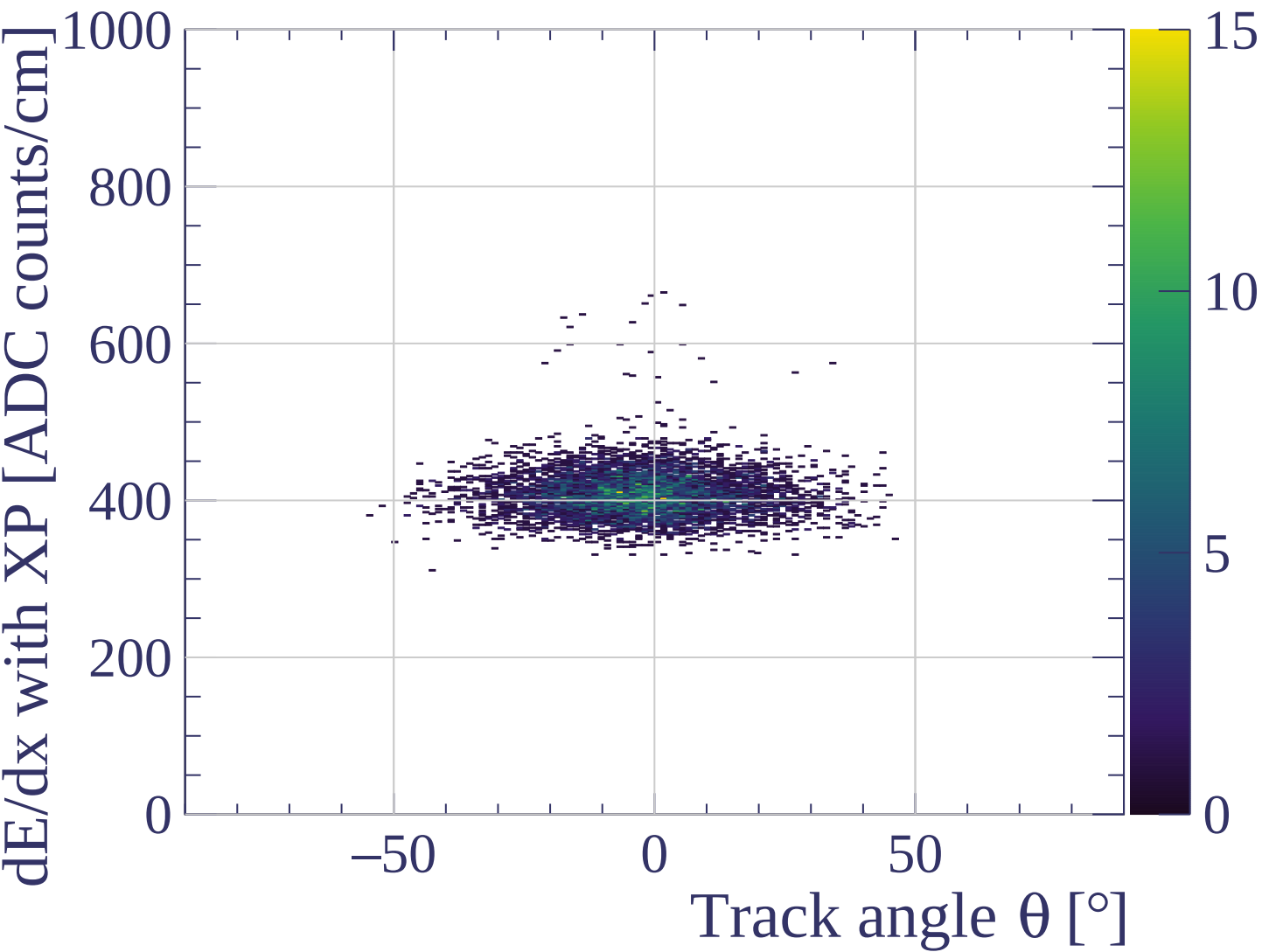
Track angle ϕ [°] angle

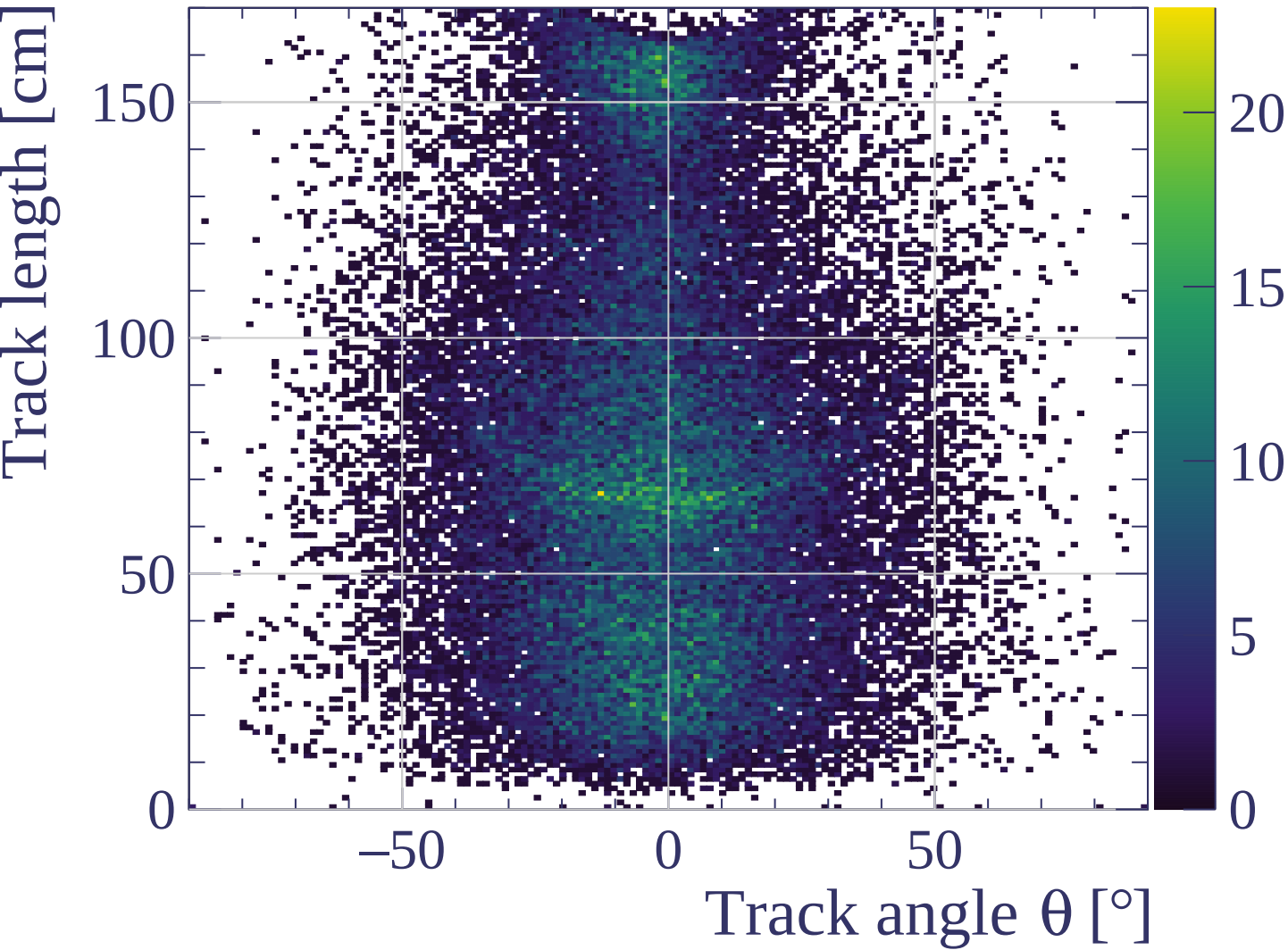


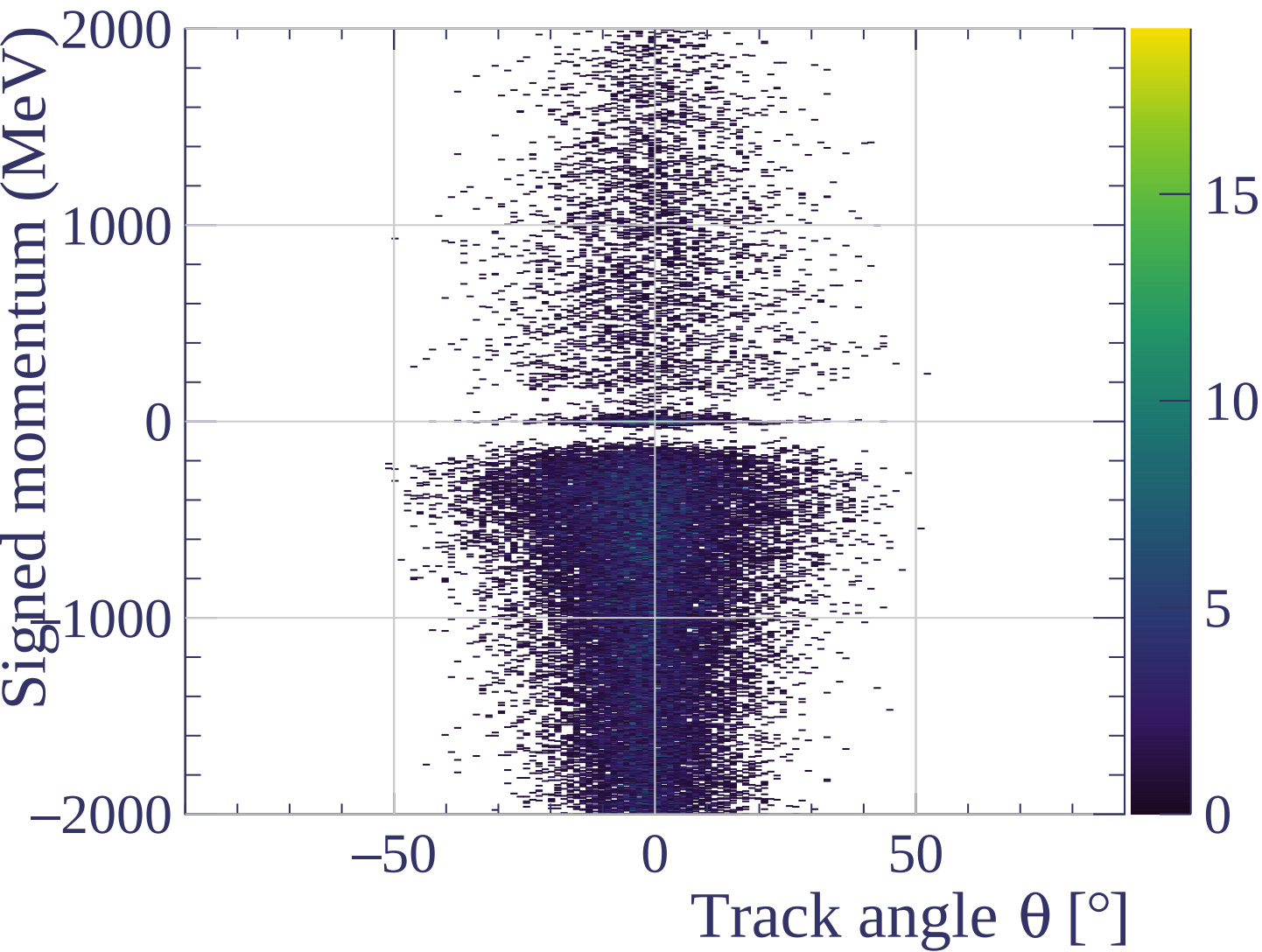




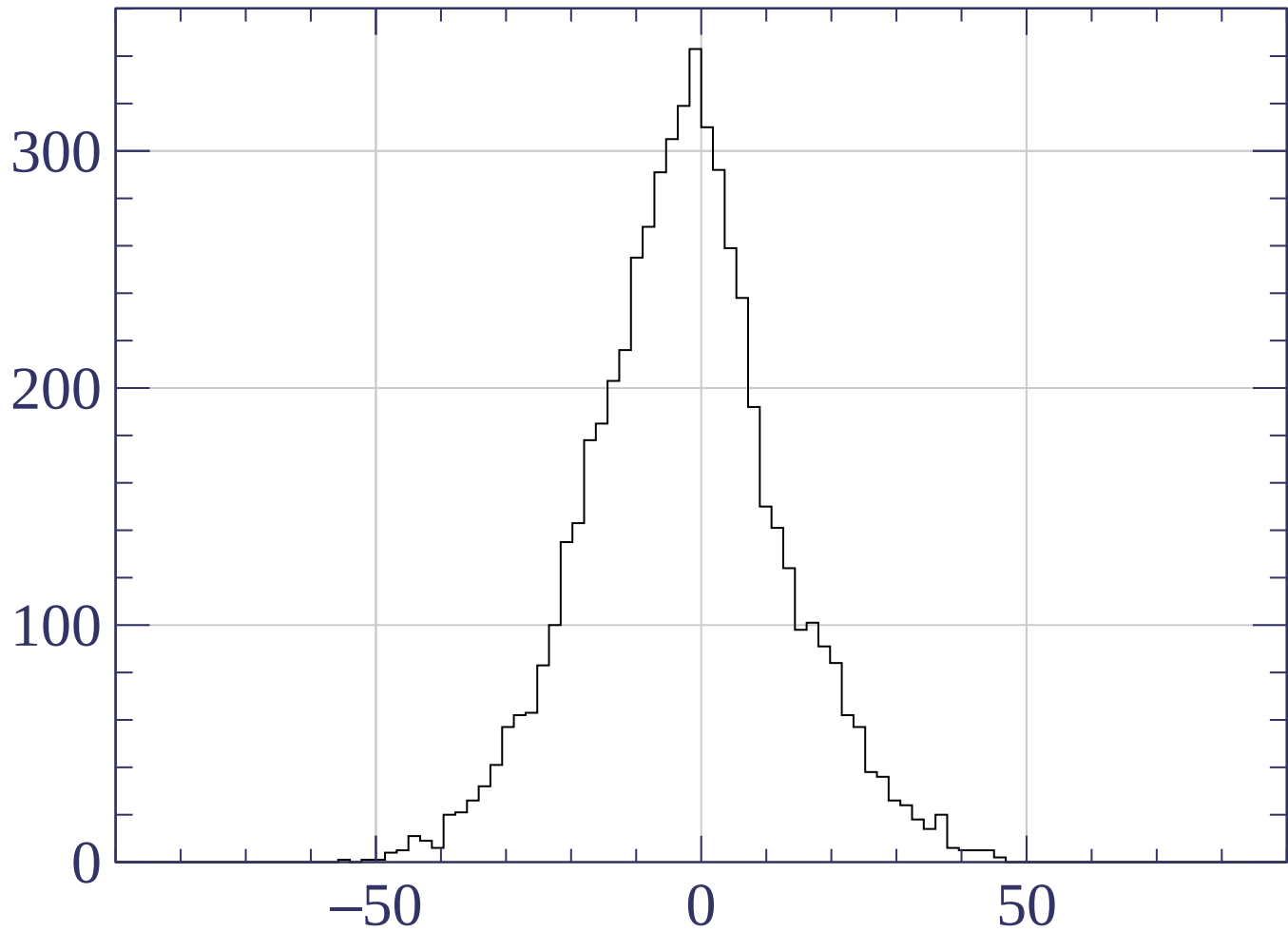






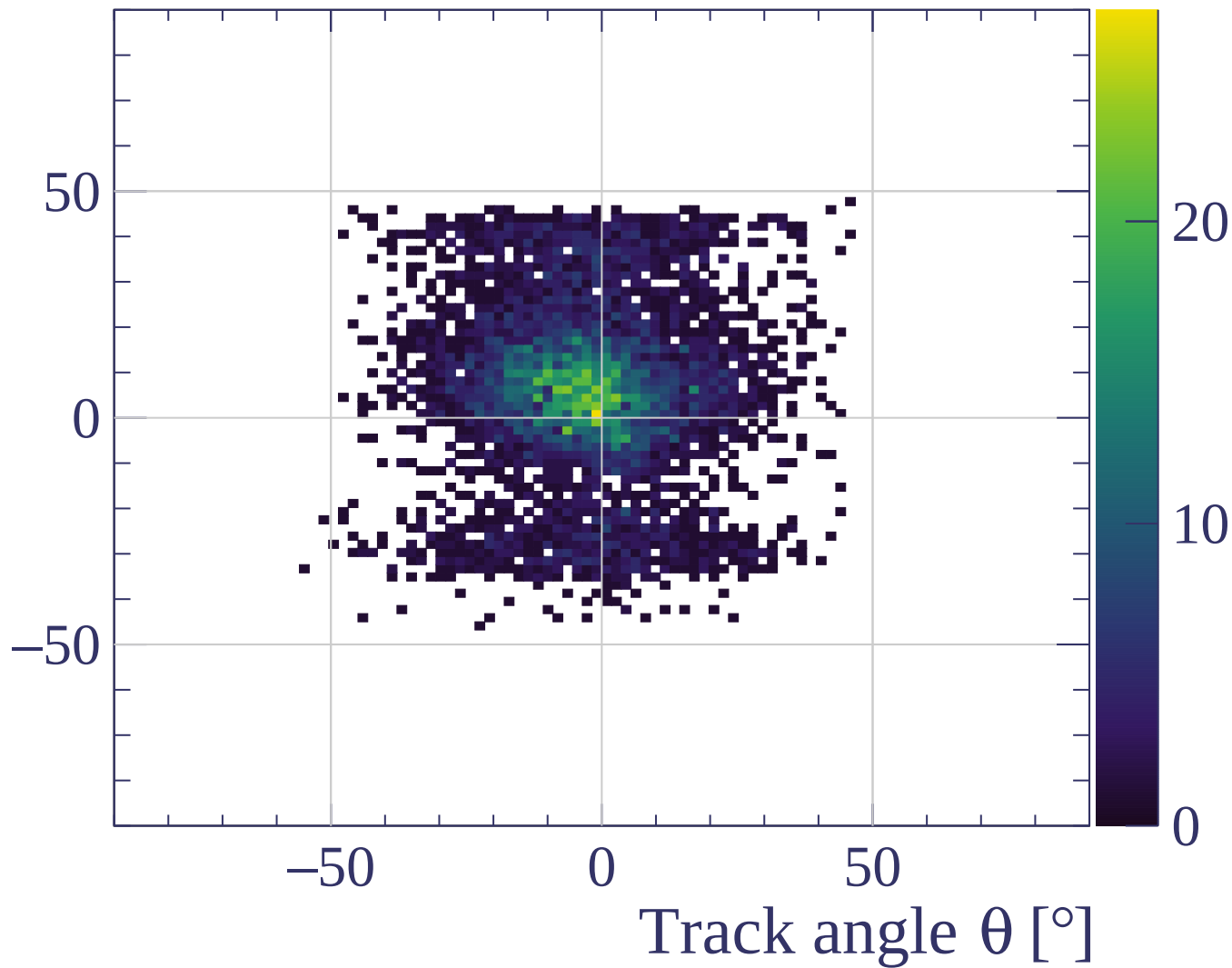


Count

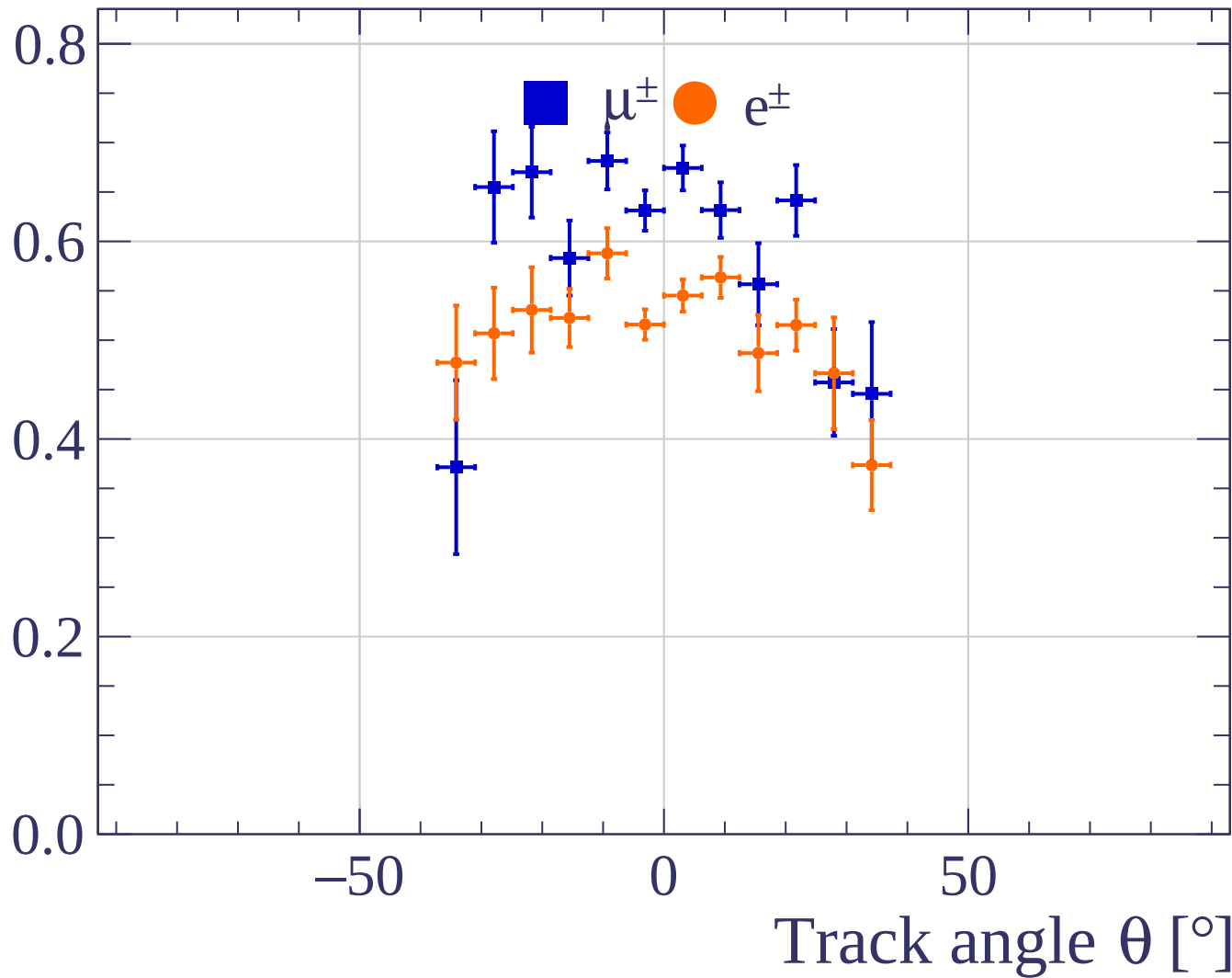


Track angle θ [°]

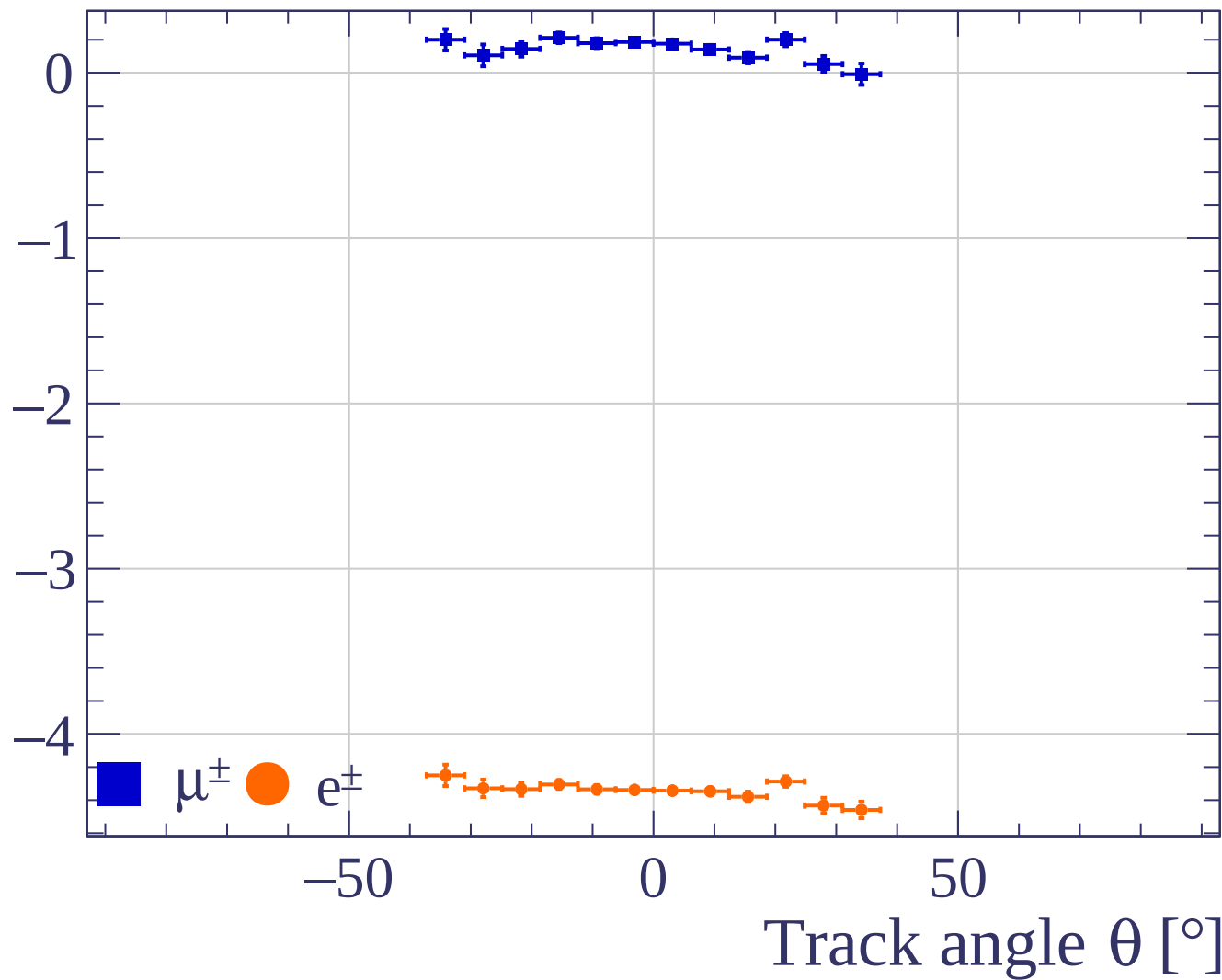
Track angle ϕ [°]



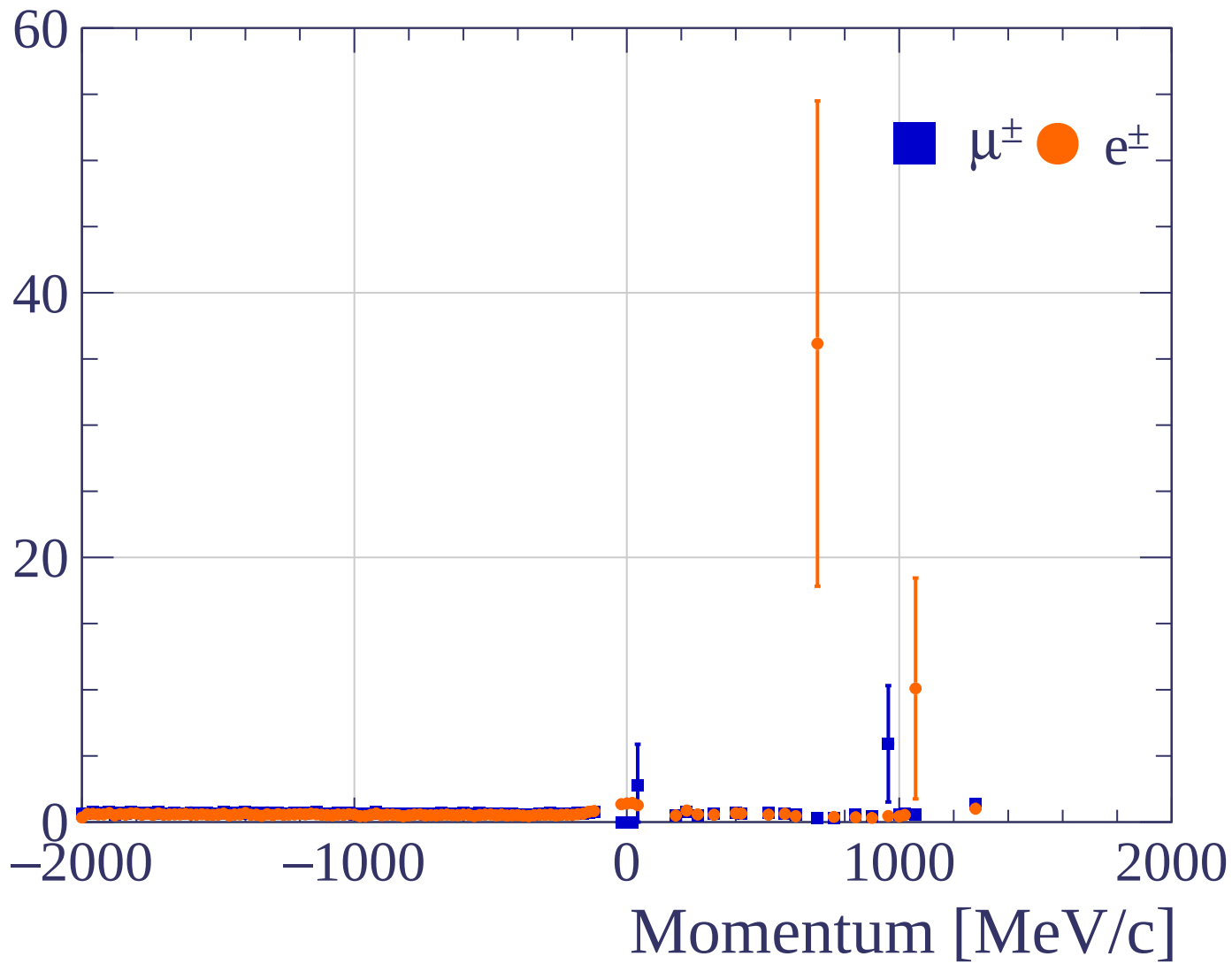
Pulls standard deviation



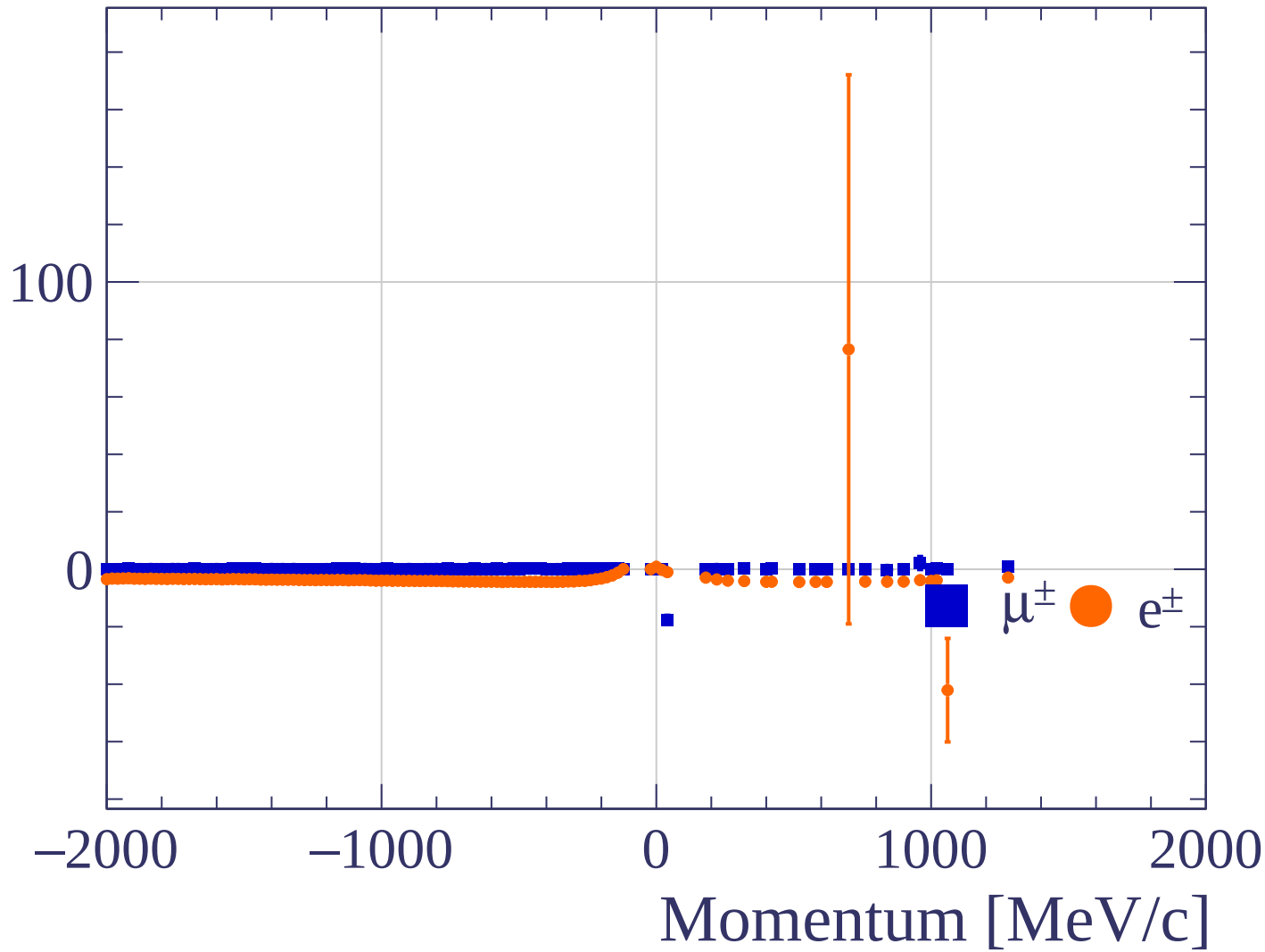
Pulls mean



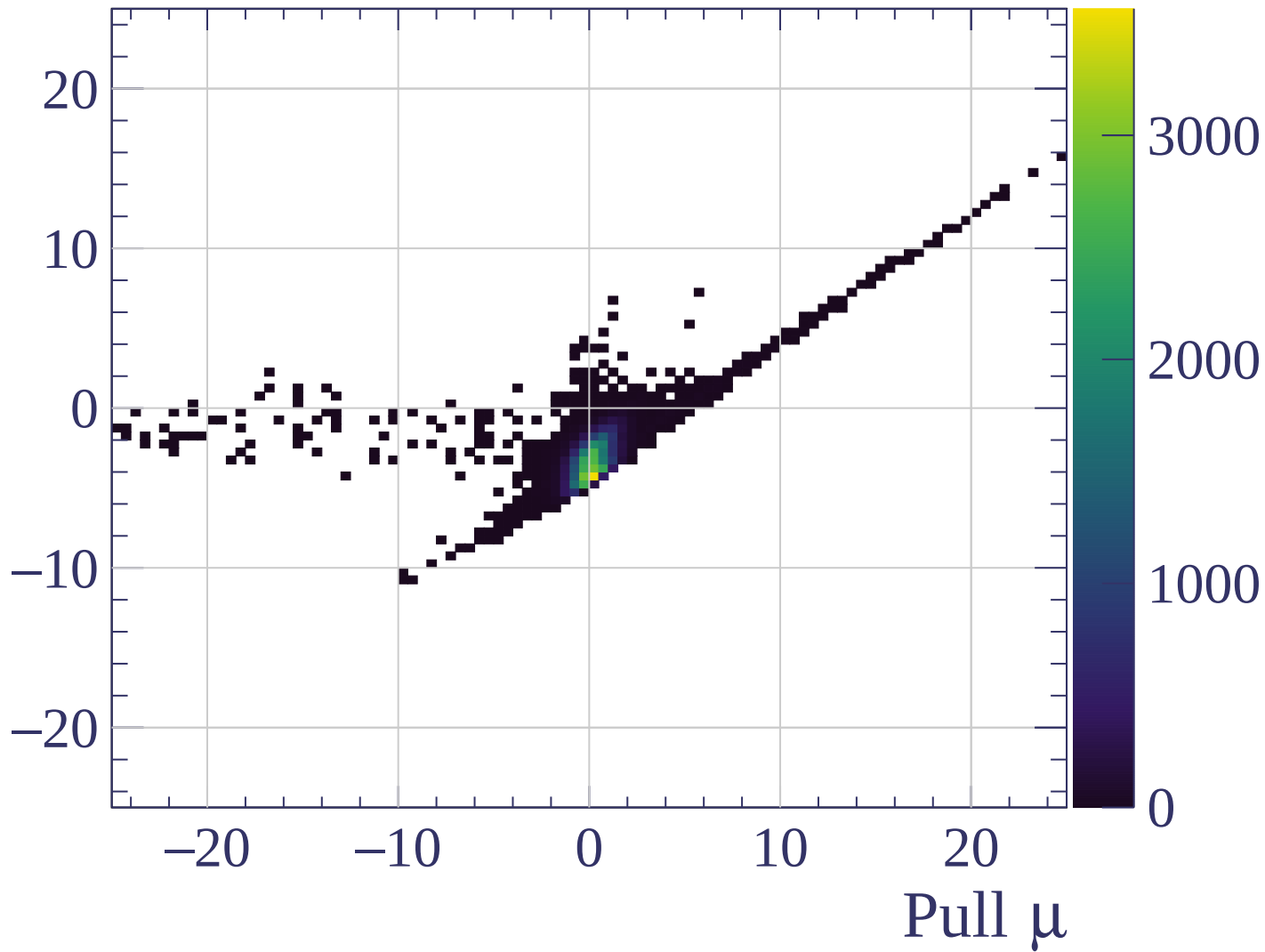
Pulls standard deviation

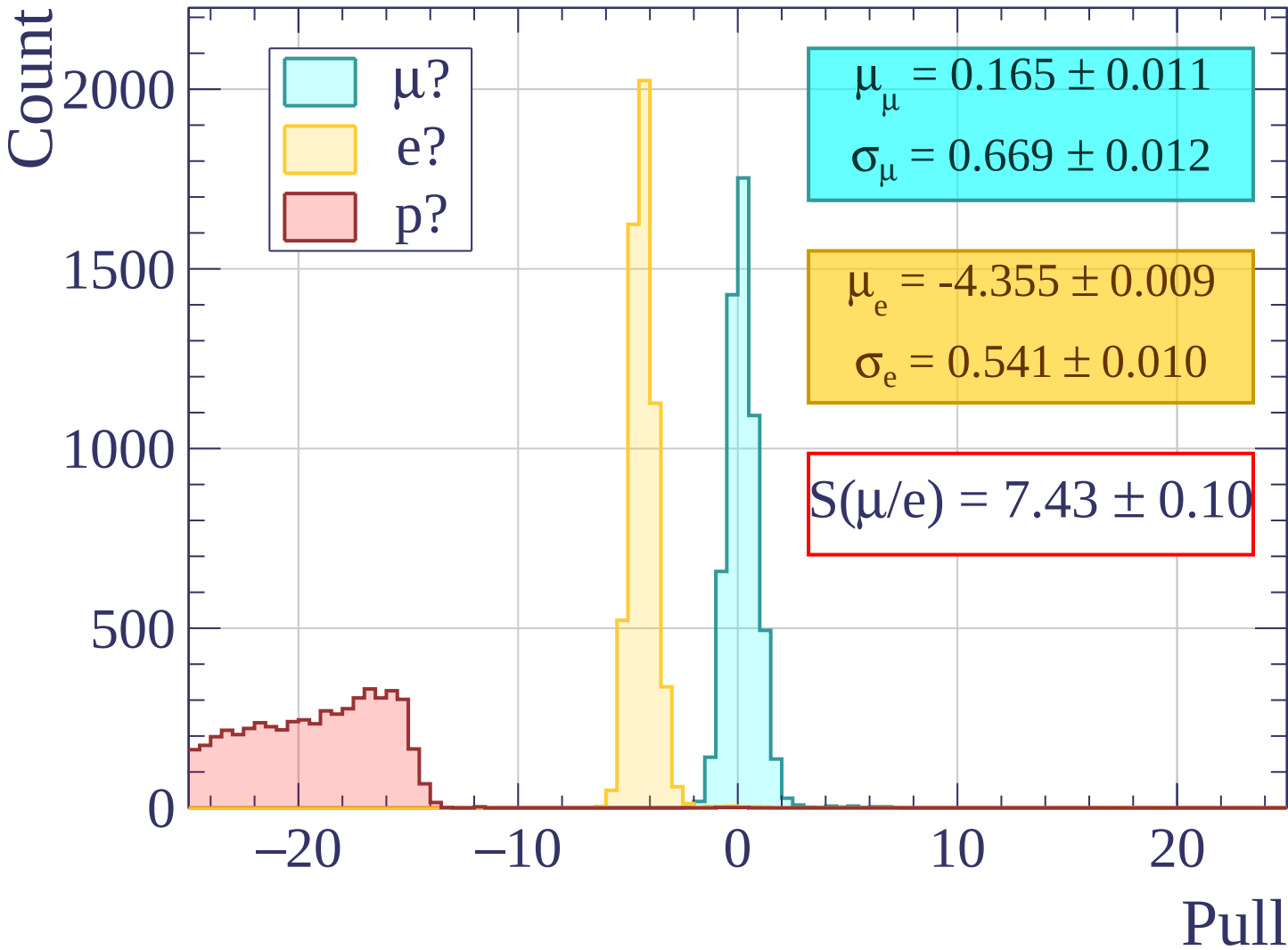


Pulls mean

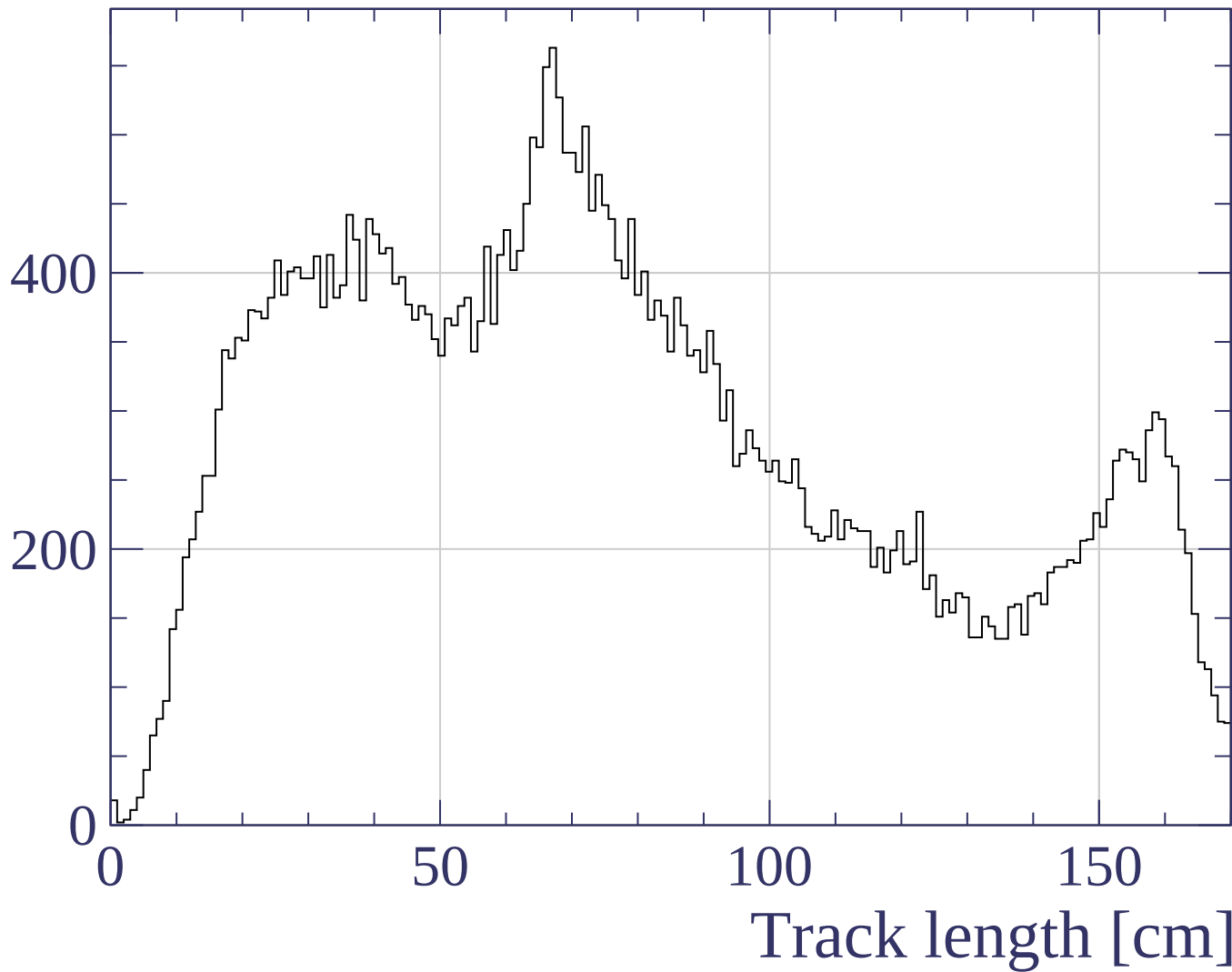


Pull e

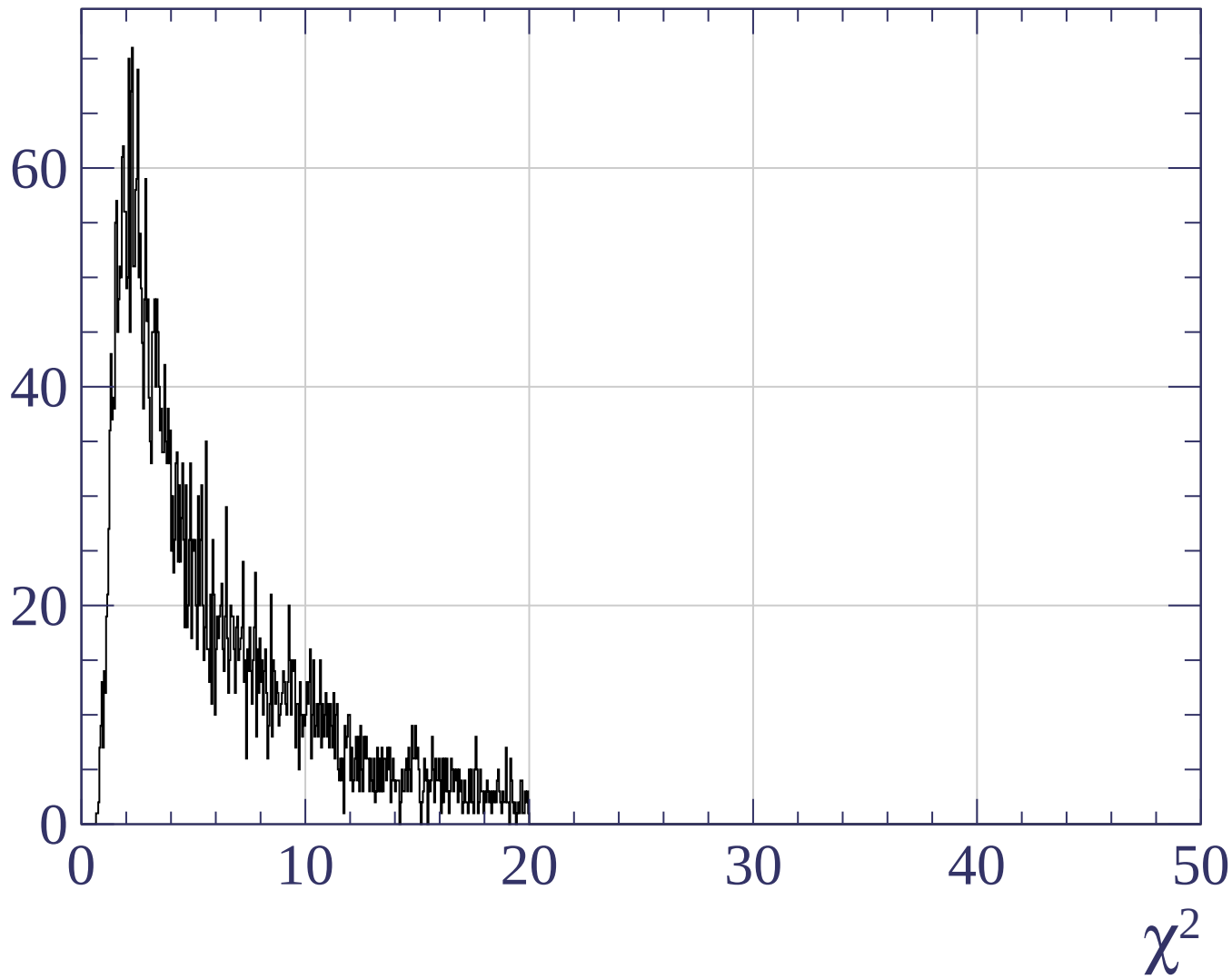




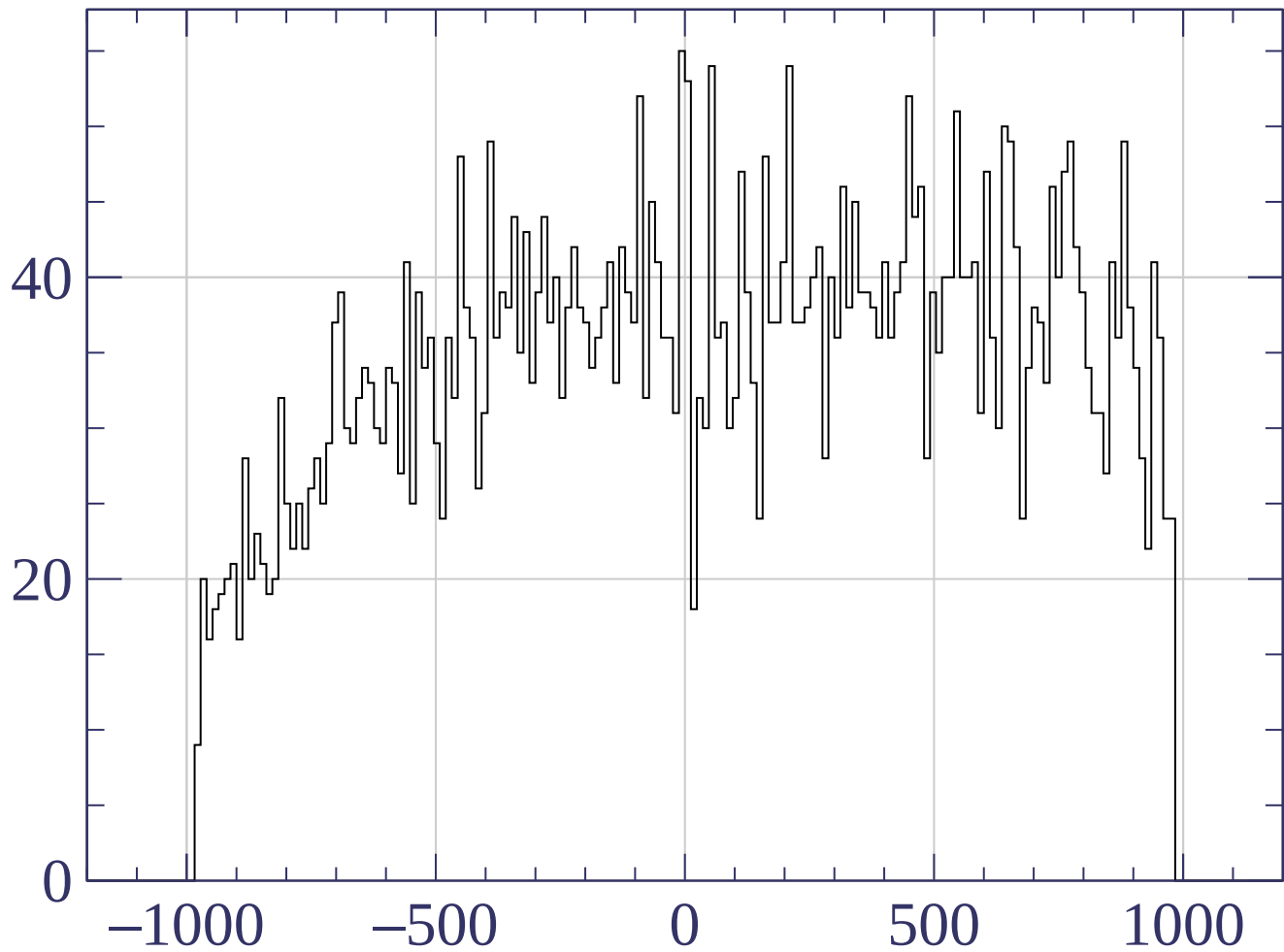
Count



Count

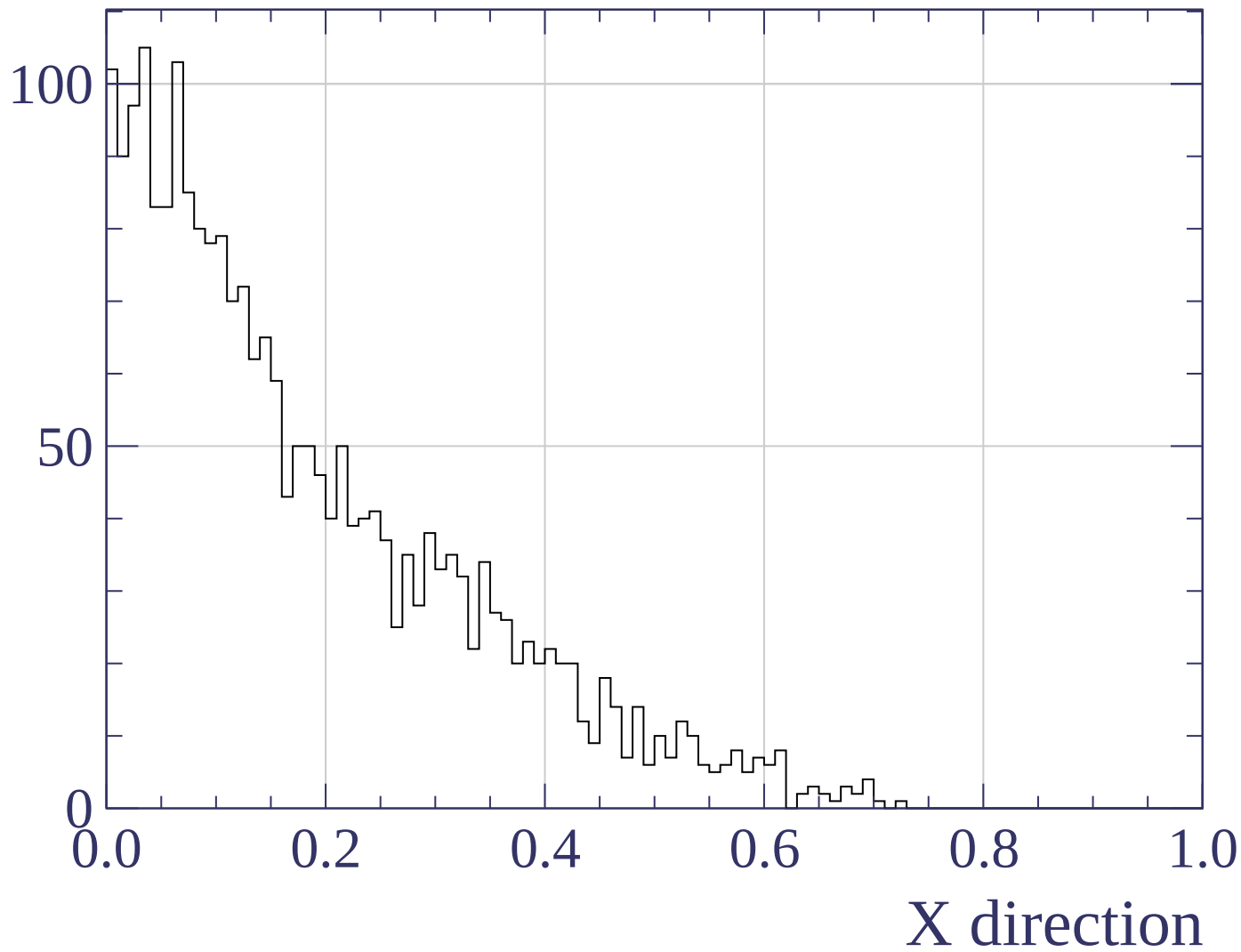


Count

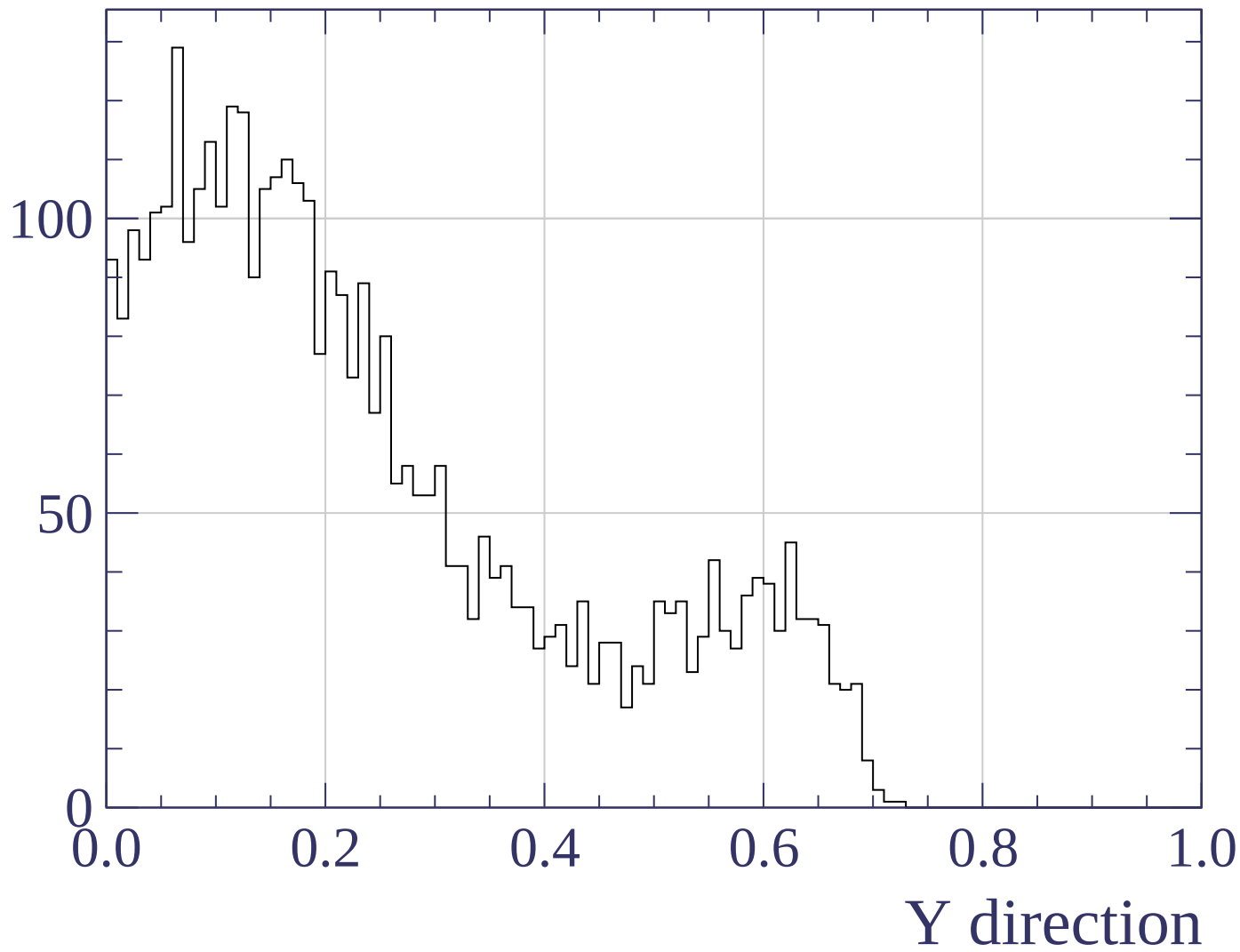


Track X position [mm]

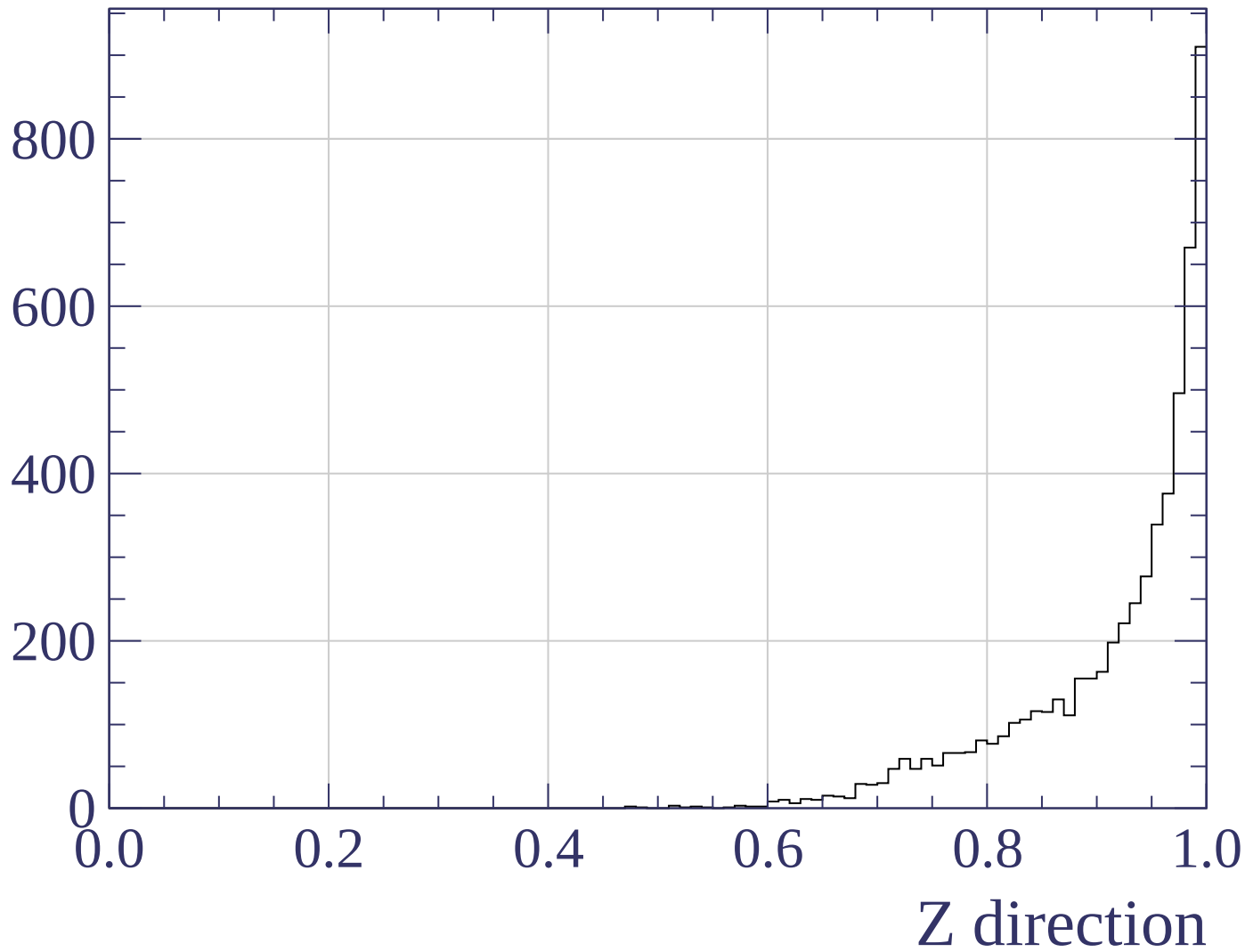
Counts

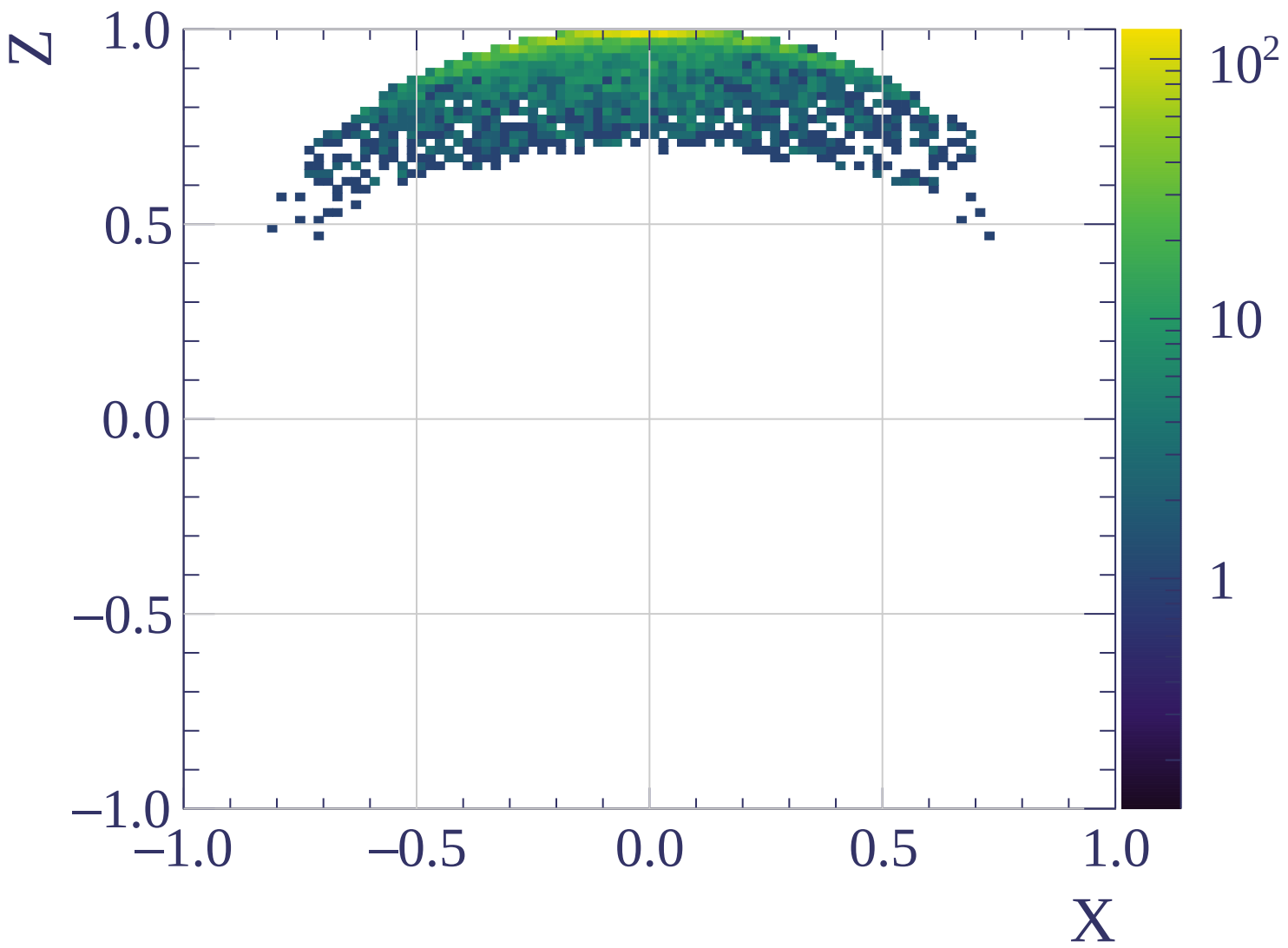


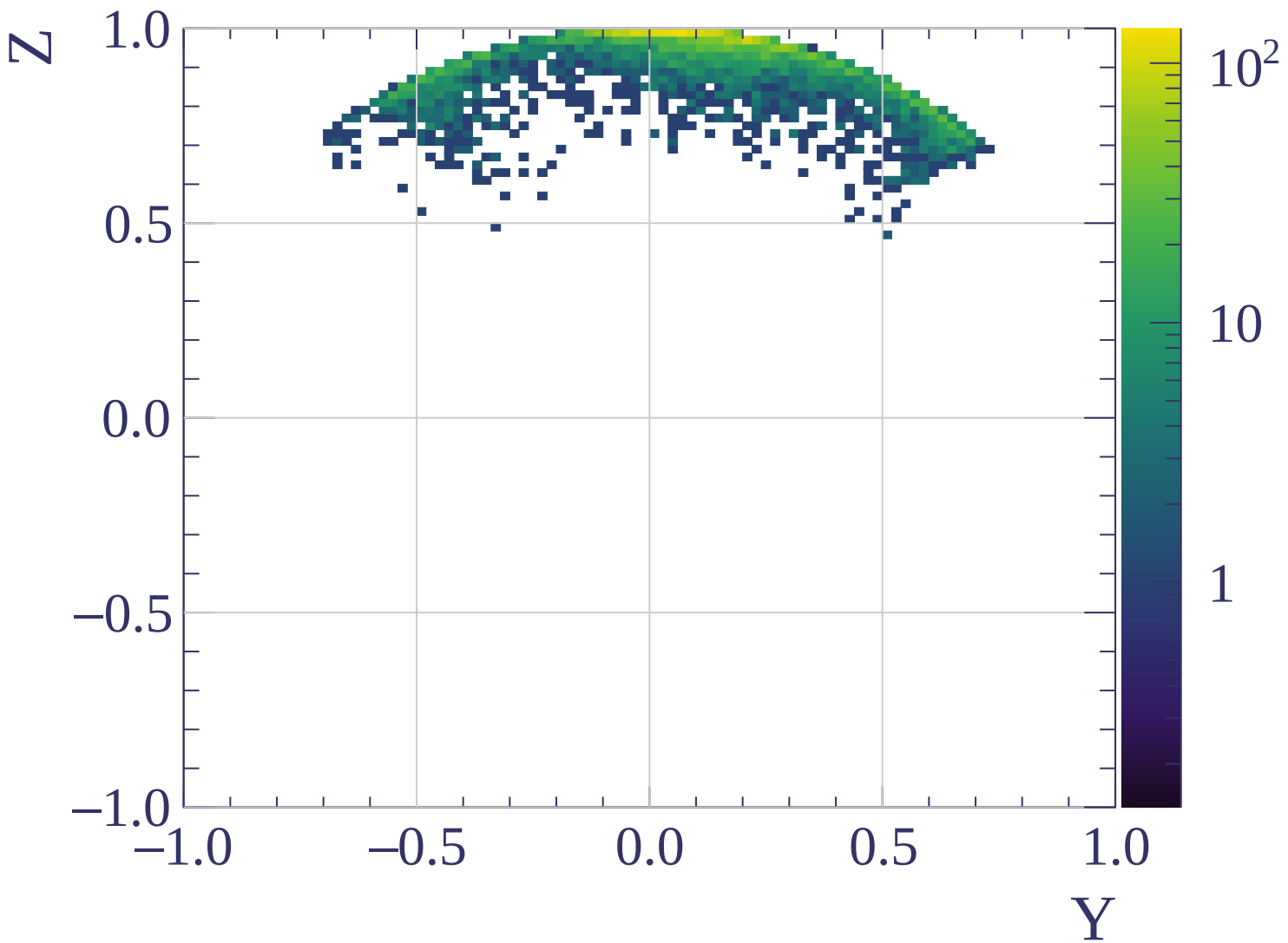
Counts

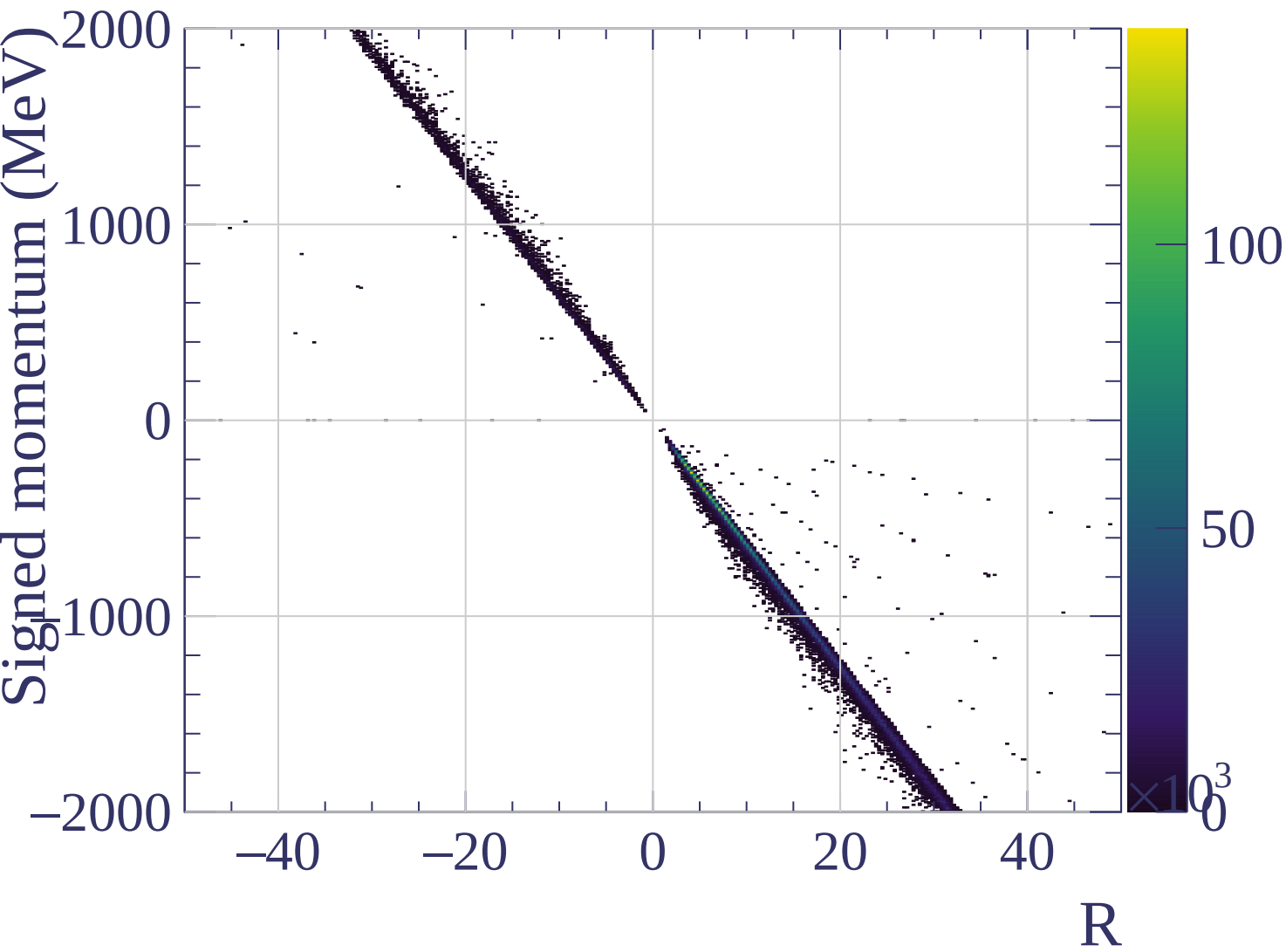


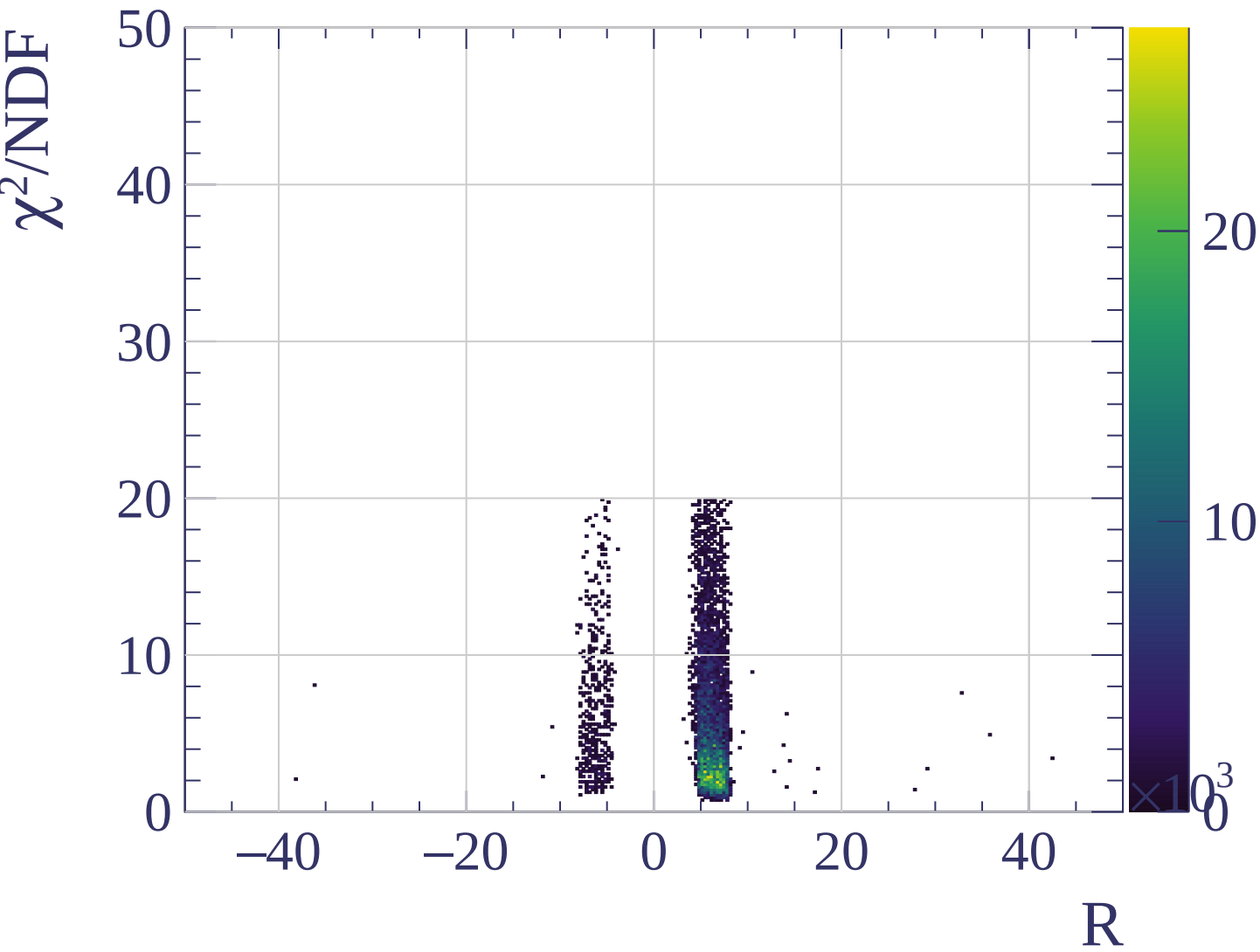
Counts



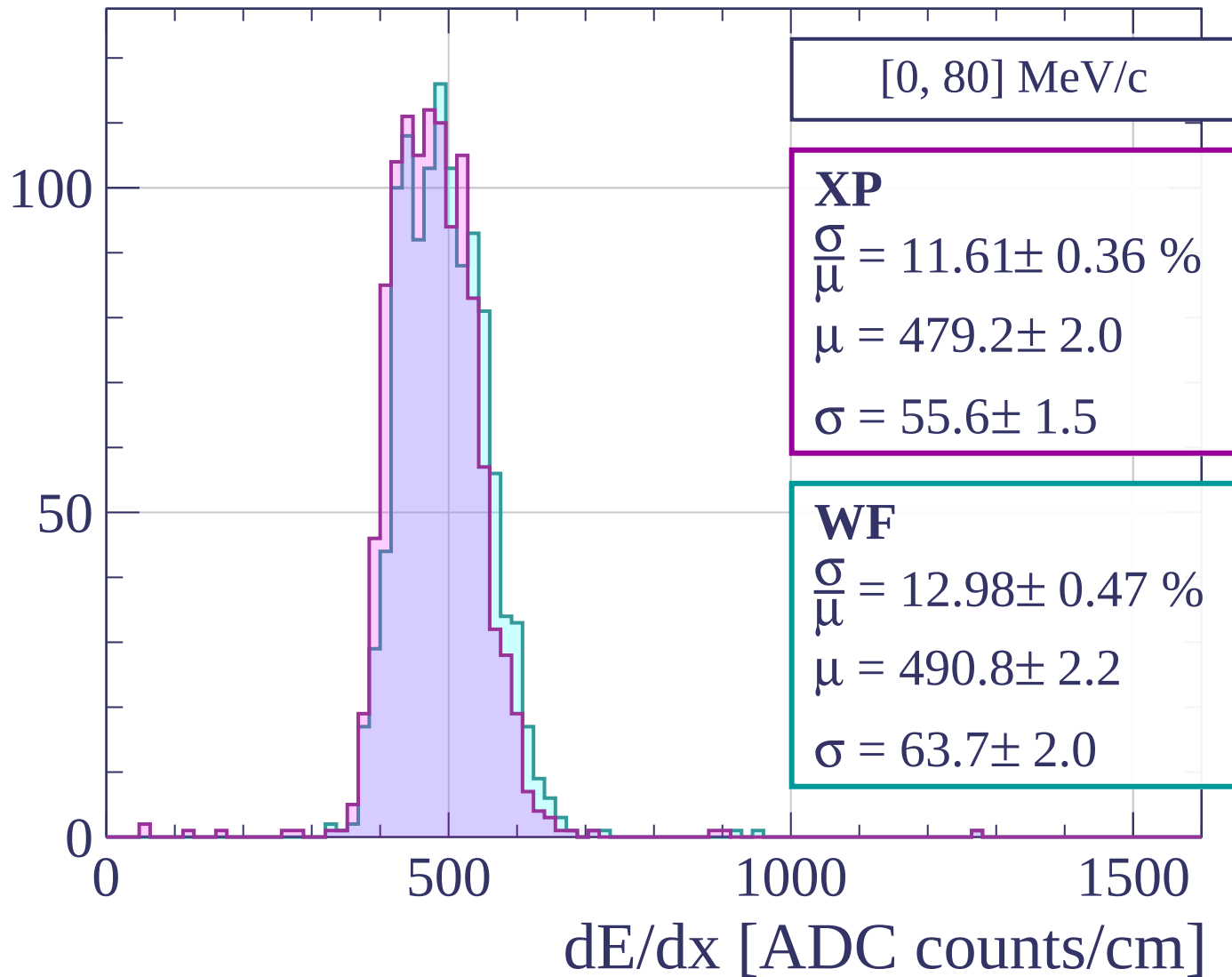




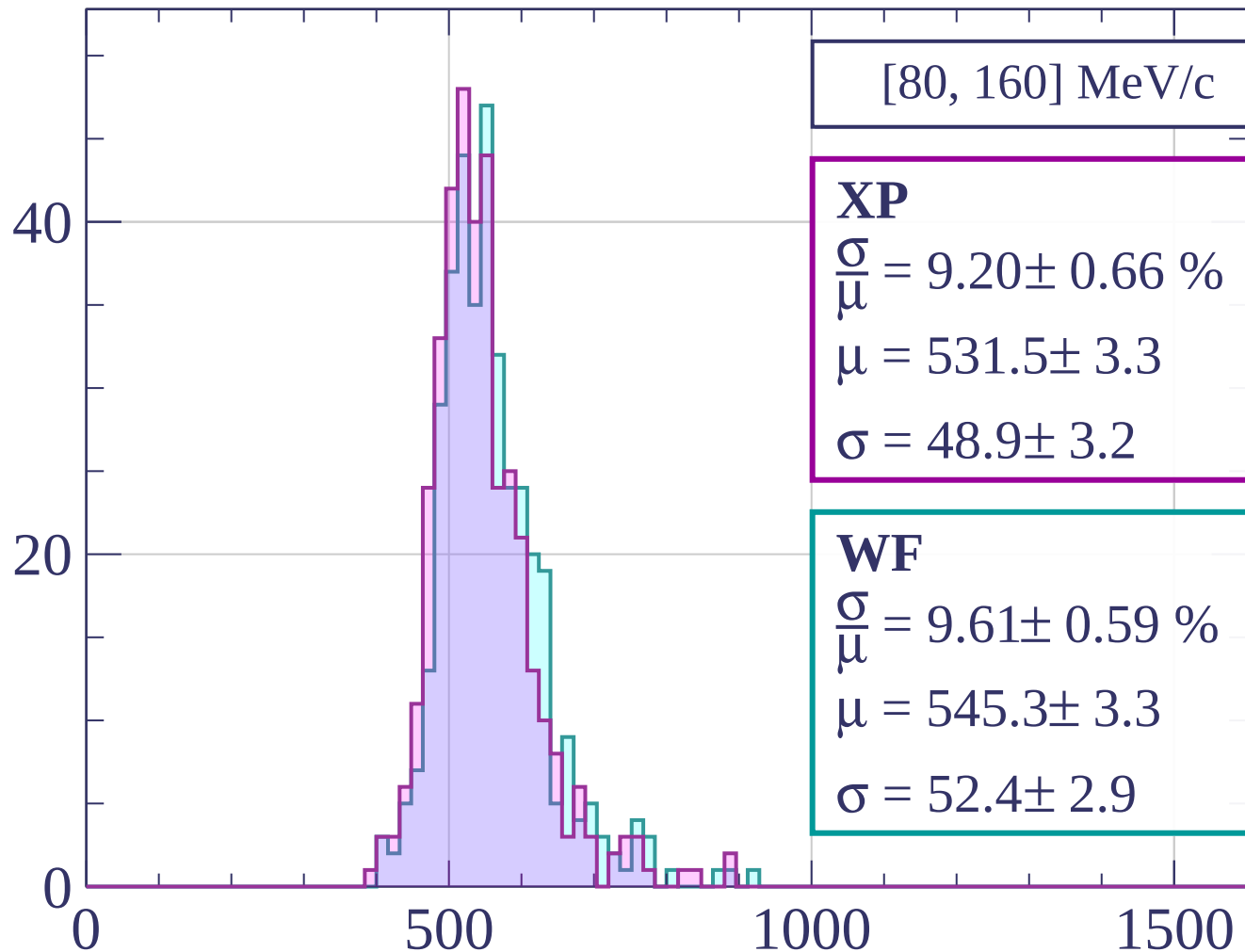




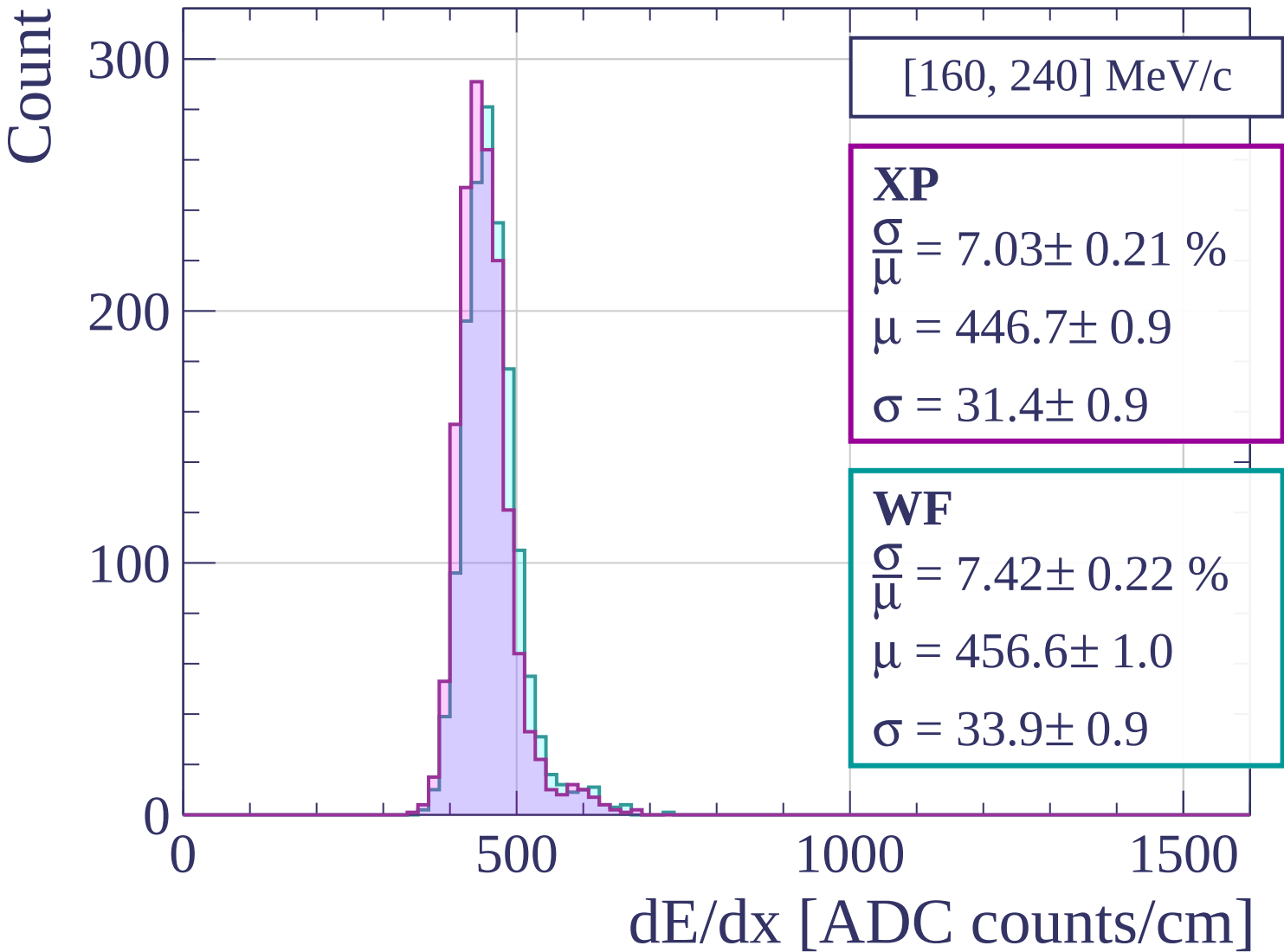
Count



Count



dE/dx [ADC counts/cm]



Count

[240, 320] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.05 \pm 0.18 \%$$

$$\mu = 413.5 \pm 0.7$$

$$\sigma = 25.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.56 \pm 0.17 \%$$

$$\mu = 422.0 \pm 0.7$$

$$\sigma = 27.7 \pm 0.7$$

400

200

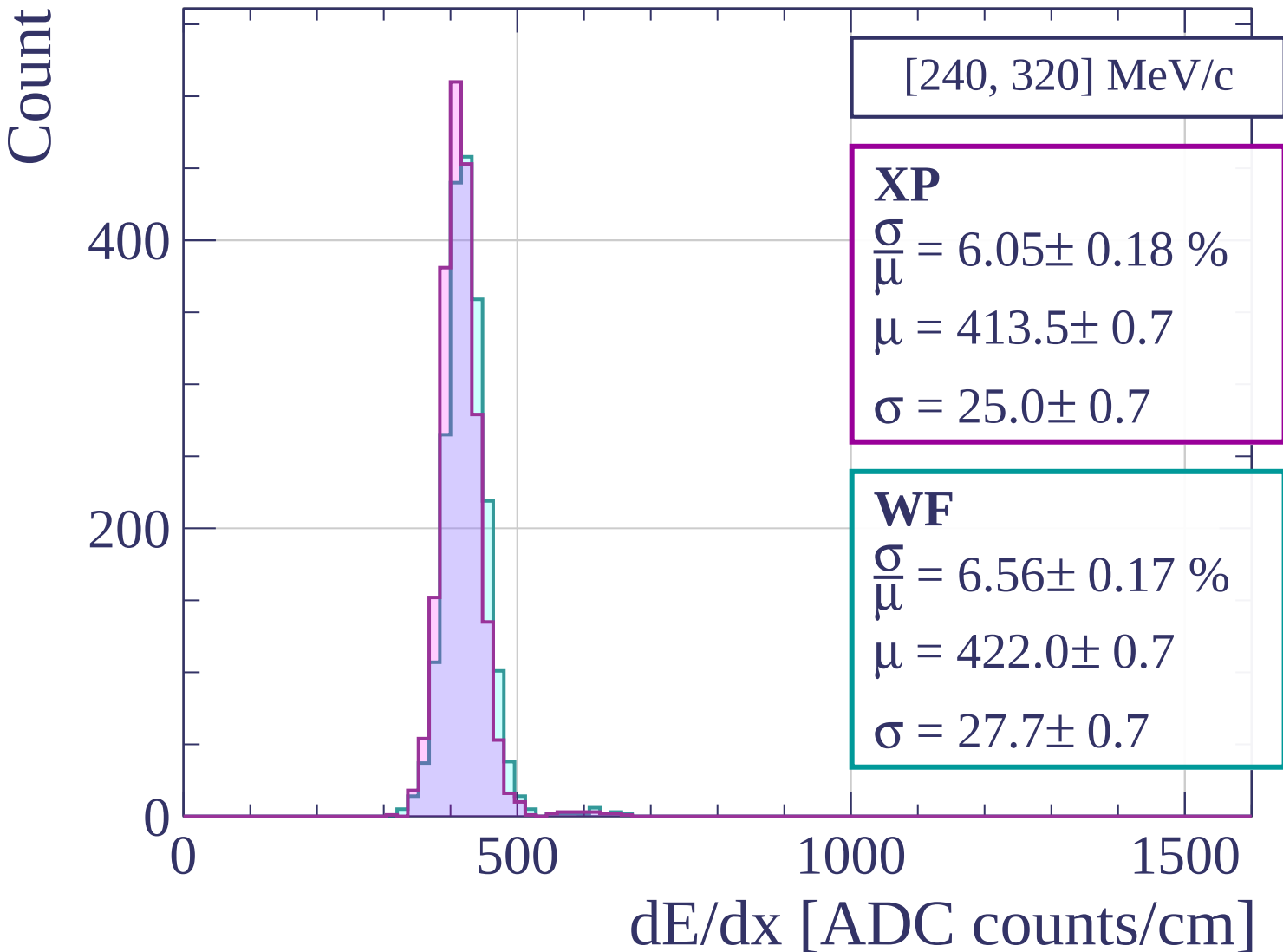
0

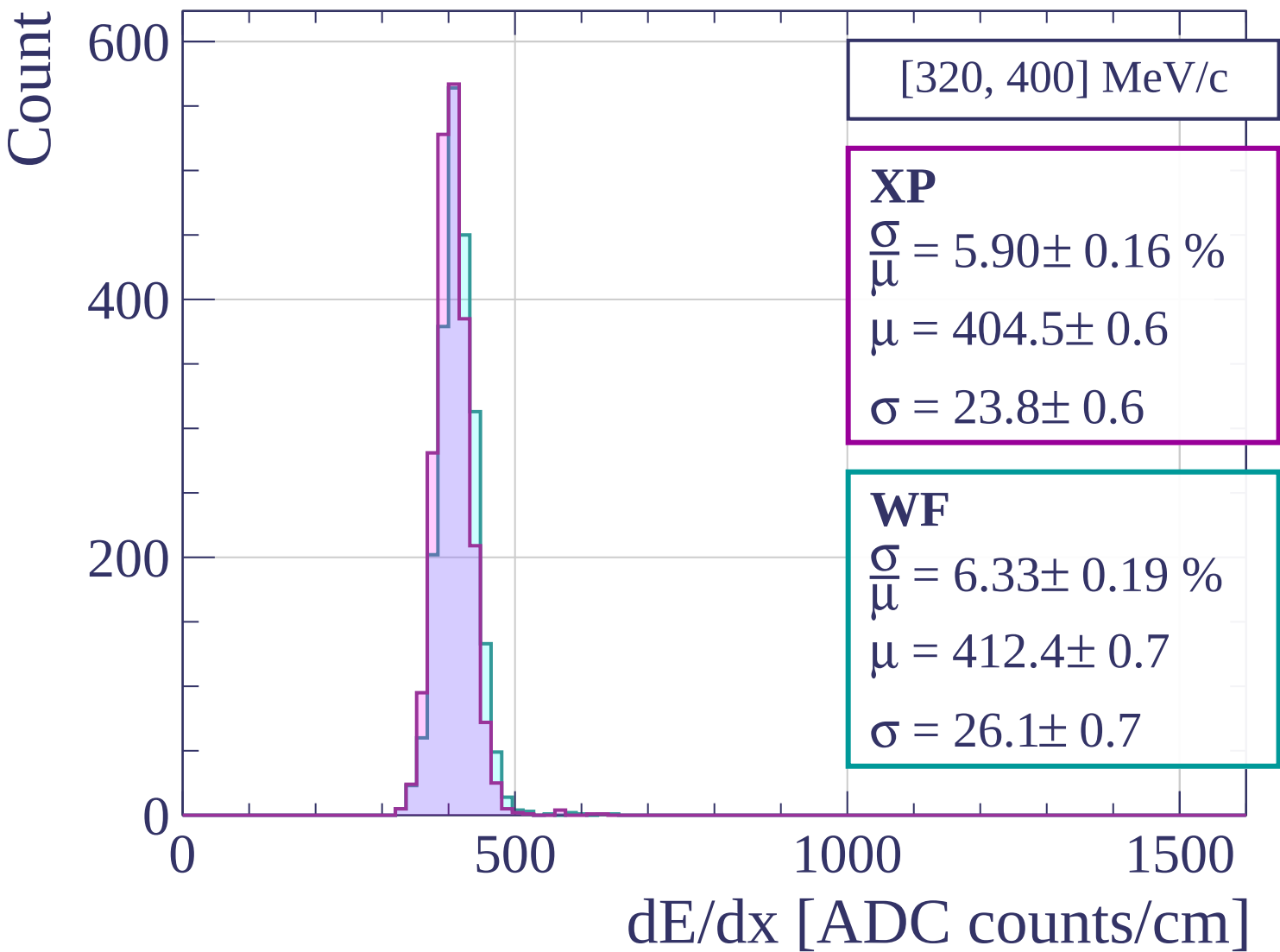
500

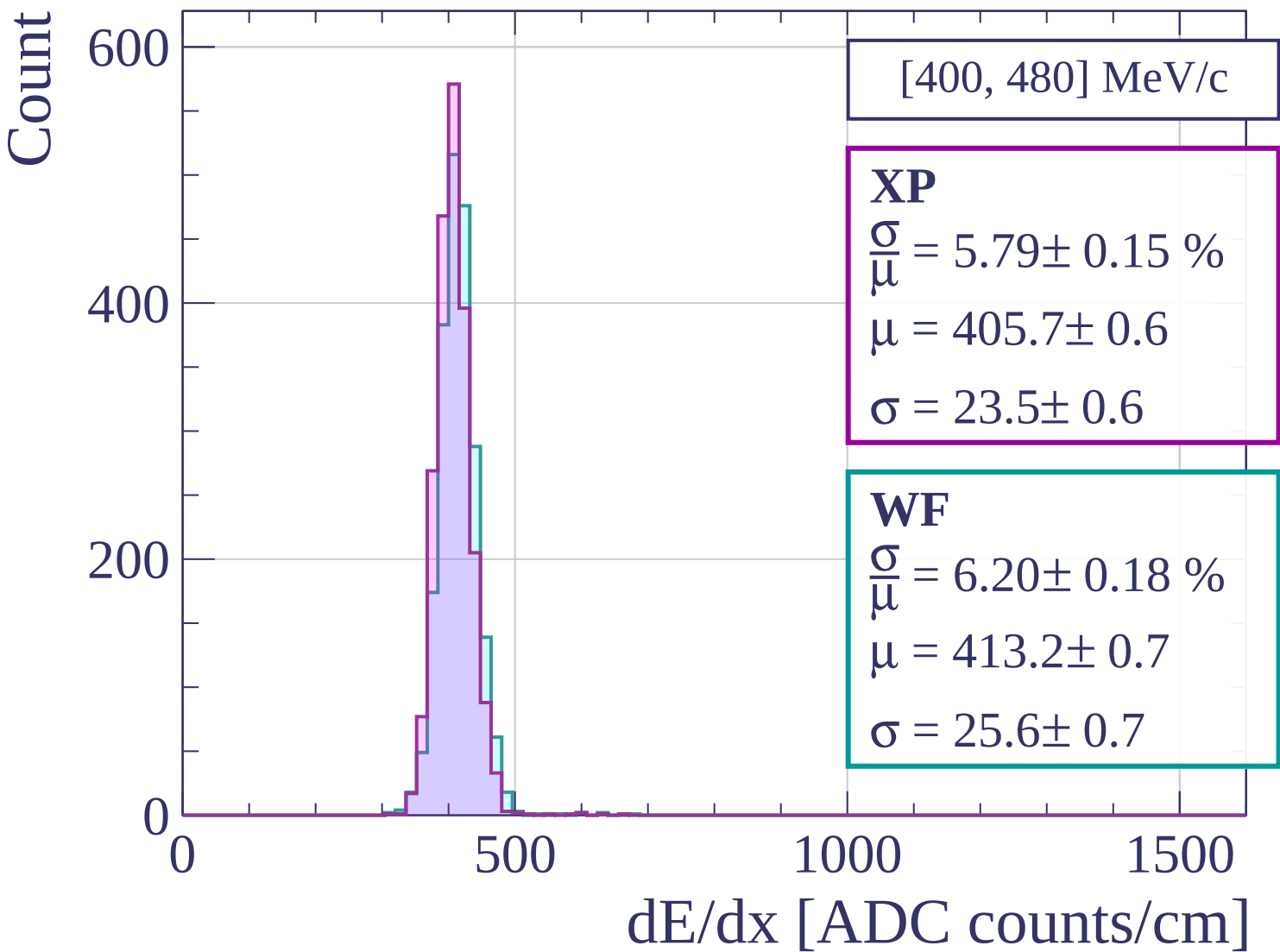
1000

1500

dE/dx [ADC counts/cm]







Count

[480, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.11 \pm 0.14 \%$$

$$\mu = 409.4 \pm 0.6$$

$$\sigma = 25.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 6.41 \pm 0.20 \%$$

$$\mu = 416.9 \pm 0.7$$

$$\sigma = 26.7 \pm 0.8$$

400

200

0

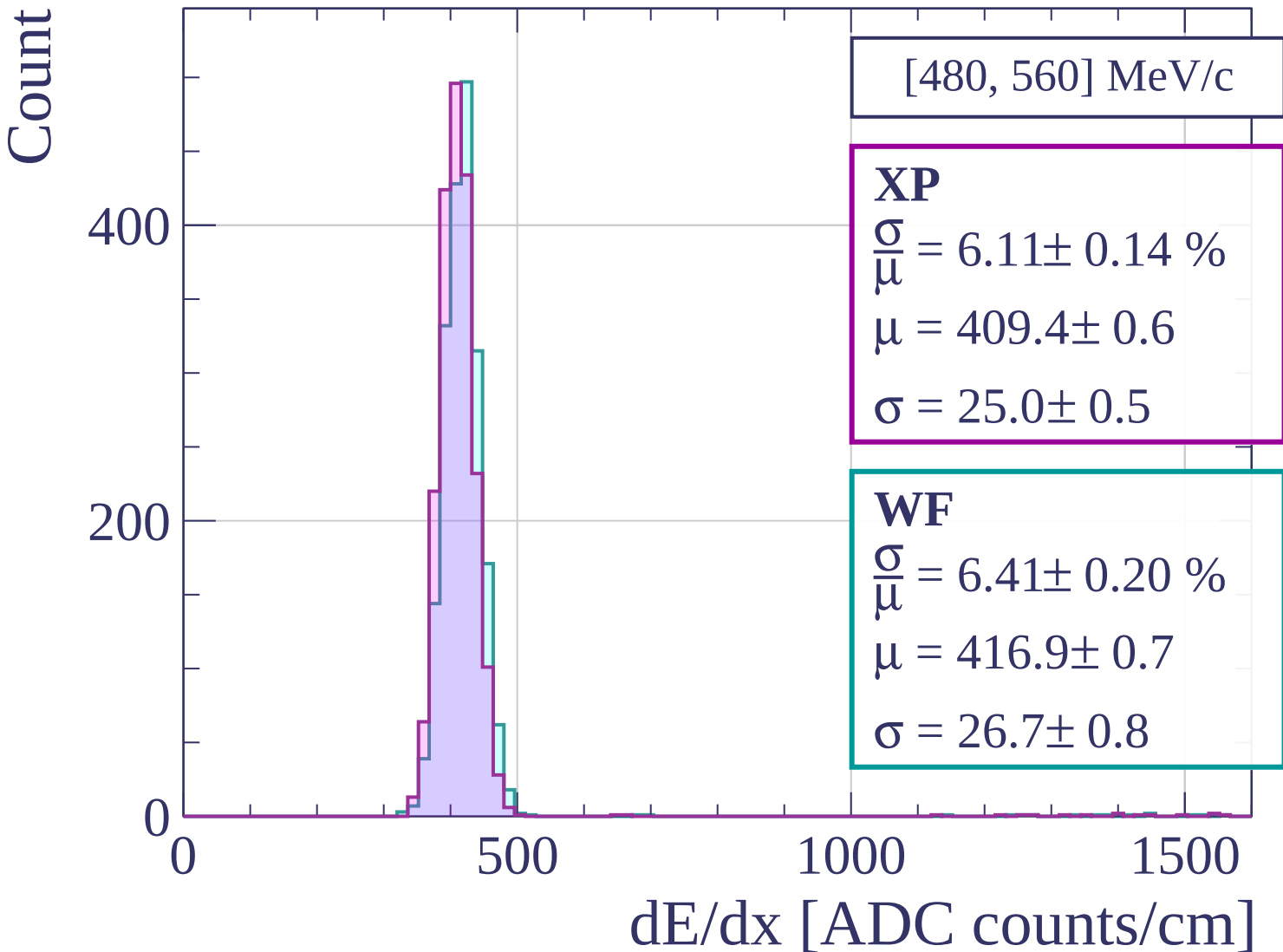
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500

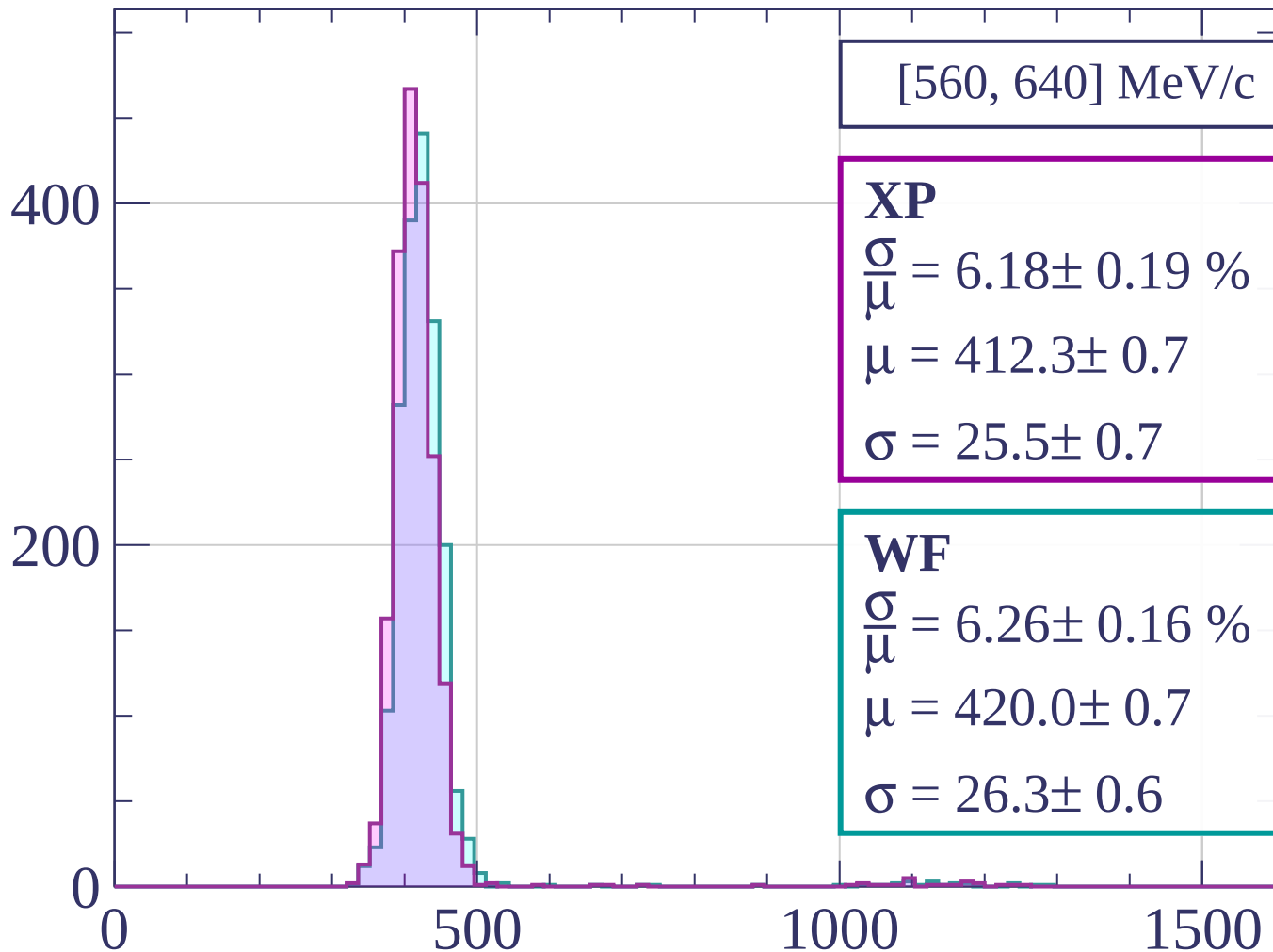
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[640, 720] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.94 \pm 0.18 \%$$

$$\mu = 417.5 \pm 0.7$$

$$\sigma = 24.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.20 \pm 0.16 \%$$

$$\mu = 425.7 \pm 0.7$$

$$\sigma = 26.4 \pm 0.6$$

400

300

200

100

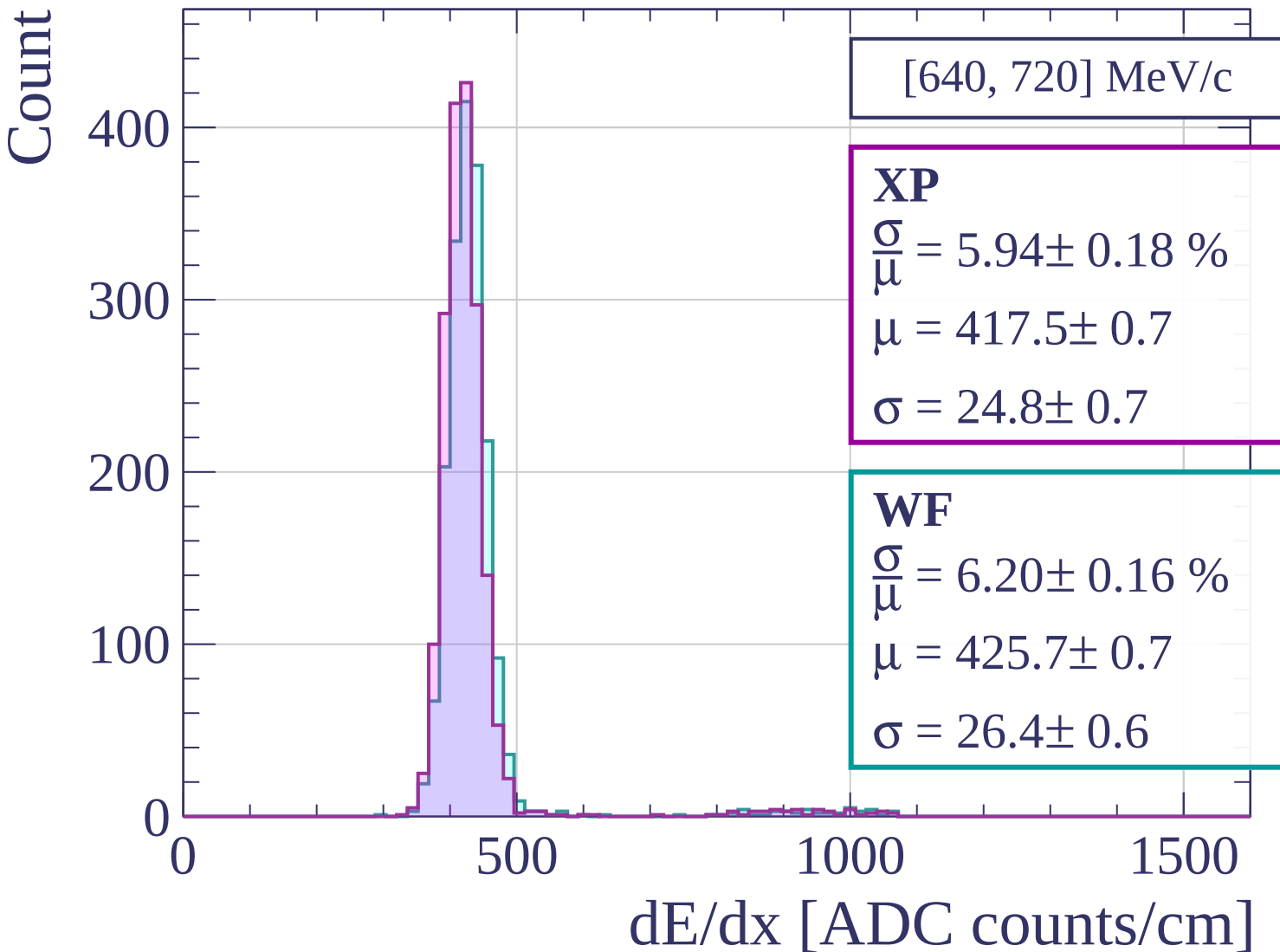
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500

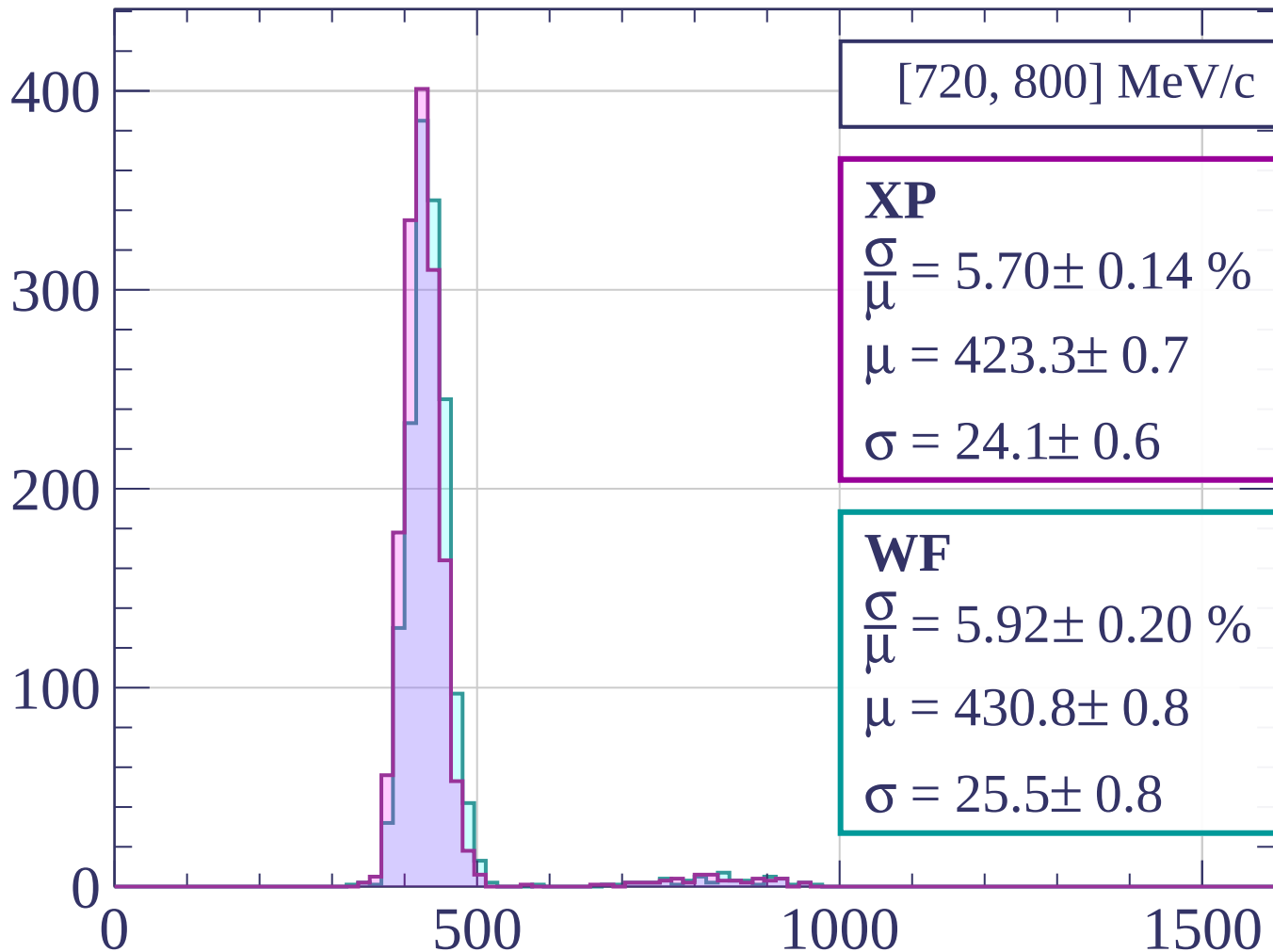
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[800, 880] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.90 \pm 0.21 \%$$

$$\mu = 427.3 \pm 0.8$$

$$\sigma = 25.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.03 \pm 0.18 \%$$

$$\mu = 435.3 \pm 0.8$$

$$\sigma = 26.2 \pm 0.7$$

300

200

100

0

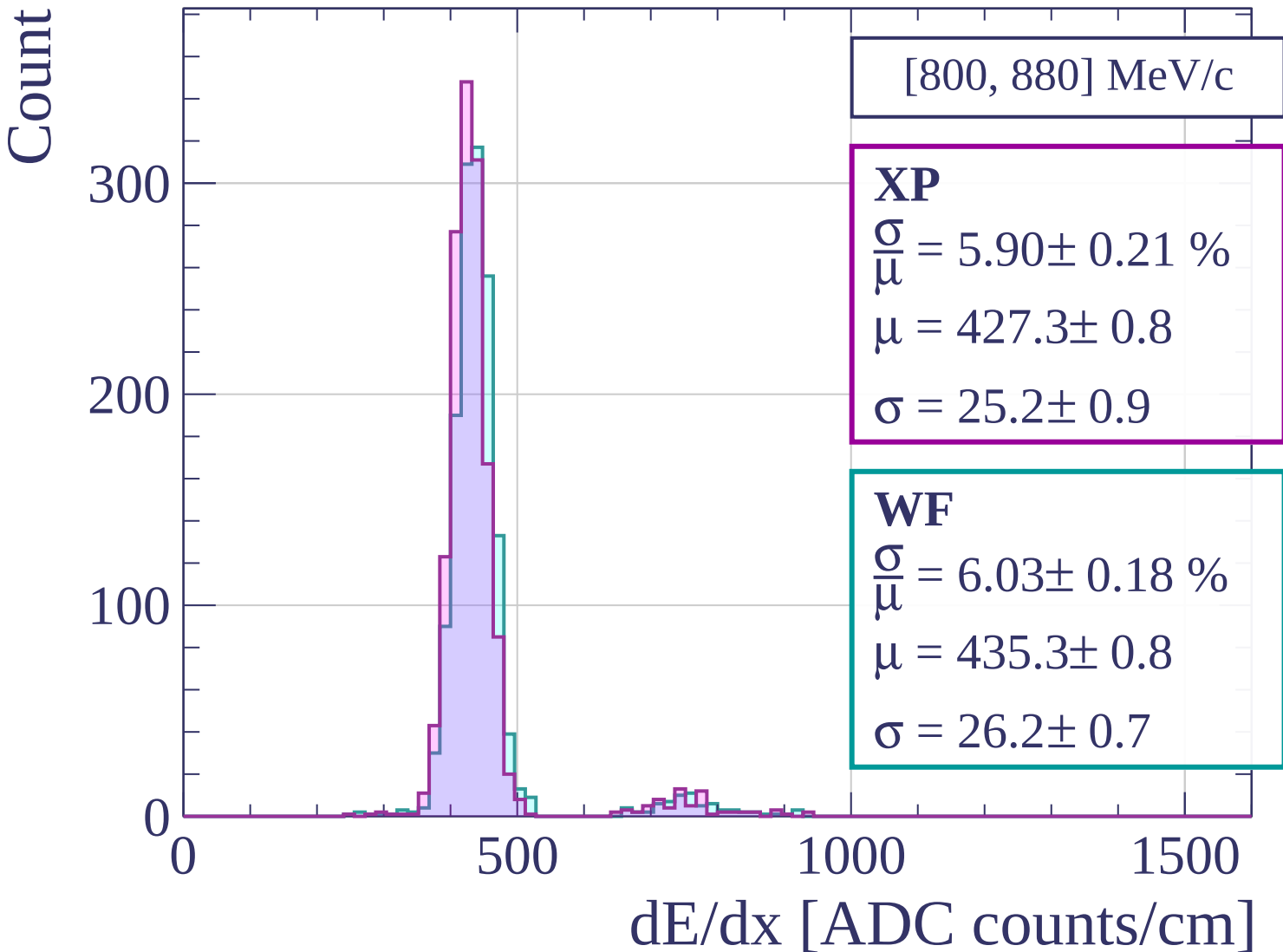
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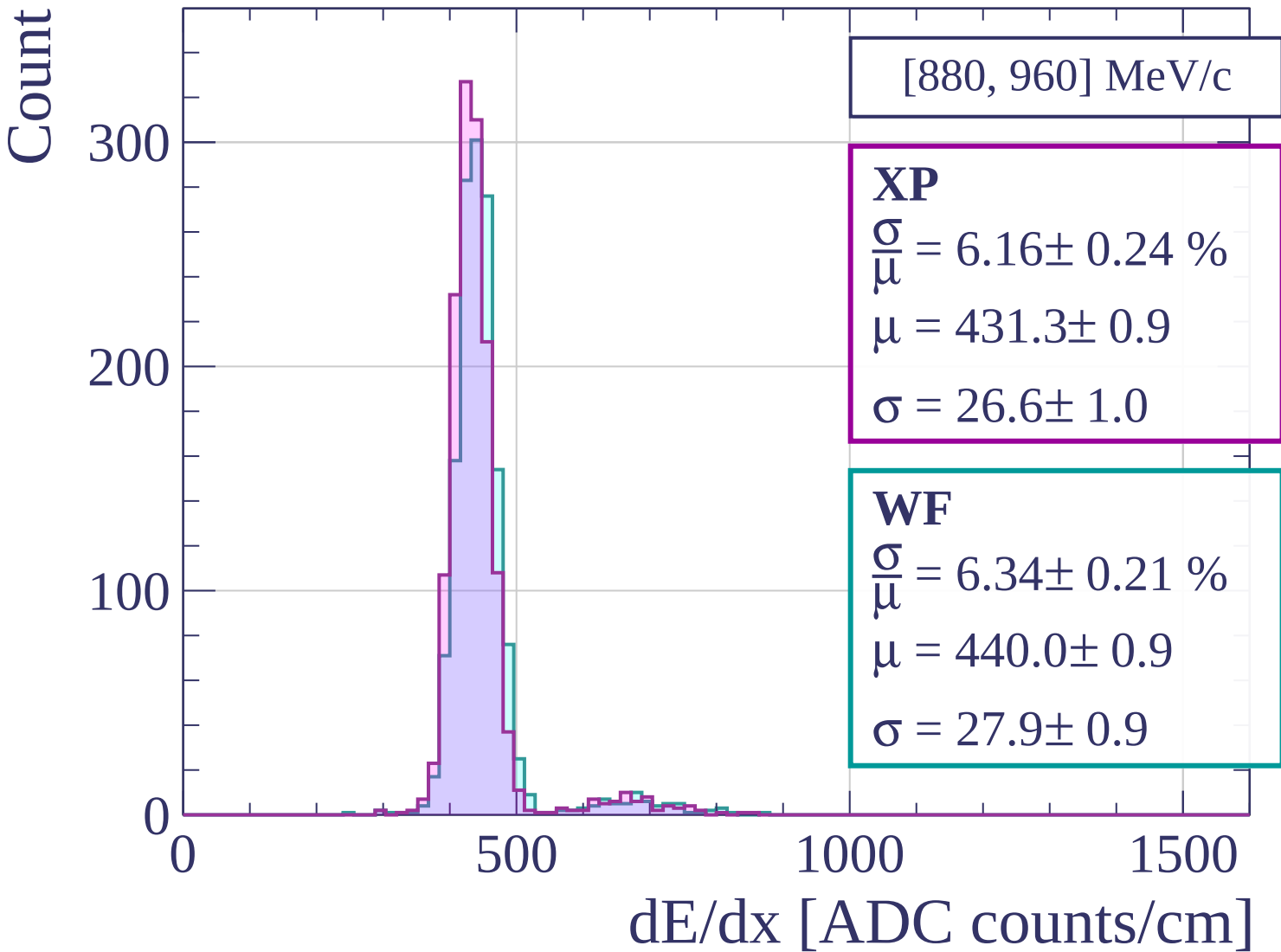
500

1000

1500

dE/dx [ADC counts/cm]





Count

[960, 1040] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.67 \pm 0.20 \%$$

$$\mu = 436.6 \pm 0.8$$

$$\sigma = 24.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.87 \pm 0.23 \%$$

$$\mu = 445.6 \pm 0.9$$

$$\sigma = 26.1 \pm 1.0$$

300

200

100

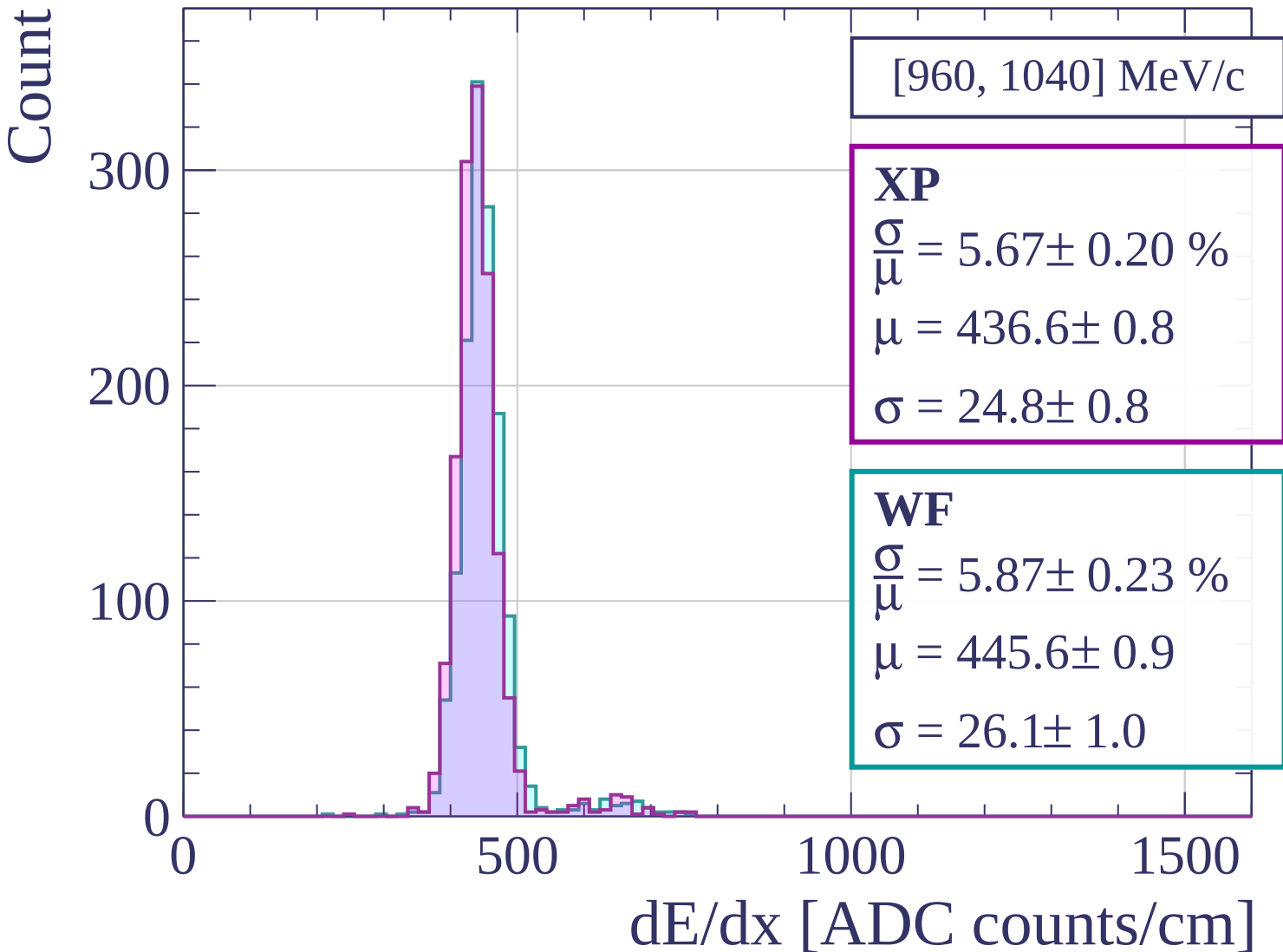
0

500

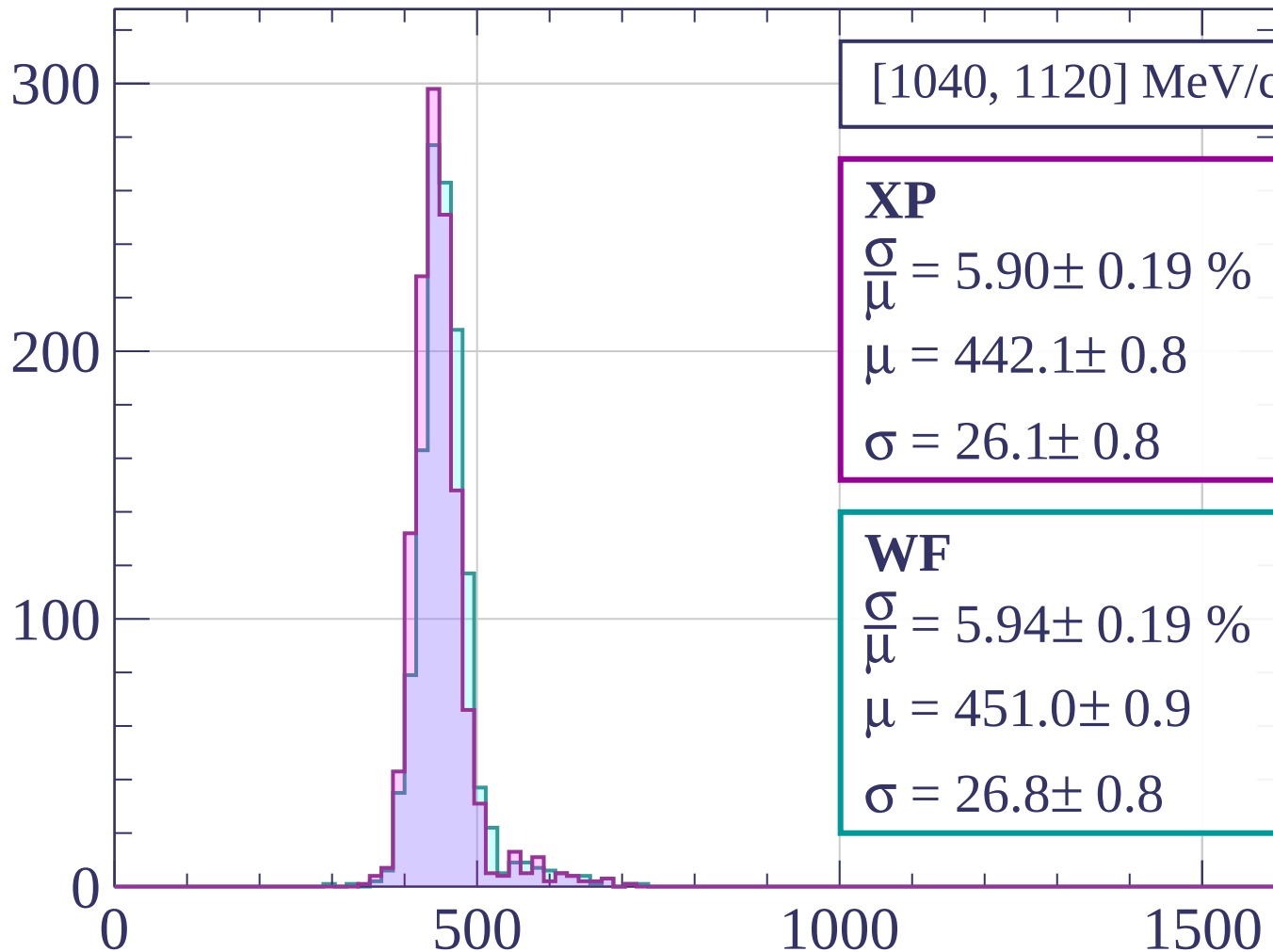
1000

1500

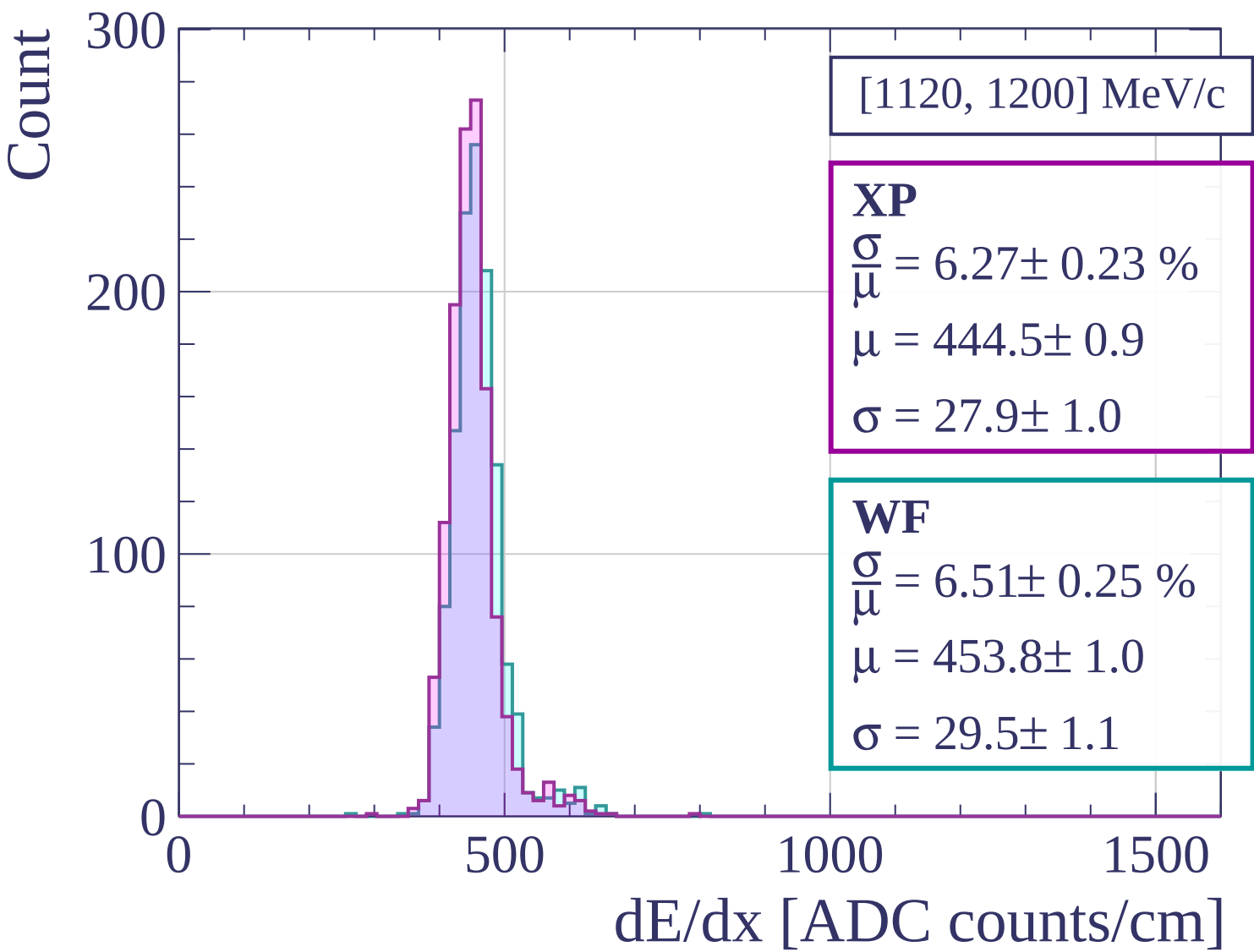
dE/dx [ADC counts/cm]



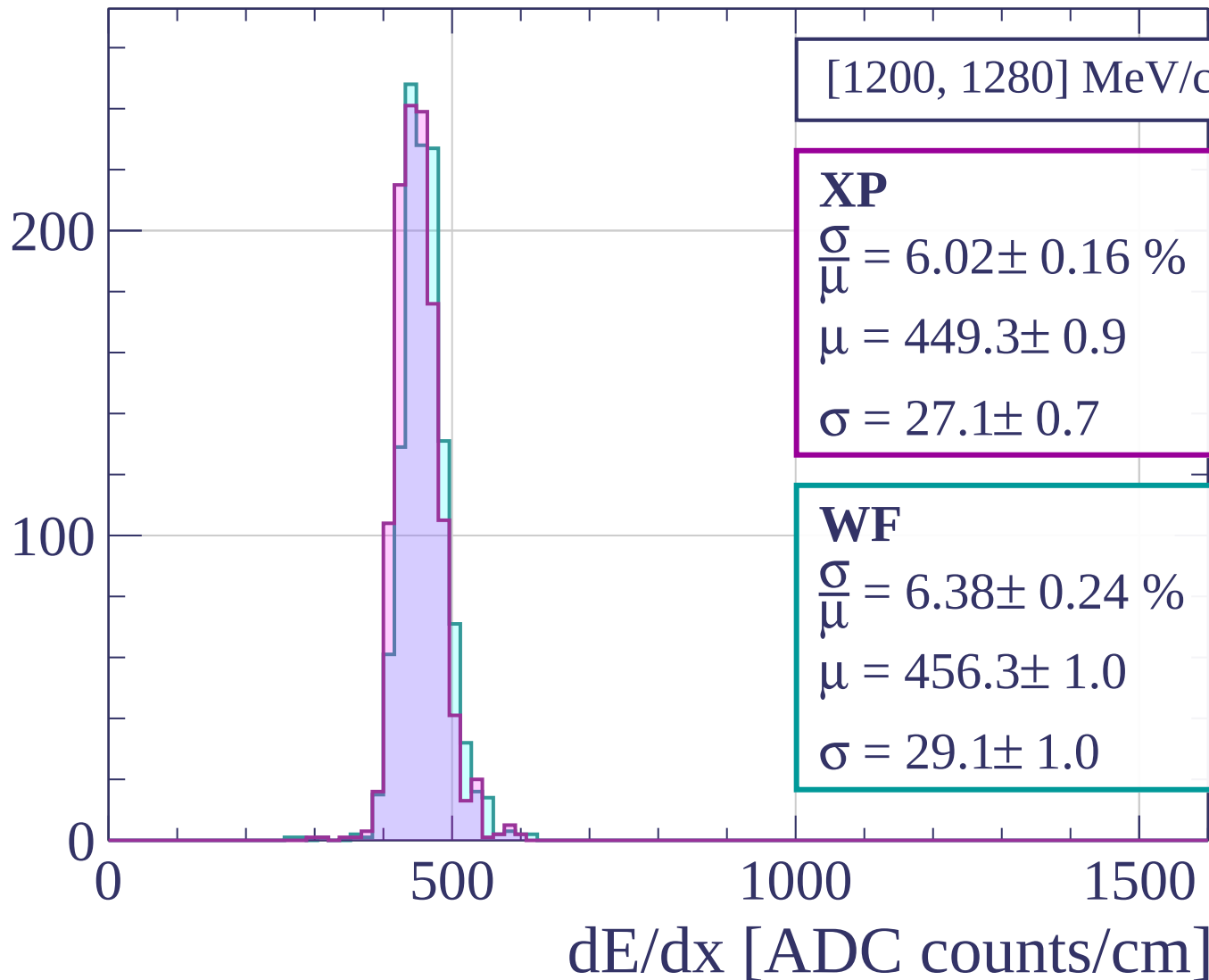
Count



dE/dx [ADC counts/cm]



Count



Count

[1280, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.23 \pm 0.24 \%$$

$$\mu = 452.6 \pm 1.0$$

$$\sigma = 28.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 0.21 \%$$

$$\mu = 462.2 \pm 1.0$$

$$\sigma = 29.9 \pm 0.9$$

200

100

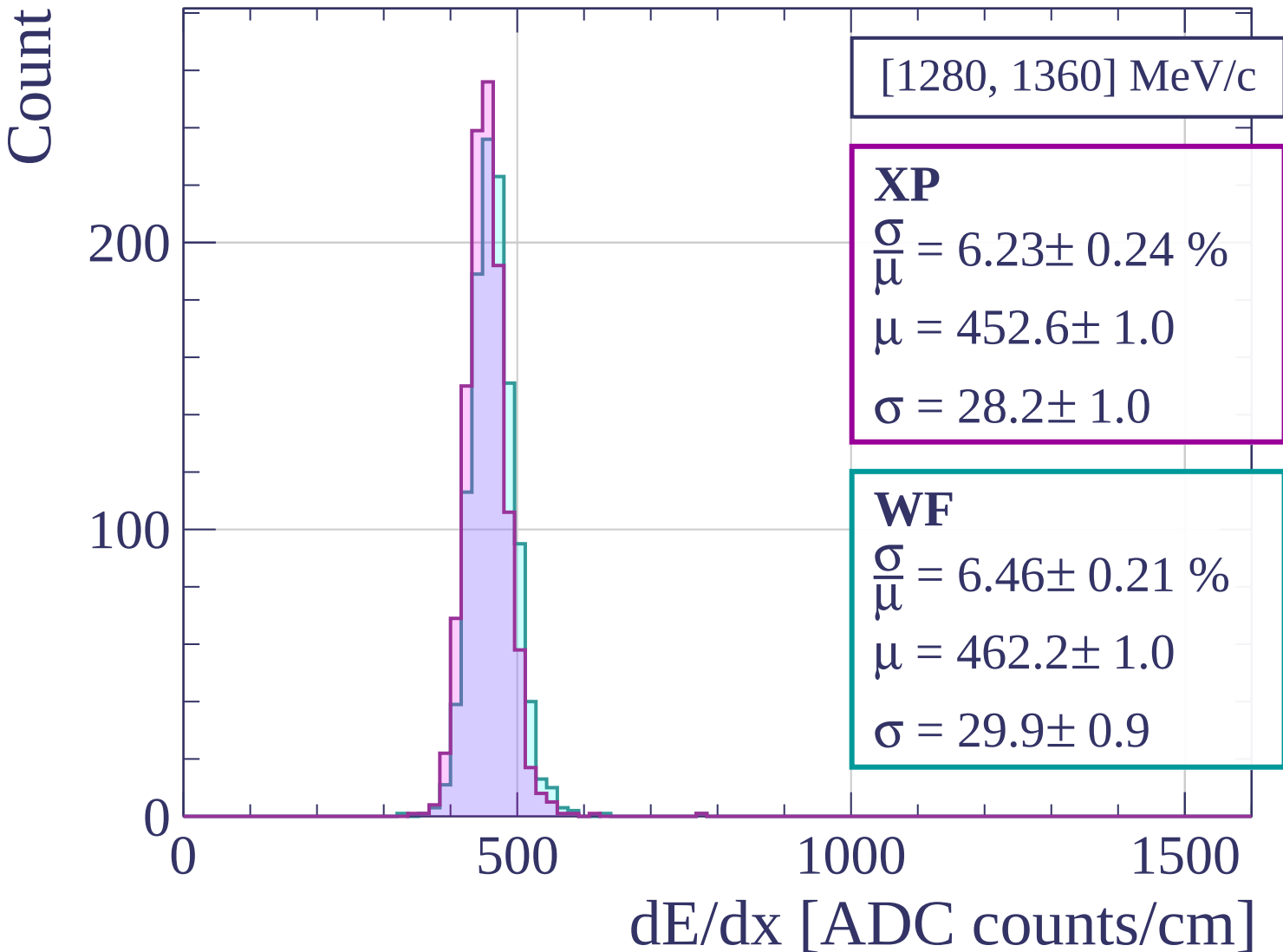
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1360, 1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.22 \pm 0.23 \%$$

$$\mu = 455.1 \pm 1.0$$

$$\sigma = 28.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.28 \pm 0.20 \%$$

$$\mu = 464.8 \pm 1.0$$

$$\sigma = 29.2 \pm 0.9$$

200

100

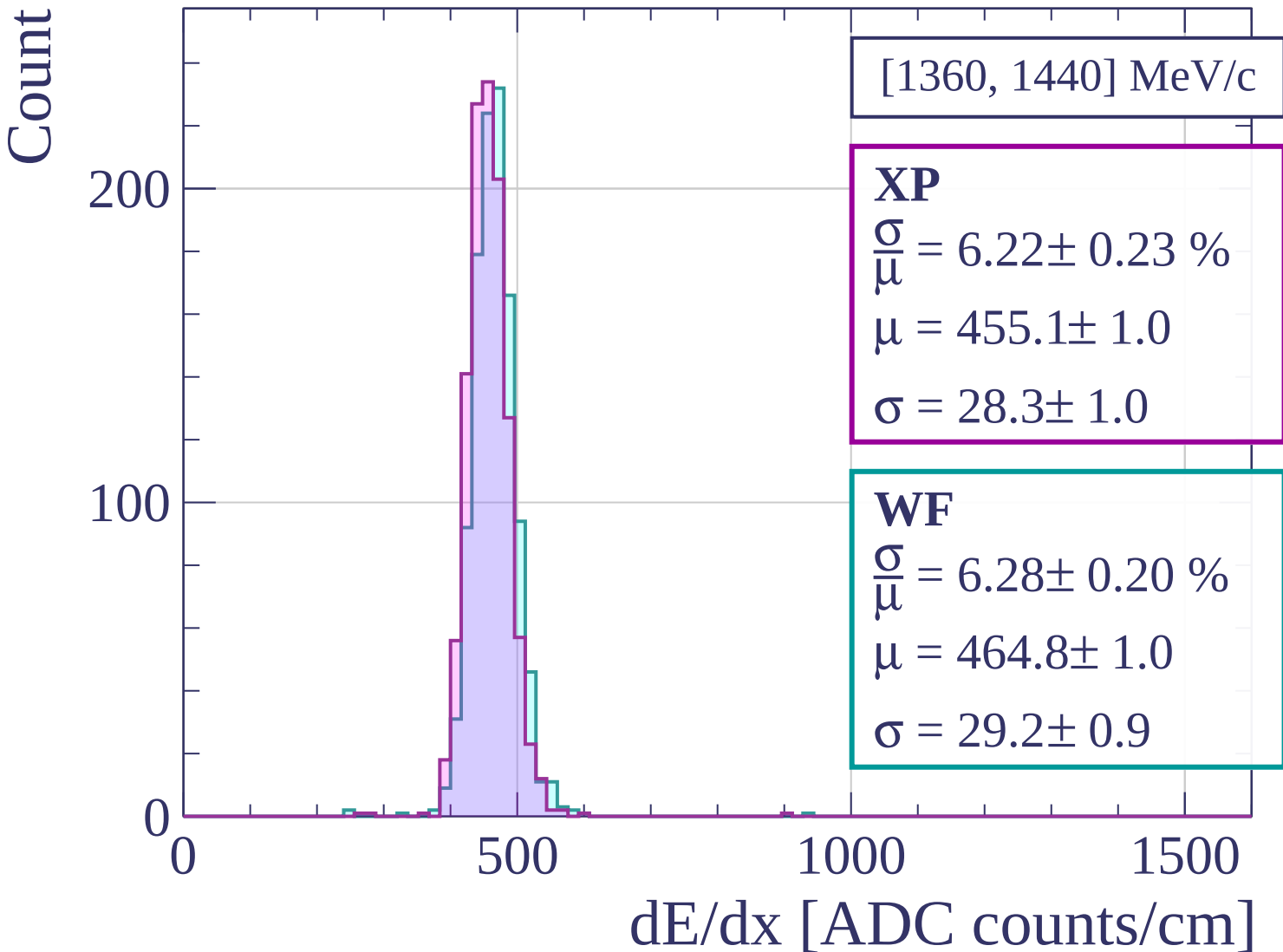
0

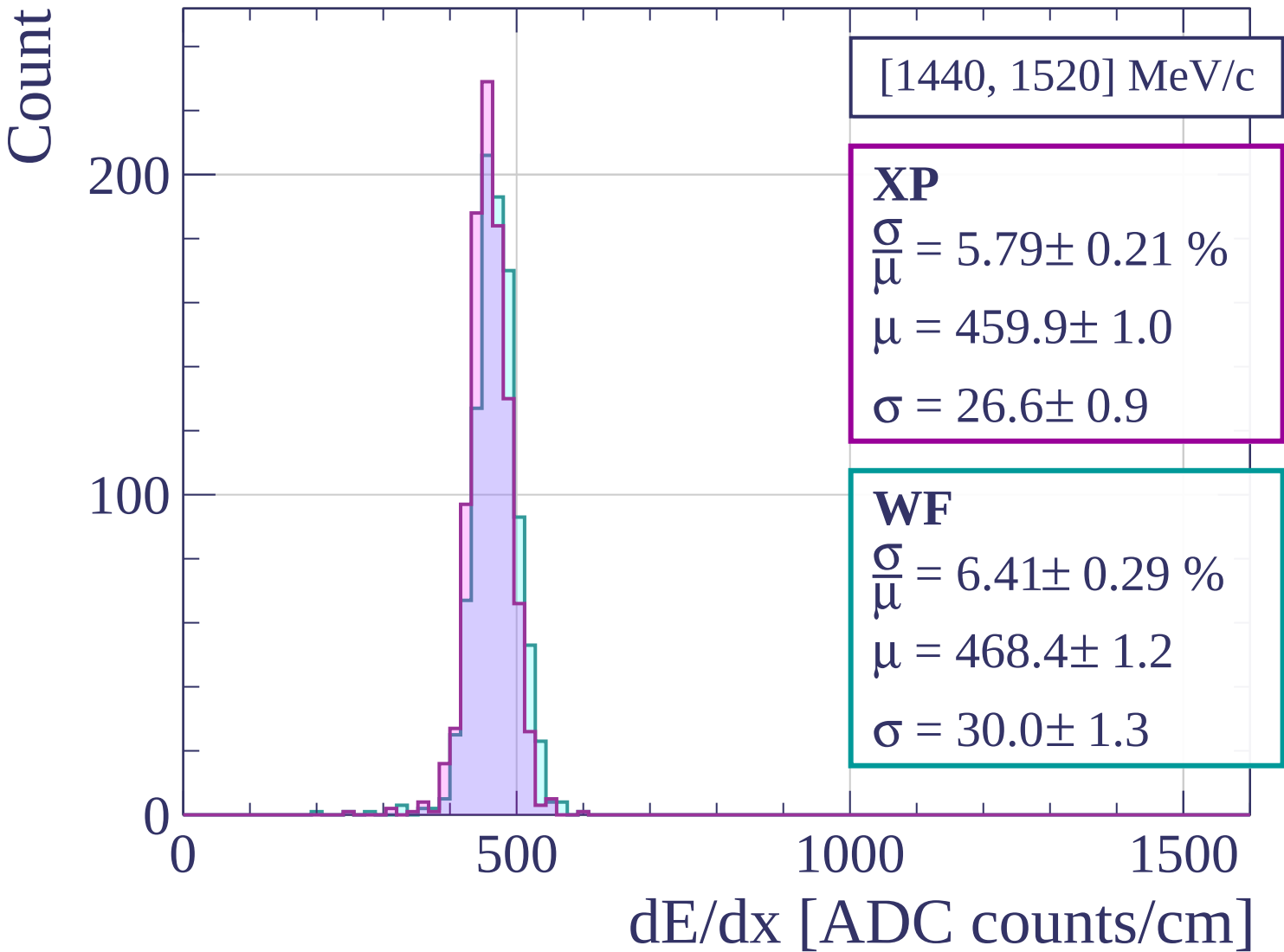
500

1000

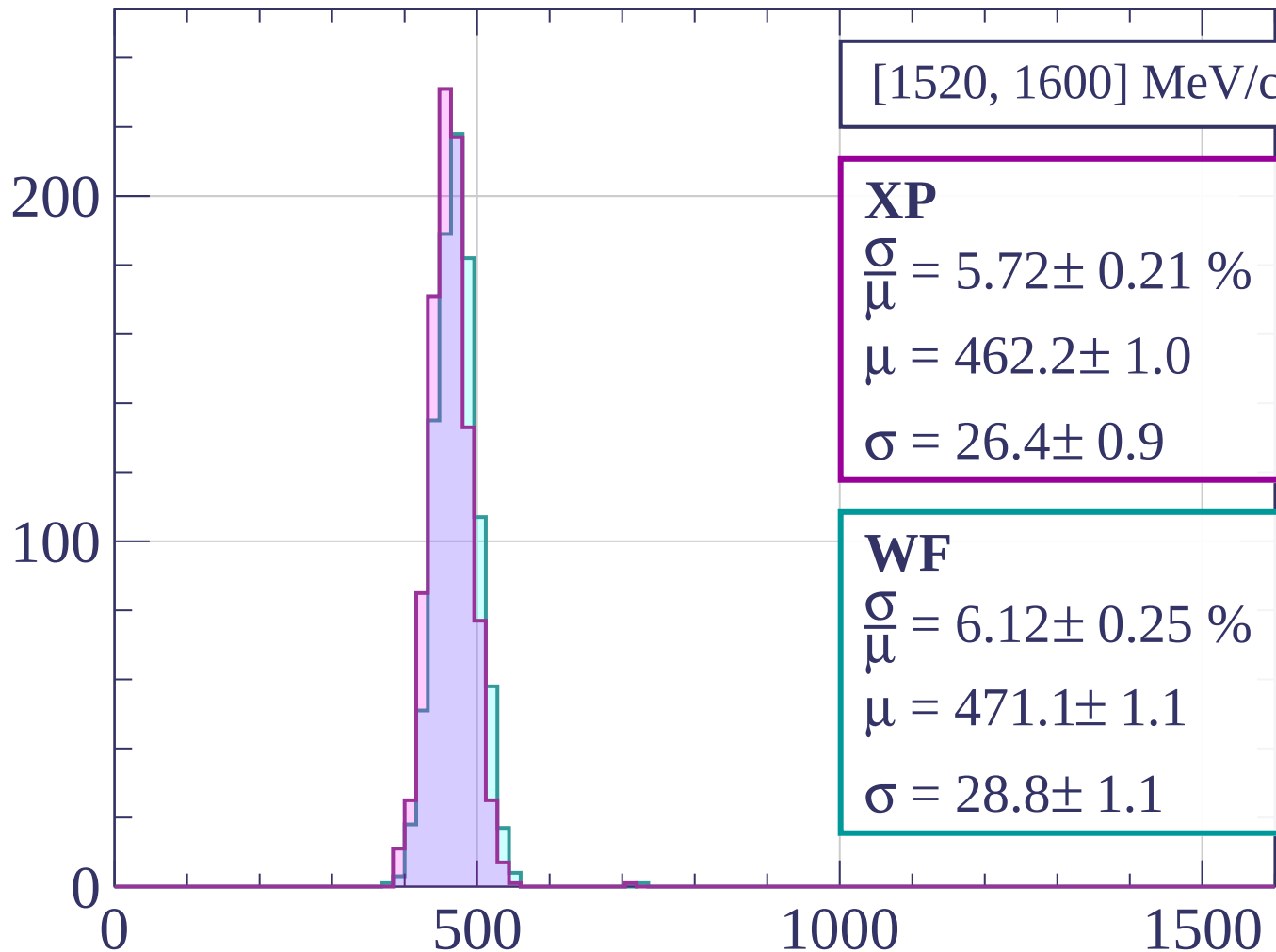
1500

dE/dx [ADC counts/cm]

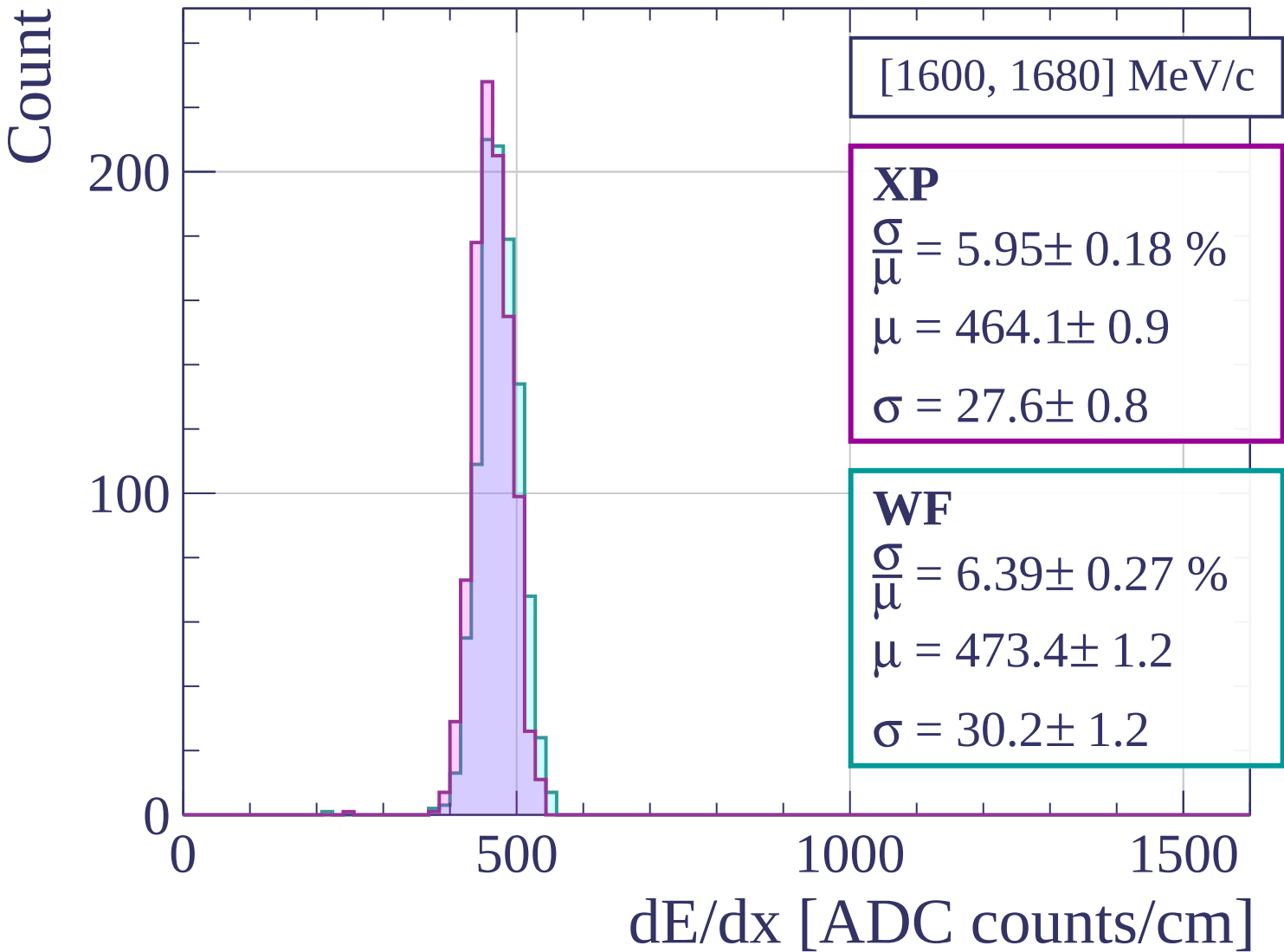




Count



dE/dx [ADC counts/cm]



Count

[1680, 1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.18 \pm 0.26 \%$$

$$\mu = 466.1 \pm 1.1$$

$$\sigma = 28.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.21 \pm 0.28 \%$$

$$\mu = 475.9 \pm 1.2$$

$$\sigma = 29.6 \pm 1.3$$

200

100

0

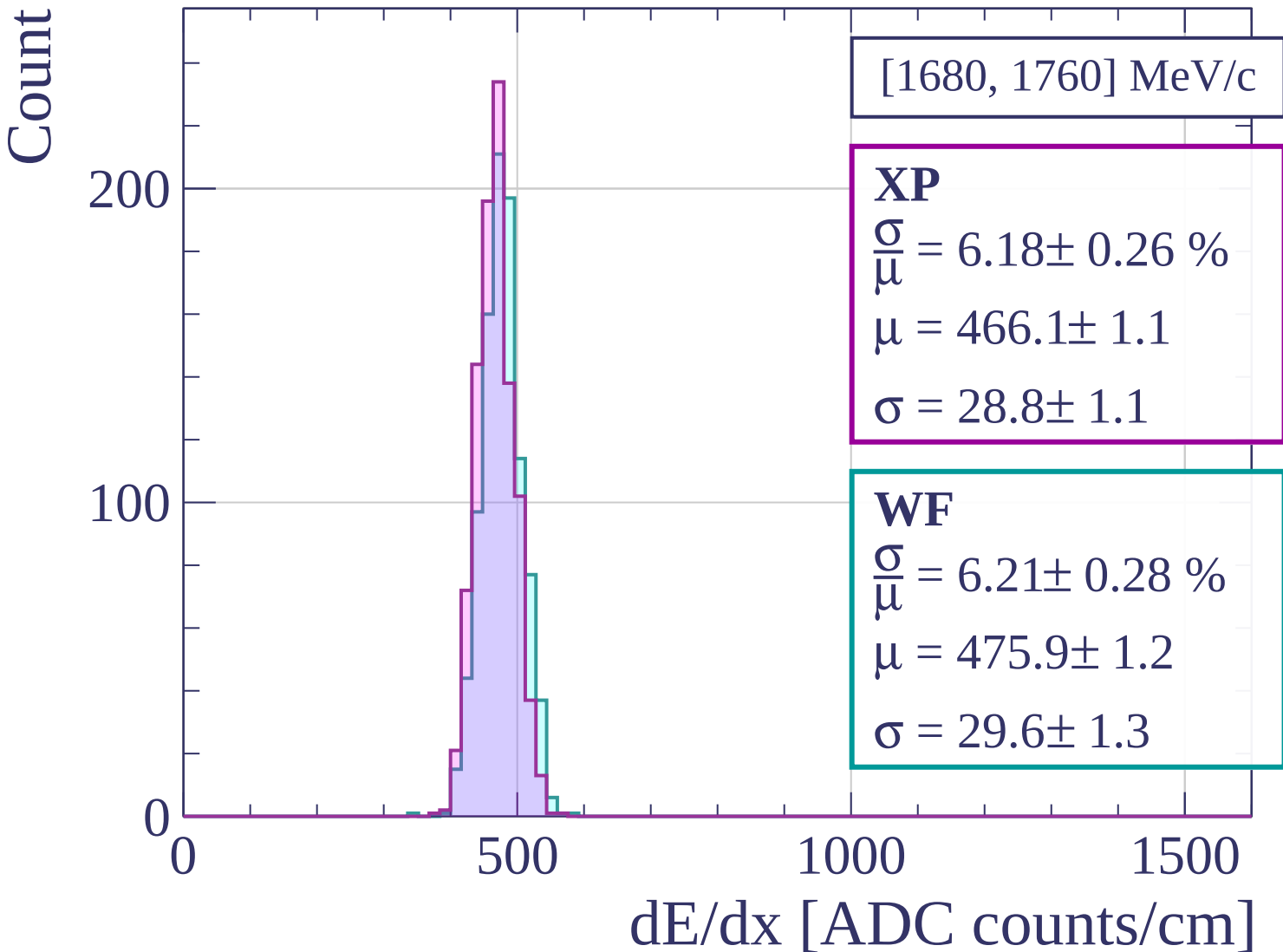
0

500

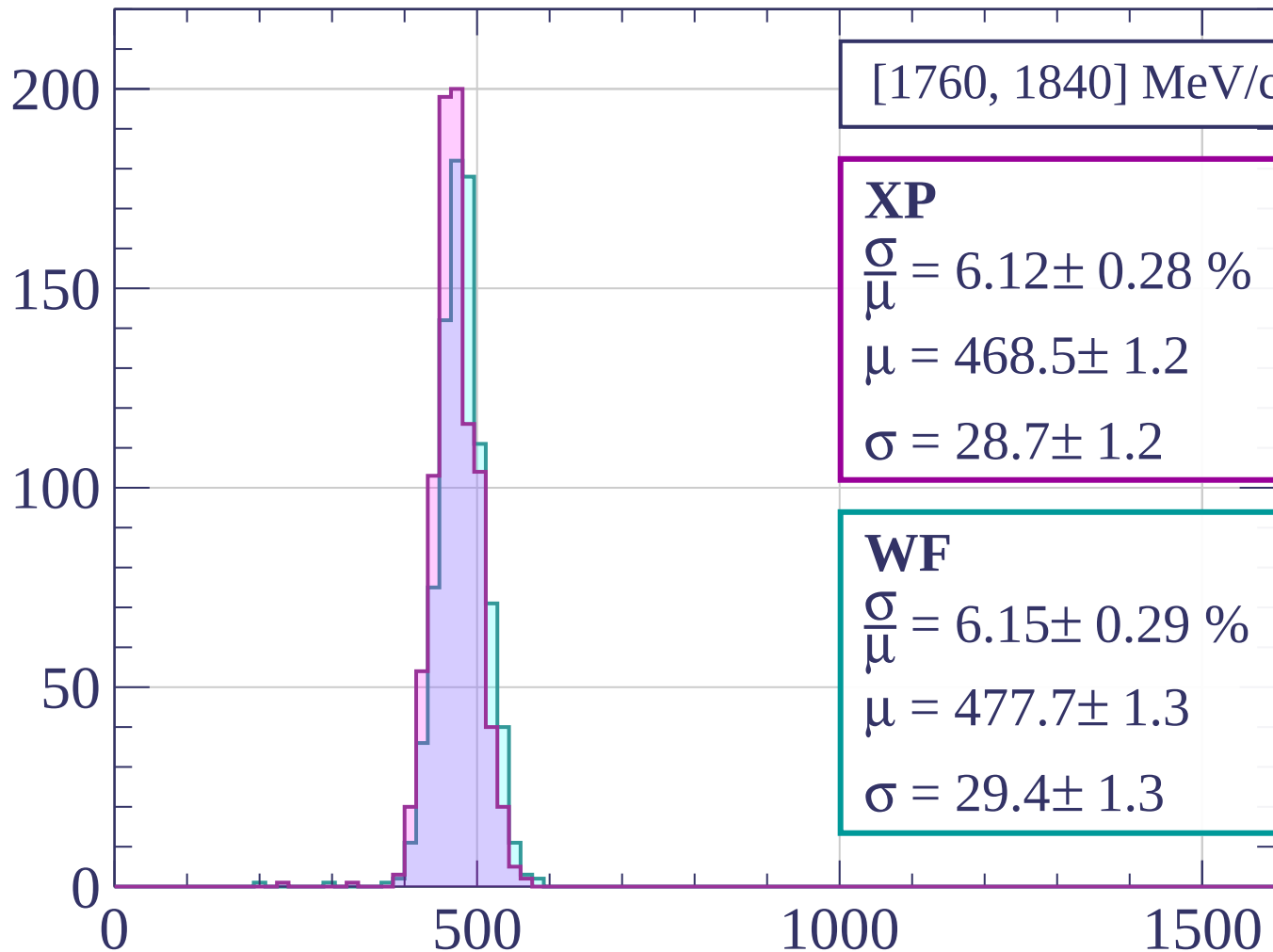
1000

1500

dE/dx [ADC counts/cm]



Count



[1760, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.12 \pm 0.28 \%$$

$$\mu = 468.5 \pm 1.2$$

$$\sigma = 28.7 \pm 1.2$$

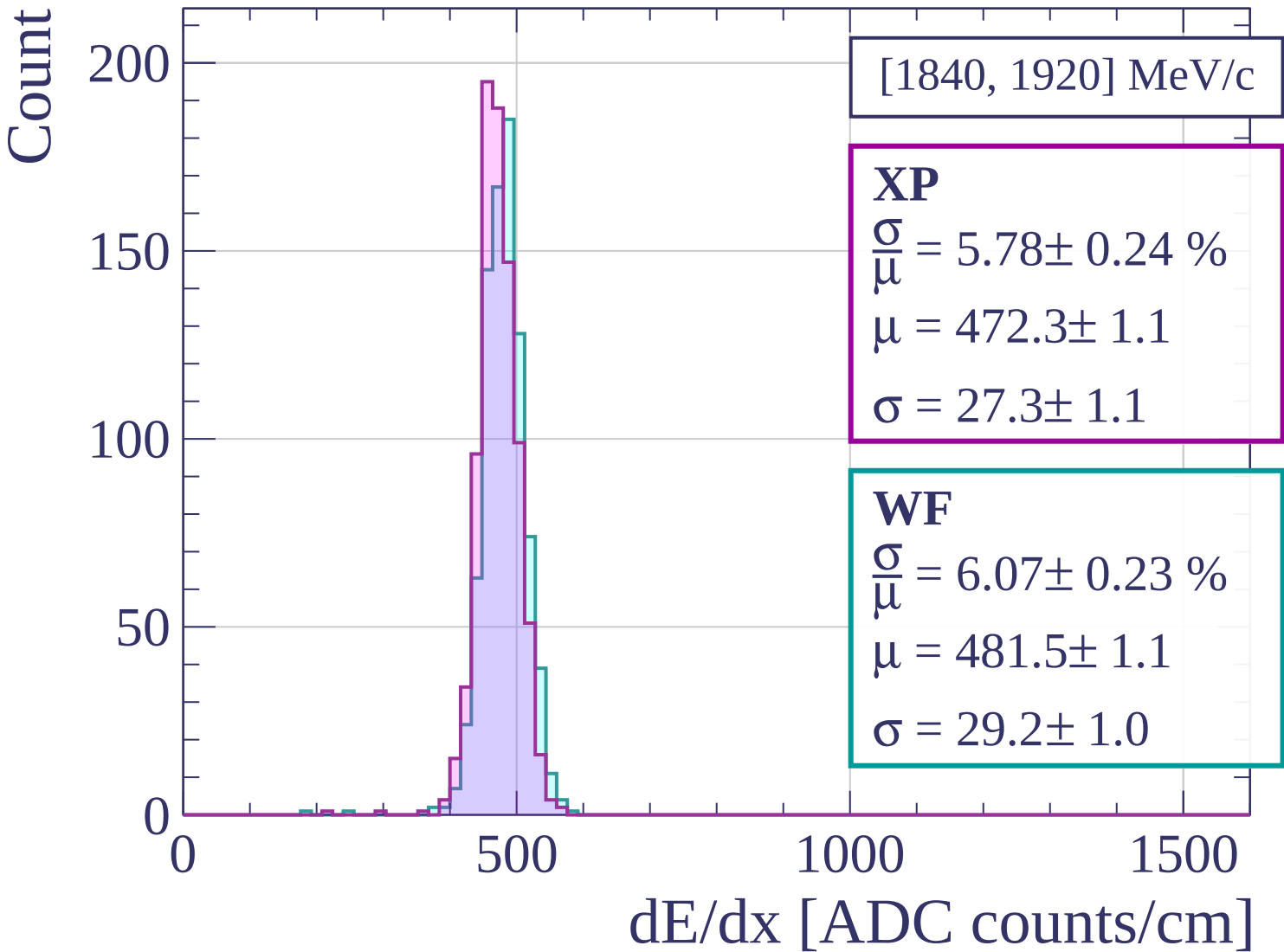
WF

$$\frac{\sigma}{\mu} = 6.15 \pm 0.29 \%$$

$$\mu = 477.7 \pm 1.3$$

$$\sigma = 29.4 \pm 1.3$$

dE/dx [ADC counts/cm]



Count

[1920, 2000] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.71 \pm 0.22 \%$$

$$\mu = 474.3 \pm 1.1$$

$$\sigma = 27.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 5.88 \pm 0.20 \%$$

$$\mu = 484.7 \pm 1.1$$

$$\sigma = 28.5 \pm 0.9$$

150

100

50

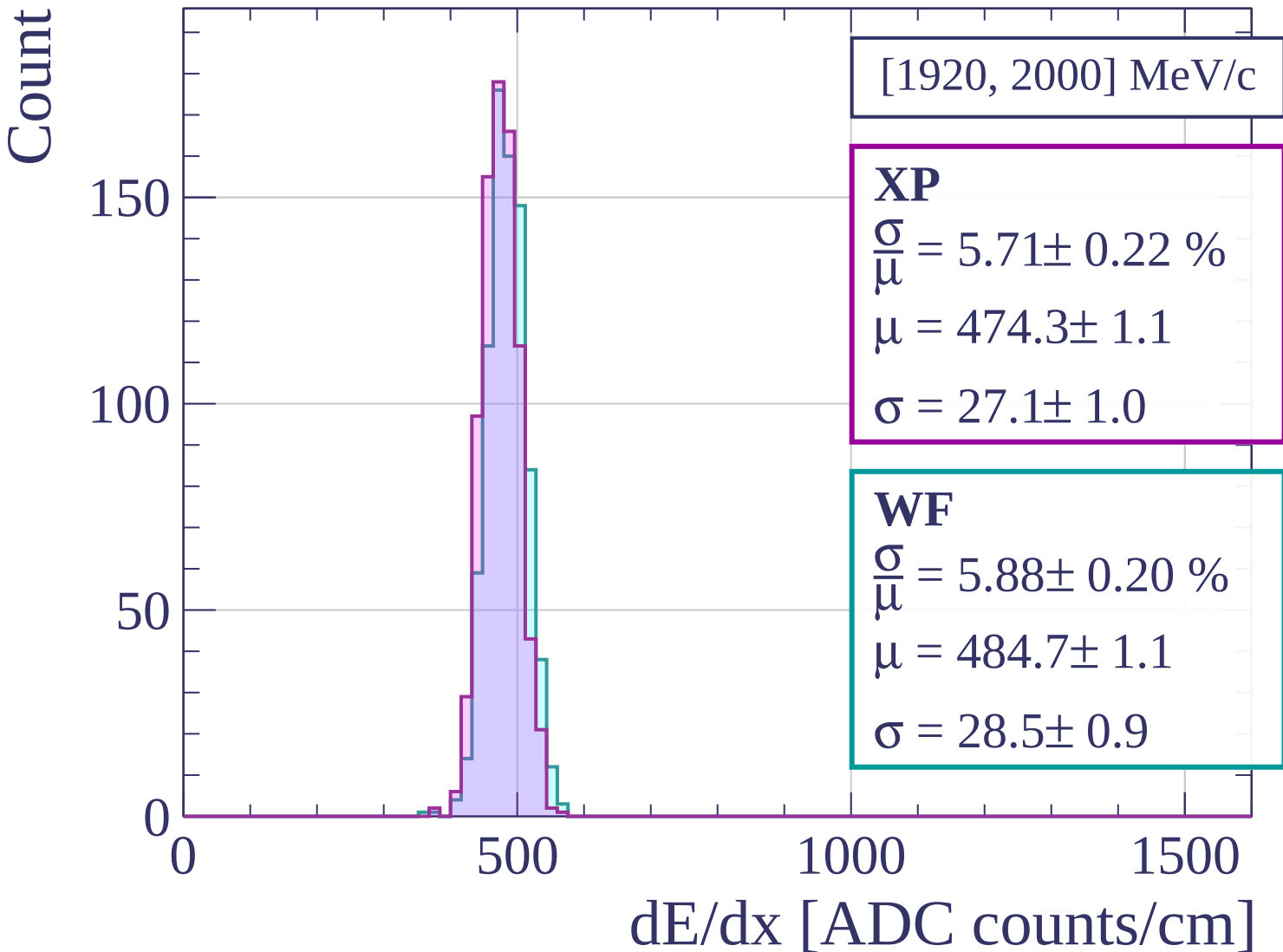
0

500

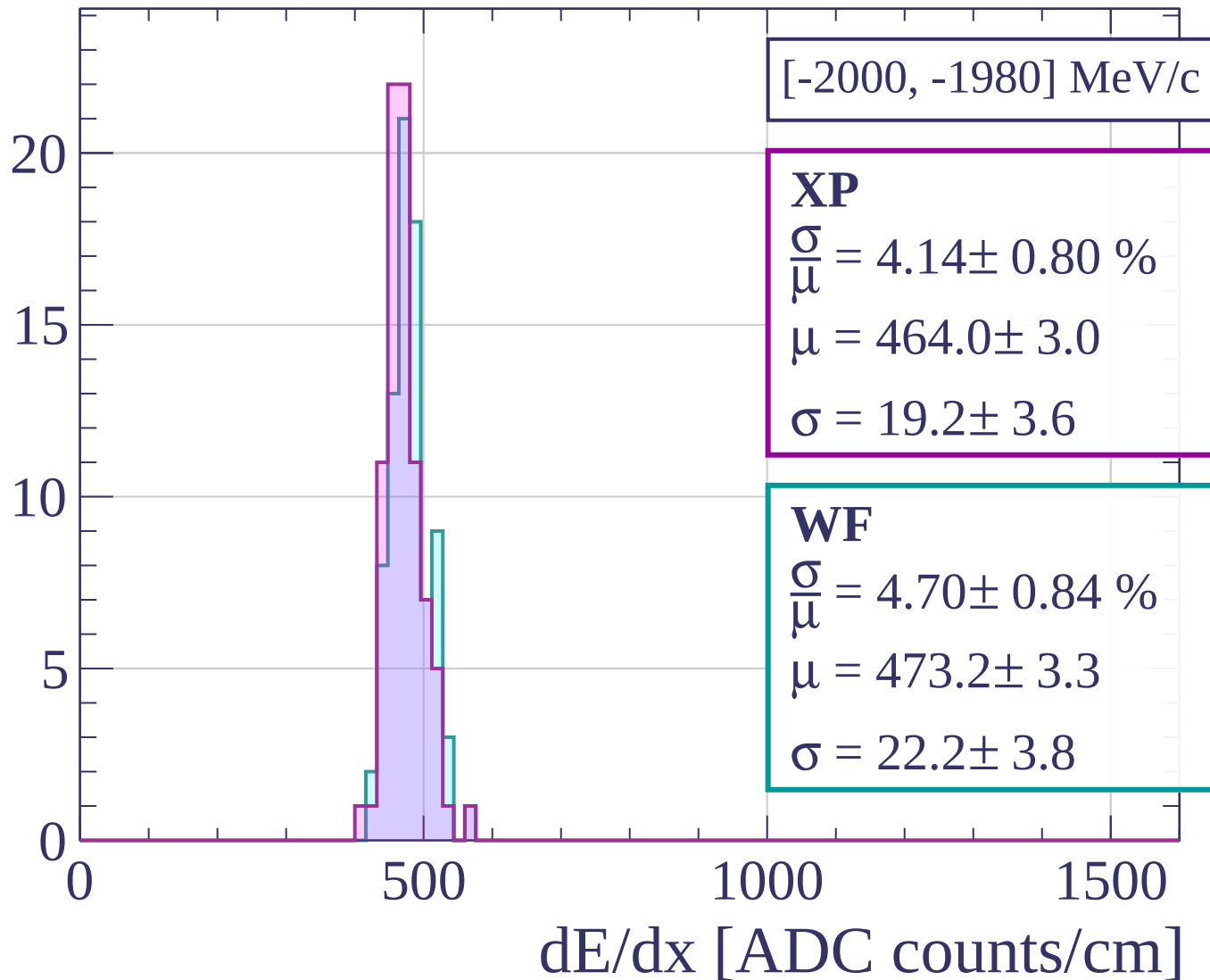
1000

1500

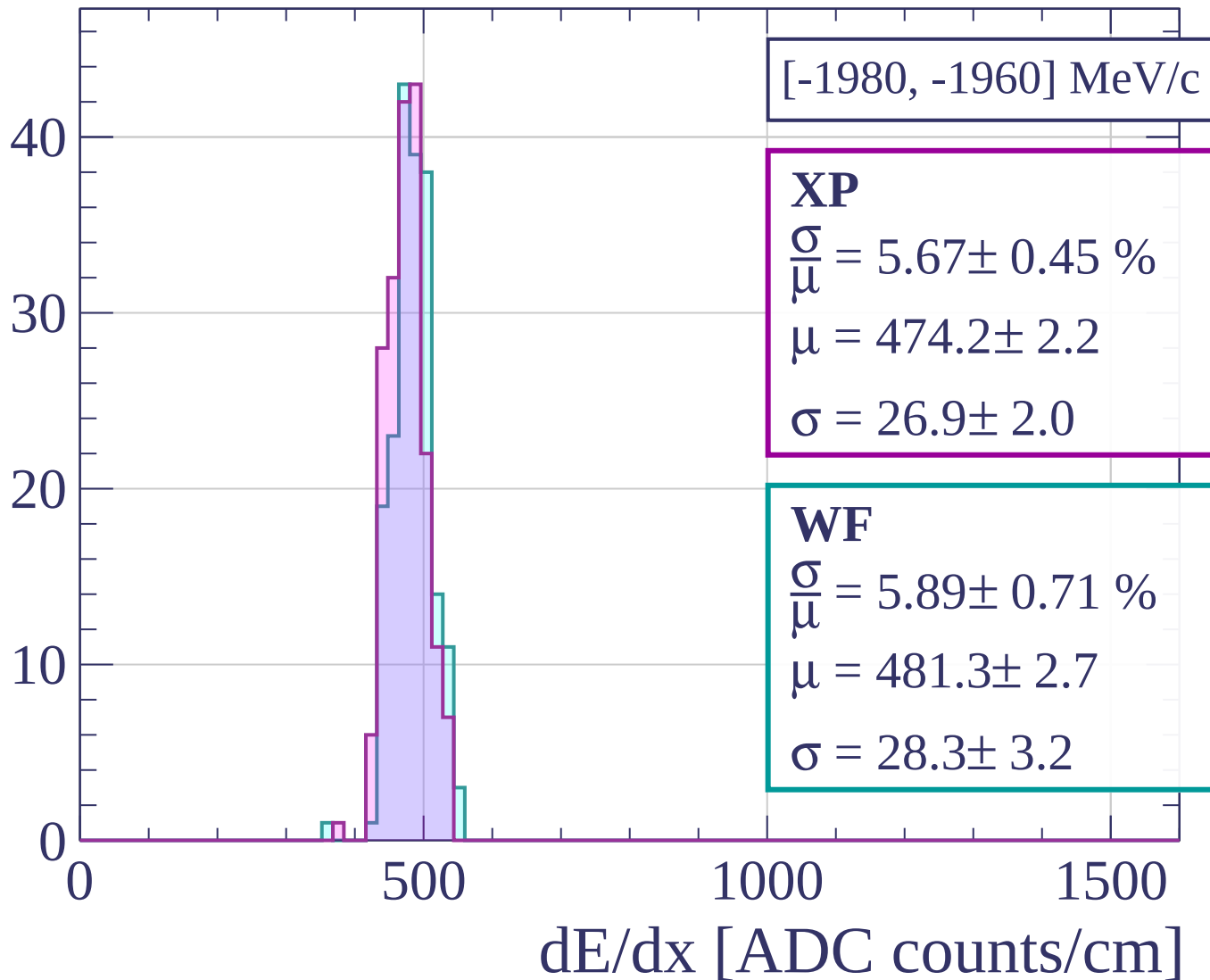
dE/dx [ADC counts/cm]



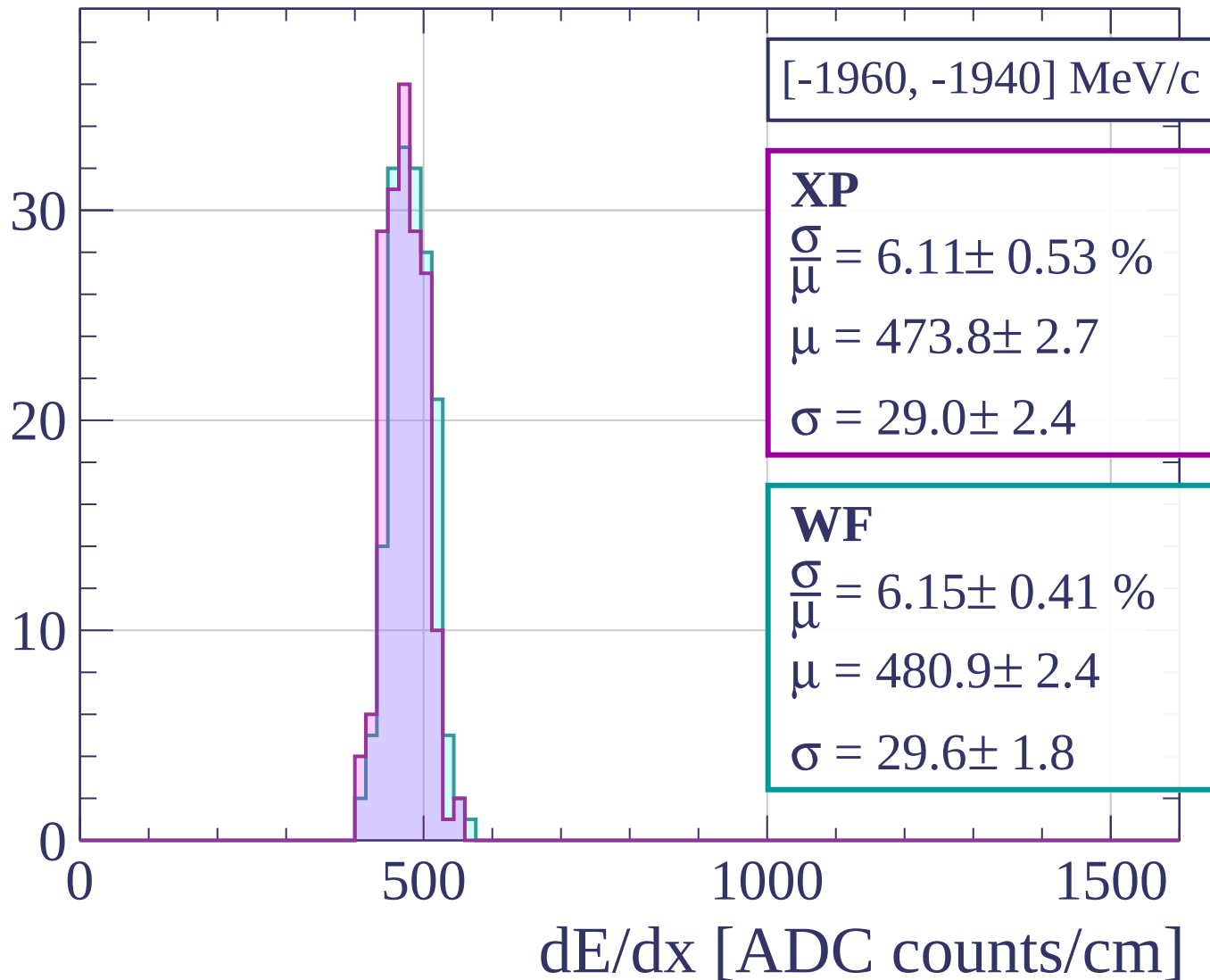
Count



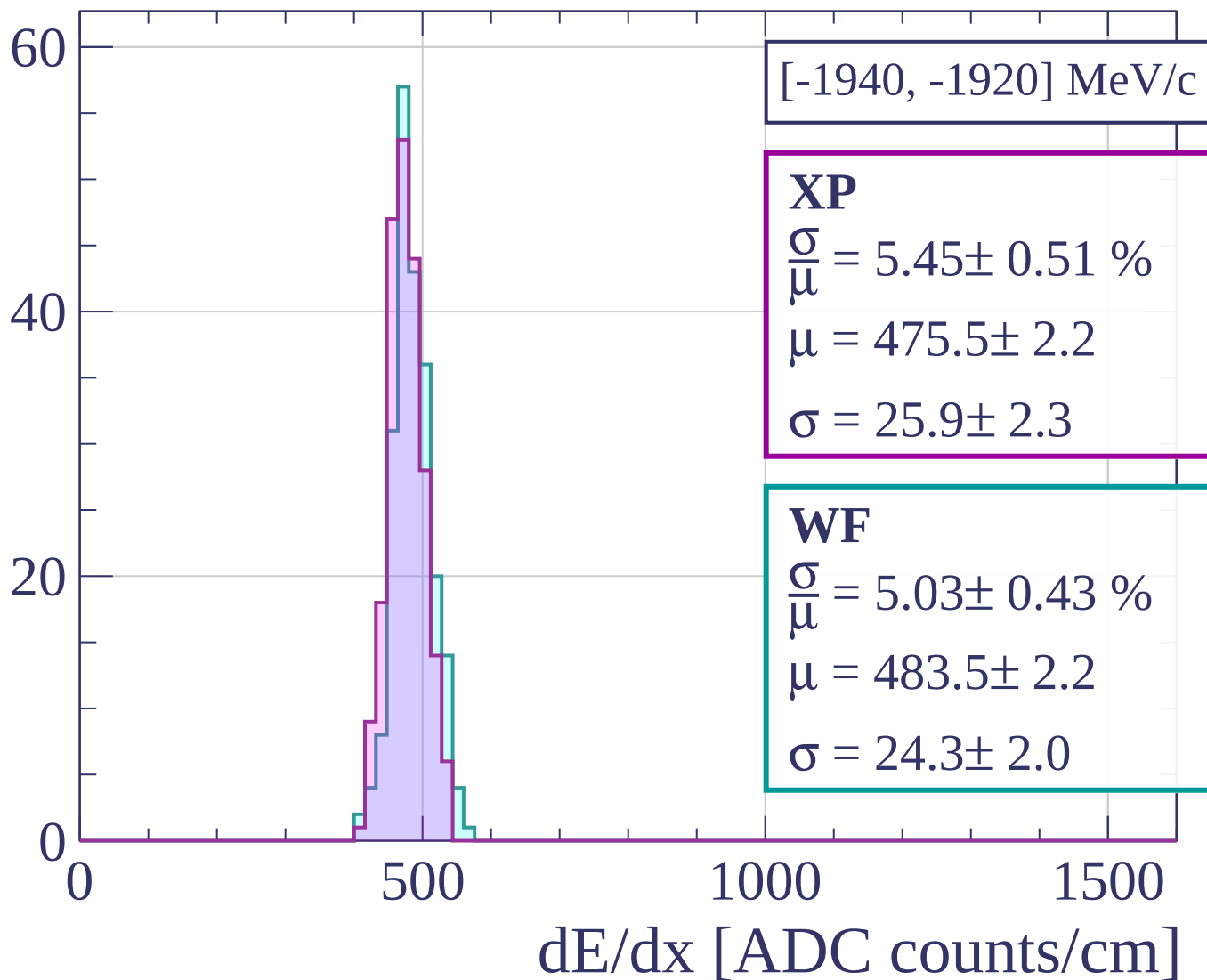
Count



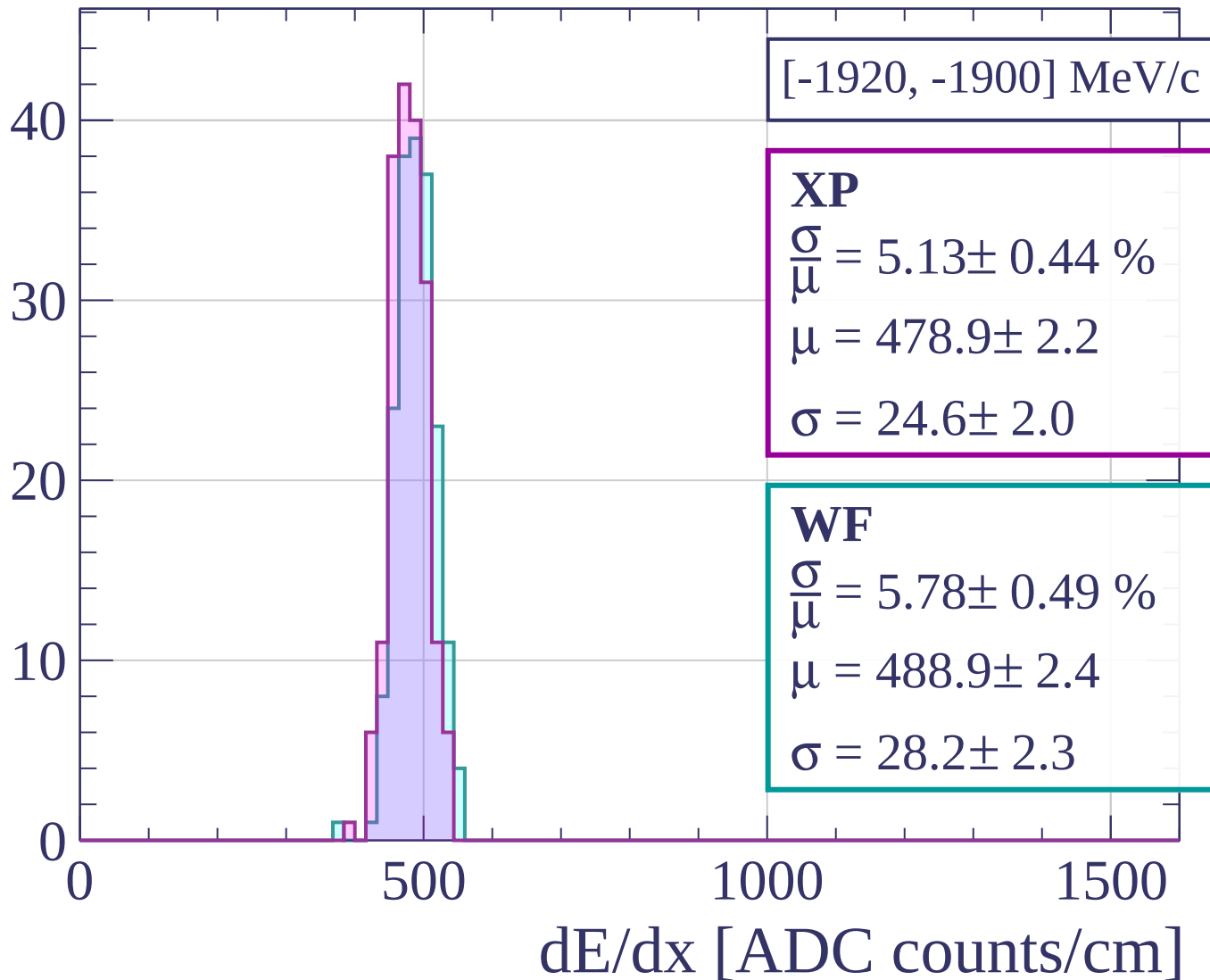
Count



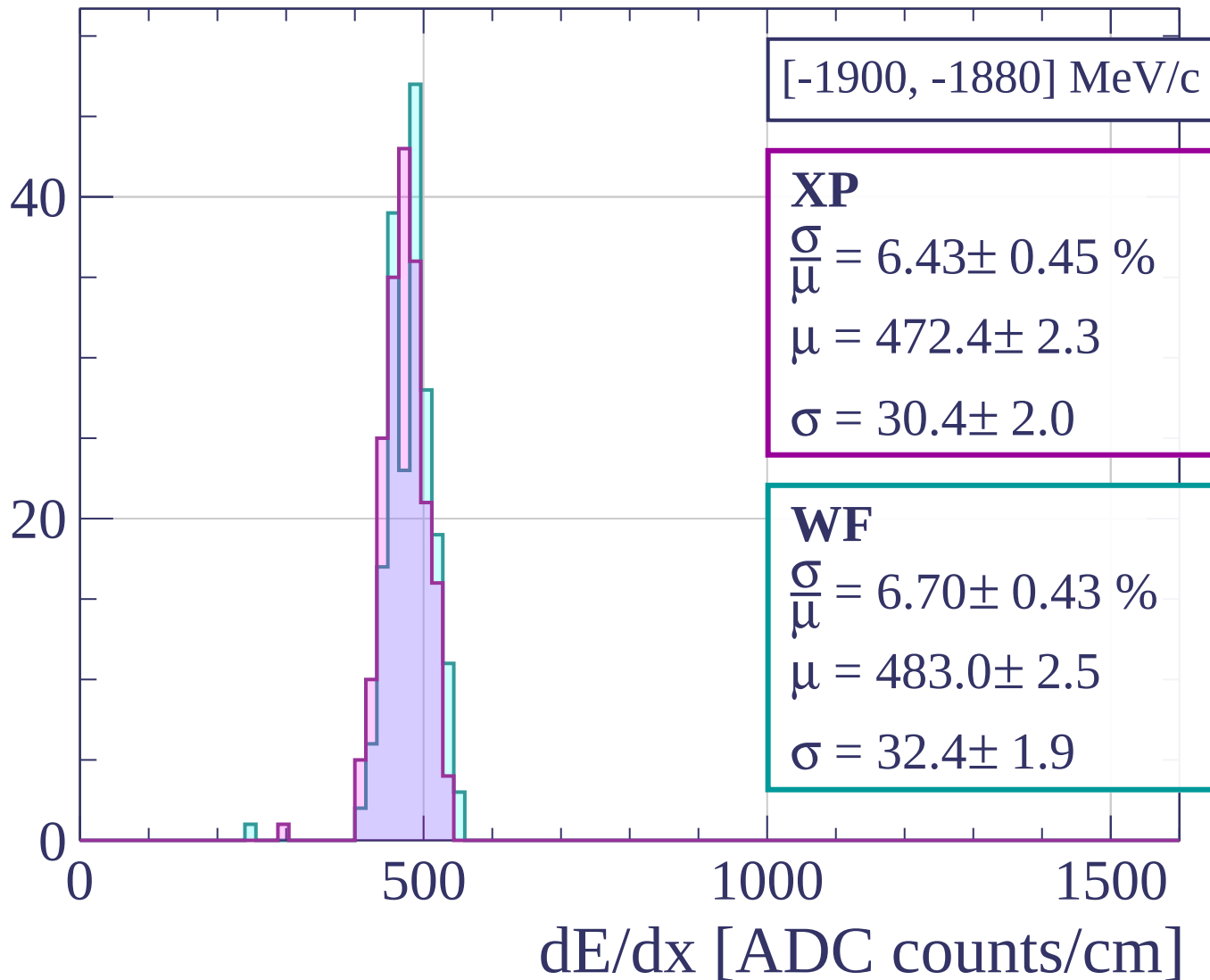
Count



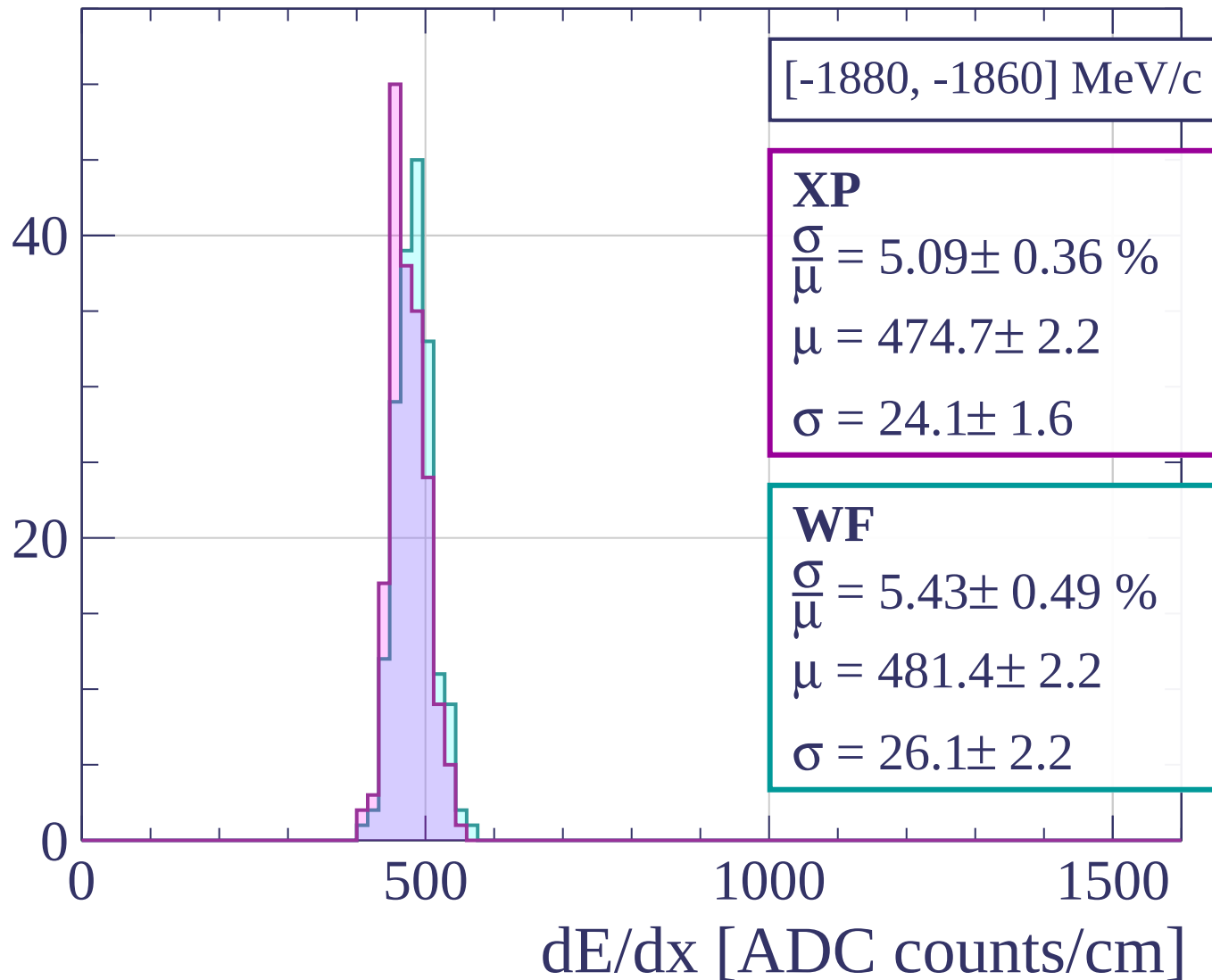
Count



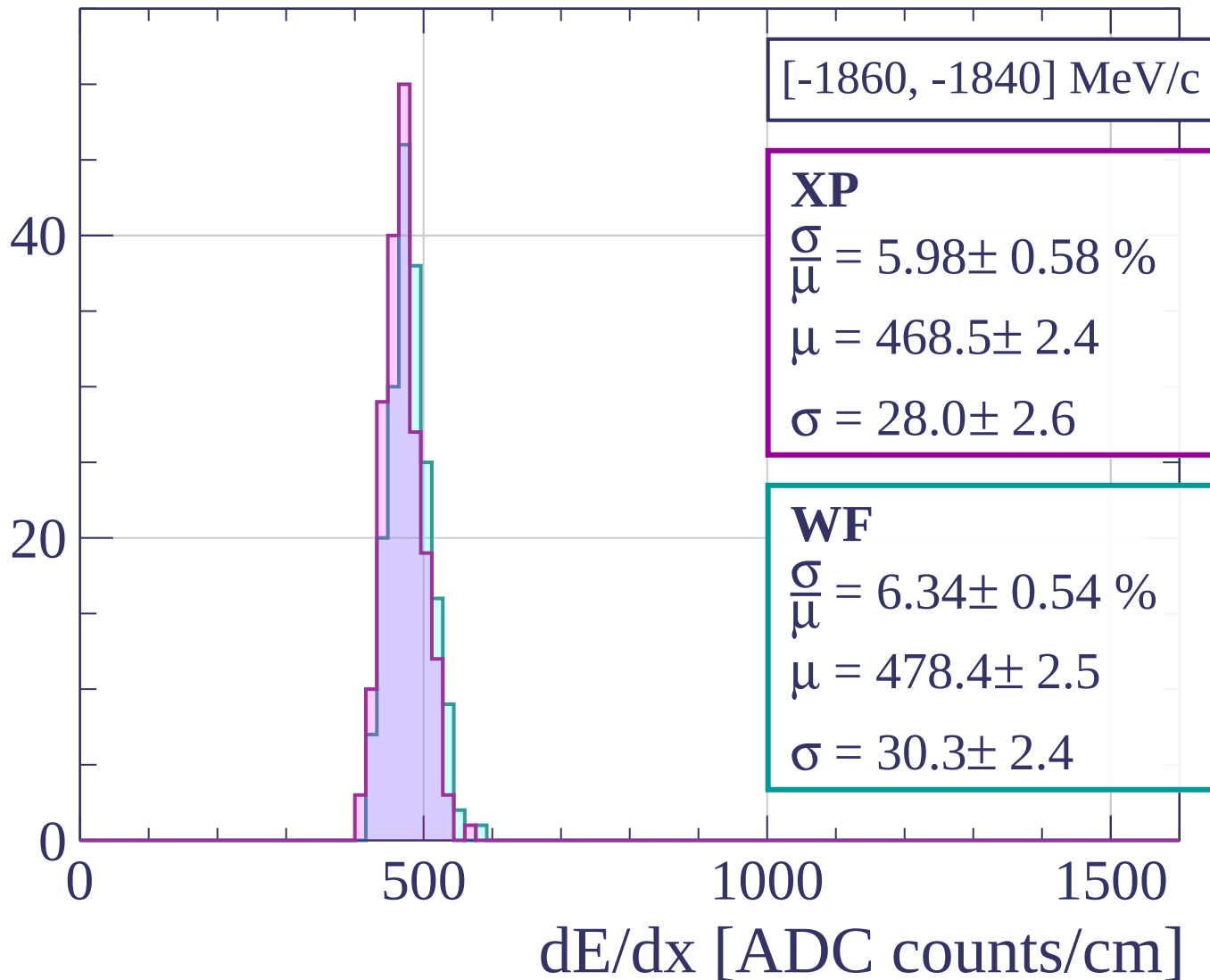
Count



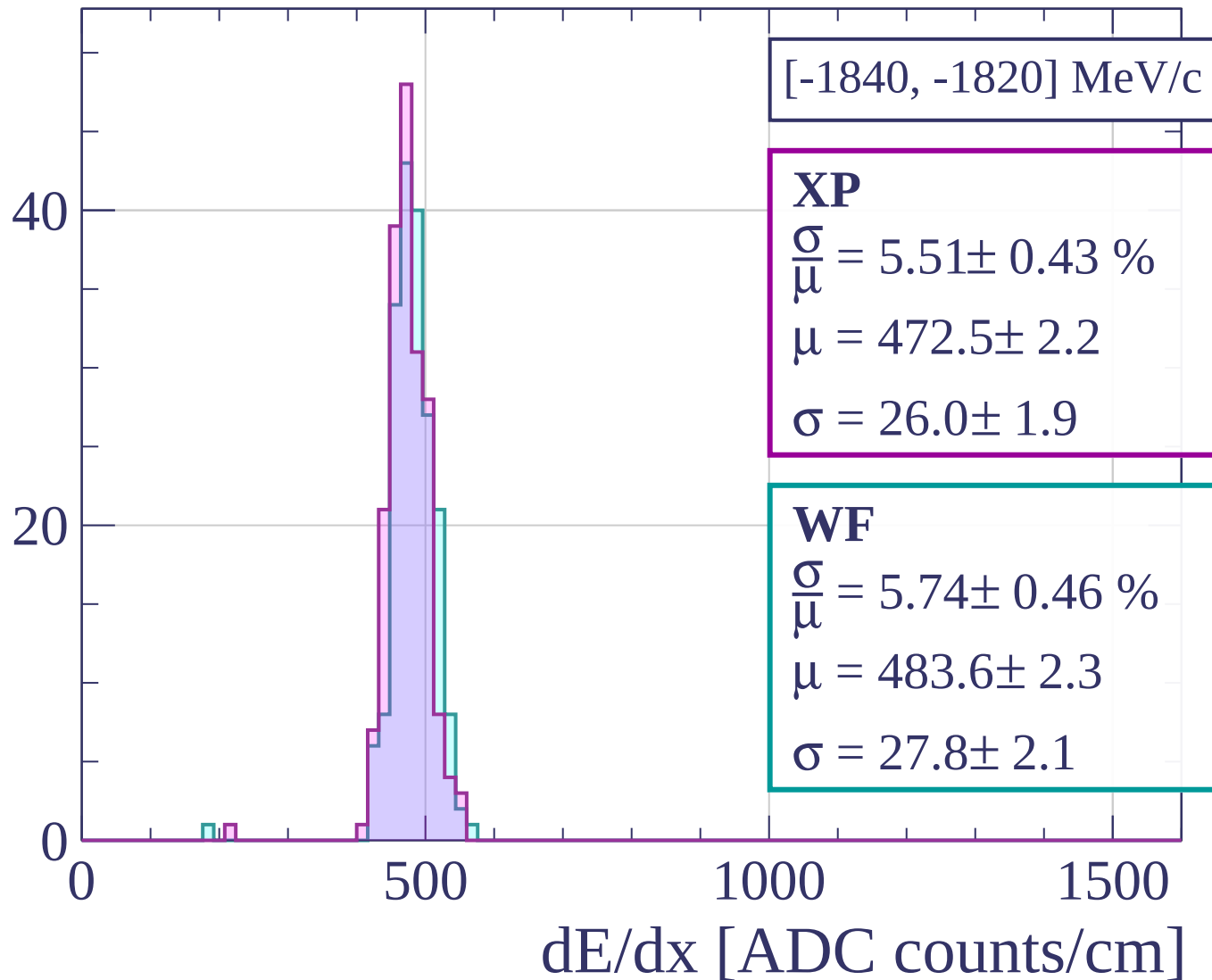
Count



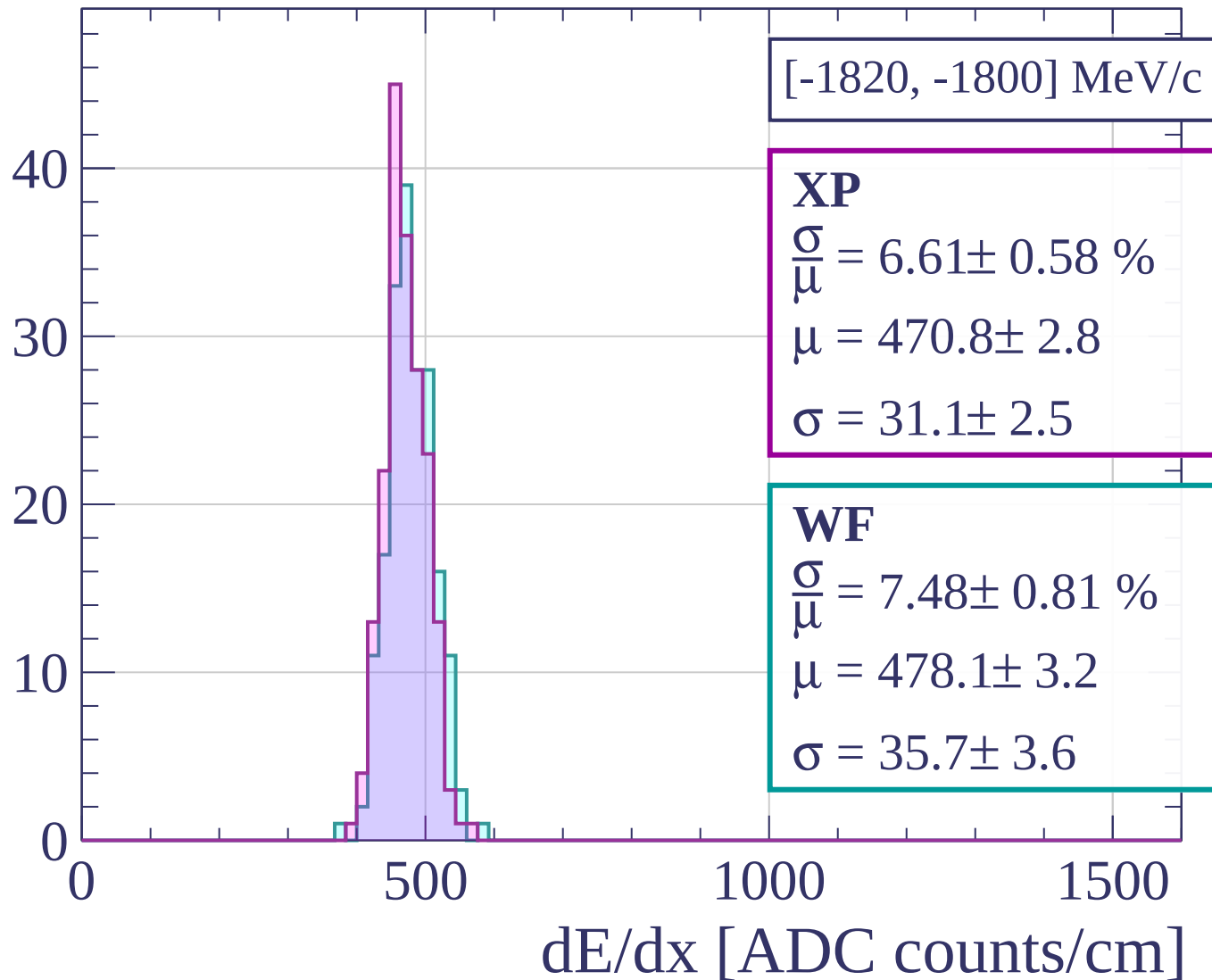
Count



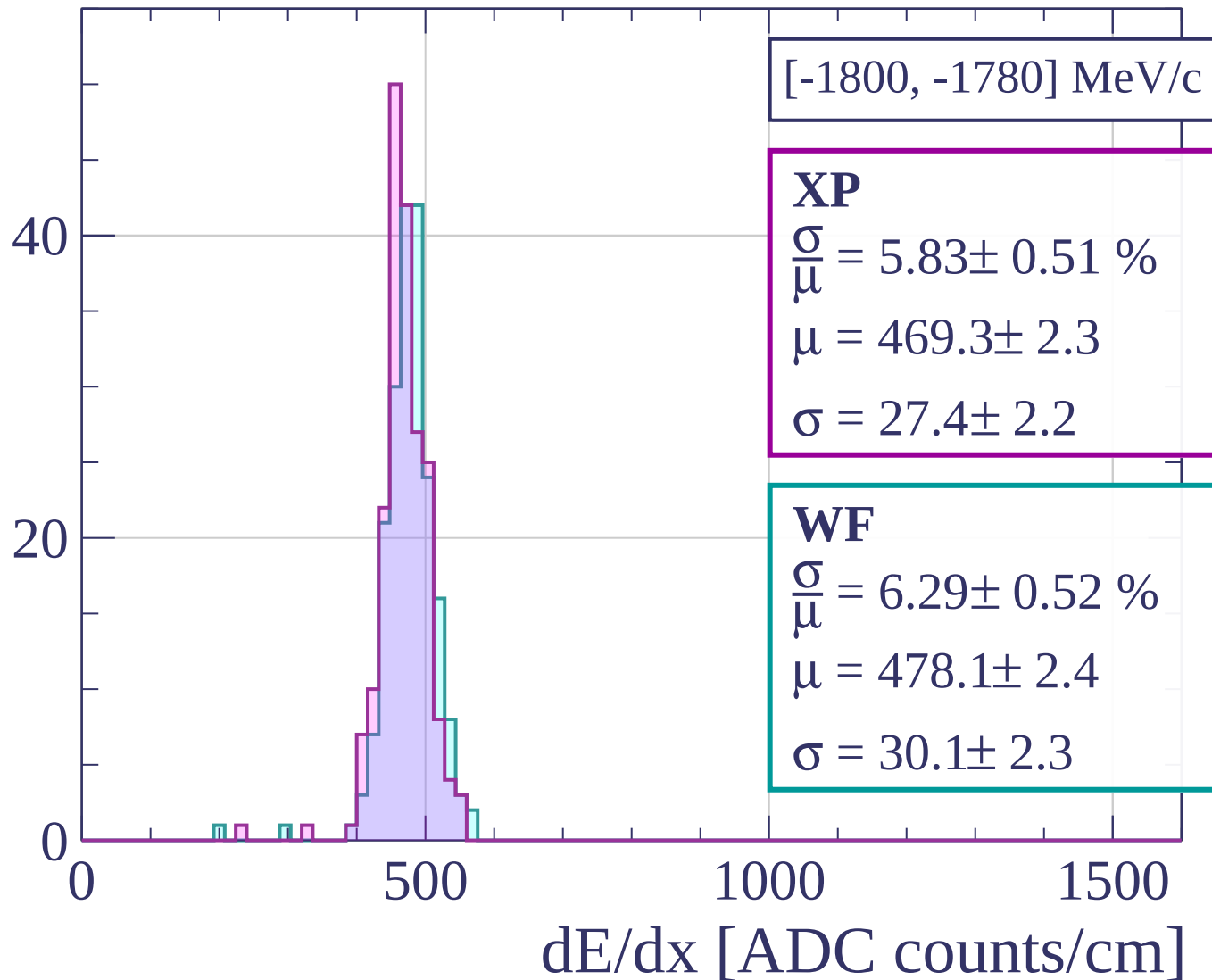
Count



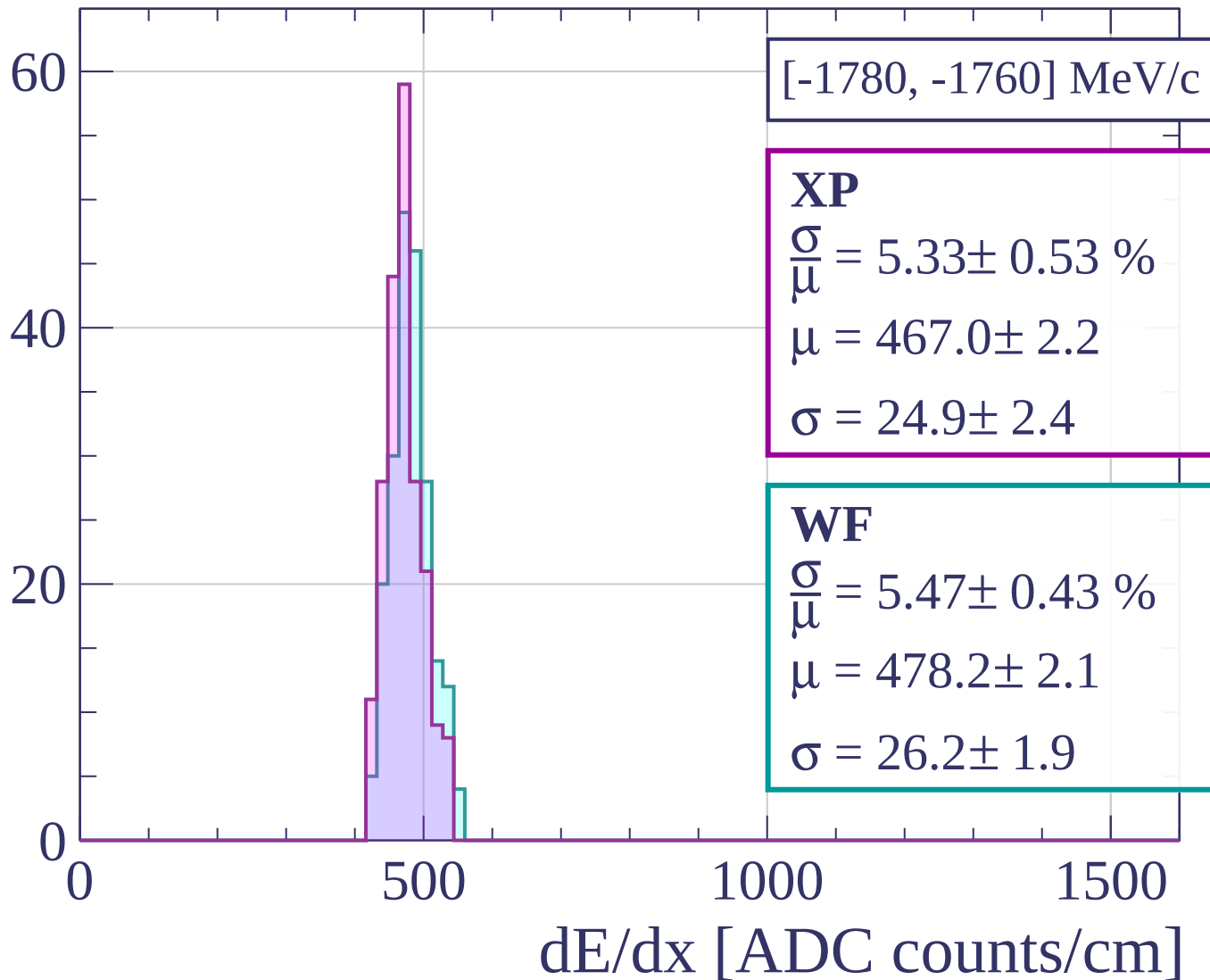
Count



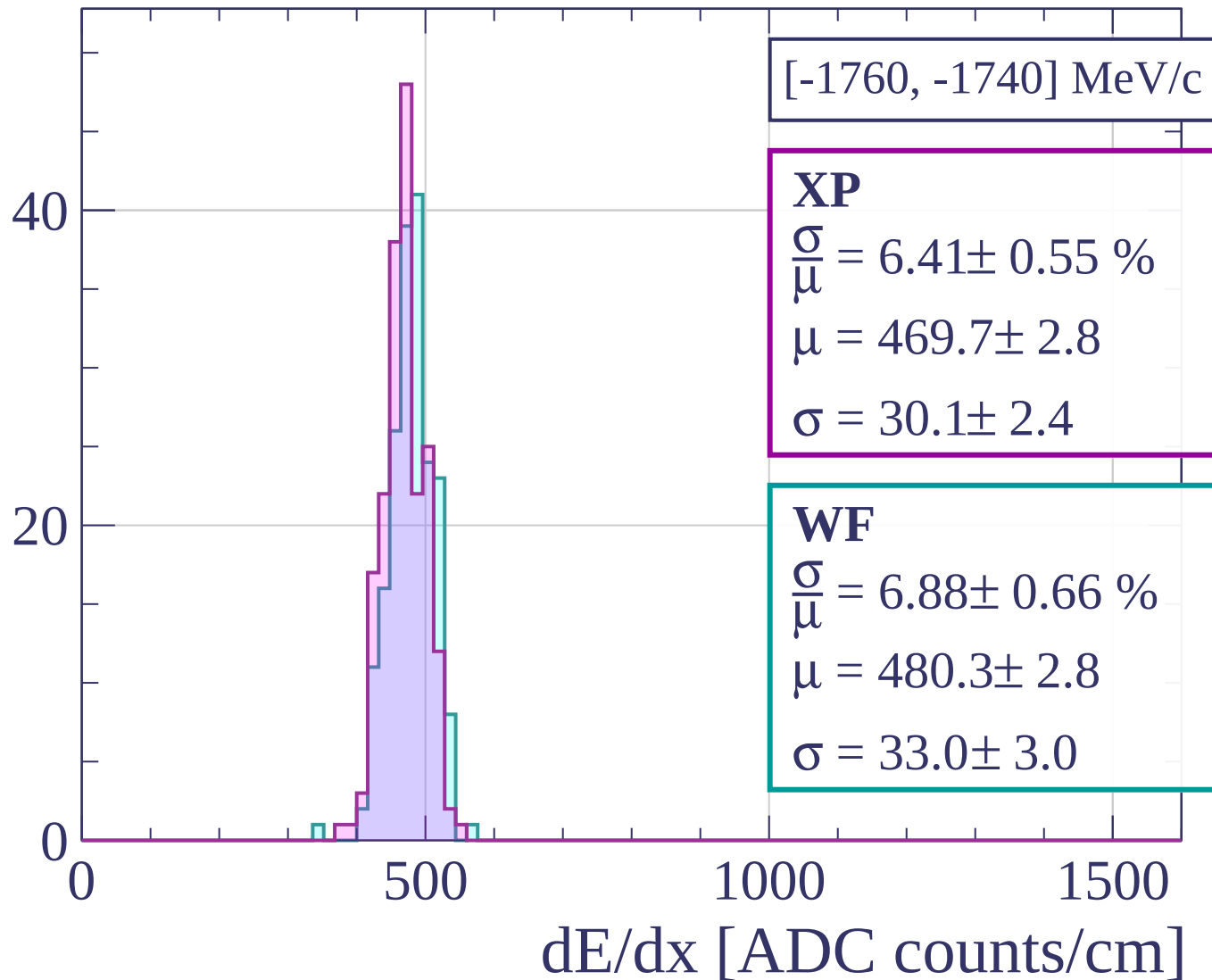
Count



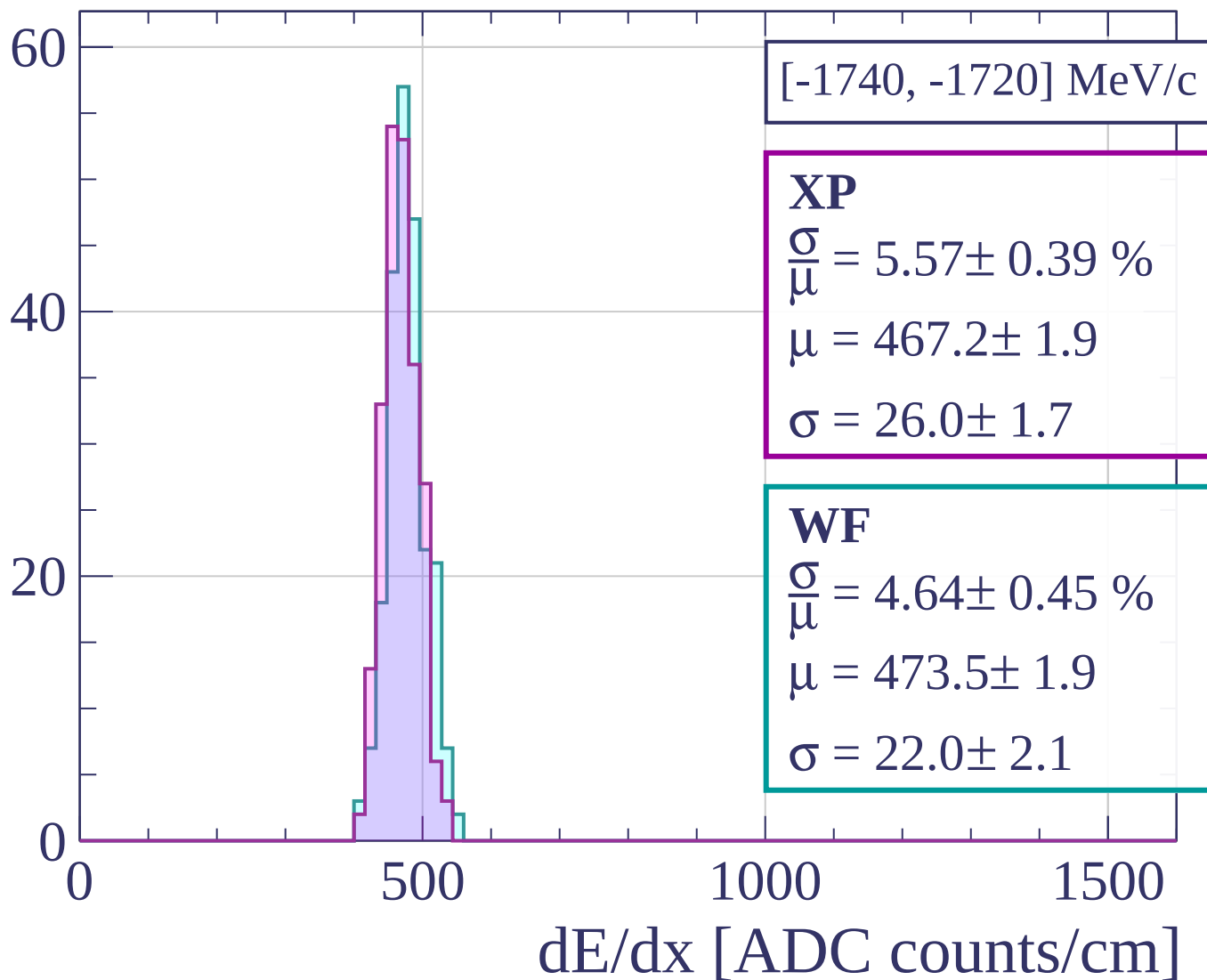
Count



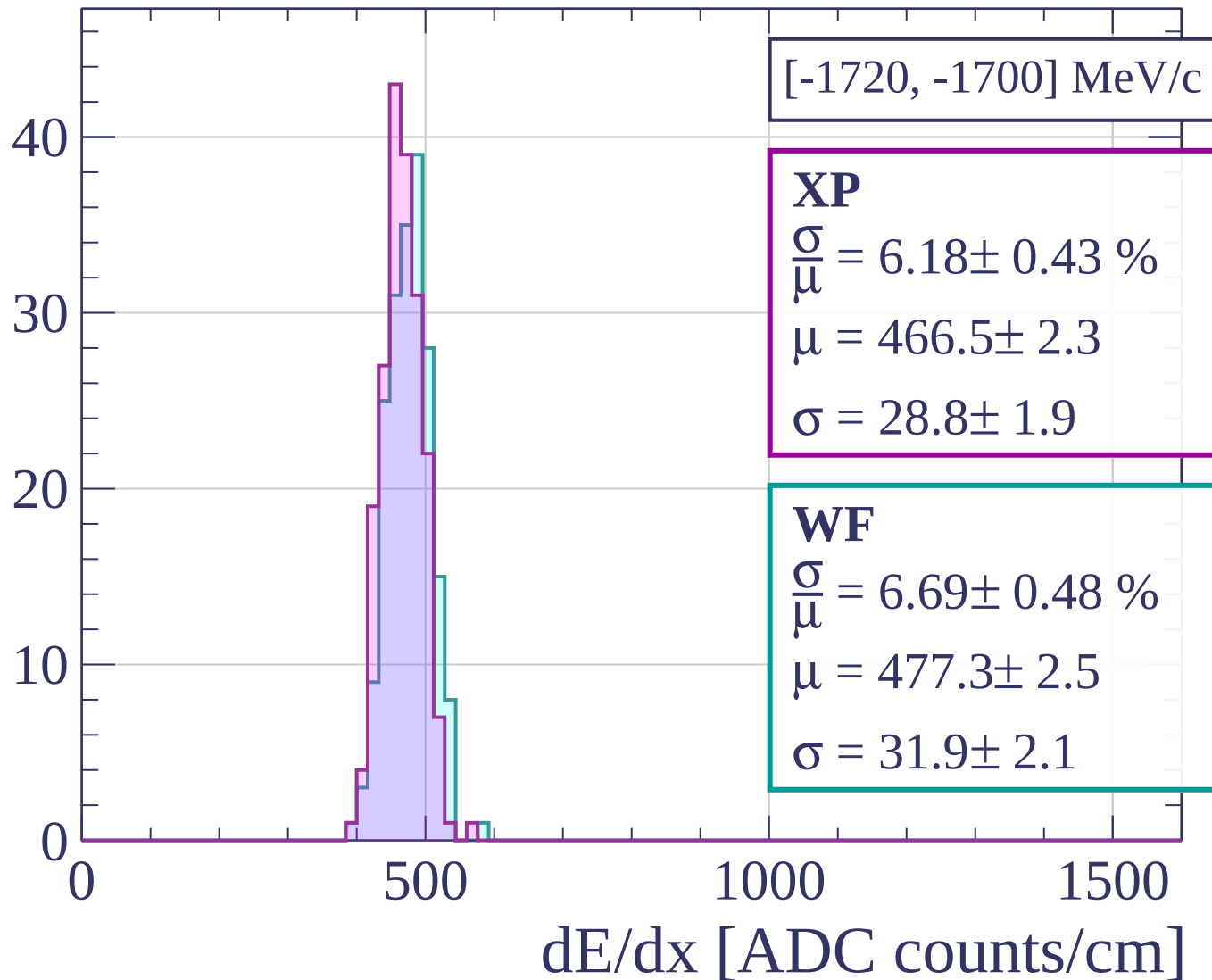
Count



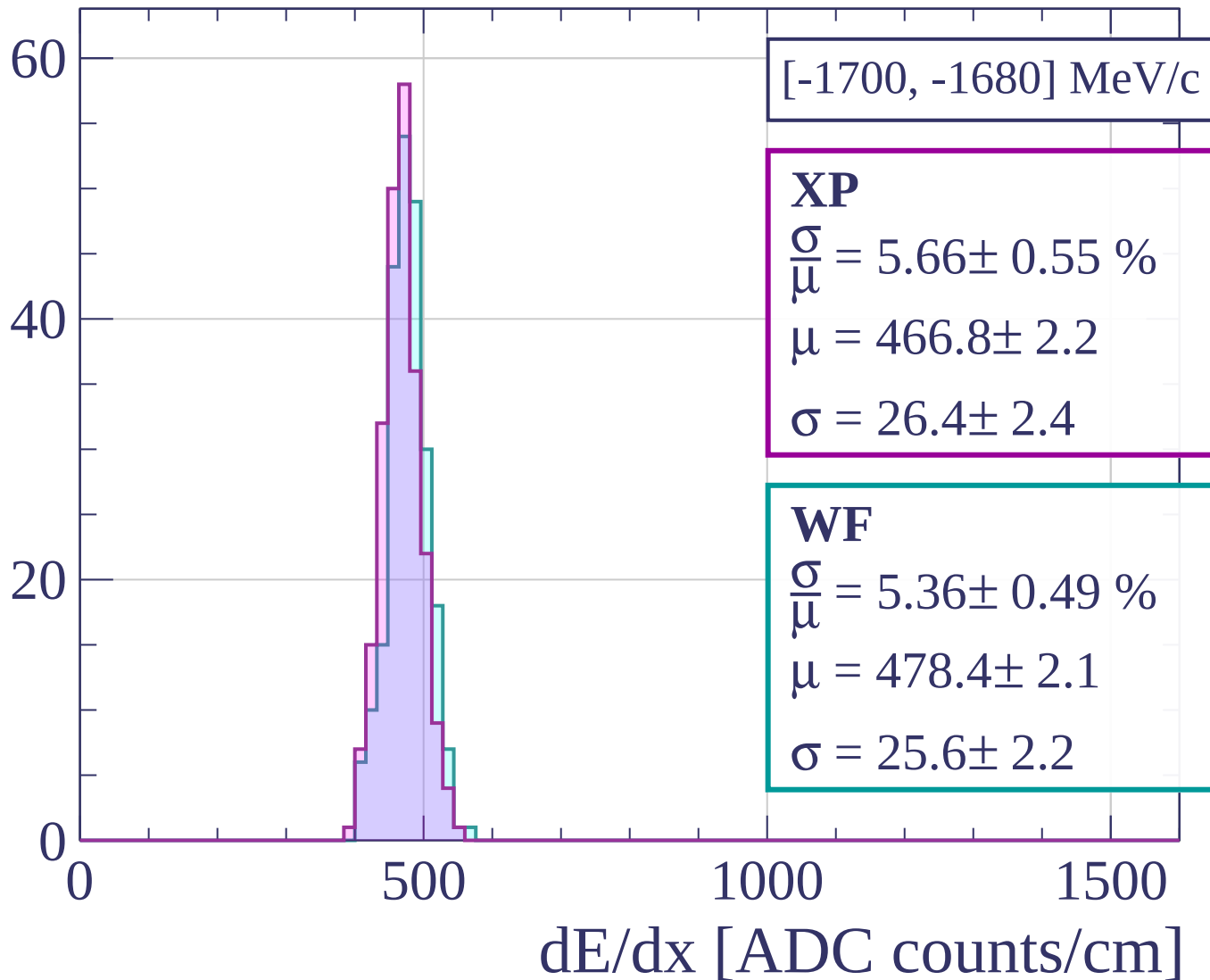
Count



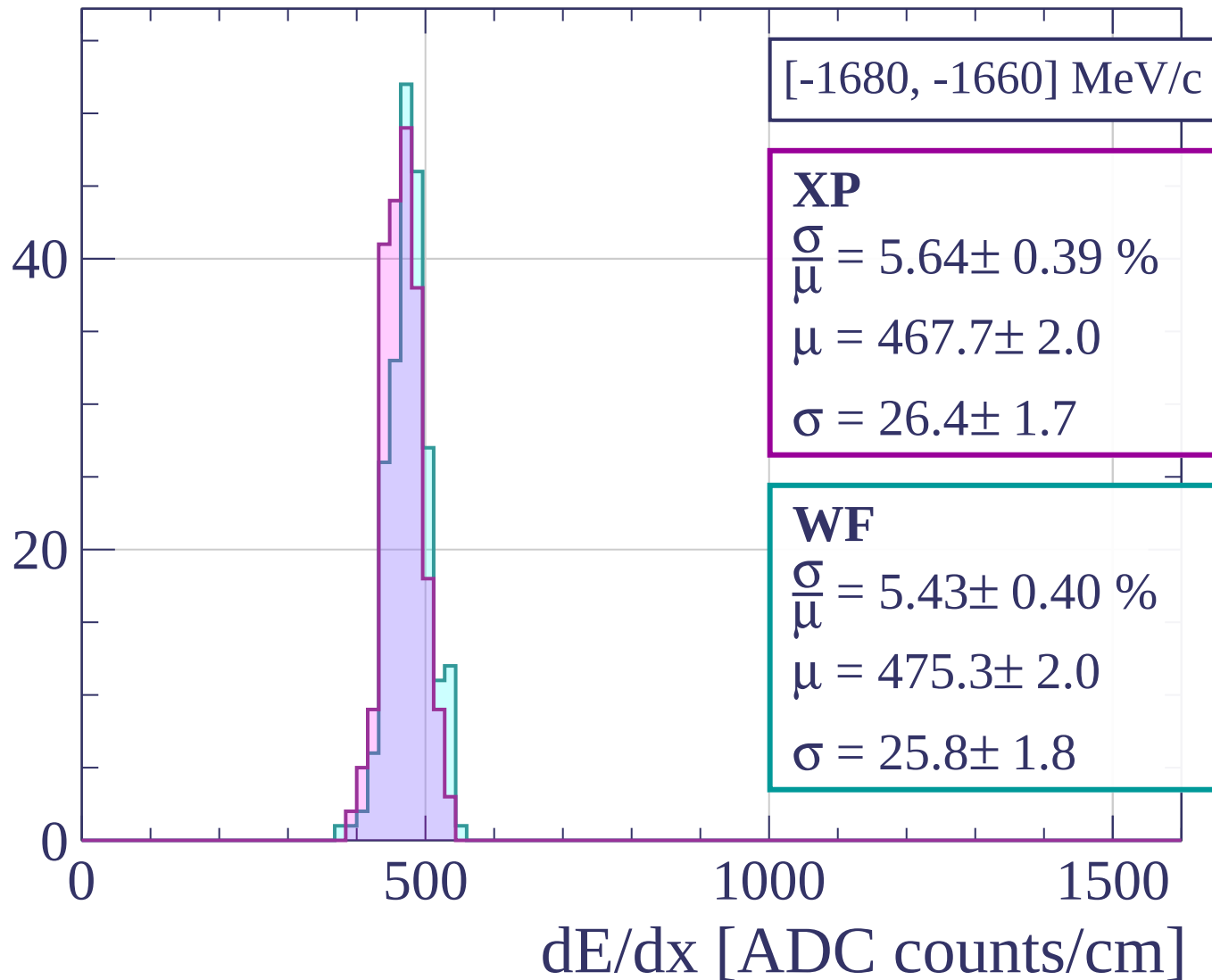
Count



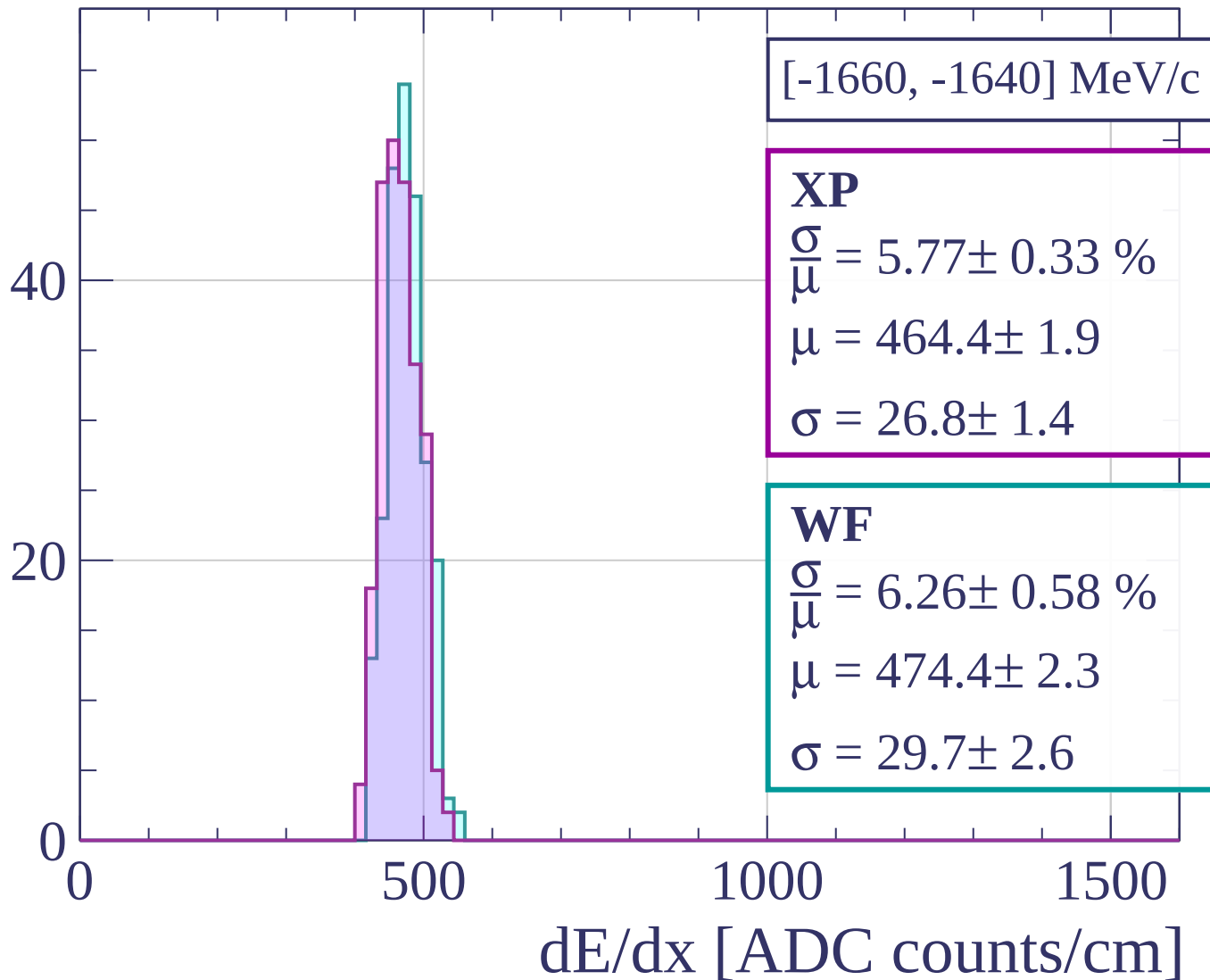
Count



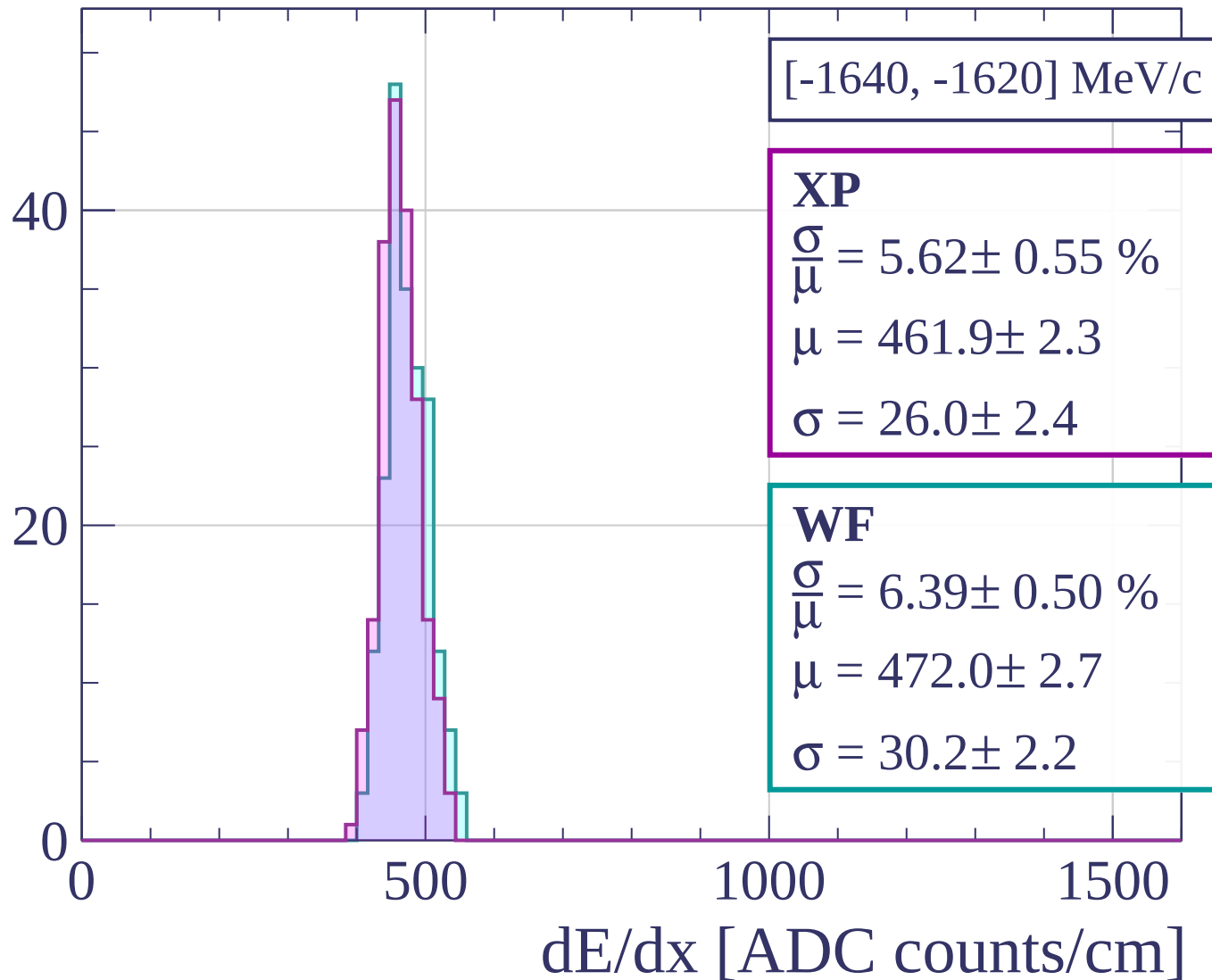
Count



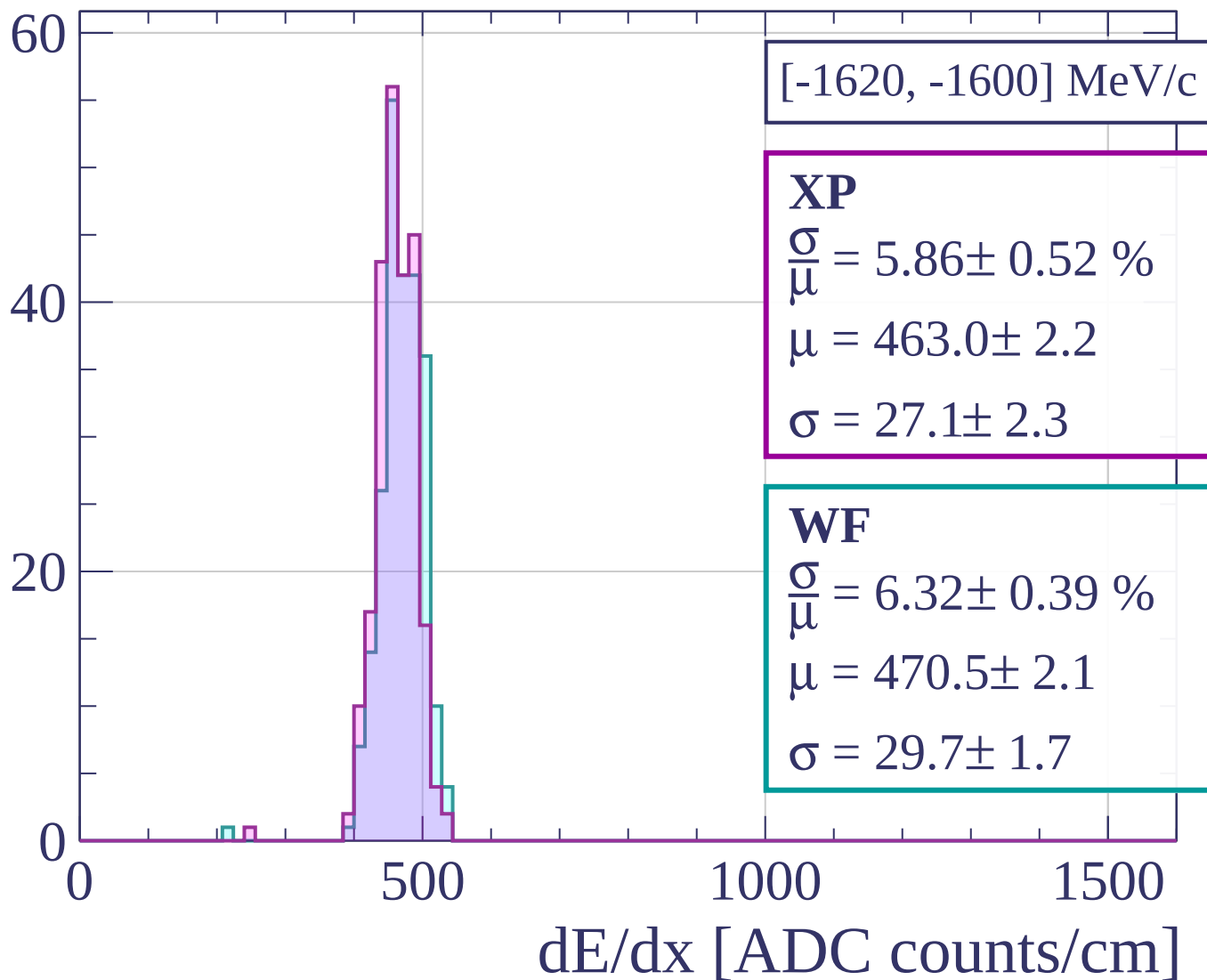
Count



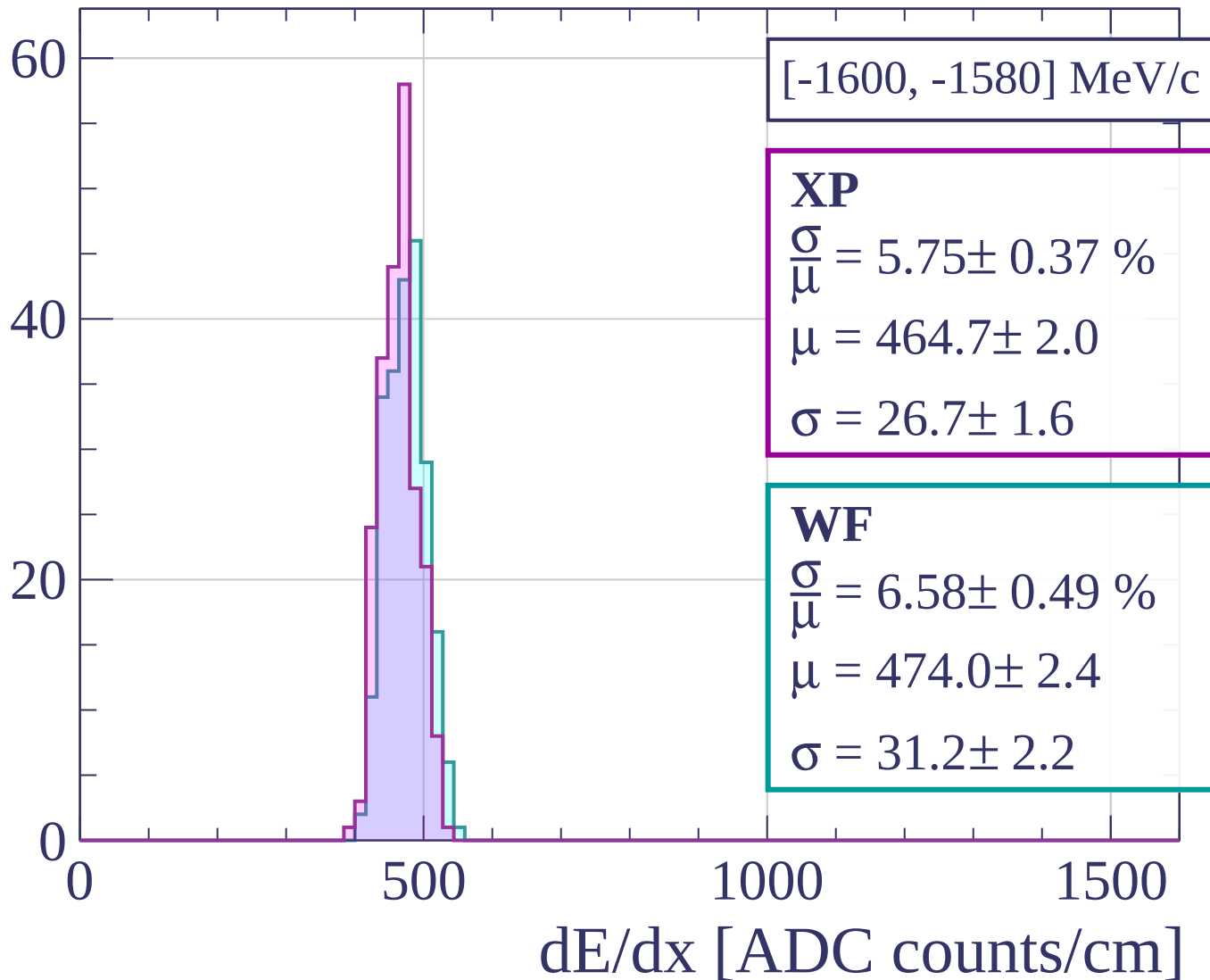
Count



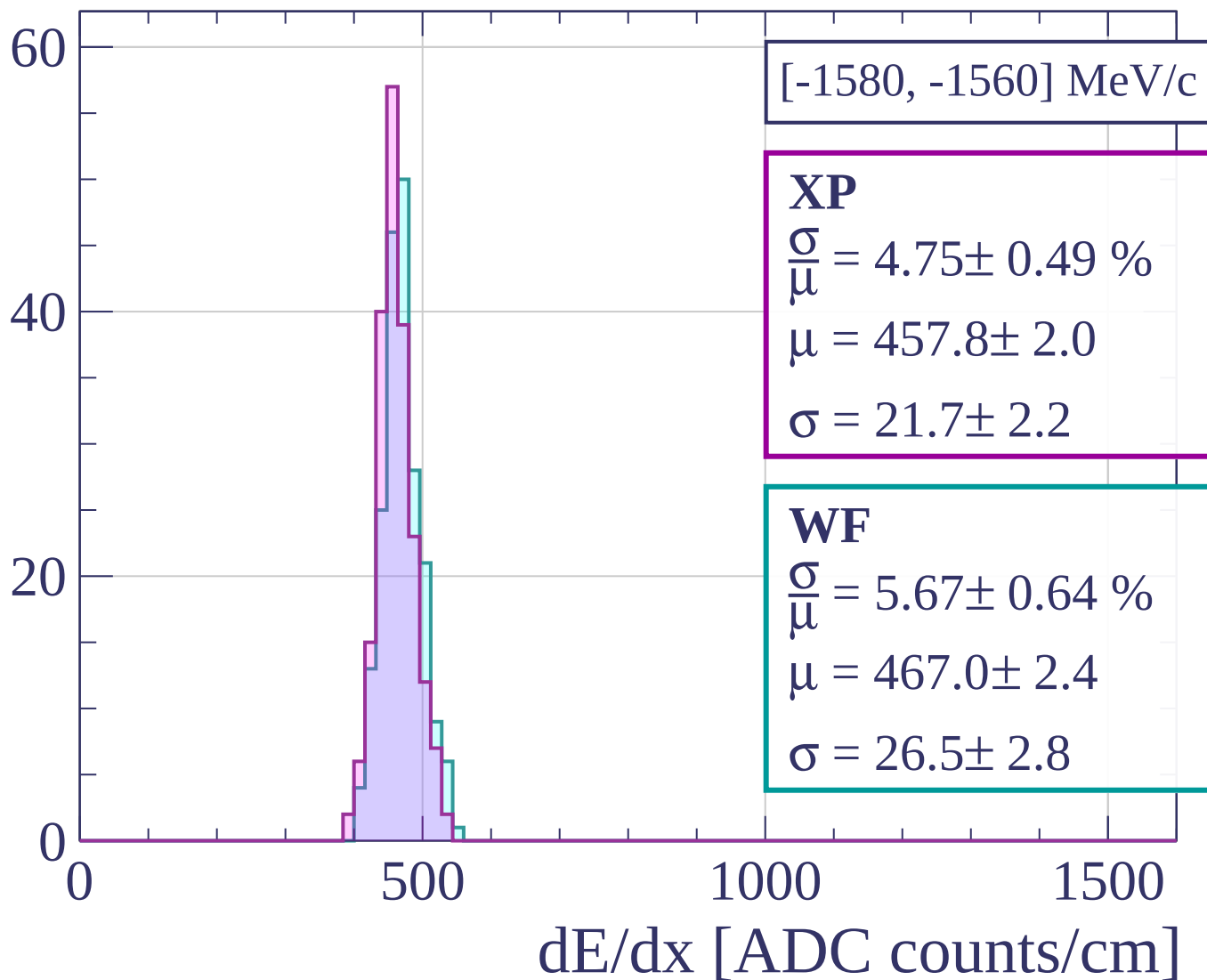
Count



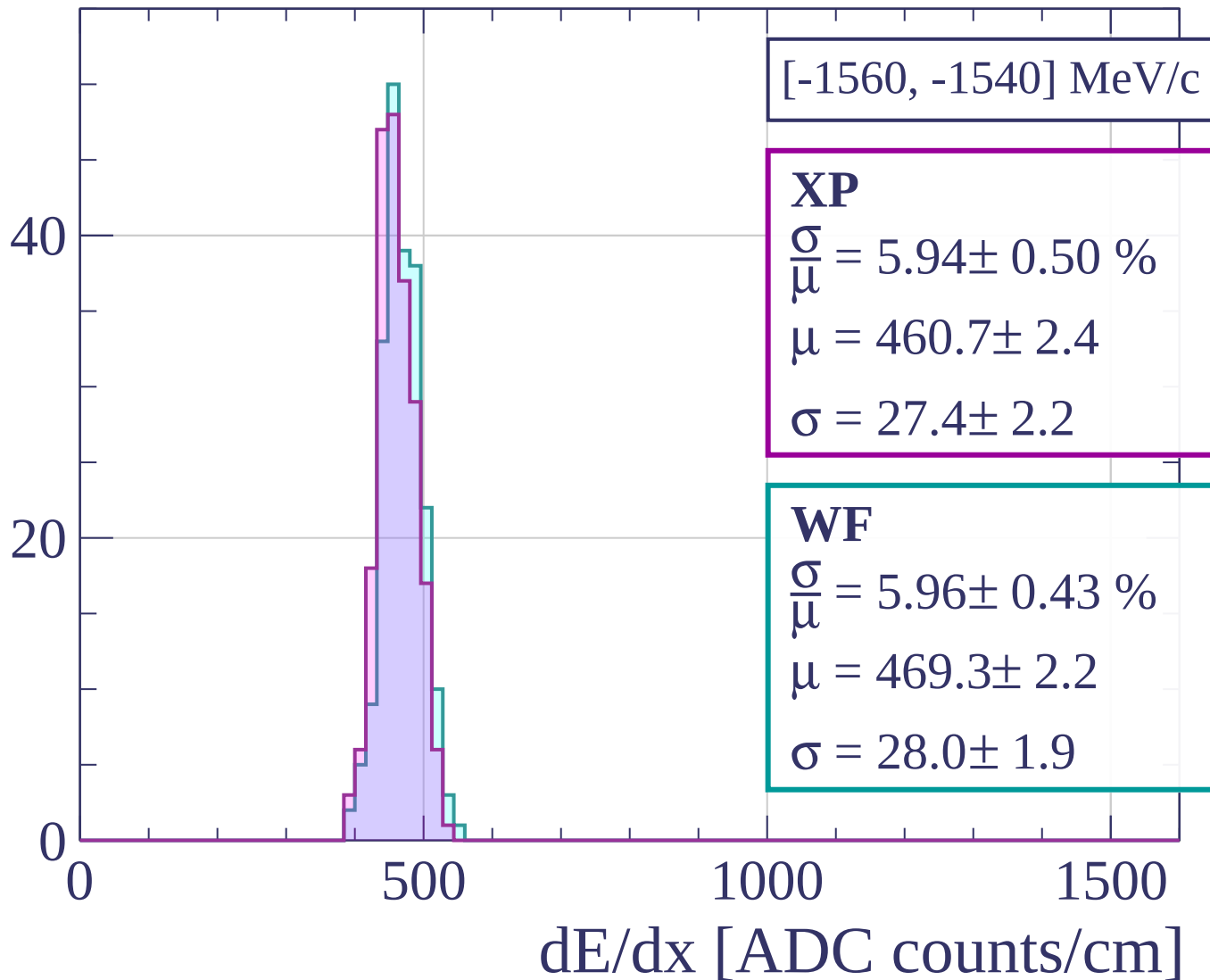
Count



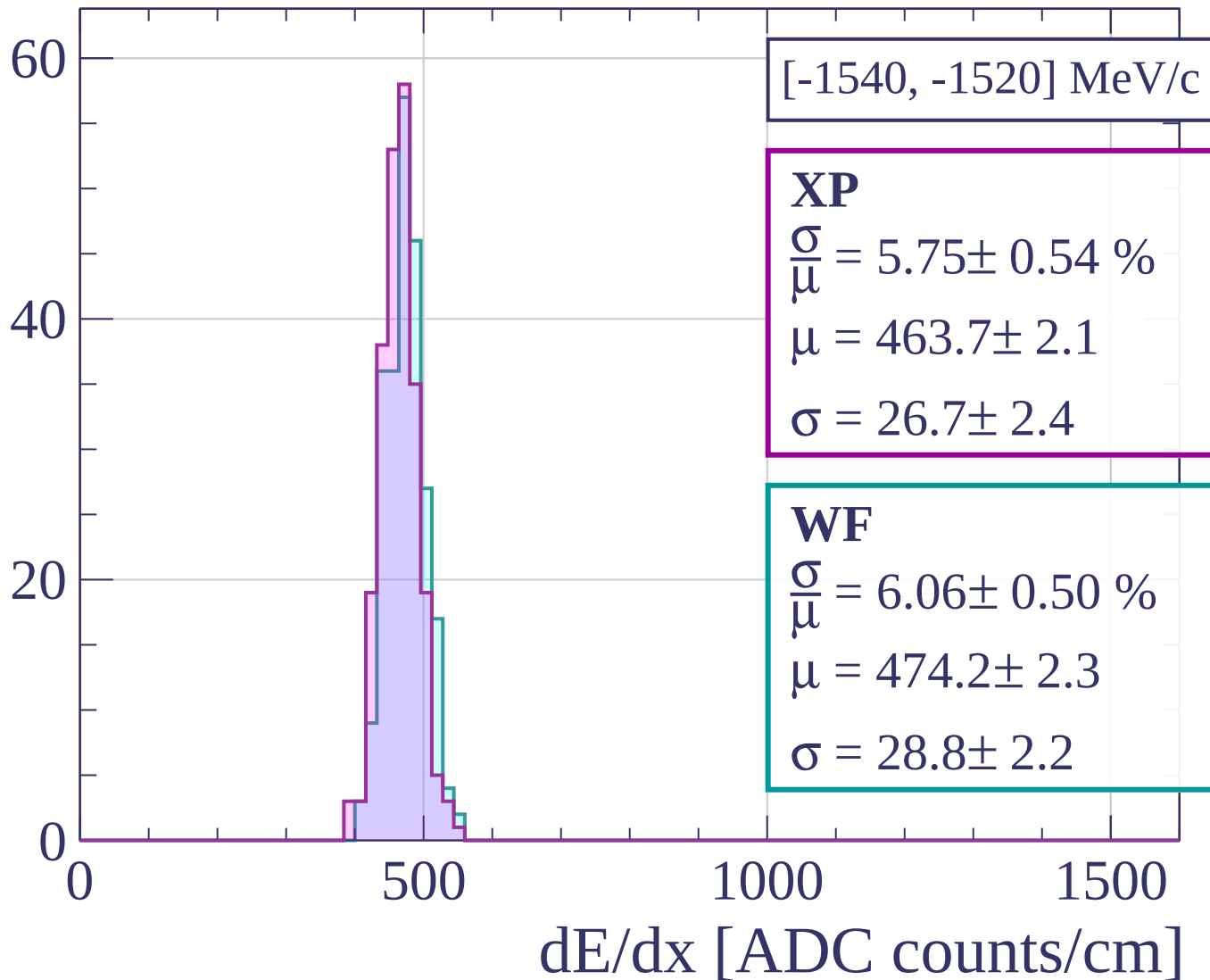
Count



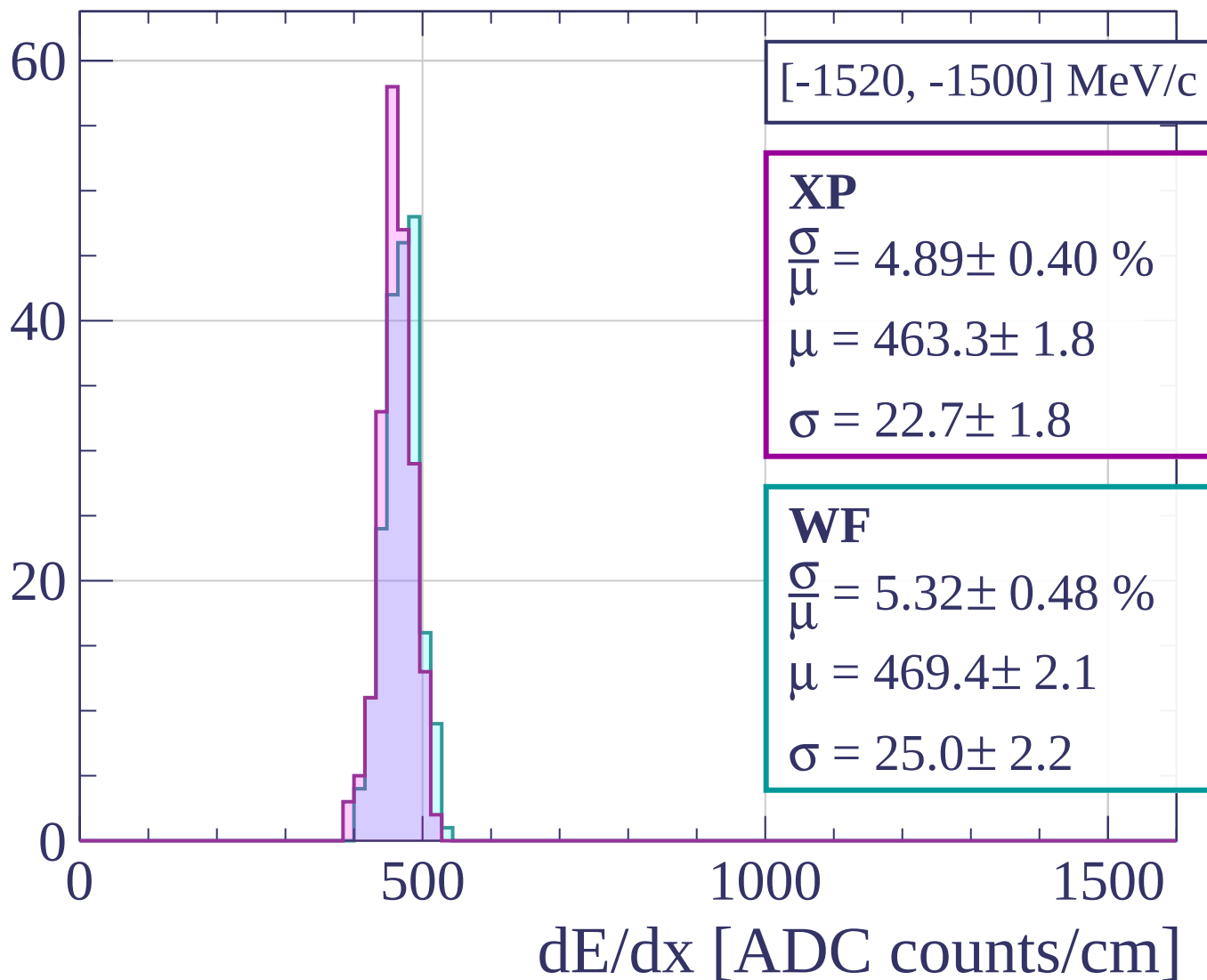
Count

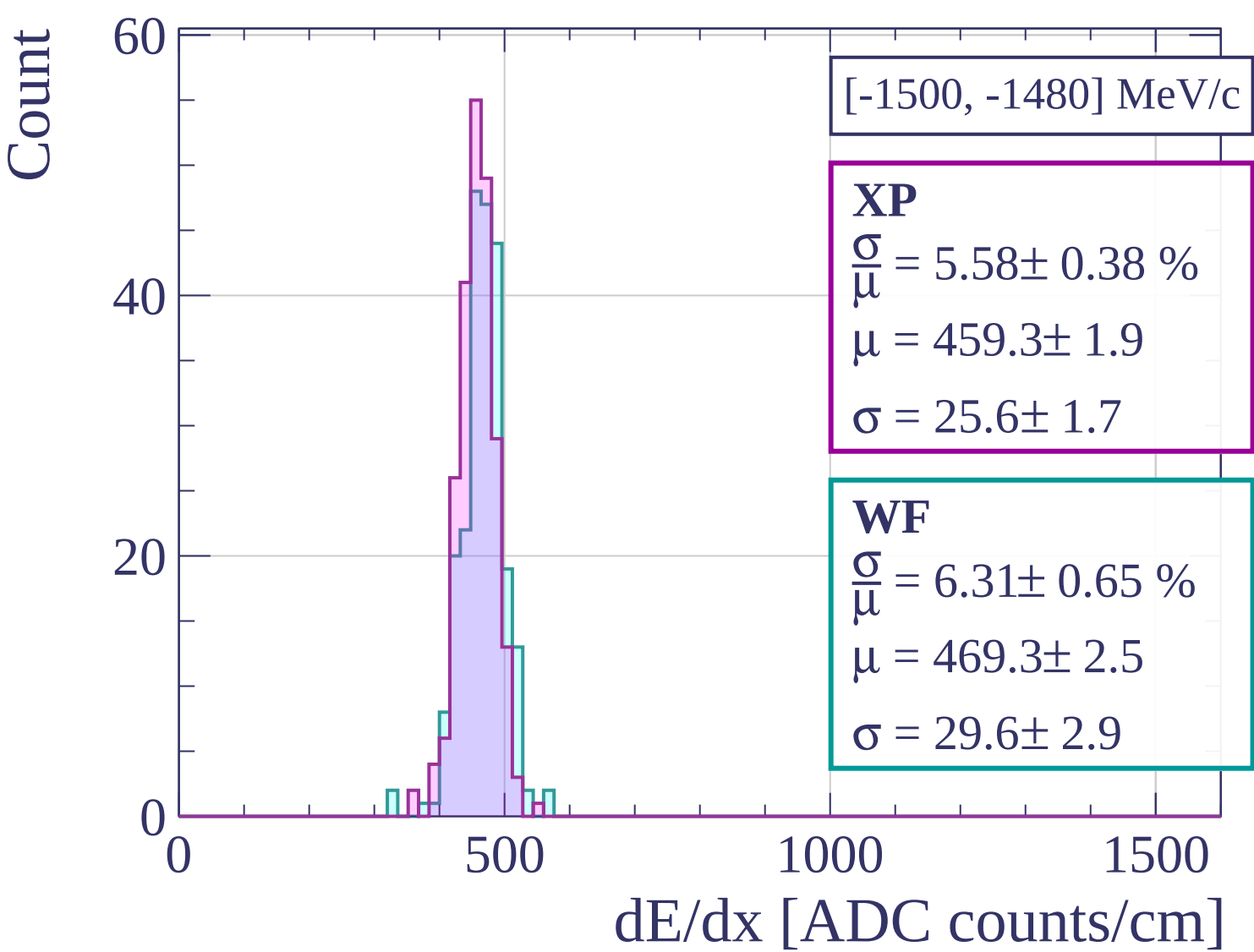


Count

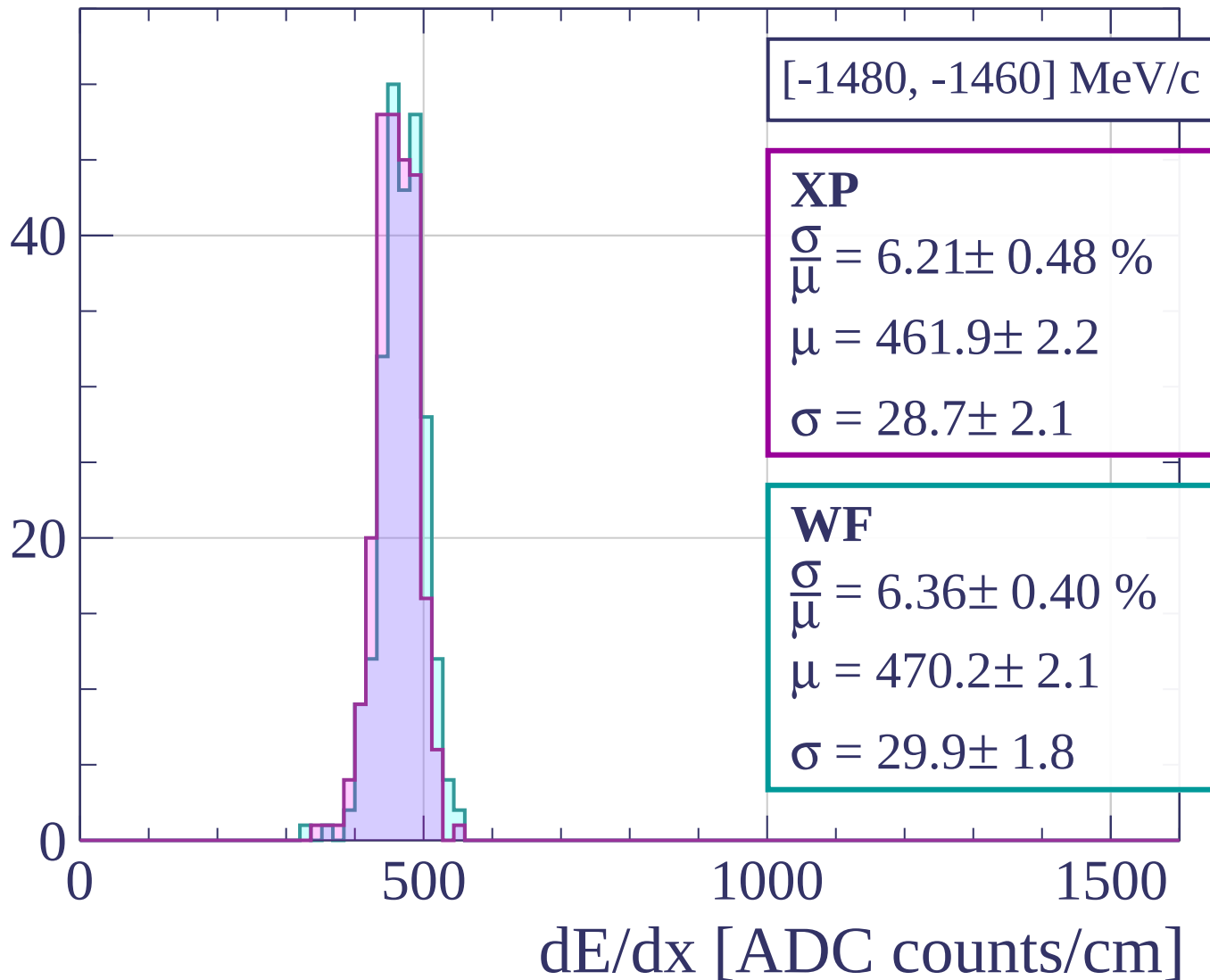


Count

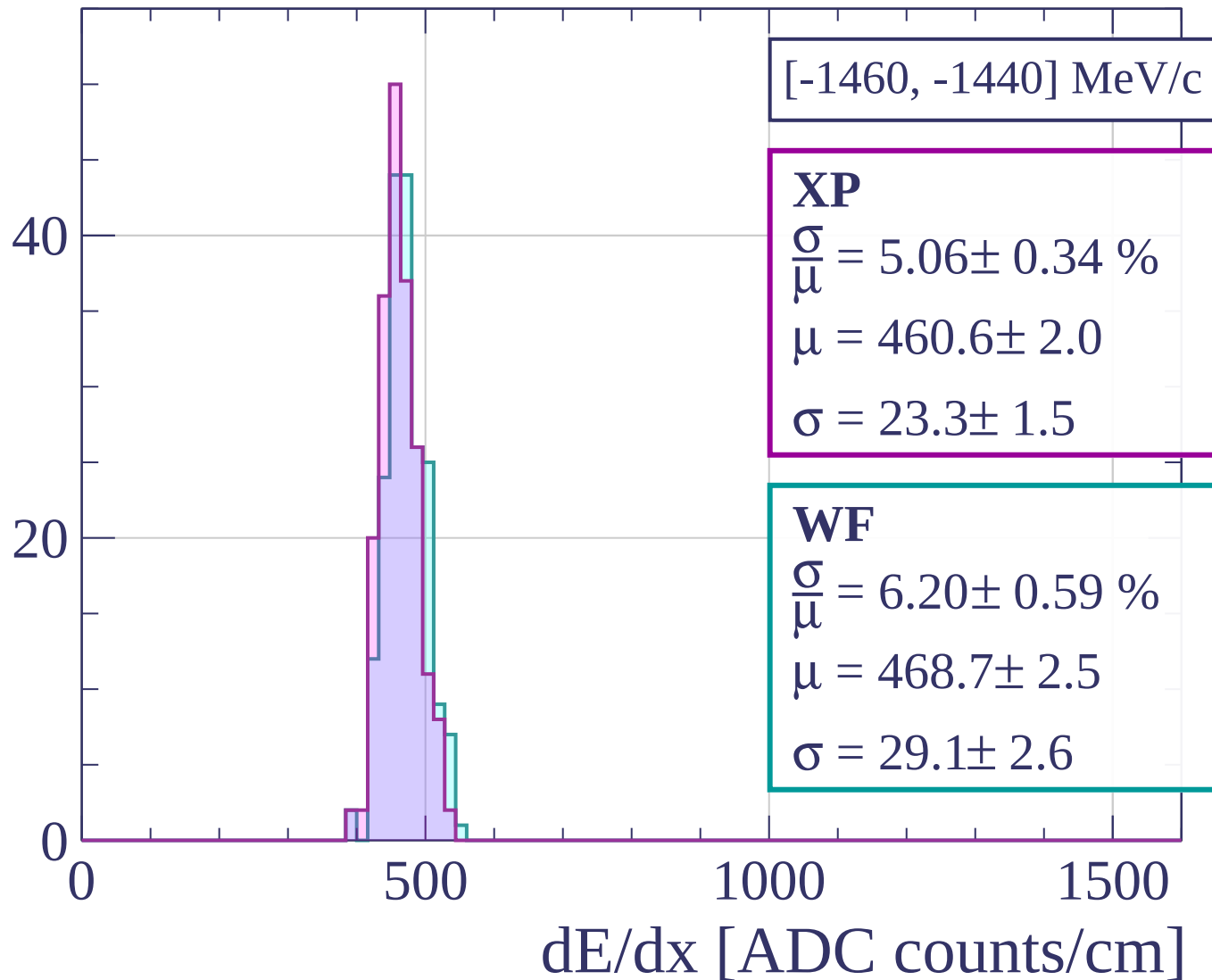




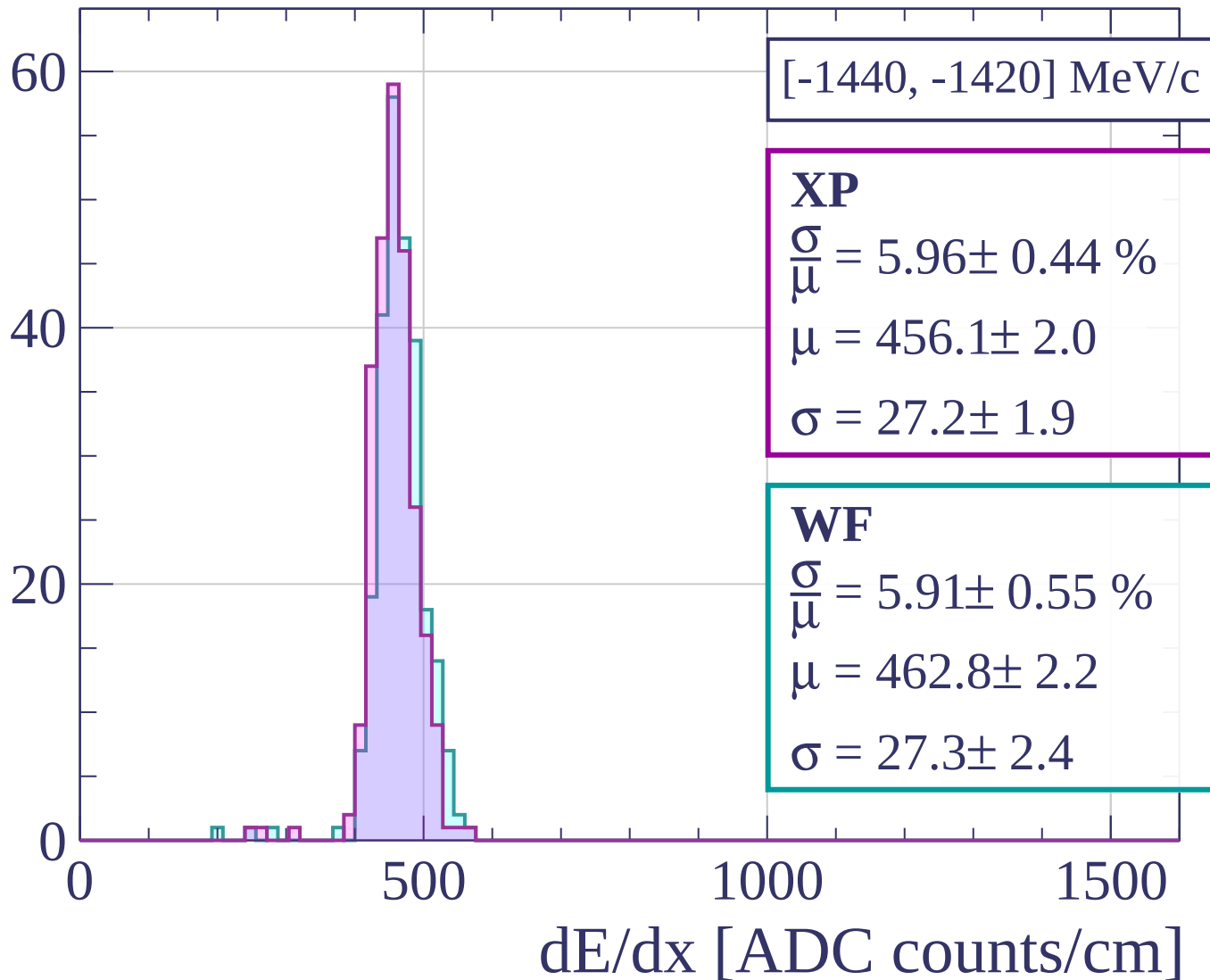
Count

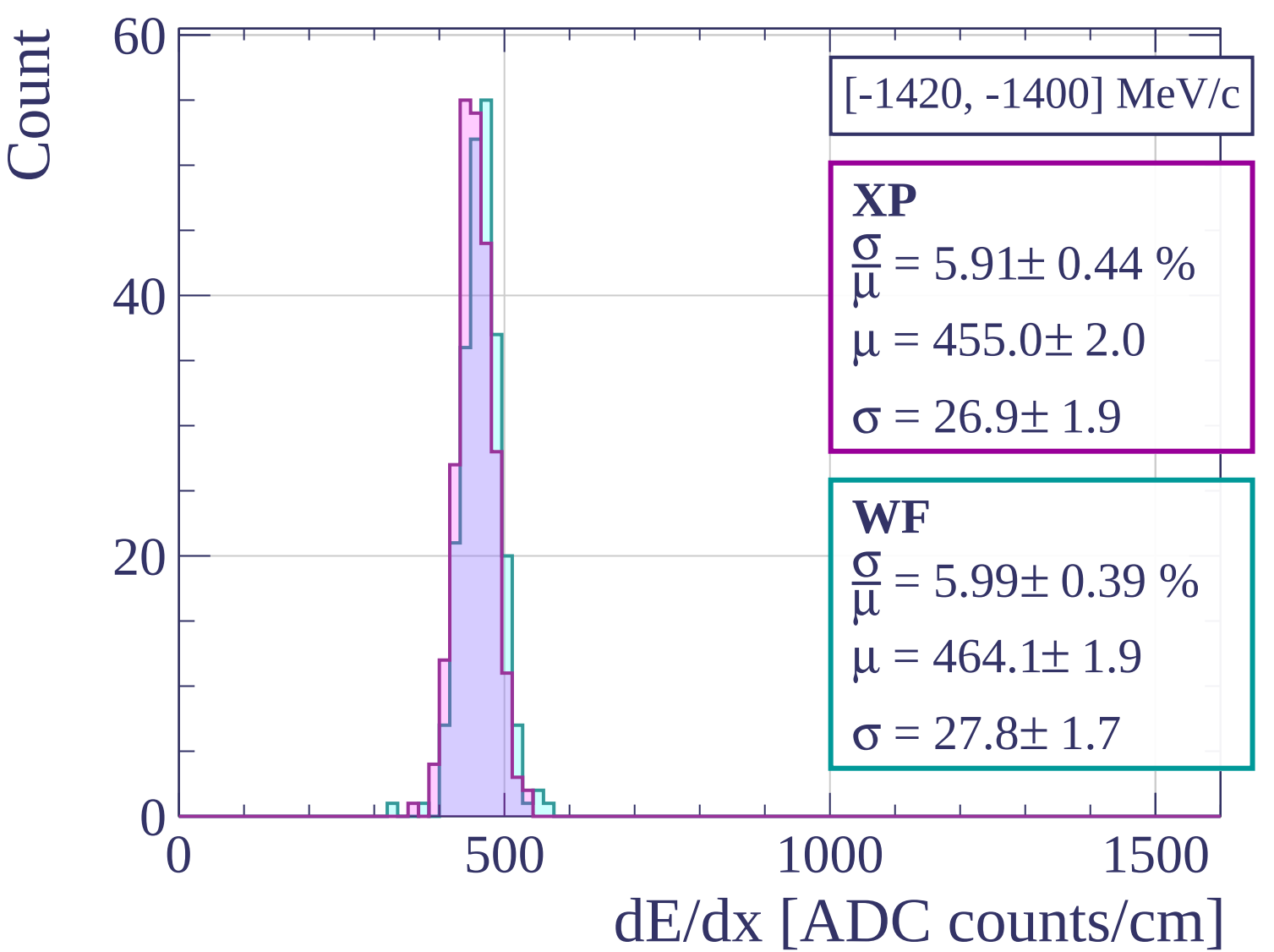


Count

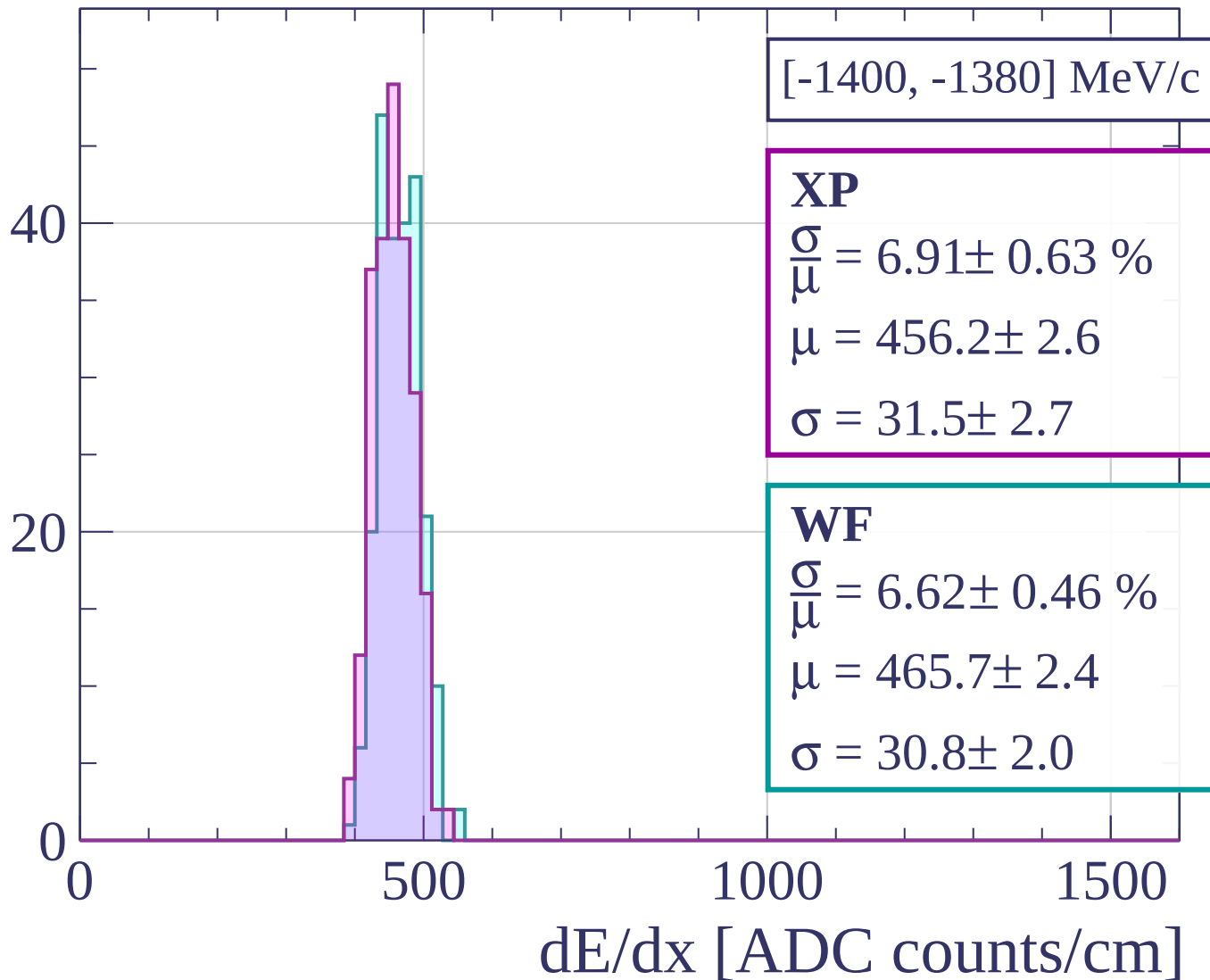


Count

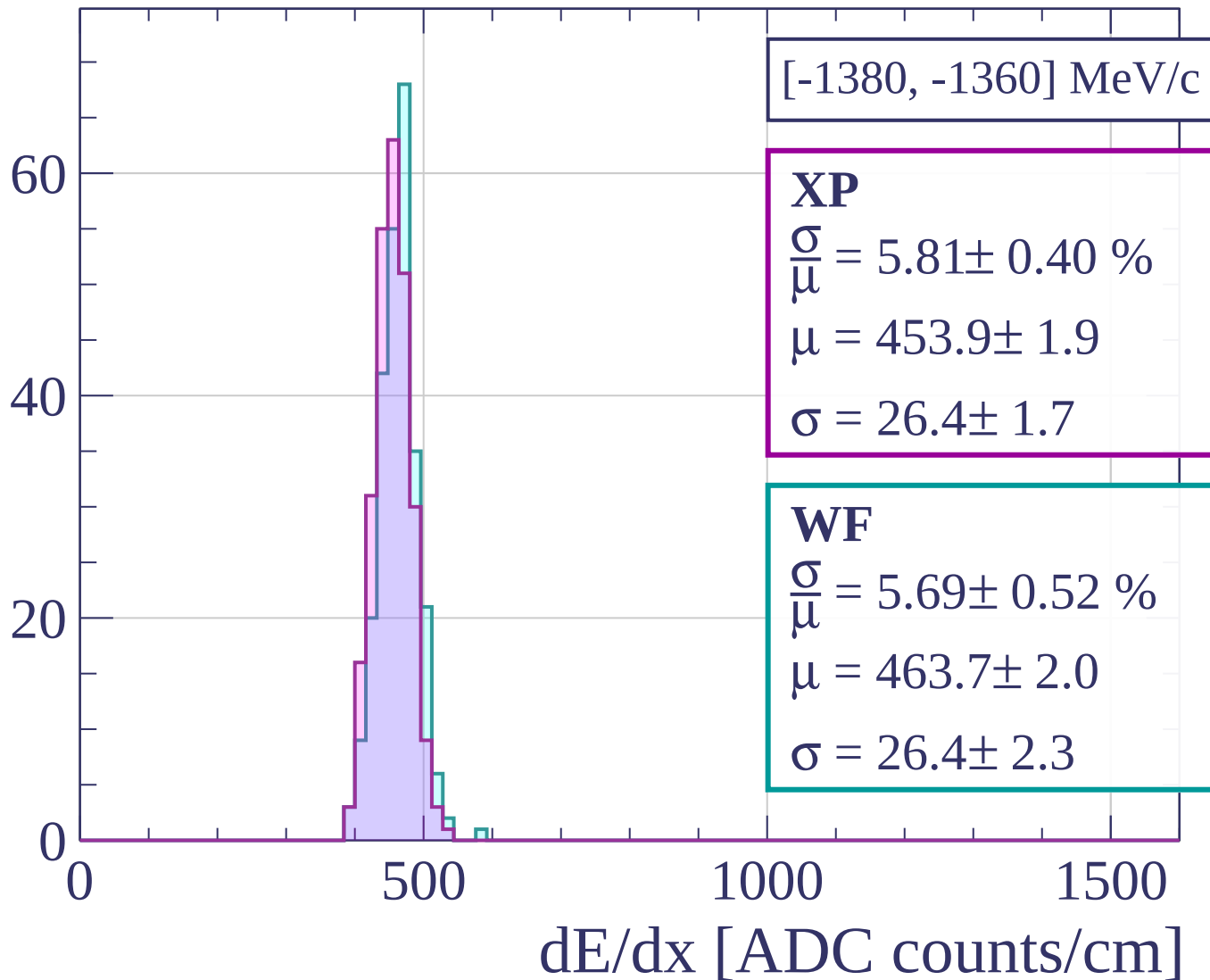




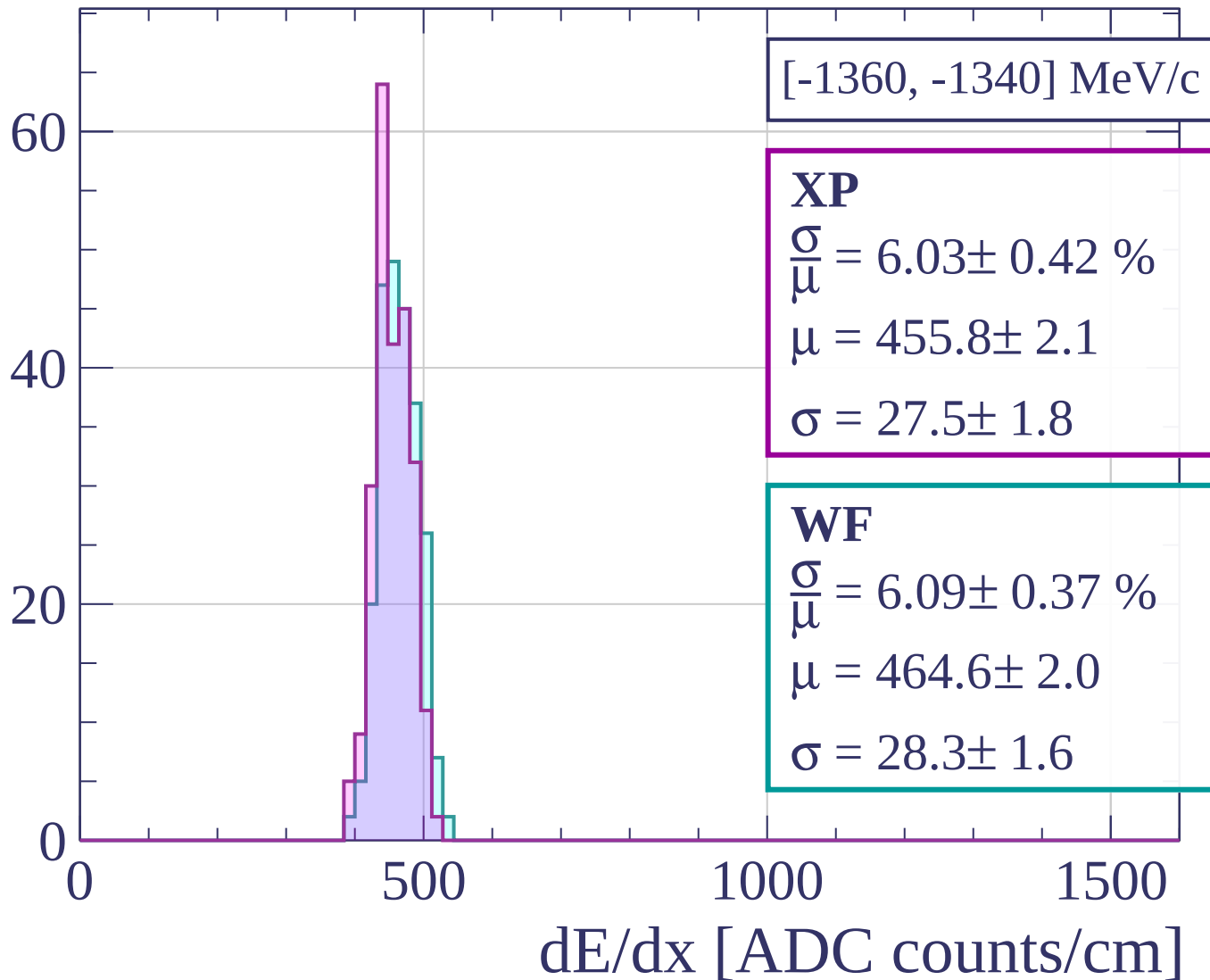
Count



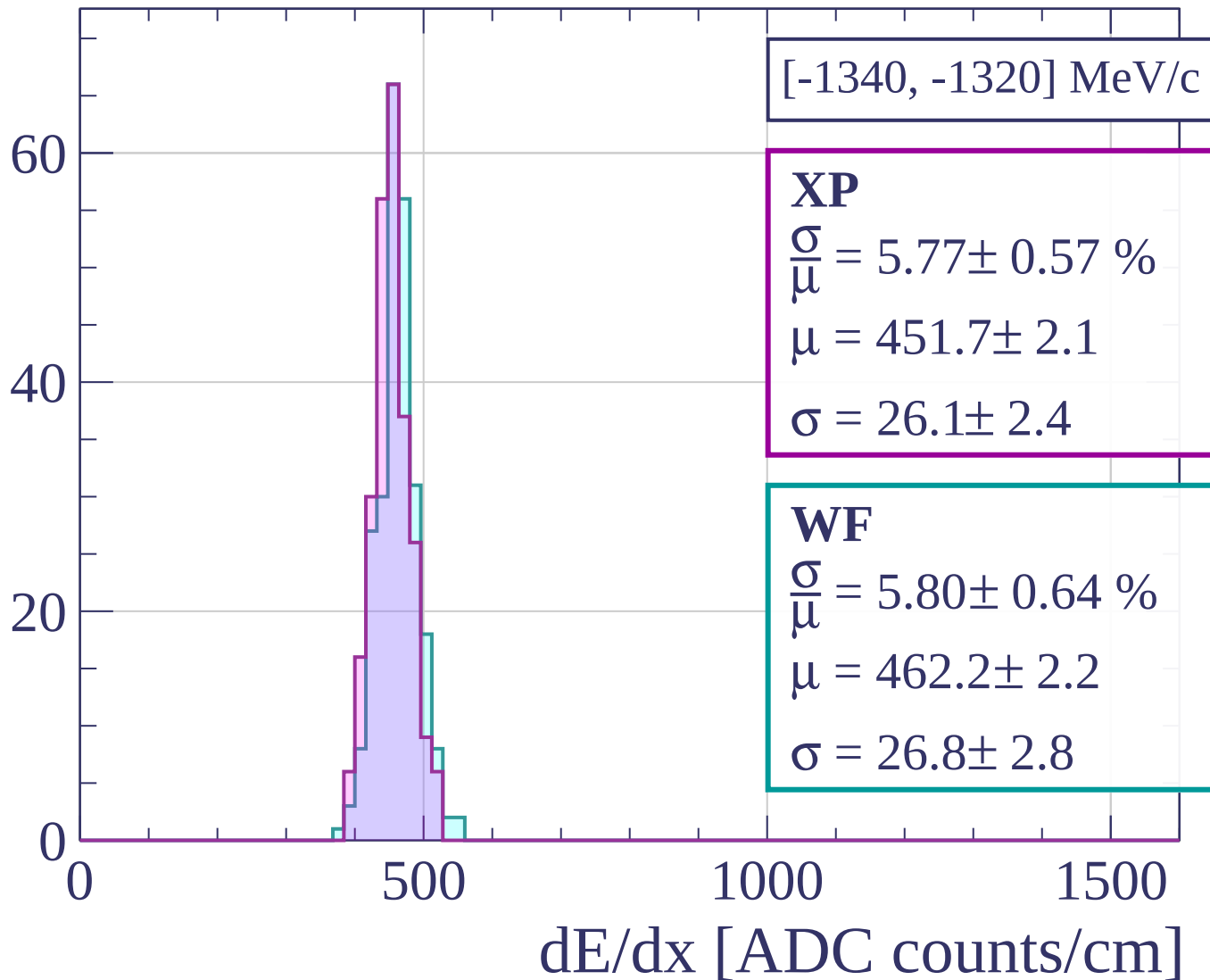
Count



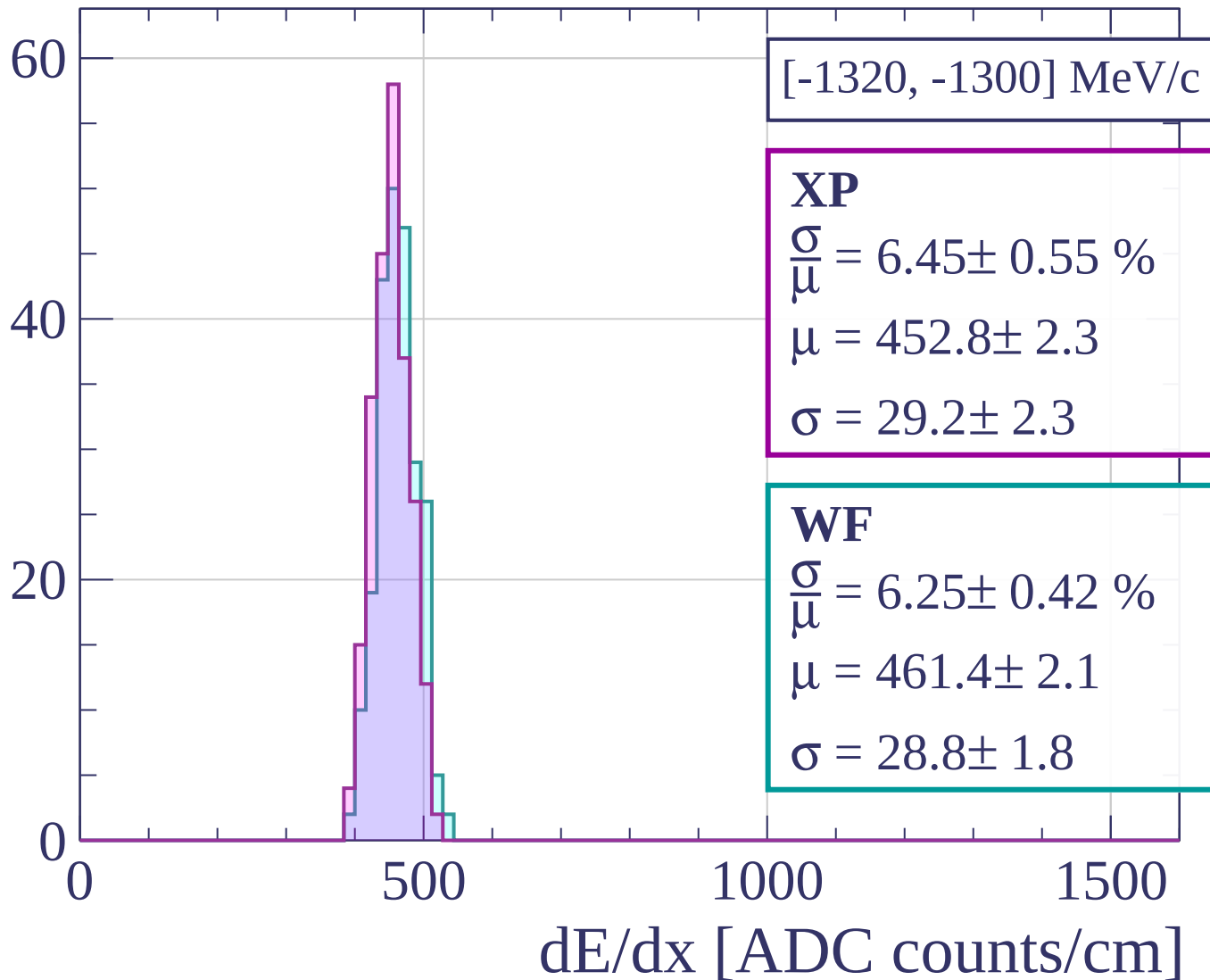
Count



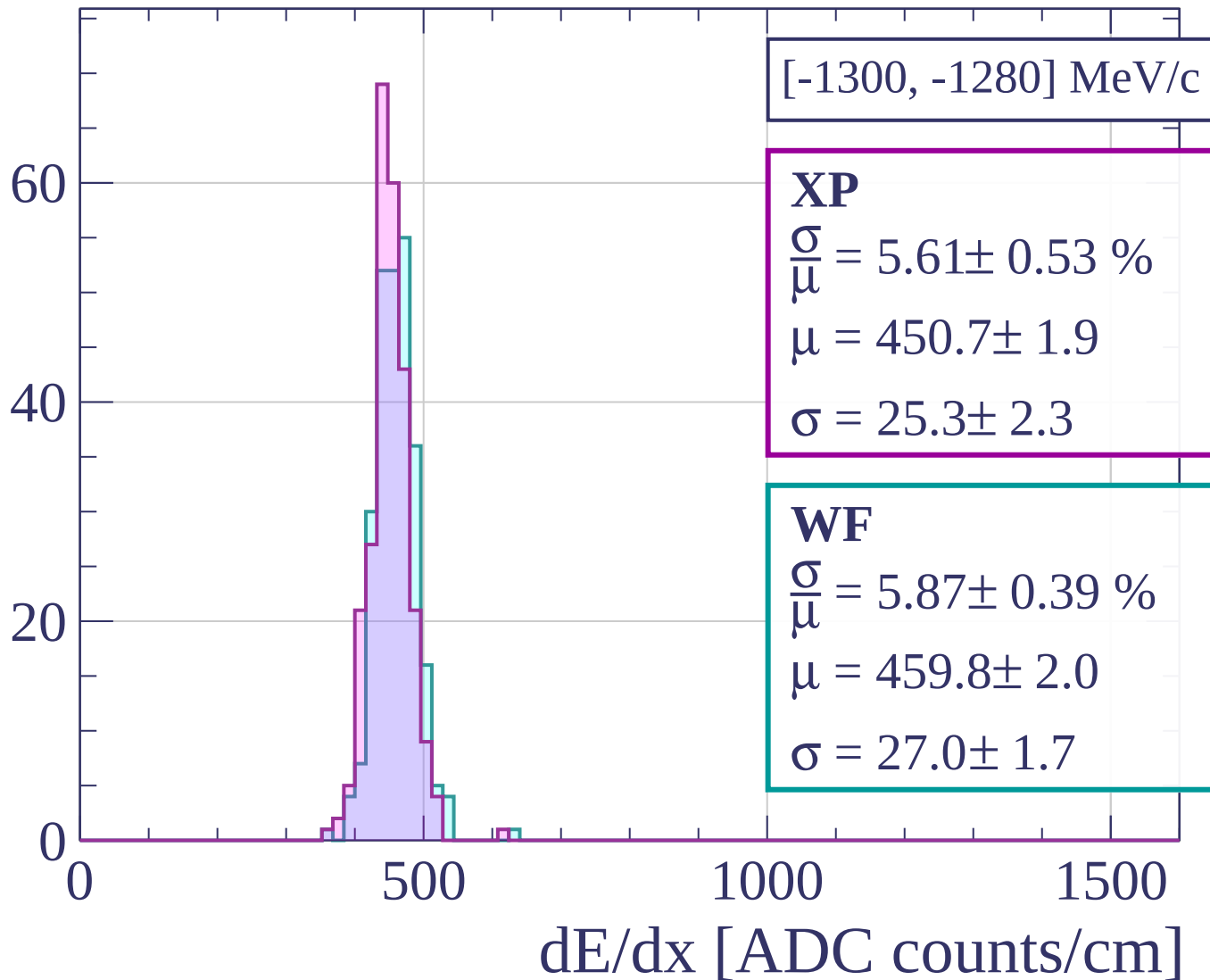
Count



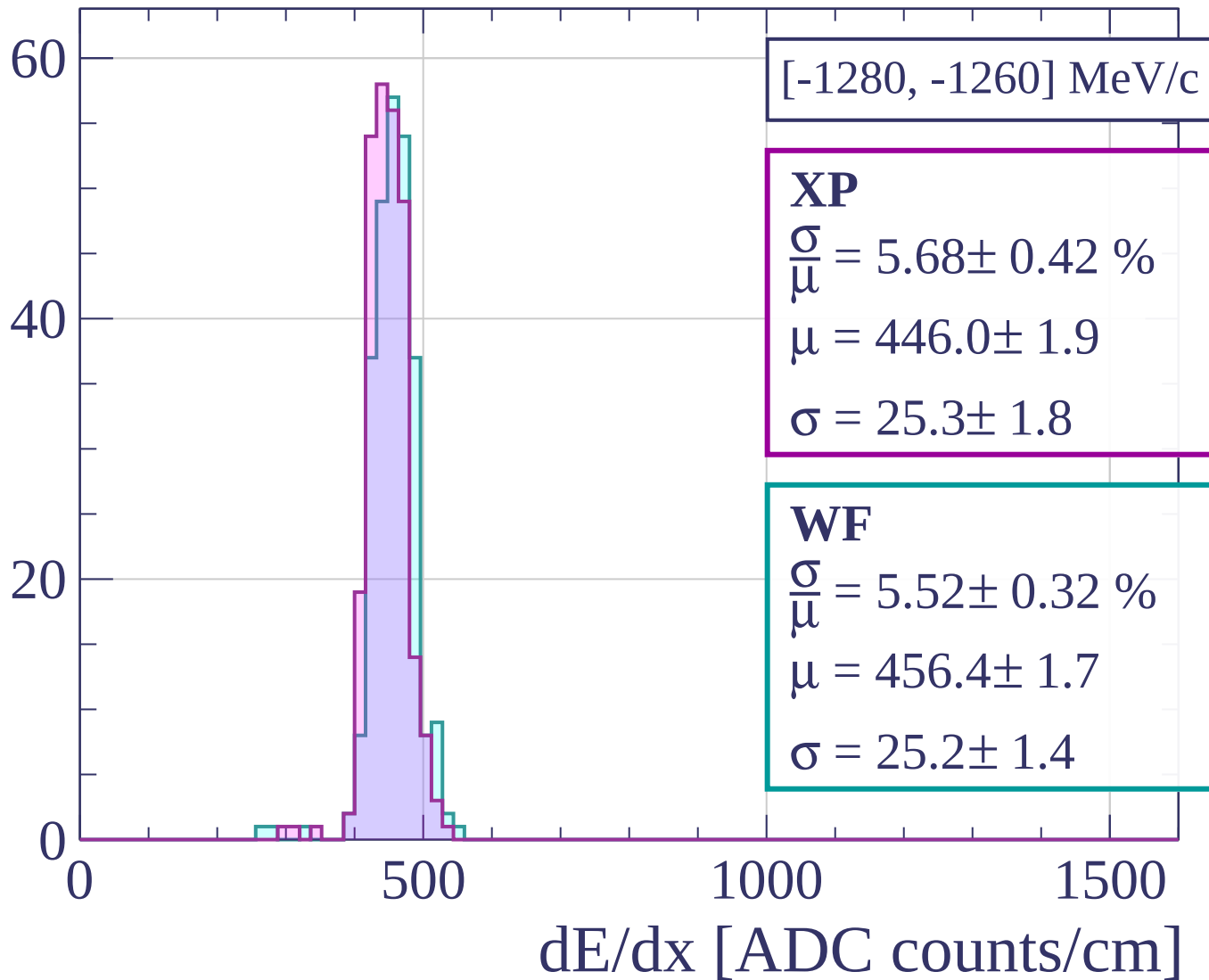
Count



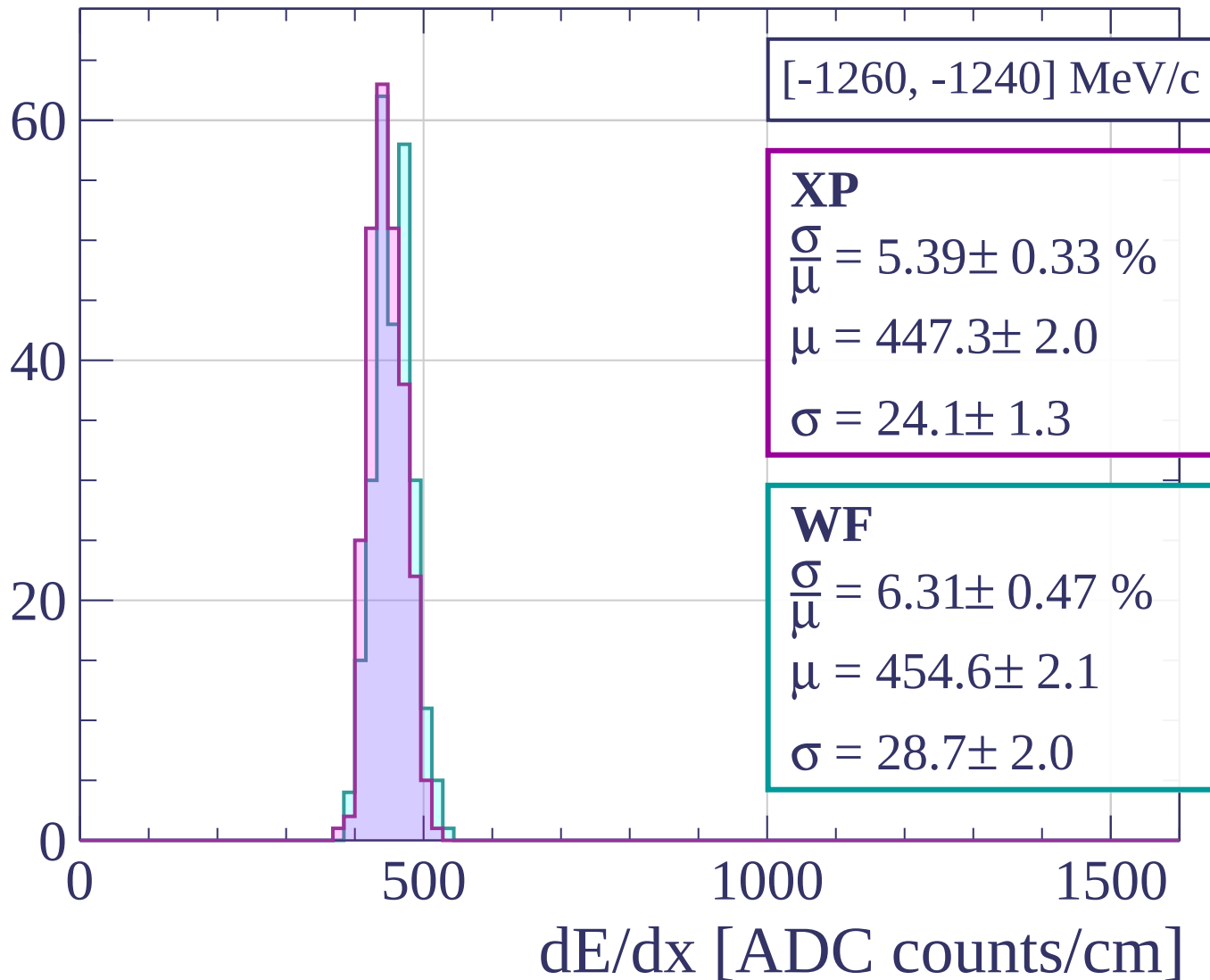
Count



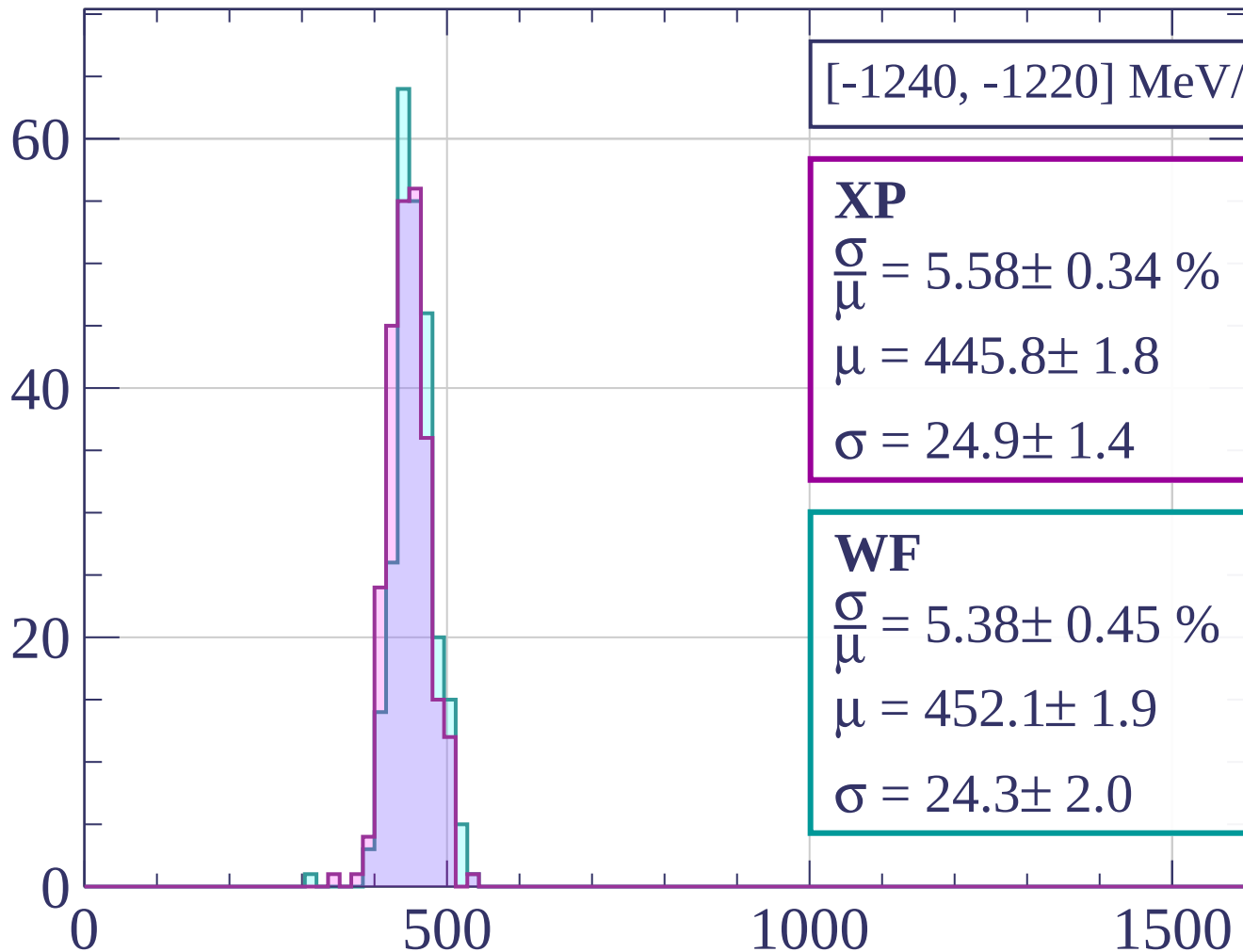
Count



Count



Count



[-1240, -1220] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.58 \pm 0.34 \%$$

$$\mu = 445.8 \pm 1.8$$

$$\sigma = 24.9 \pm 1.4$$

WF

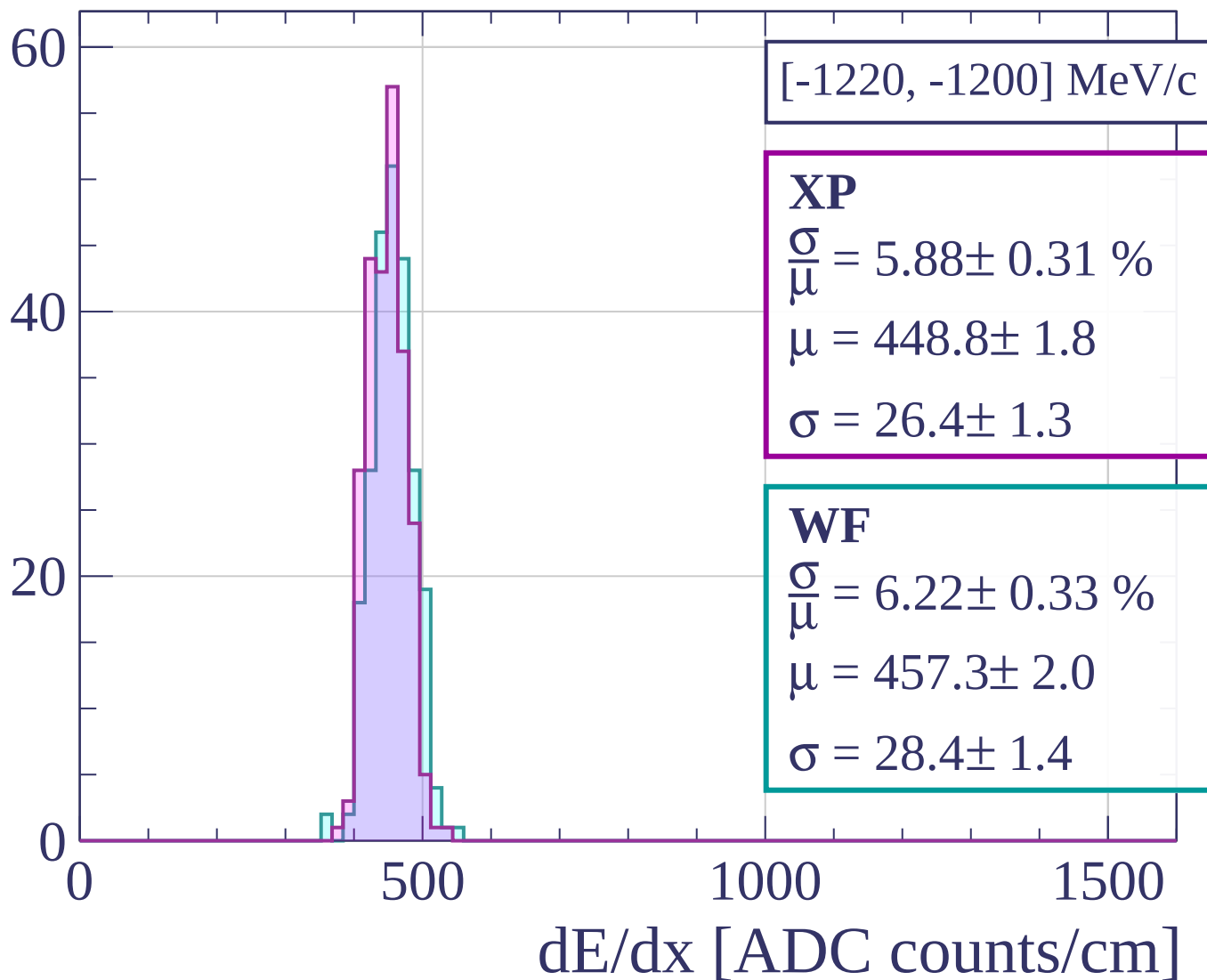
$$\frac{\sigma}{\mu} = 5.38 \pm 0.45 \%$$

$$\mu = 452.1 \pm 1.9$$

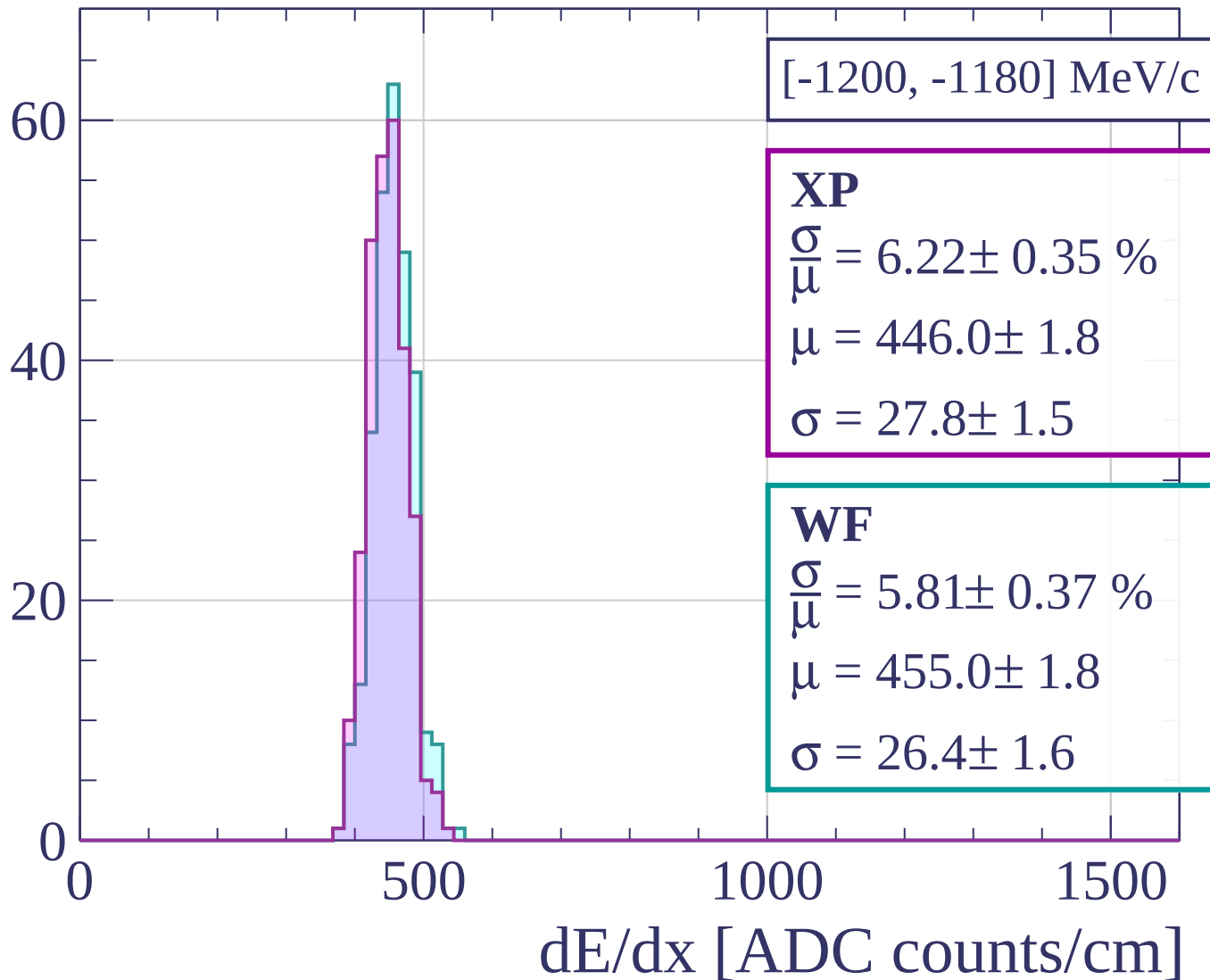
$$\sigma = 24.3 \pm 2.0$$

dE/dx [ADC counts/cm]

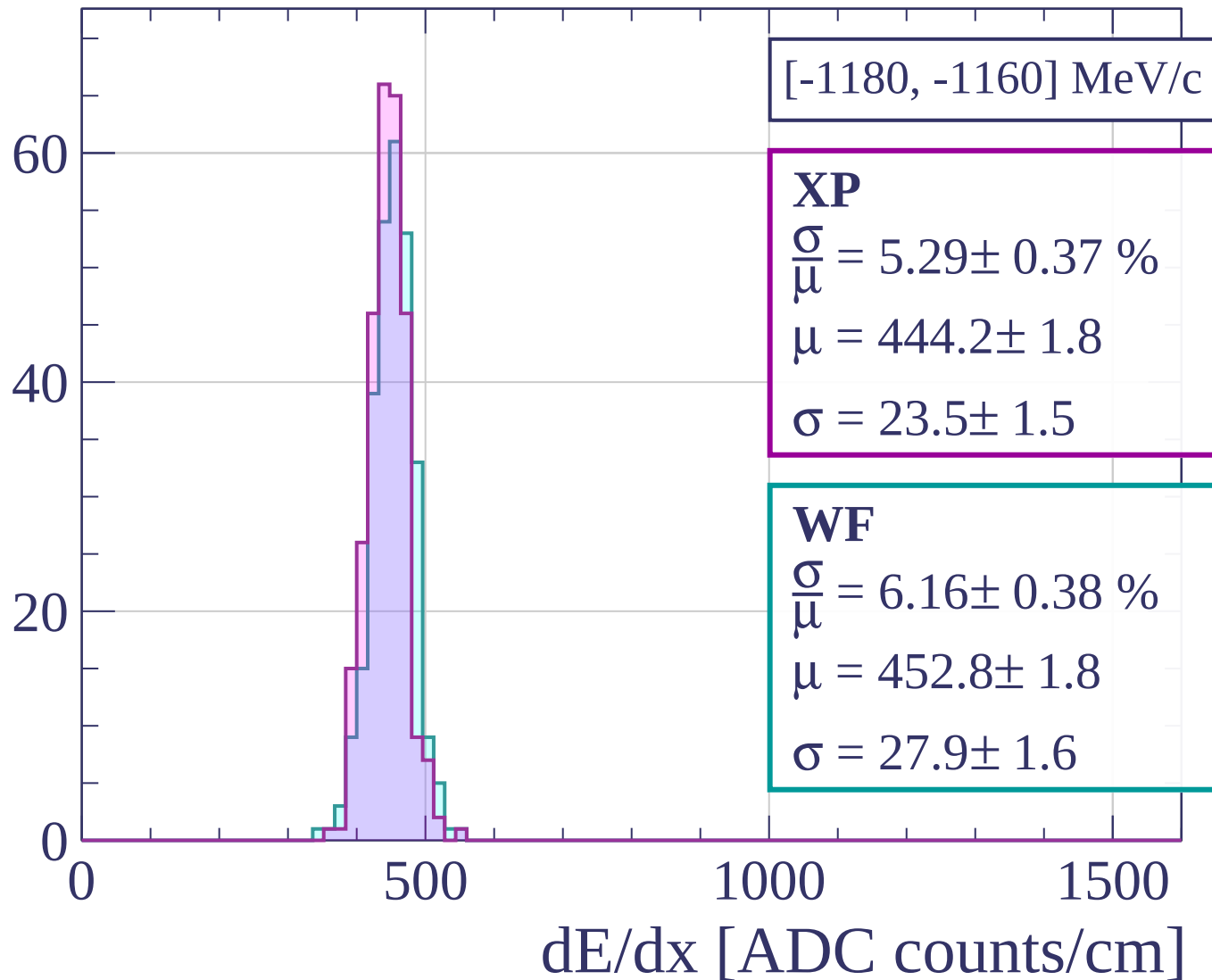
Count



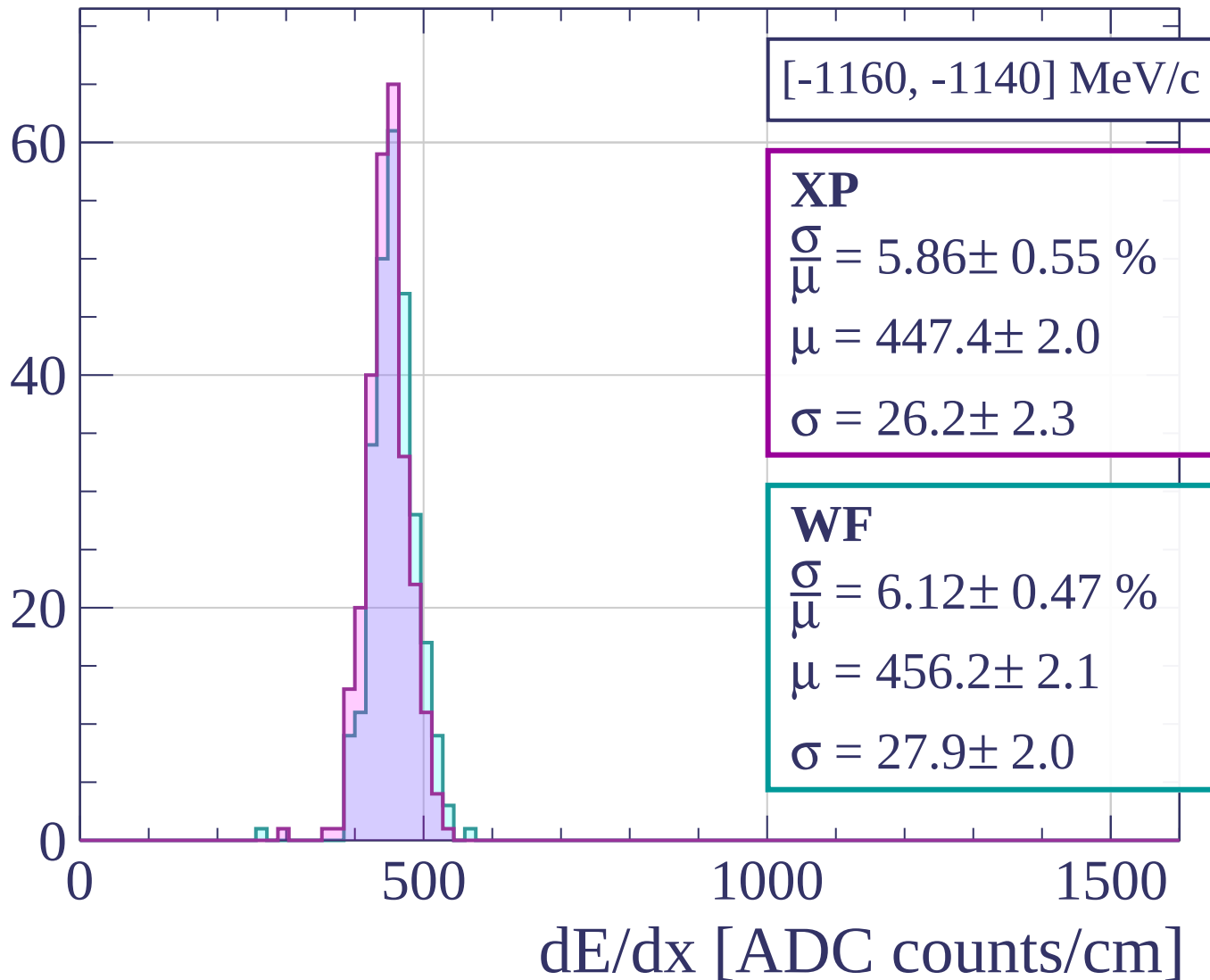
Count



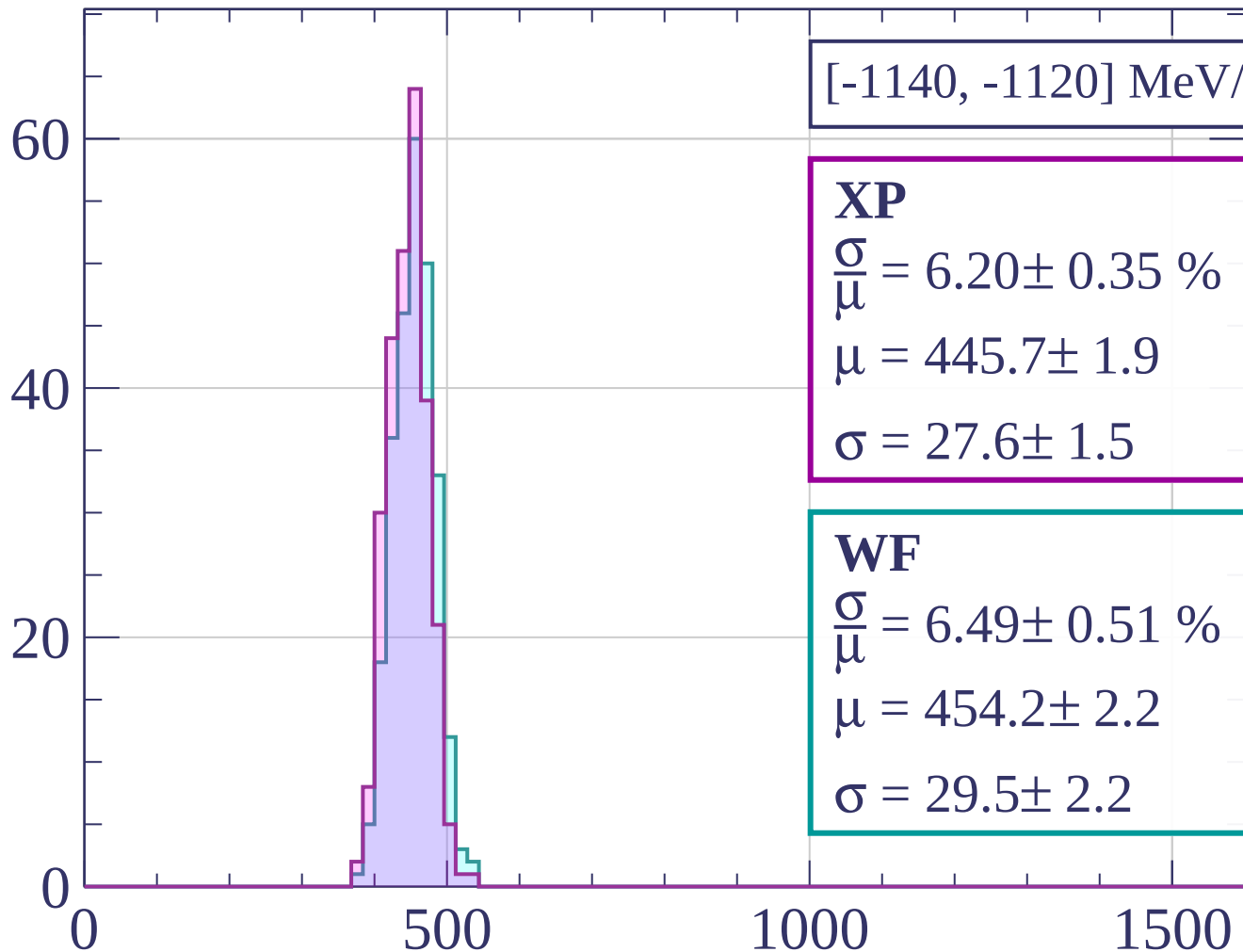
Count



Count



Count



[-1140, -1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.20 \pm 0.35 \%$$

$$\mu = 445.7 \pm 1.9$$

$$\sigma = 27.6 \pm 1.5$$

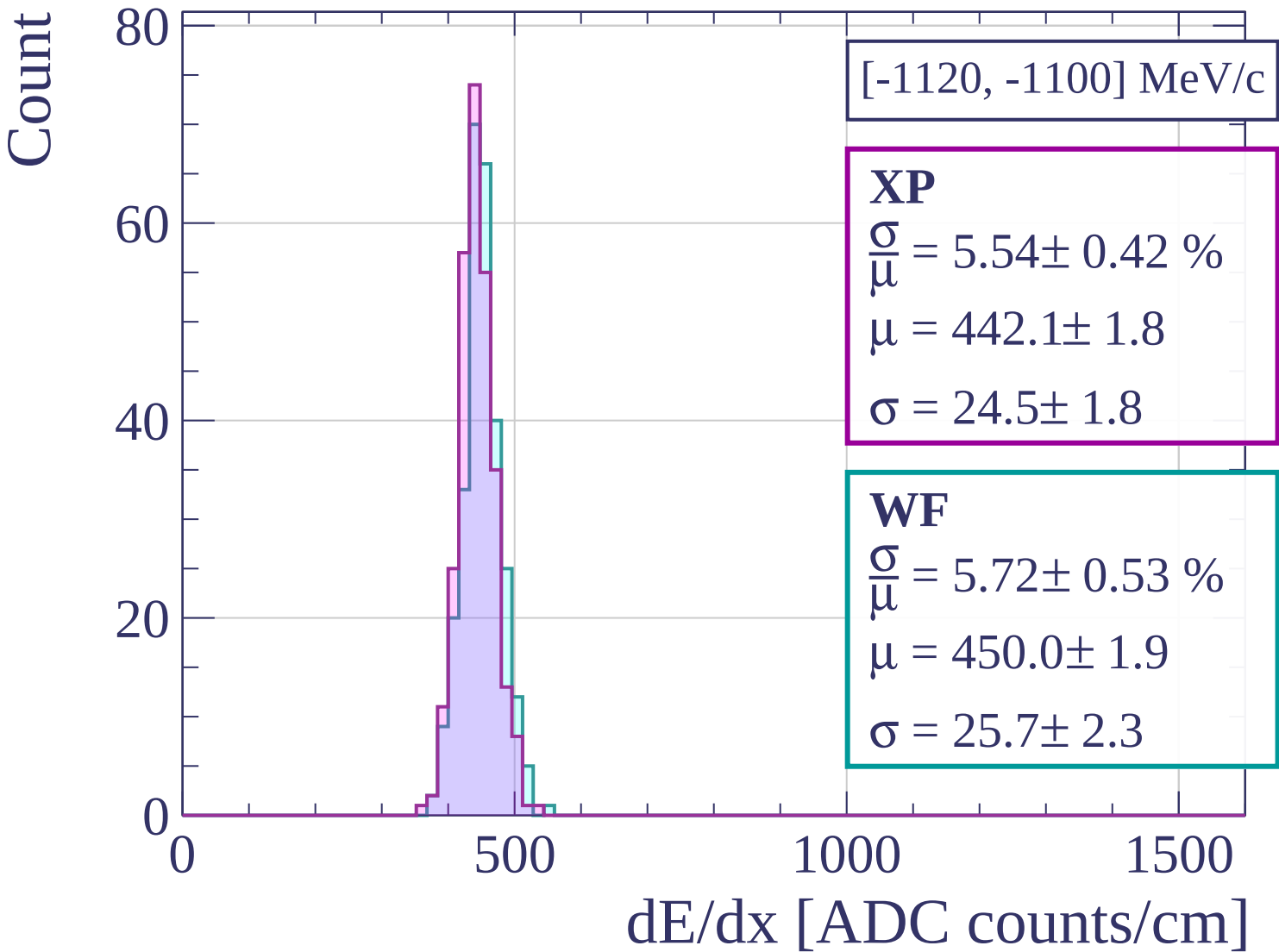
WF

$$\frac{\sigma}{\mu} = 6.49 \pm 0.51 \%$$

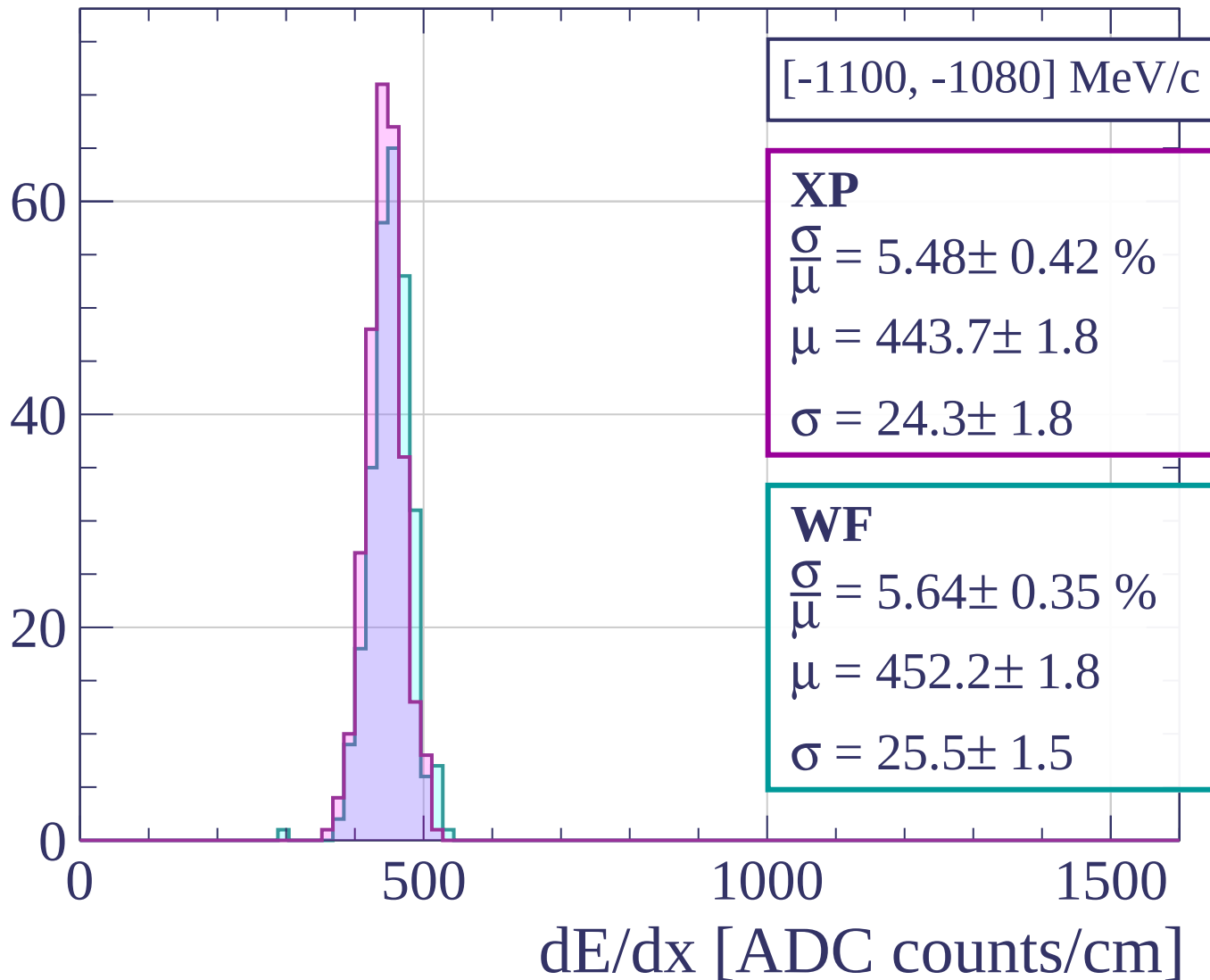
$$\mu = 454.2 \pm 2.2$$

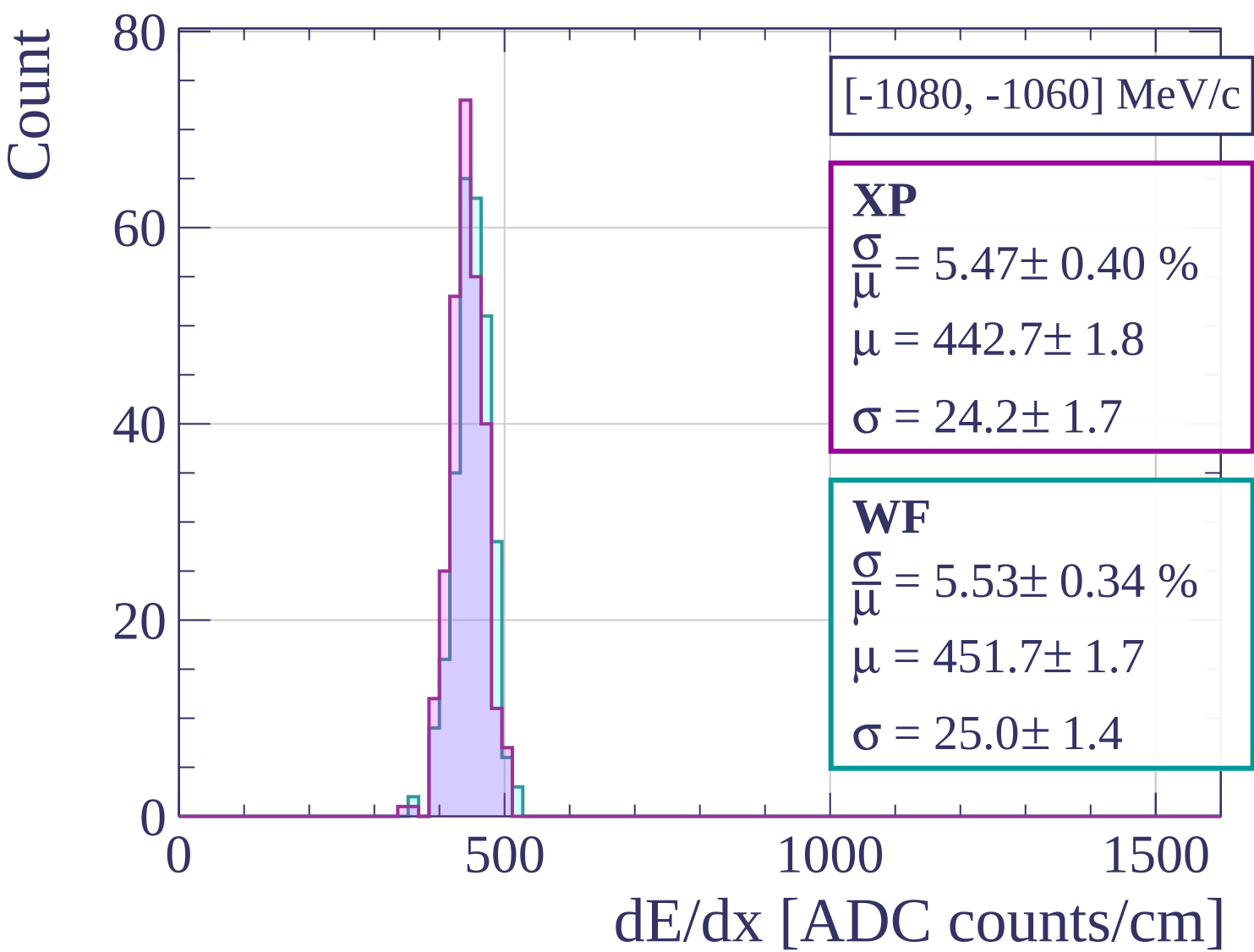
$$\sigma = 29.5 \pm 2.2$$

dE/dx [ADC counts/cm]

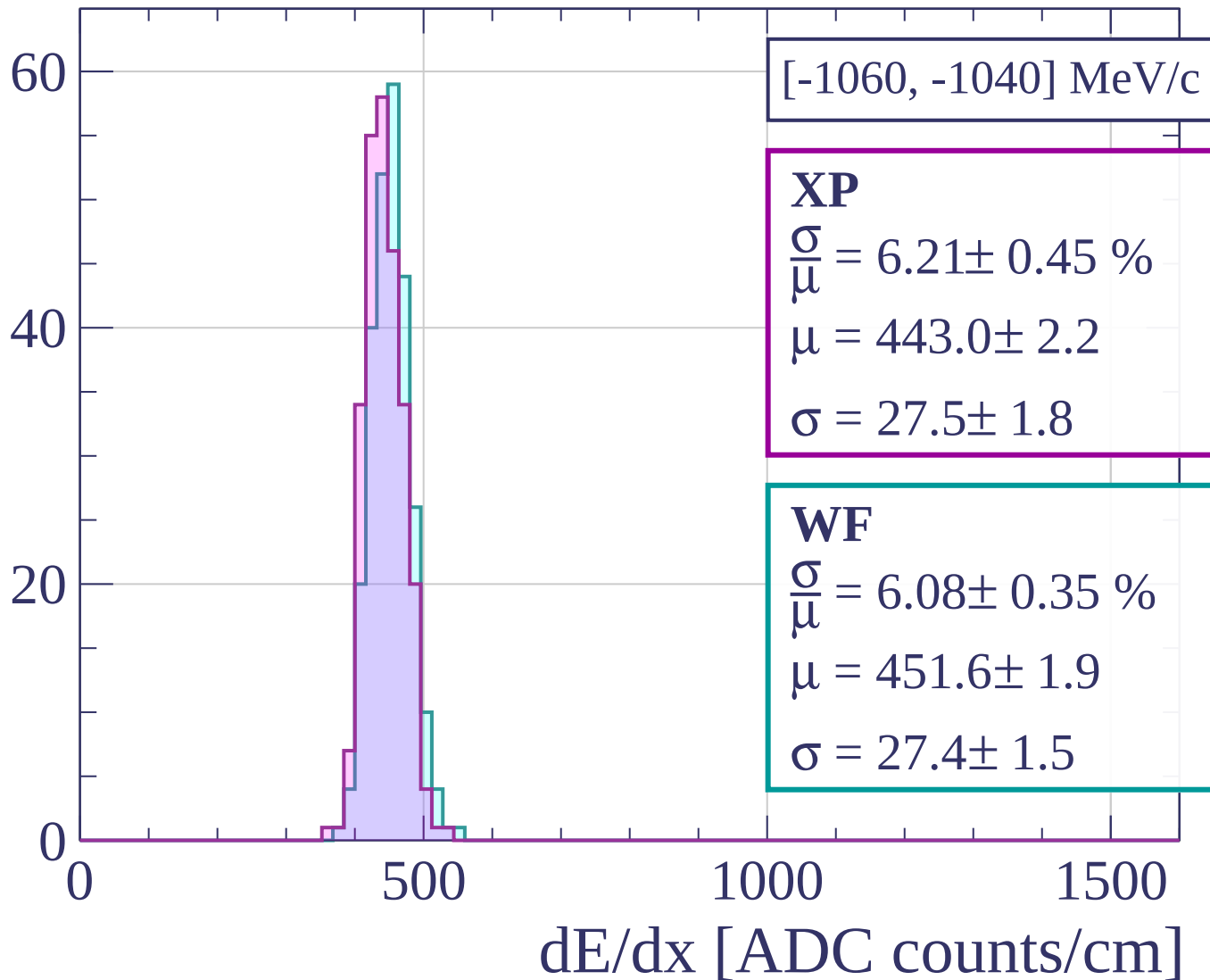


Count

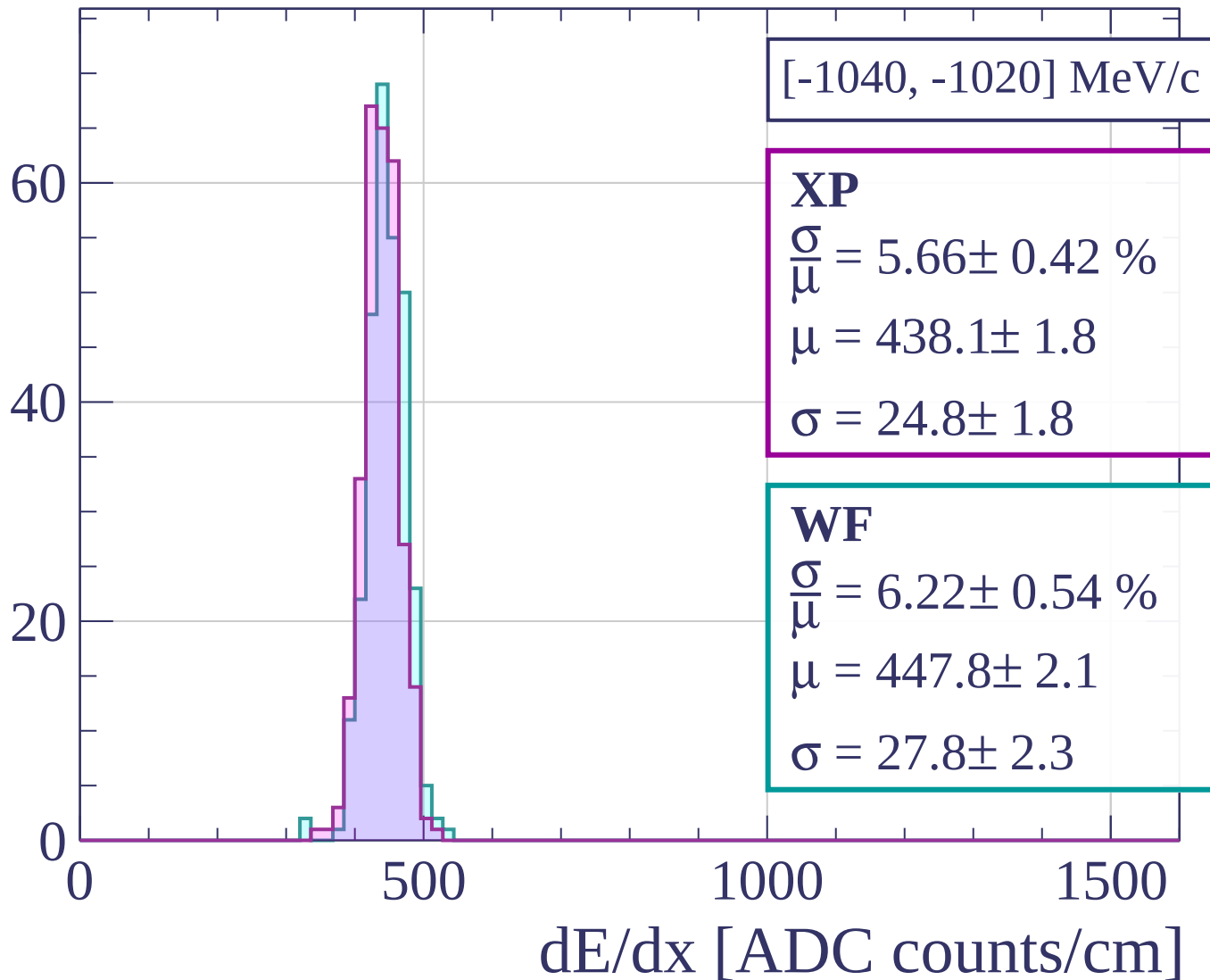




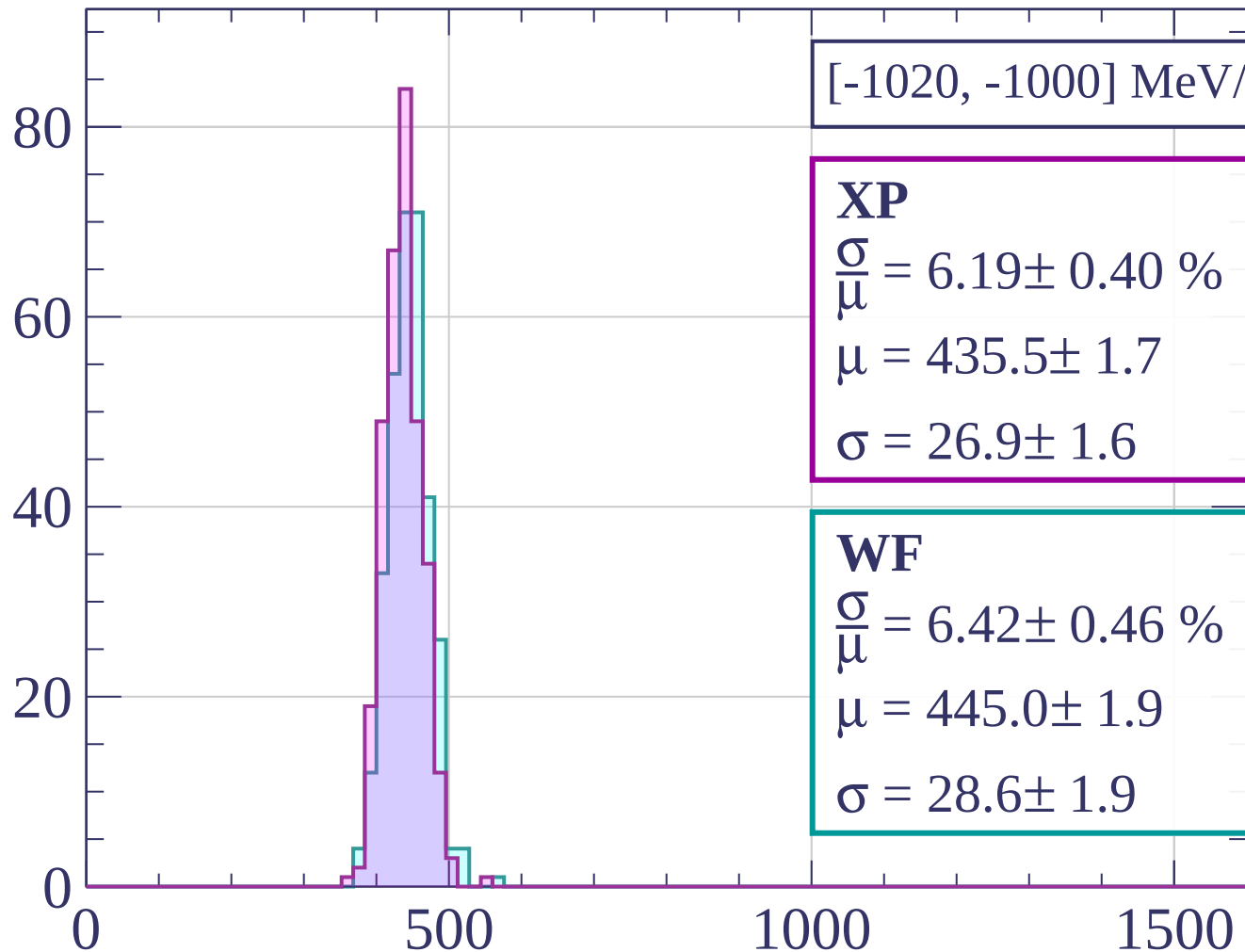
Count



Count



Count



[-1020, -1000] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.19 \pm 0.40 \%$$

$$\mu = 435.5 \pm 1.7$$

$$\sigma = 26.9 \pm 1.6$$

WF

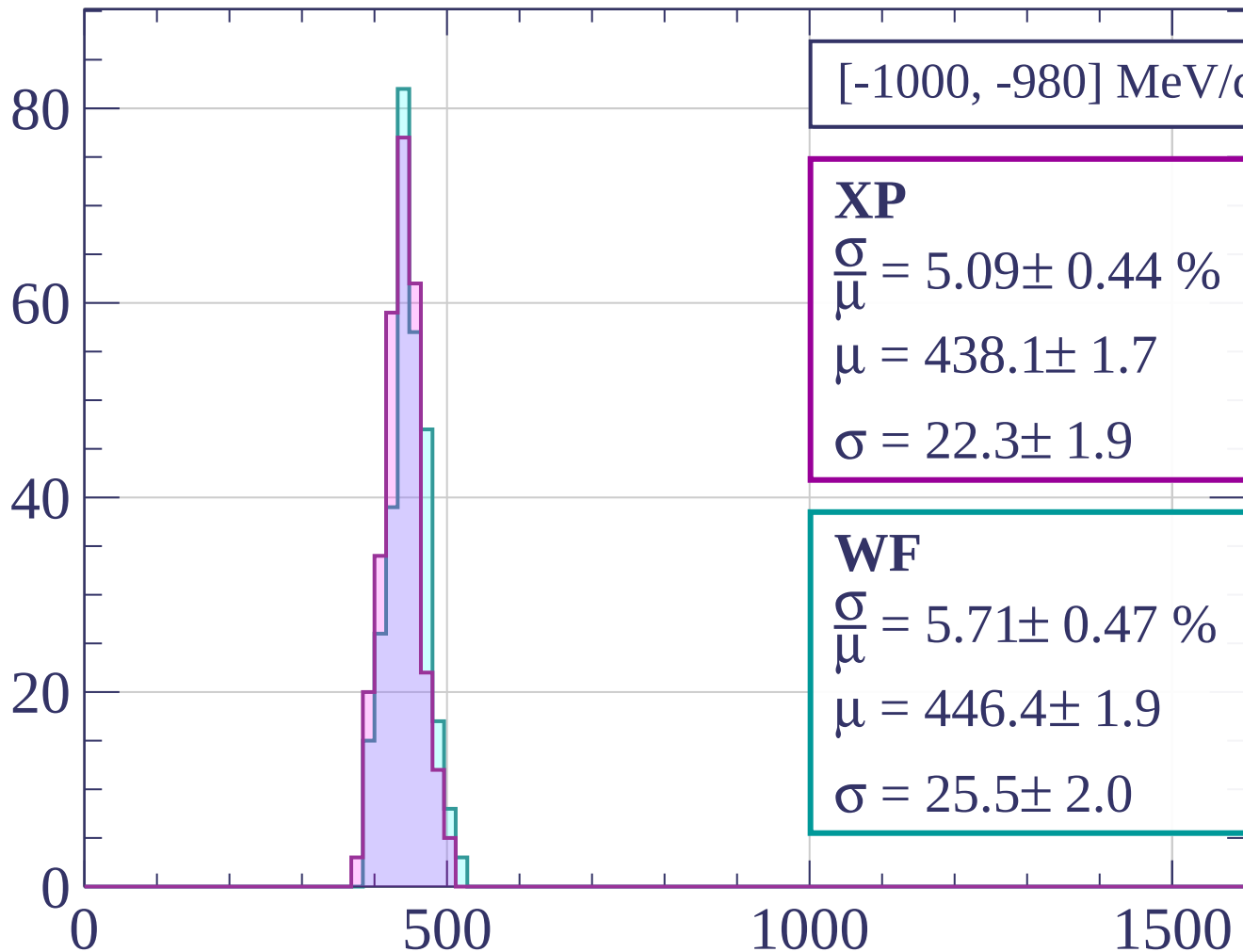
$$\frac{\sigma}{\mu} = 6.42 \pm 0.46 \%$$

$$\mu = 445.0 \pm 1.9$$

$$\sigma = 28.6 \pm 1.9$$

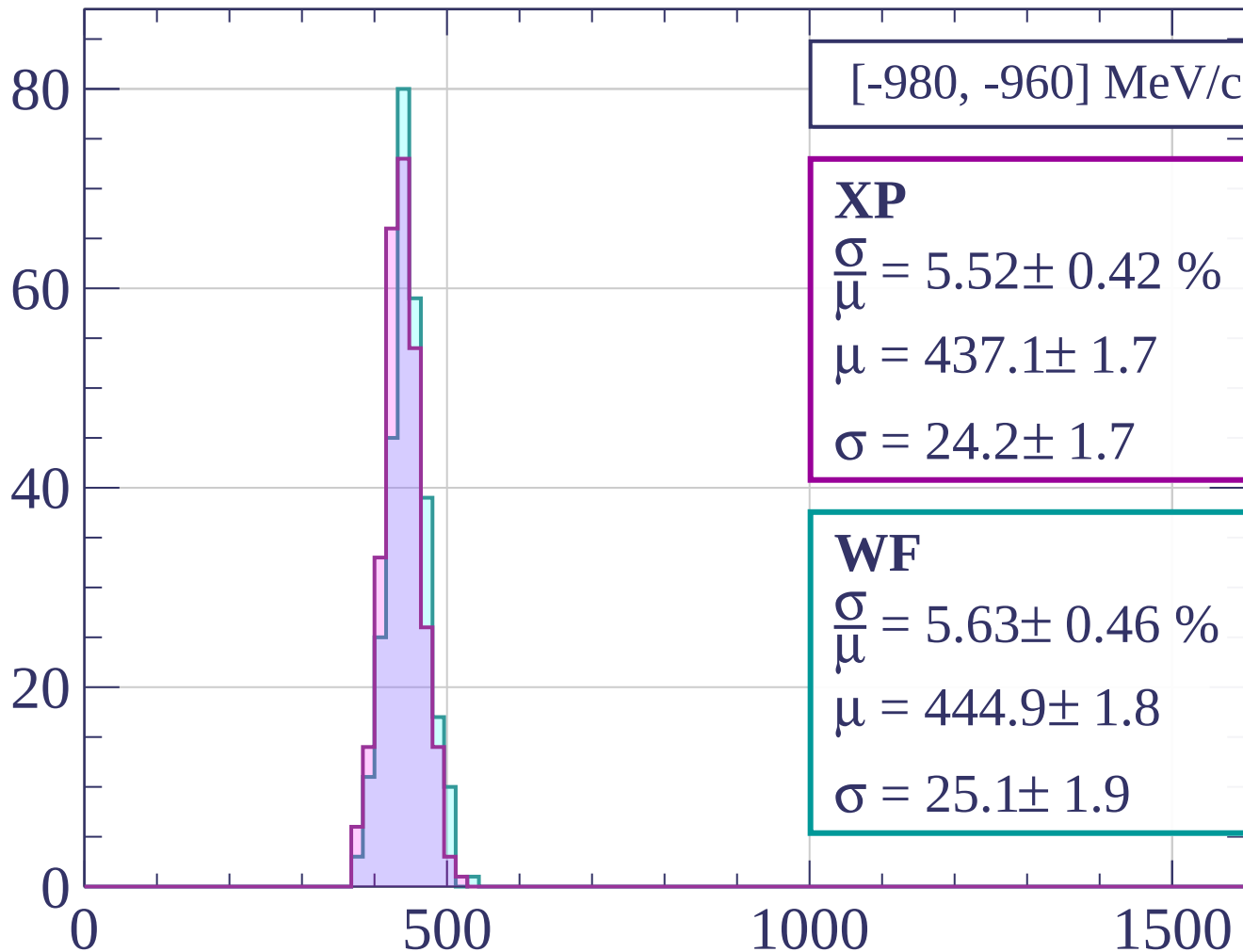
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



$[-980, -960]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 5.52 \pm 0.42 \%$$

$$\mu = 437.1 \pm 1.7$$

$$\sigma = 24.2 \pm 1.7$$

WF

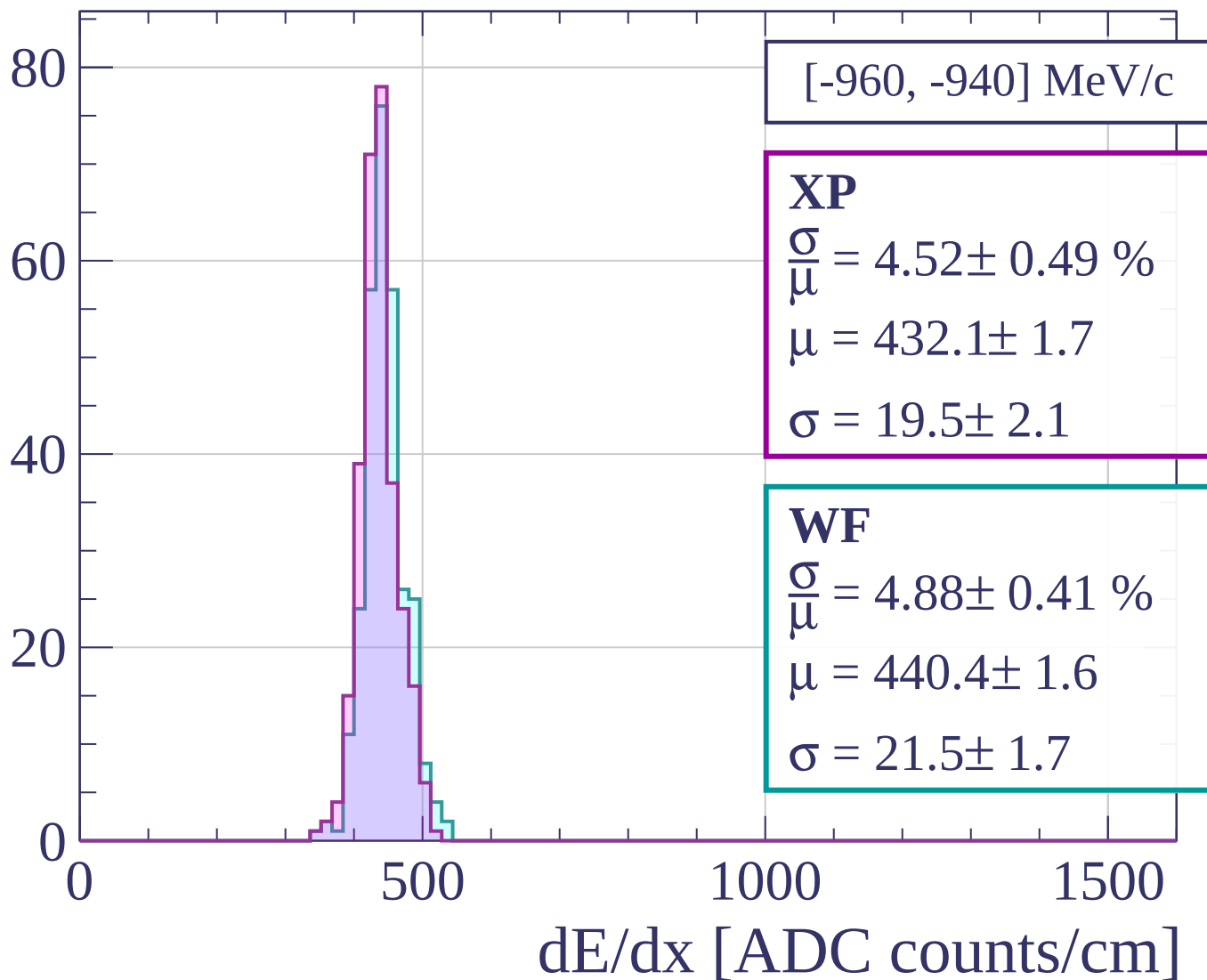
$$\frac{\sigma}{\mu} = 5.63 \pm 0.46 \%$$

$$\mu = 444.9 \pm 1.8$$

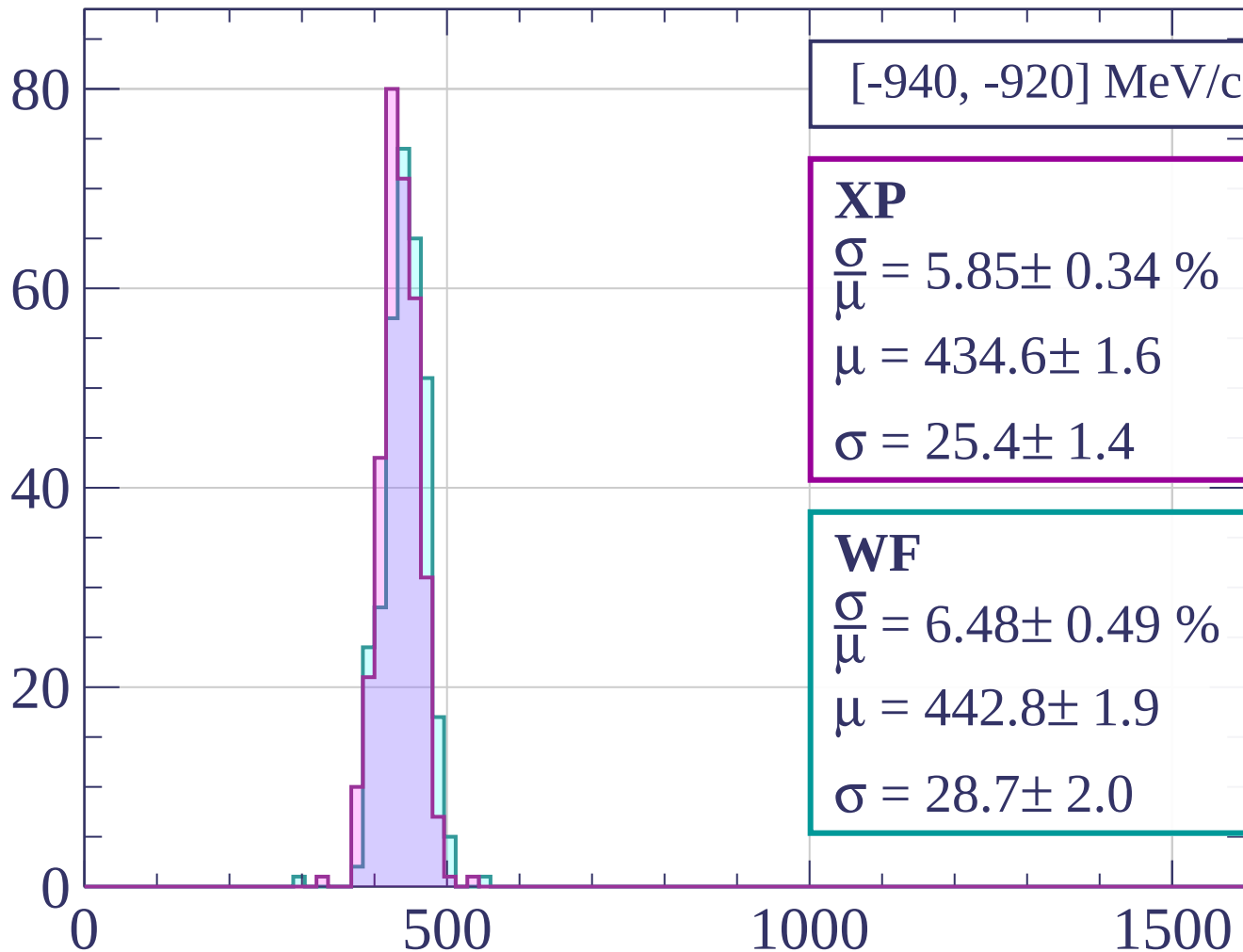
$$\sigma = 25.1 \pm 1.9$$

dE/dx [ADC counts/cm]

Count



Count



$[-940, -920] \text{ MeV/c}$

XP

$$\frac{\sigma}{\bar{\mu}} = 5.85 \pm 0.34 \%$$

$$\mu = 434.6 \pm 1.6$$

$$\sigma = 25.4 \pm 1.4$$

WF

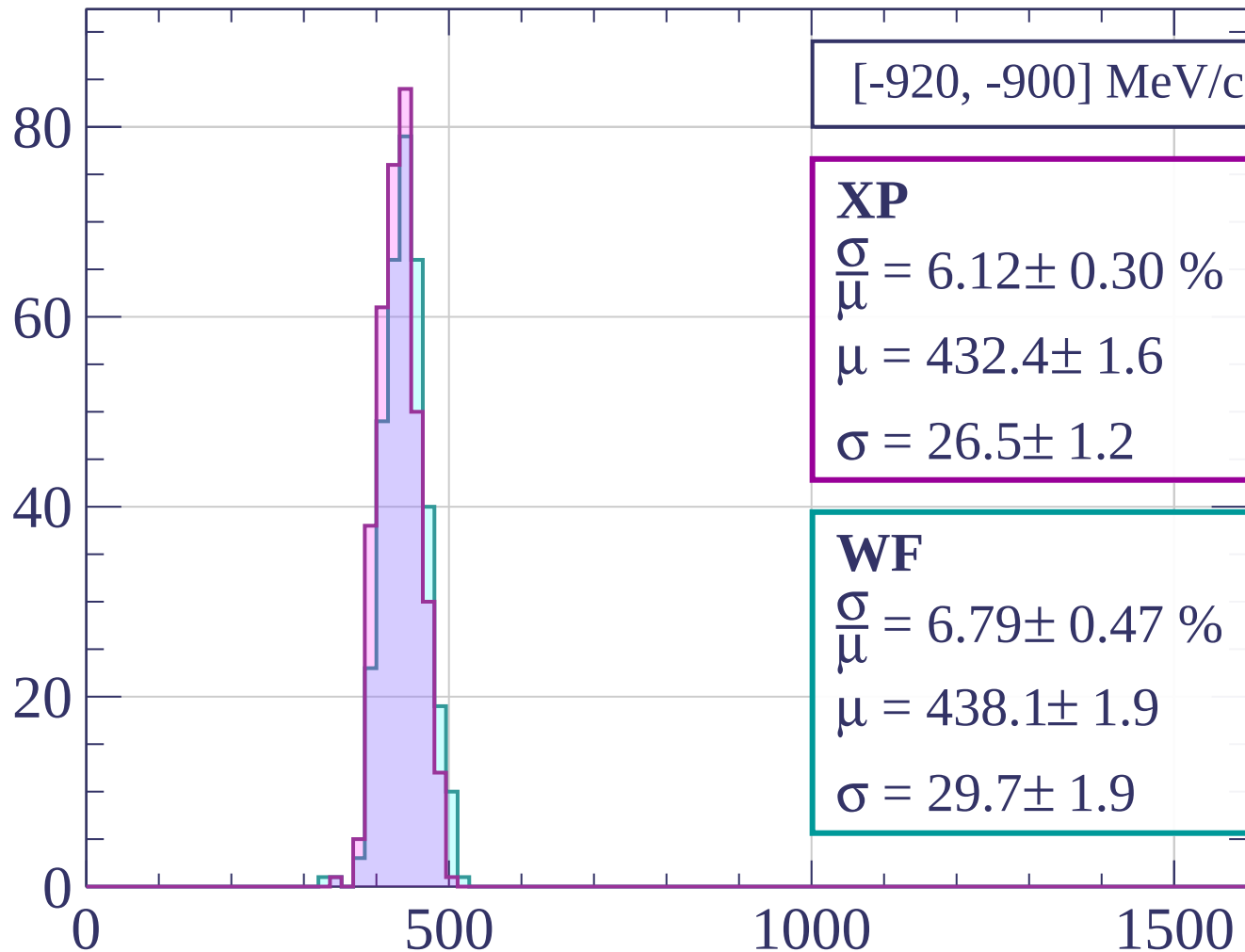
$$\frac{\sigma}{\bar{\mu}} = 6.48 \pm 0.49 \%$$

$$\mu = 442.8 \pm 1.9$$

$$\sigma = 28.7 \pm 2.0$$

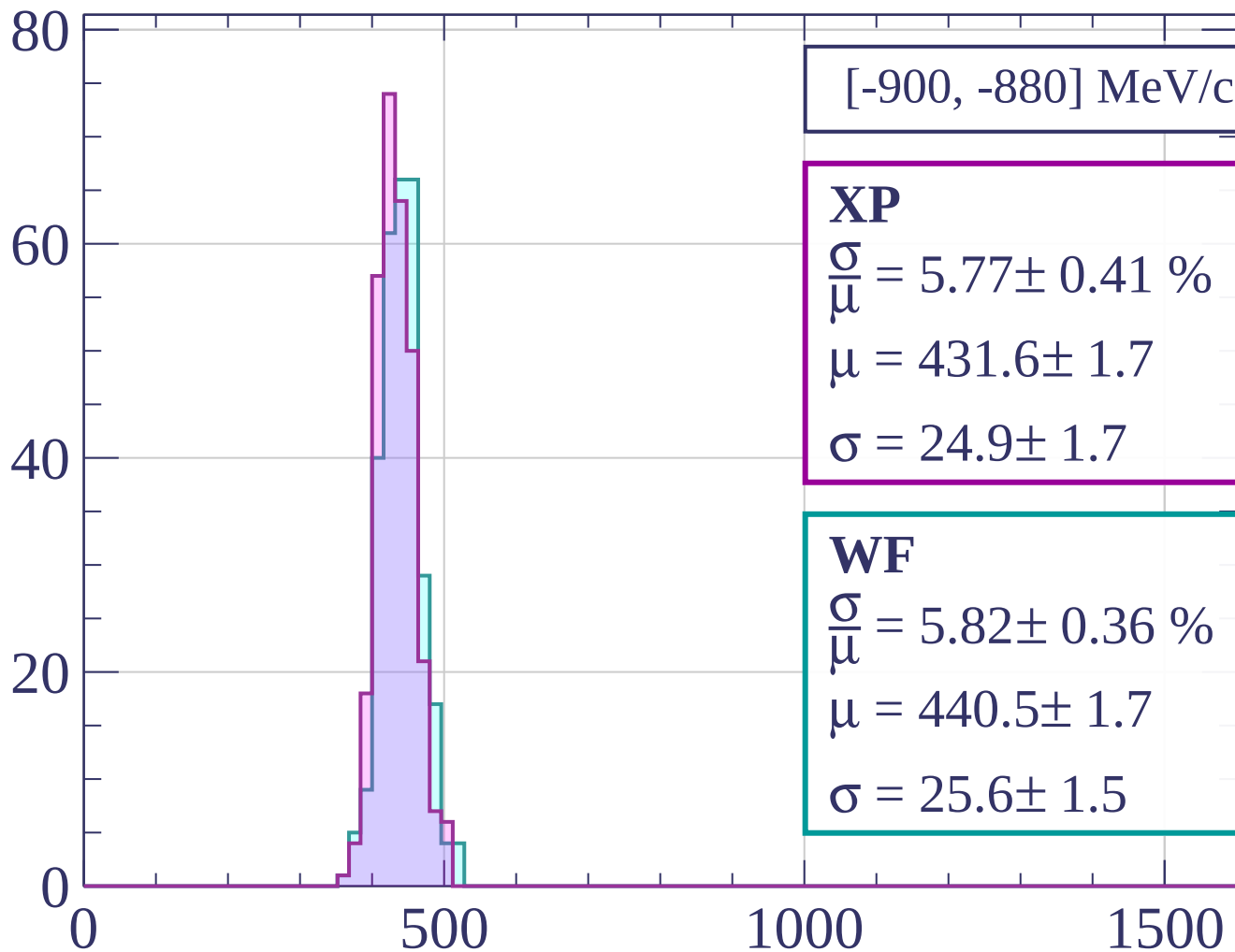
dE/dx [ADC counts/cm]

Count



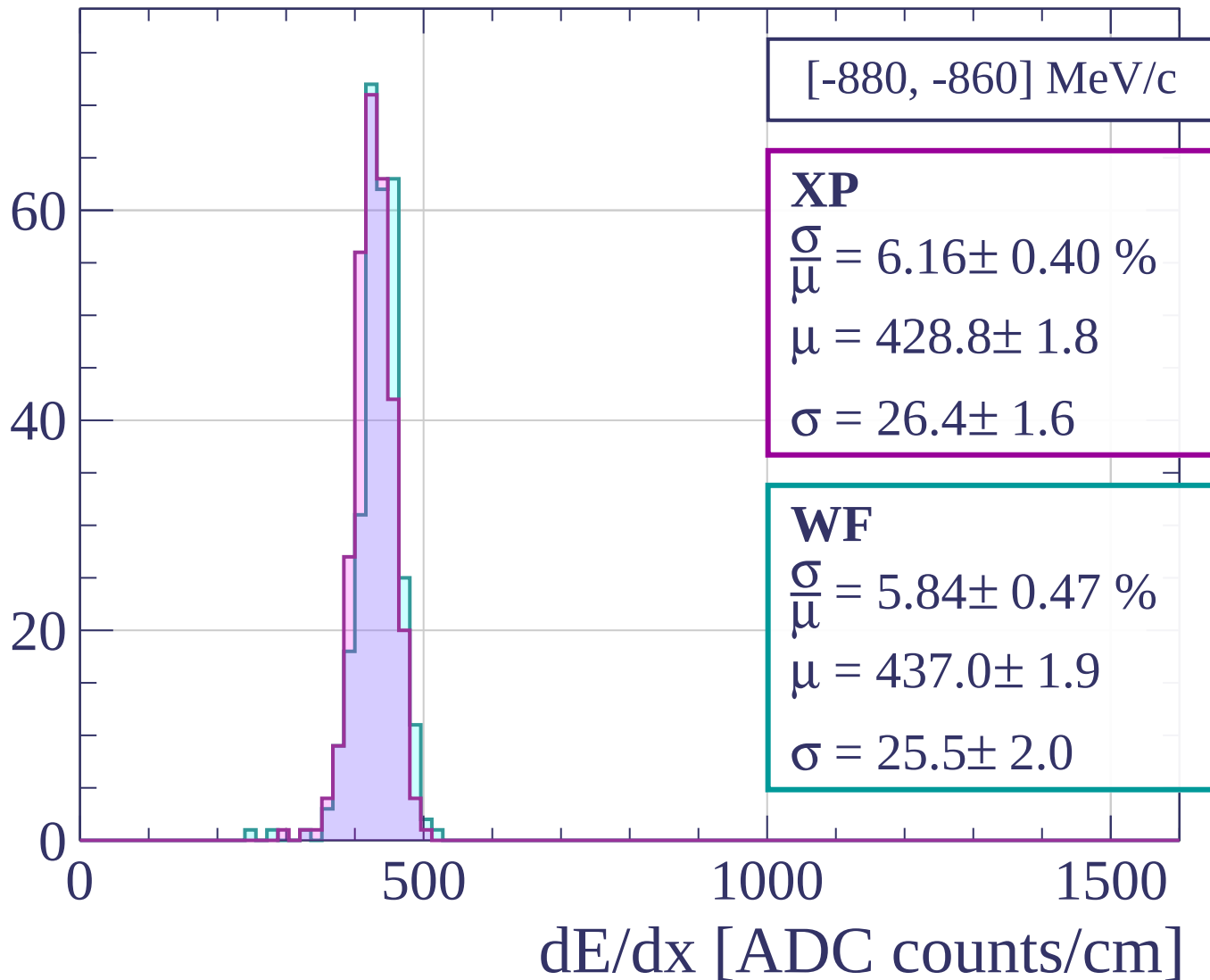
dE/dx [ADC counts/cm]

Count

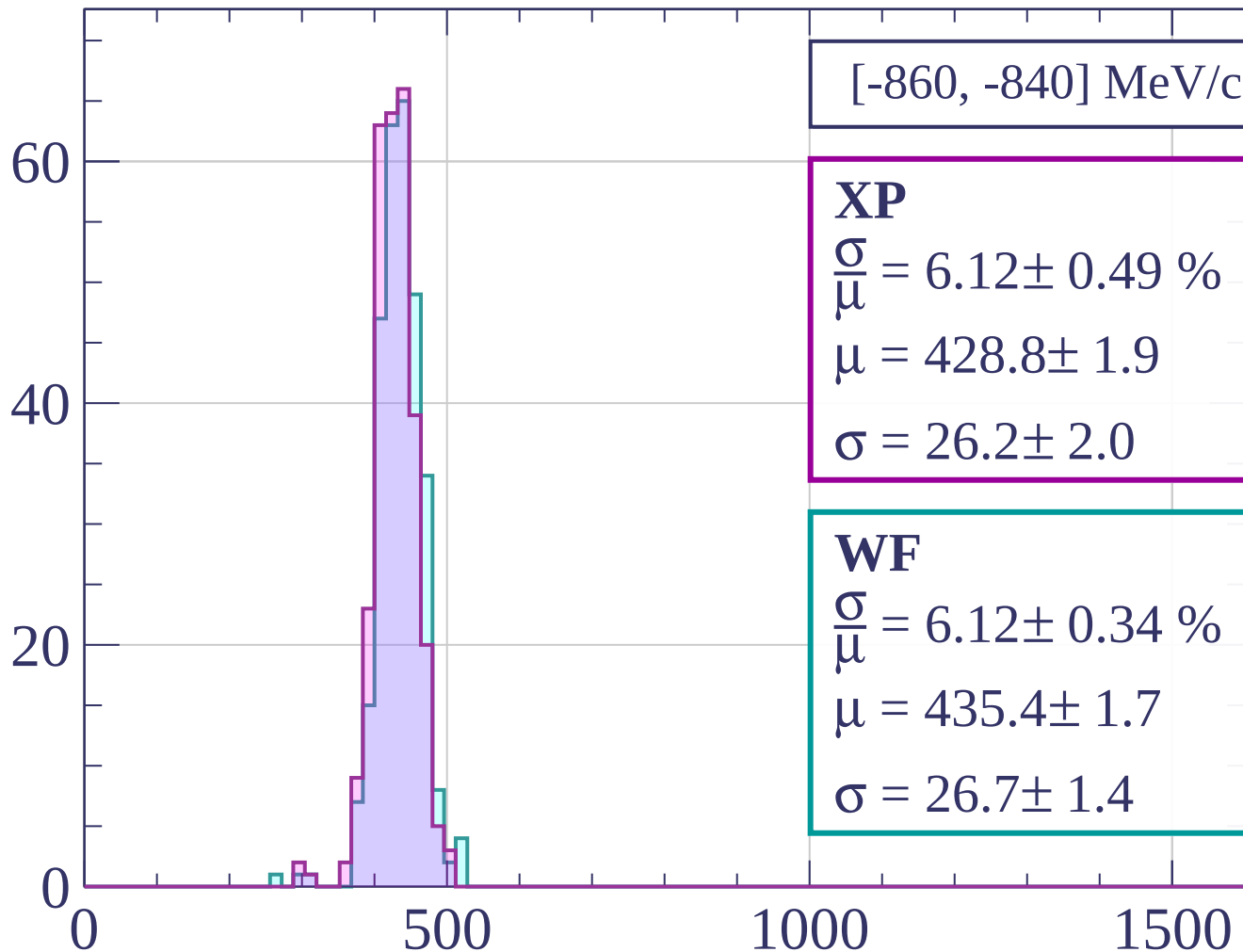


dE/dx [ADC counts/cm]

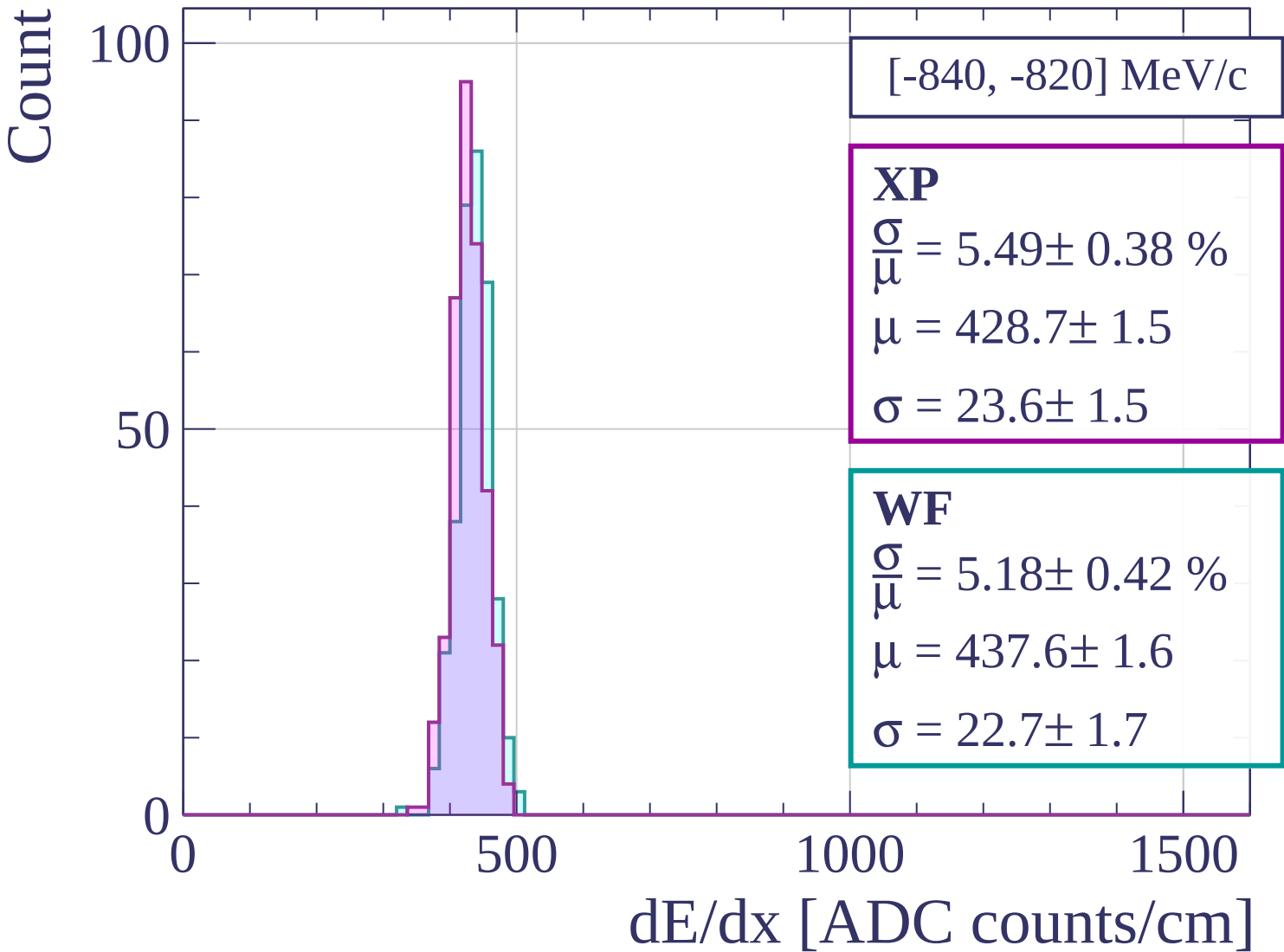
Count



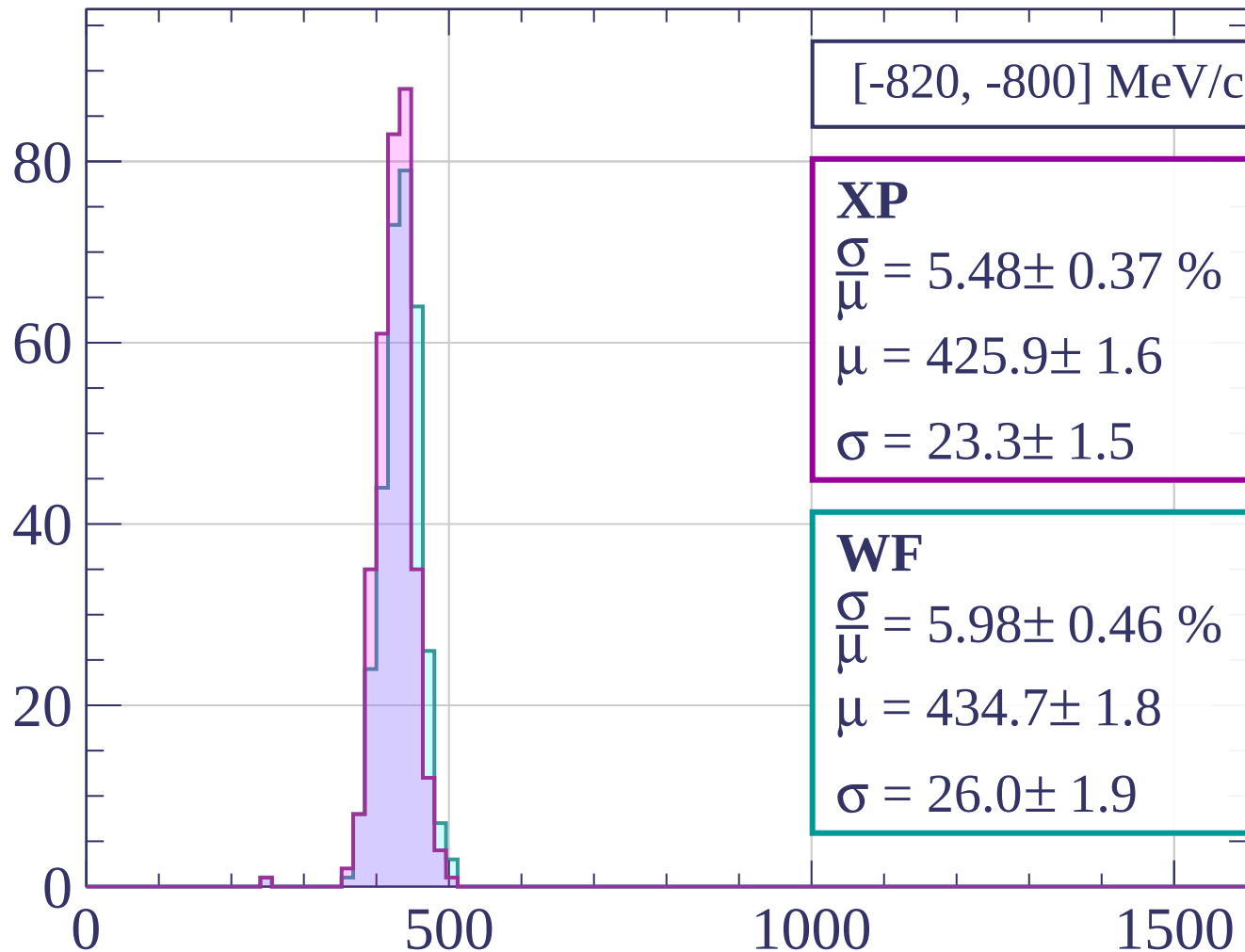
Count



dE/dx [ADC counts/cm]

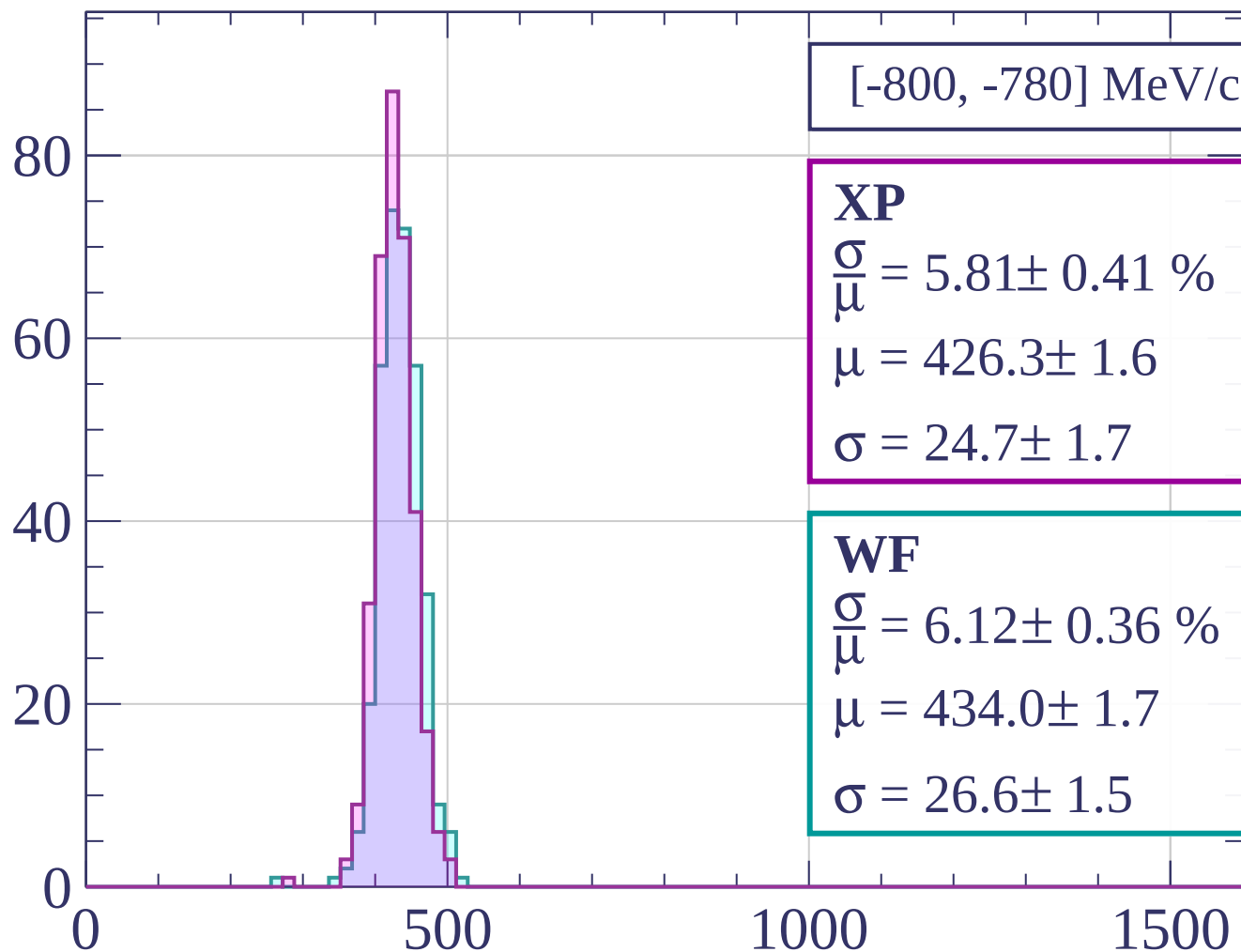


Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[-780, -760] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.64 \pm 0.52 \%$$

$$\mu = 424.1 \pm 1.8$$

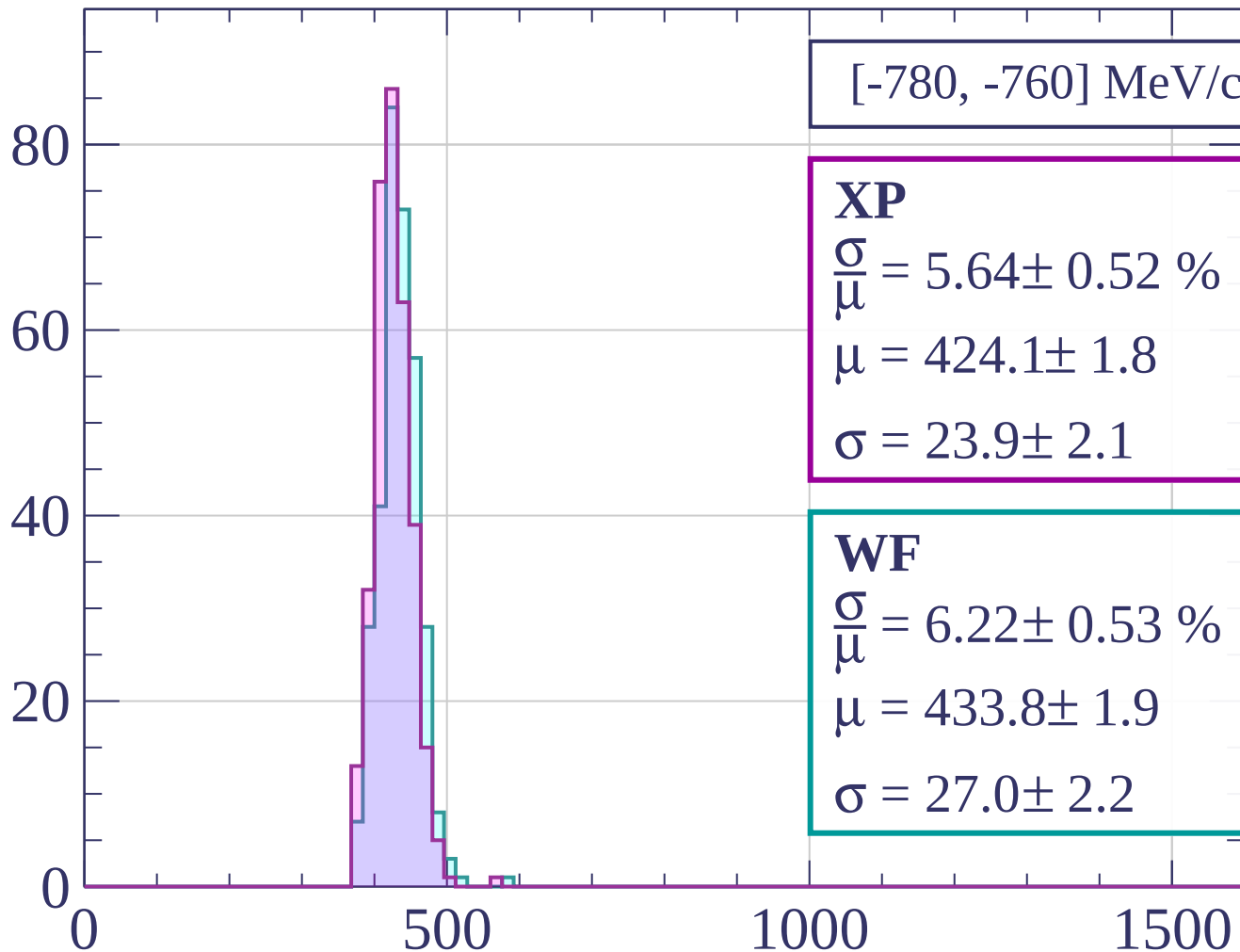
$$\sigma = 23.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 6.22 \pm 0.53 \%$$

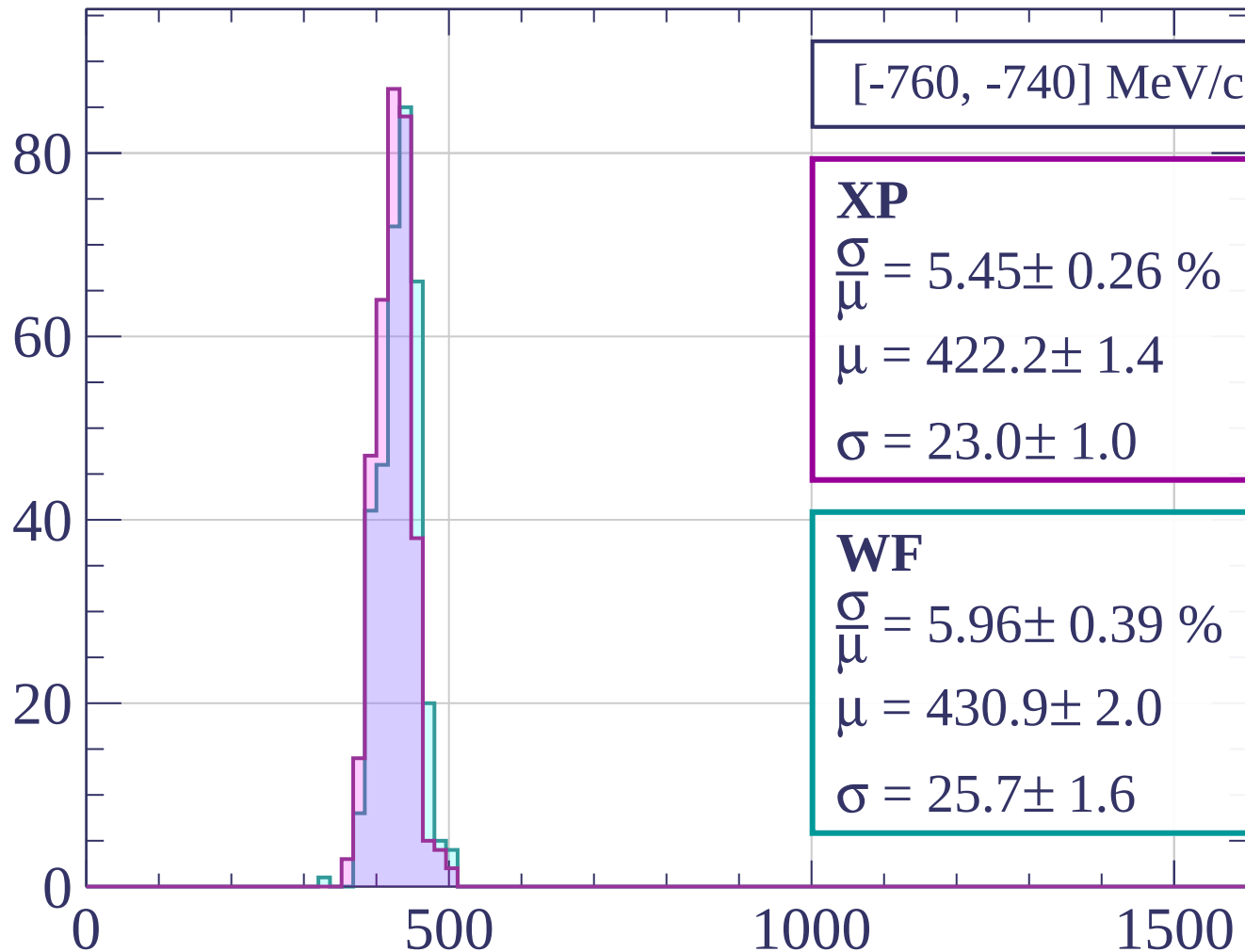
$$\mu = 433.8 \pm 1.9$$

$$\sigma = 27.0 \pm 2.2$$



dE/dx [ADC counts/cm]

Count



[-760, -740] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.45 \pm 0.26 \%$$

$$\mu = 422.2 \pm 1.4$$

$$\sigma = 23.0 \pm 1.0$$

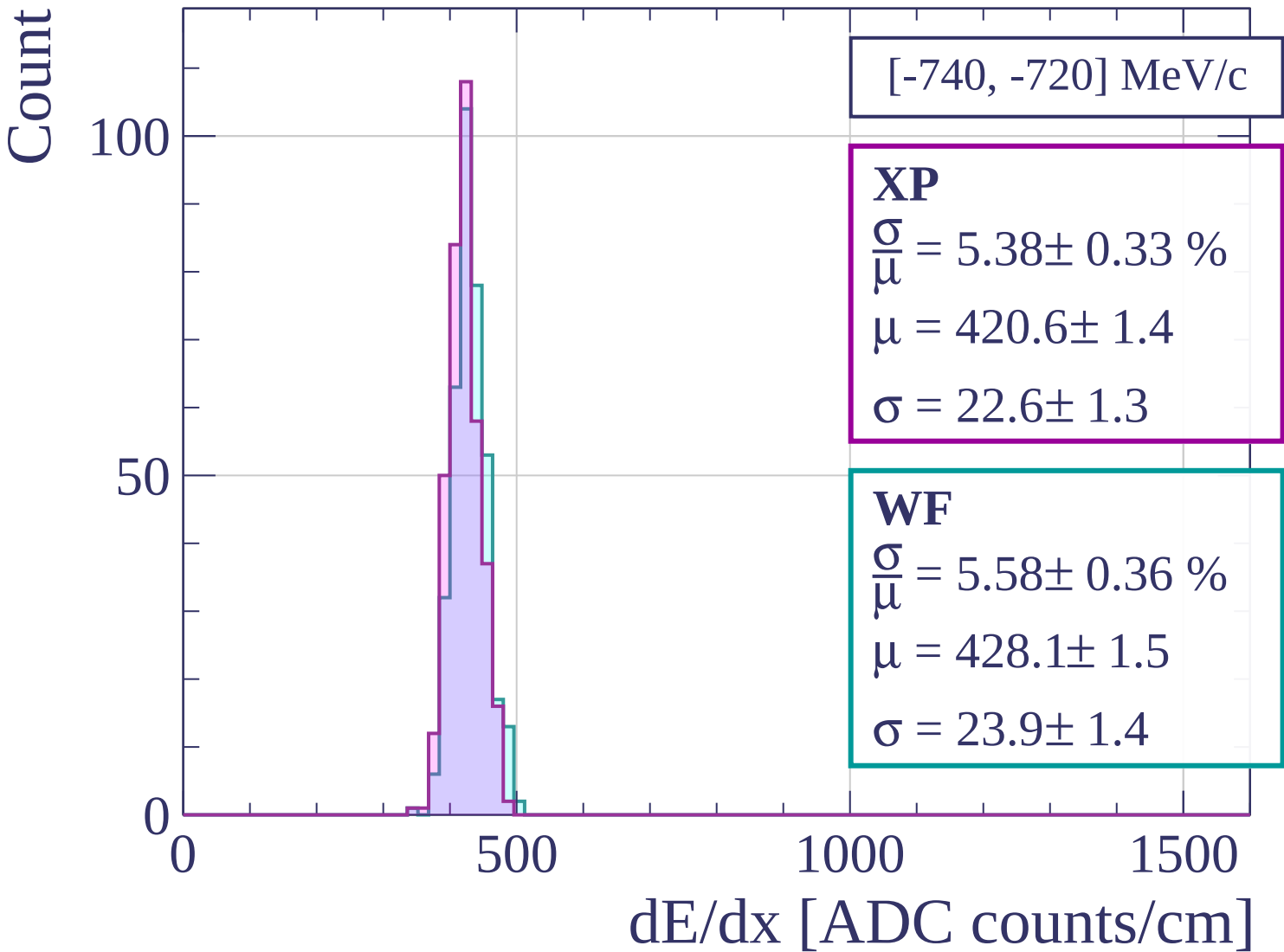
WF

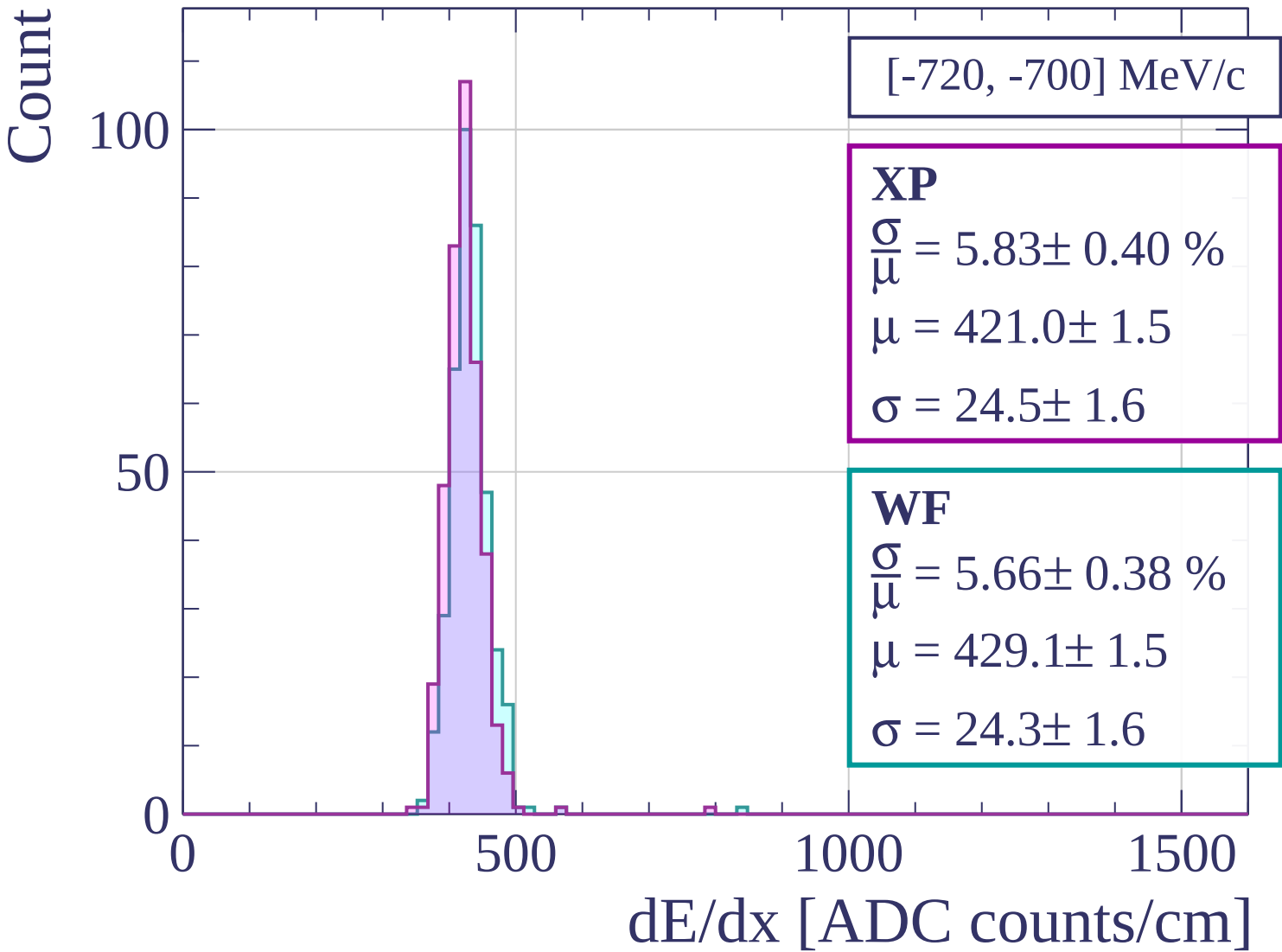
$$\frac{\sigma}{\mu} = 5.96 \pm 0.39 \%$$

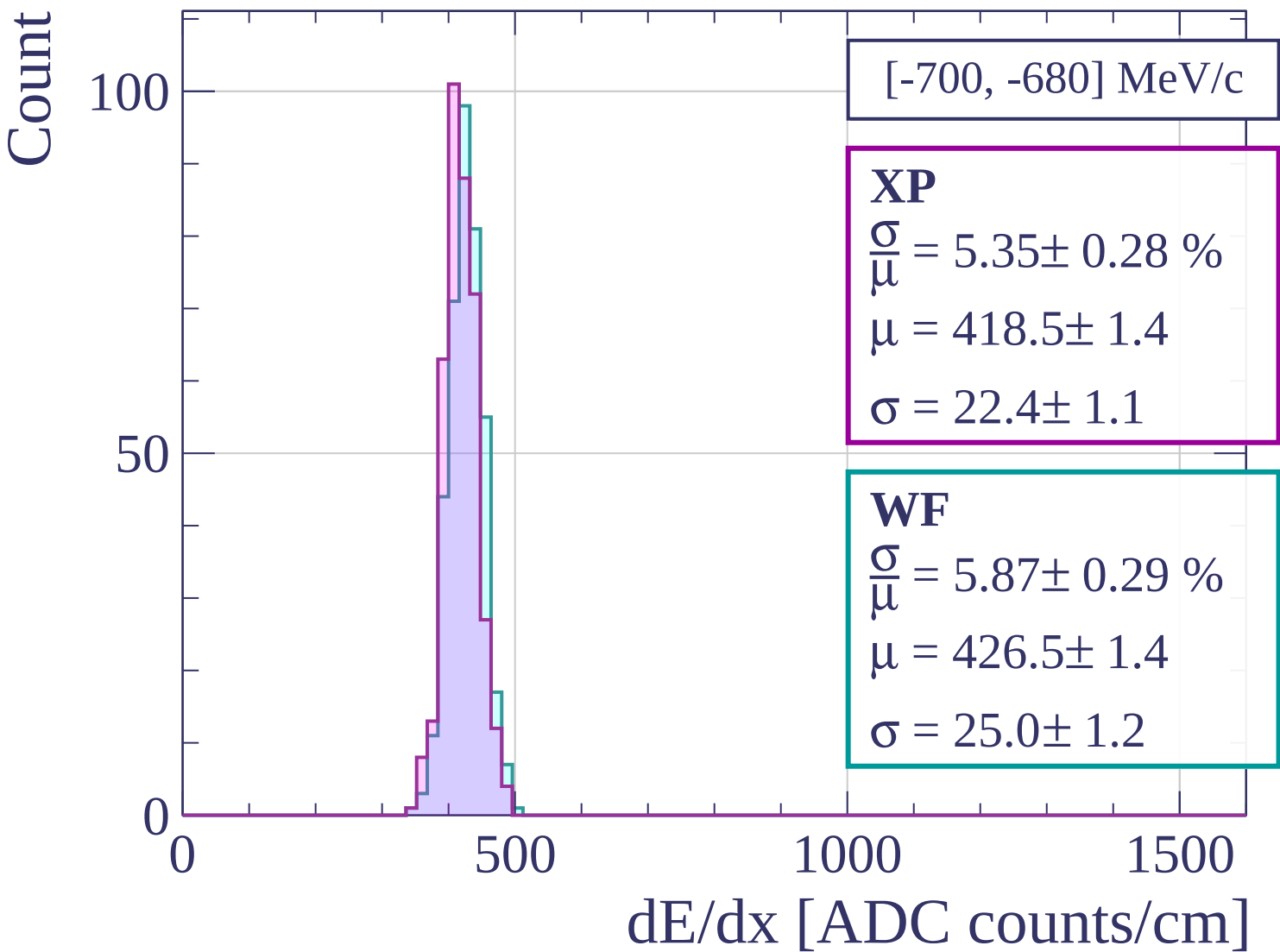
$$\mu = 430.9 \pm 2.0$$

$$\sigma = 25.7 \pm 1.6$$

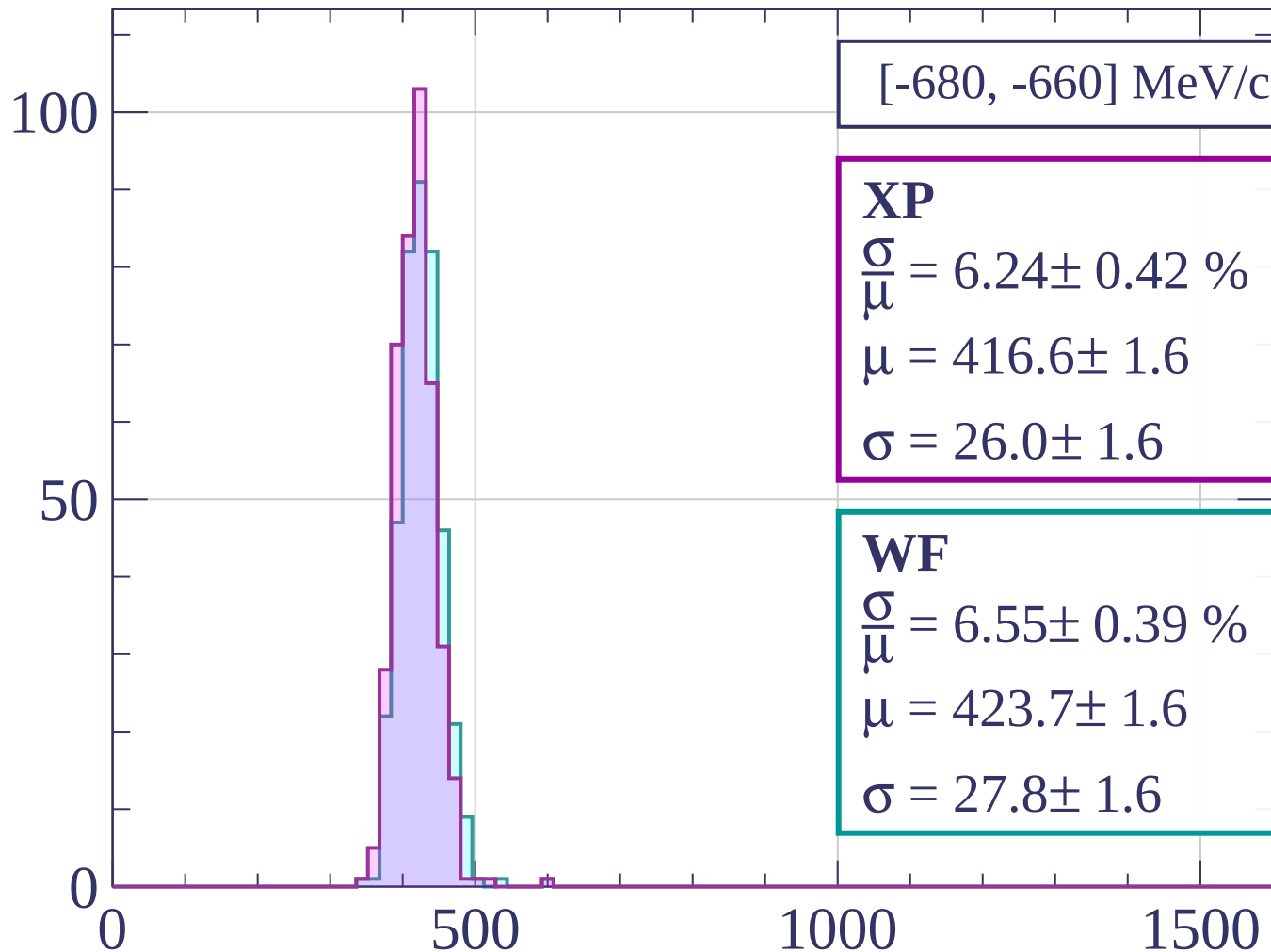
dE/dx [ADC counts/cm]







Count



[-680, -660] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.24 \pm 0.42 \%$$

$$\mu = 416.6 \pm 1.6$$

$$\sigma = 26.0 \pm 1.6$$

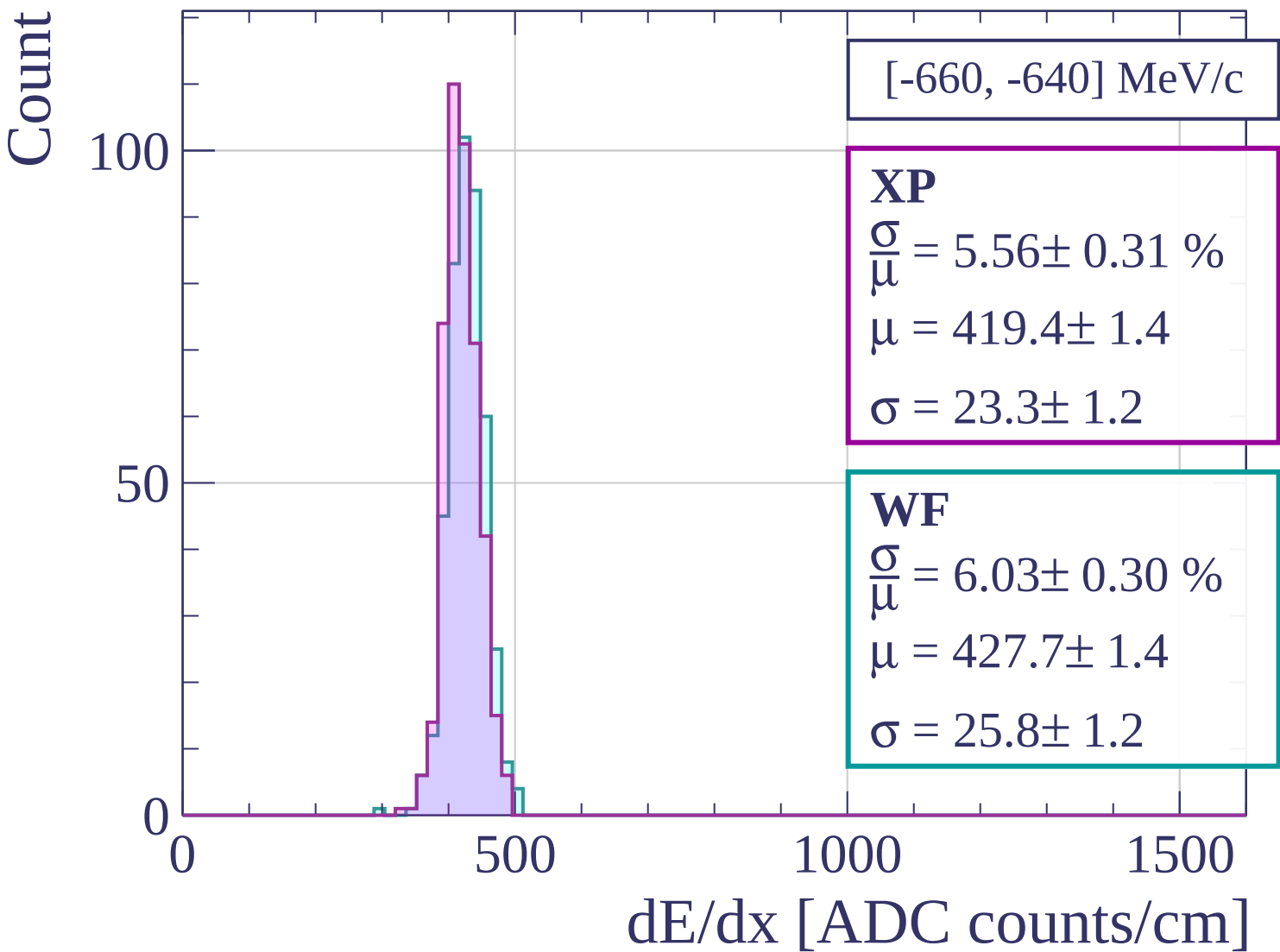
WF

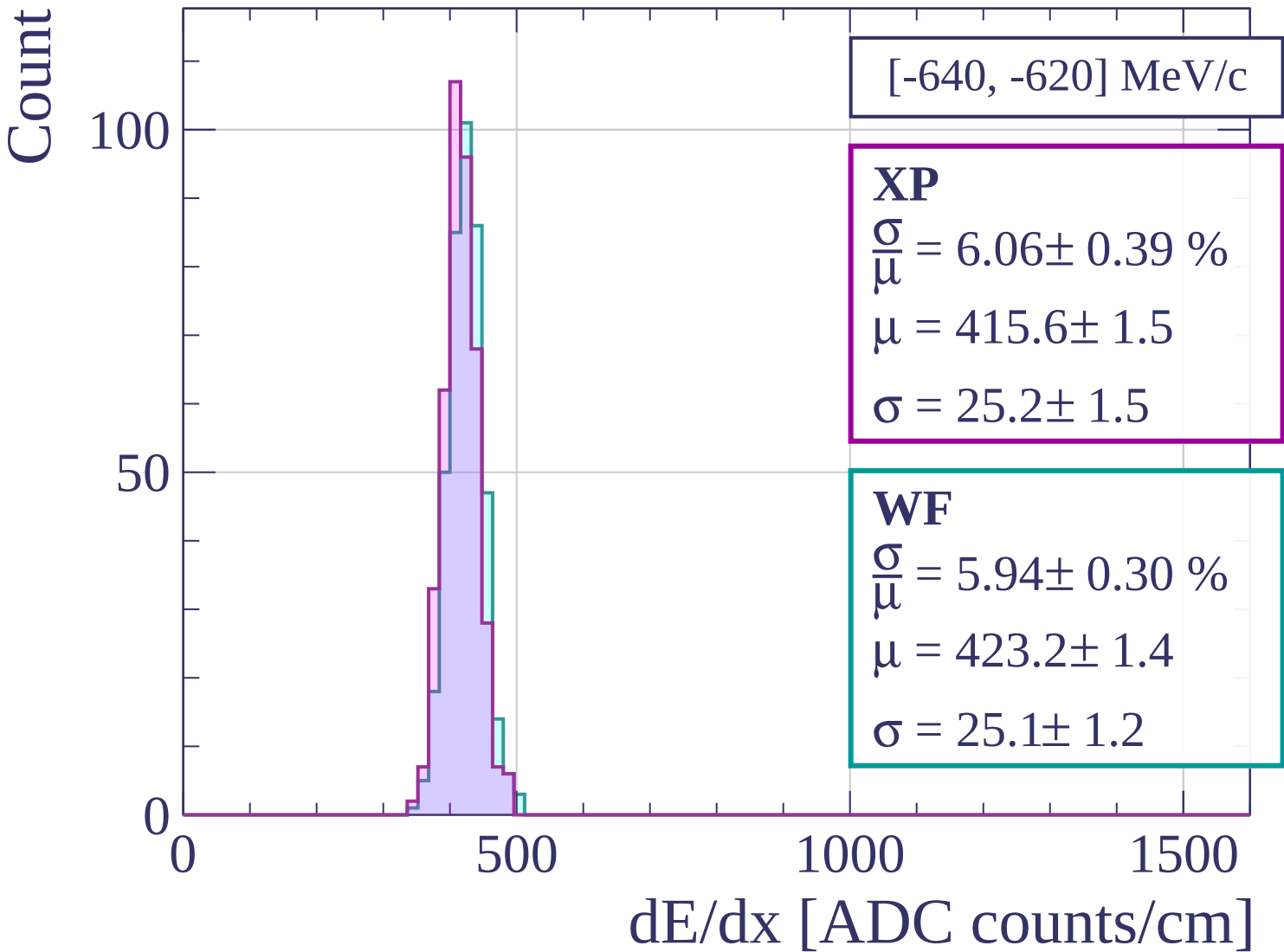
$$\frac{\sigma}{\mu} = 6.55 \pm 0.39 \%$$

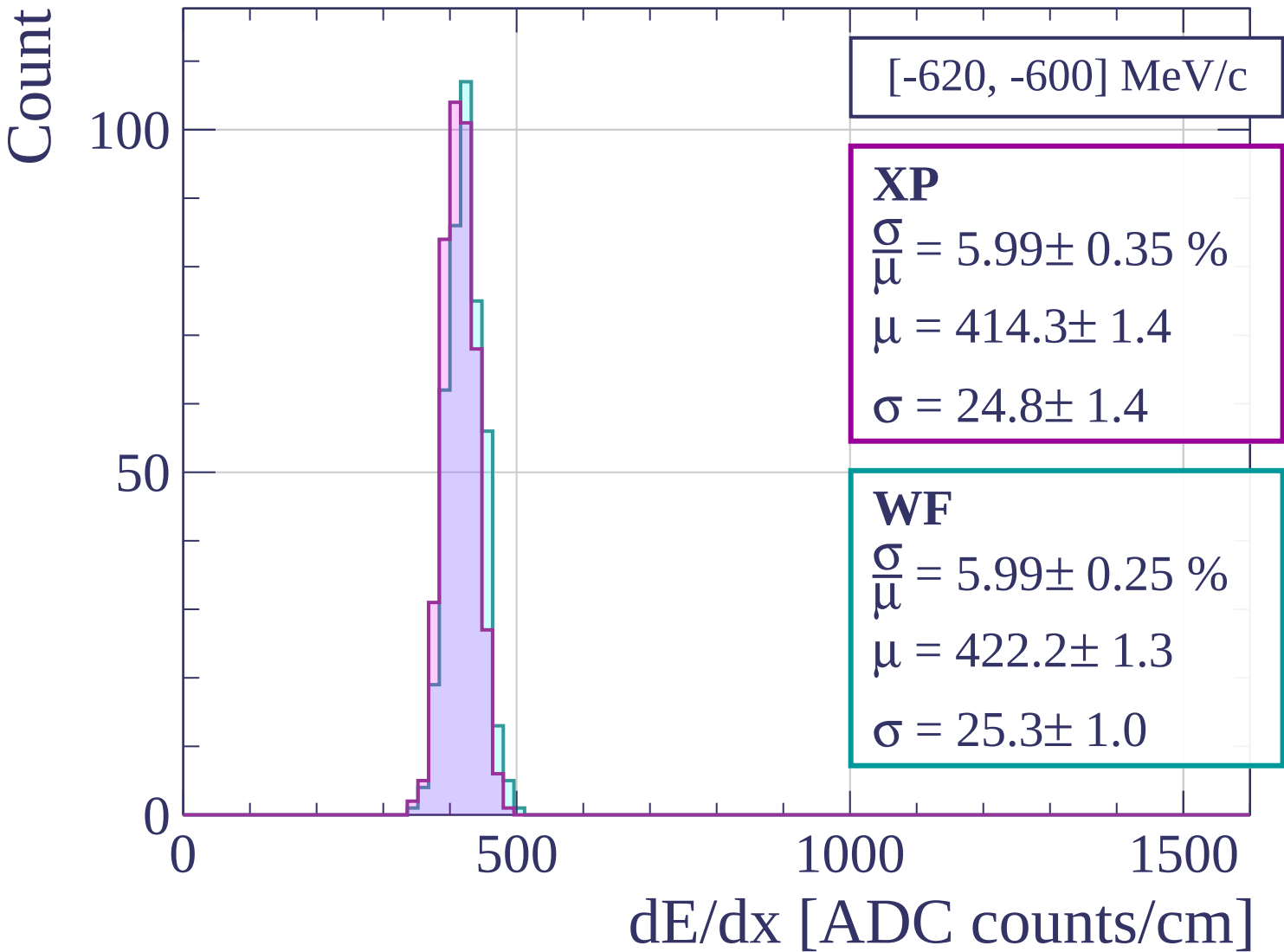
$$\mu = 423.7 \pm 1.6$$

$$\sigma = 27.8 \pm 1.6$$

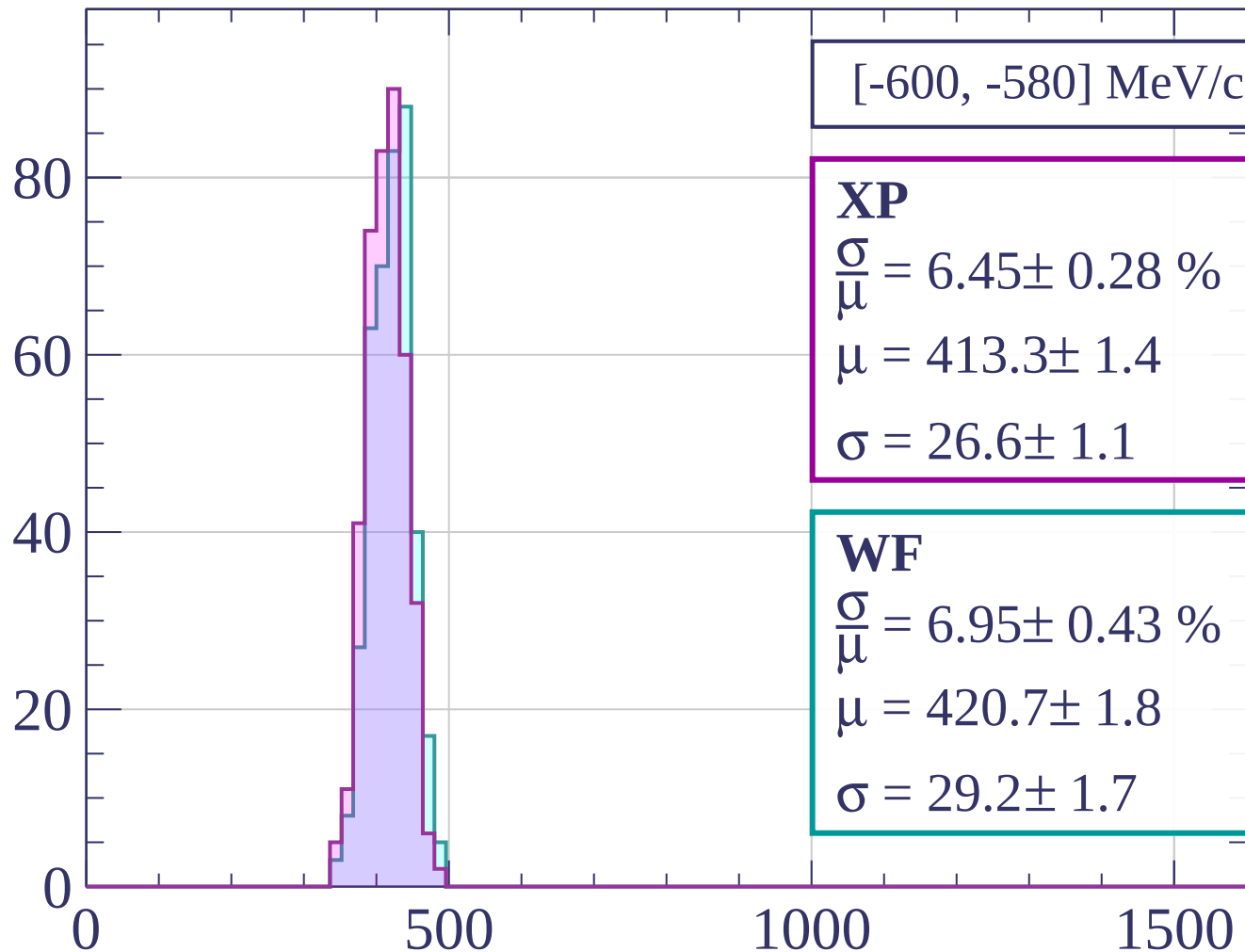
dE/dx [ADC counts/cm]







Count



[-600, -580] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.45 \pm 0.28 \%$$

$$\mu = 413.3 \pm 1.4$$

$$\sigma = 26.6 \pm 1.1$$

WF

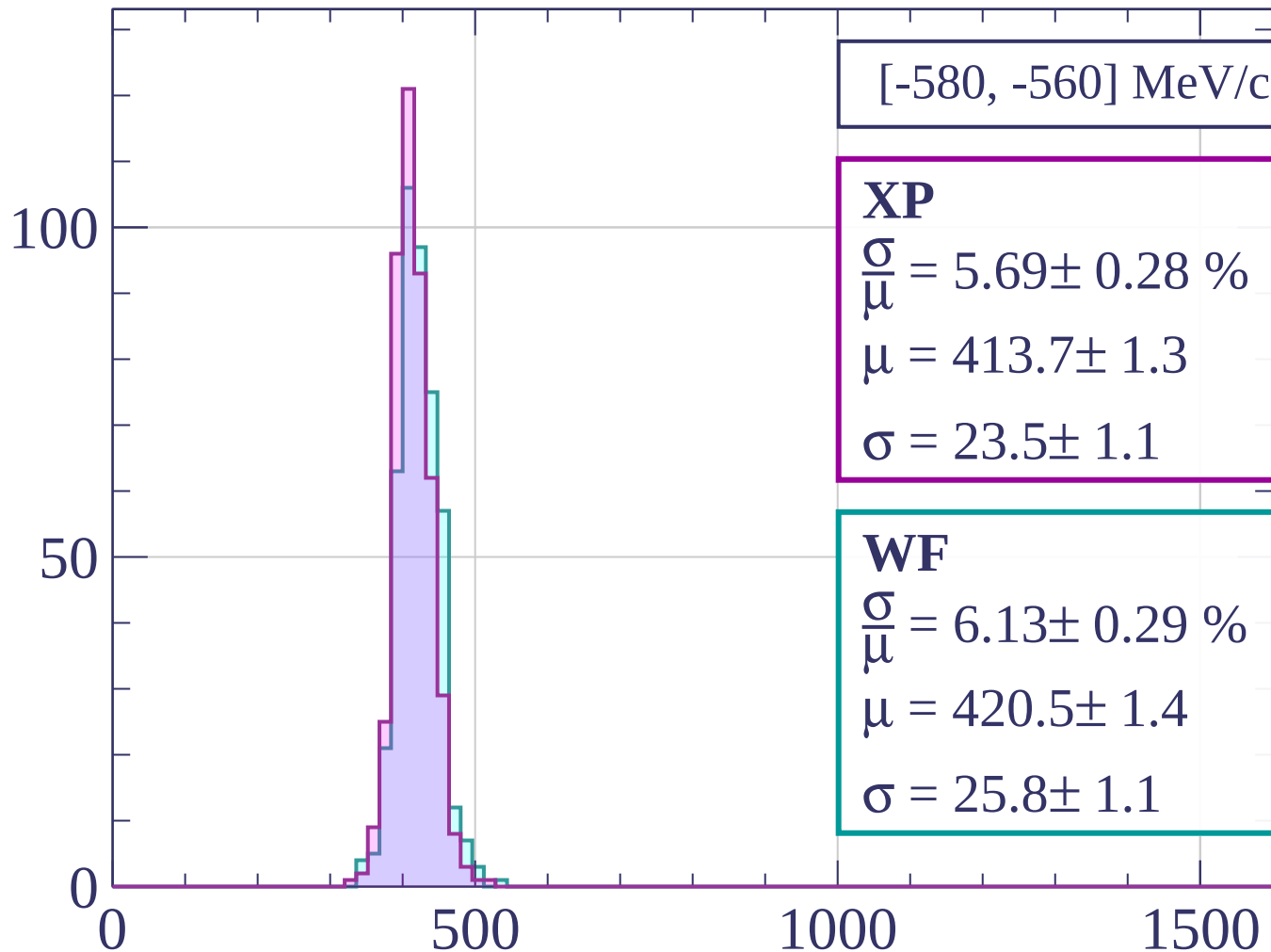
$$\frac{\sigma}{\mu} = 6.95 \pm 0.43 \%$$

$$\mu = 420.7 \pm 1.8$$

$$\sigma = 29.2 \pm 1.7$$

dE/dx [ADC counts/cm]

Count



$[-580, -560] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 5.69 \pm 0.28 \%$$

$$\mu = 413.7 \pm 1.3$$

$$\sigma = 23.5 \pm 1.1$$

WF

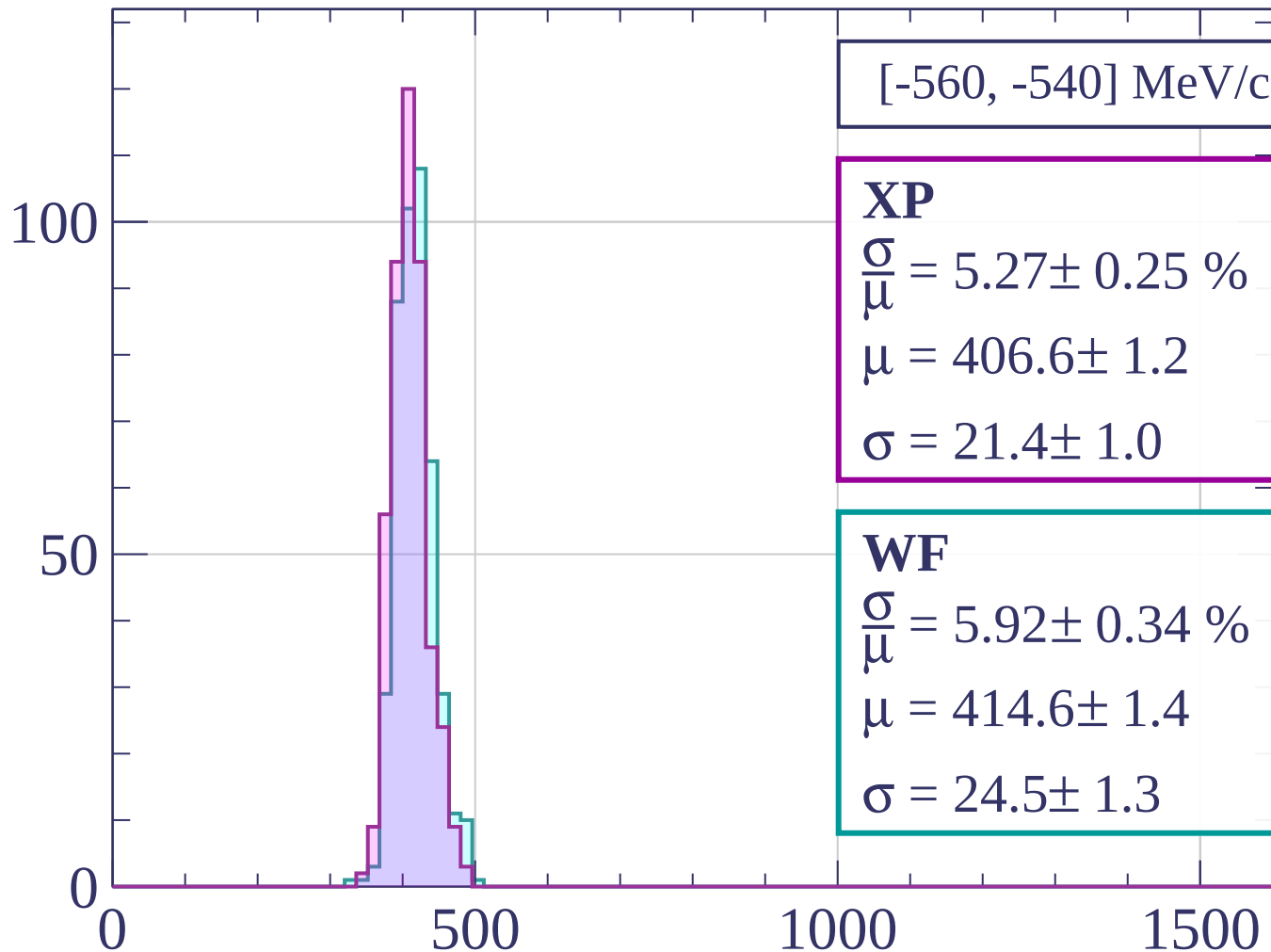
$$\frac{\sigma}{\mu} = 6.13 \pm 0.29 \%$$

$$\mu = 420.5 \pm 1.4$$

$$\sigma = 25.8 \pm 1.1$$

dE/dx [ADC counts/cm]

Count



[-560, -540] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.27 \pm 0.25 \%$$

$$\mu = 406.6 \pm 1.2$$

$$\sigma = 21.4 \pm 1.0$$

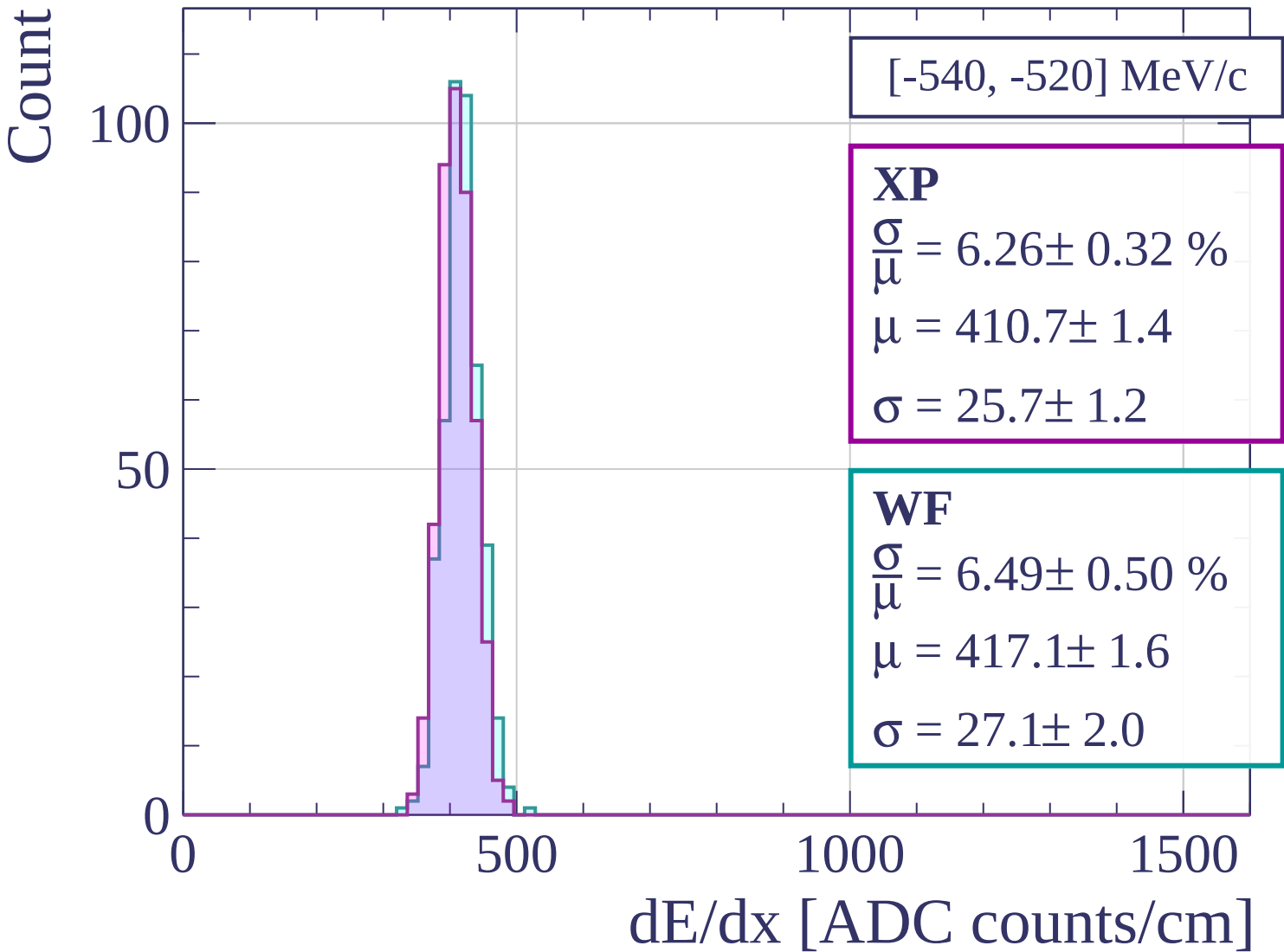
WF

$$\frac{\sigma}{\mu} = 5.92 \pm 0.34 \%$$

$$\mu = 414.6 \pm 1.4$$

$$\sigma = 24.5 \pm 1.3$$

dE/dx [ADC counts/cm]



Count

[-520, -500] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.91 \pm 0.26 \%$$

$$\mu = 411.4 \pm 1.3$$

$$\sigma = 24.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.52 \pm 0.36 \%$$

$$\mu = 419.0 \pm 1.5$$

$$\sigma = 27.3 \pm 1.4$$

100

50

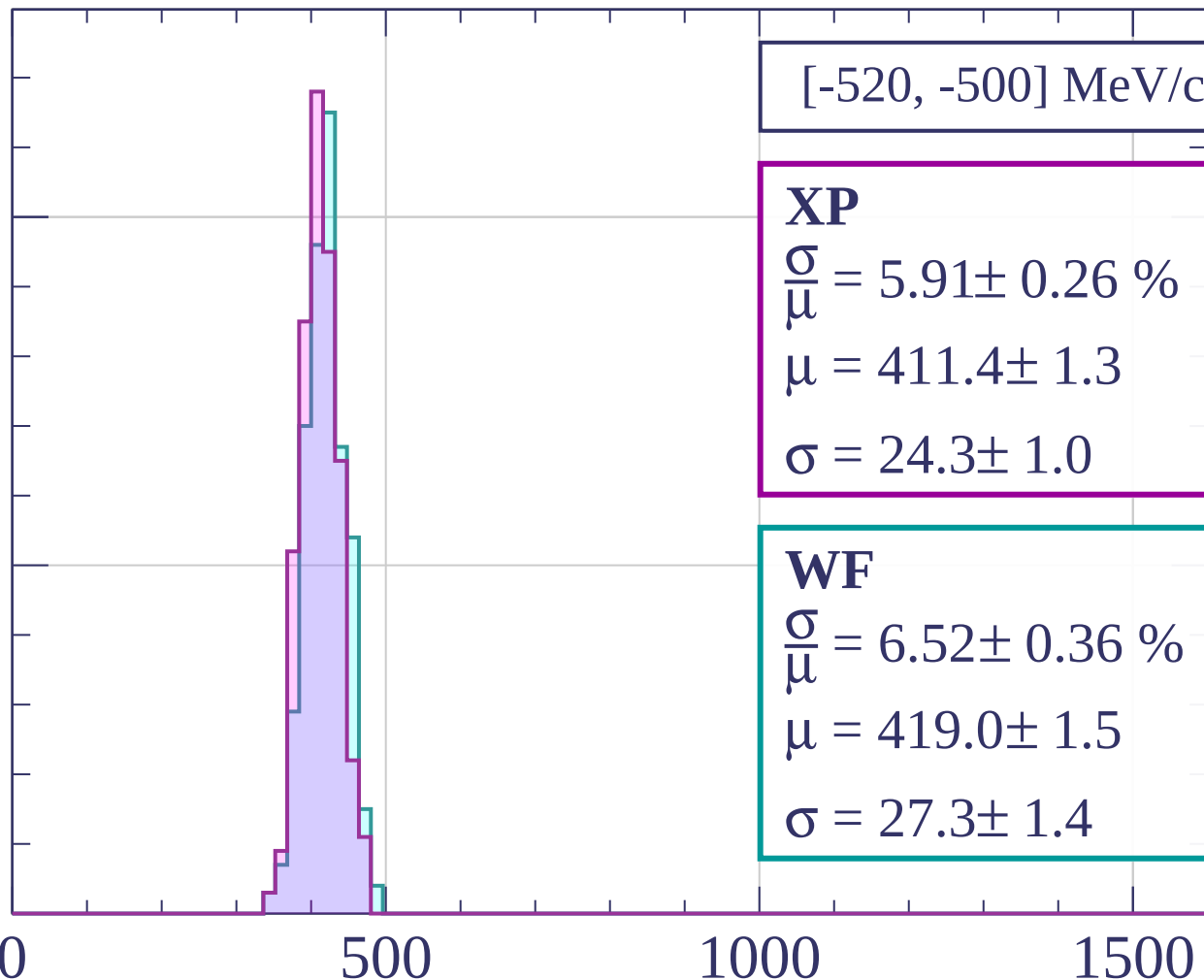
0

500

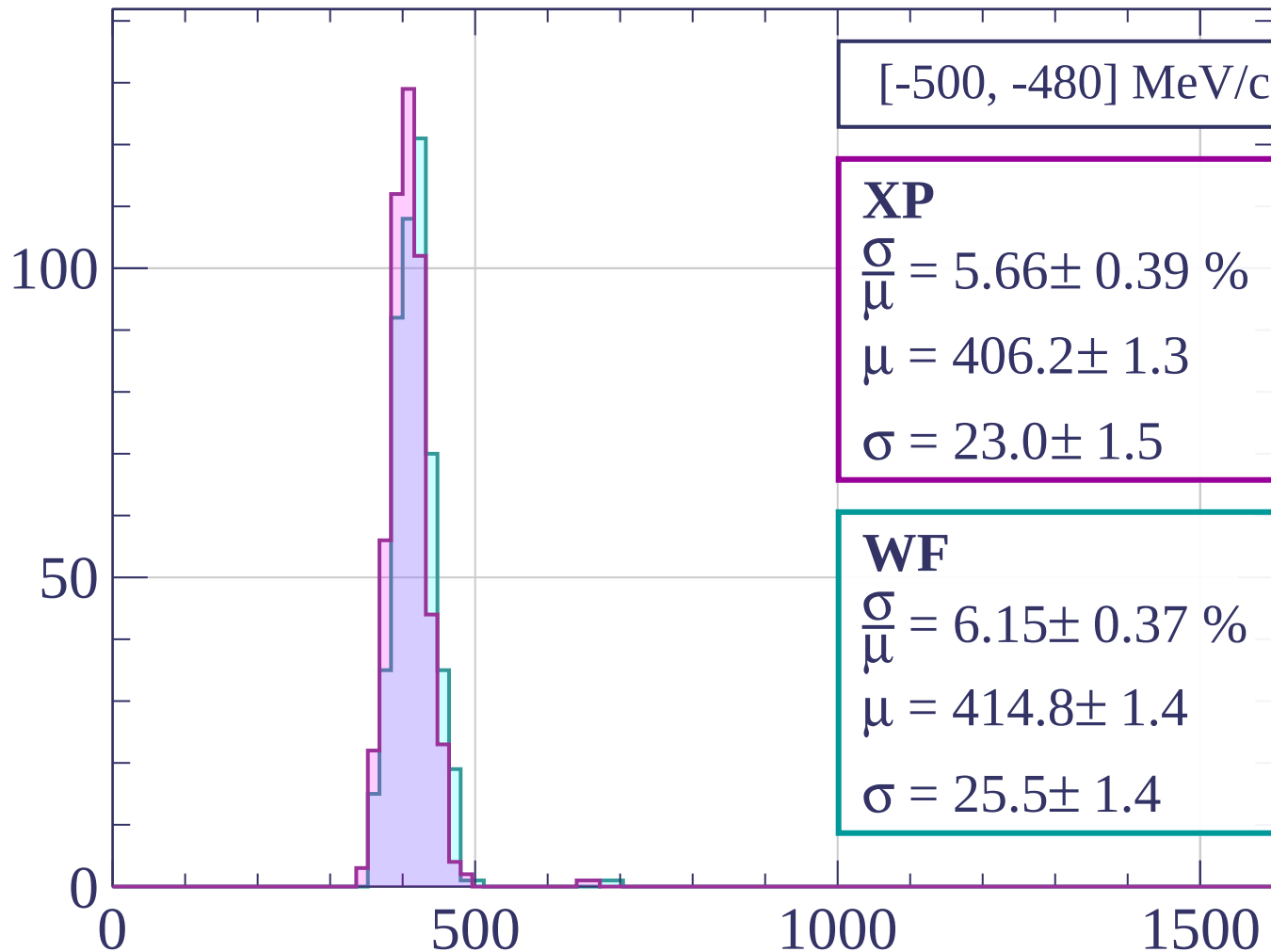
1000

1500

dE/dx [ADC counts/cm]



Count



[-500, -480] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.66 \pm 0.39 \%$$

$$\mu = 406.2 \pm 1.3$$

$$\sigma = 23.0 \pm 1.5$$

WF

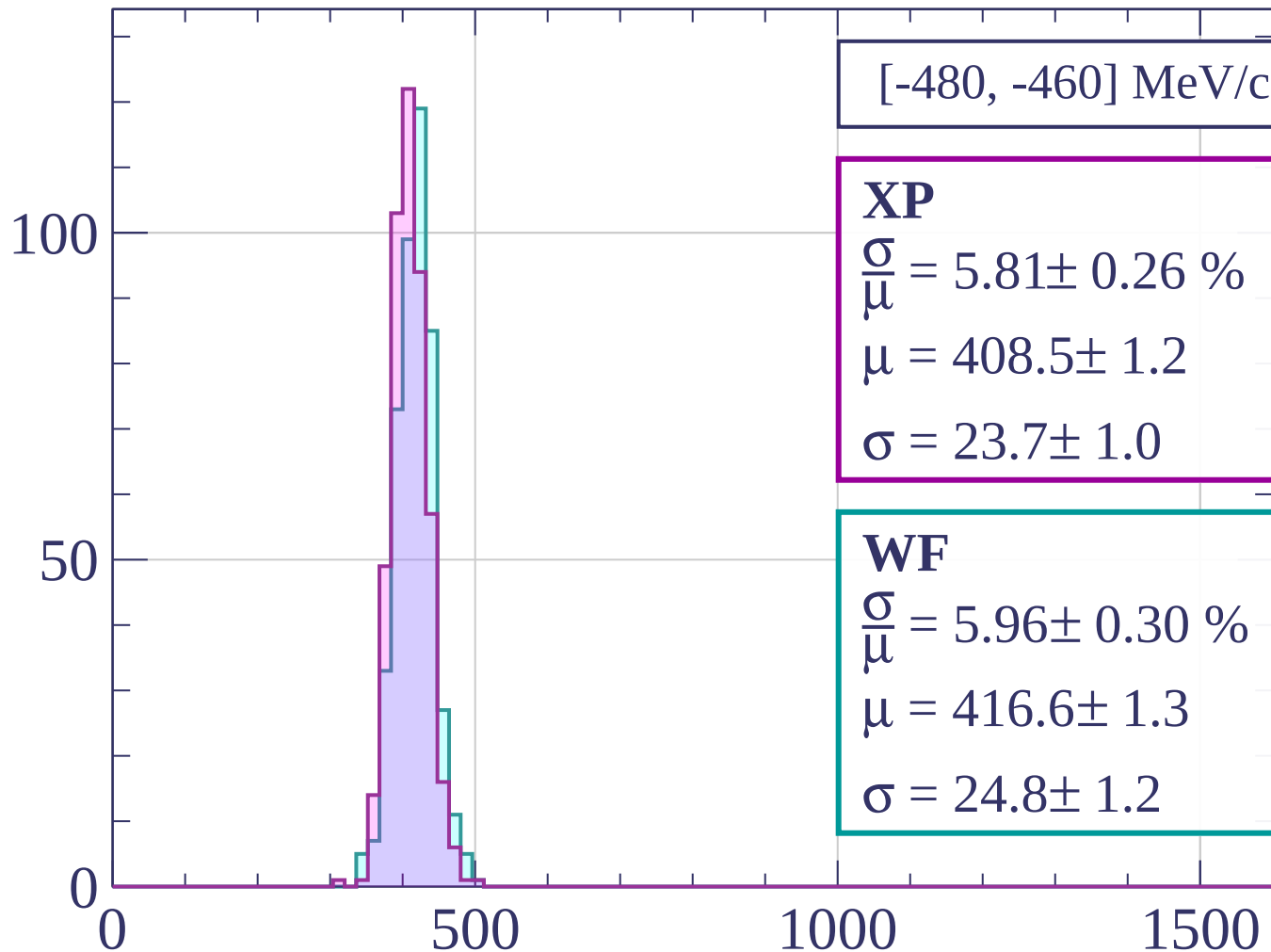
$$\frac{\sigma}{\mu} = 6.15 \pm 0.37 \%$$

$$\mu = 414.8 \pm 1.4$$

$$\sigma = 25.5 \pm 1.4$$

dE/dx [ADC counts/cm]

Count



[-480, -460] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.81 \pm 0.26 \%$$

$$\mu = 408.5 \pm 1.2$$

$$\sigma = 23.7 \pm 1.0$$

WF

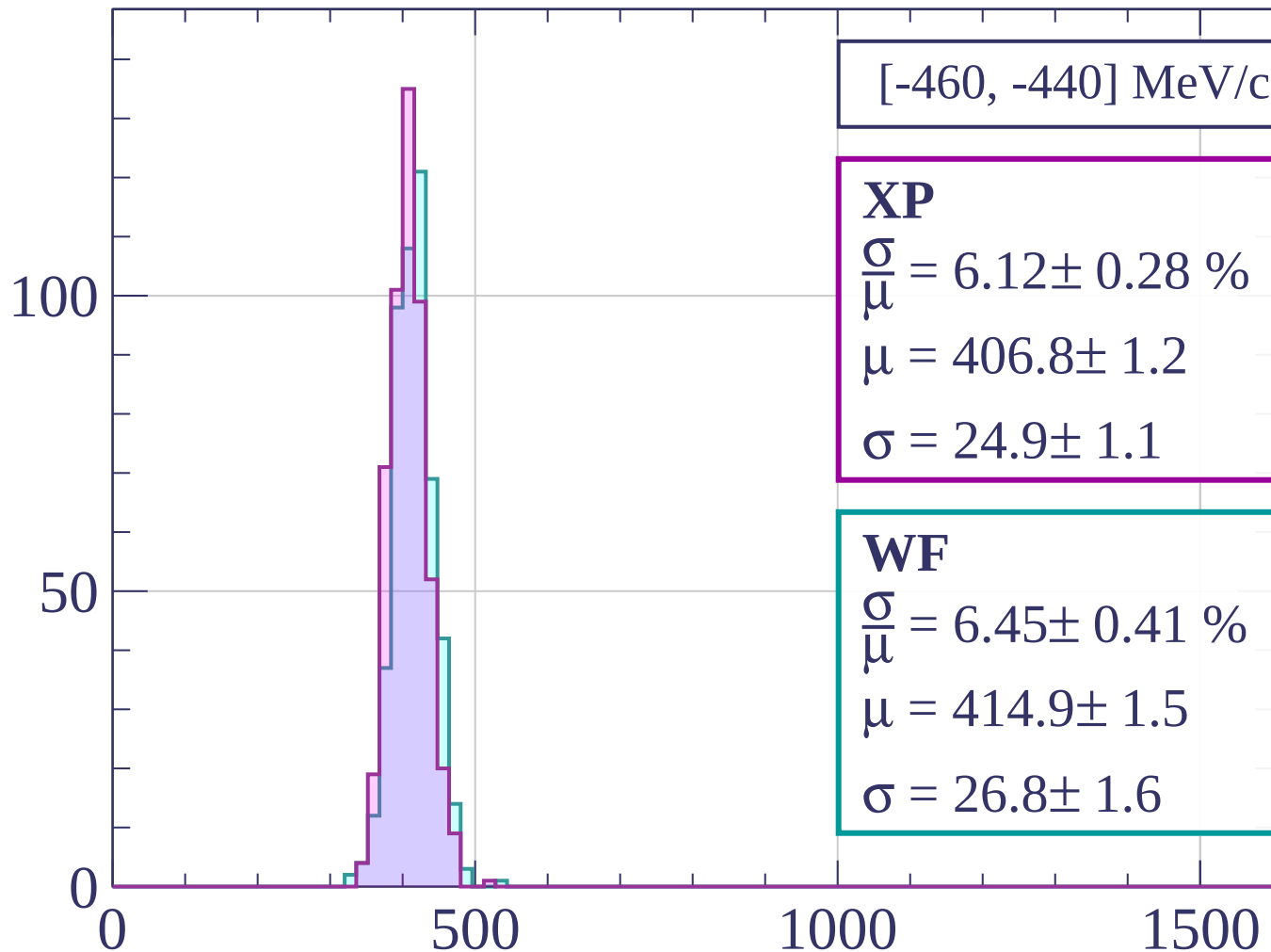
$$\frac{\sigma}{\mu} = 5.96 \pm 0.30 \%$$

$$\mu = 416.6 \pm 1.3$$

$$\sigma = 24.8 \pm 1.2$$

dE/dx [ADC counts/cm]

Count



[-460, -440] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.12 \pm 0.28 \%$$

$$\mu = 406.8 \pm 1.2$$

$$\sigma = 24.9 \pm 1.1$$

WF

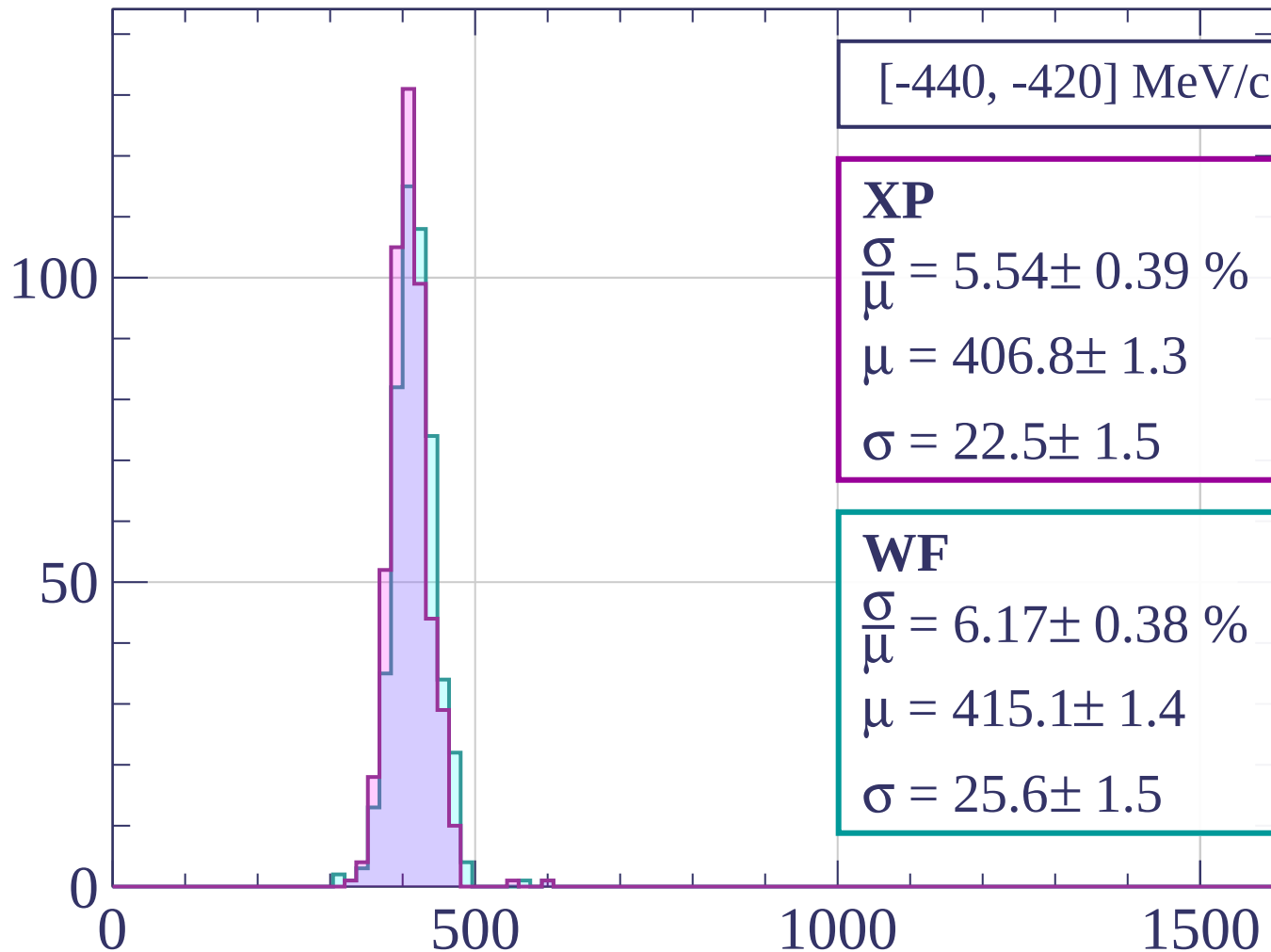
$$\frac{\sigma}{\mu} = 6.45 \pm 0.41 \%$$

$$\mu = 414.9 \pm 1.5$$

$$\sigma = 26.8 \pm 1.6$$

dE/dx [ADC counts/cm]

Count



[-440, -420] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.54 \pm 0.39 \%$$

$$\mu = 406.8 \pm 1.3$$

$$\sigma = 22.5 \pm 1.5$$

WF

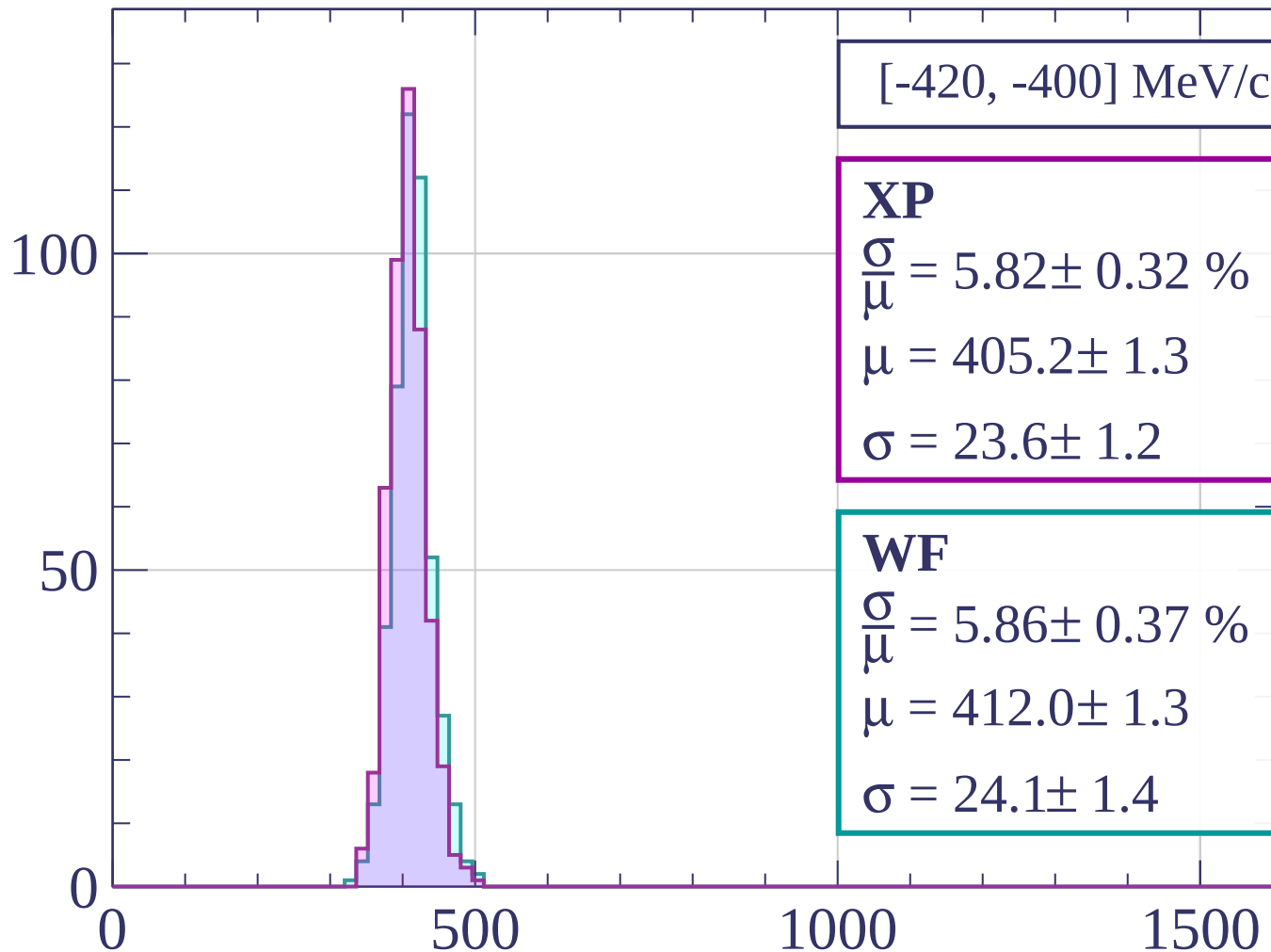
$$\frac{\sigma}{\mu} = 6.17 \pm 0.38 \%$$

$$\mu = 415.1 \pm 1.4$$

$$\sigma = 25.6 \pm 1.5$$

dE/dx [ADC counts/cm]

Count



[-420, -400] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.82 \pm 0.32 \%$$

$$\mu = 405.2 \pm 1.3$$

$$\sigma = 23.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.86 \pm 0.37 \%$$

$$\mu = 412.0 \pm 1.3$$

$$\sigma = 24.1 \pm 1.4$$

dE/dx [ADC counts/cm]

Count

[-400, -380] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.77 \pm 0.31 \%$$

$$\mu = 403.5 \pm 1.3$$

$$\sigma = 23.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.09 \pm 0.36 \%$$

$$\mu = 411.6 \pm 1.4$$

$$\sigma = 25.1 \pm 1.4$$

100

50

0

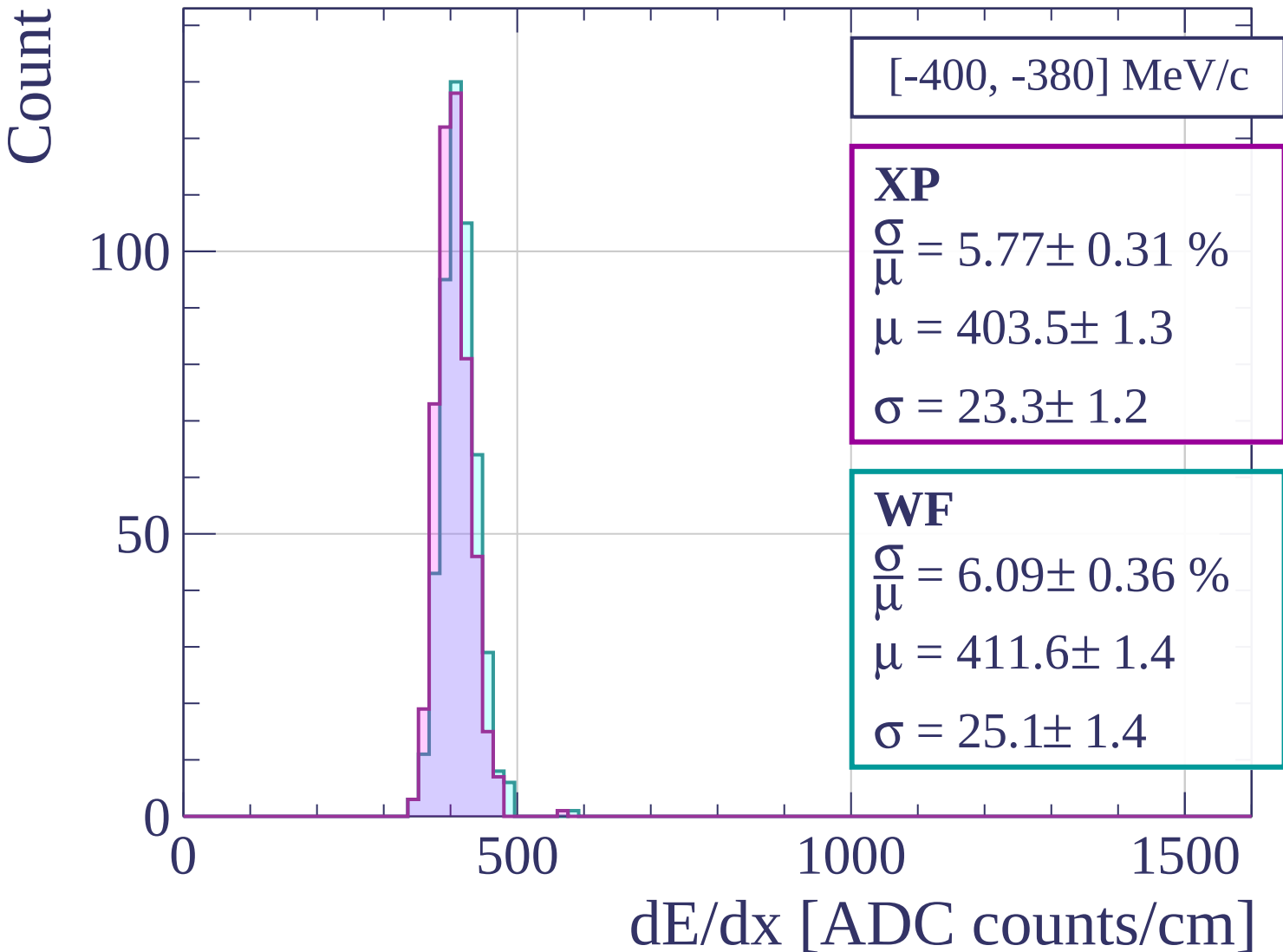
0

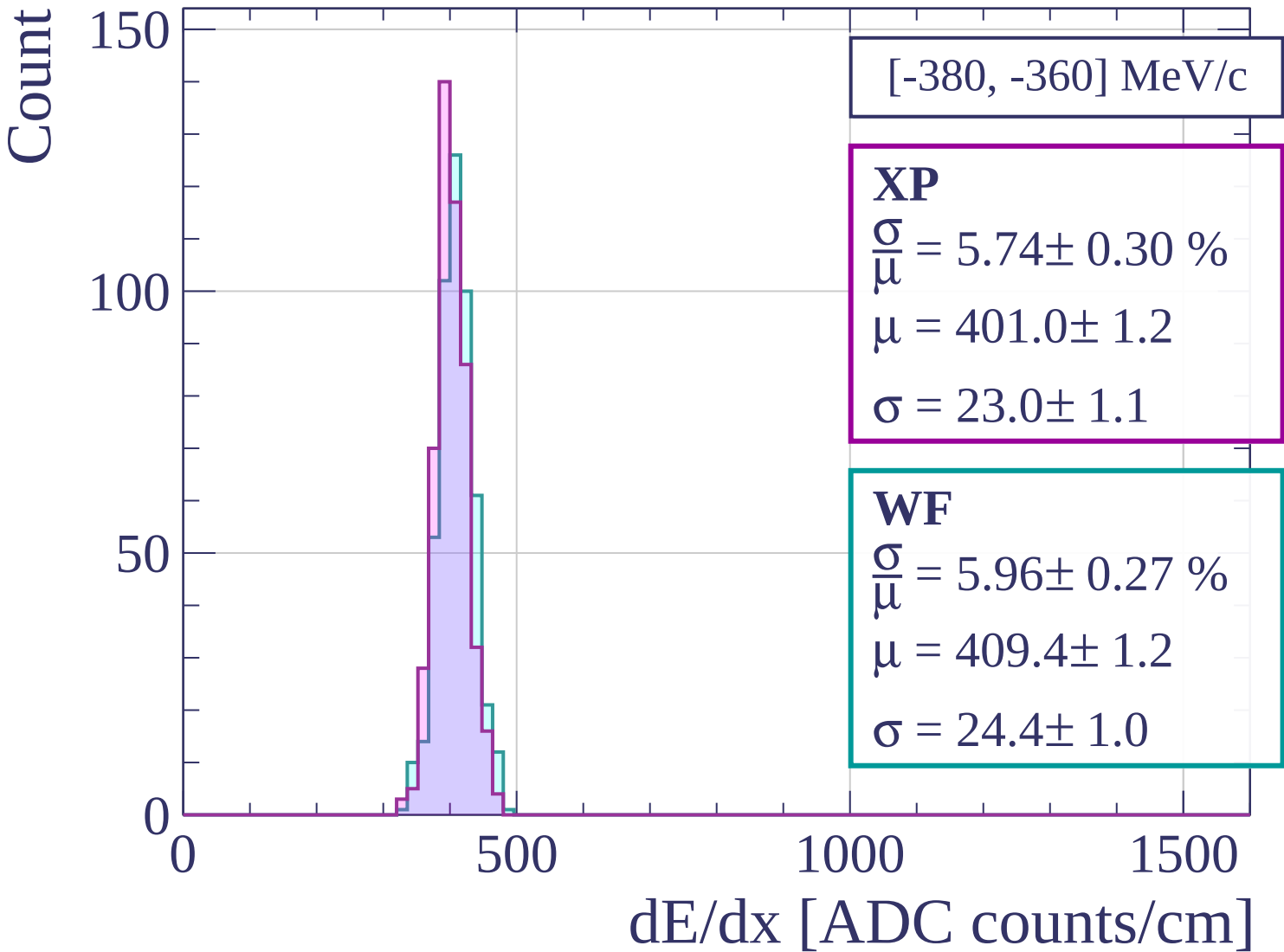
500

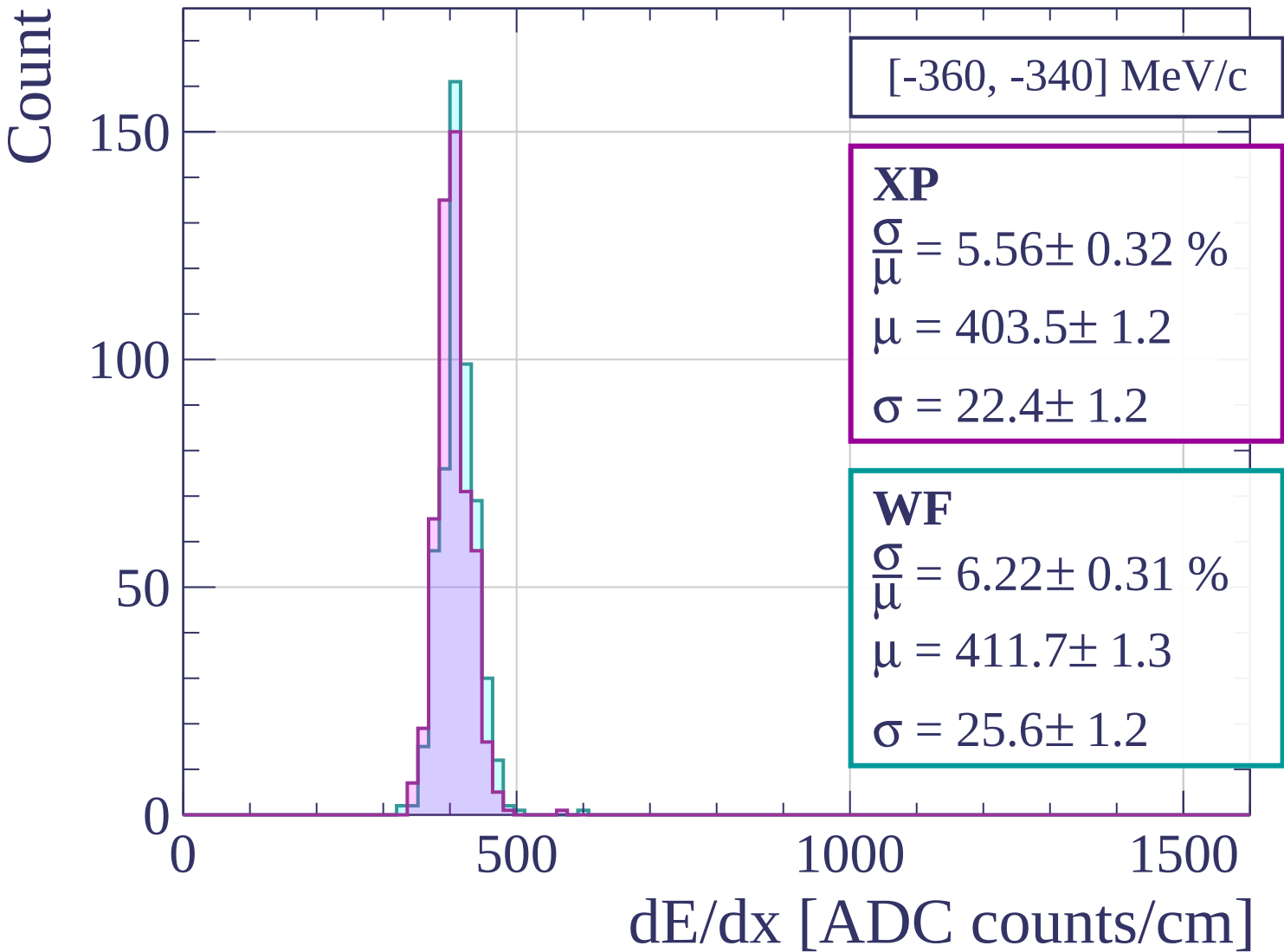
1000

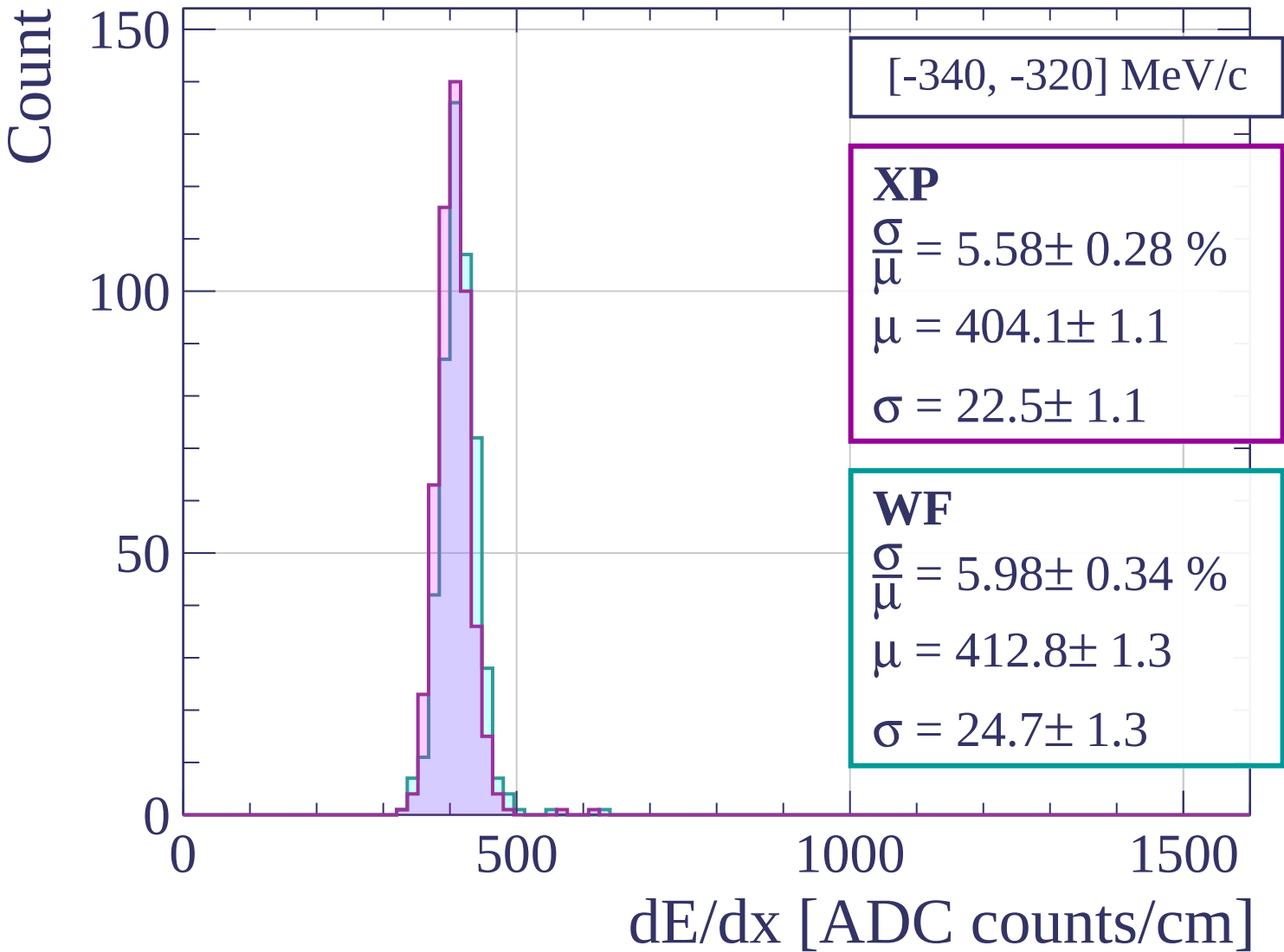
1500

dE/dx [ADC counts/cm]

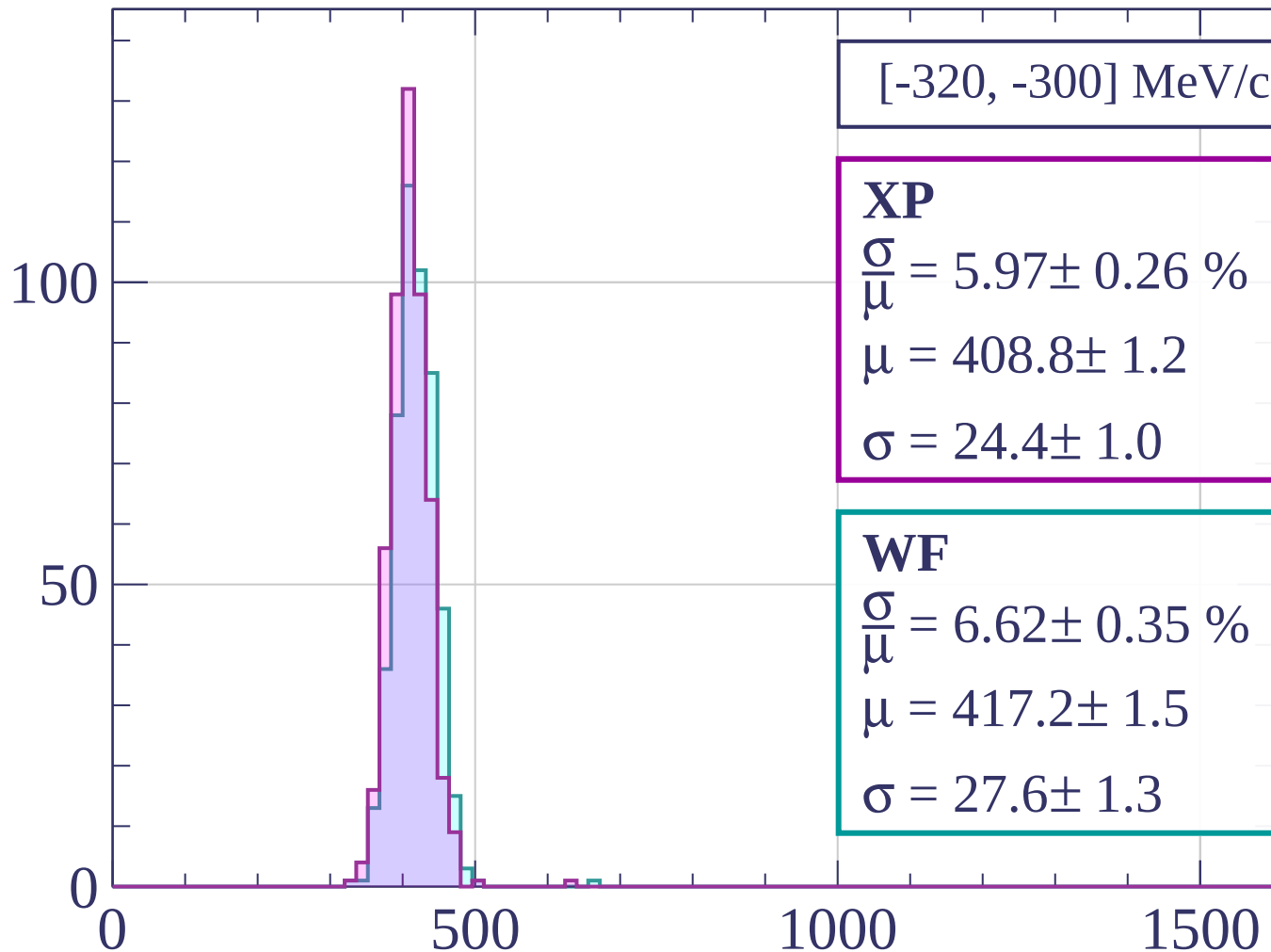






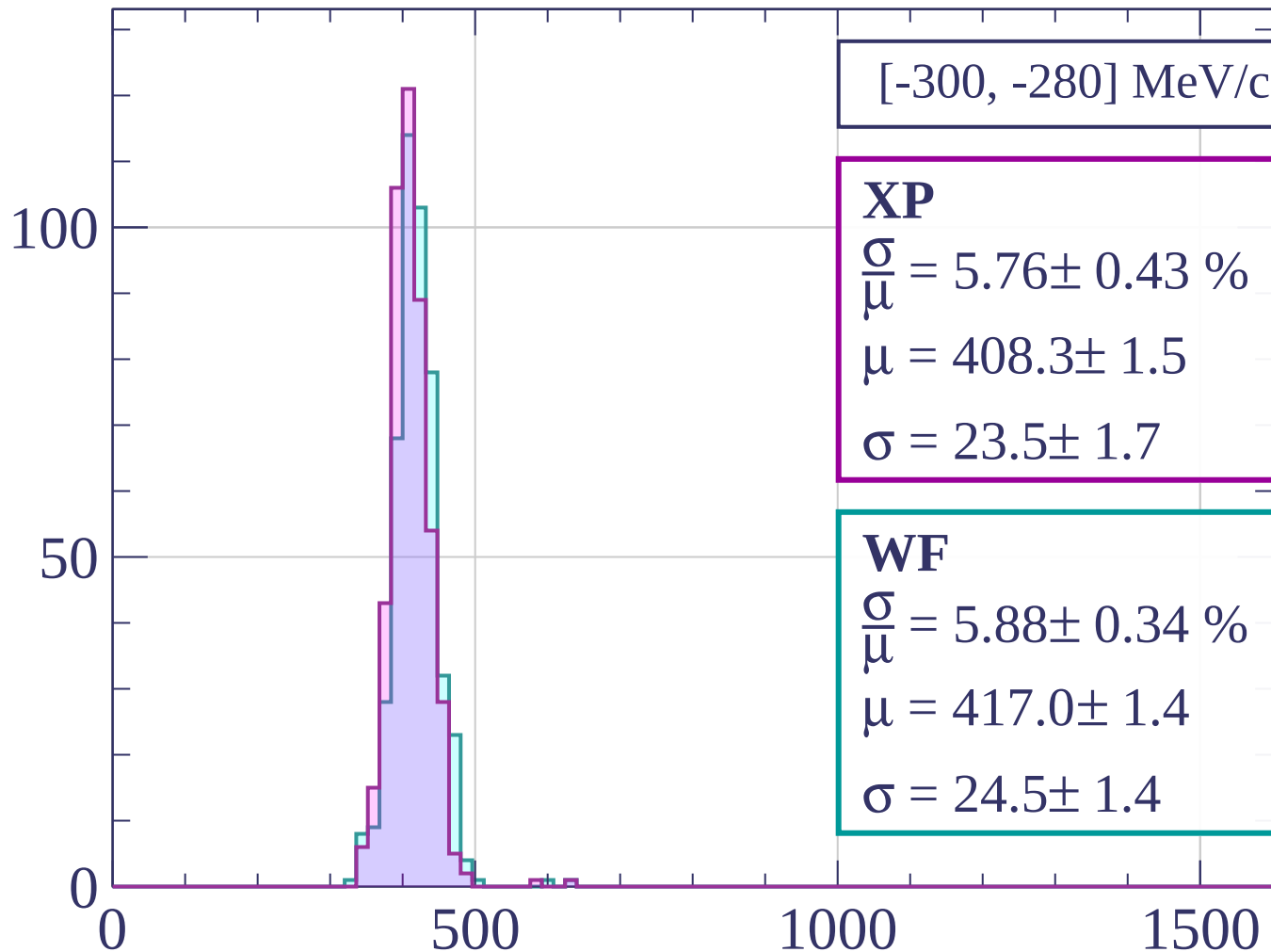


Count



dE/dx [ADC counts/cm]

Count



[-300, -280] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.76 \pm 0.43 \%$$

$$\mu = 408.3 \pm 1.5$$

$$\sigma = 23.5 \pm 1.7$$

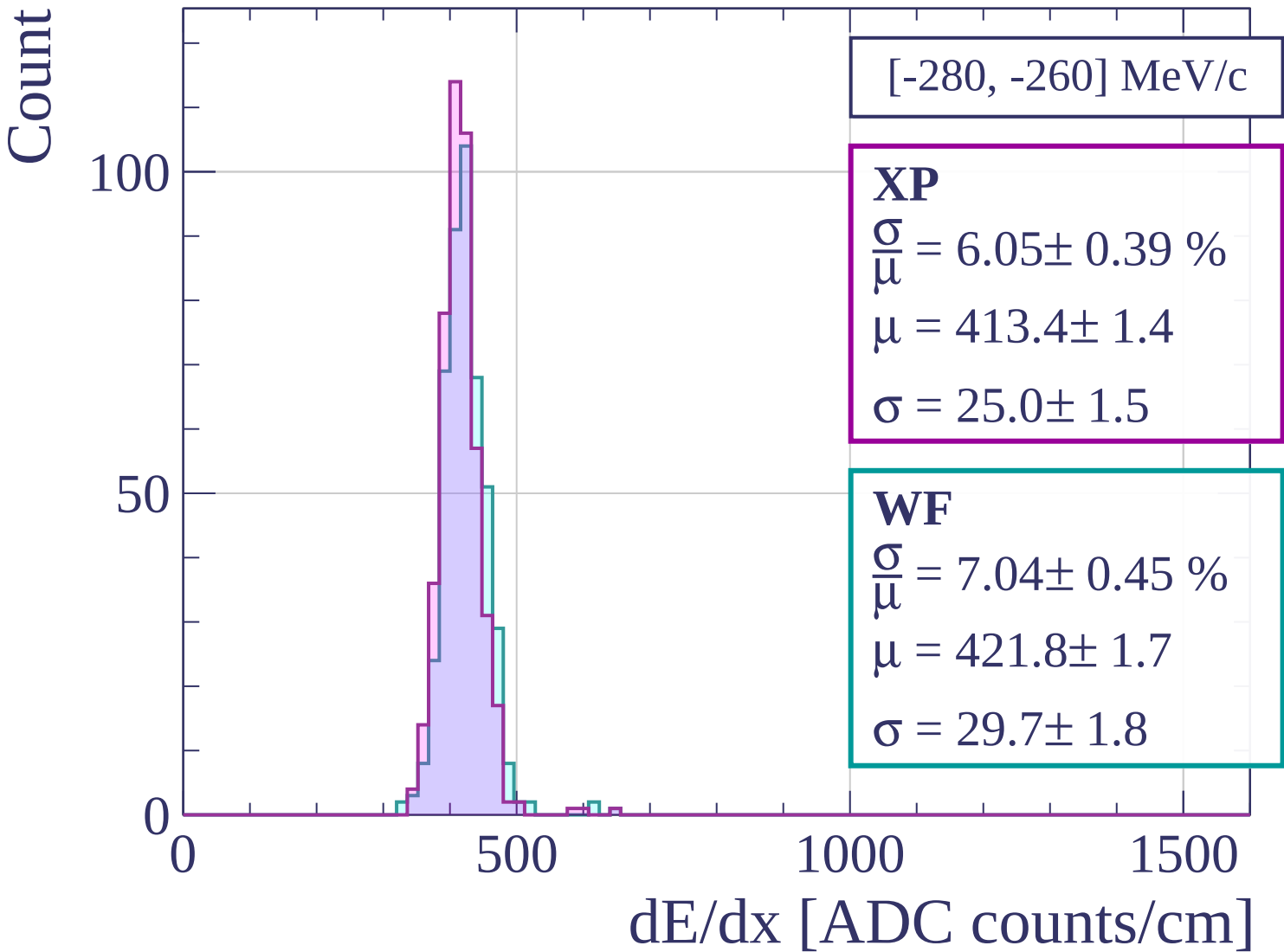
WF

$$\frac{\sigma}{\mu} = 5.88 \pm 0.34 \%$$

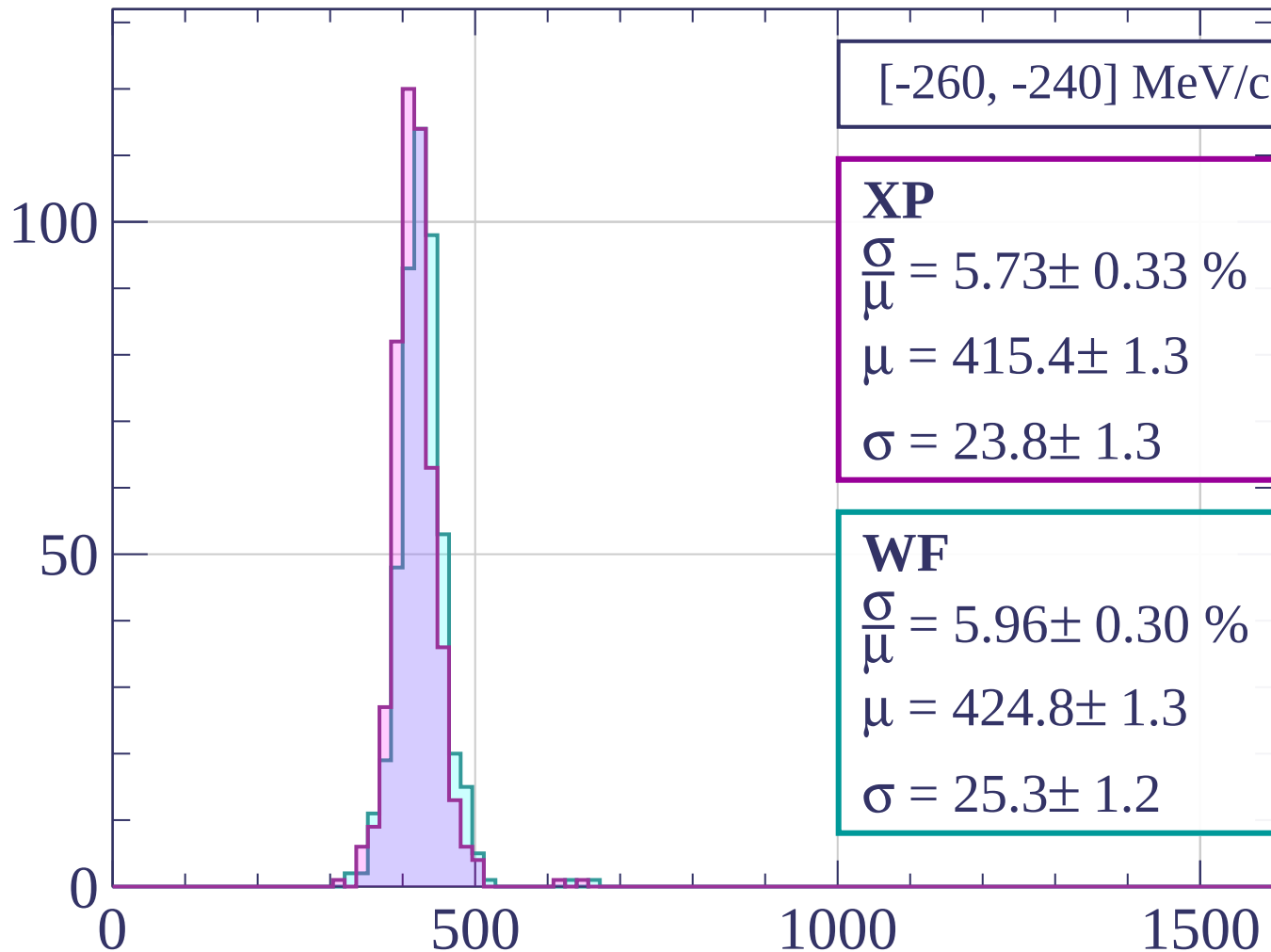
$$\mu = 417.0 \pm 1.4$$

$$\sigma = 24.5 \pm 1.4$$

dE/dx [ADC counts/cm]



Count



[-260, -240] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.73 \pm 0.33 \%$$

$$\mu = 415.4 \pm 1.3$$

$$\sigma = 23.8 \pm 1.3$$

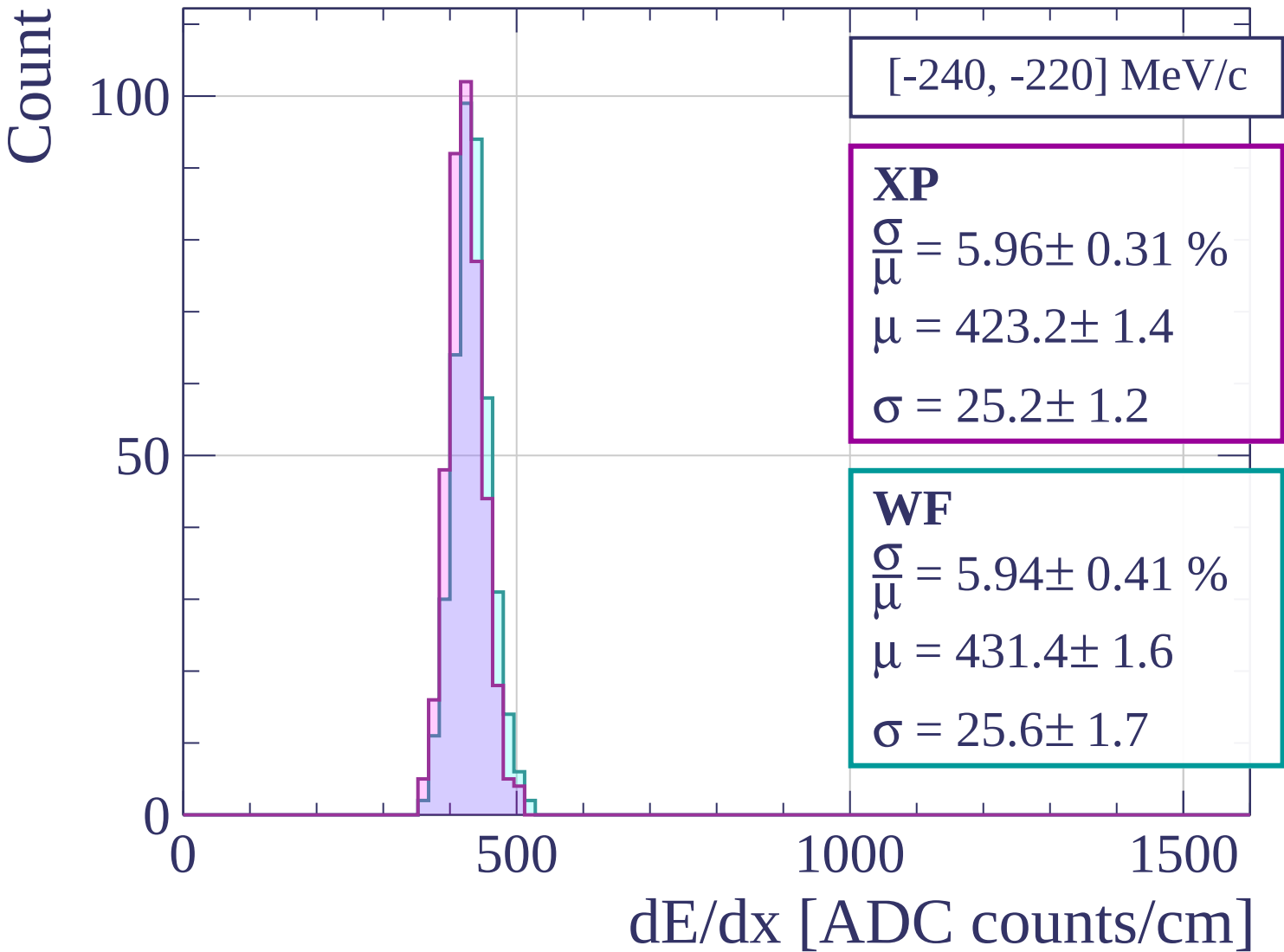
WF

$$\frac{\sigma}{\mu} = 5.96 \pm 0.30 \%$$

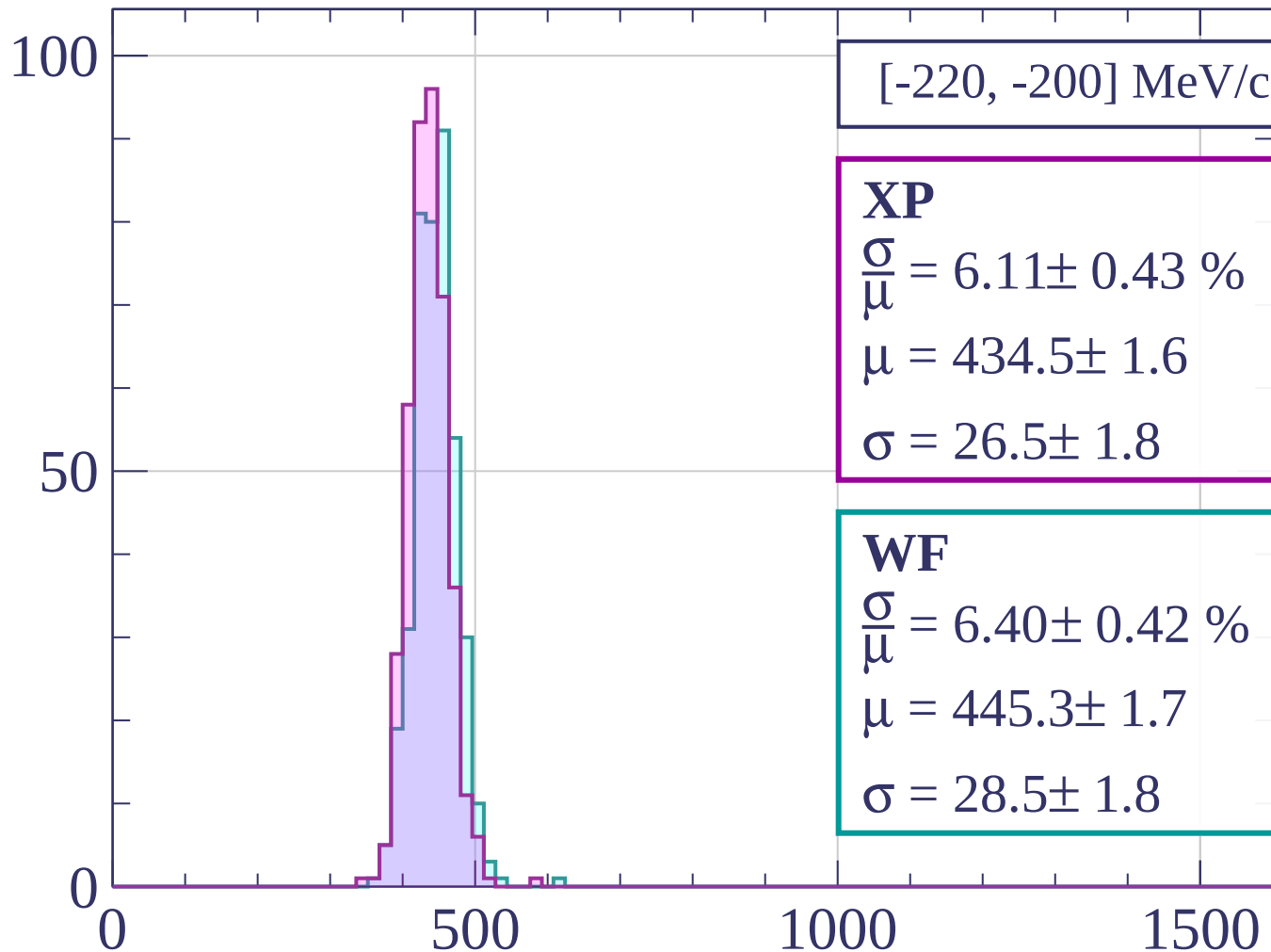
$$\mu = 424.8 \pm 1.3$$

$$\sigma = 25.3 \pm 1.2$$

dE/dx [ADC counts/cm]



Count



[-220, -200] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.11 \pm 0.43 \%$$

$$\mu = 434.5 \pm 1.6$$

$$\sigma = 26.5 \pm 1.8$$

WF

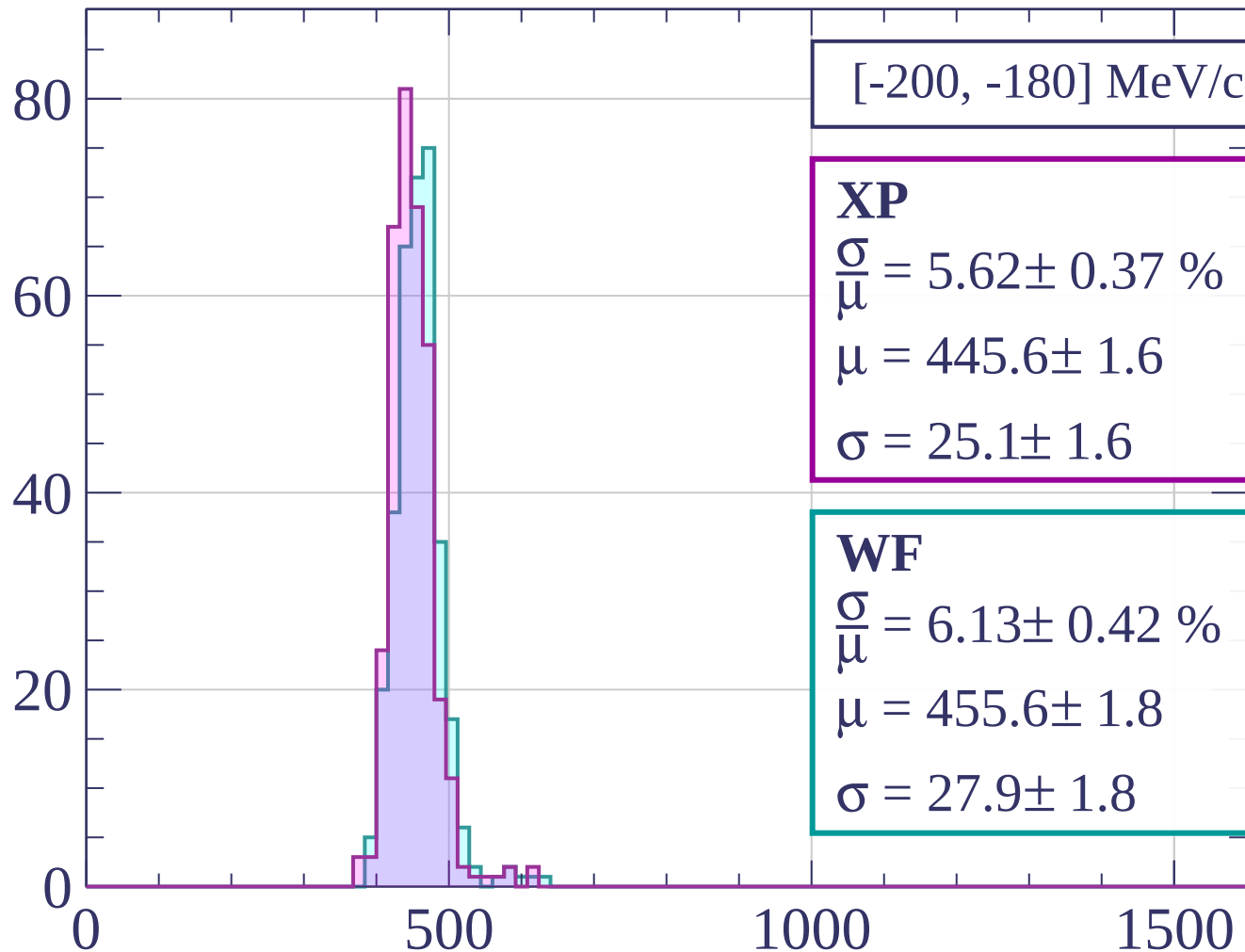
$$\frac{\sigma}{\mu} = 6.40 \pm 0.42 \%$$

$$\mu = 445.3 \pm 1.7$$

$$\sigma = 28.5 \pm 1.8$$

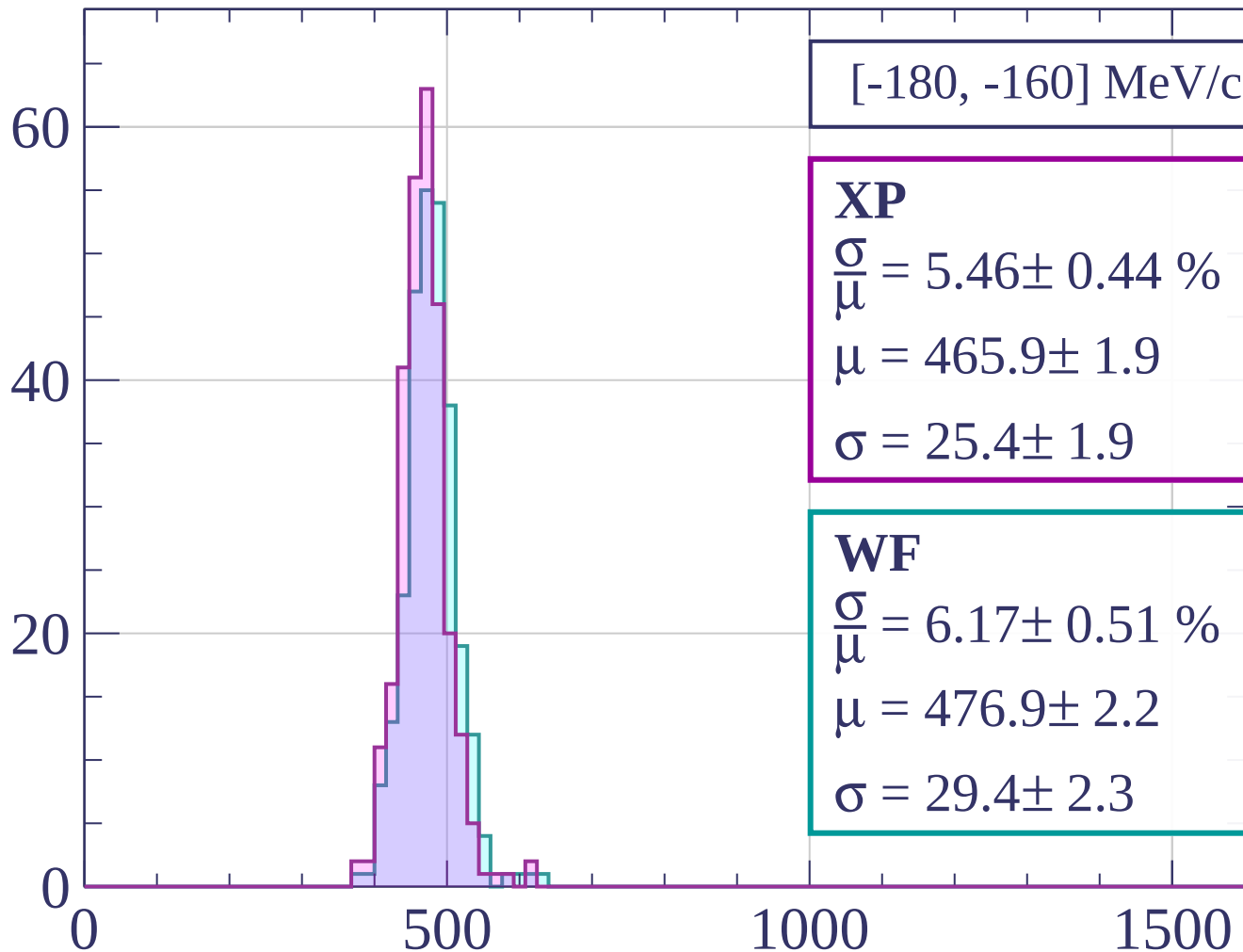
dE/dx [ADC counts/cm]

Count



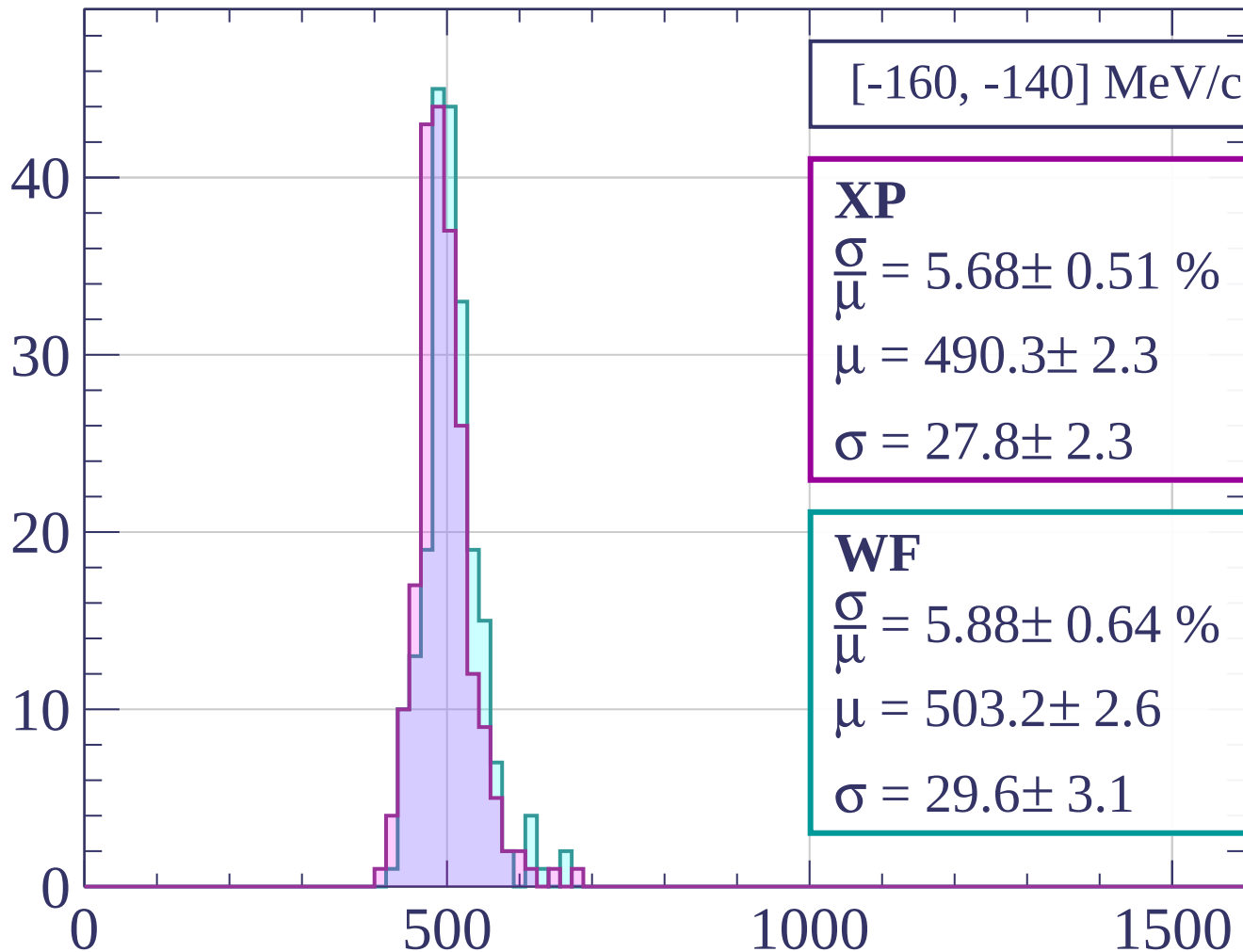
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.68 \pm 0.51 \%$$

$$\mu = 490.3 \pm 2.3$$

$$\sigma = 27.8 \pm 2.3$$

WF

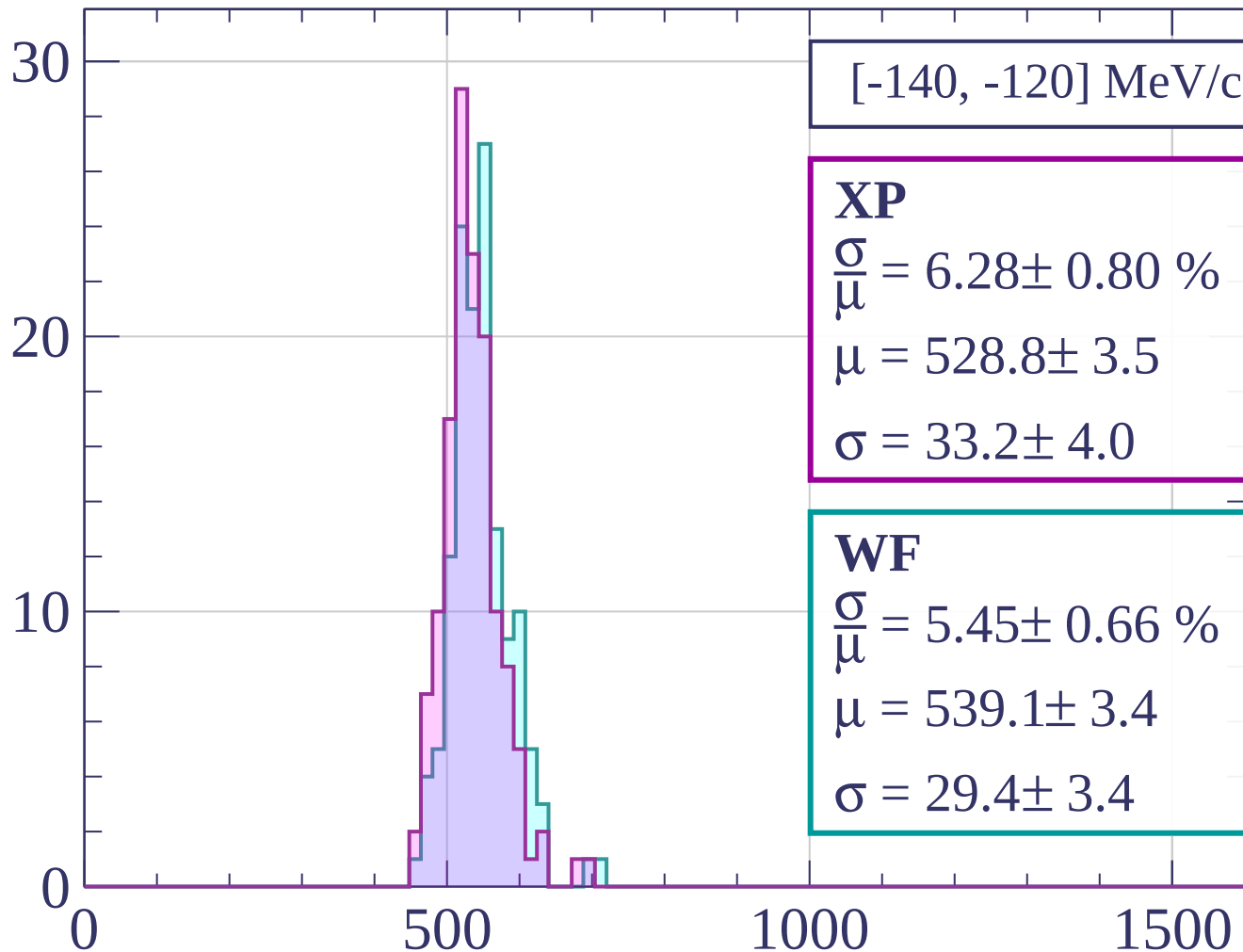
$$\frac{\sigma}{\mu} = 5.88 \pm 0.64 \%$$

$$\mu = 503.2 \pm 2.6$$

$$\sigma = 29.6 \pm 3.1$$

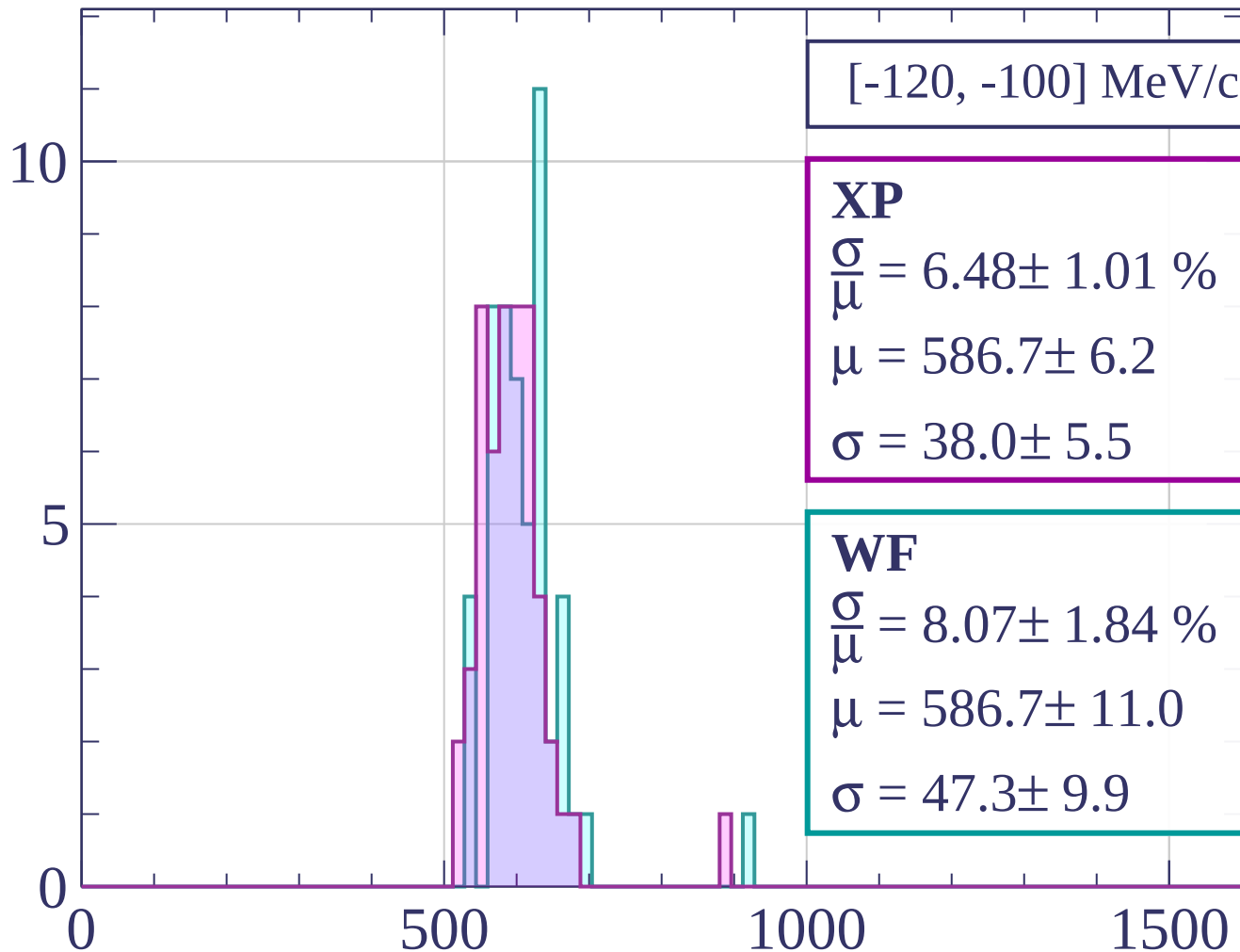
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



$[-120, -100]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 6.48 \pm 1.01 \%$$

$$\mu = 586.7 \pm 6.2$$

$$\sigma = 38.0 \pm 5.5$$

WF

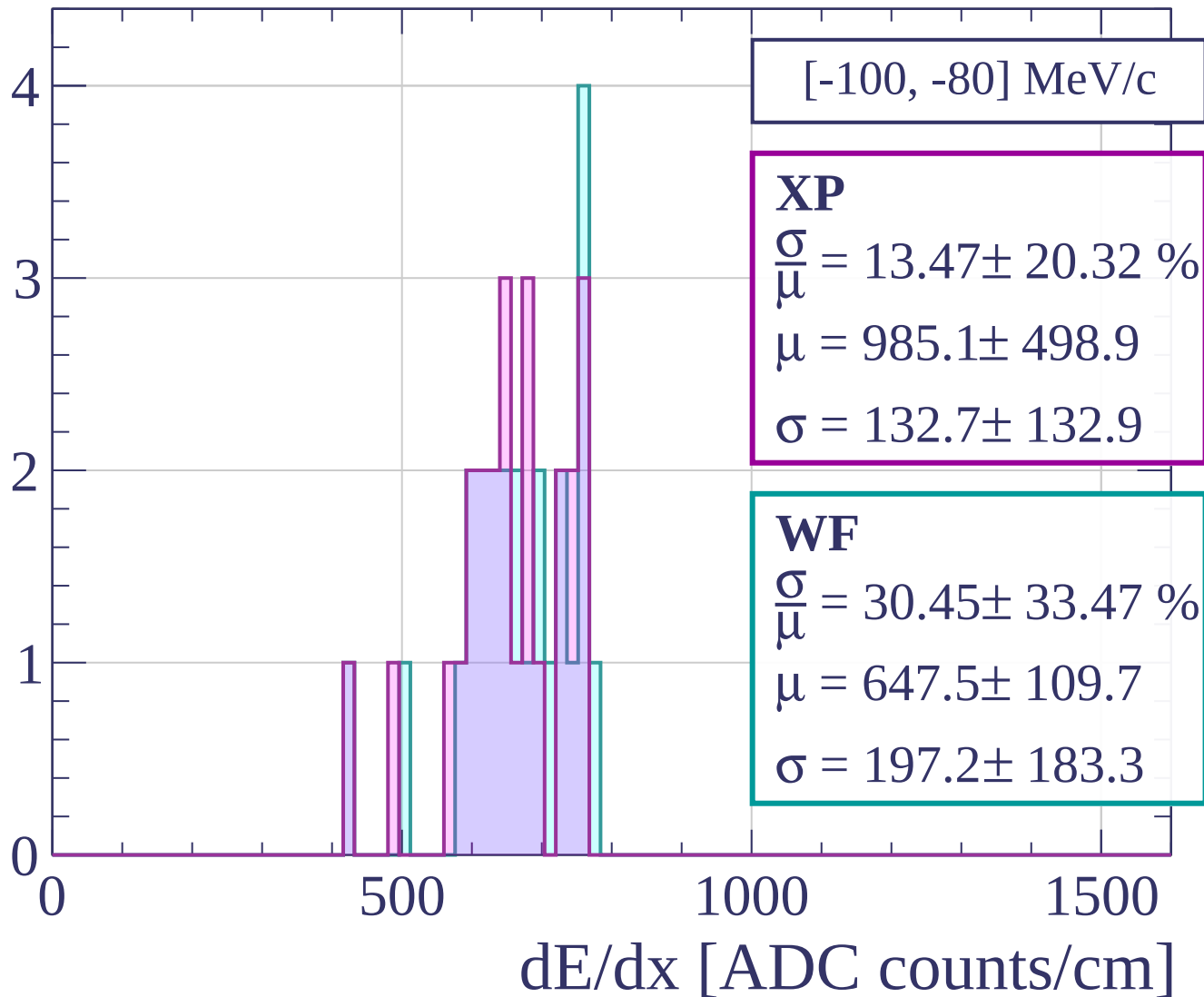
$$\frac{\sigma}{\mu} = 8.07 \pm 1.84 \%$$

$$\mu = 586.7 \pm 11.0$$

$$\sigma = 47.3 \pm 9.9$$

dE/dx [ADC counts/cm]

Count



Count

[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 104.34 \pm 485.54 \%$$

$$\mu = 482.8 \pm 1359.1$$

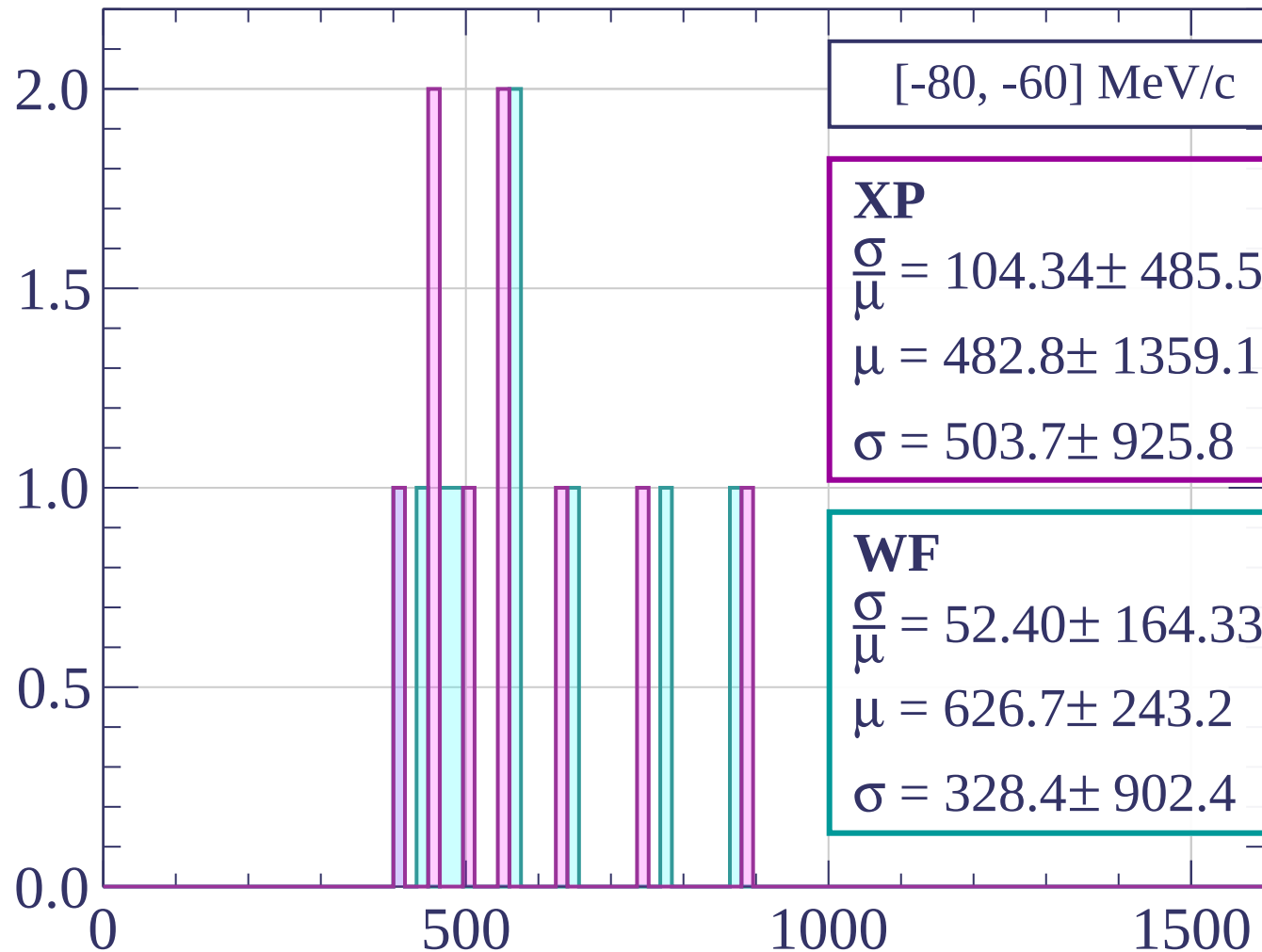
$$\sigma = 503.7 \pm 925.8$$

WF

$$\frac{\sigma}{\mu} = 52.40 \pm 164.33 \%$$

$$\mu = 626.7 \pm 243.2$$

$$\sigma = 328.4 \pm 902.4$$



dE/dx [ADC counts/cm]

Count

[-60, -40] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.84 \pm 45.54 \%$$

$$\mu = 496.0 \pm 76.4$$

$$\sigma = 24.0 \pm 222.2$$

WF

$$\frac{\sigma}{\mu} = 4.69 \pm 44.13 \%$$

$$\mu = 512.0 \pm 76.4$$

$$\sigma = 24.0 \pm 222.3$$

1.0

0.5

0.0

0

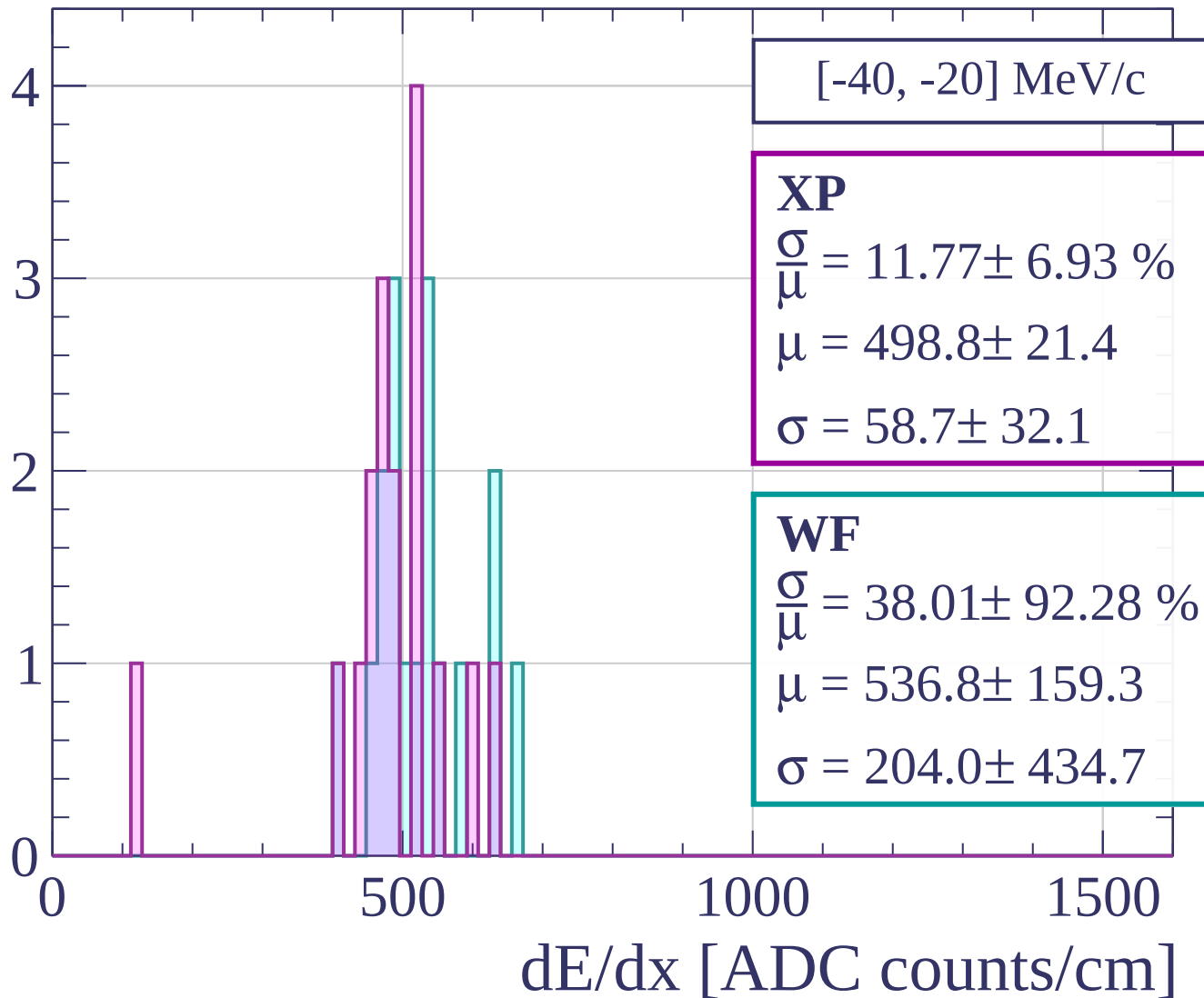
500

1000

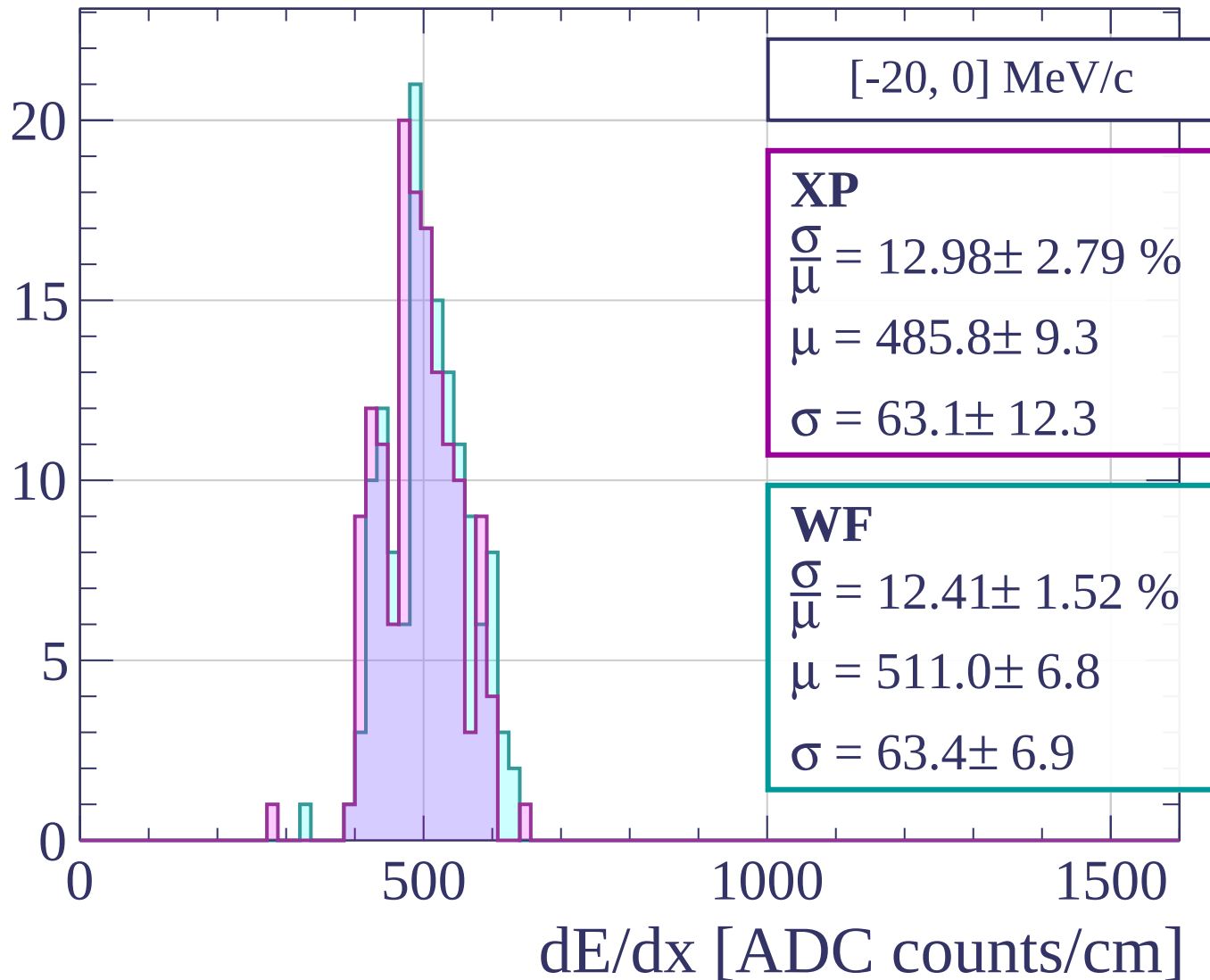
1500

dE/dx [ADC counts/cm]

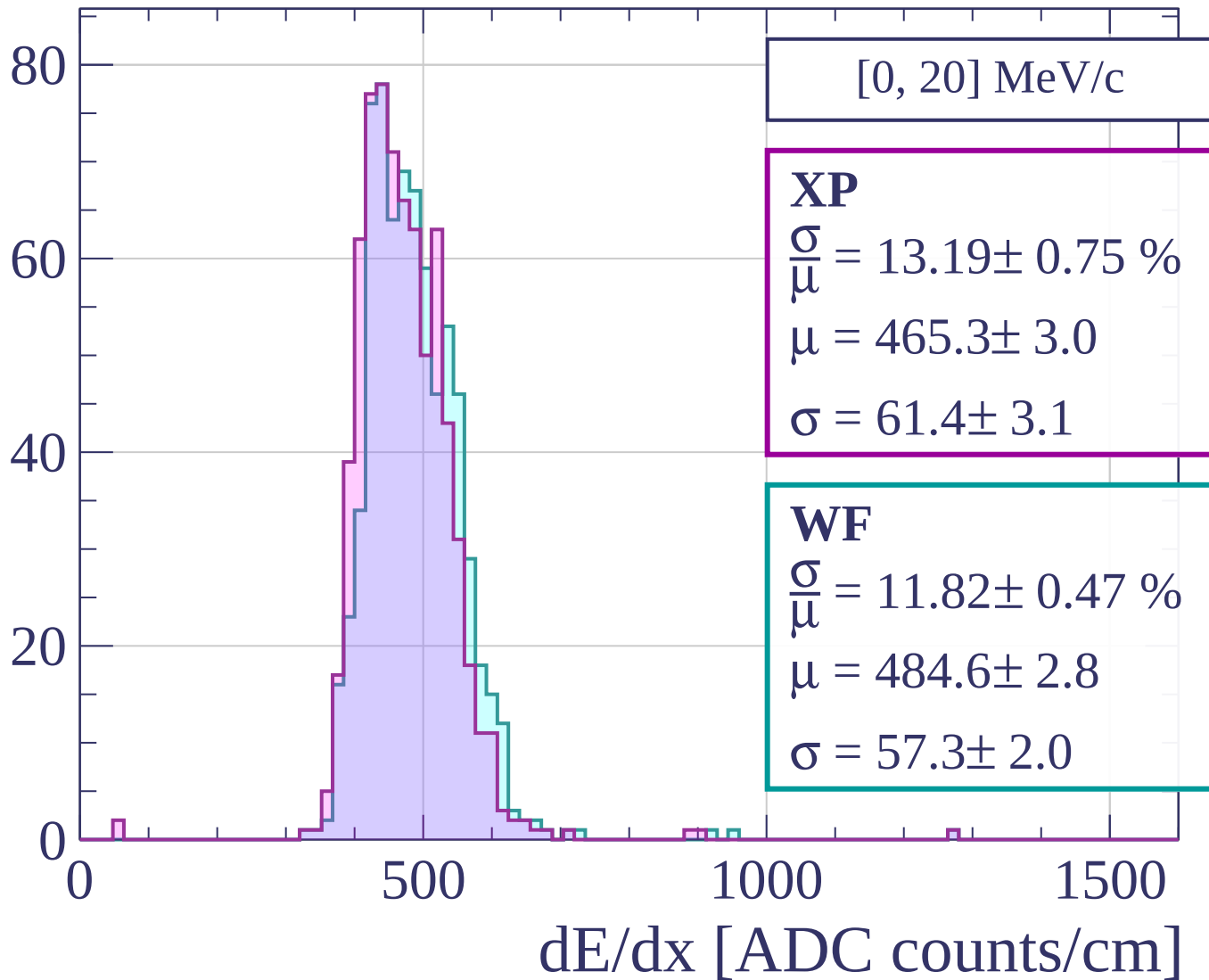
Count



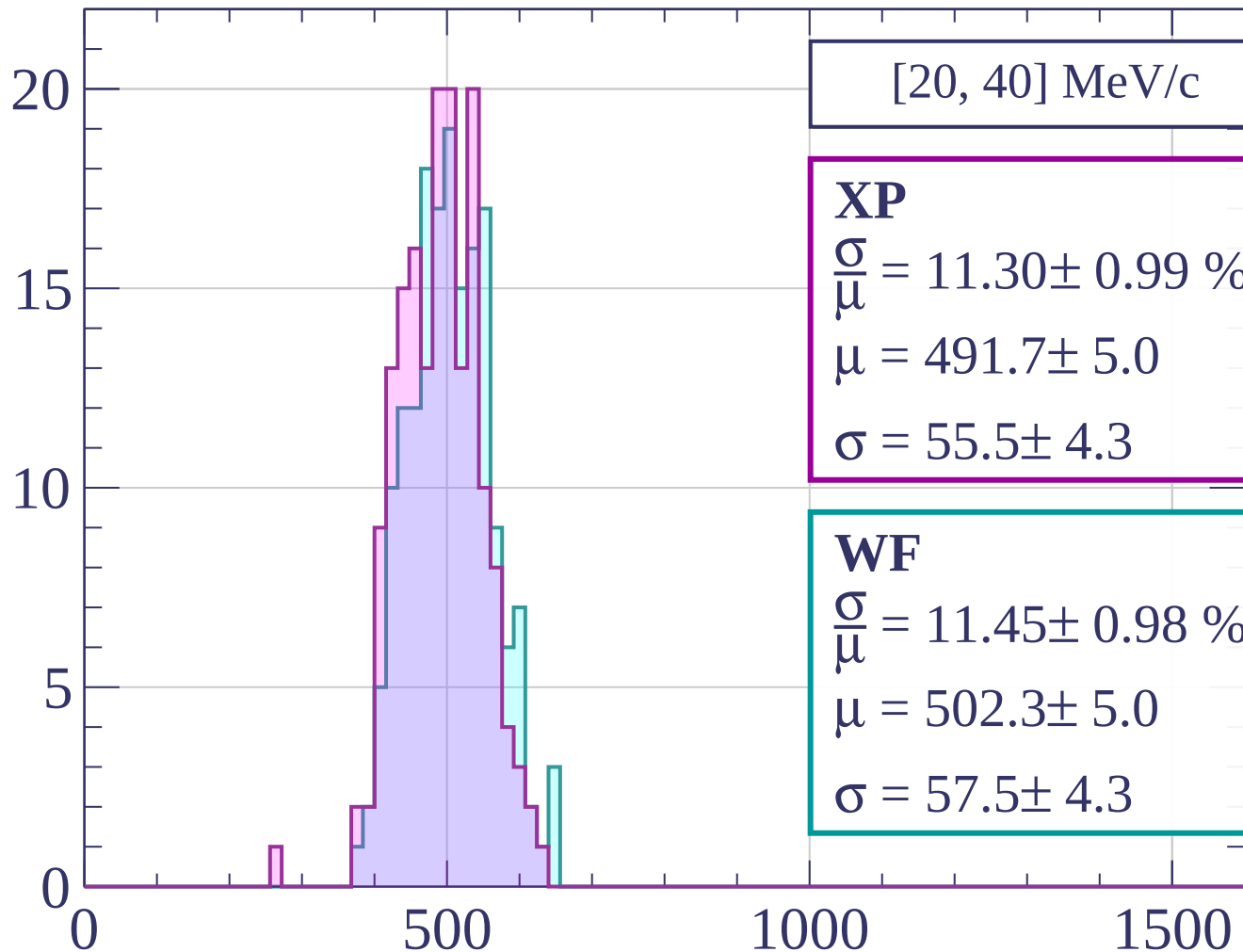
Count



Count

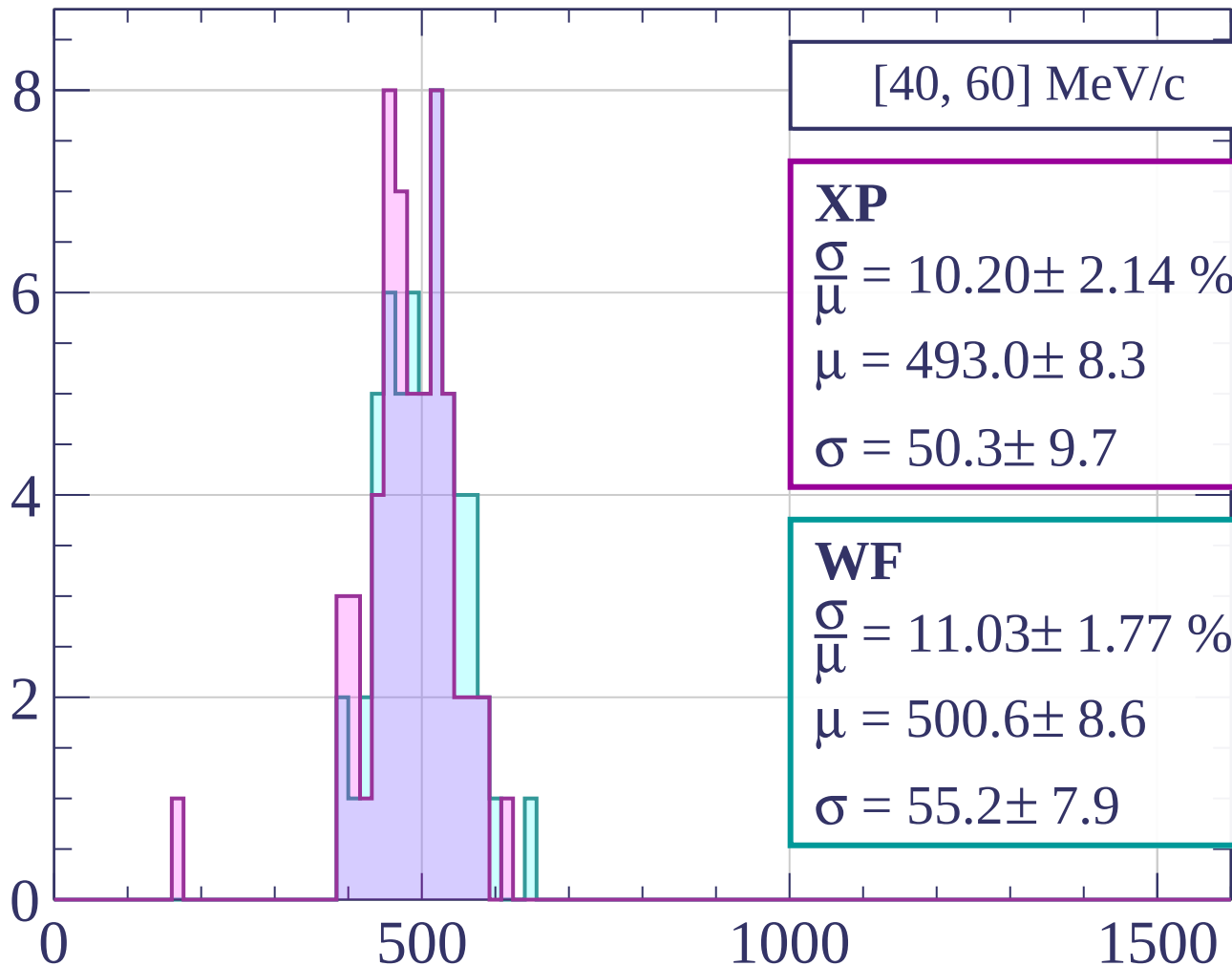


Count



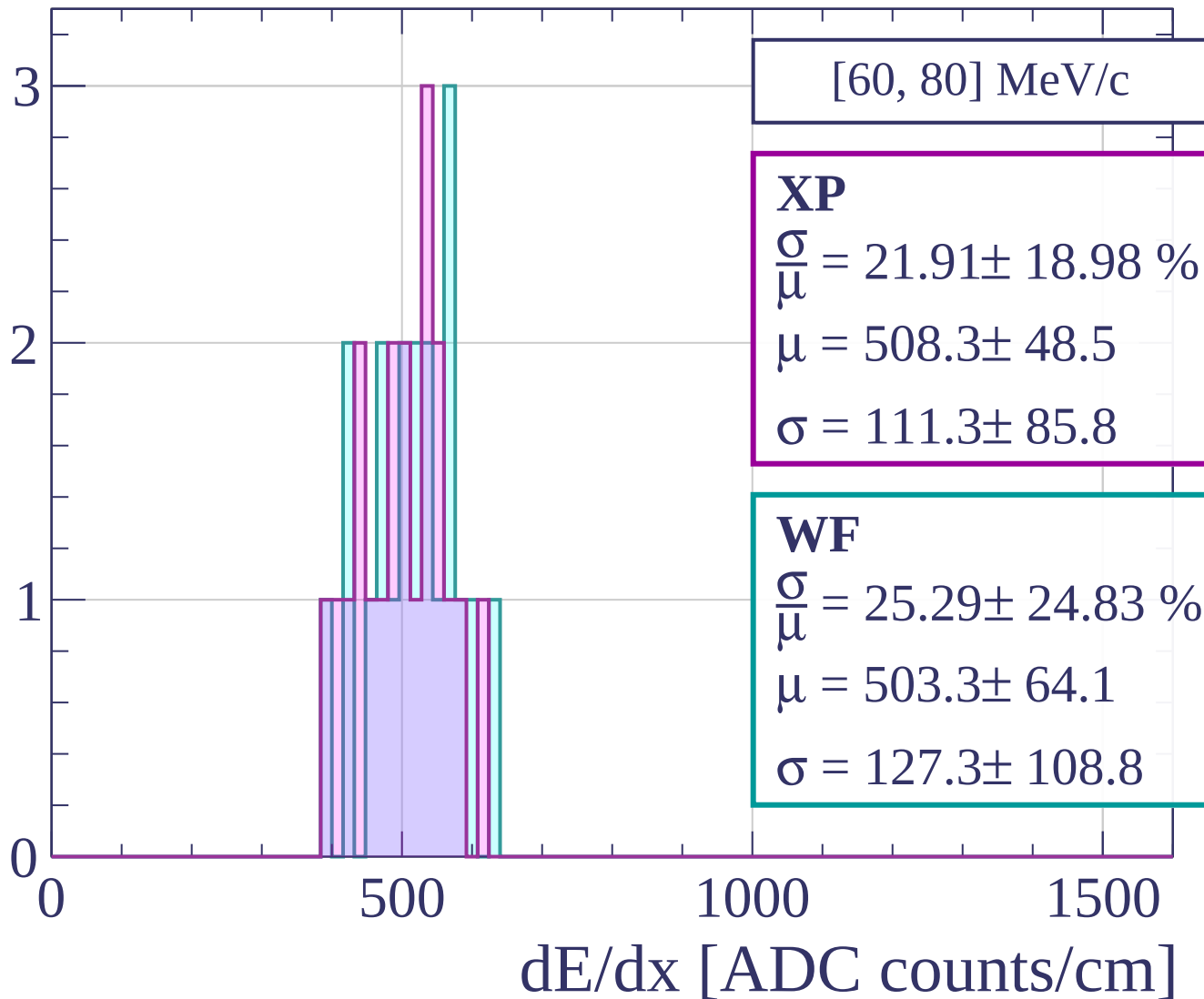
dE/dx [ADC counts/cm]

Count

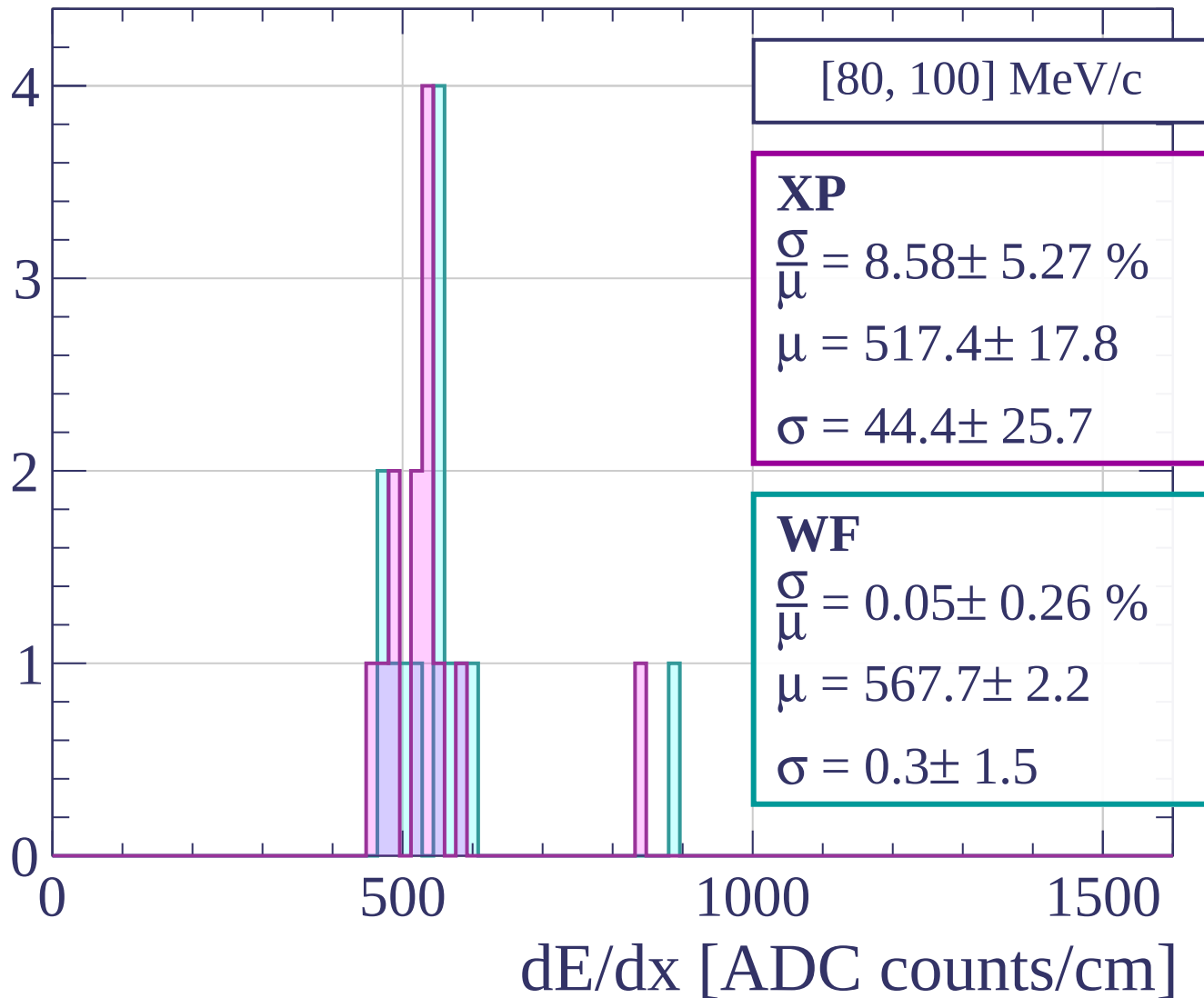


dE/dx [ADC counts/cm]

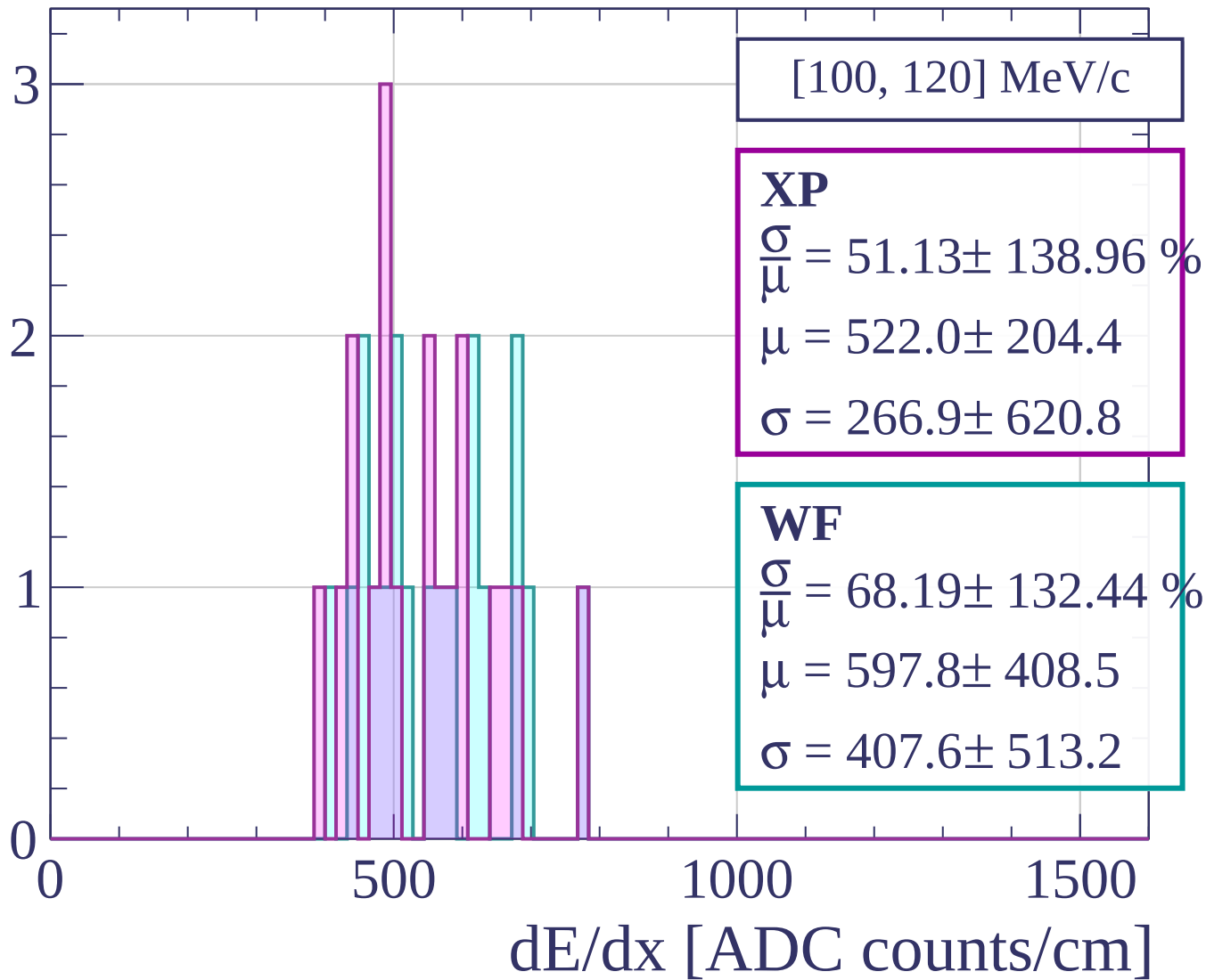
Count



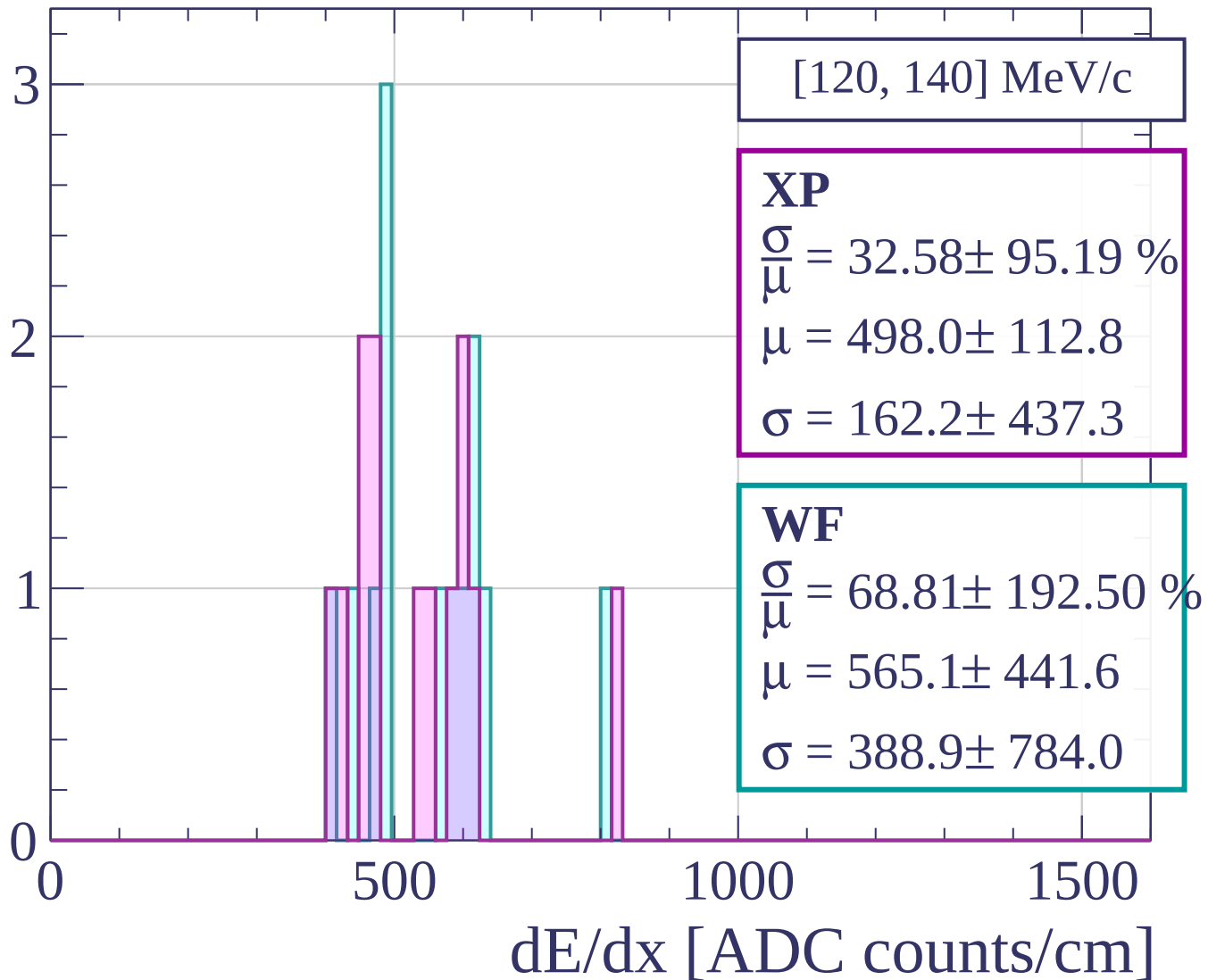
Count



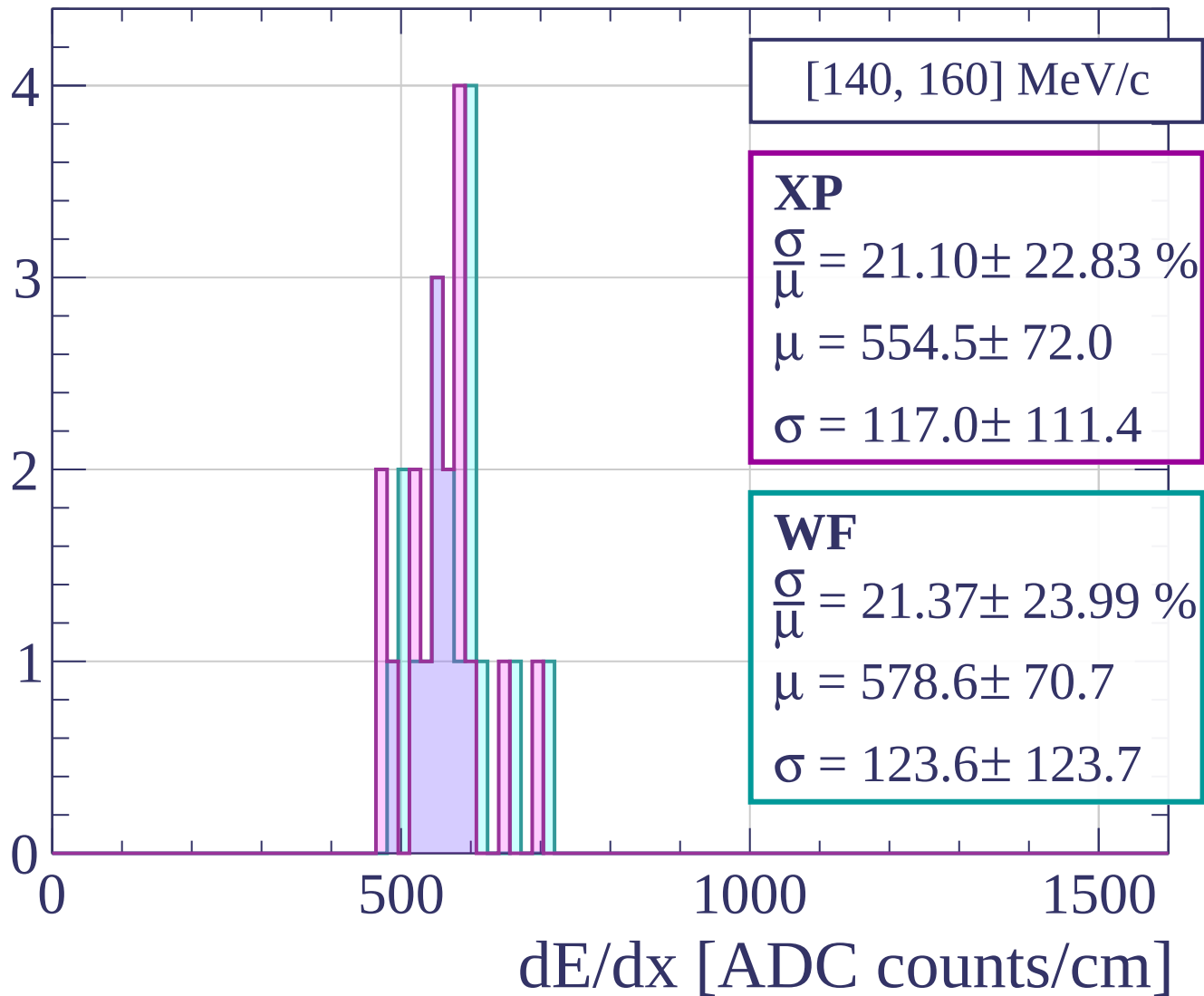
Count



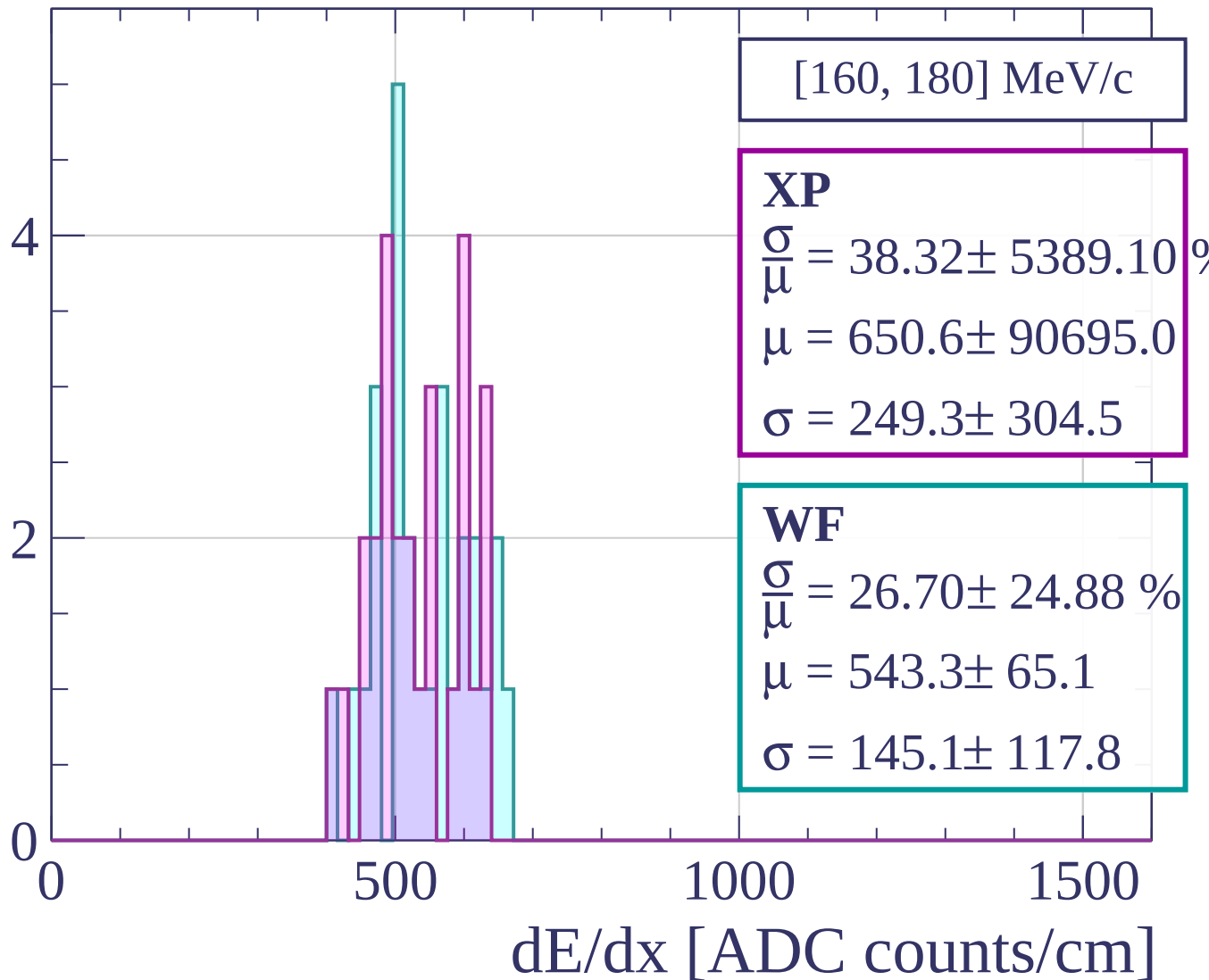
Count



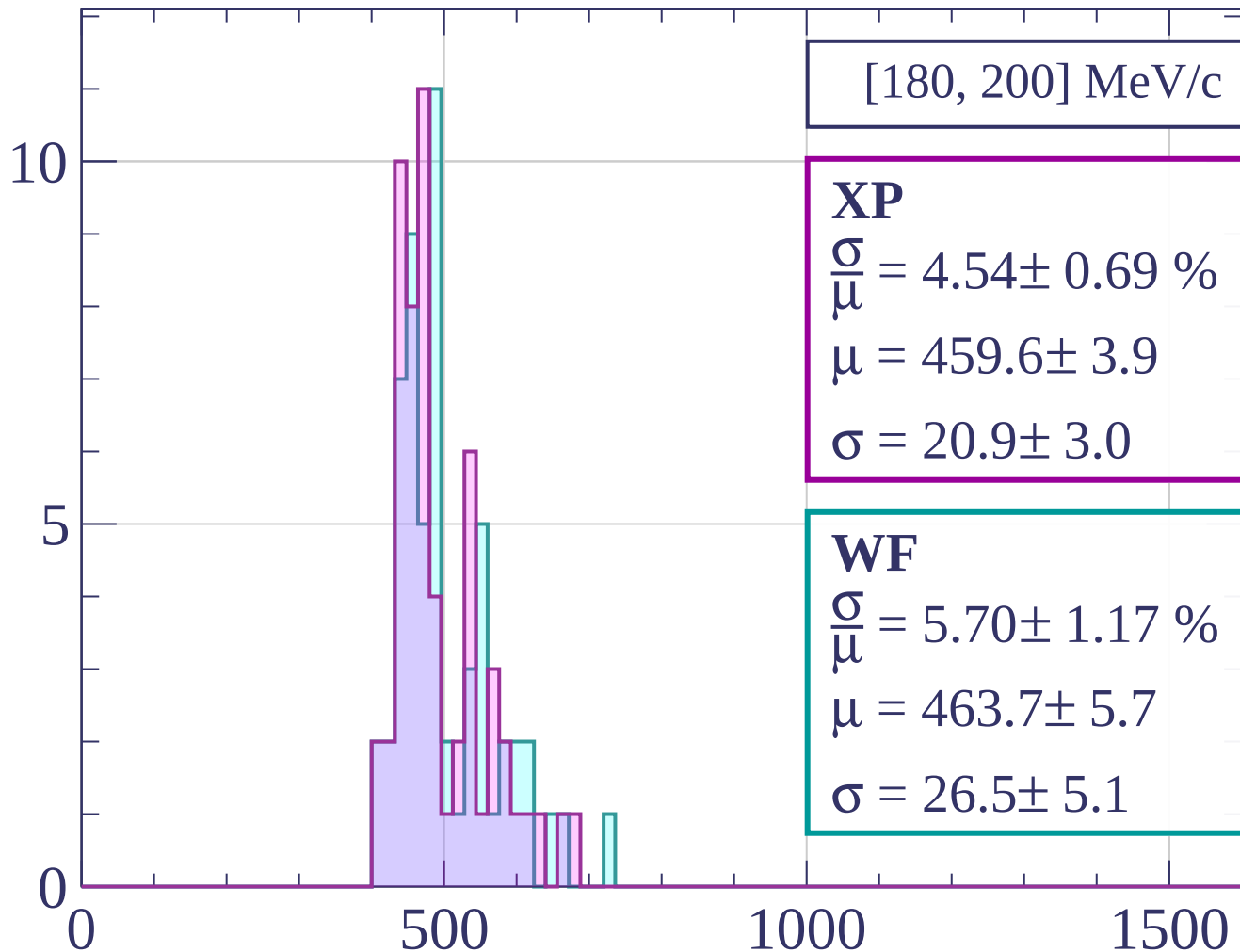
Count



Count



Count



[180, 200] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.54 \pm 0.69 \%$$

$$\mu = 459.6 \pm 3.9$$

$$\sigma = 20.9 \pm 3.0$$

WF

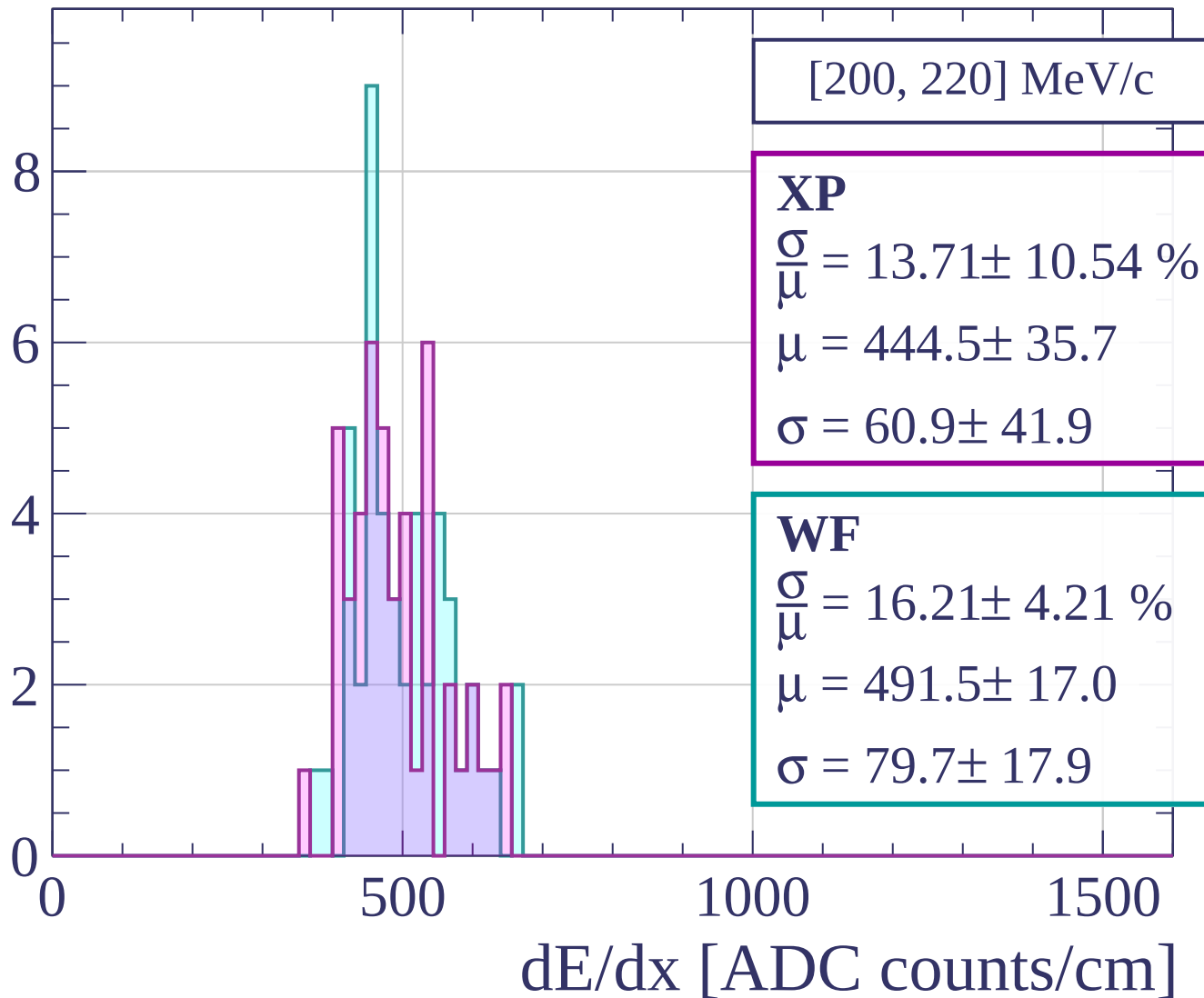
$$\frac{\sigma}{\mu} = 5.70 \pm 1.17 \%$$

$$\mu = 463.7 \pm 5.7$$

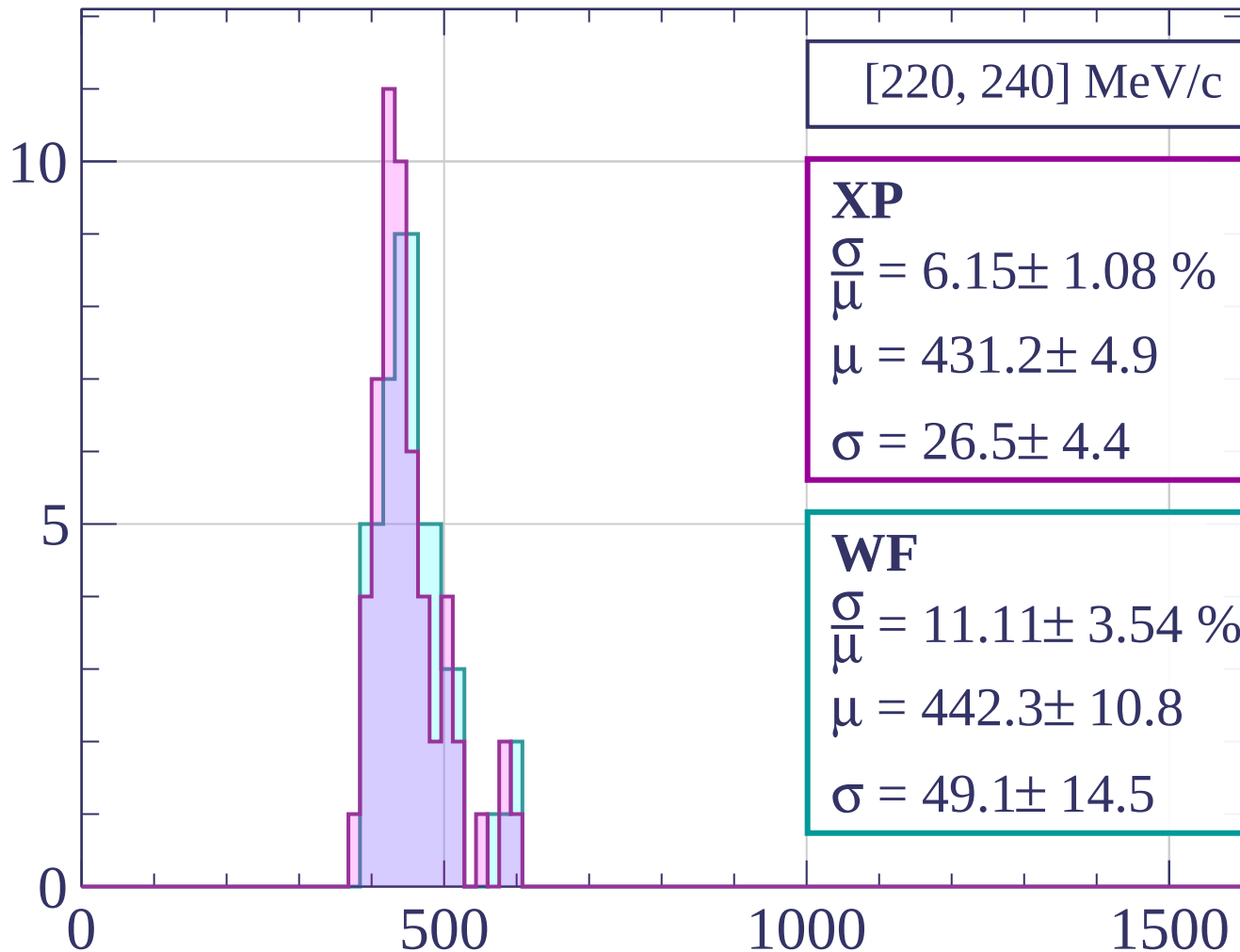
$$\sigma = 26.5 \pm 5.1$$

dE/dx [ADC counts/cm]

Count



Count



$[220, 240] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 6.15 \pm 1.08 \%$$

$$\mu = 431.2 \pm 4.9$$

$$\sigma = 26.5 \pm 4.4$$

WF

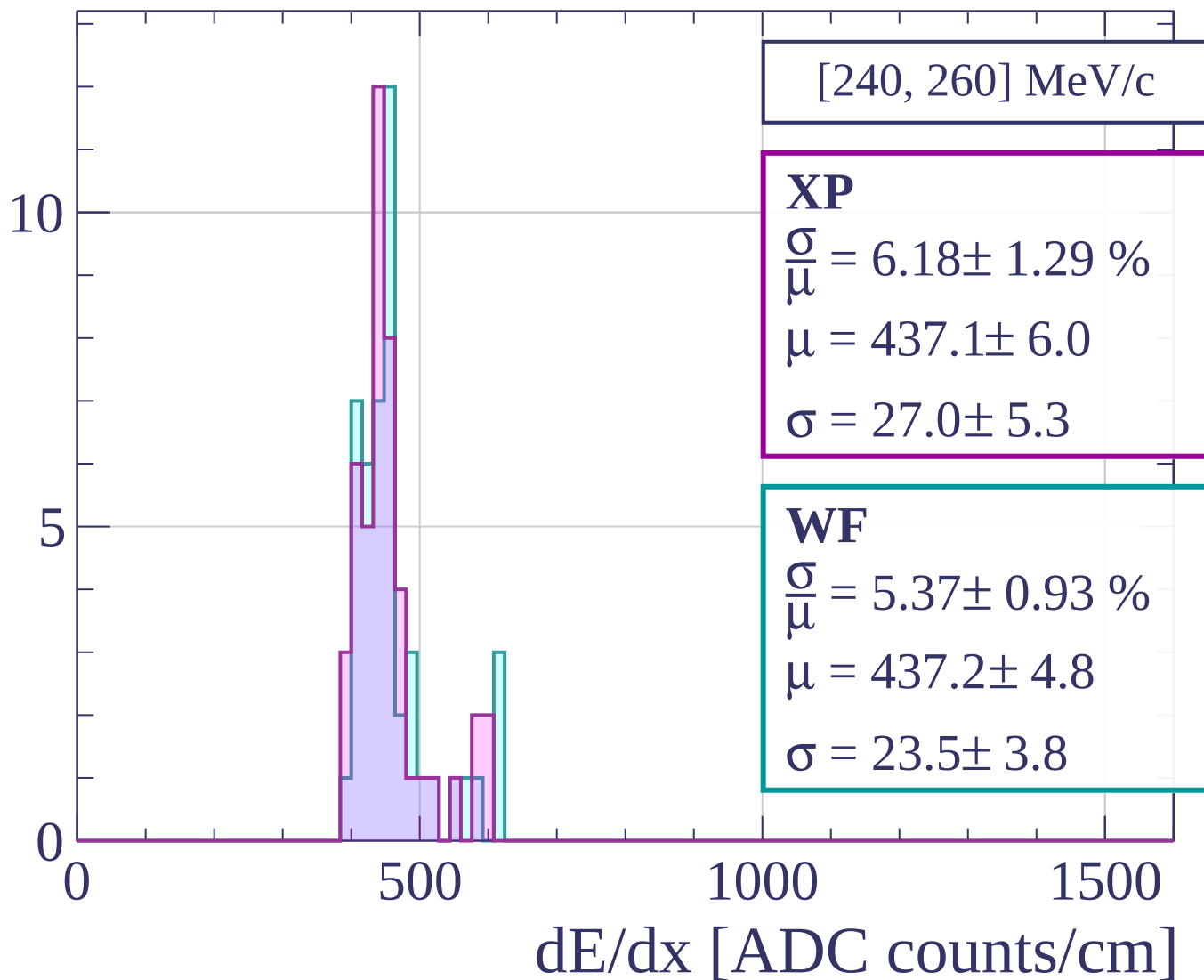
$$\frac{\sigma}{\mu} = 11.11 \pm 3.54 \%$$

$$\mu = 442.3 \pm 10.8$$

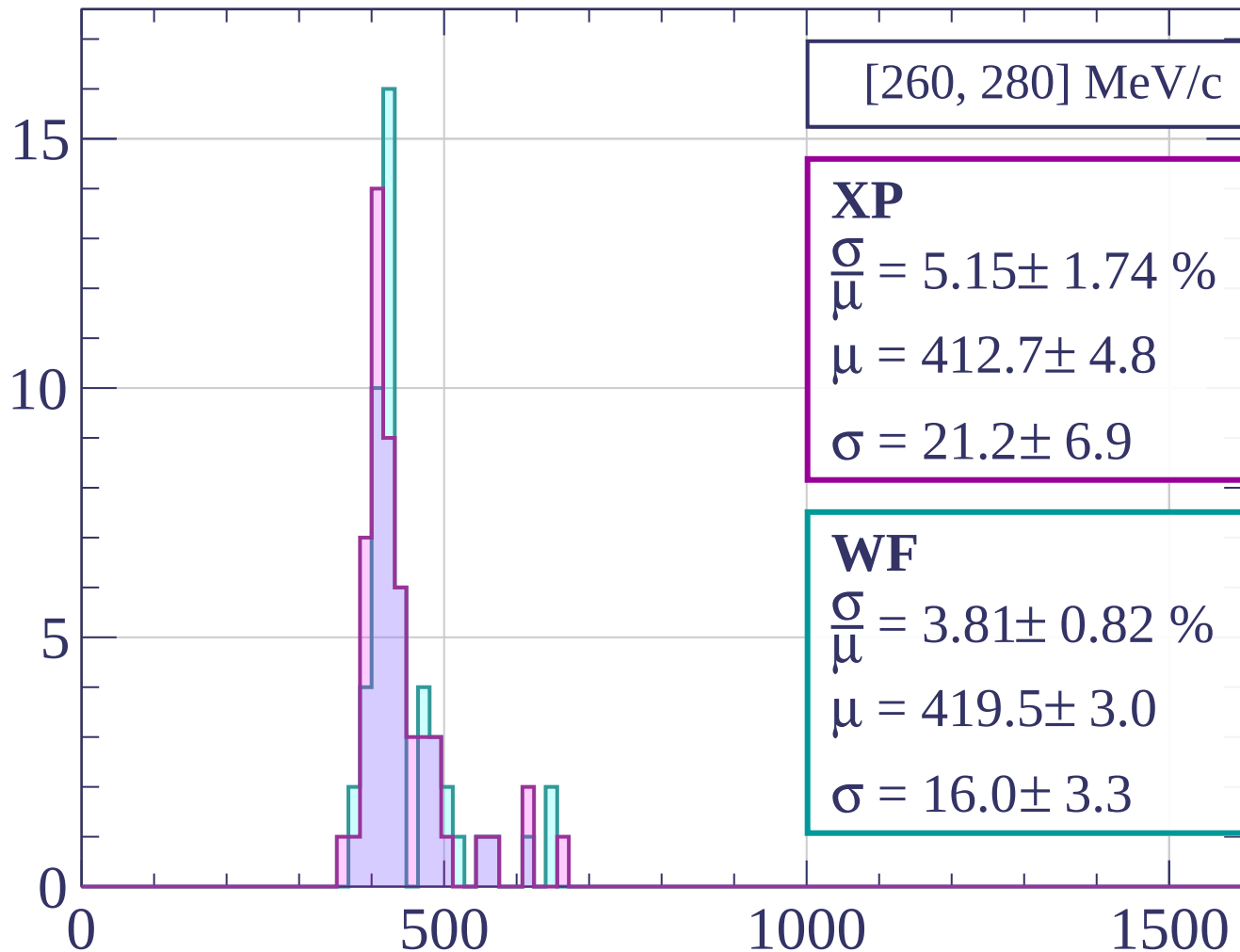
$$\sigma = 49.1 \pm 14.5$$

dE/dx [ADC counts/cm]

Count

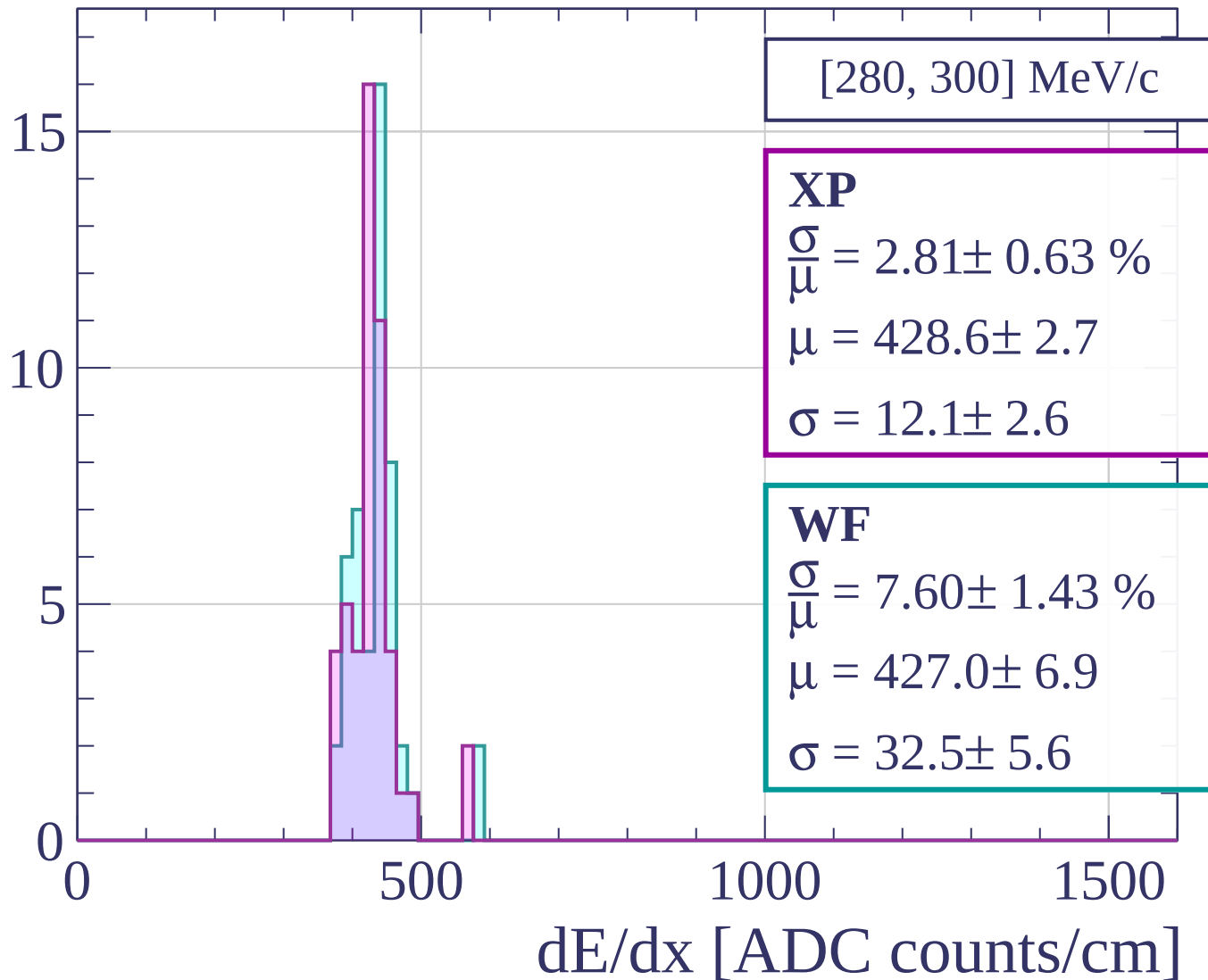


Count

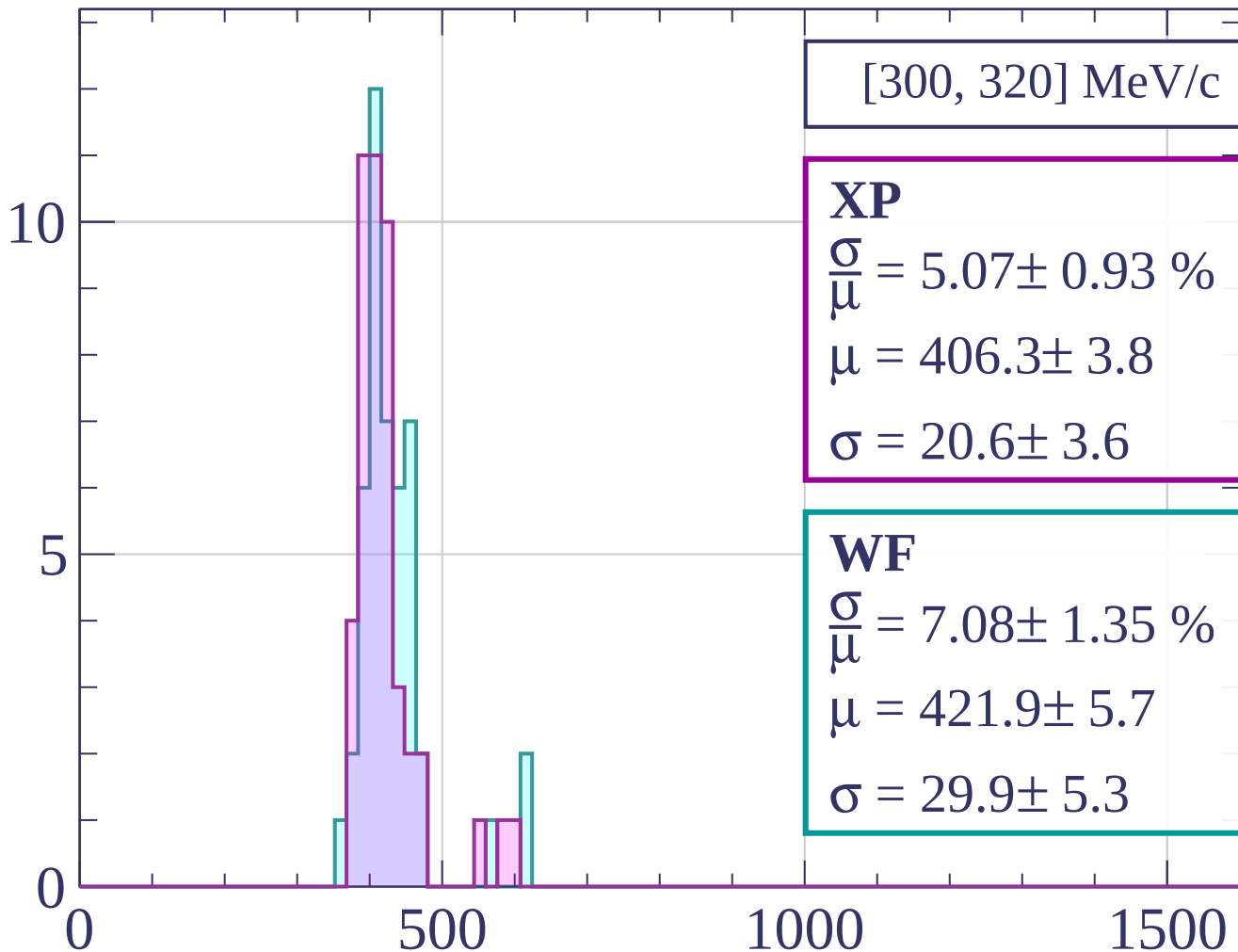


dE/dx [ADC counts/cm]

Count

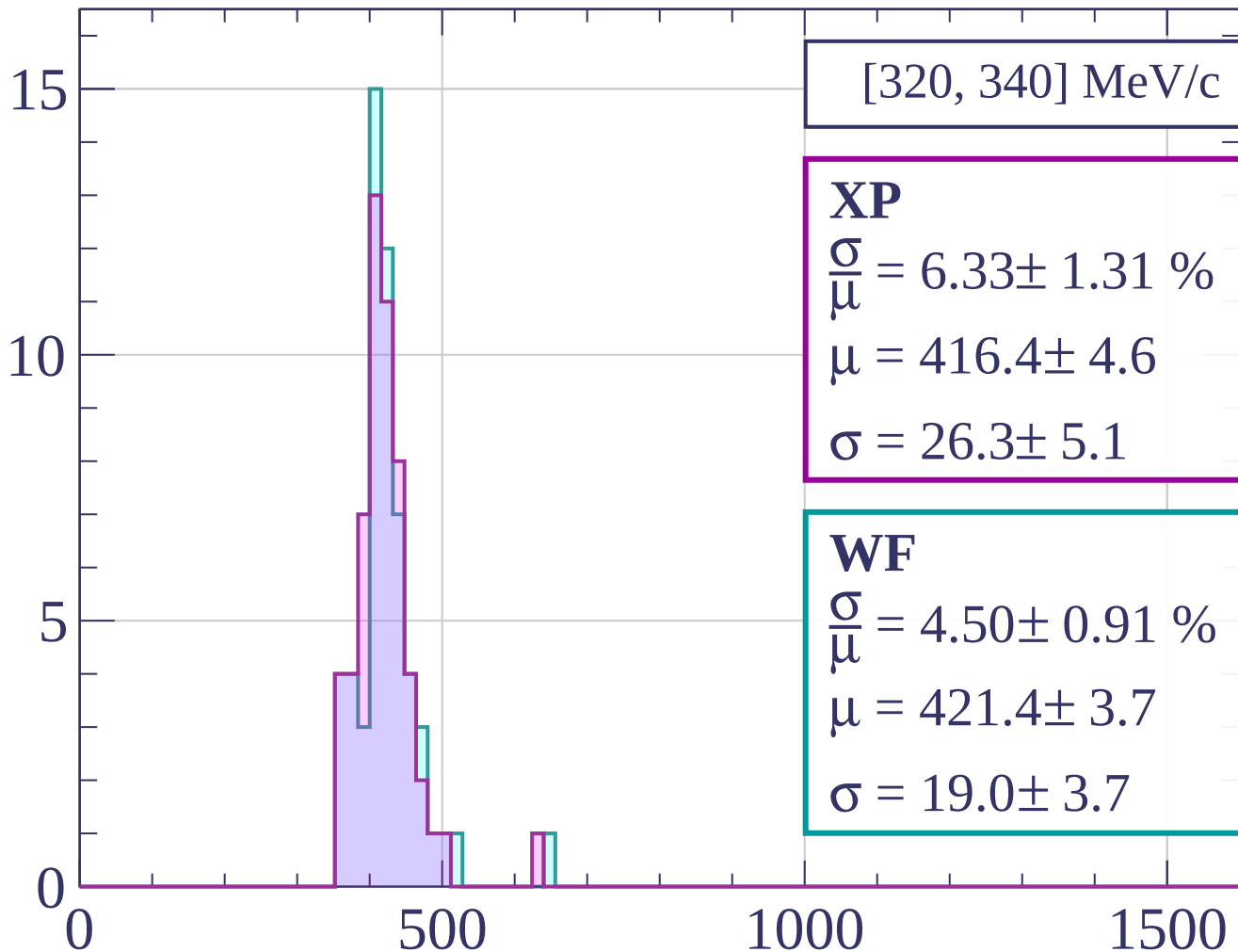


Count



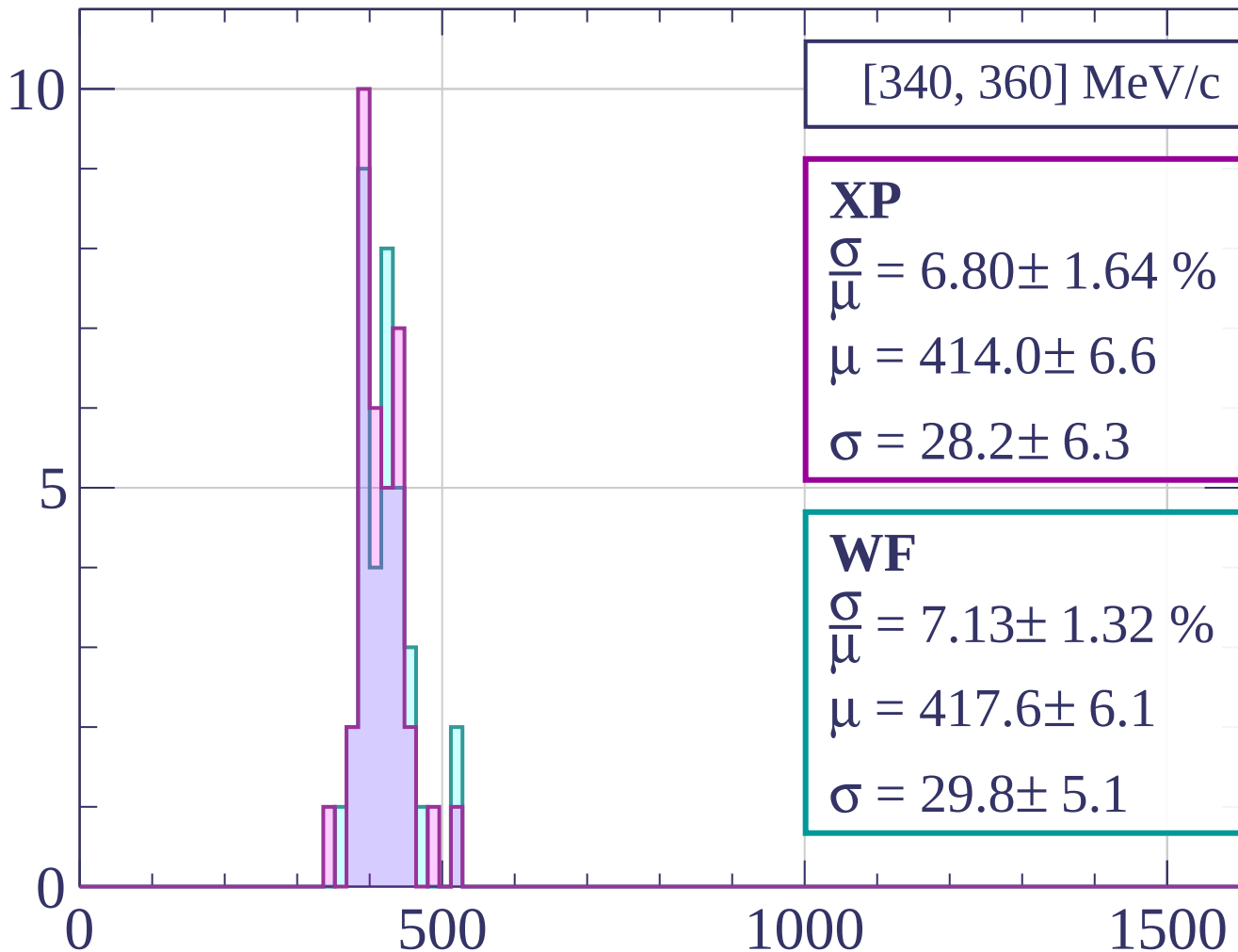
dE/dx [ADC counts/cm]

Count



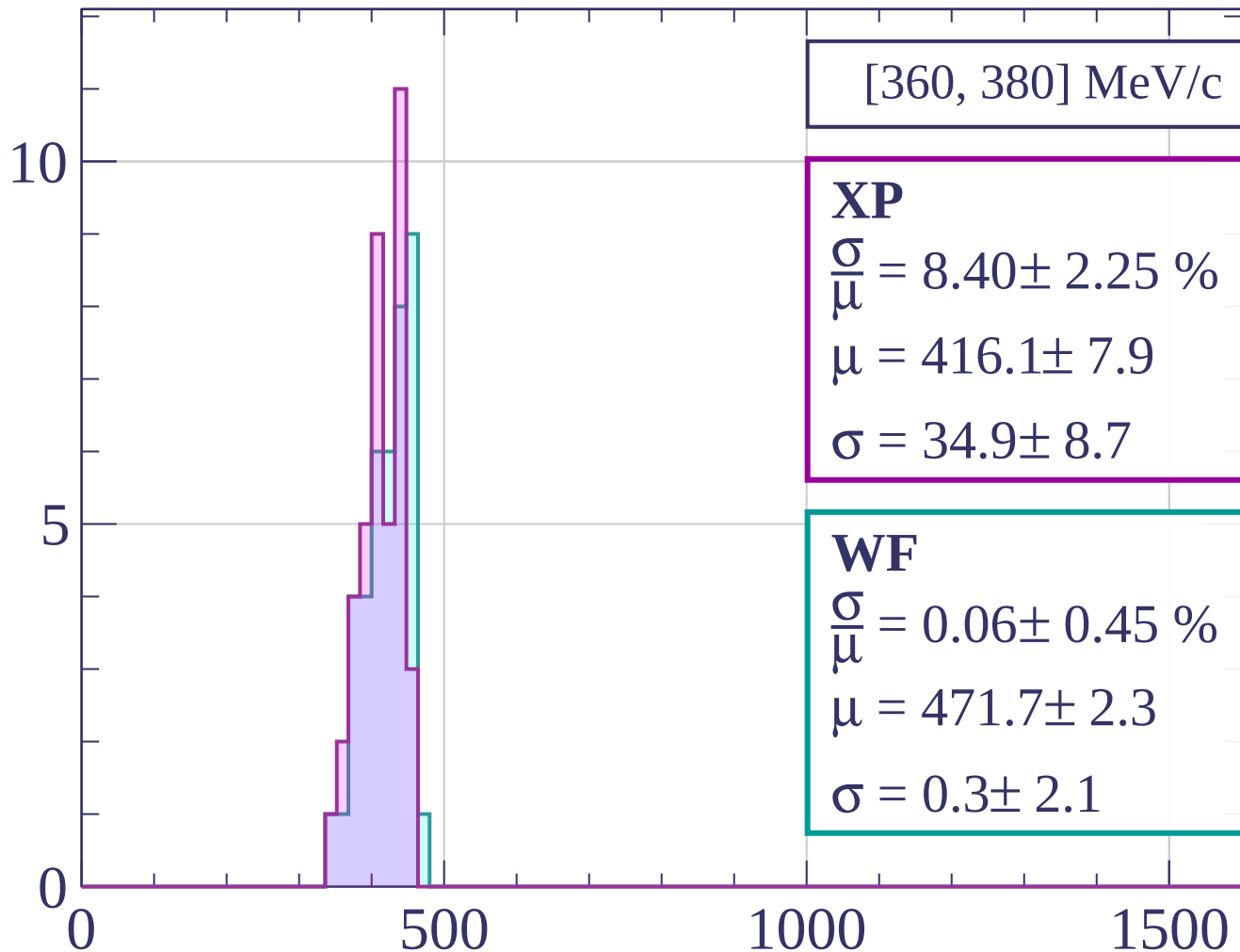
dE/dx [ADC counts/cm]

Count



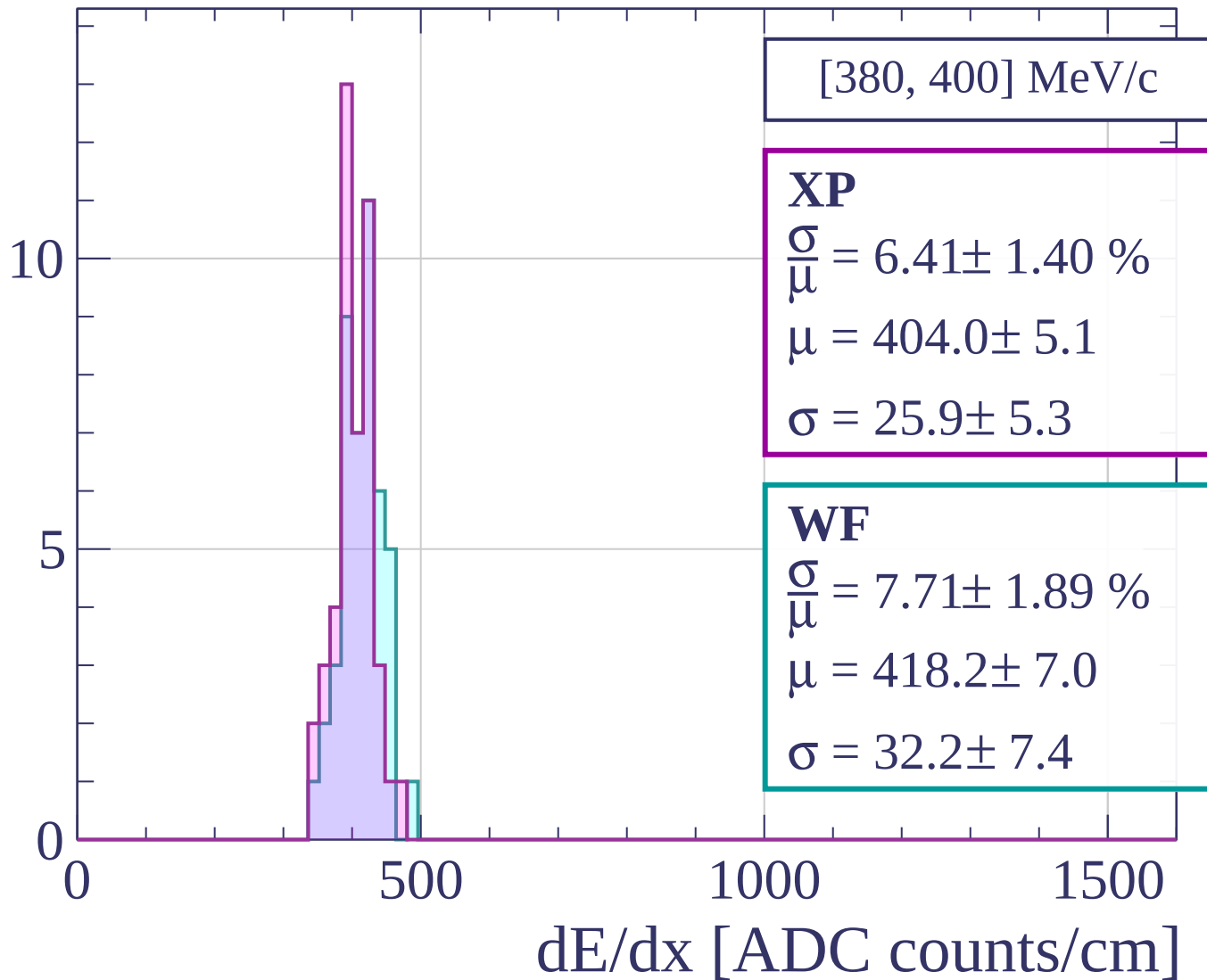
dE/dx [ADC counts/cm]

Count

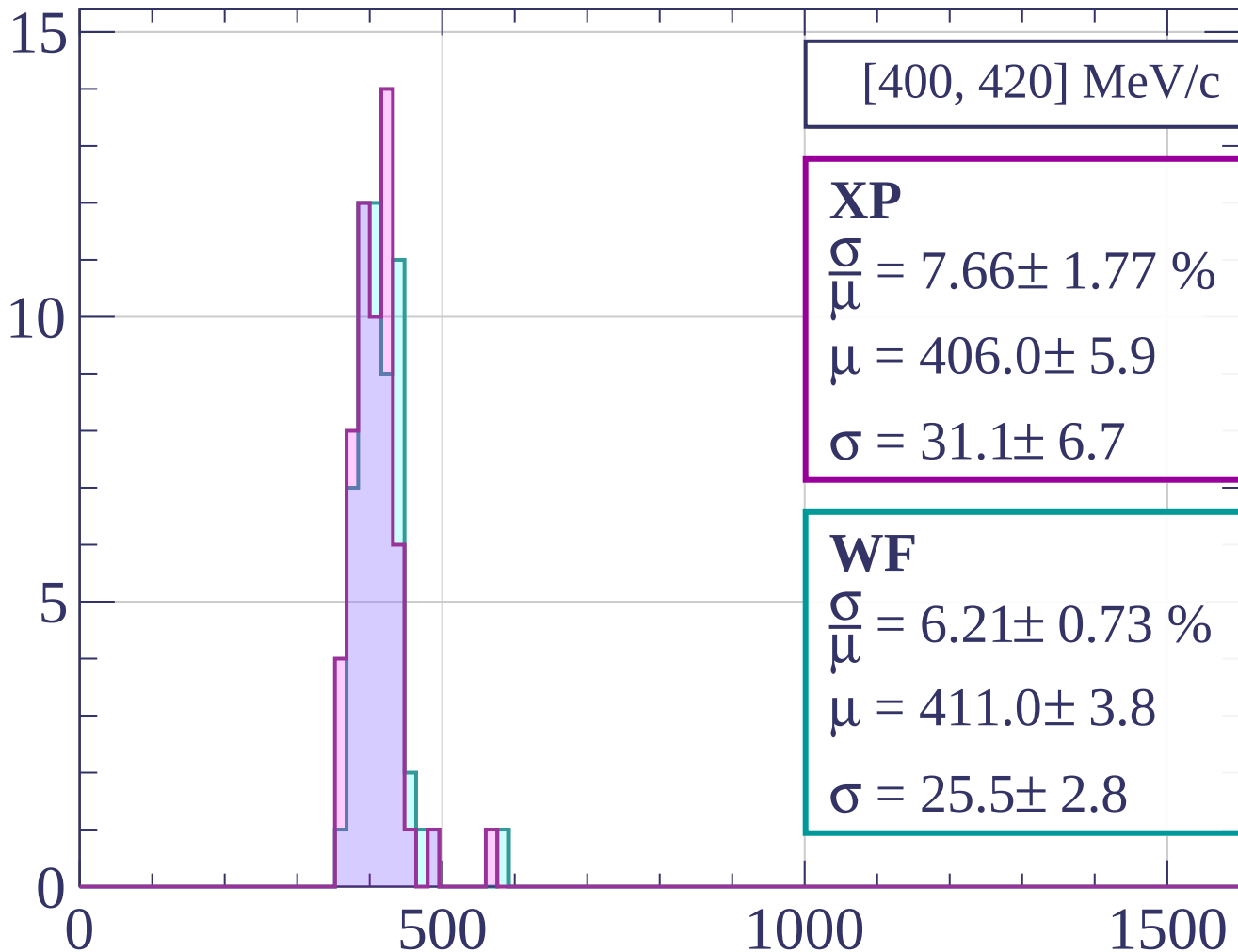


dE/dx [ADC counts/cm]

Count



Count



XP

$$\frac{\sigma}{\mu} = 7.66 \pm 1.77 \%$$

$$\mu = 406.0 \pm 5.9$$

$$\sigma = 31.1 \pm 6.7$$

WF

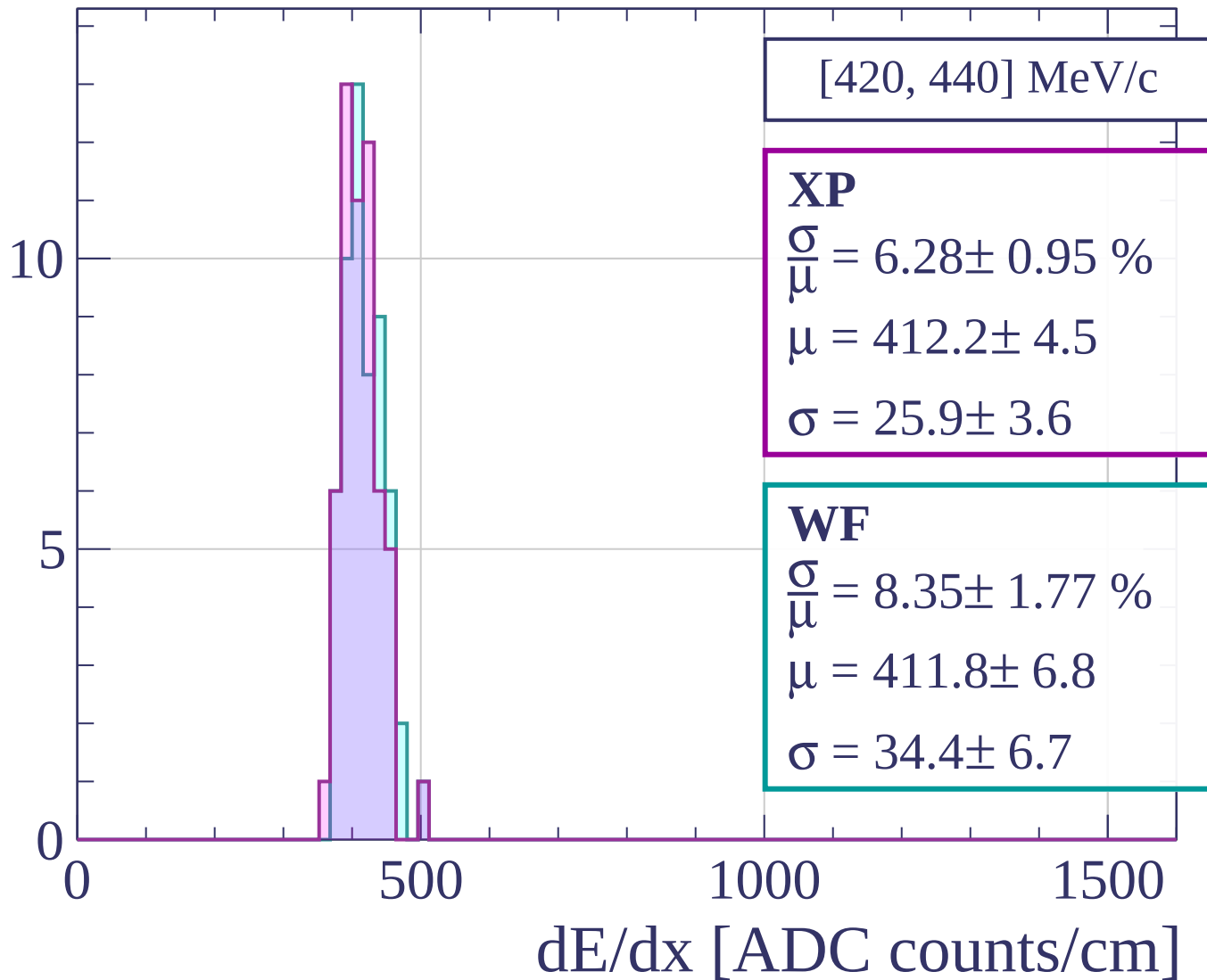
$$\frac{\sigma}{\mu} = 6.21 \pm 0.73 \%$$

$$\mu = 411.0 \pm 3.8$$

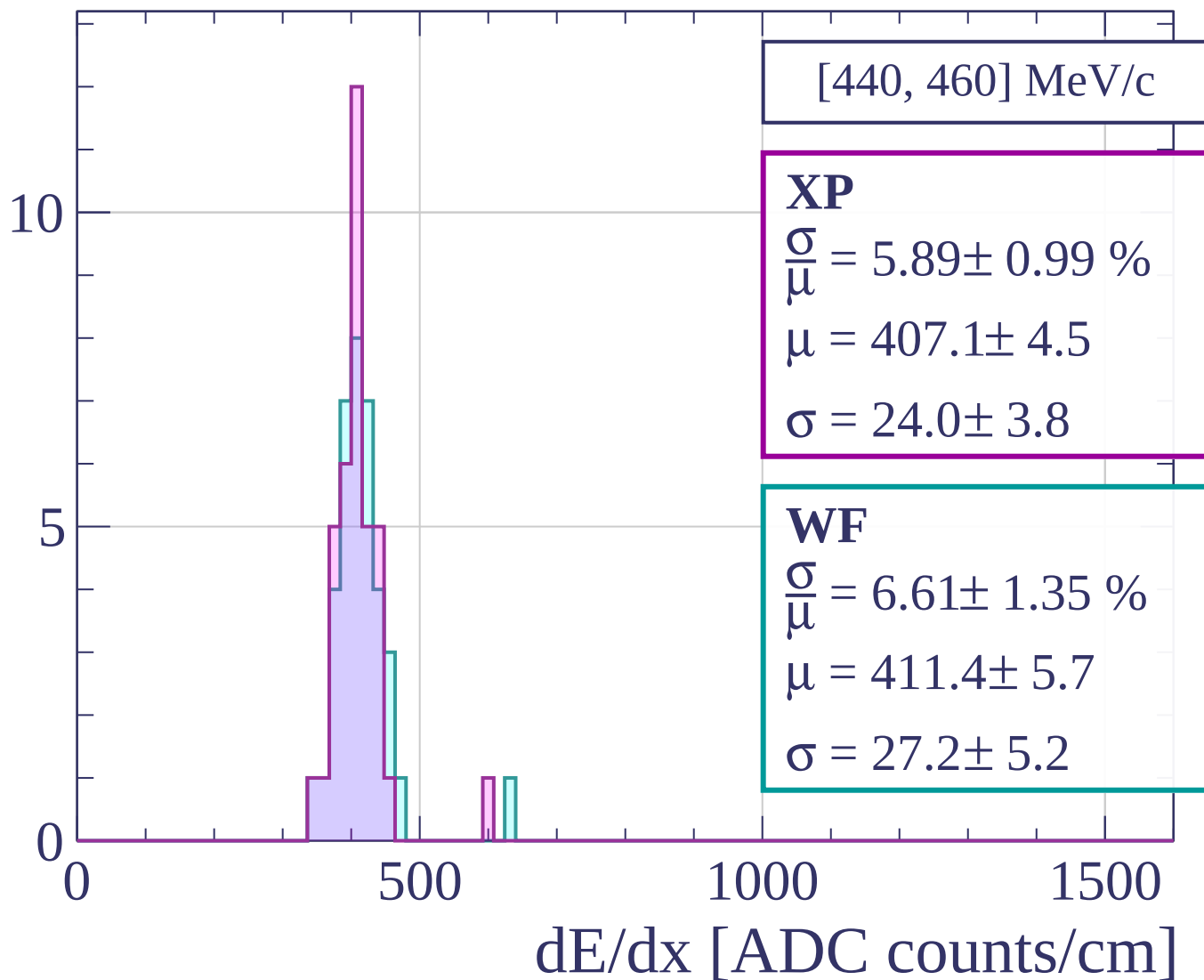
$$\sigma = 25.5 \pm 2.8$$

dE/dx [ADC counts/cm]

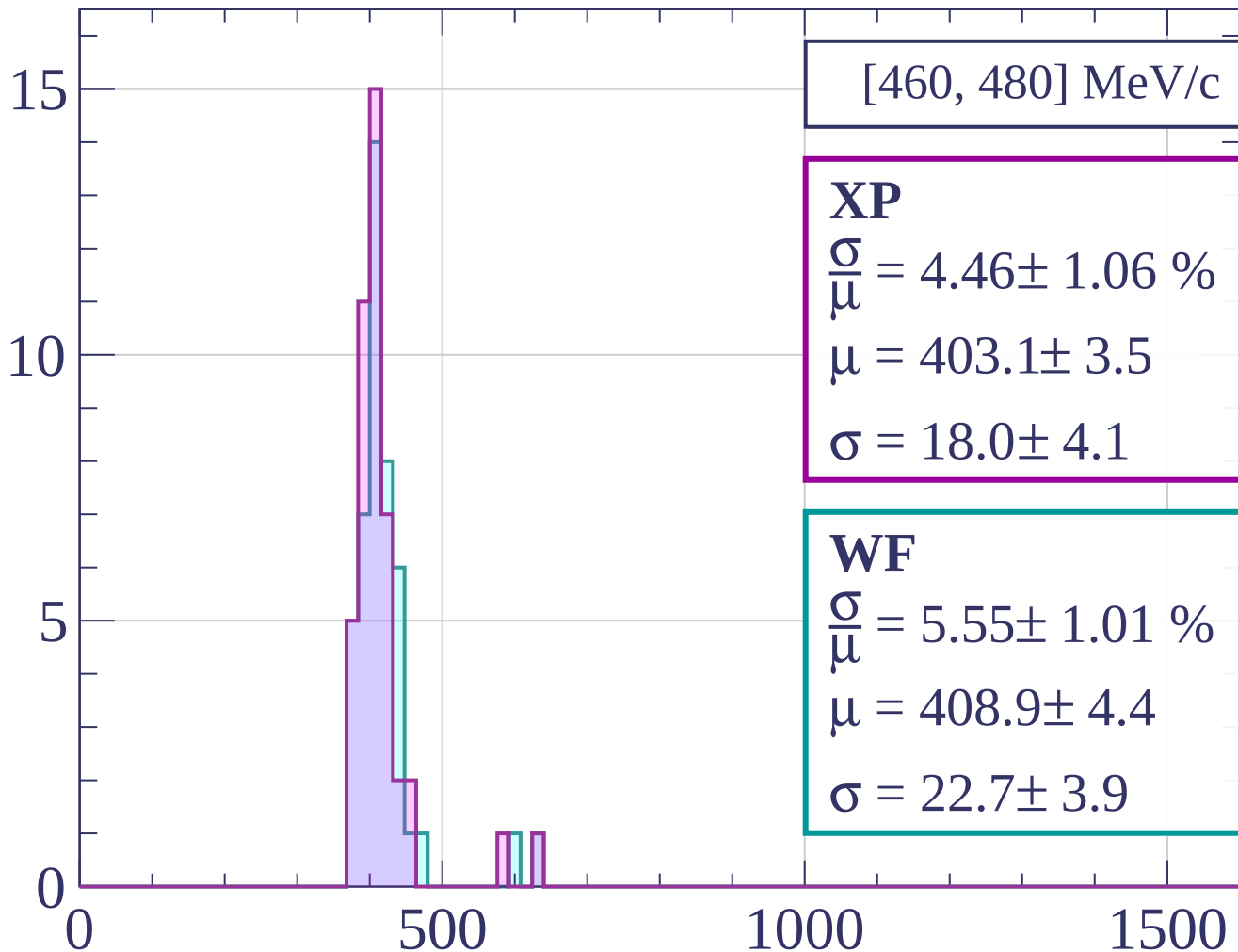
Count



Count

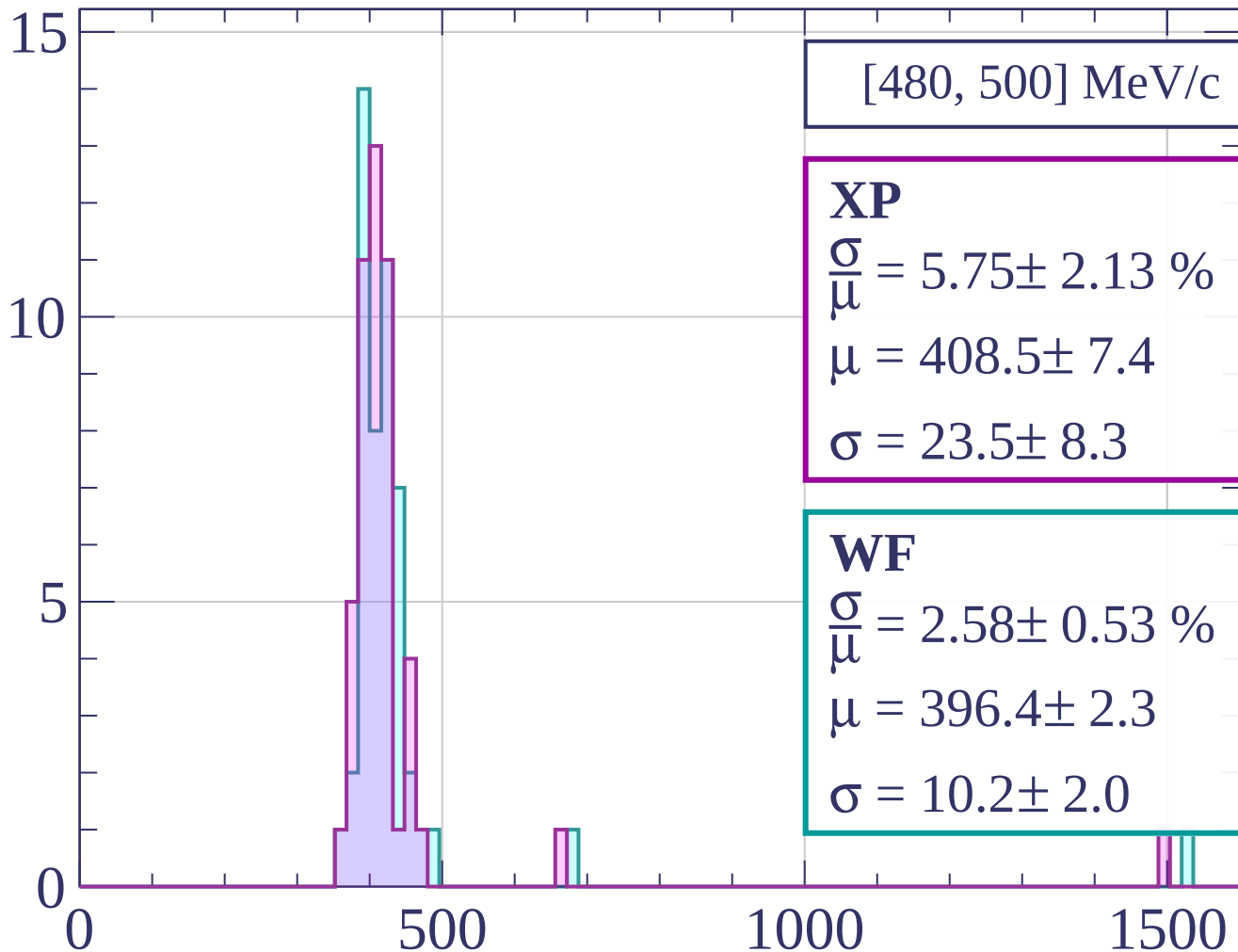


Count



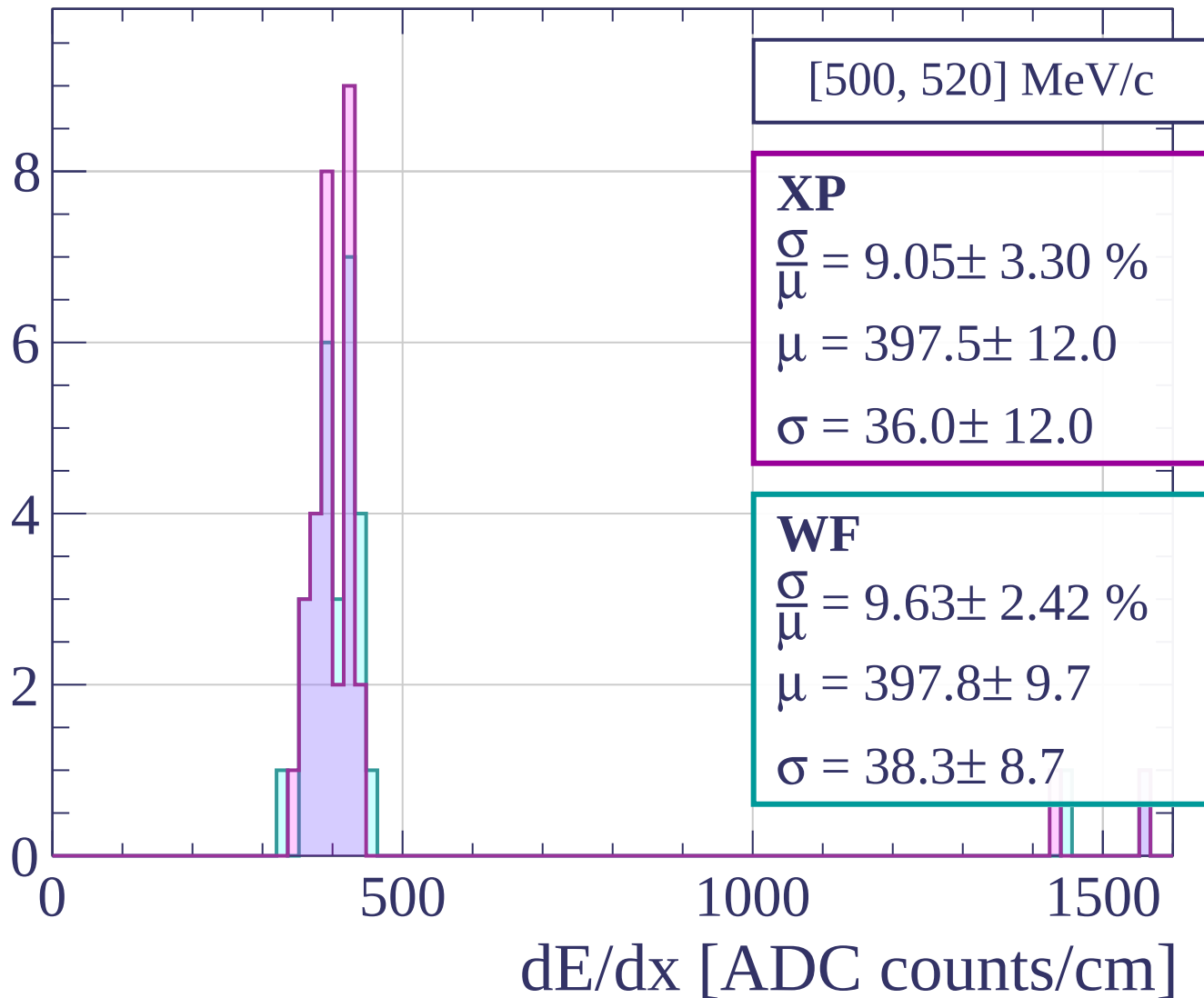
dE/dx [ADC counts/cm]

Count

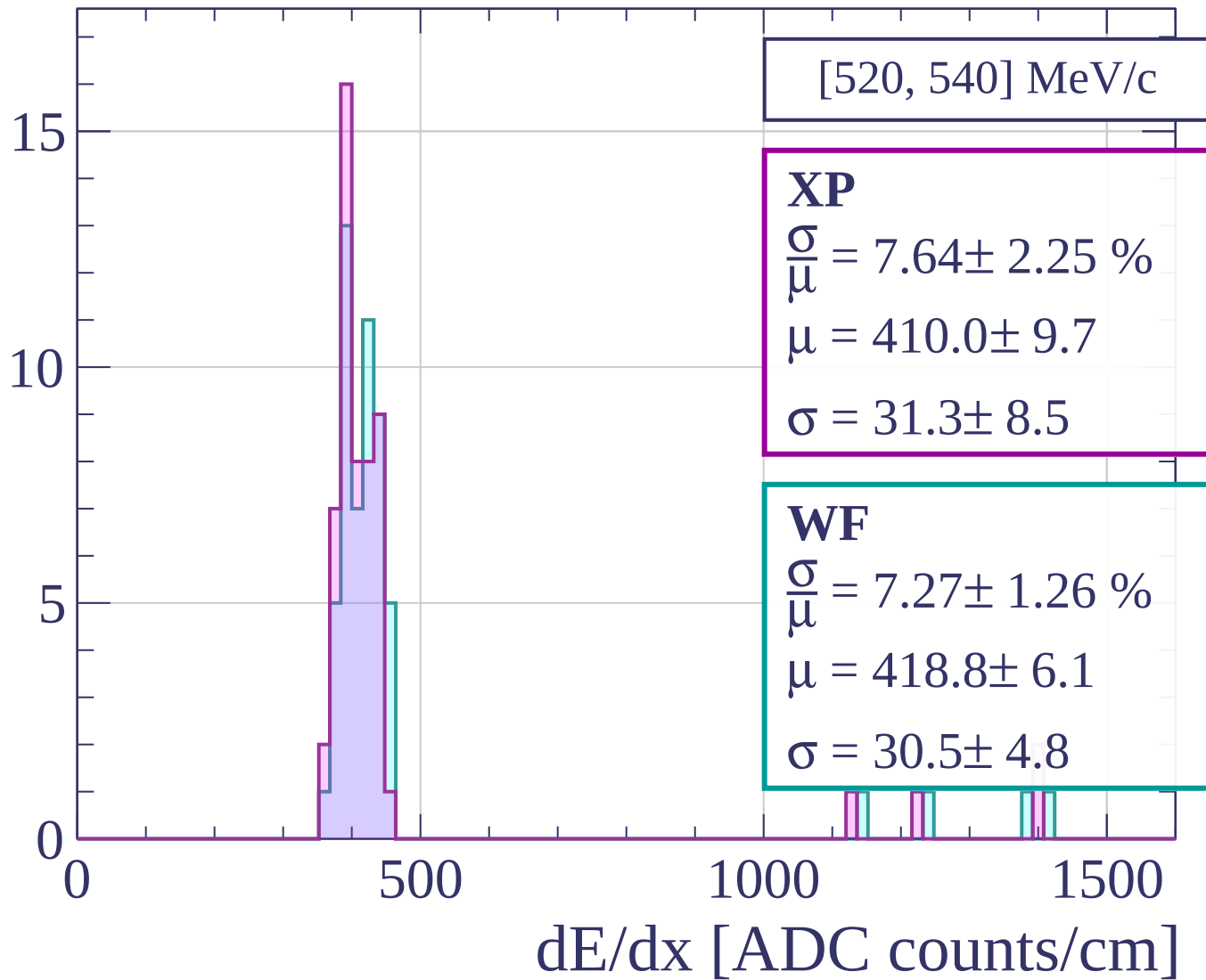


dE/dx [ADC counts/cm]

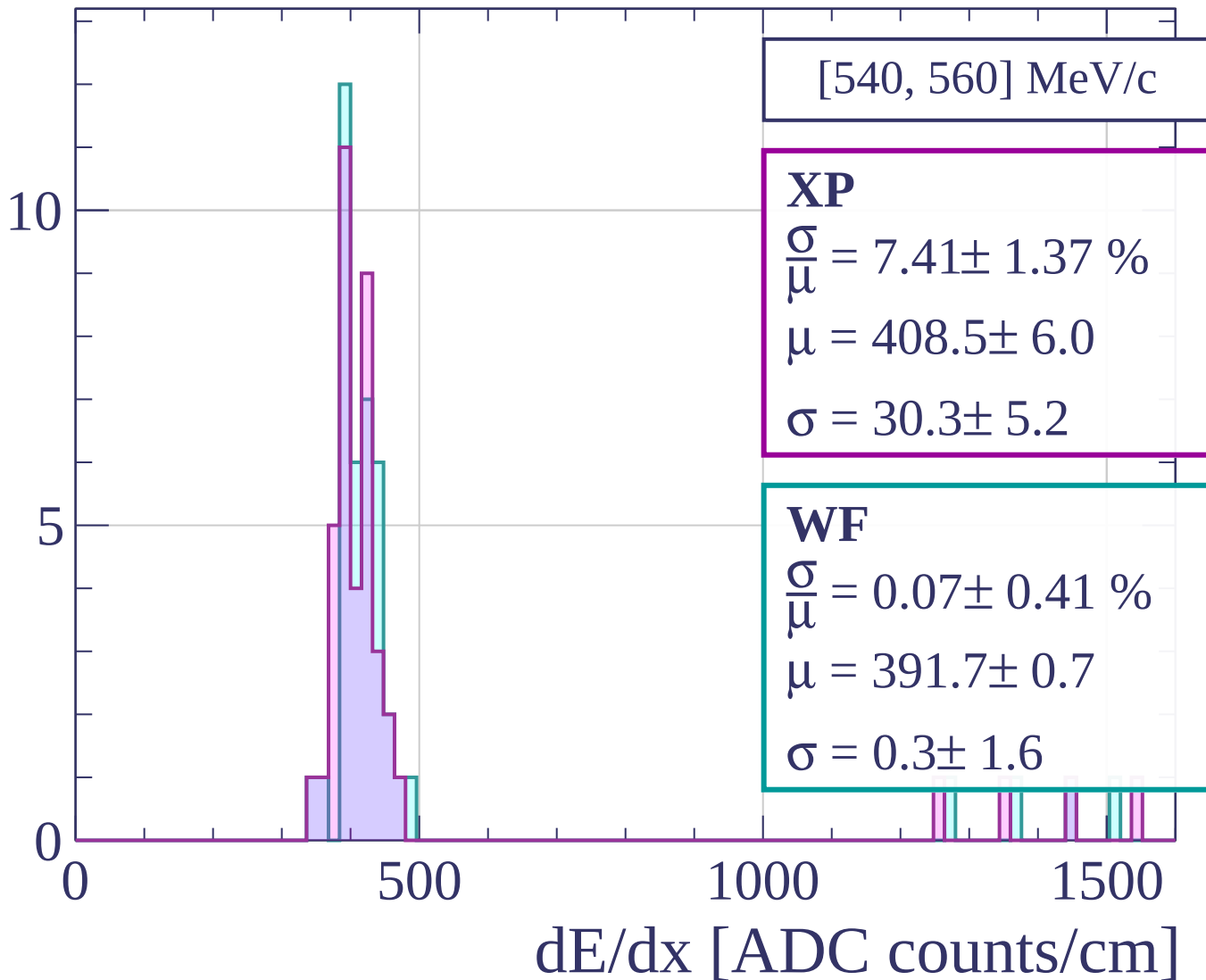
Count



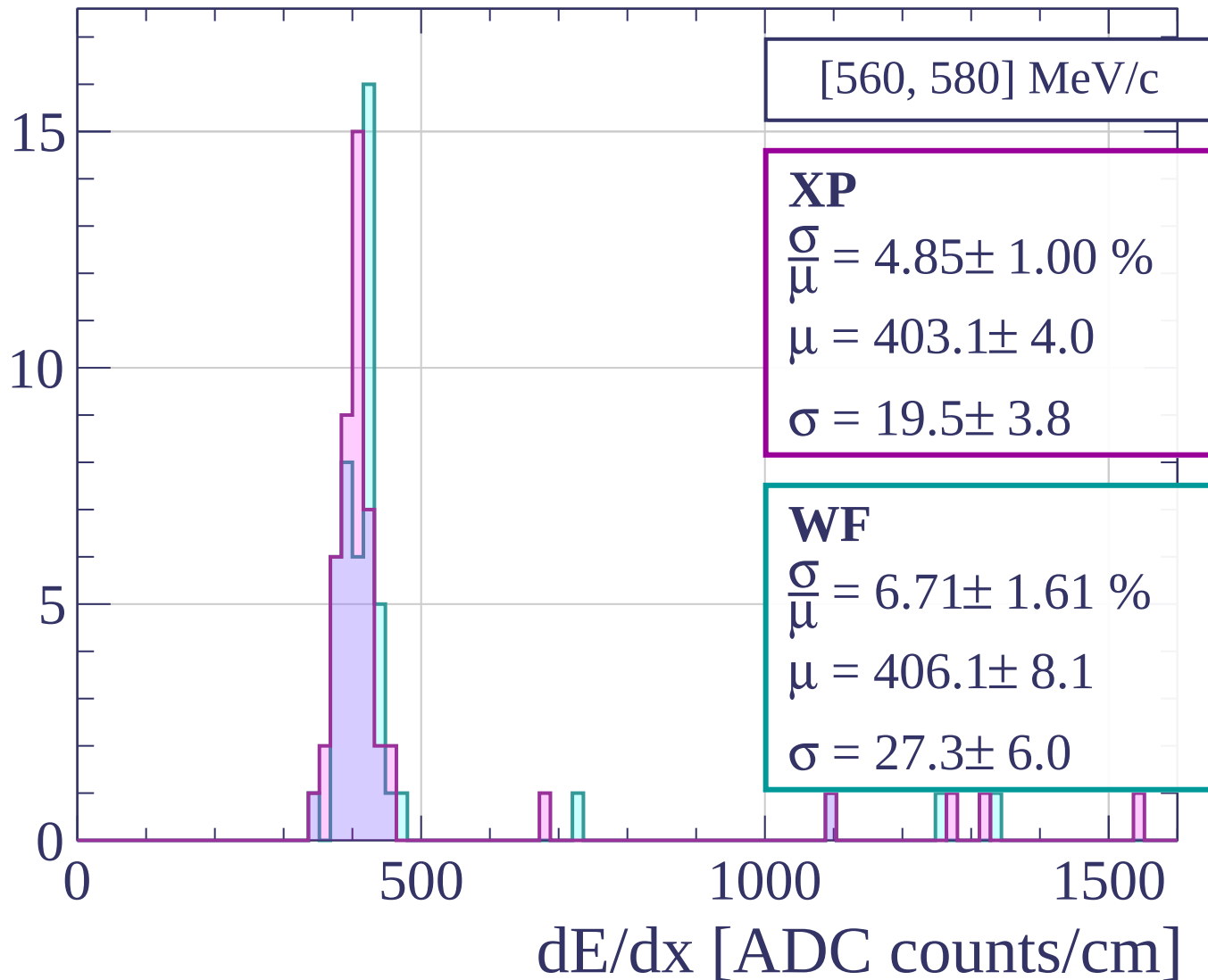
Count



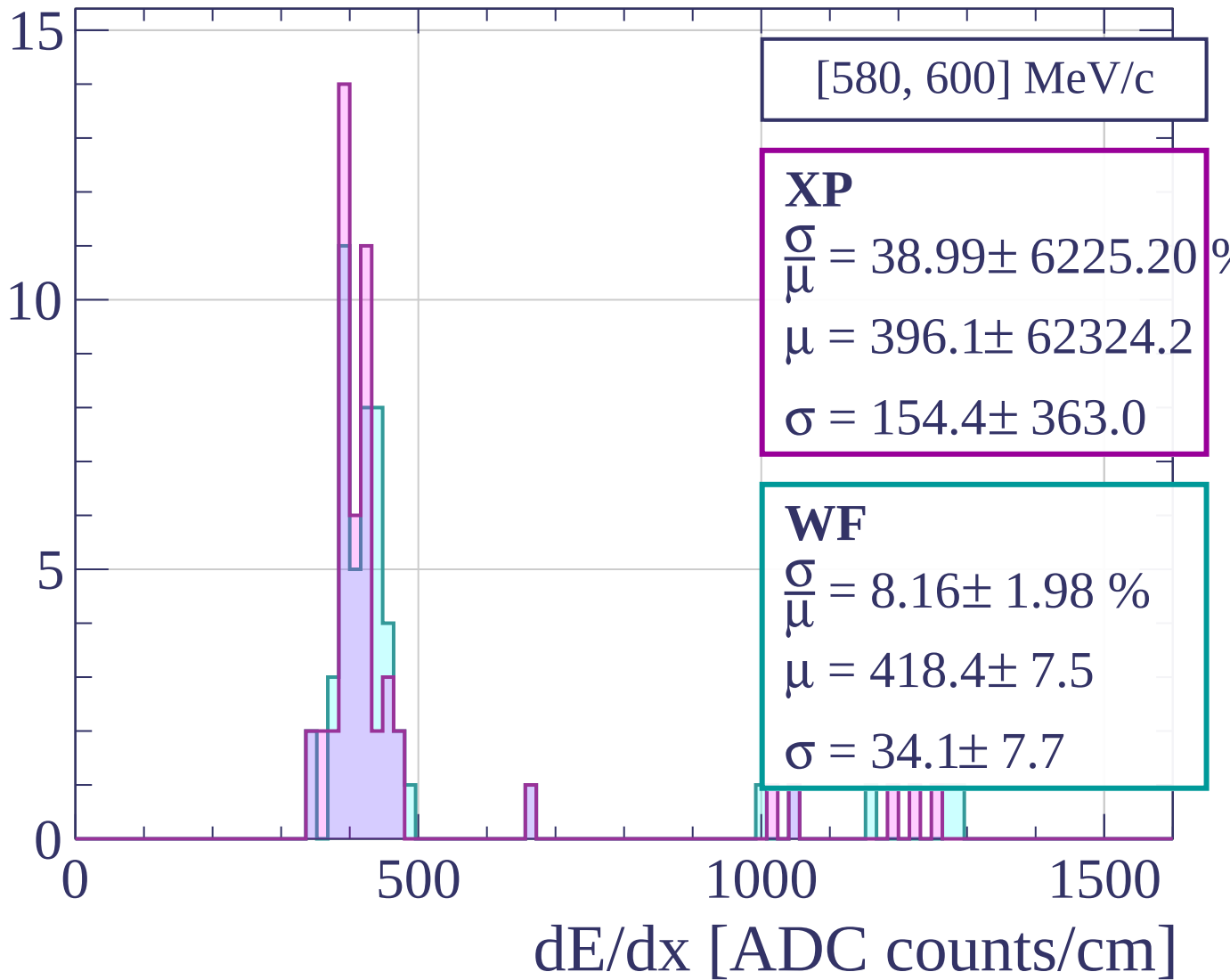
Count



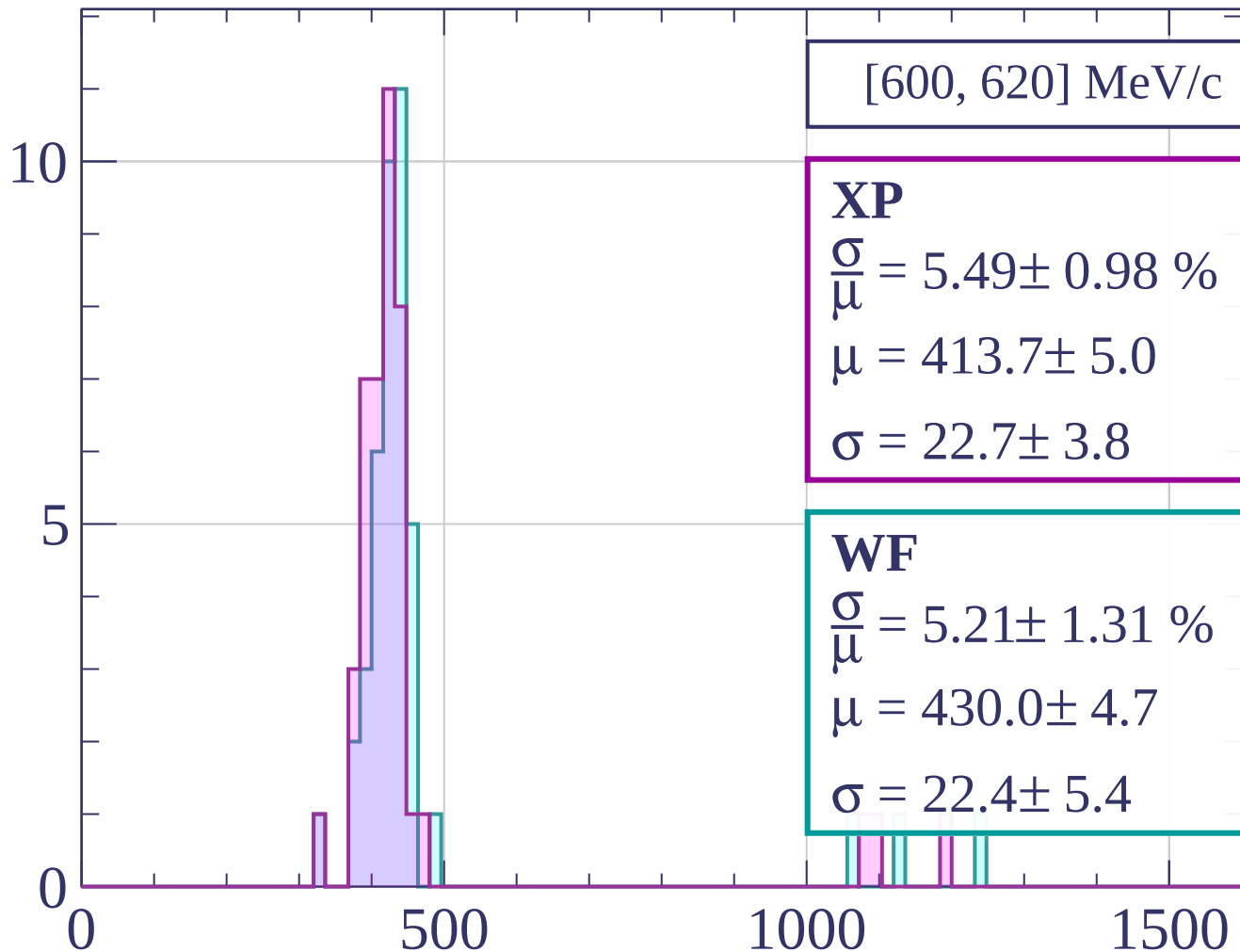
Count



Count

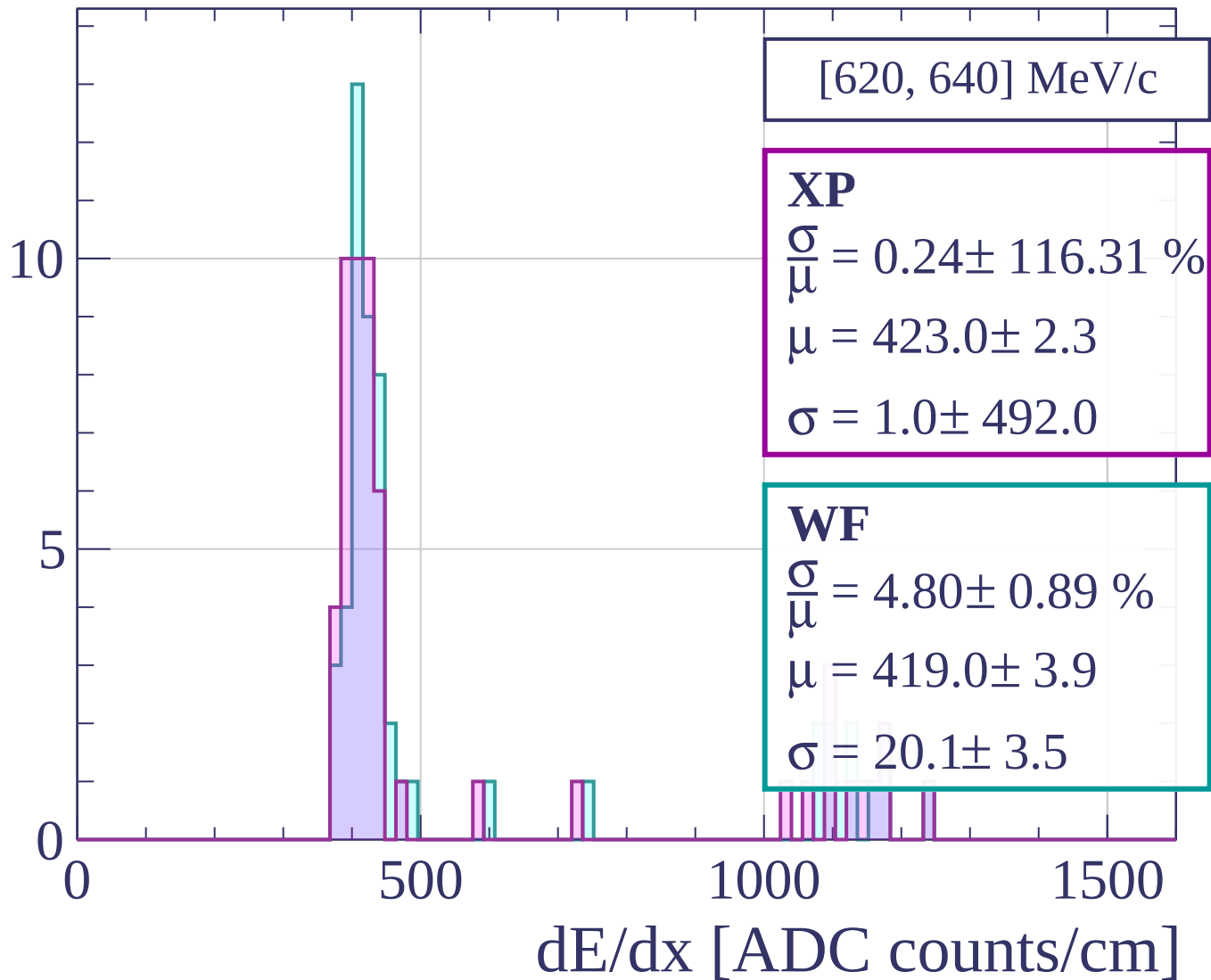


Count

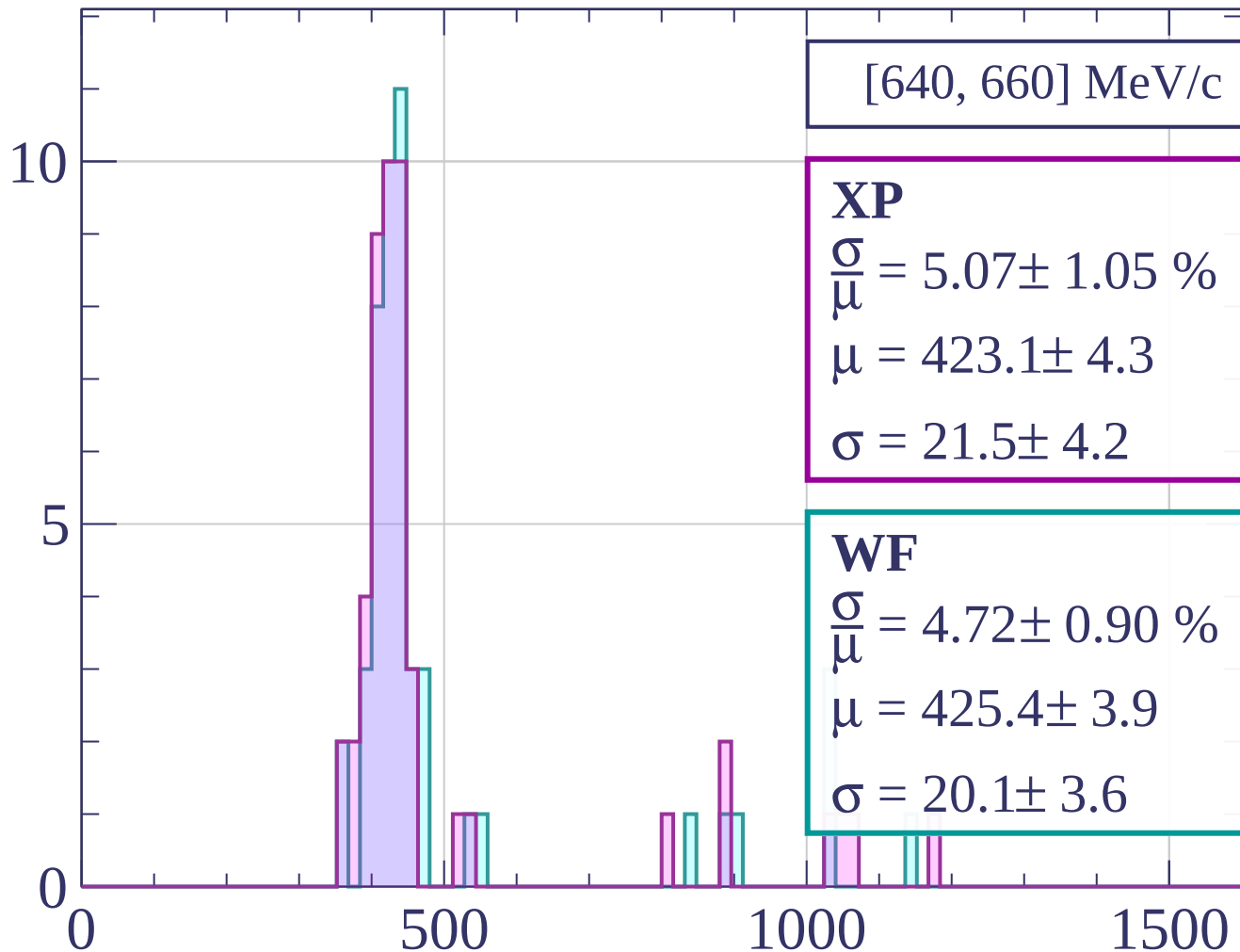


dE/dx [ADC counts/cm]

Count

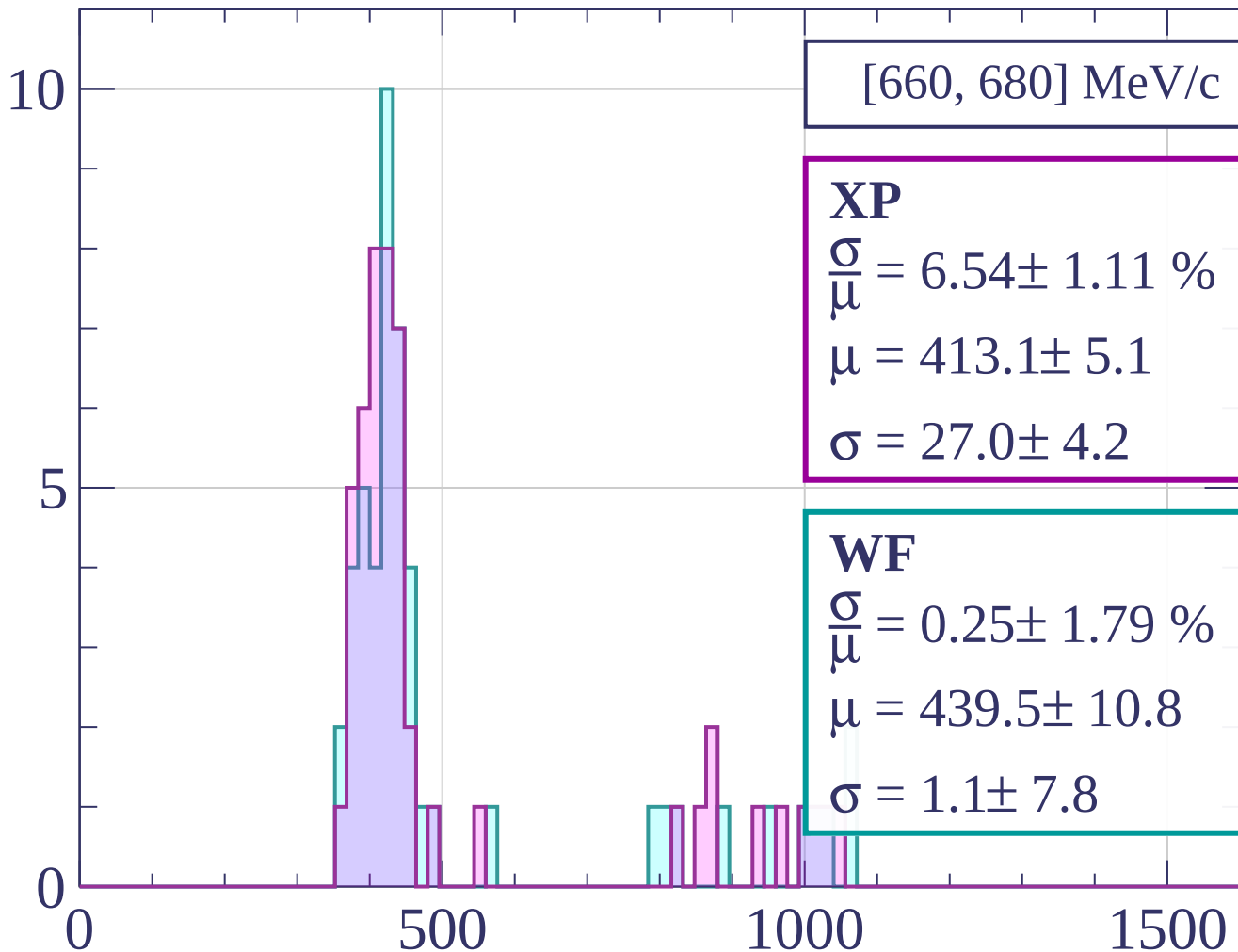


Count



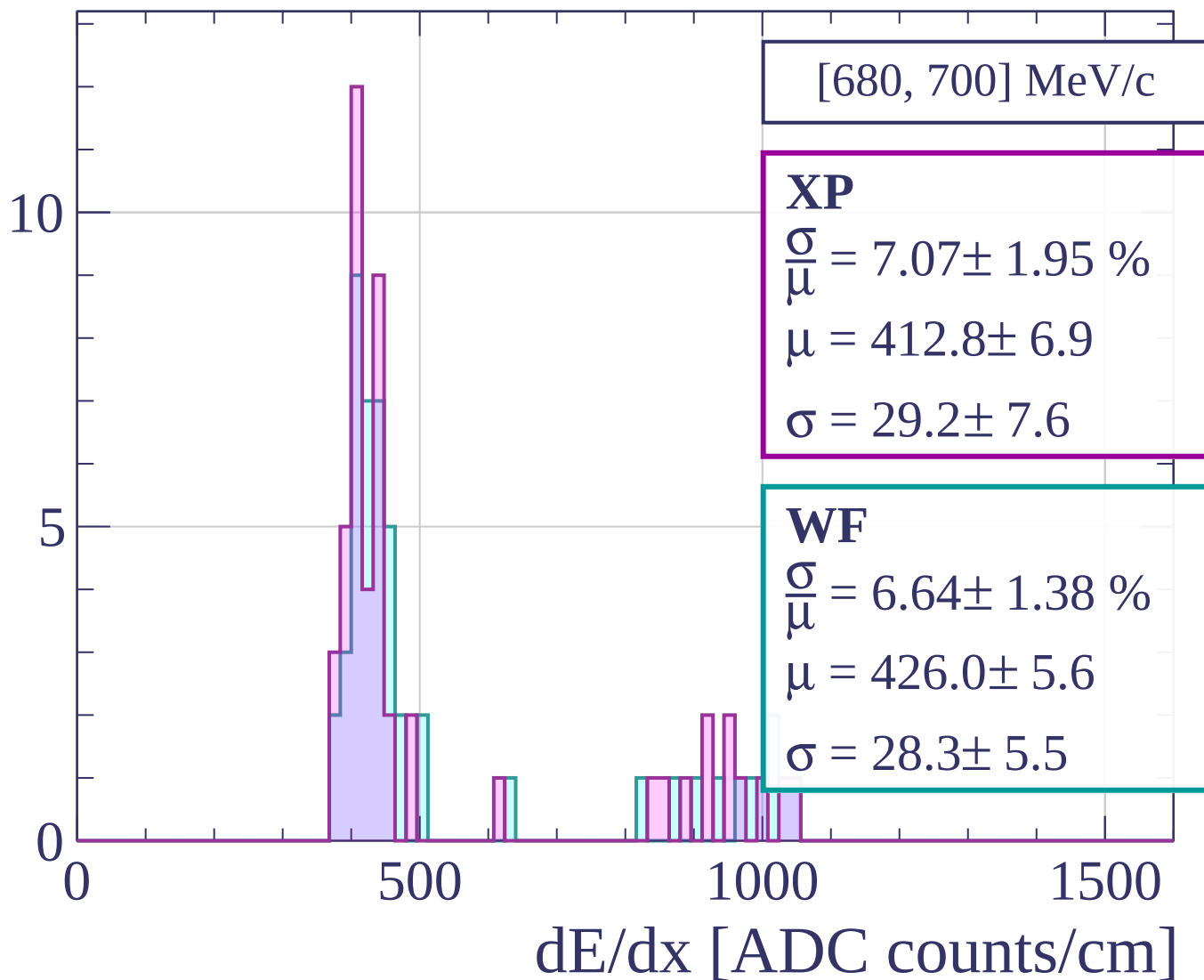
dE/dx [ADC counts/cm]

Count

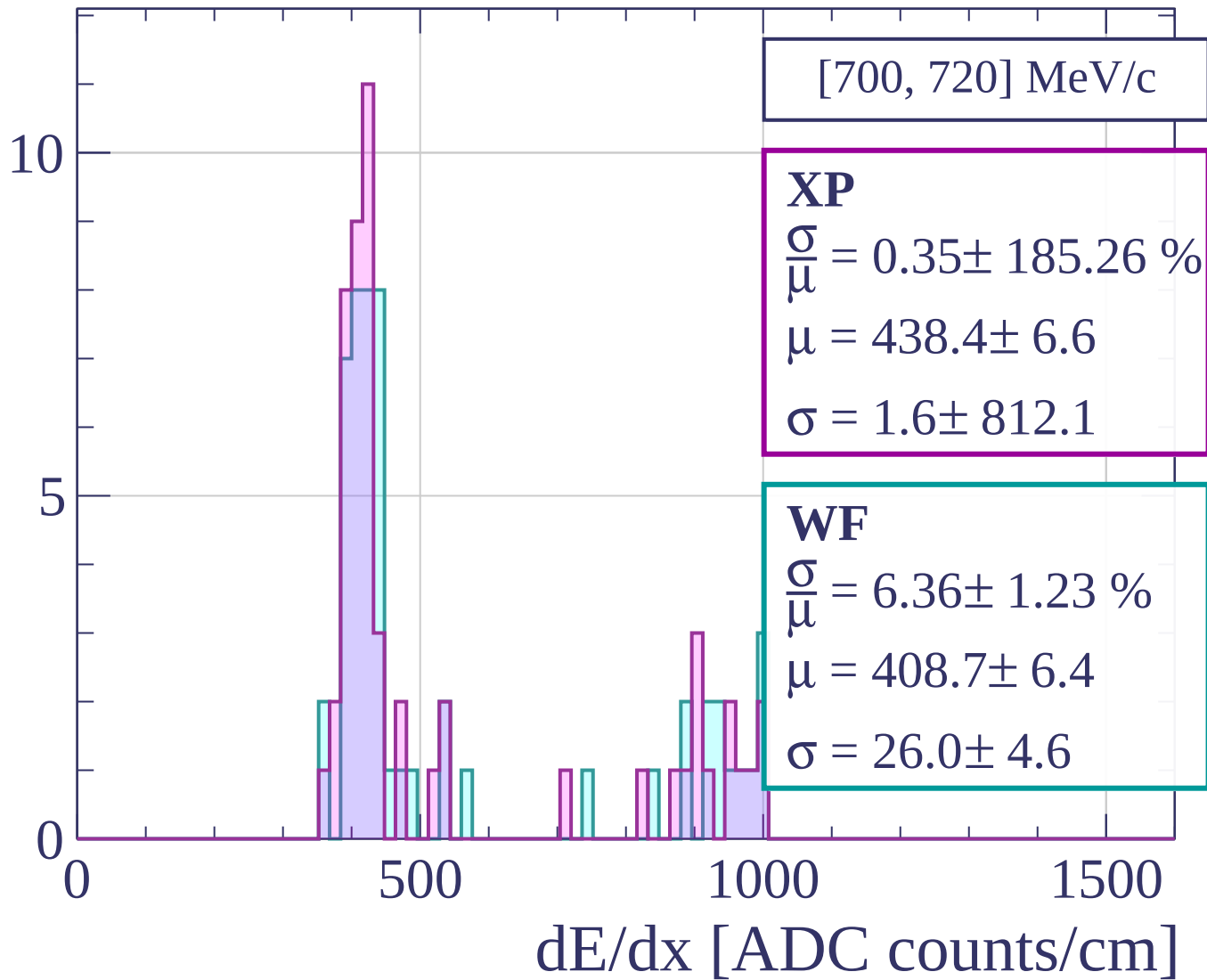


dE/dx [ADC counts/cm]

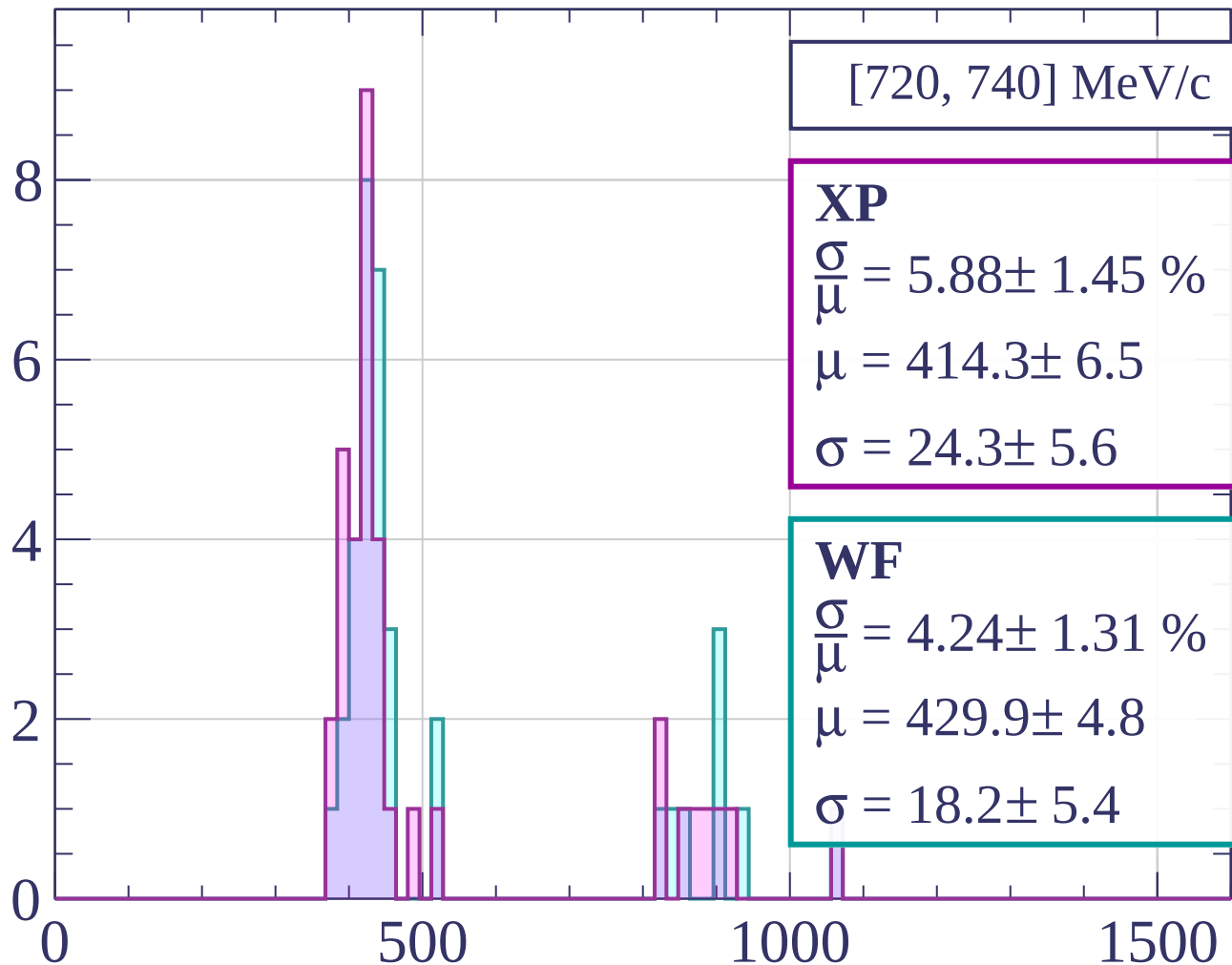
Count



Count

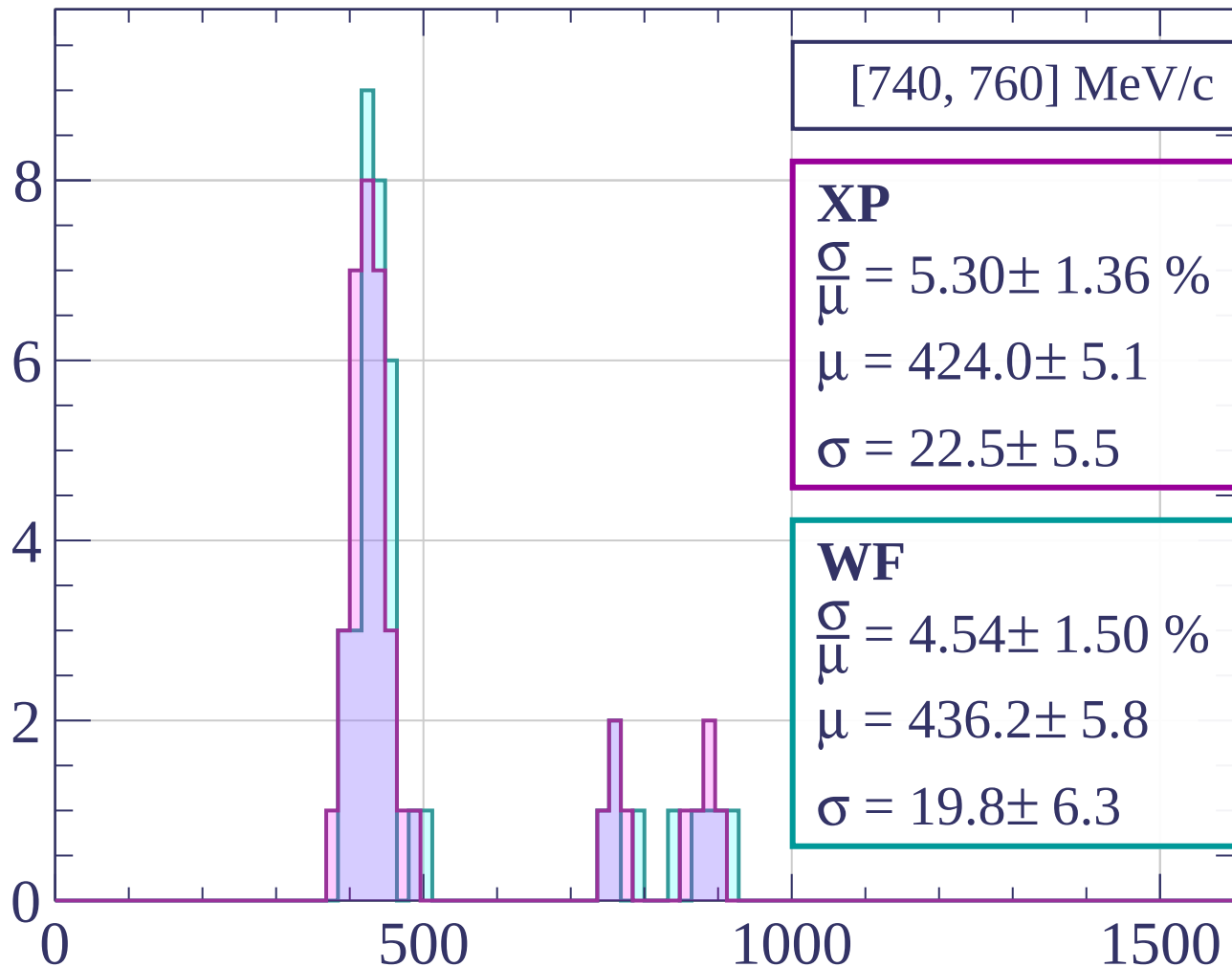


Count



dE/dx [ADC counts/cm]

Count



[740, 760] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.30 \pm 1.36 \%$$

$$\mu = 424.0 \pm 5.1$$

$$\sigma = 22.5 \pm 5.5$$

WF

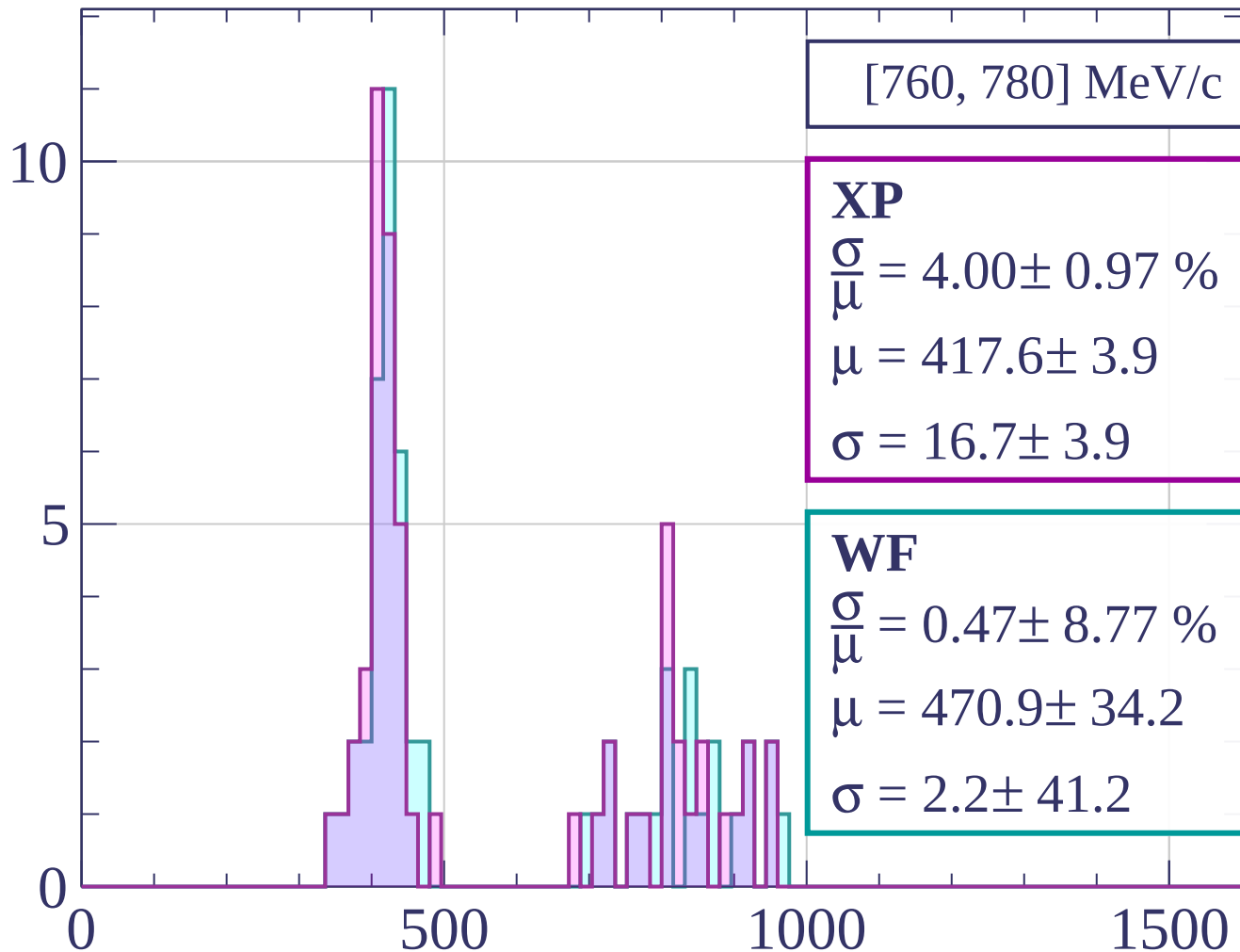
$$\frac{\sigma}{\mu} = 4.54 \pm 1.50 \%$$

$$\mu = 436.2 \pm 5.8$$

$$\sigma = 19.8 \pm 6.3$$

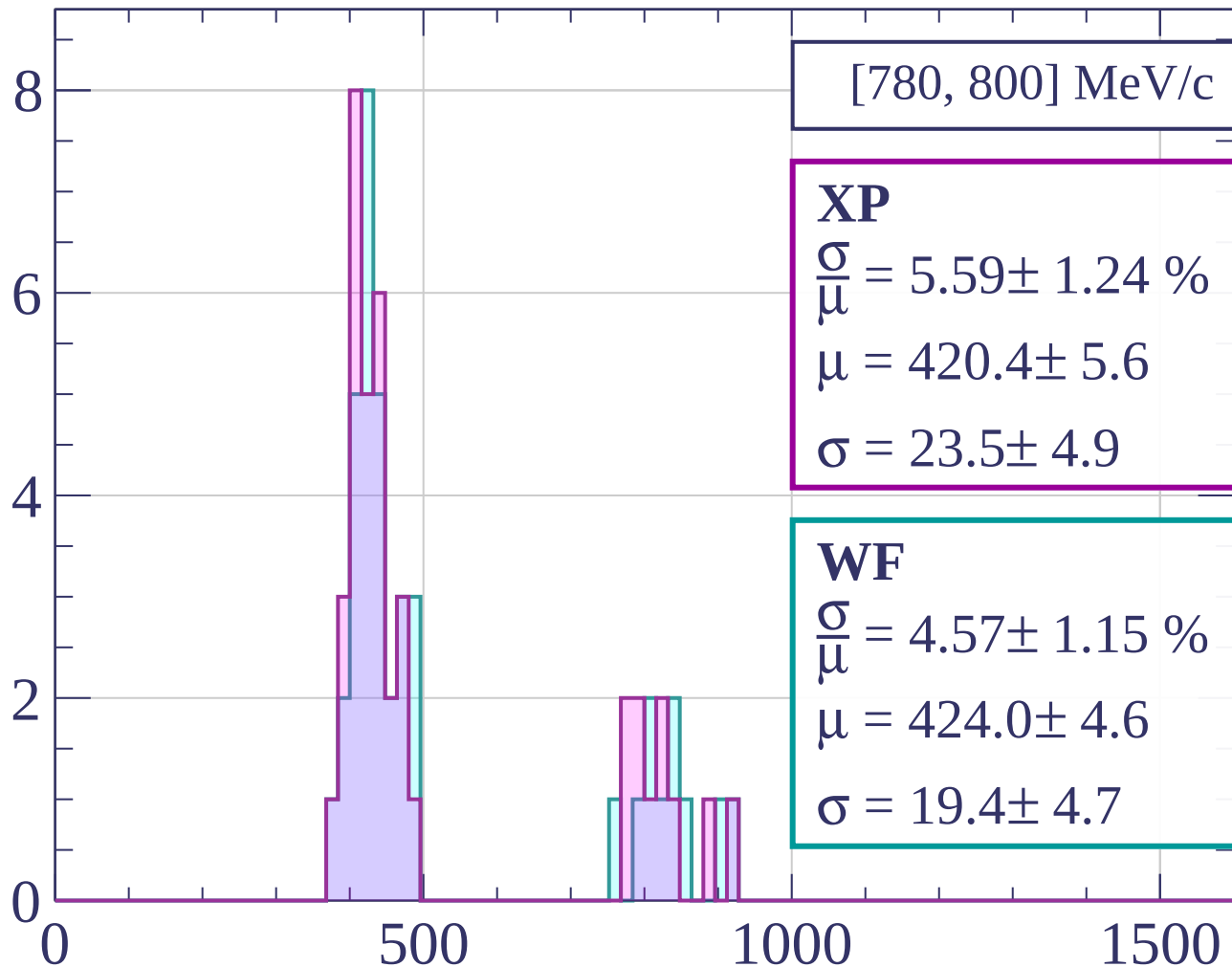
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[780, 800] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.59 \pm 1.24 \%$$

$$\mu = 420.4 \pm 5.6$$

$$\sigma = 23.5 \pm 4.9$$

WF

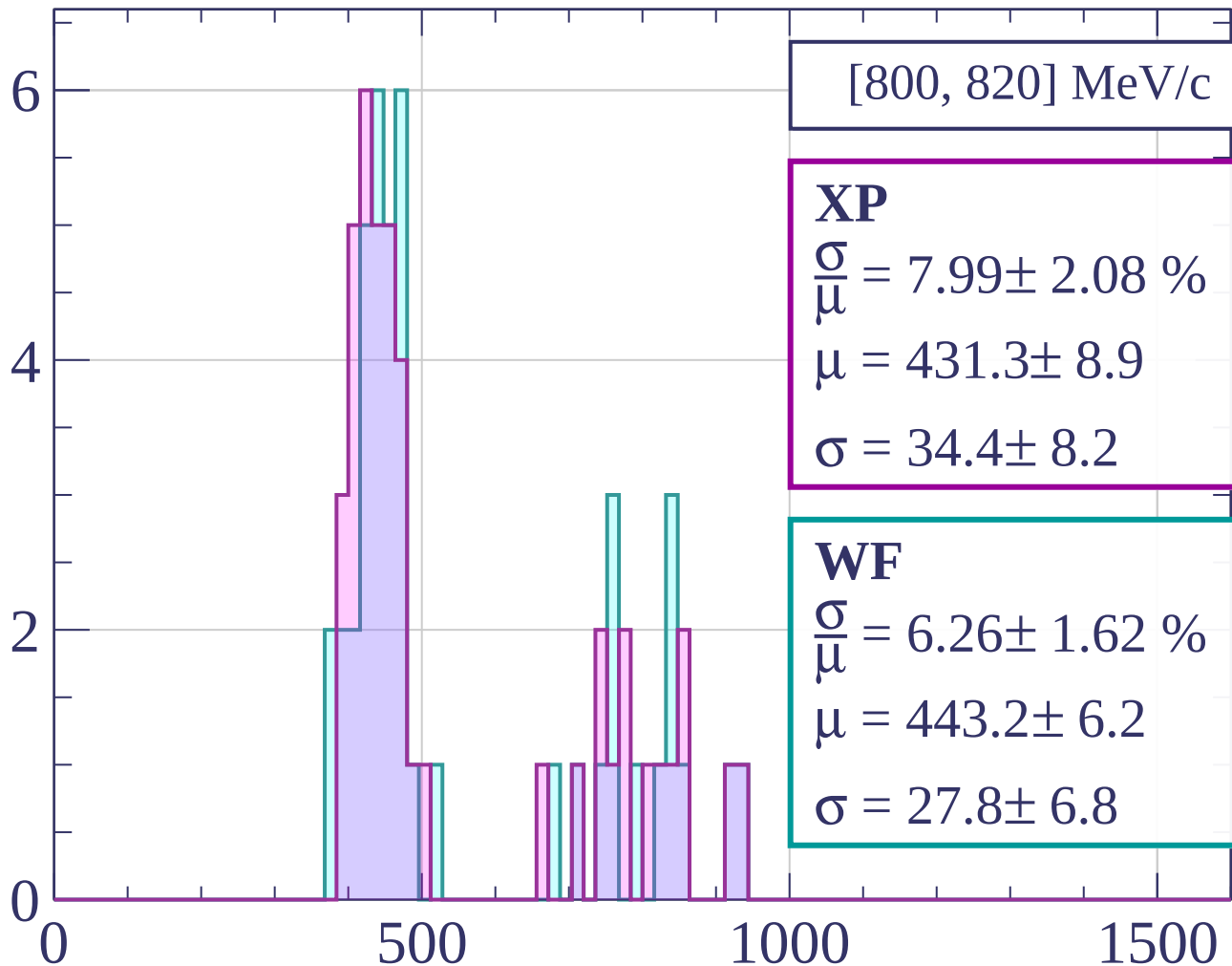
$$\frac{\sigma}{\mu} = 4.57 \pm 1.15 \%$$

$$\mu = 424.0 \pm 4.6$$

$$\sigma = 19.4 \pm 4.7$$

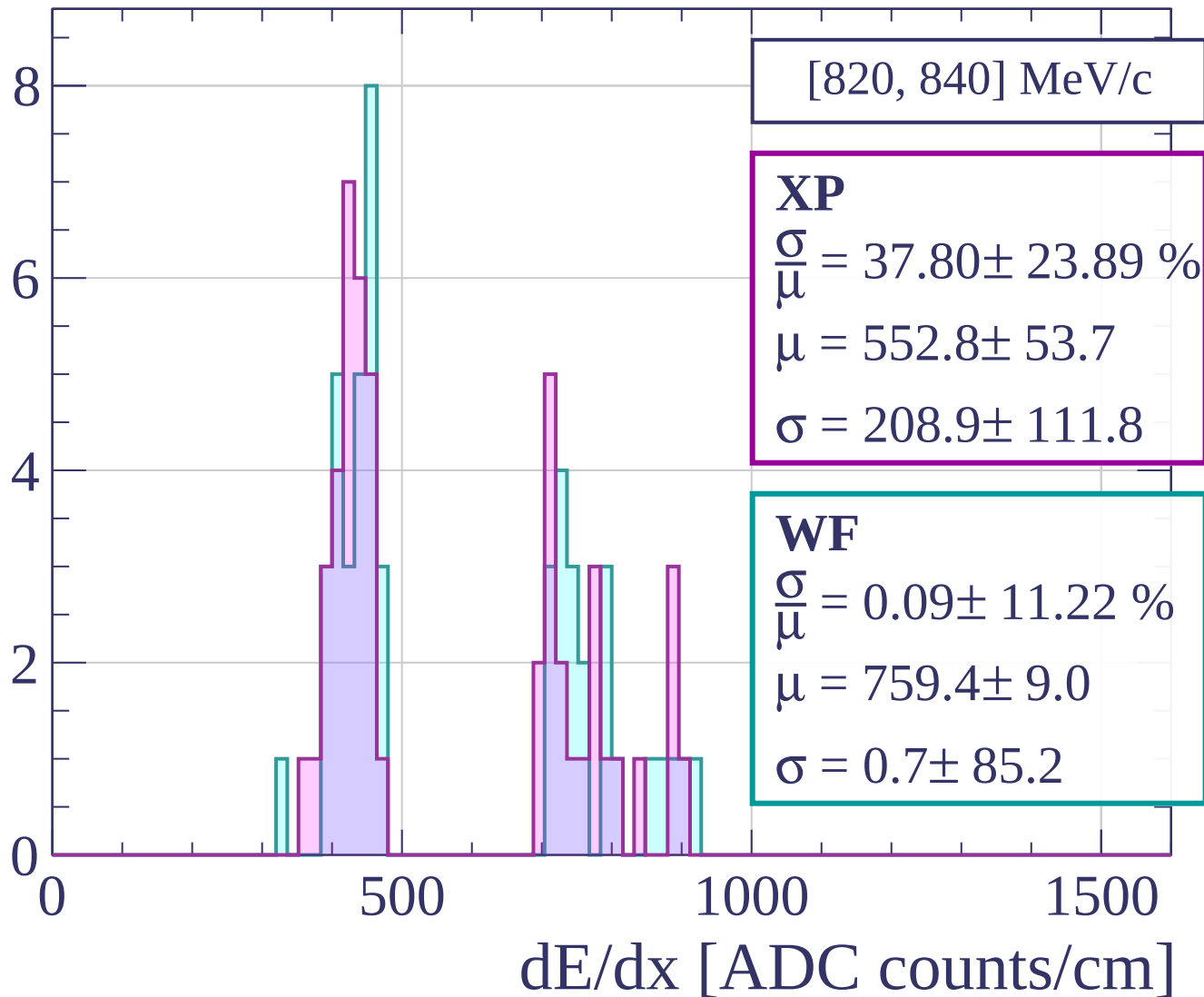
dE/dx [ADC counts/cm]

Count

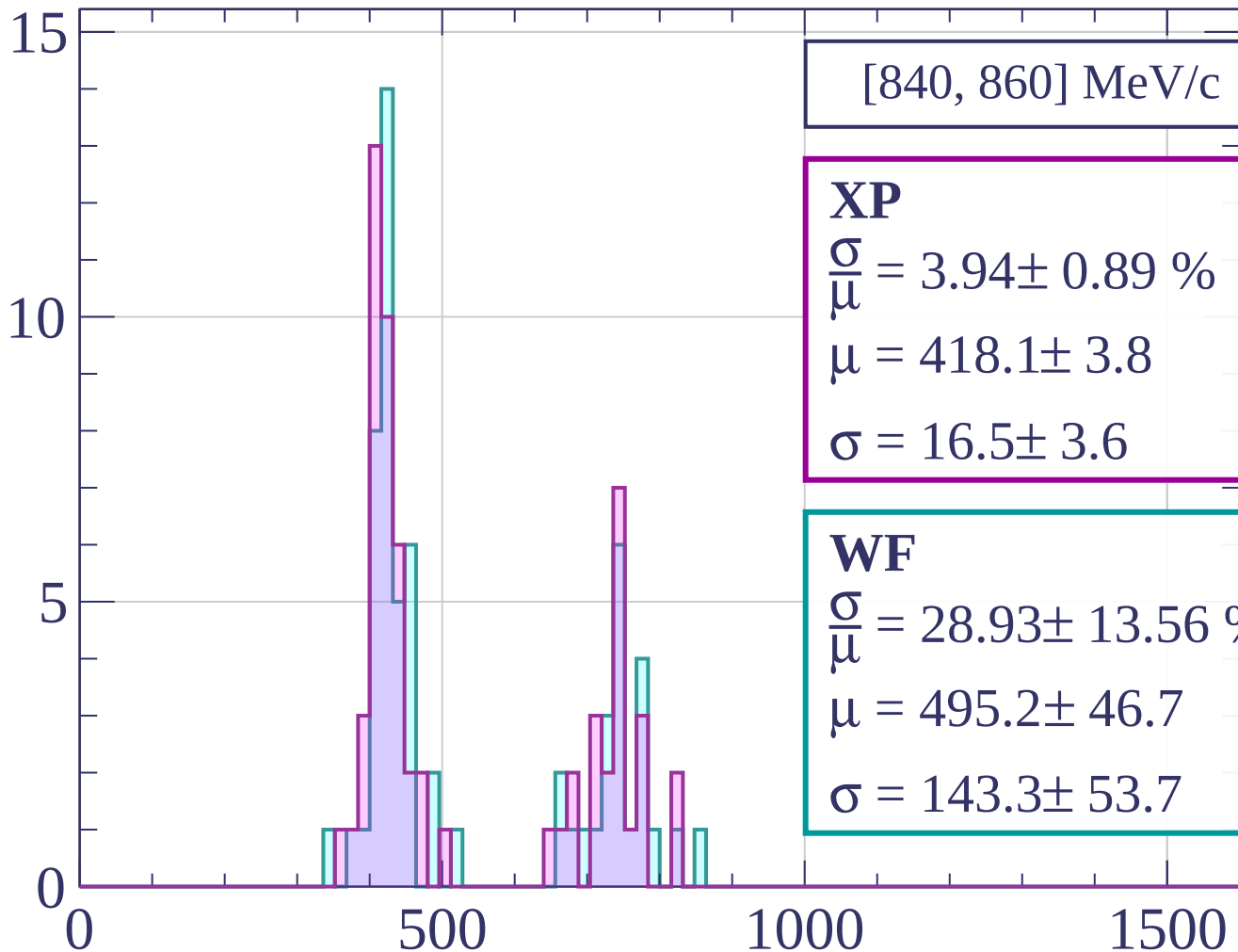


dE/dx [ADC counts/cm]

Count

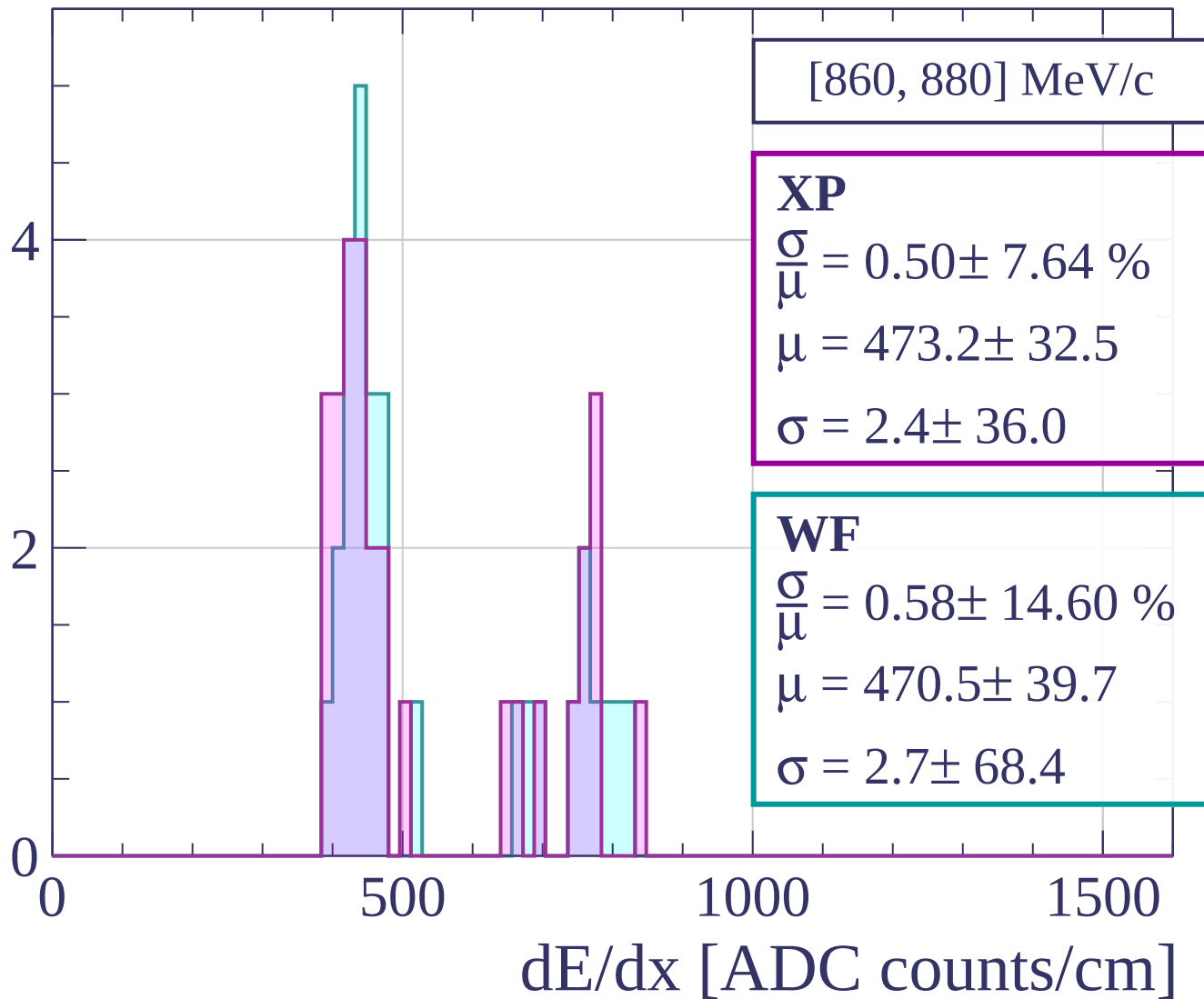


Count

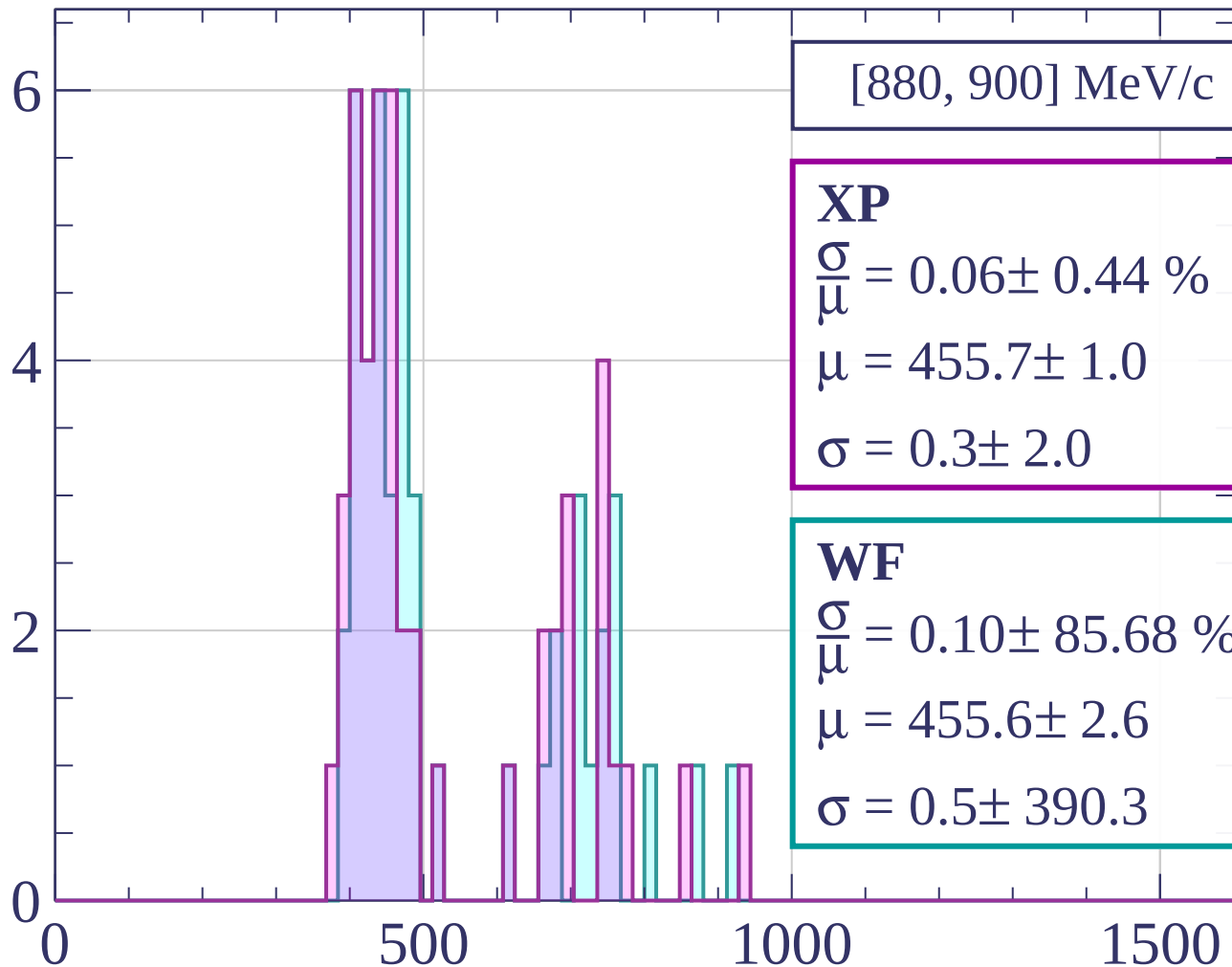


dE/dx [ADC counts/cm]

Count

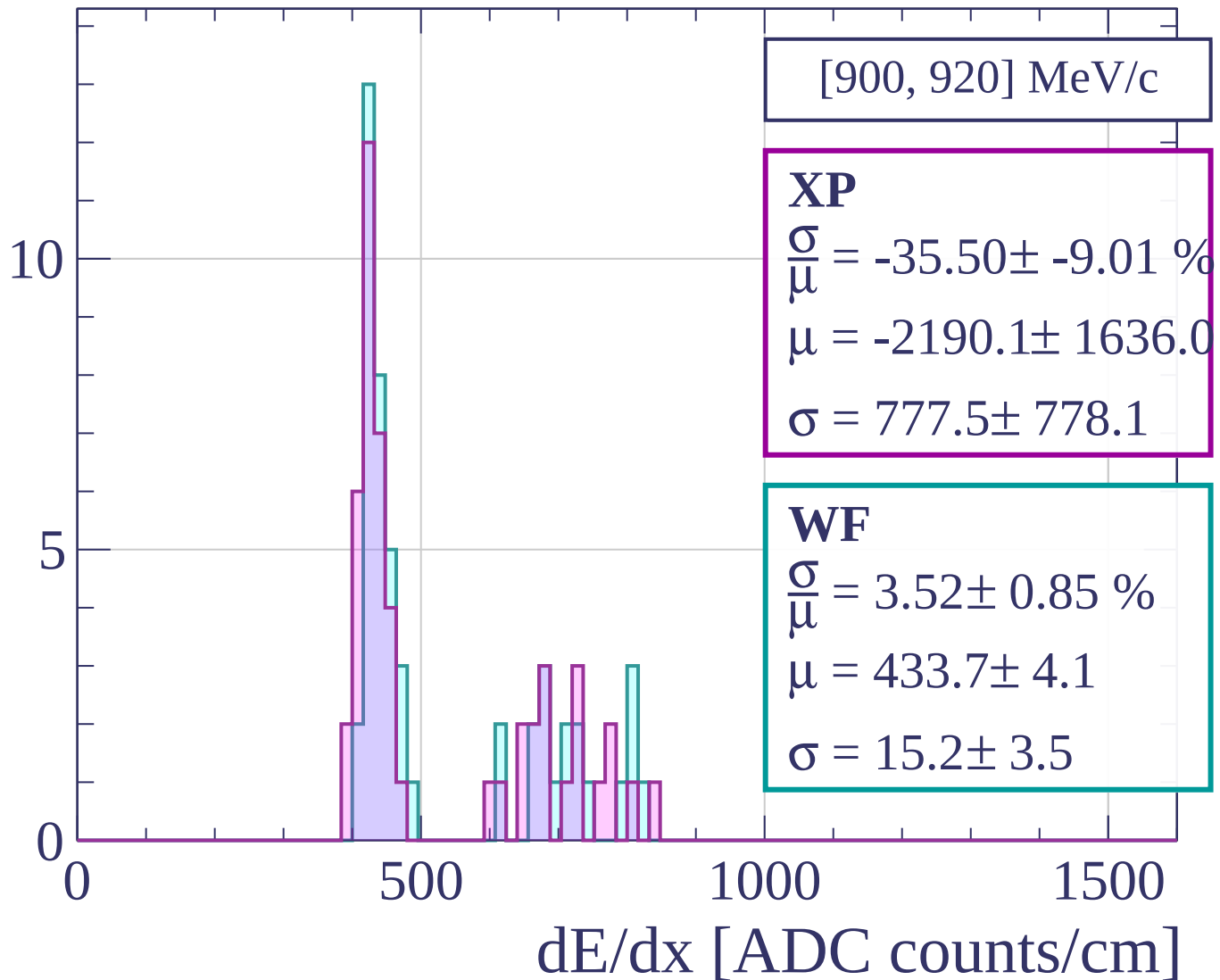


Count

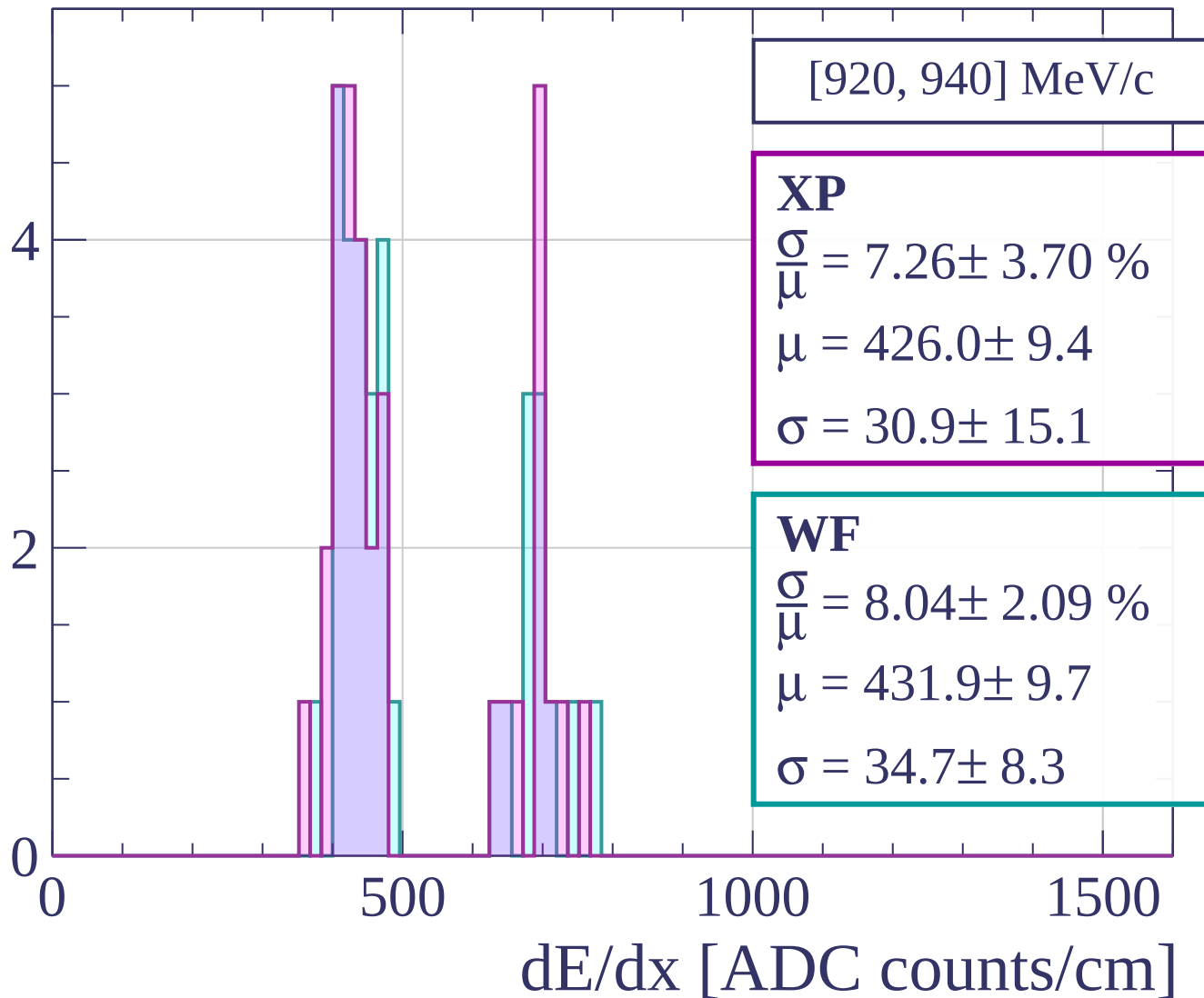


dE/dx [ADC counts/cm]

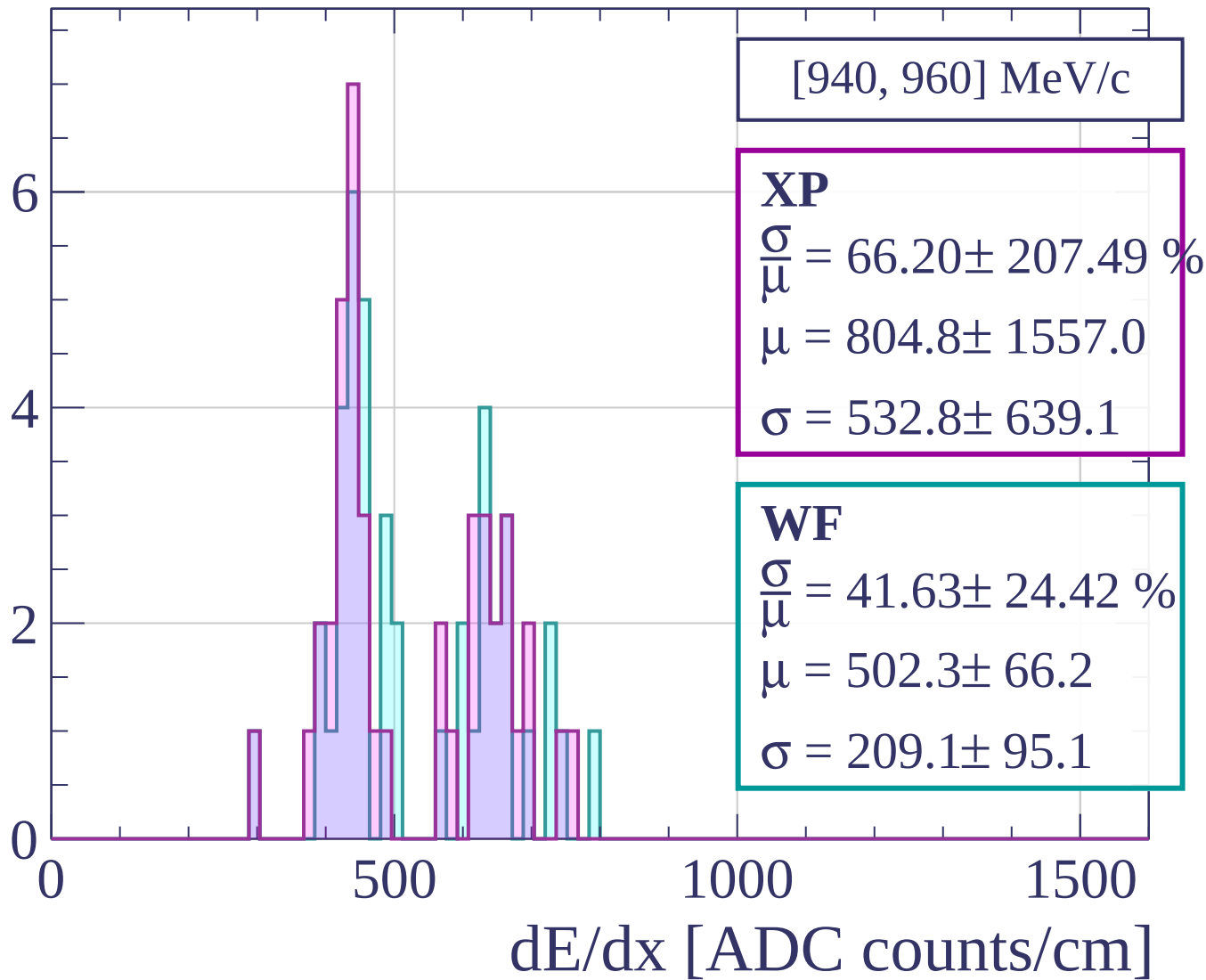
Count



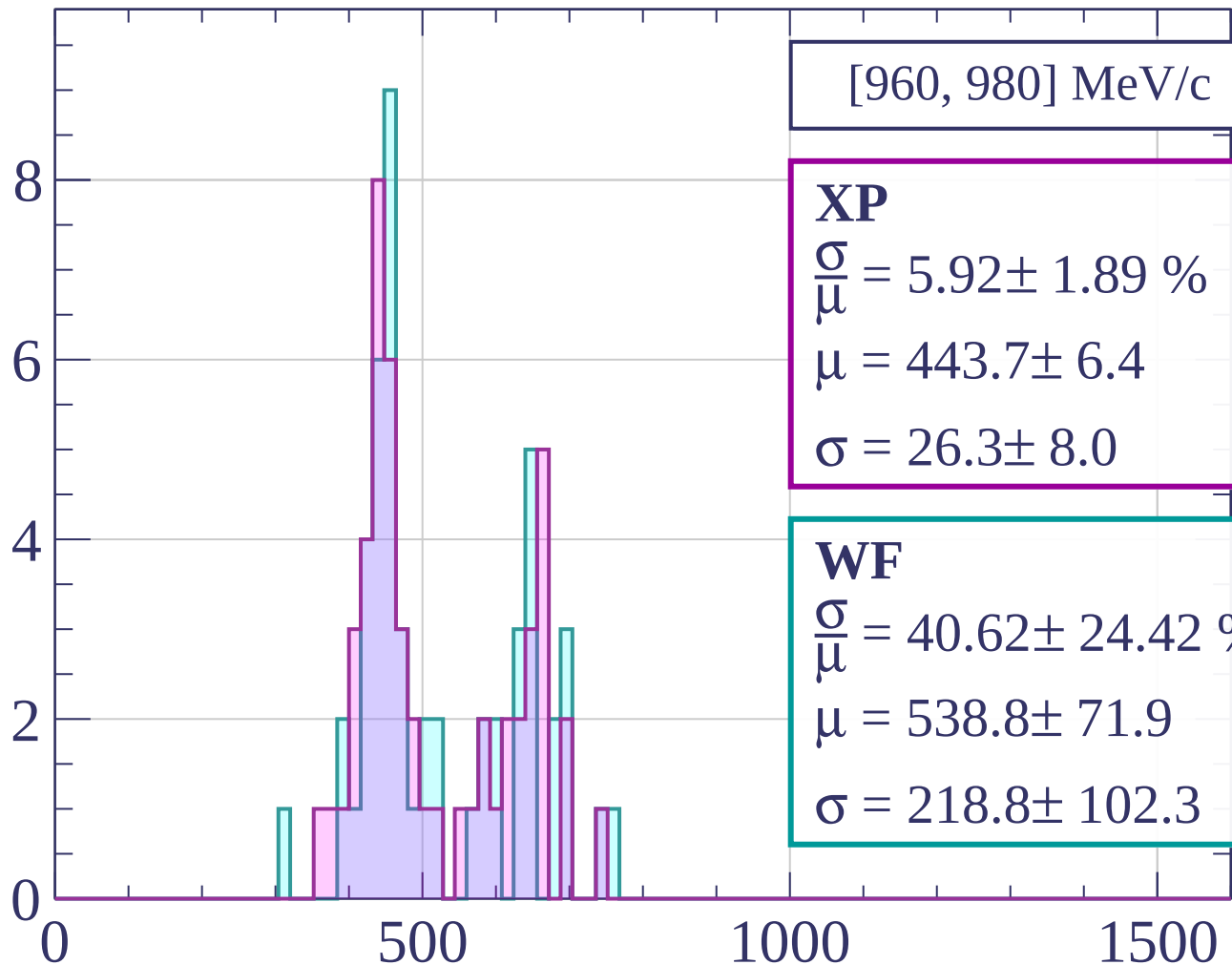
Count



Count

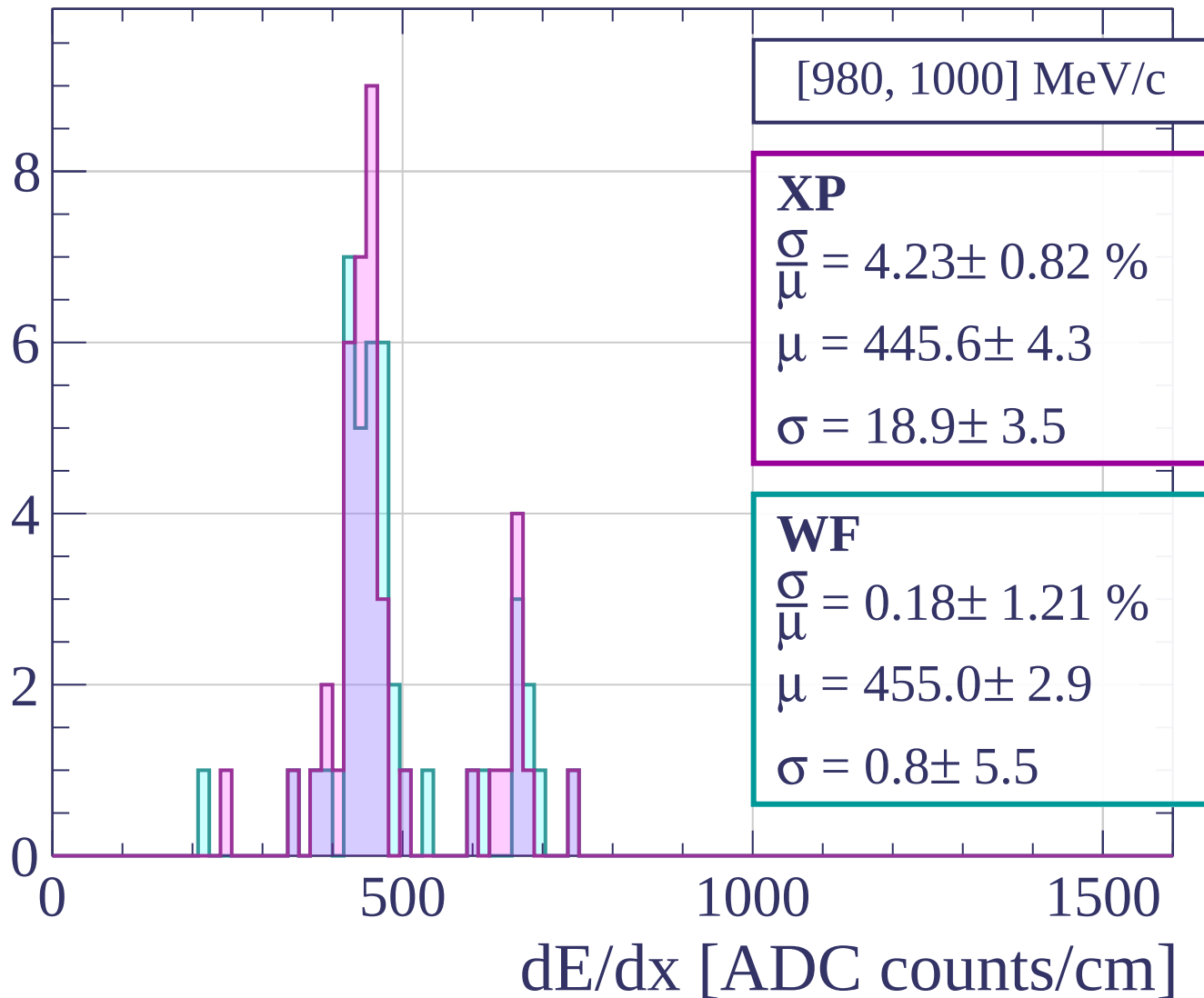


Count

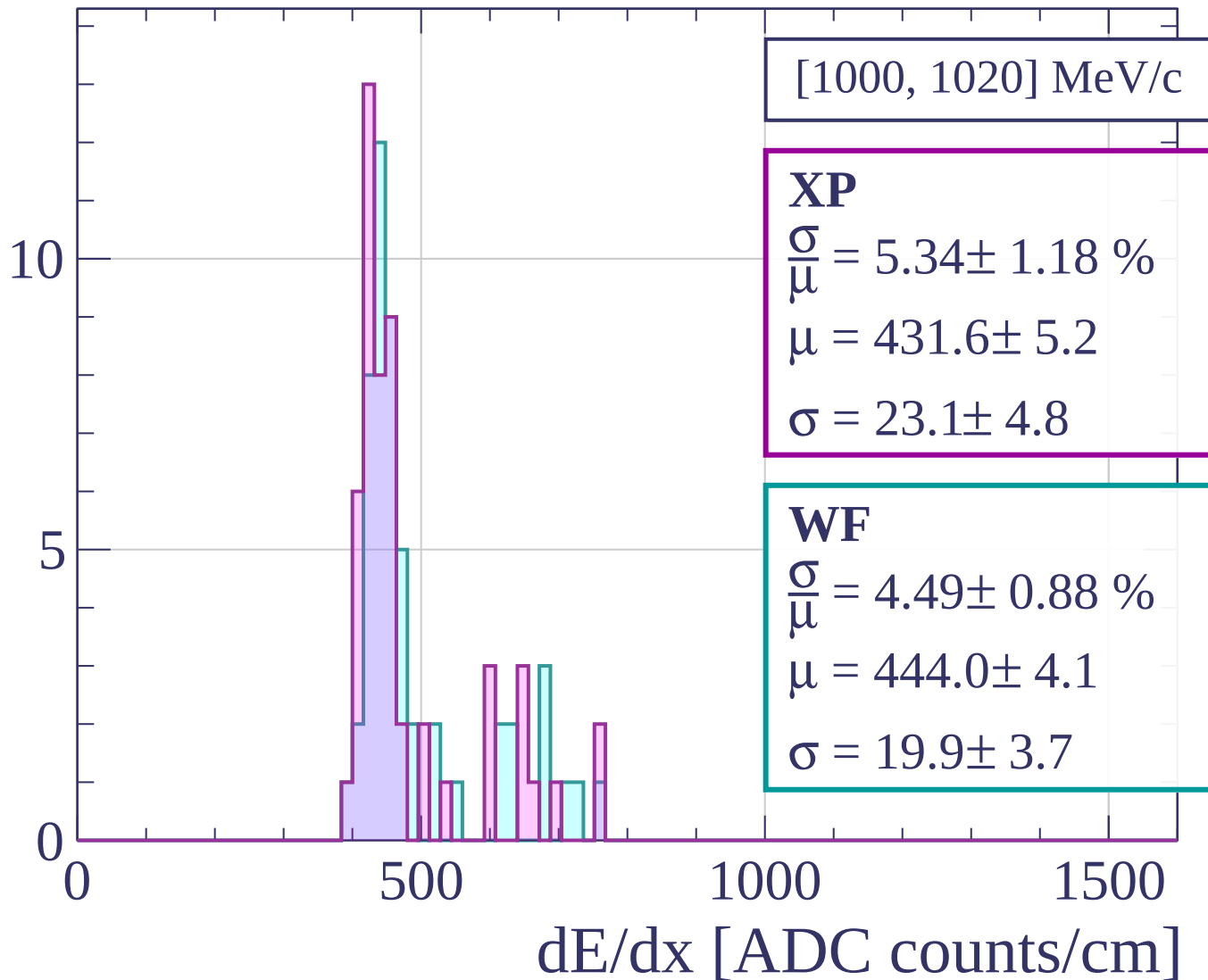


dE/dx [ADC counts/cm]

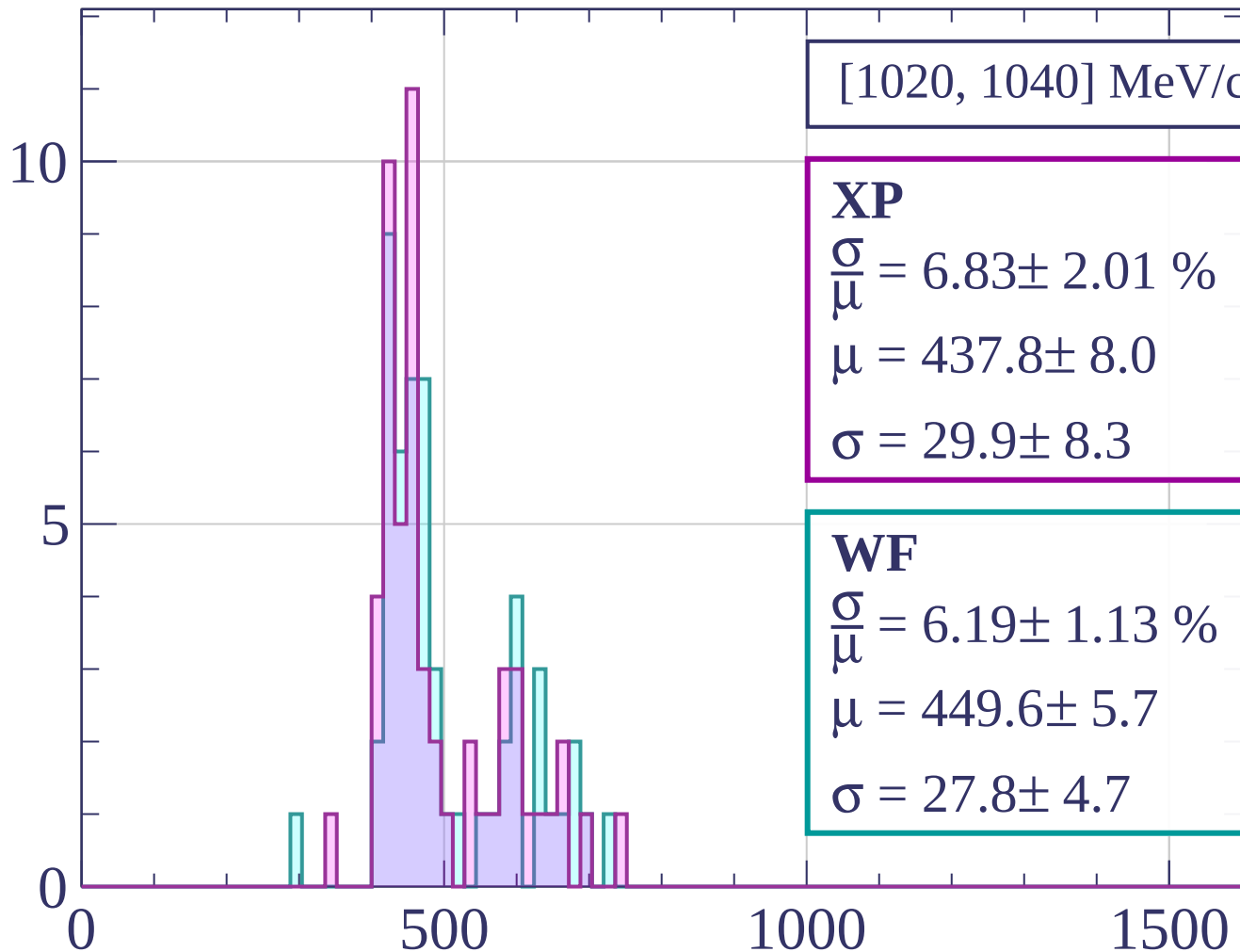
Count



Count

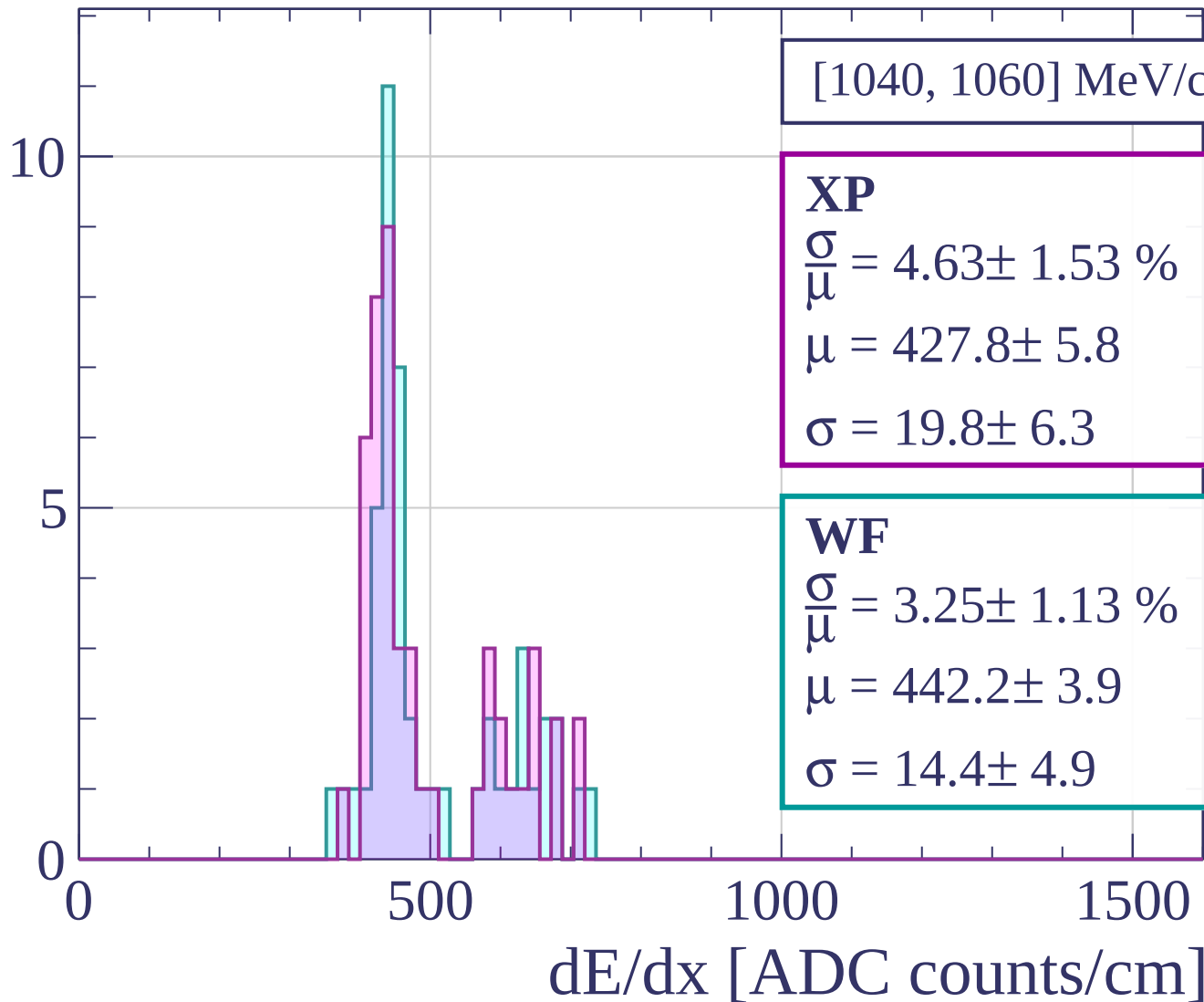


Count

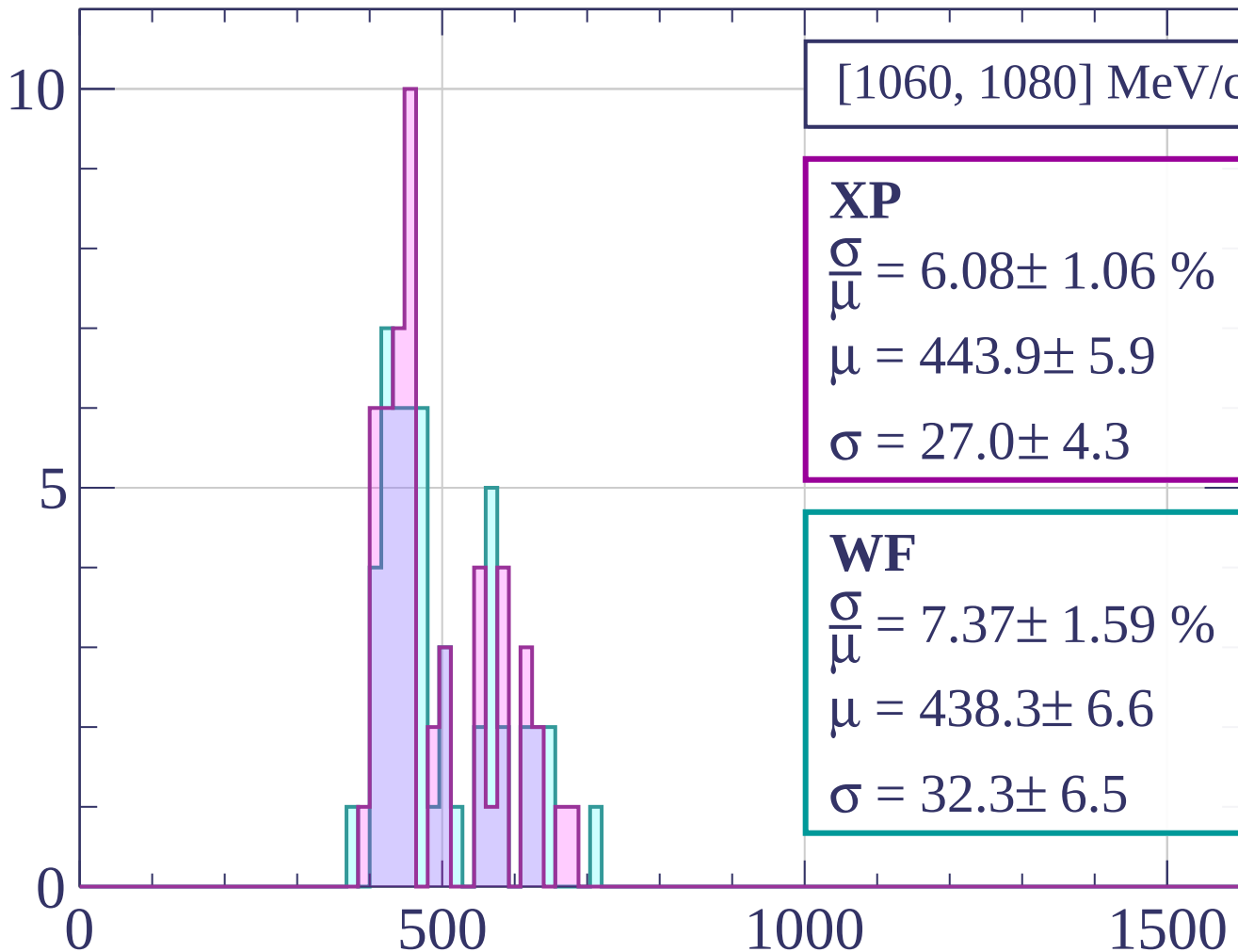


dE/dx [ADC counts/cm]

Count

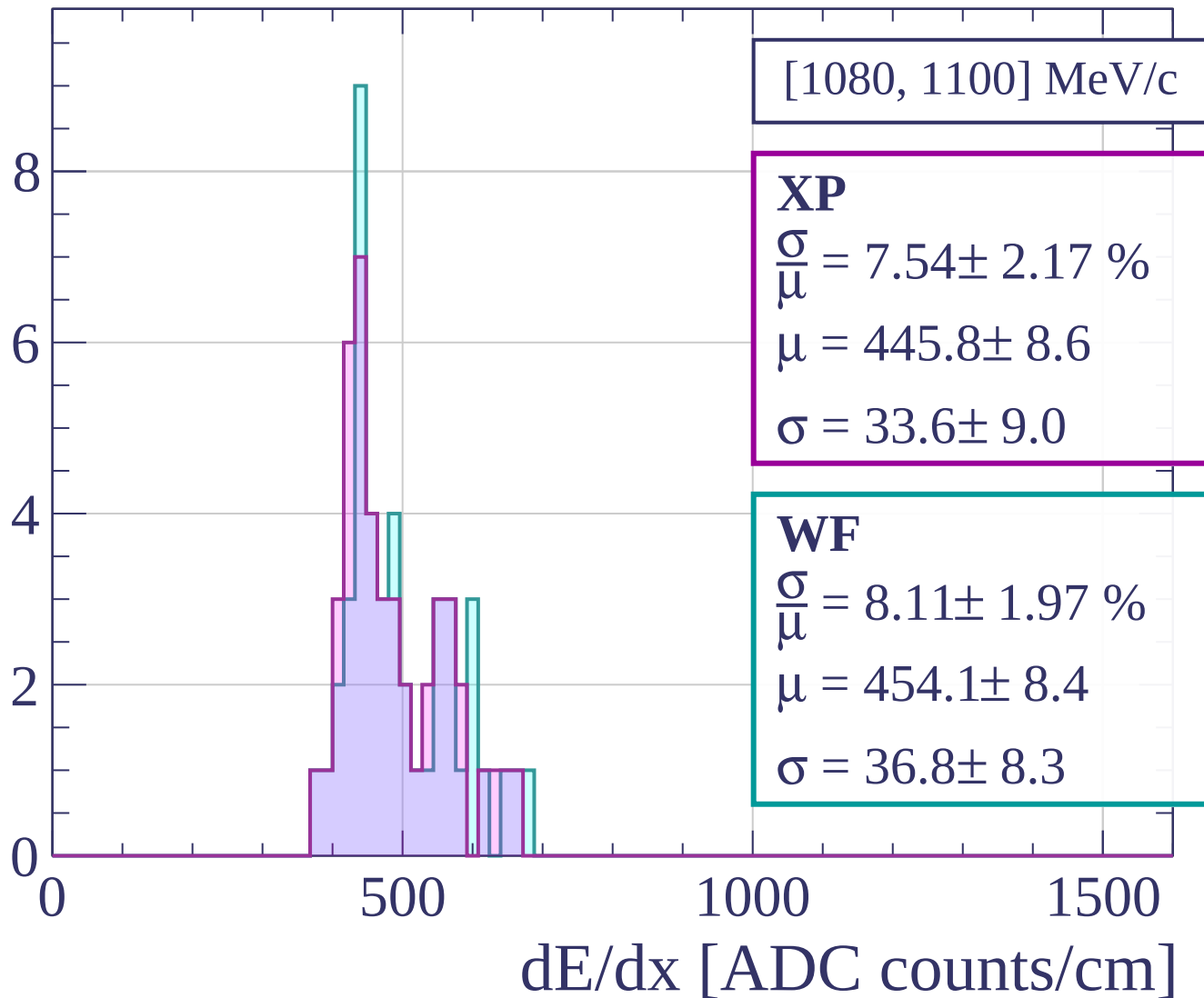


Count

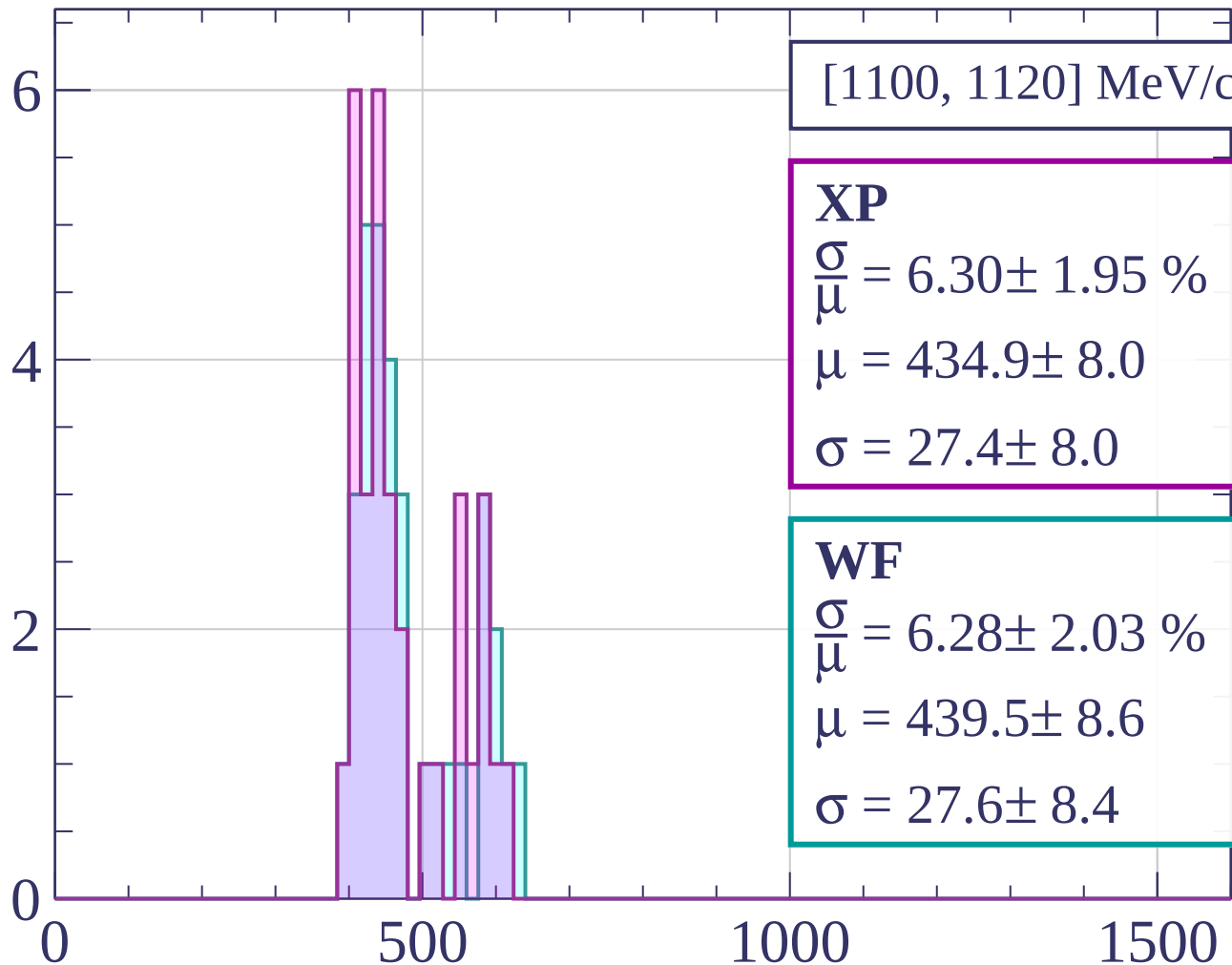


dE/dx [ADC counts/cm]

Count

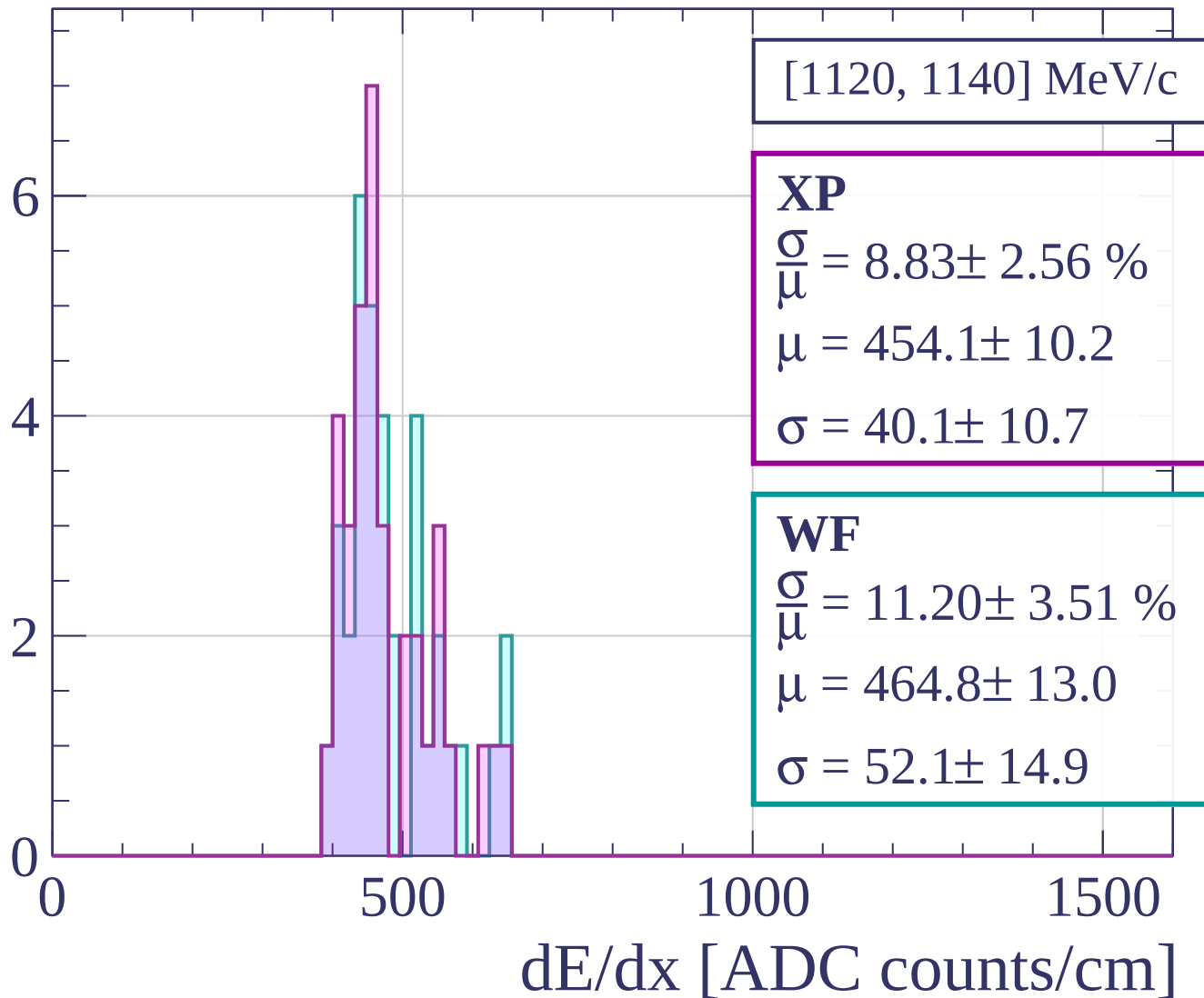


Count

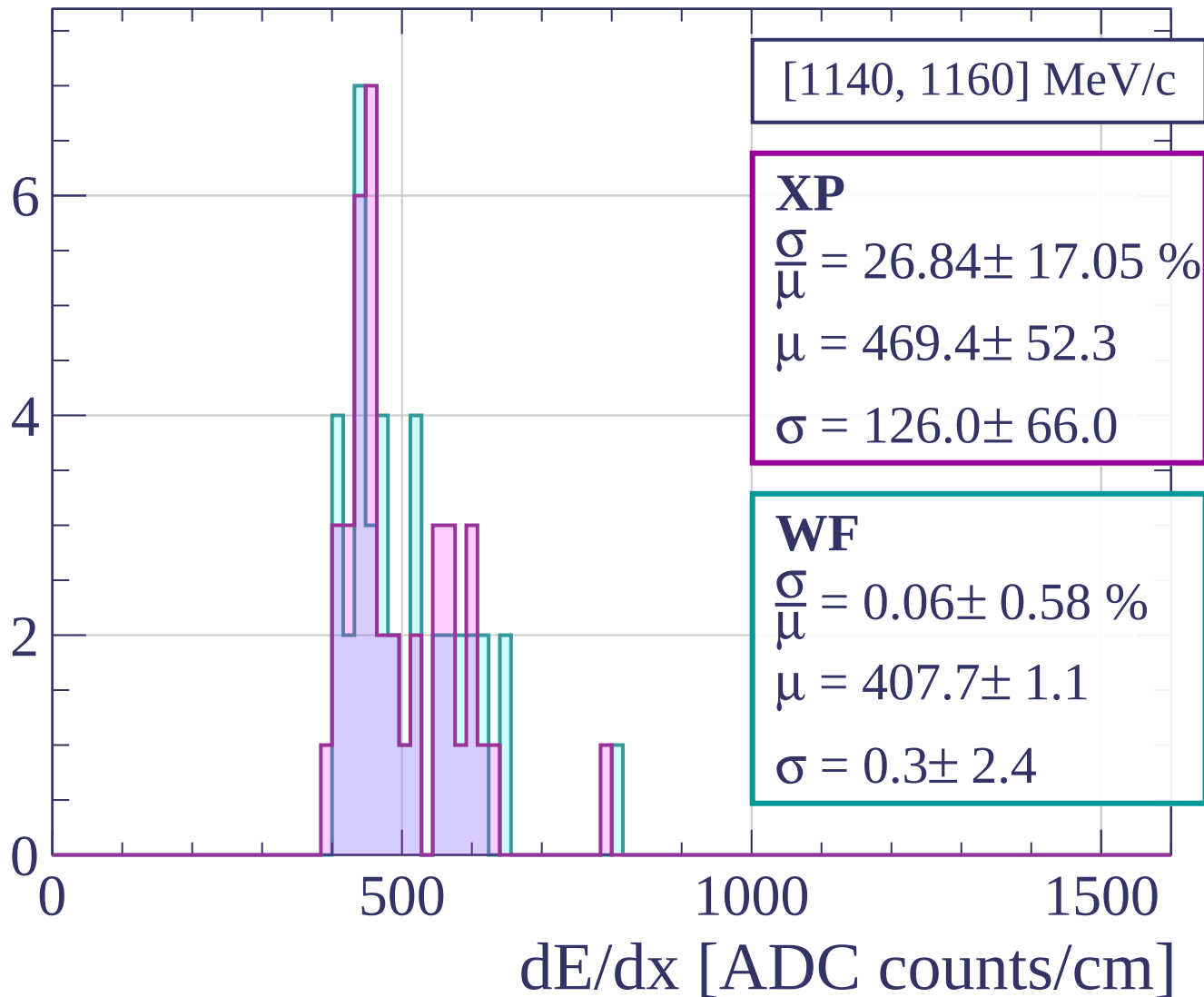


dE/dx [ADC counts/cm]

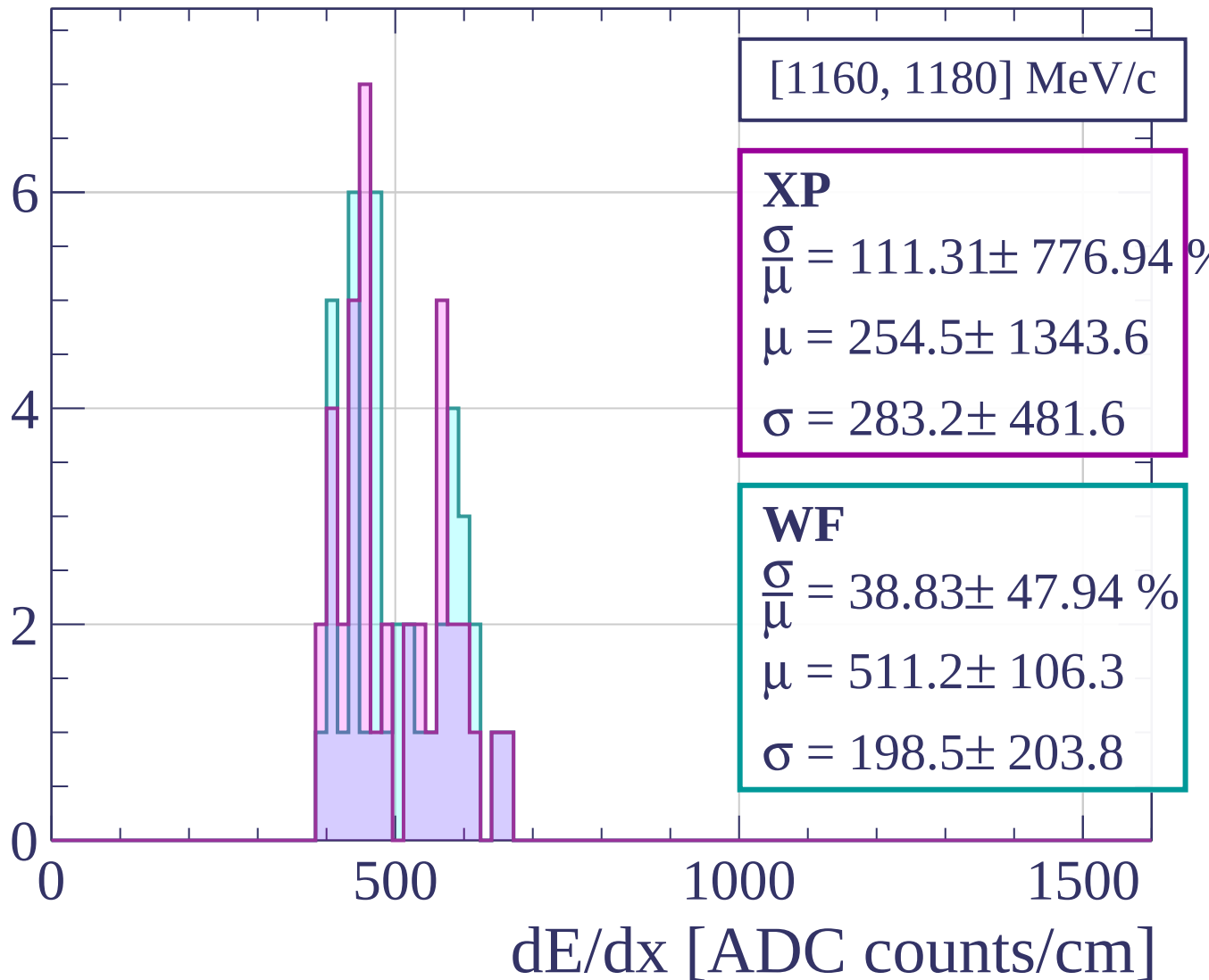
Count



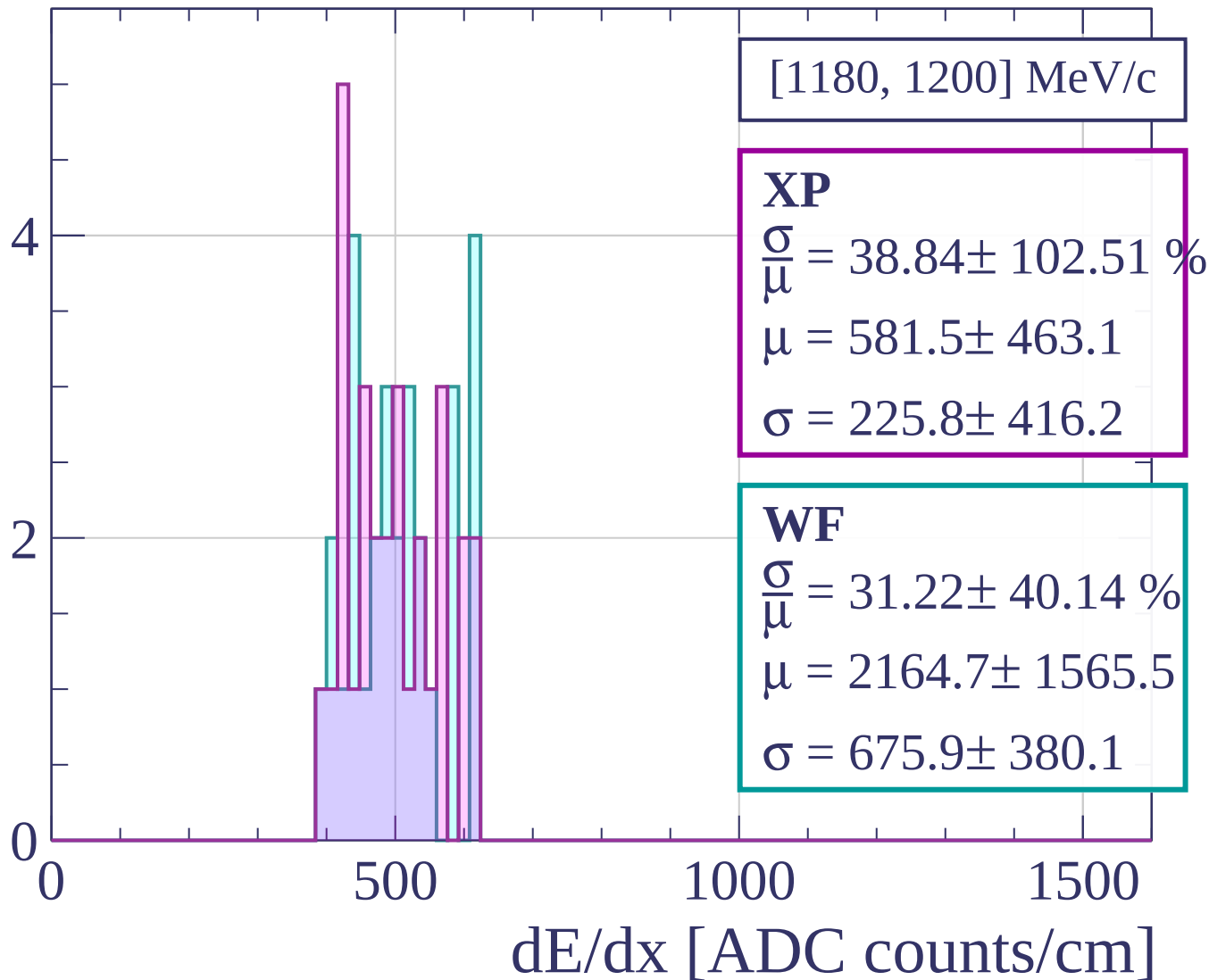
Count



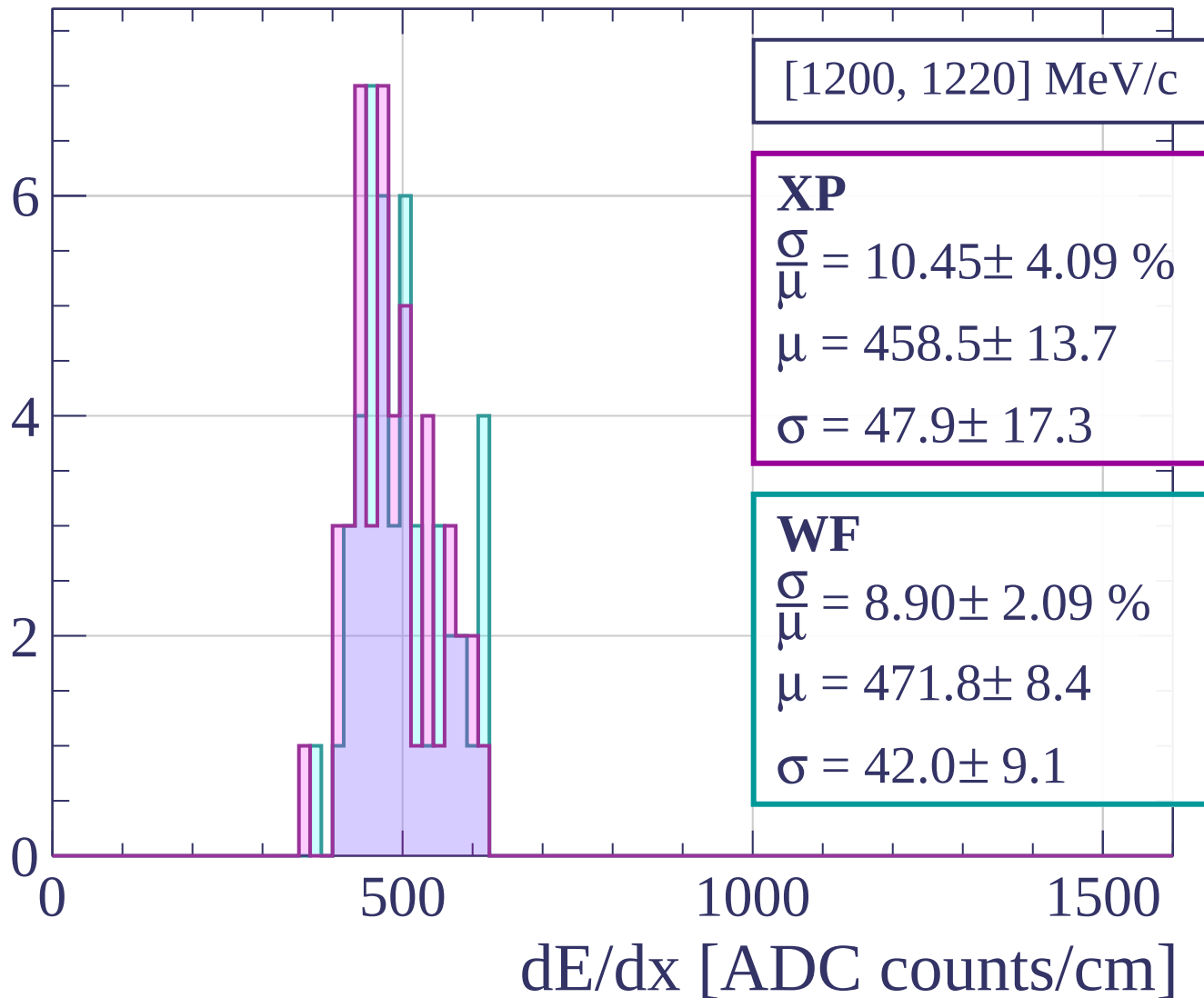
Count



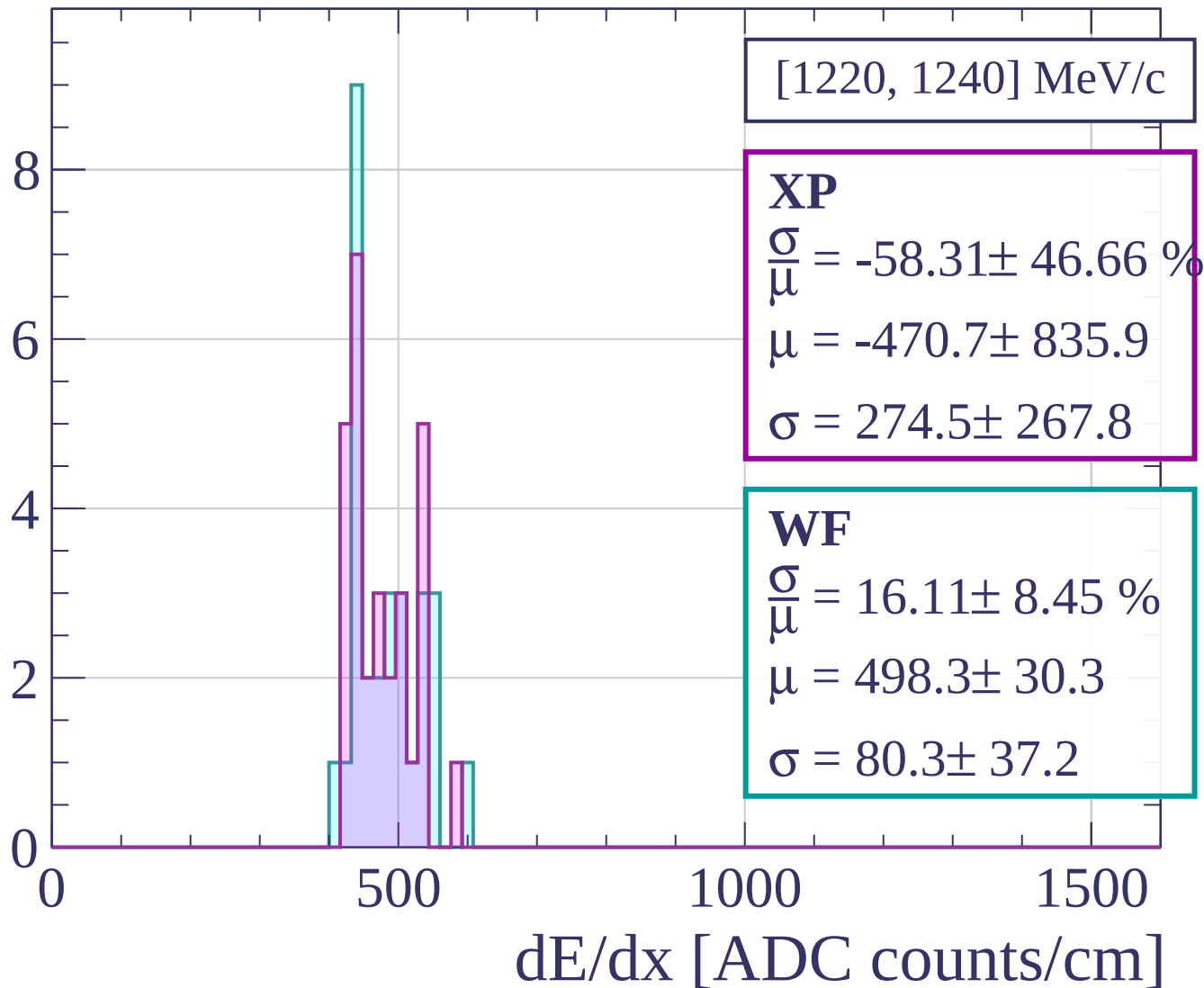
Count



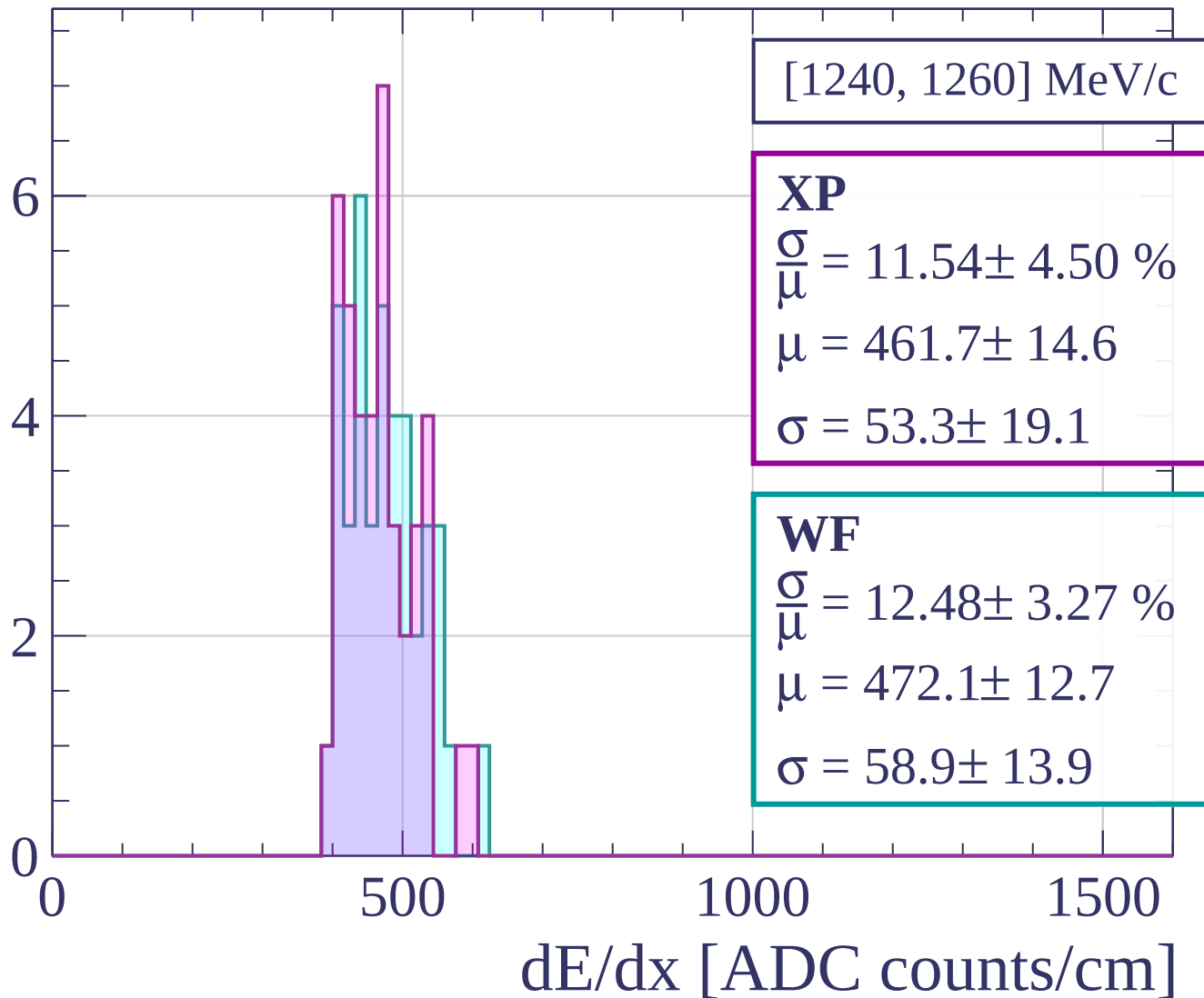
Count



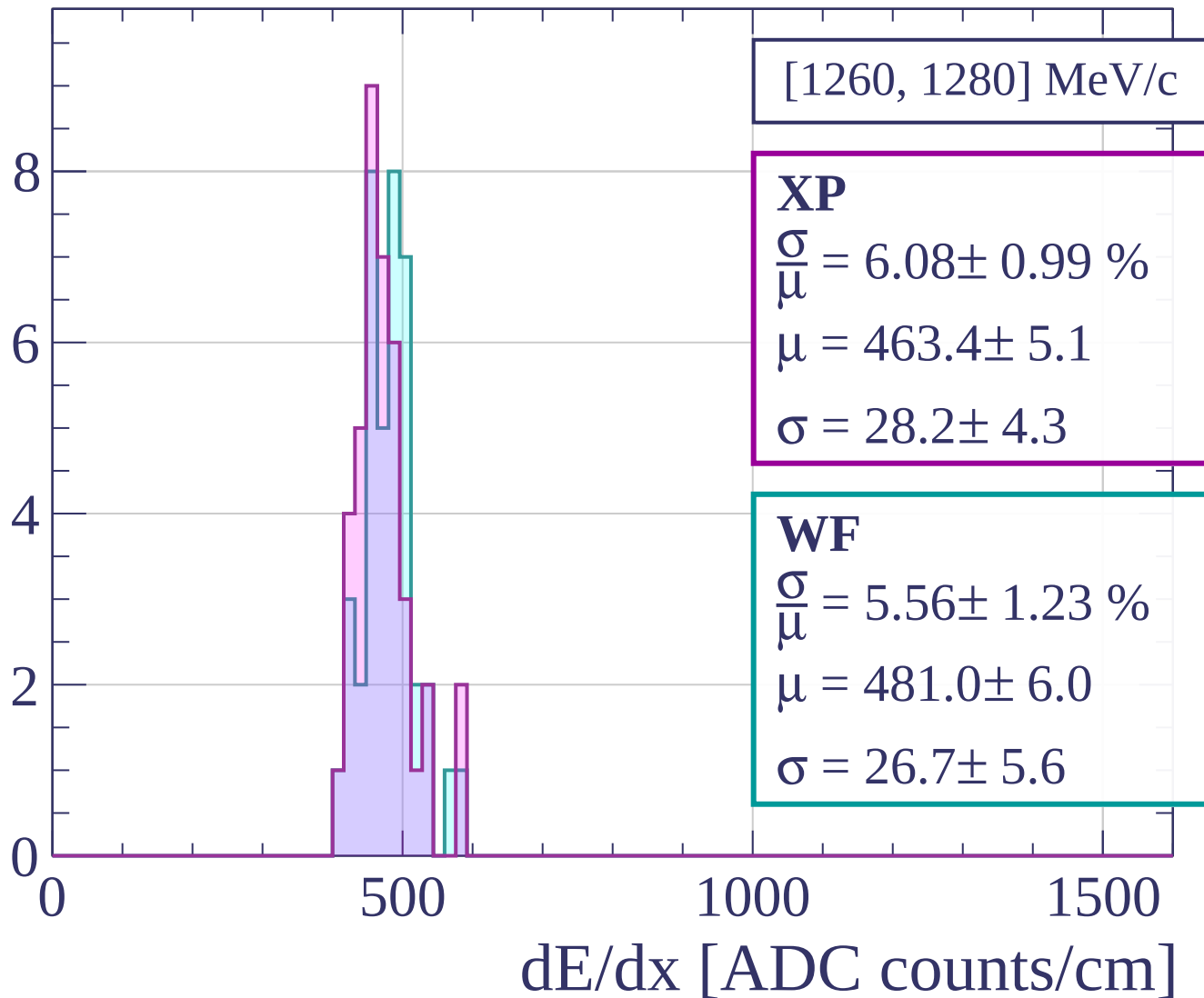
Count



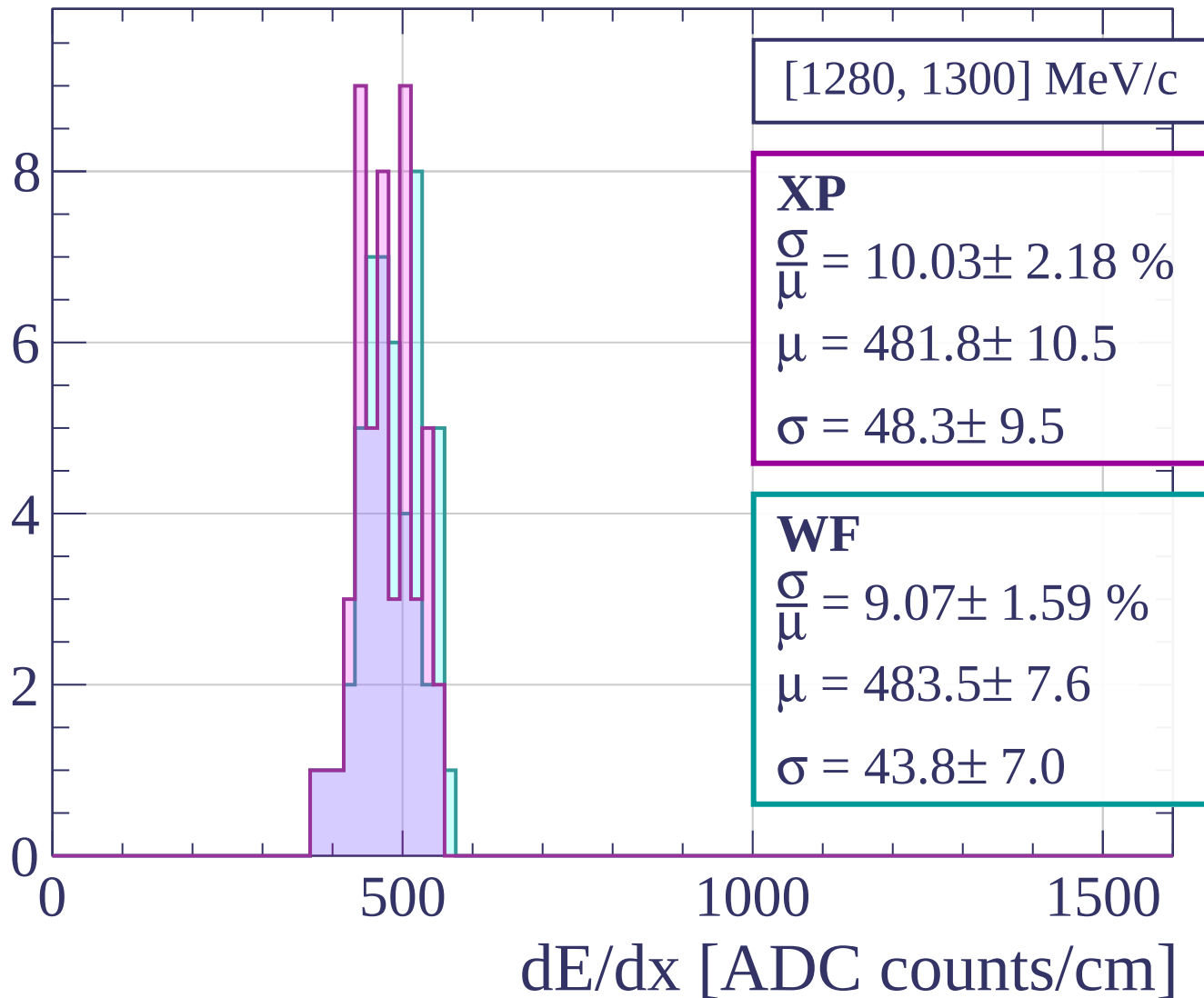
Count



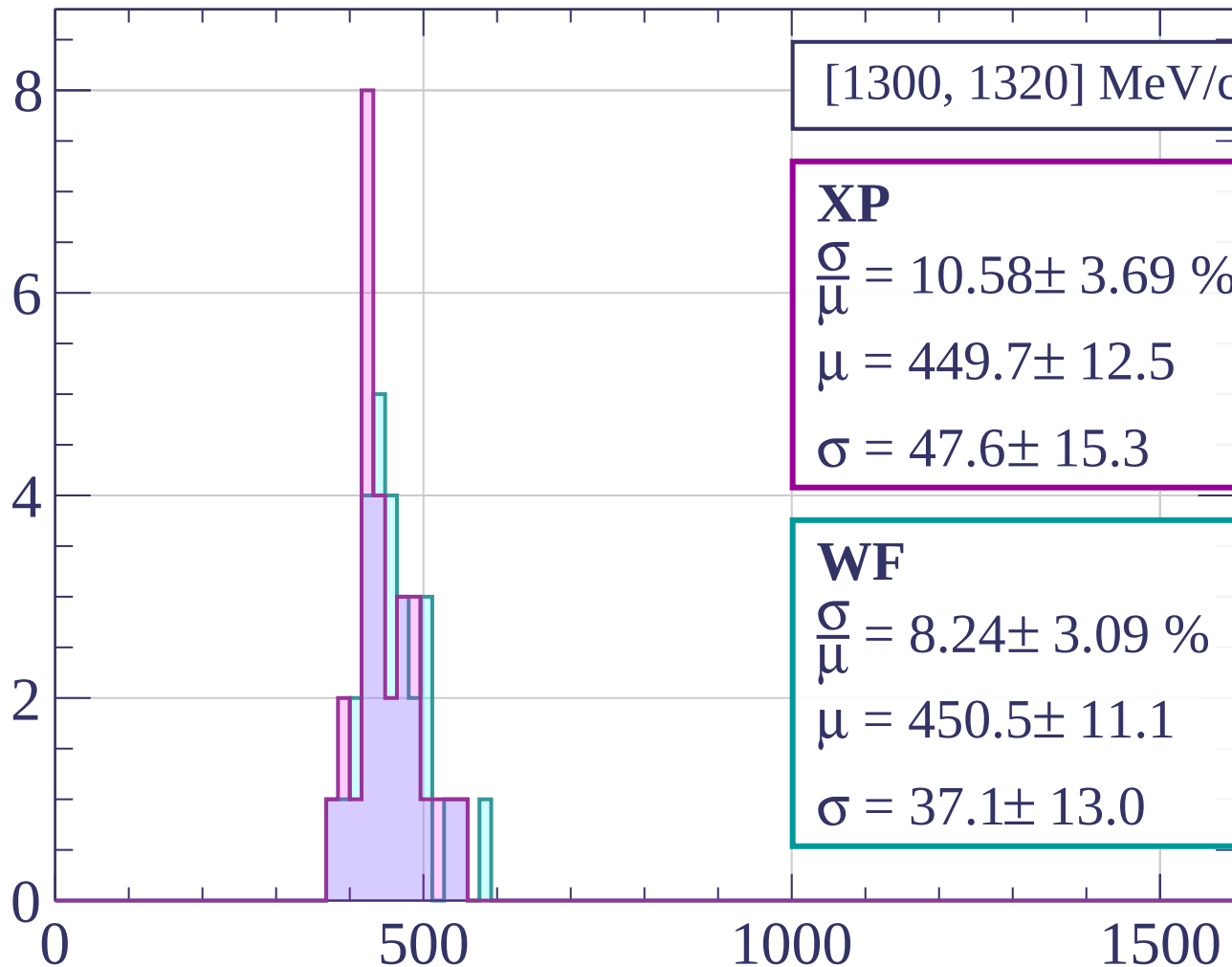
Count



Count

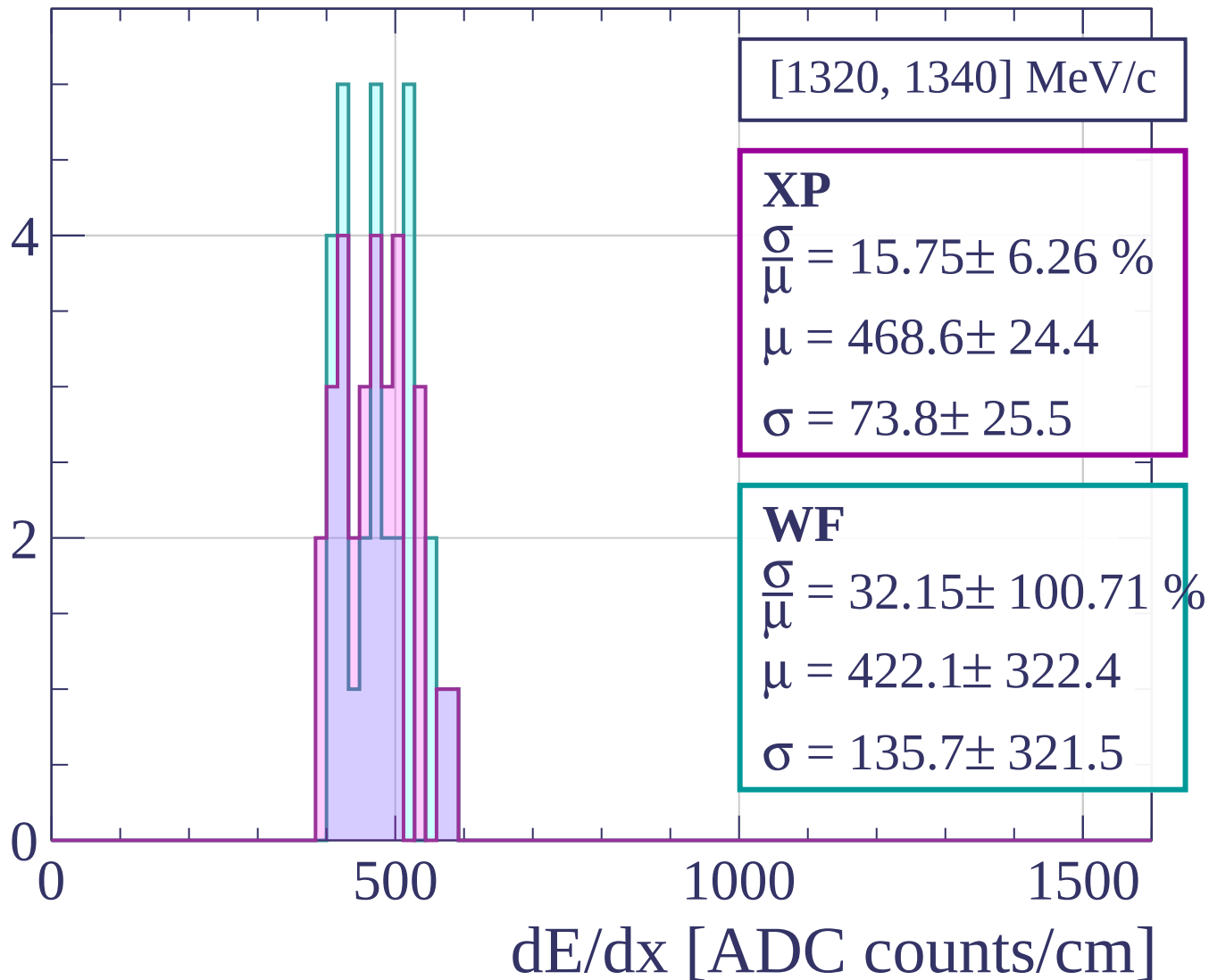


Count

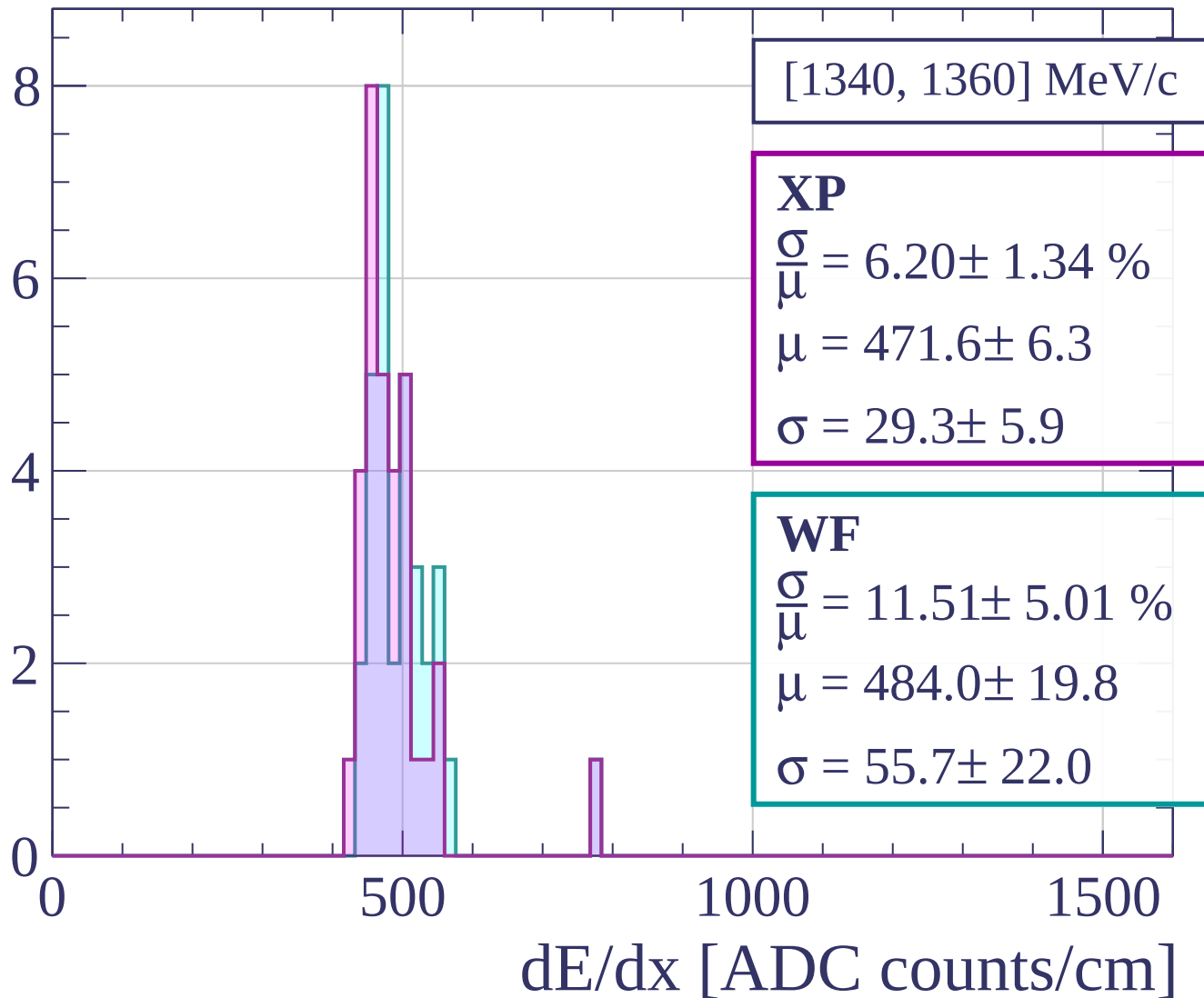


dE/dx [ADC counts/cm]

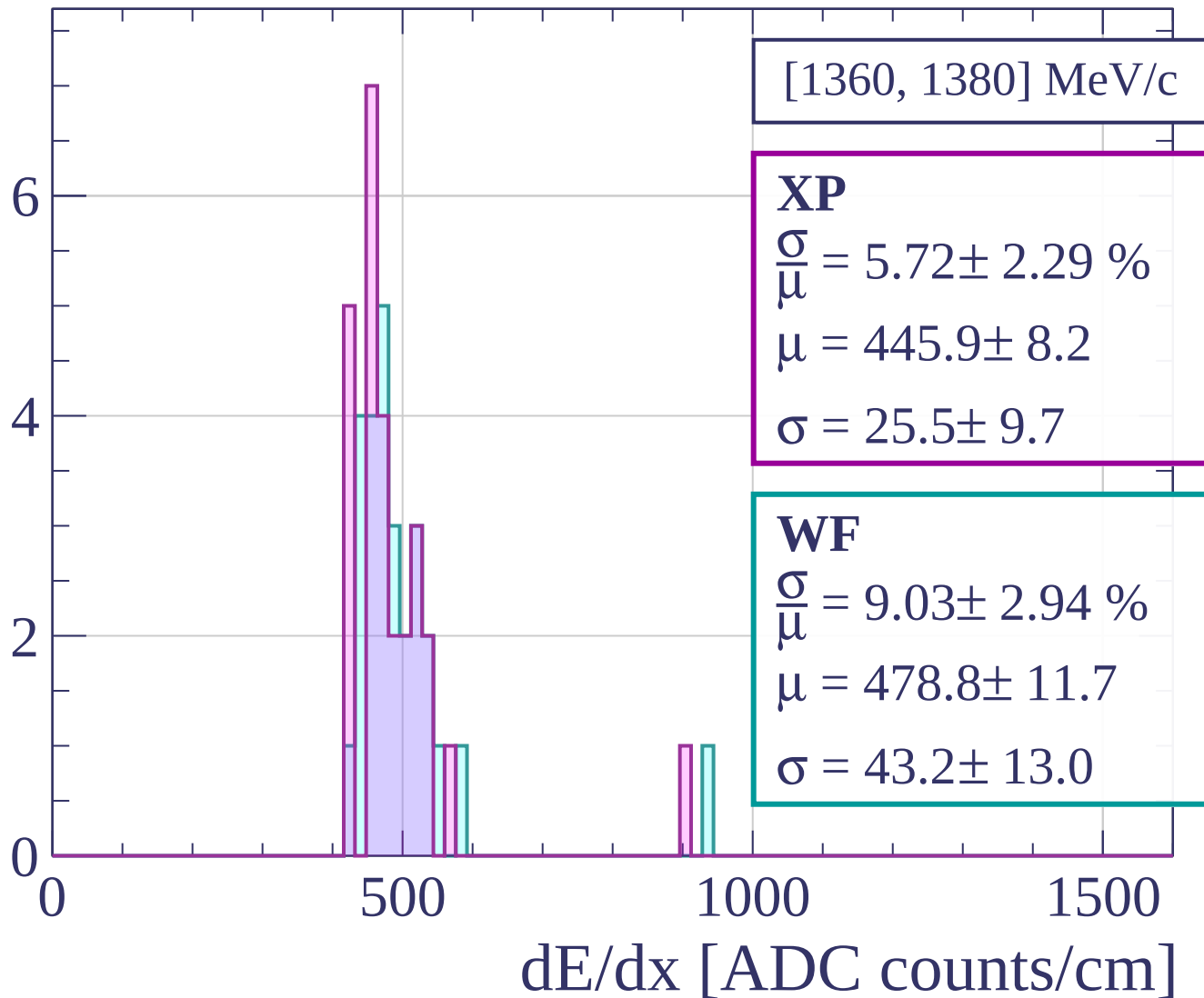
Count



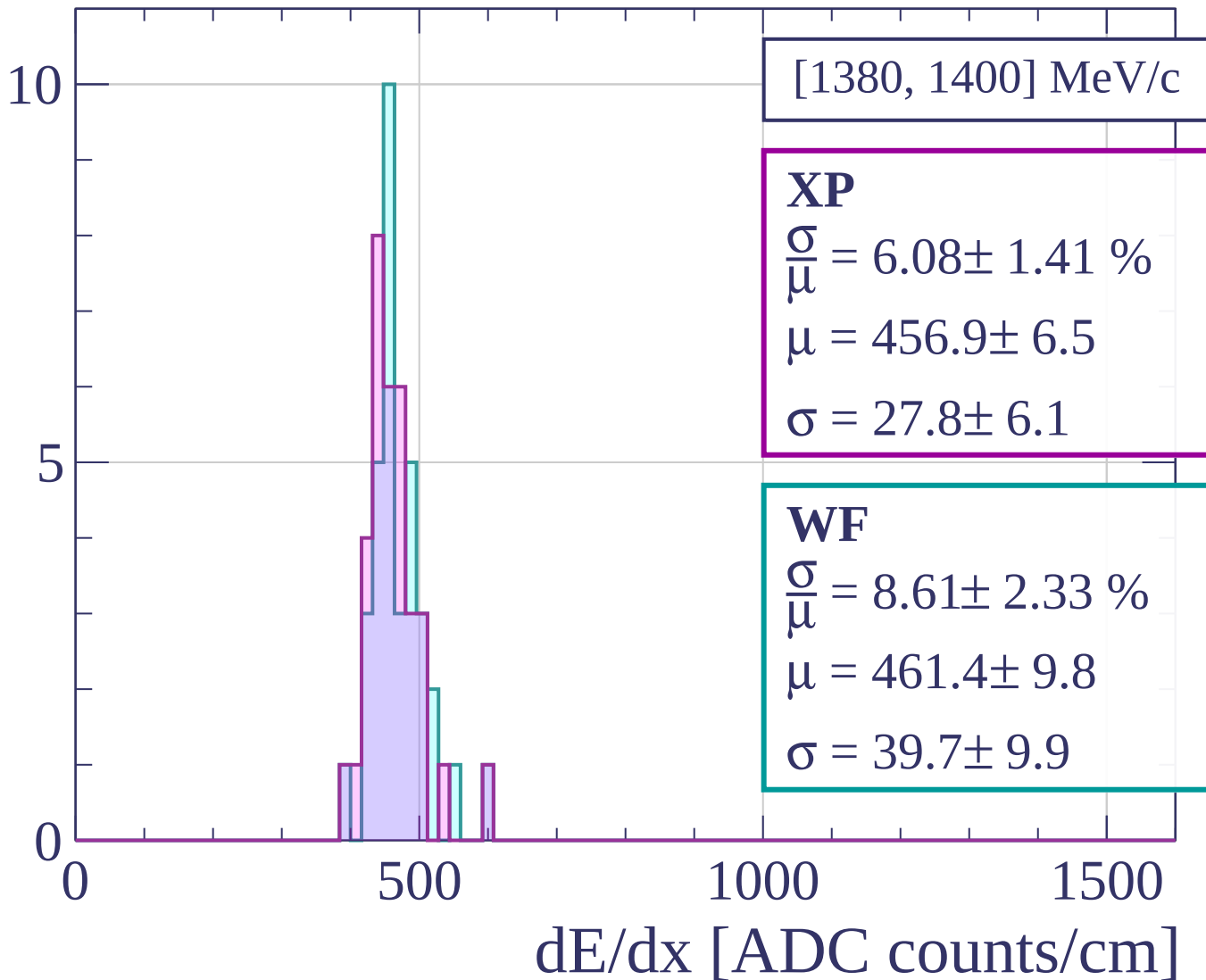
Count



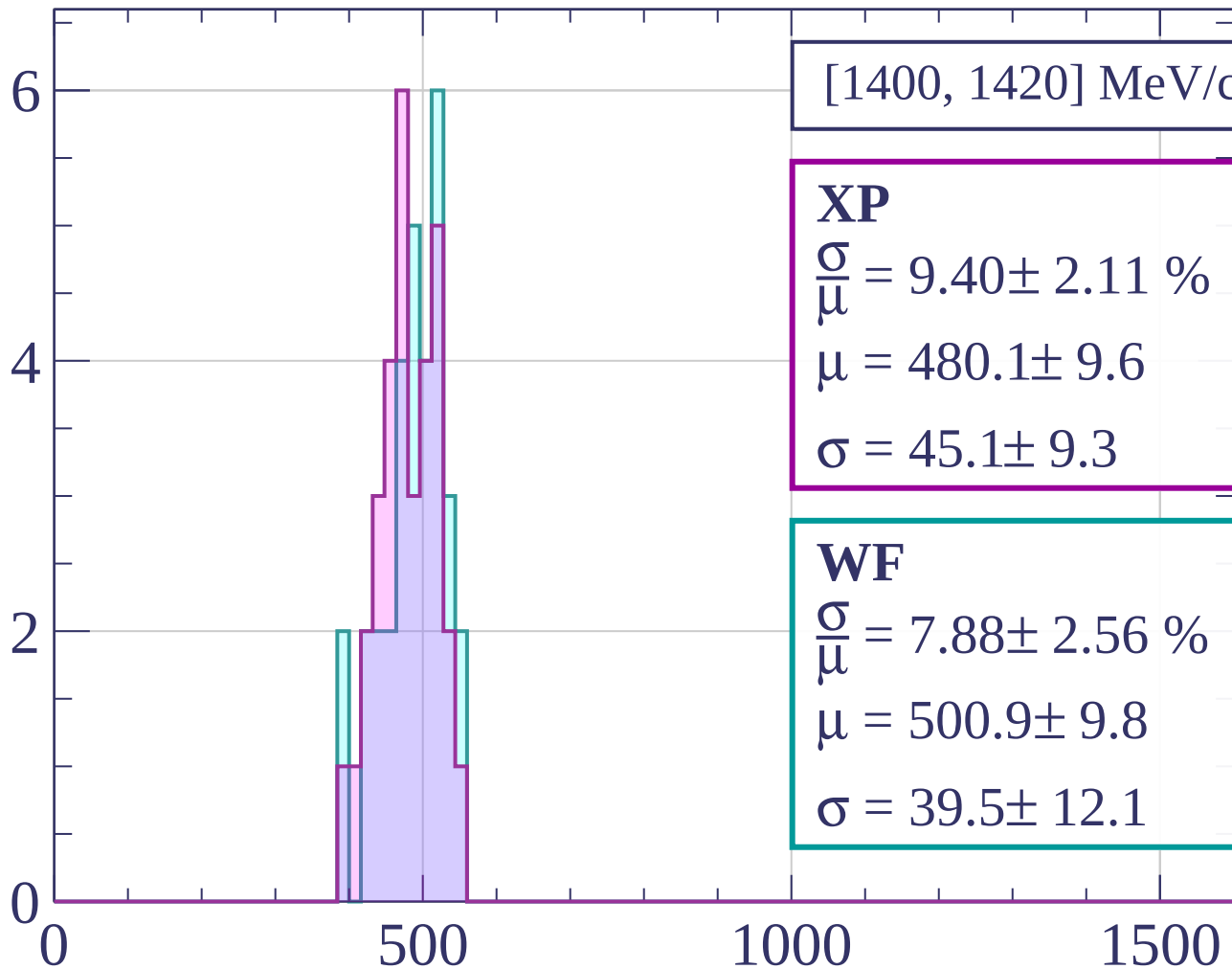
Count



Count

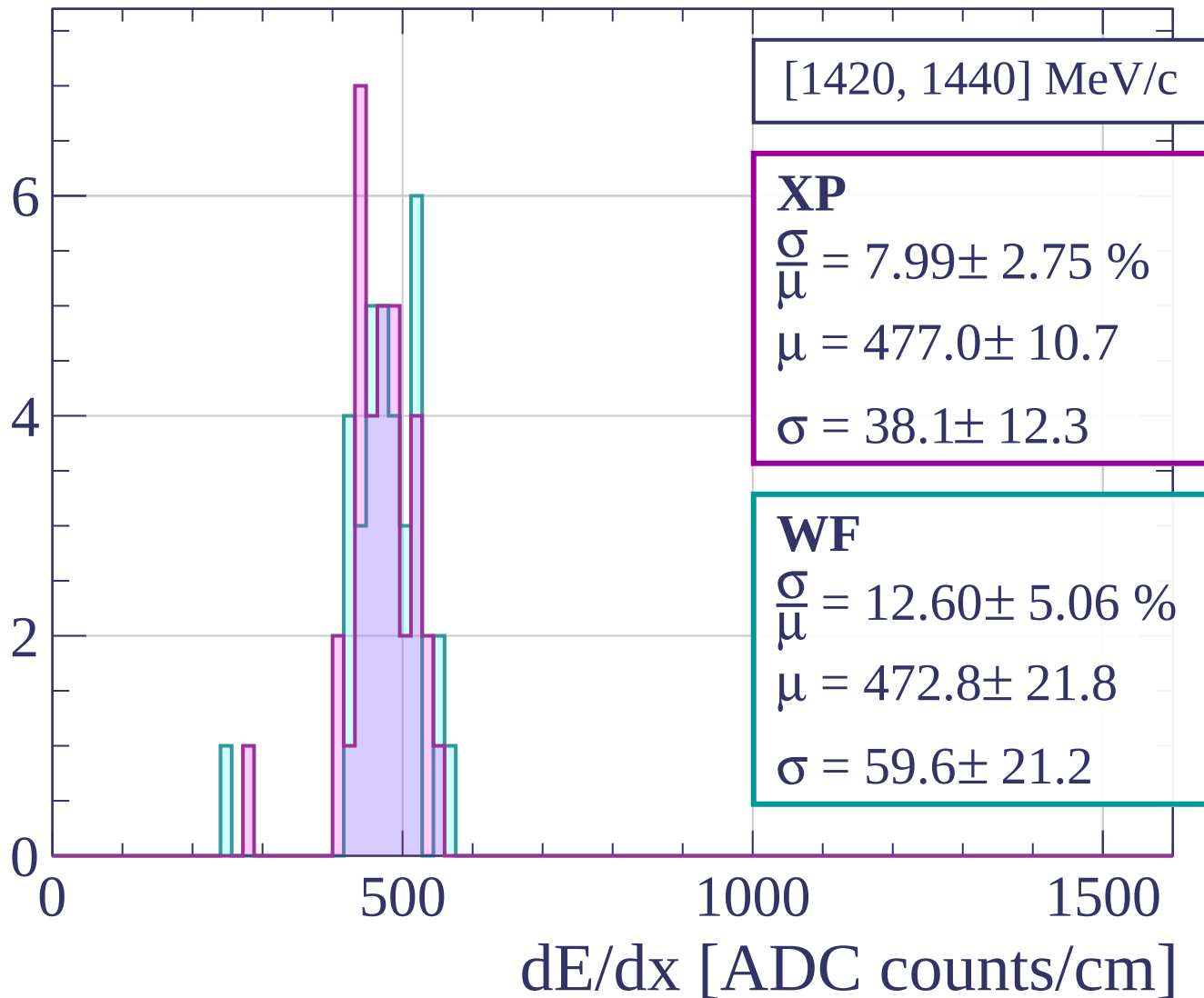


Count

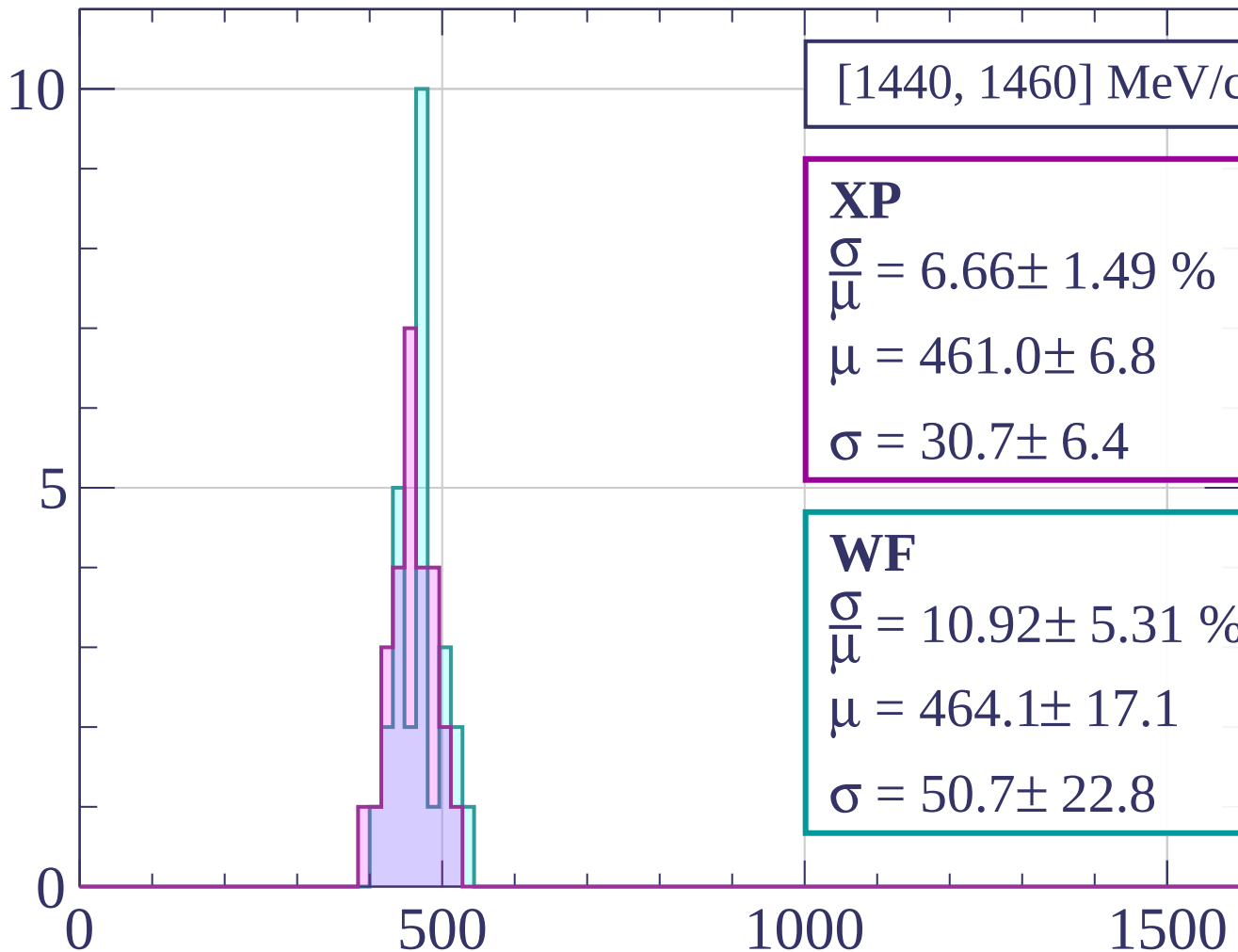


dE/dx [ADC counts/cm]

Count

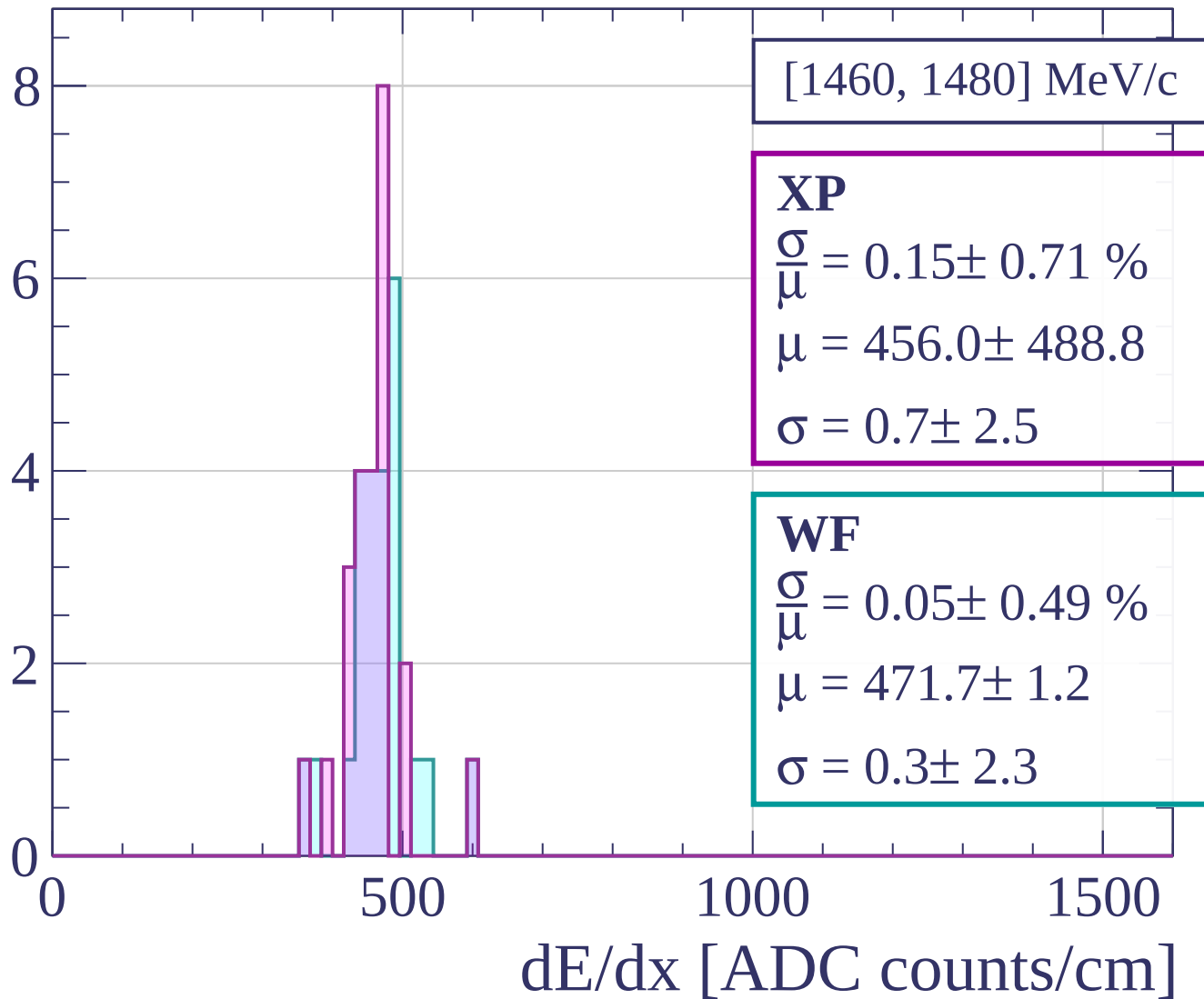


Count

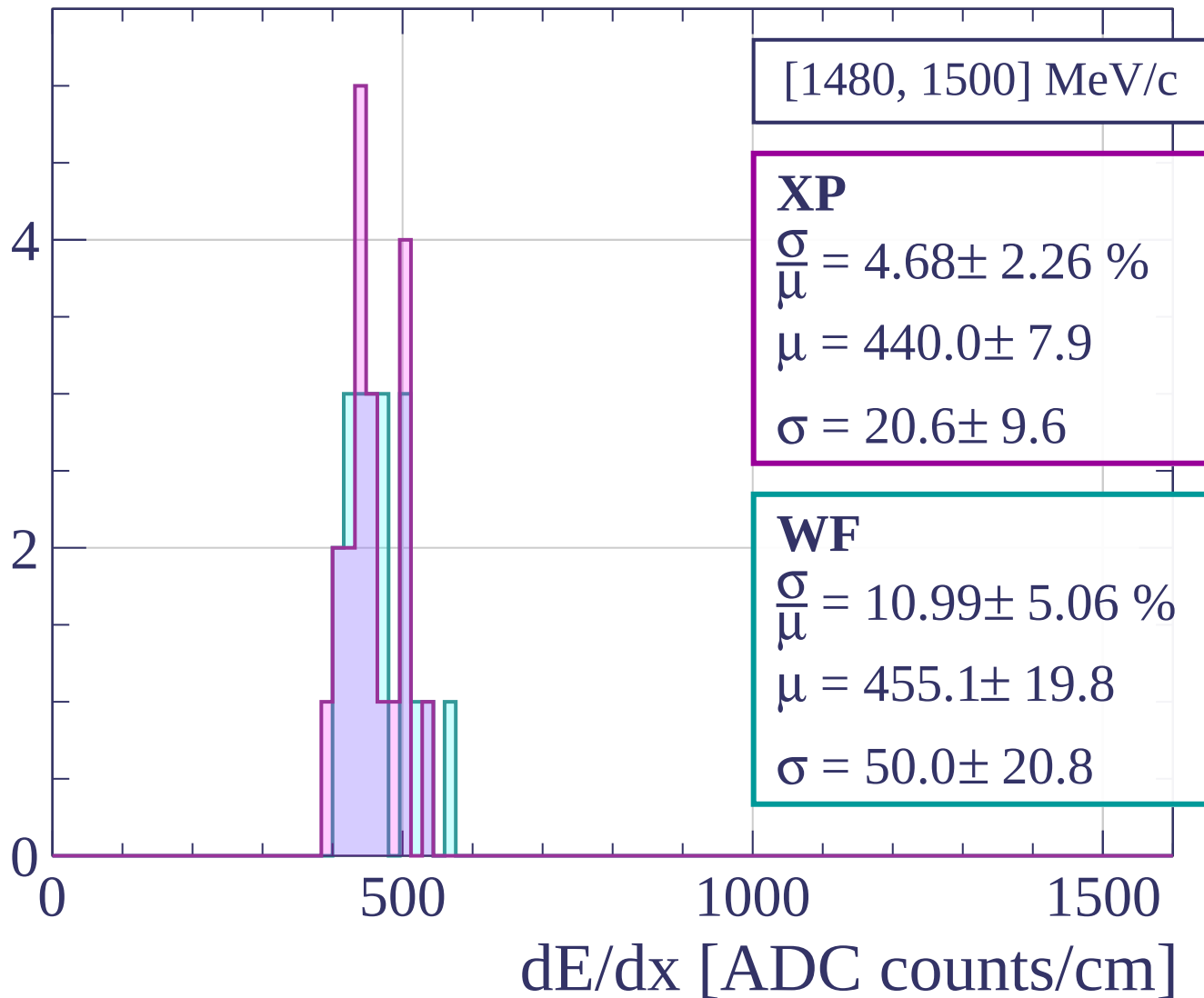


dE/dx [ADC counts/cm]

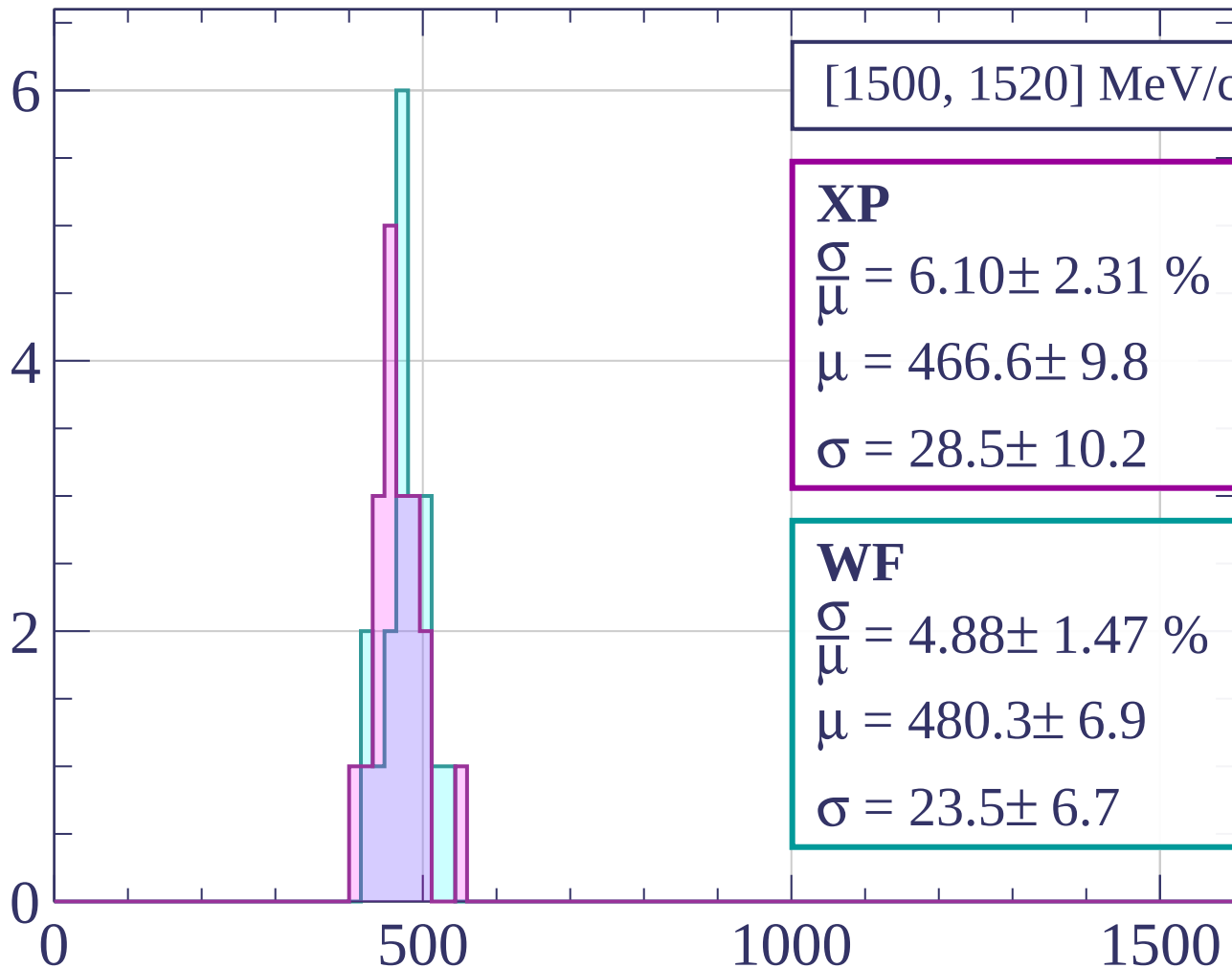
Count



Count

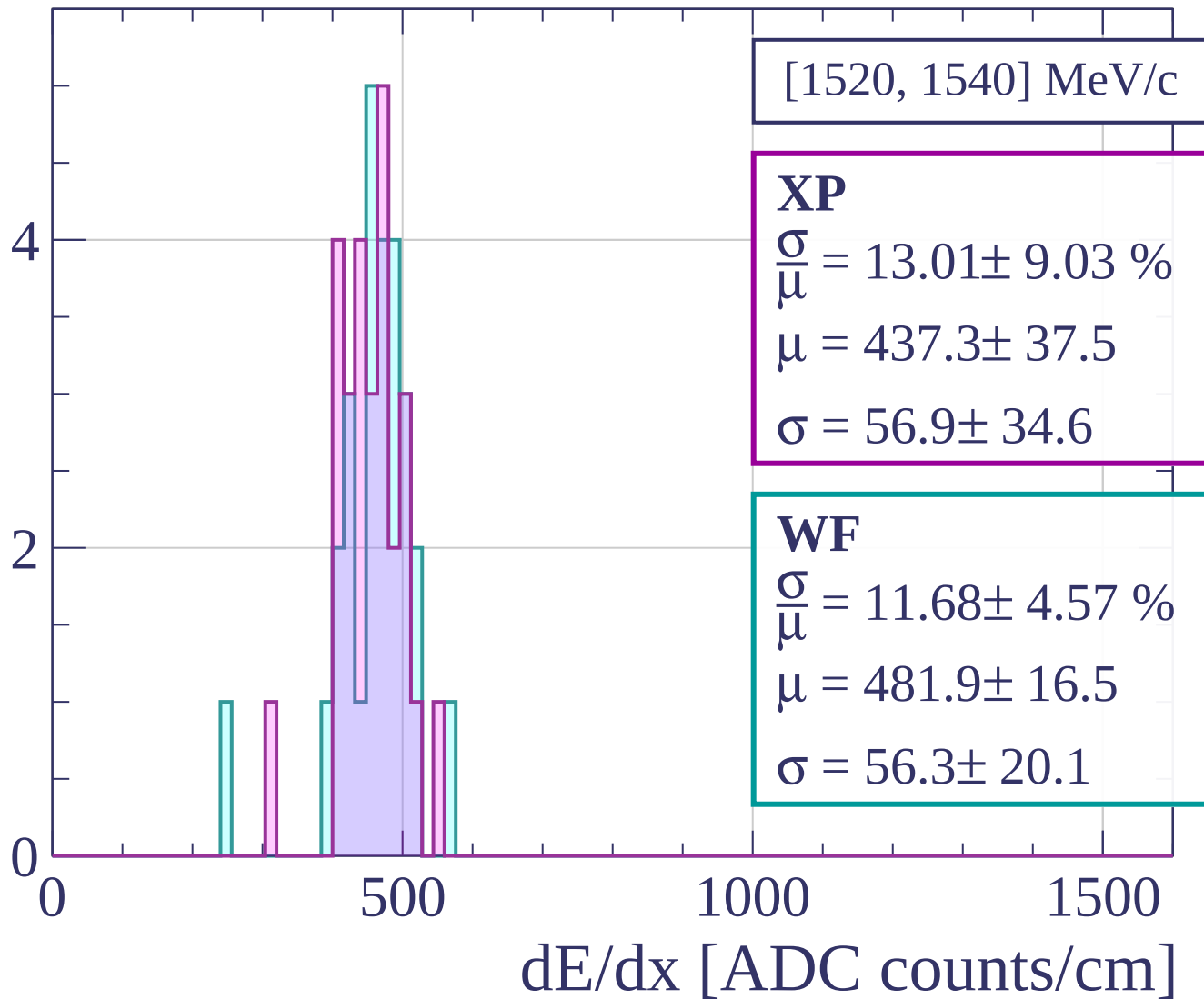


Count

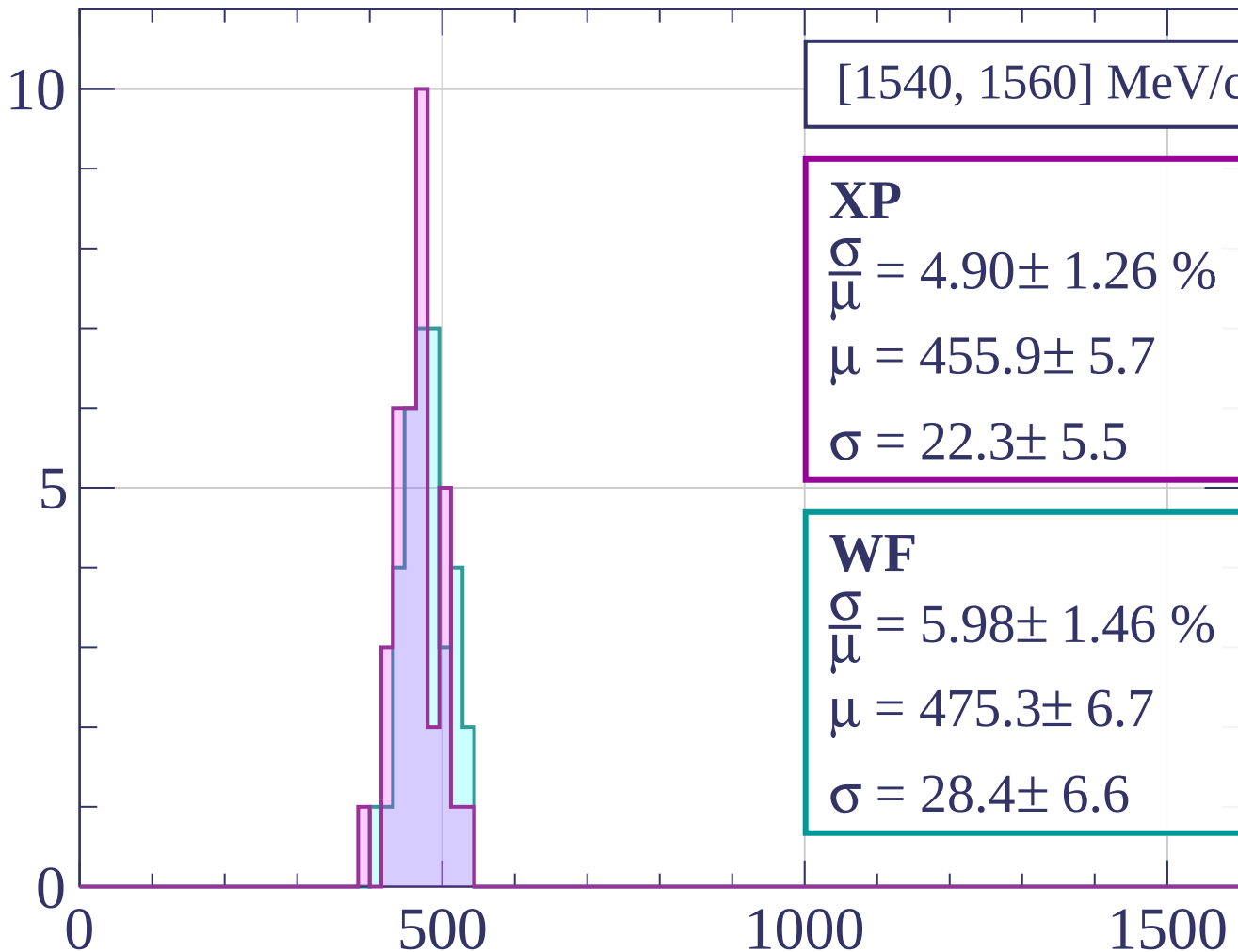


dE/dx [ADC counts/cm]

Count

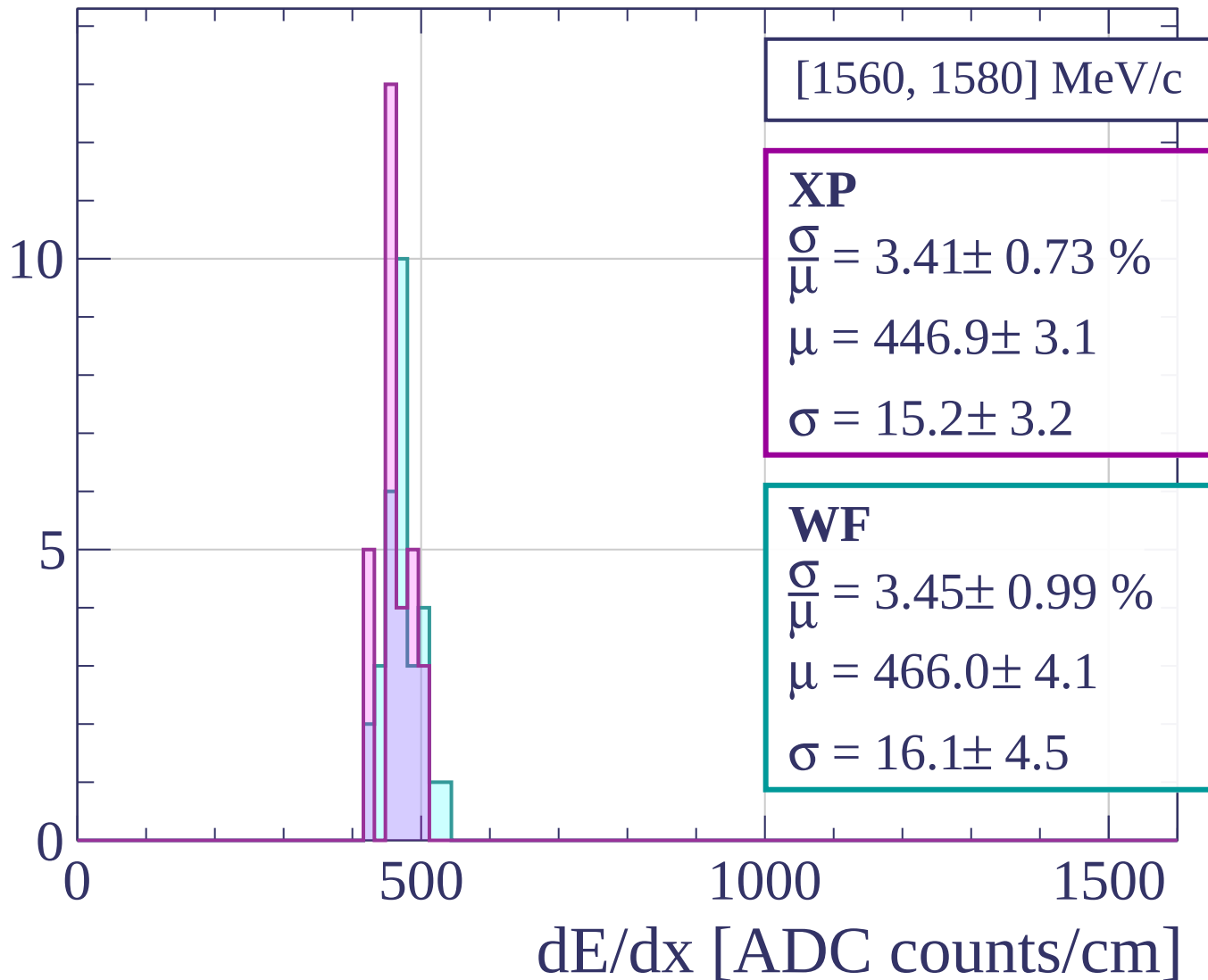


Count

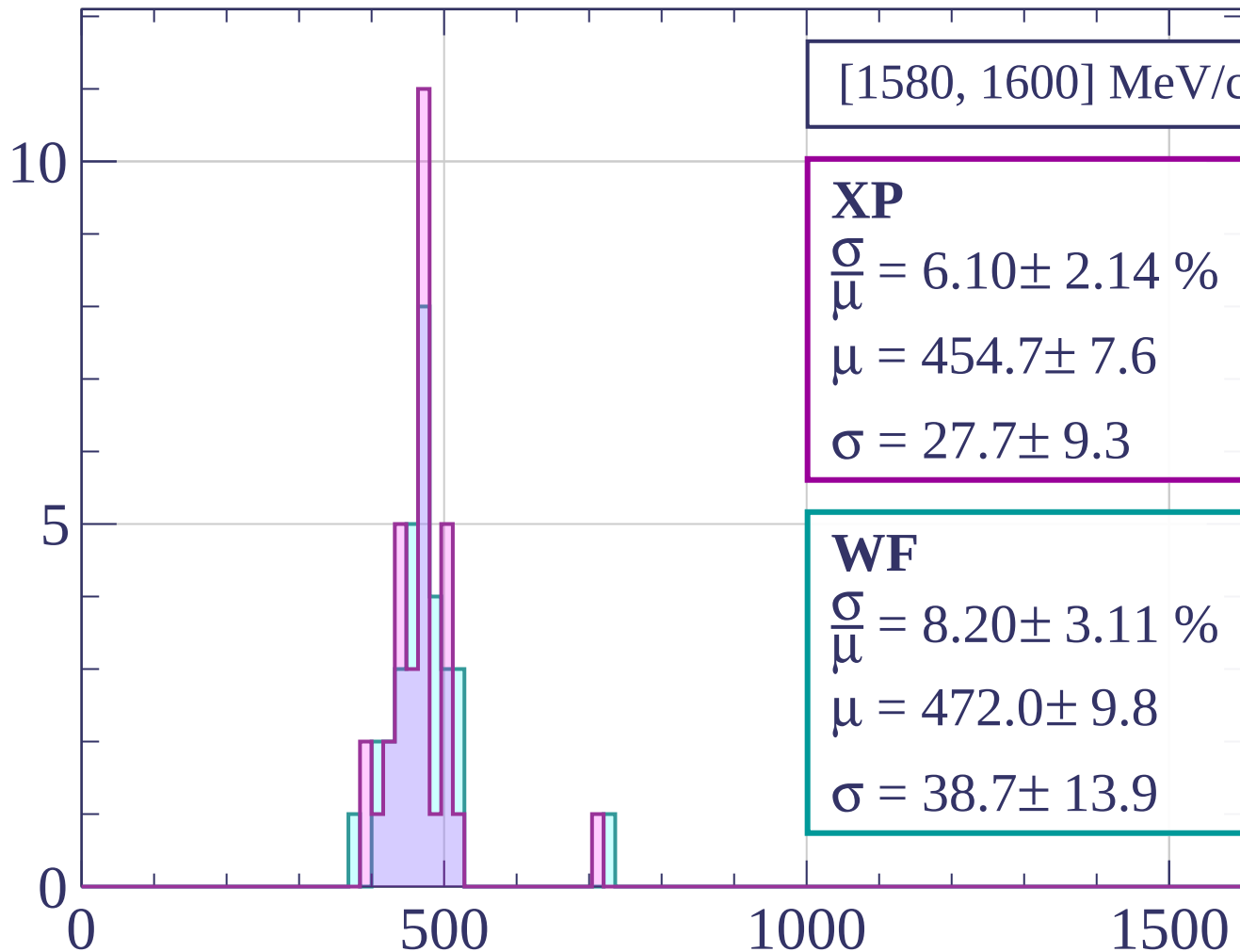


dE/dx [ADC counts/cm]

Count

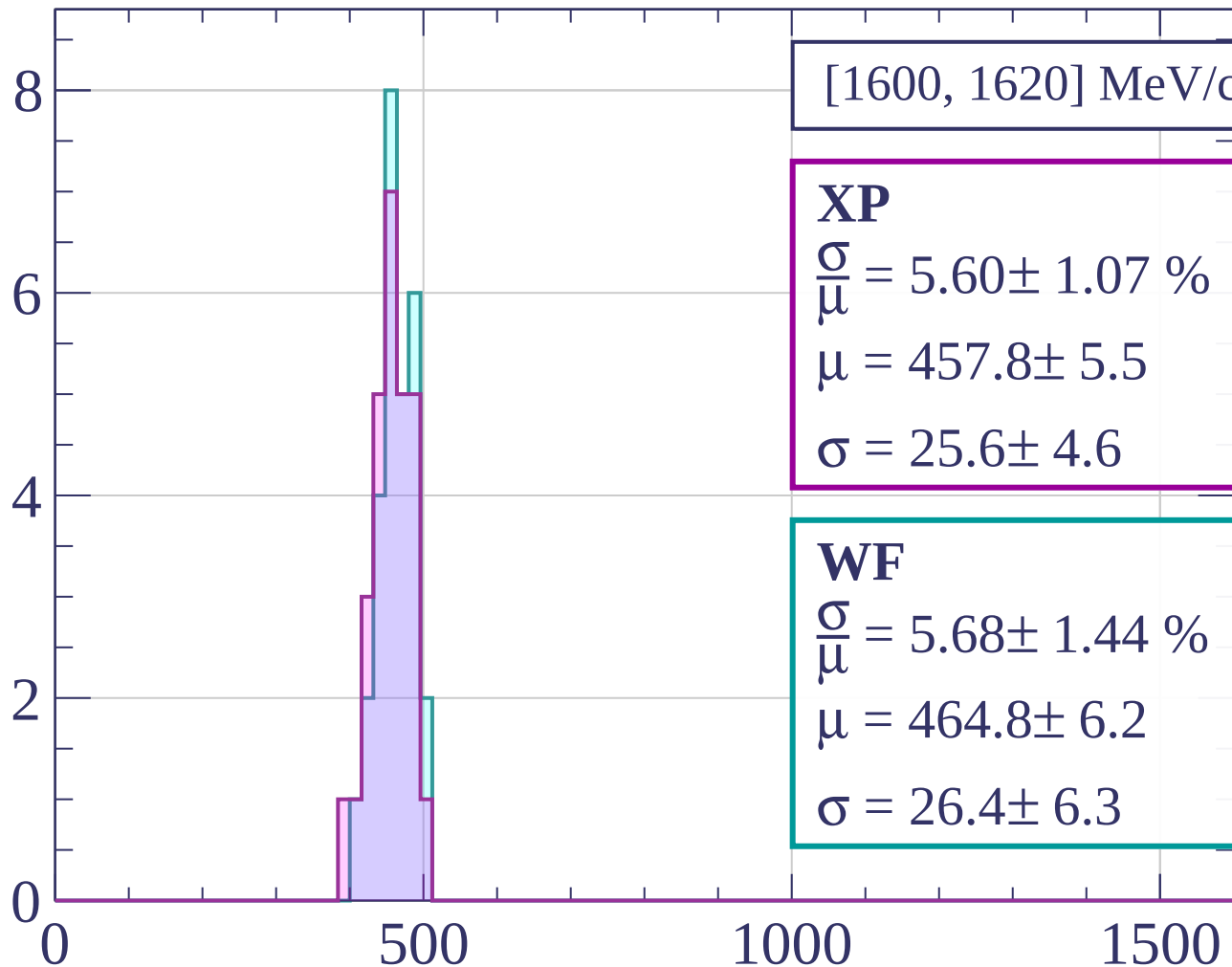


Count



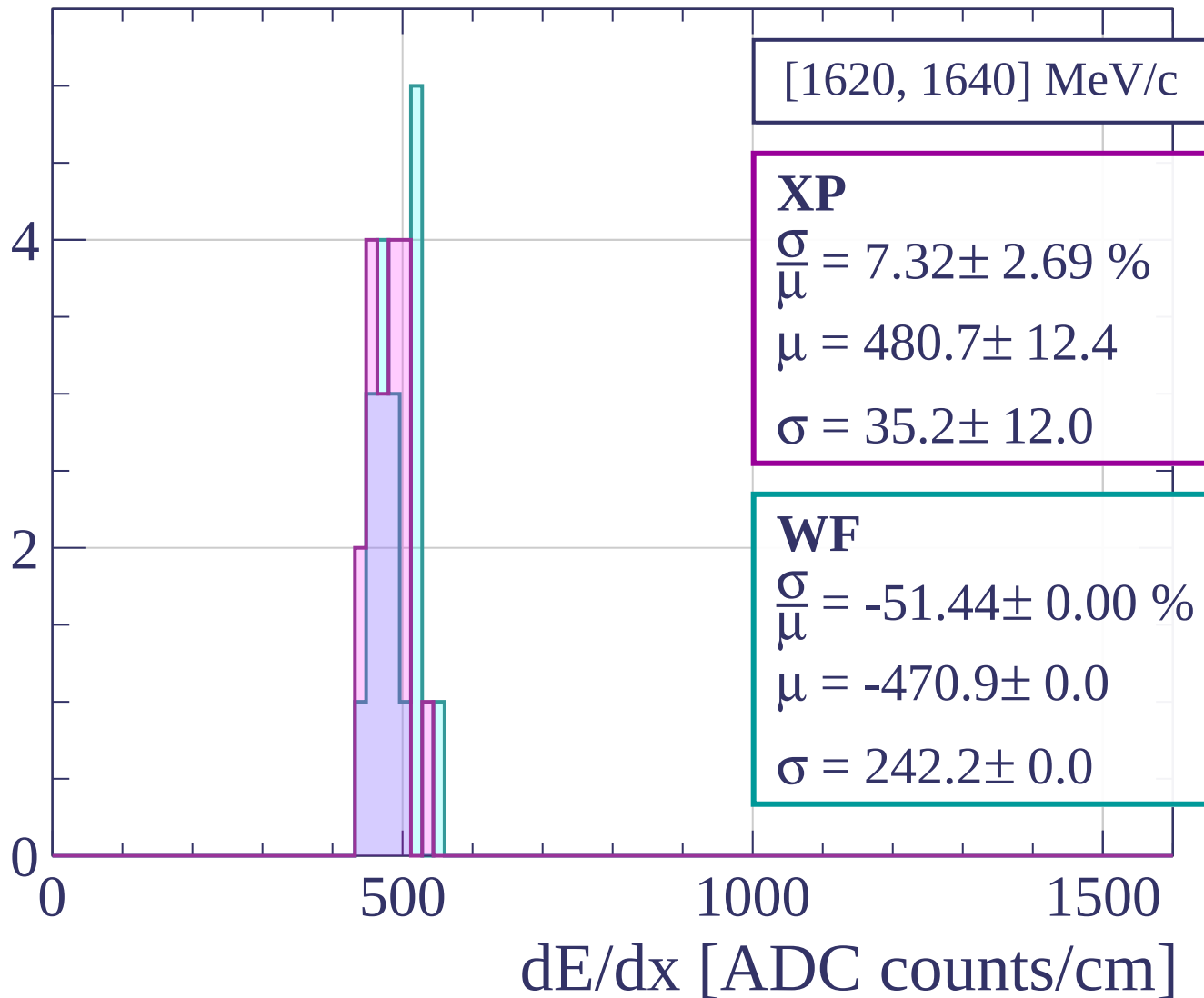
dE/dx [ADC counts/cm]

Count

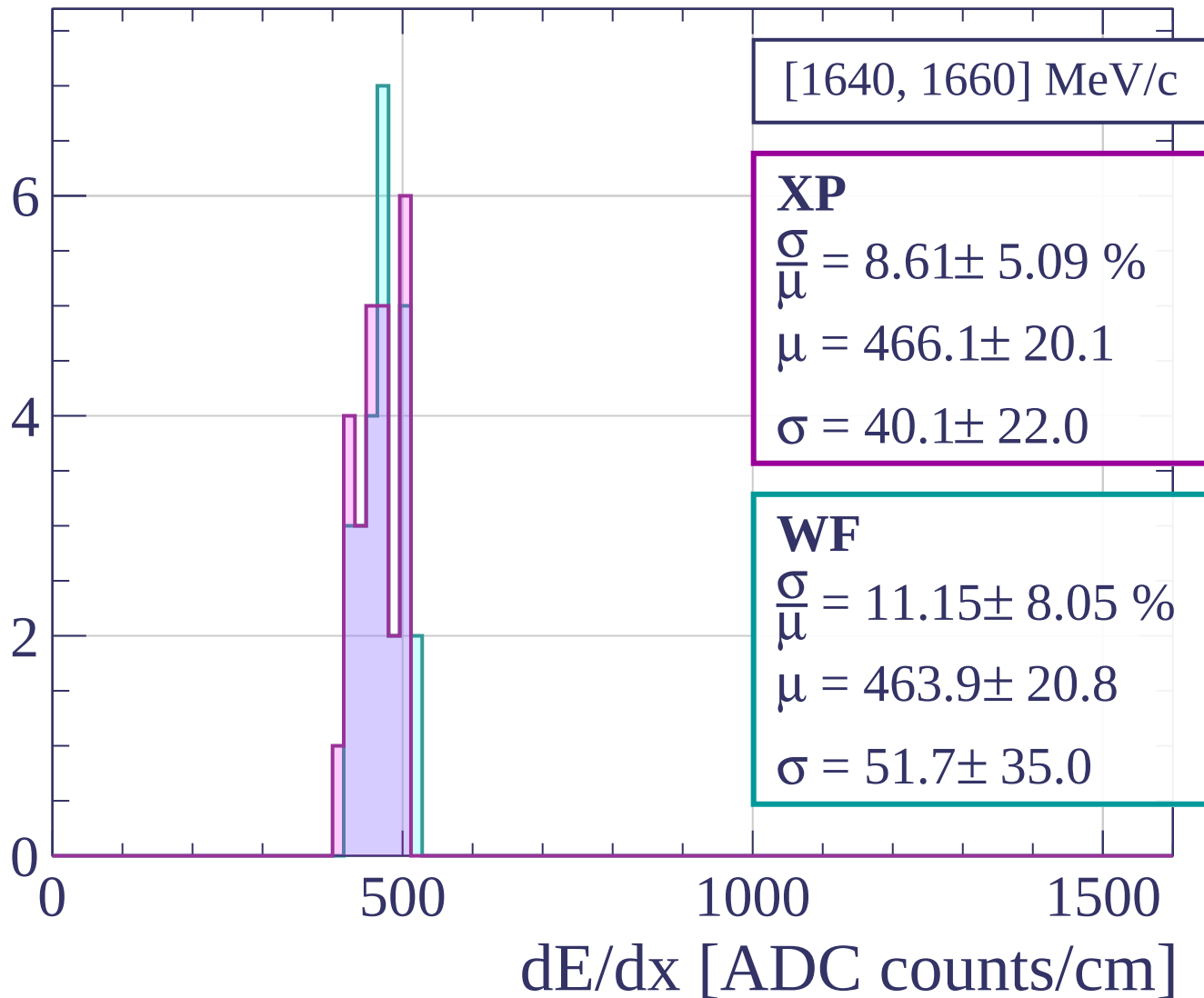


dE/dx [ADC counts/cm]

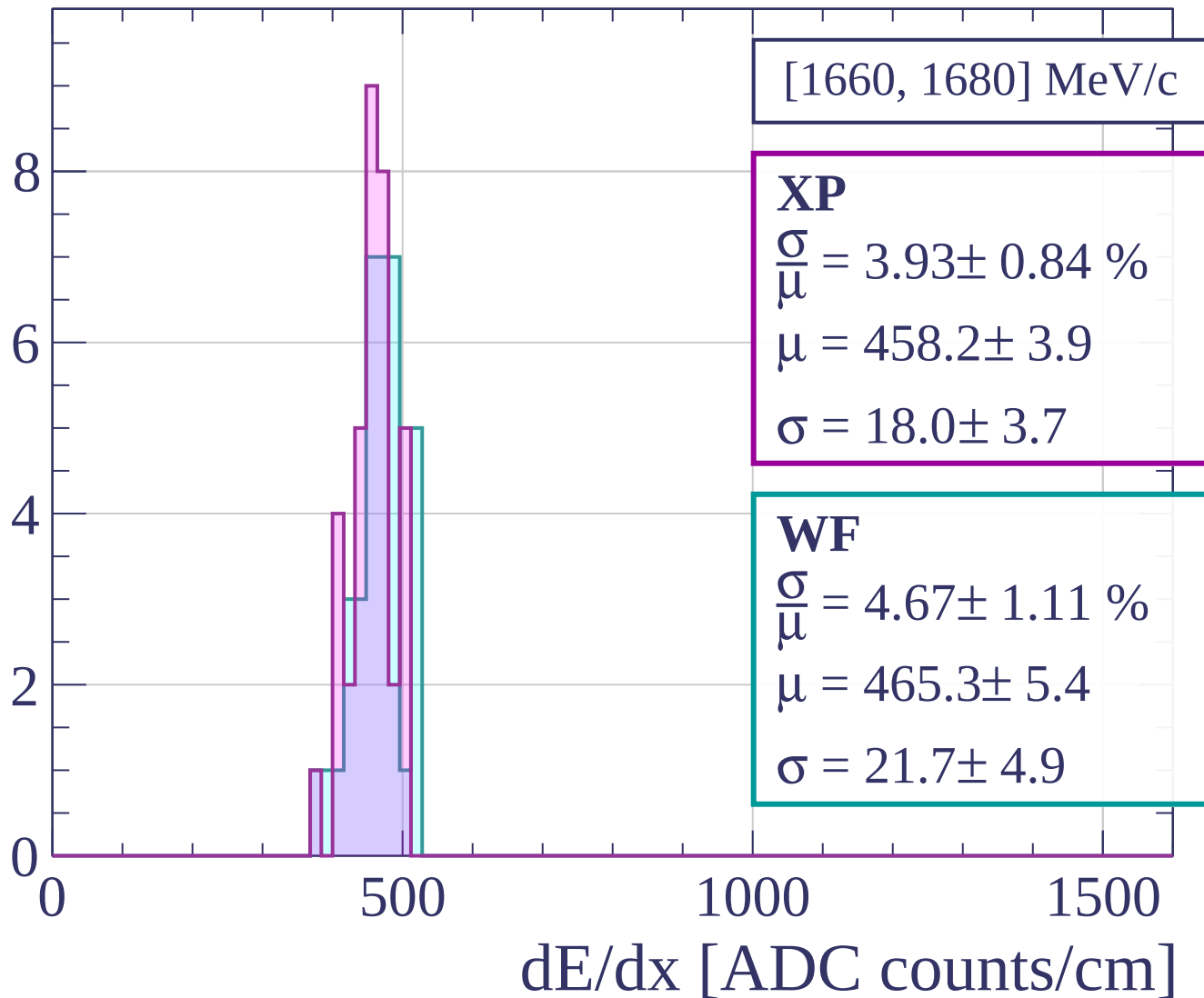
Count



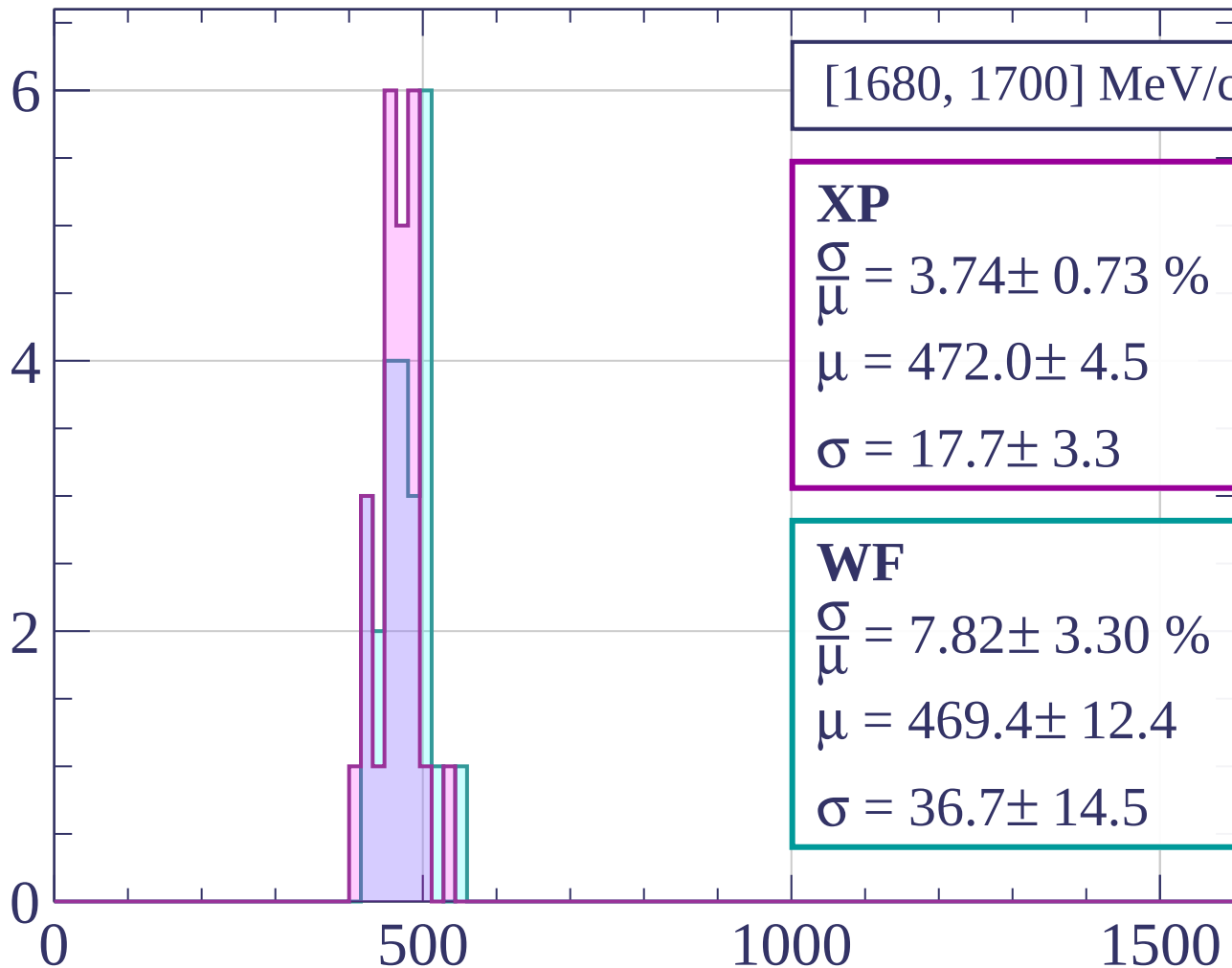
Count



Count



Count



[1680, 1700] MeV/c

XP

$$\frac{\sigma}{\mu} = 3.74 \pm 0.73 \%$$

$$\mu = 472.0 \pm 4.5$$

$$\sigma = 17.7 \pm 3.3$$

WF

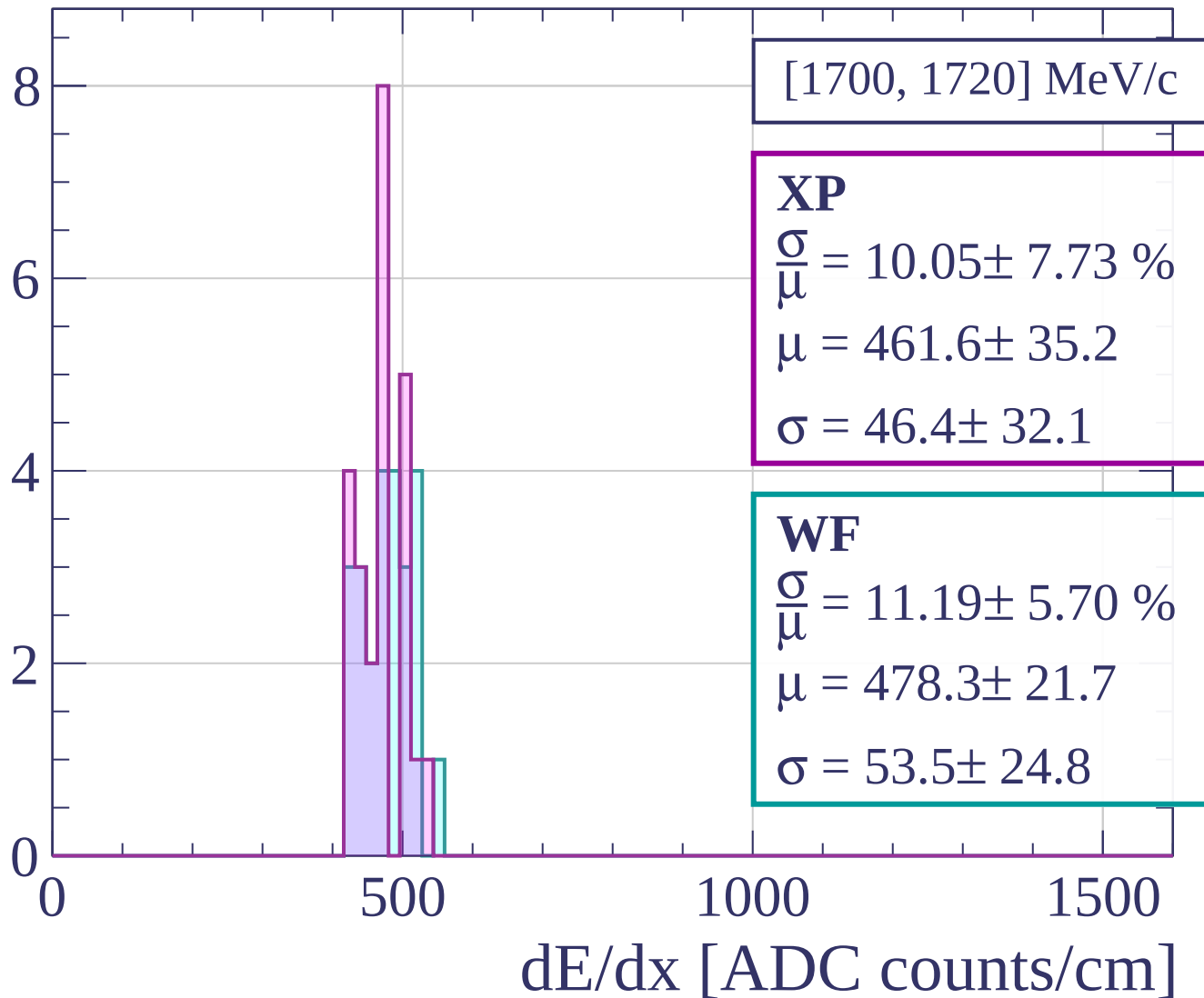
$$\frac{\sigma}{\mu} = 7.82 \pm 3.30 \%$$

$$\mu = 469.4 \pm 12.4$$

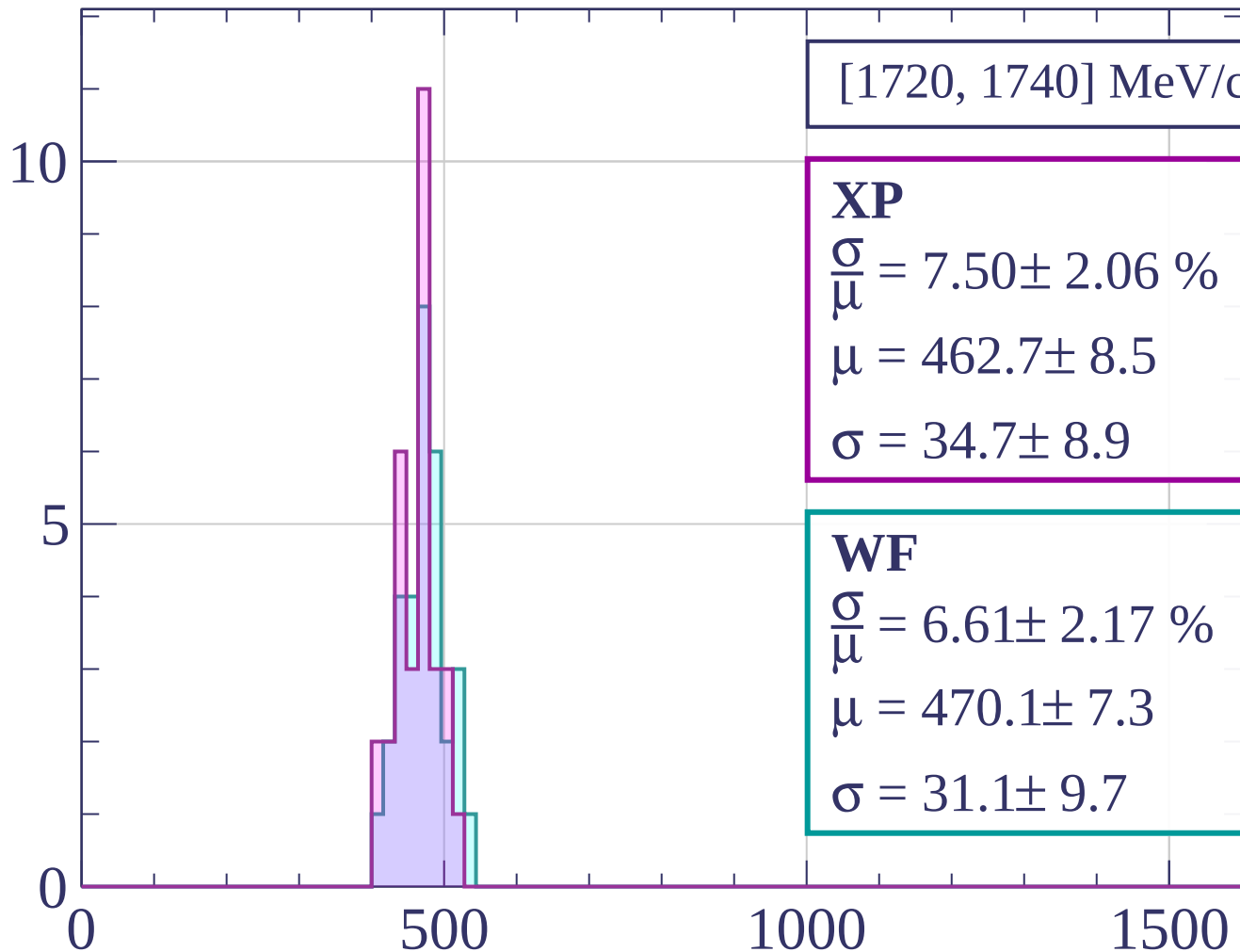
$$\sigma = 36.7 \pm 14.5$$

dE/dx [ADC counts/cm]

Count

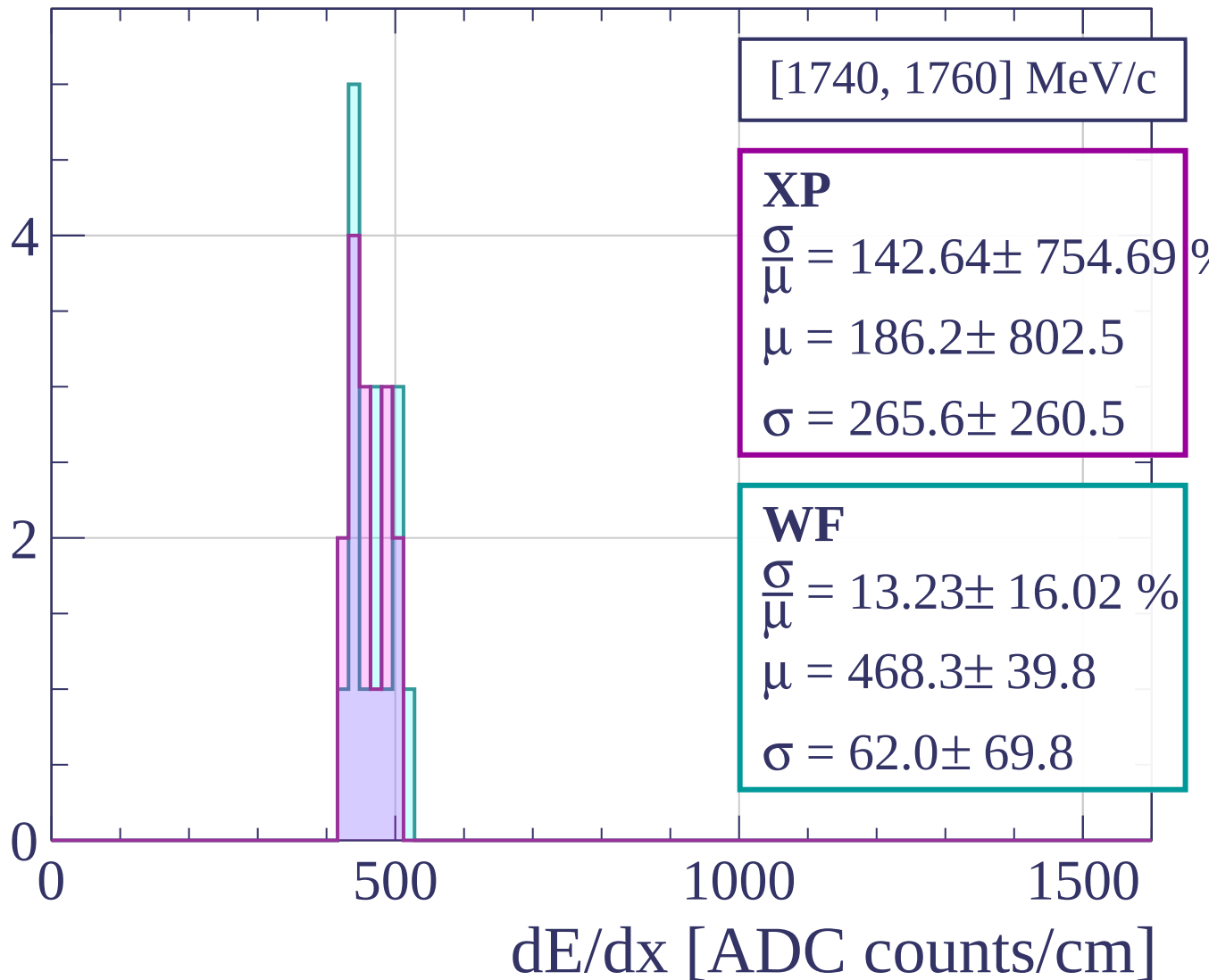


Count

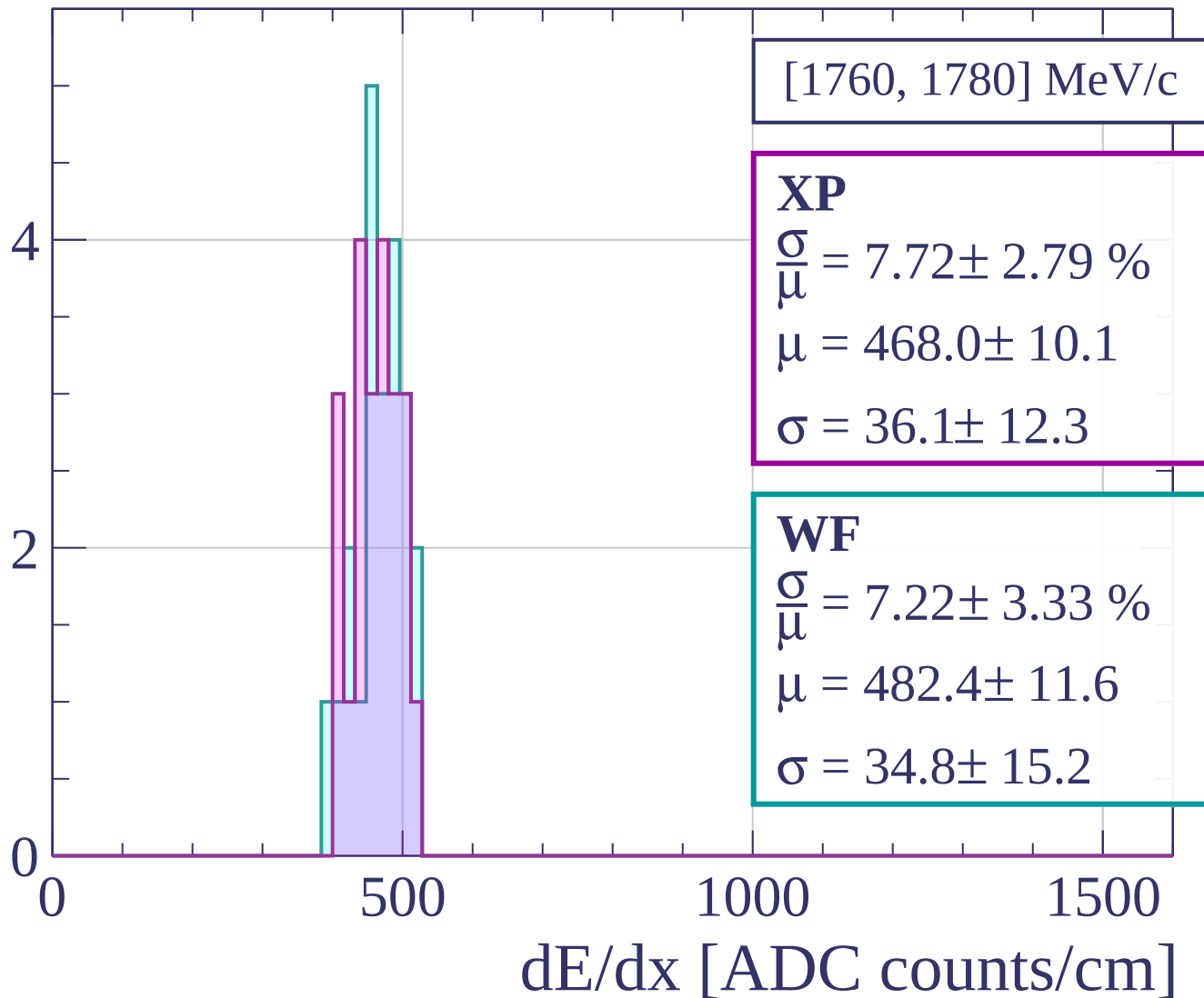


dE/dx [ADC counts/cm]

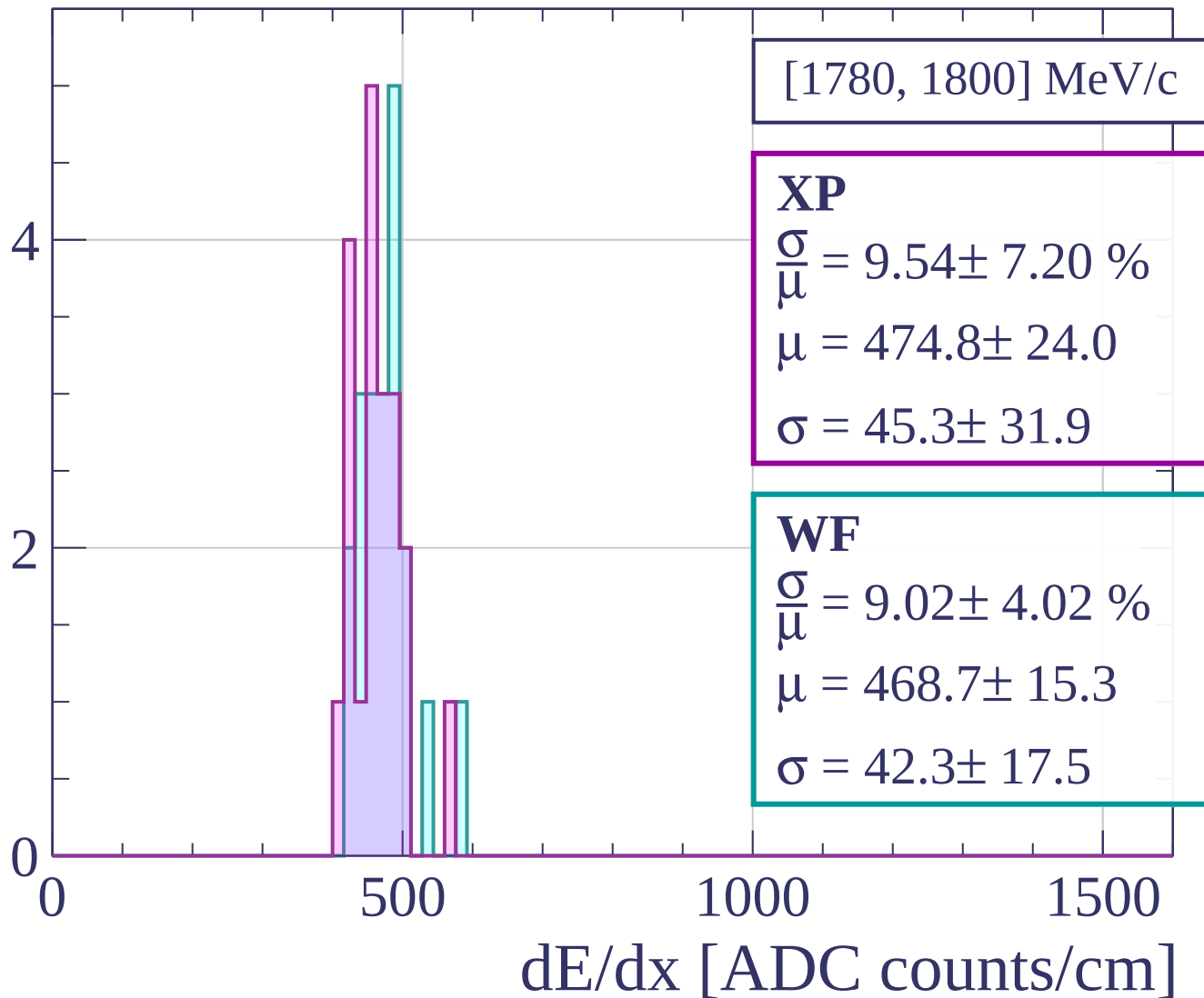
Count



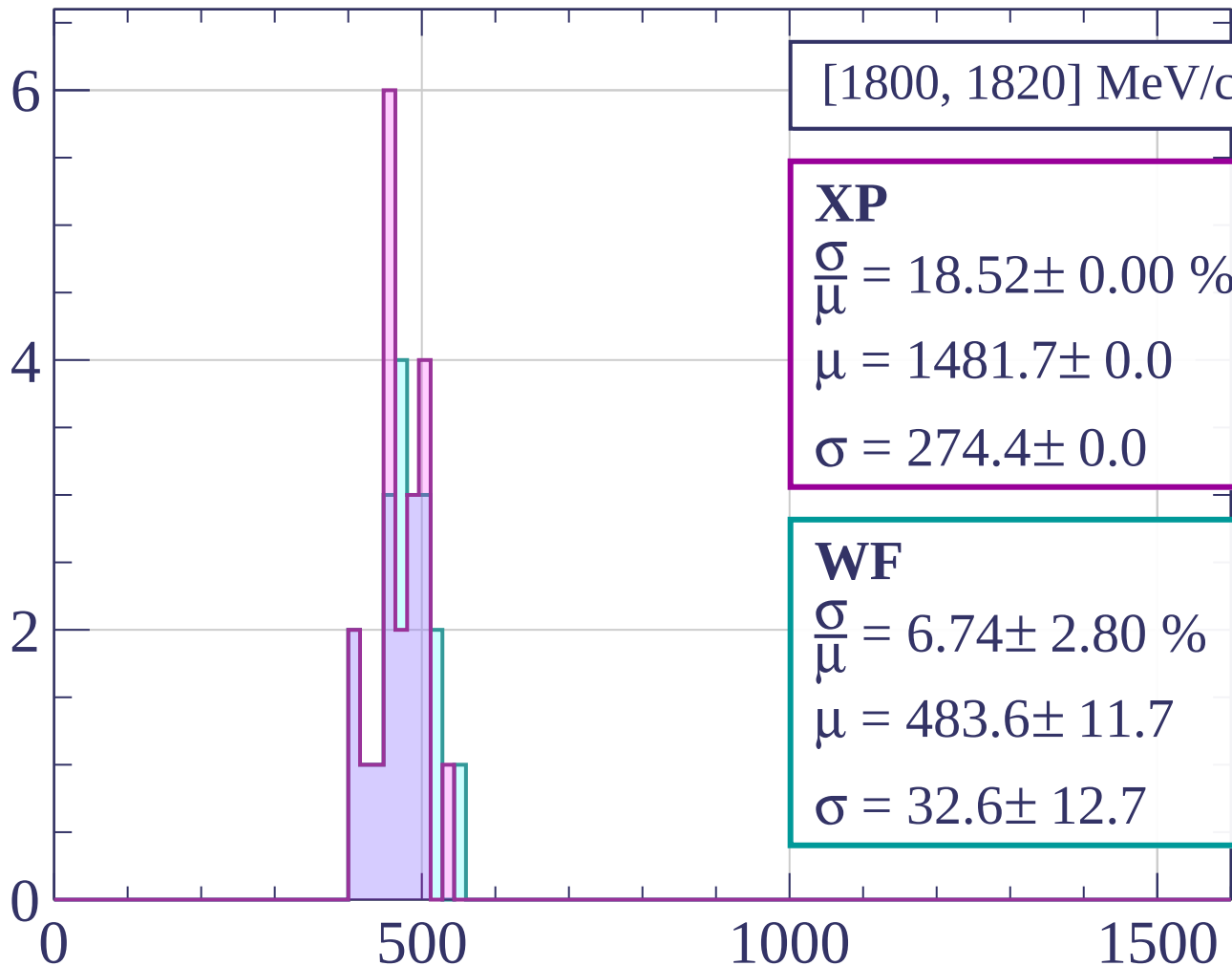
Count



Count

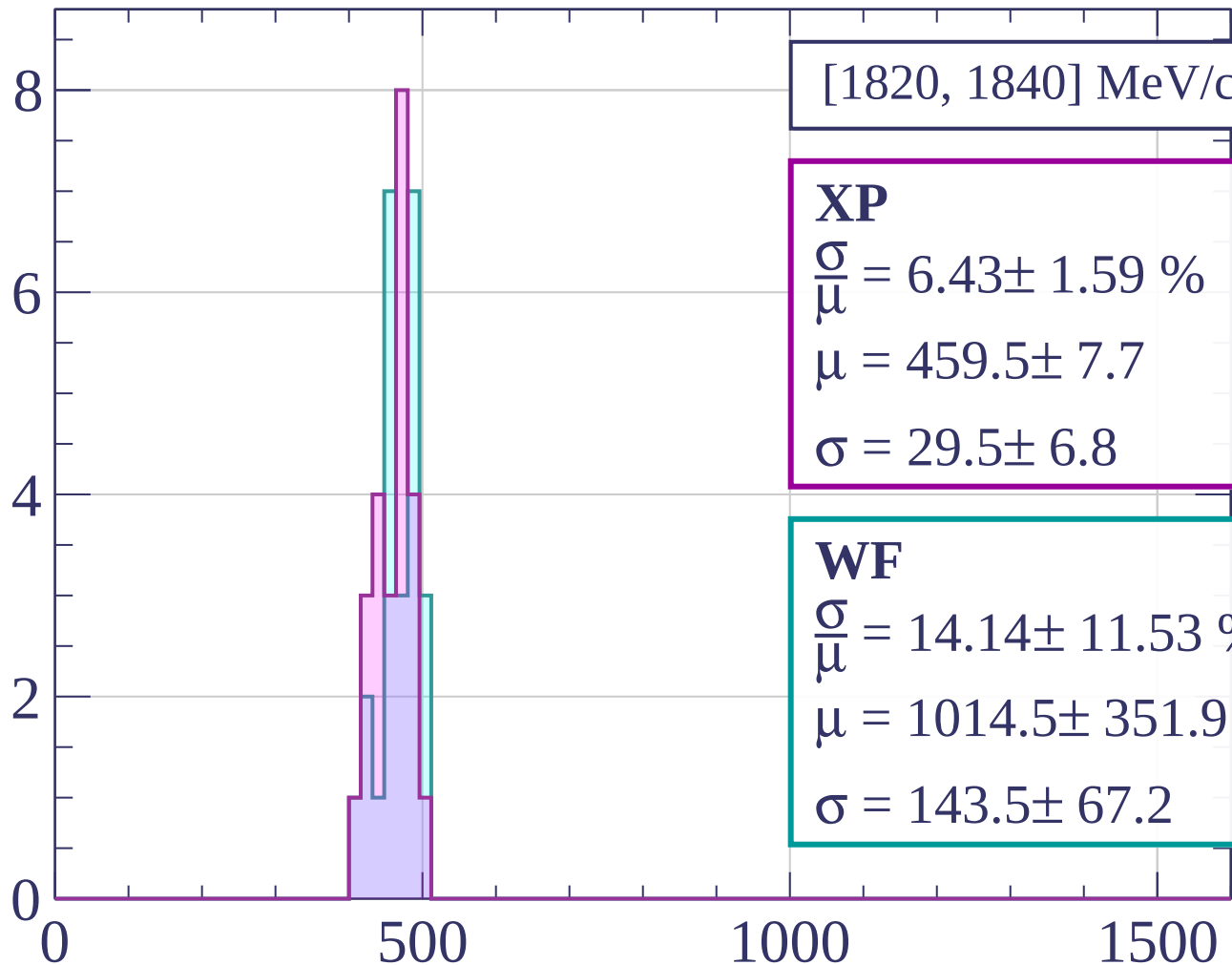


Count



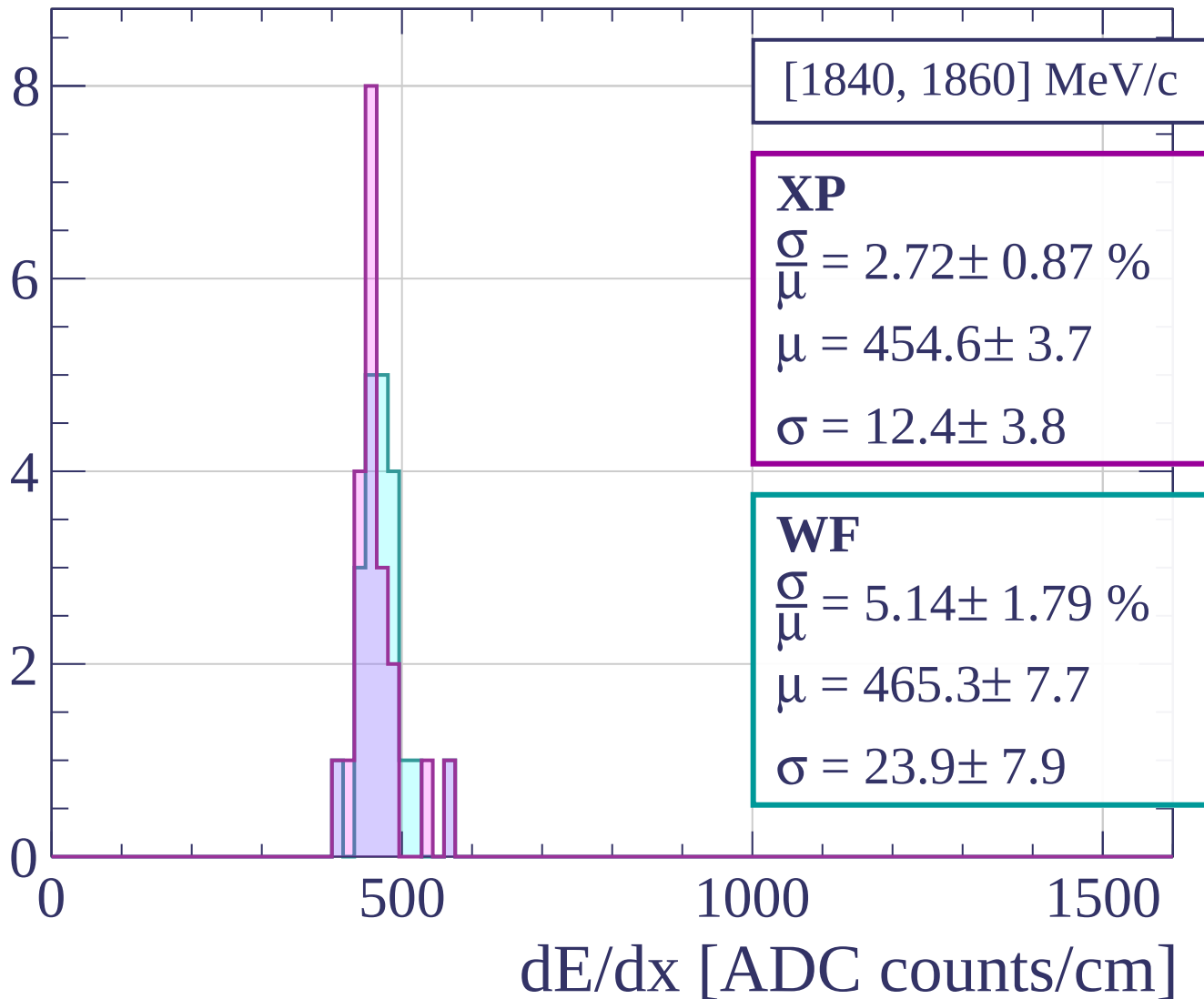
dE/dx [ADC counts/cm]

Count

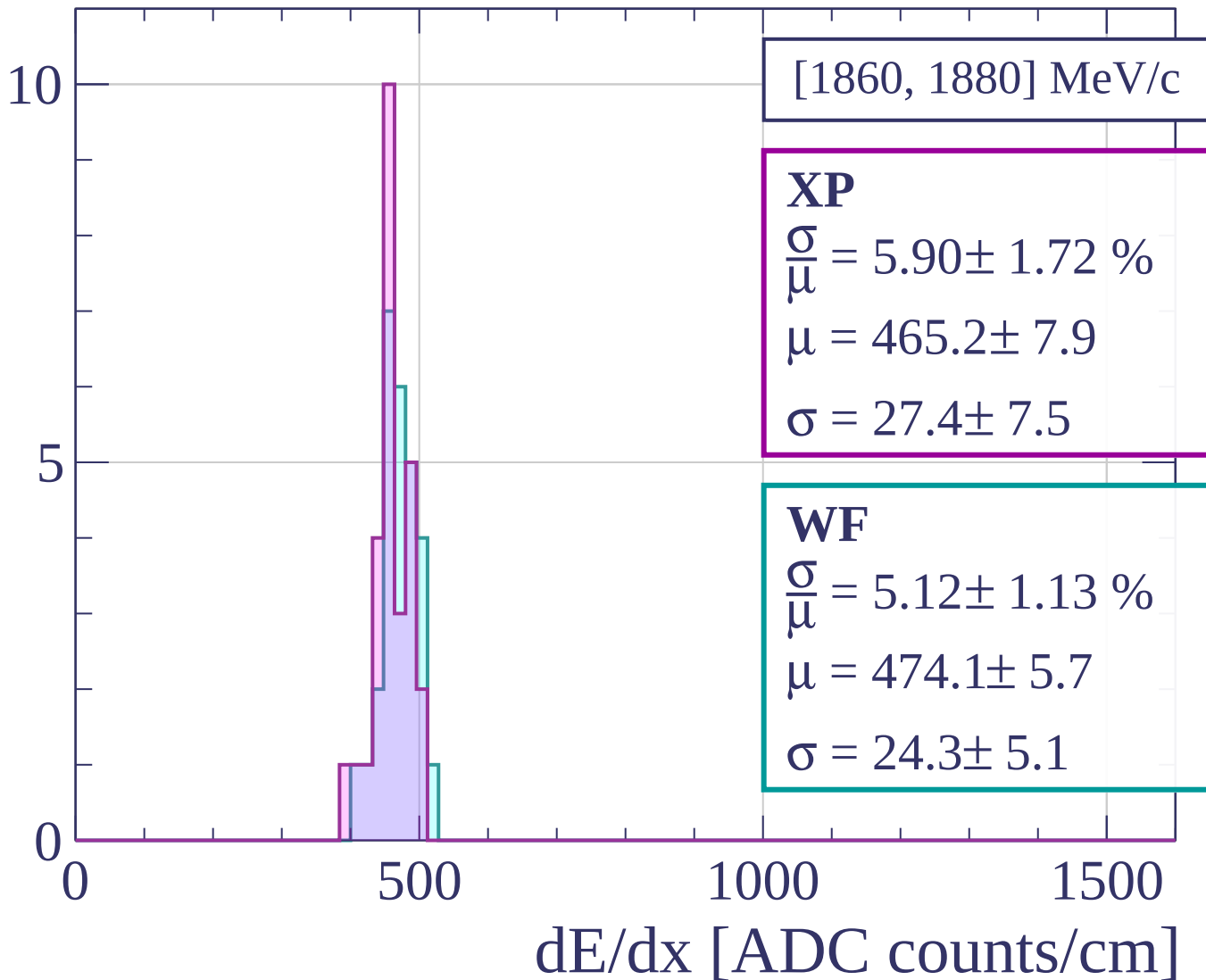


dE/dx [ADC counts/cm]

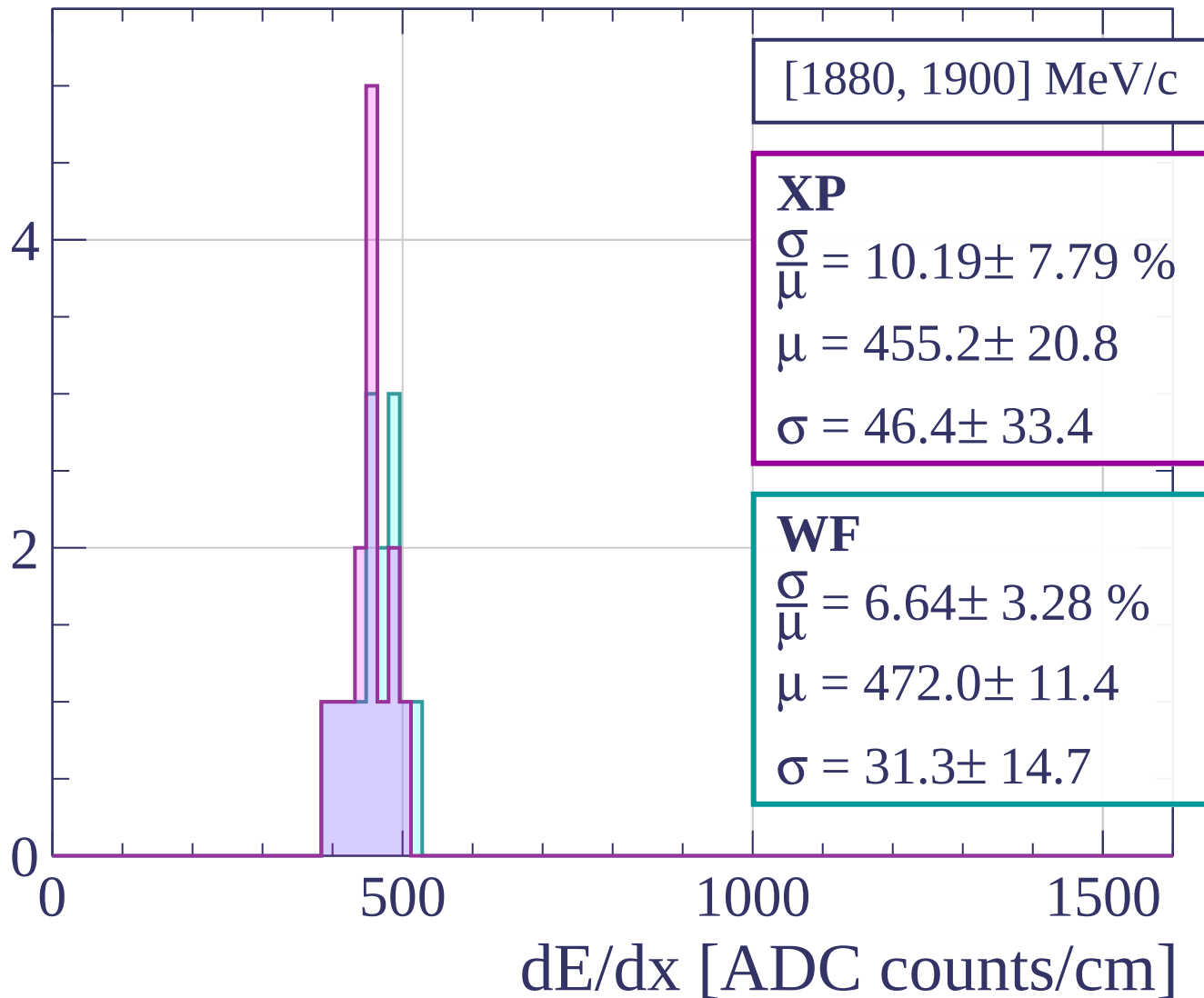
Count



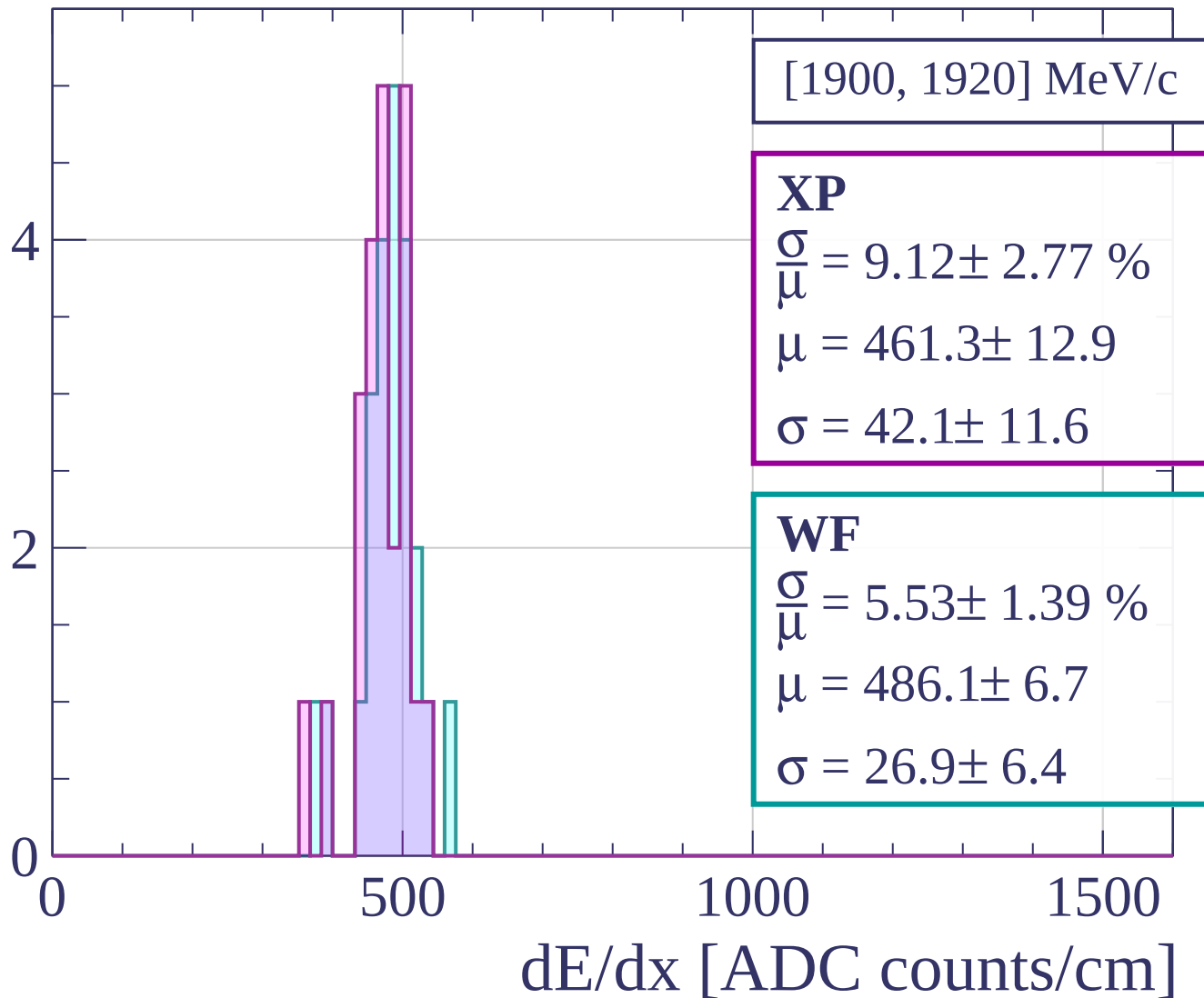
Count



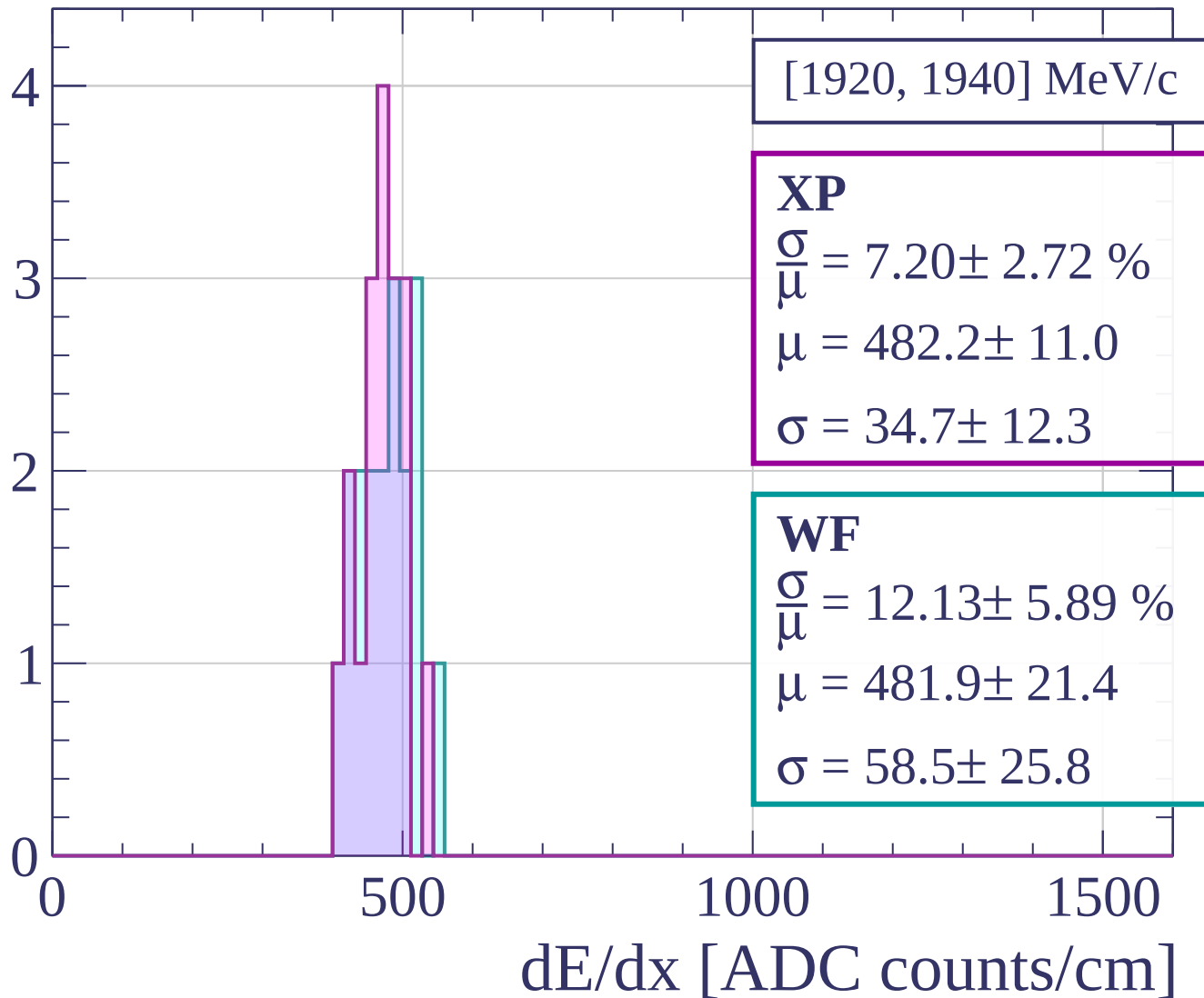
Count



Count



Count



Count

[1940, 1960] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.46 \pm 11.77 \%$$

$$\mu = 460.1 \pm 33.7$$

$$\sigma = 66.5 \pm 49.3$$

WF

$$\frac{\sigma}{\mu} = 17.46 \pm 14.74 \%$$

$$\mu = 454.0 \pm 44.4$$

$$\sigma = 79.3 \pm 59.2$$

2.0

1.5

1.0

0.5

0.0

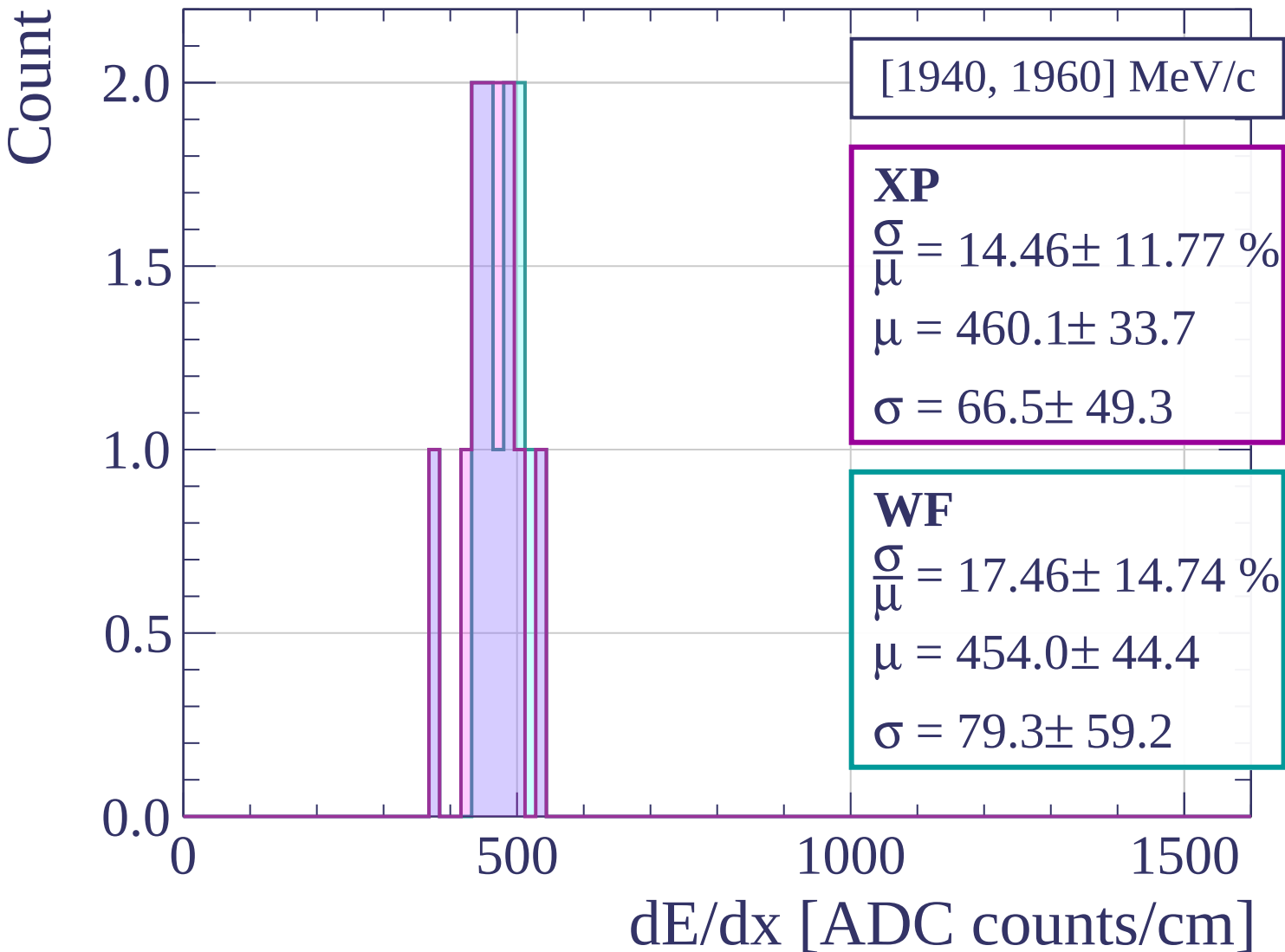
0

500

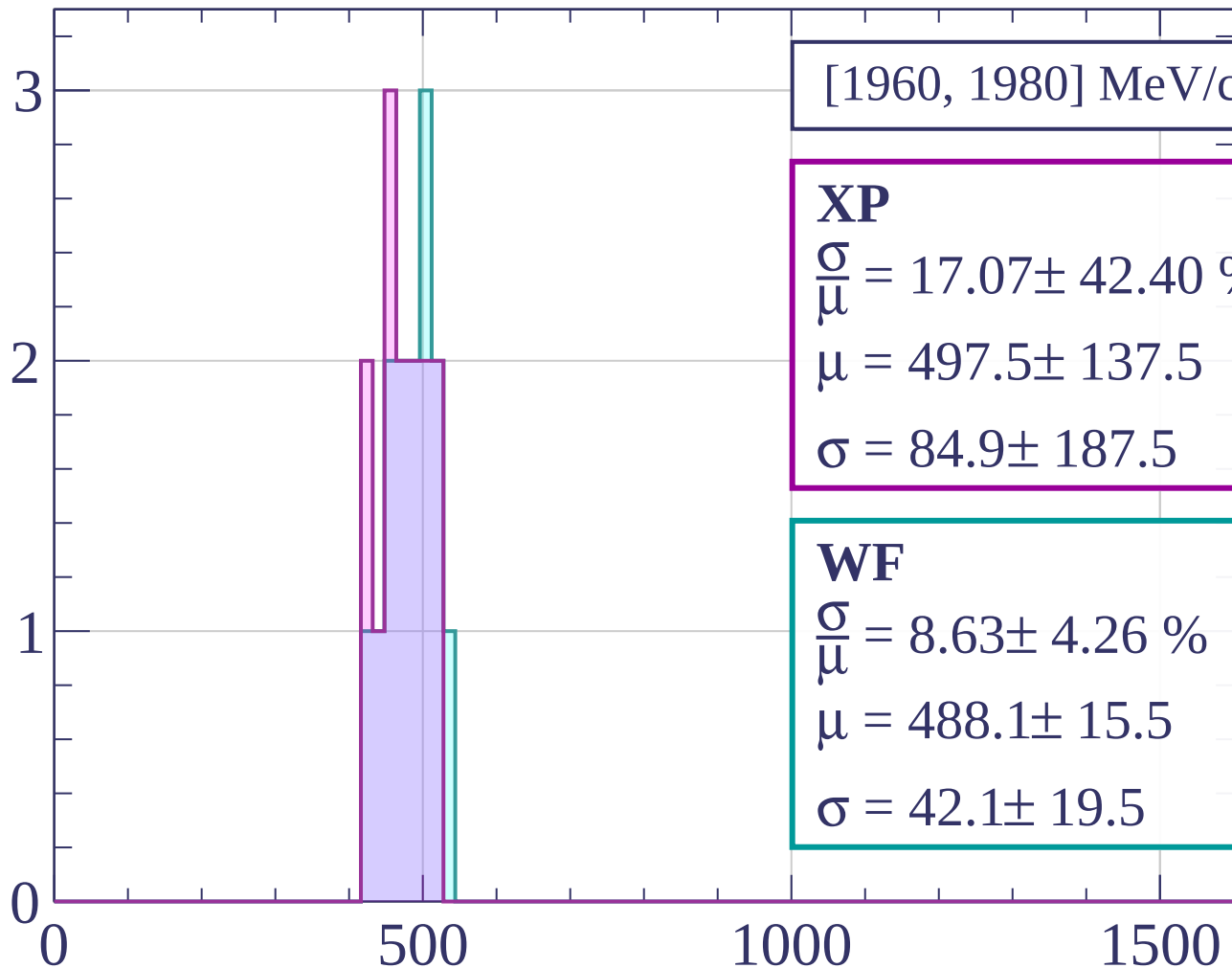
1000

1500

dE/dx [ADC counts/cm]

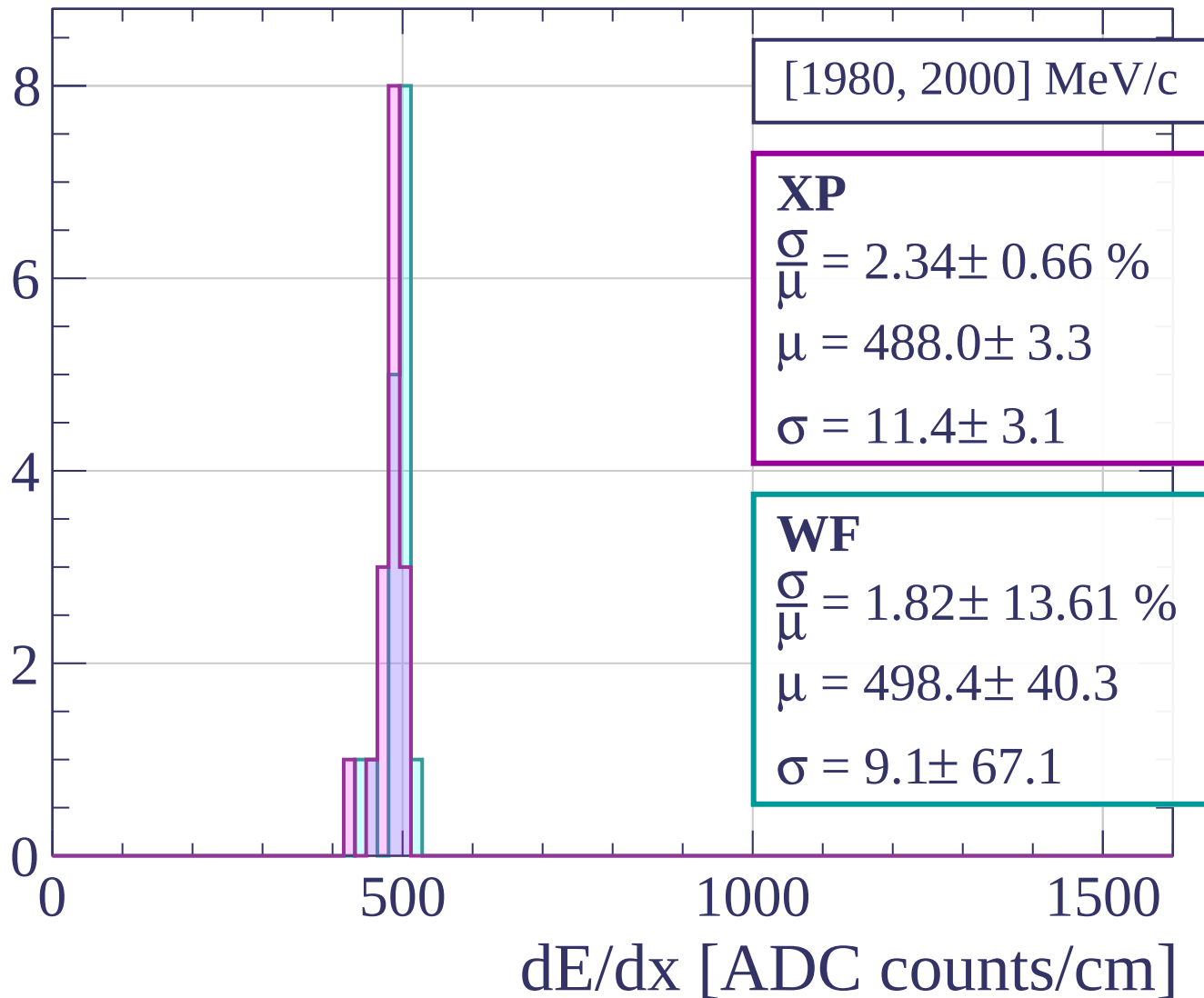


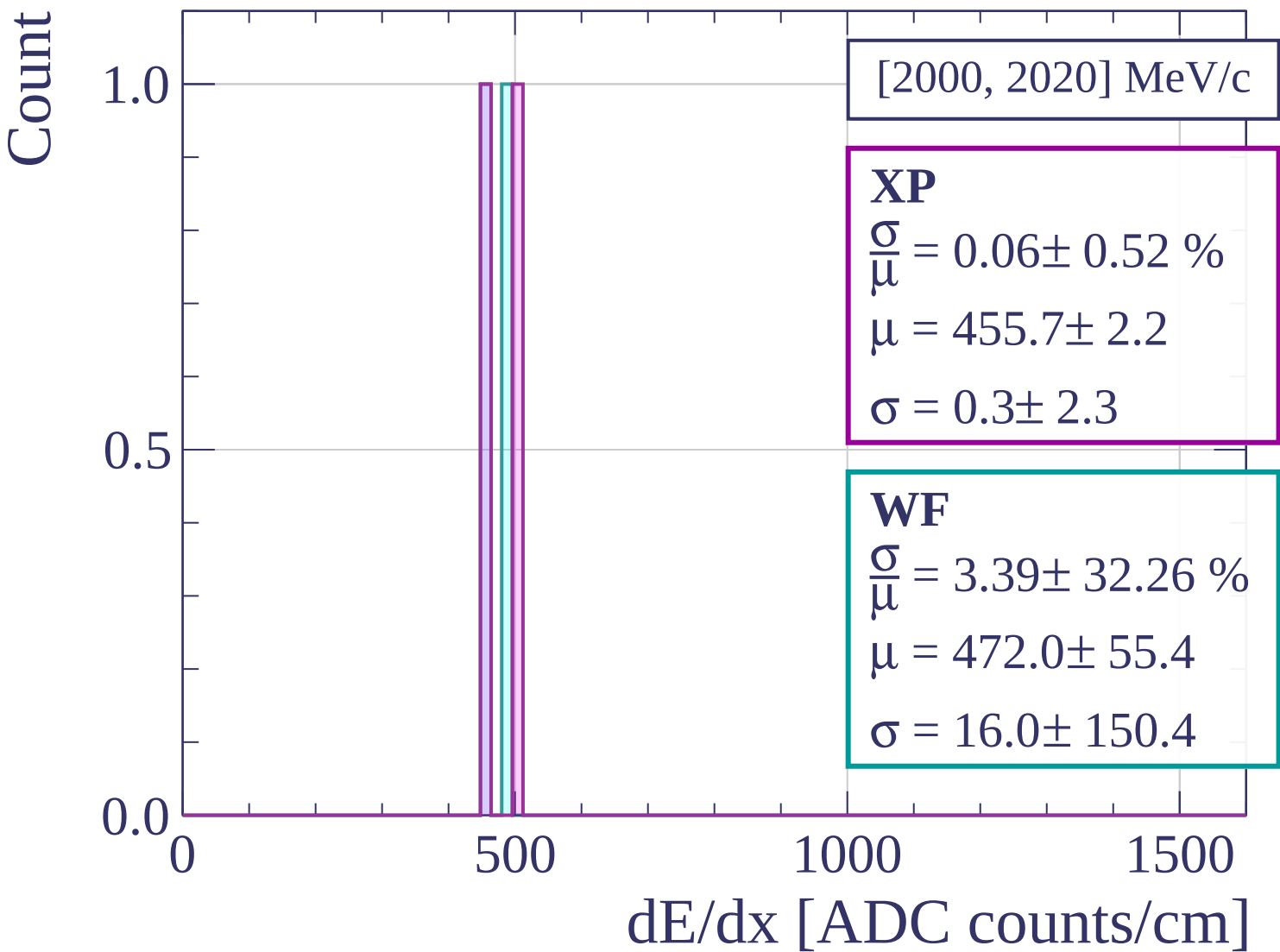
Count



dE/dx [ADC counts/cm]

Count





Count

$\varphi \in [0^\circ, 3^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.58 \pm 0.30 \%$$

$$\mu = 407.9 \pm 1.0$$

$$\sigma = 22.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.98 \pm 0.27 \%$$

$$\mu = 417.0 \pm 1.0$$

$$\sigma = 24.9 \pm 1.1$$

200

100

0

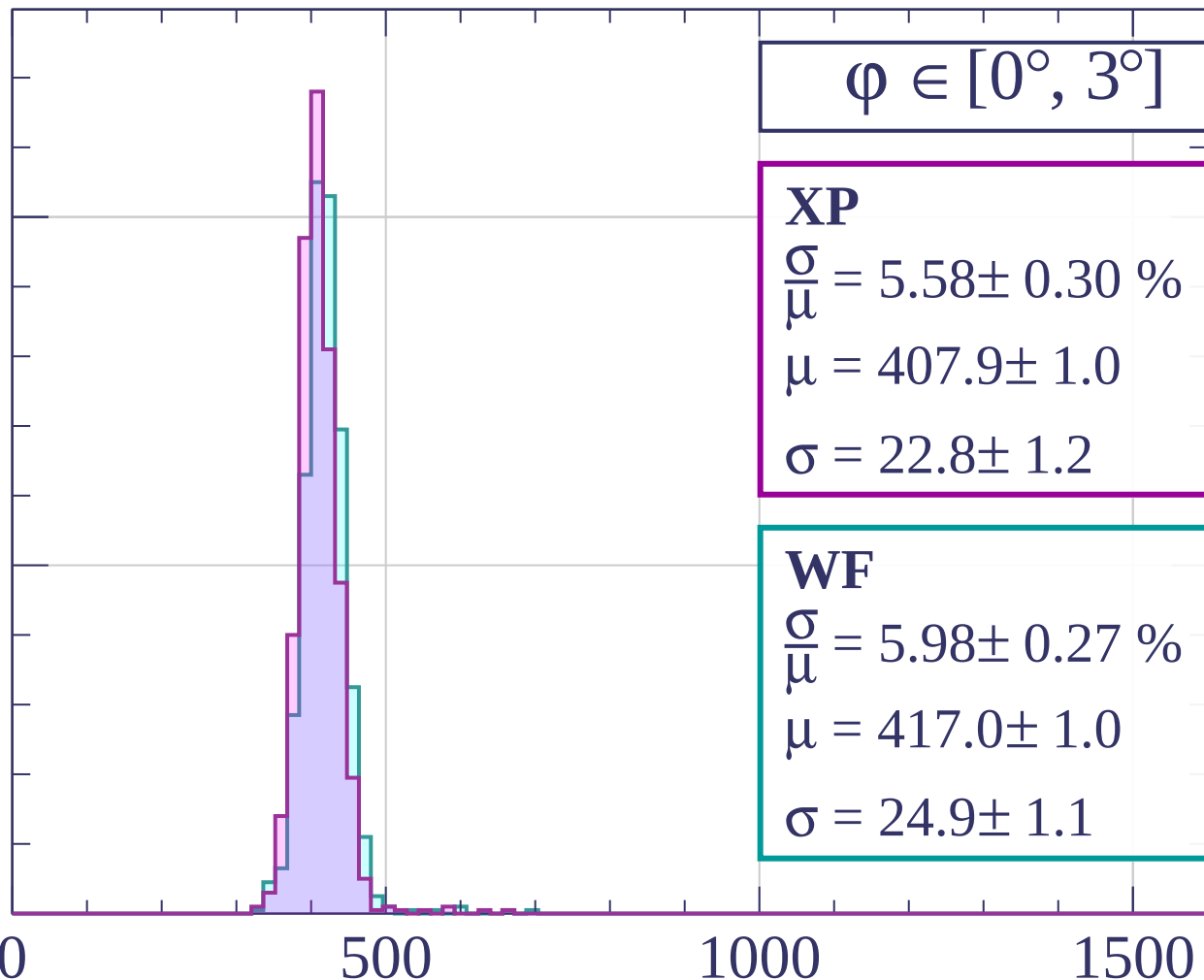
0

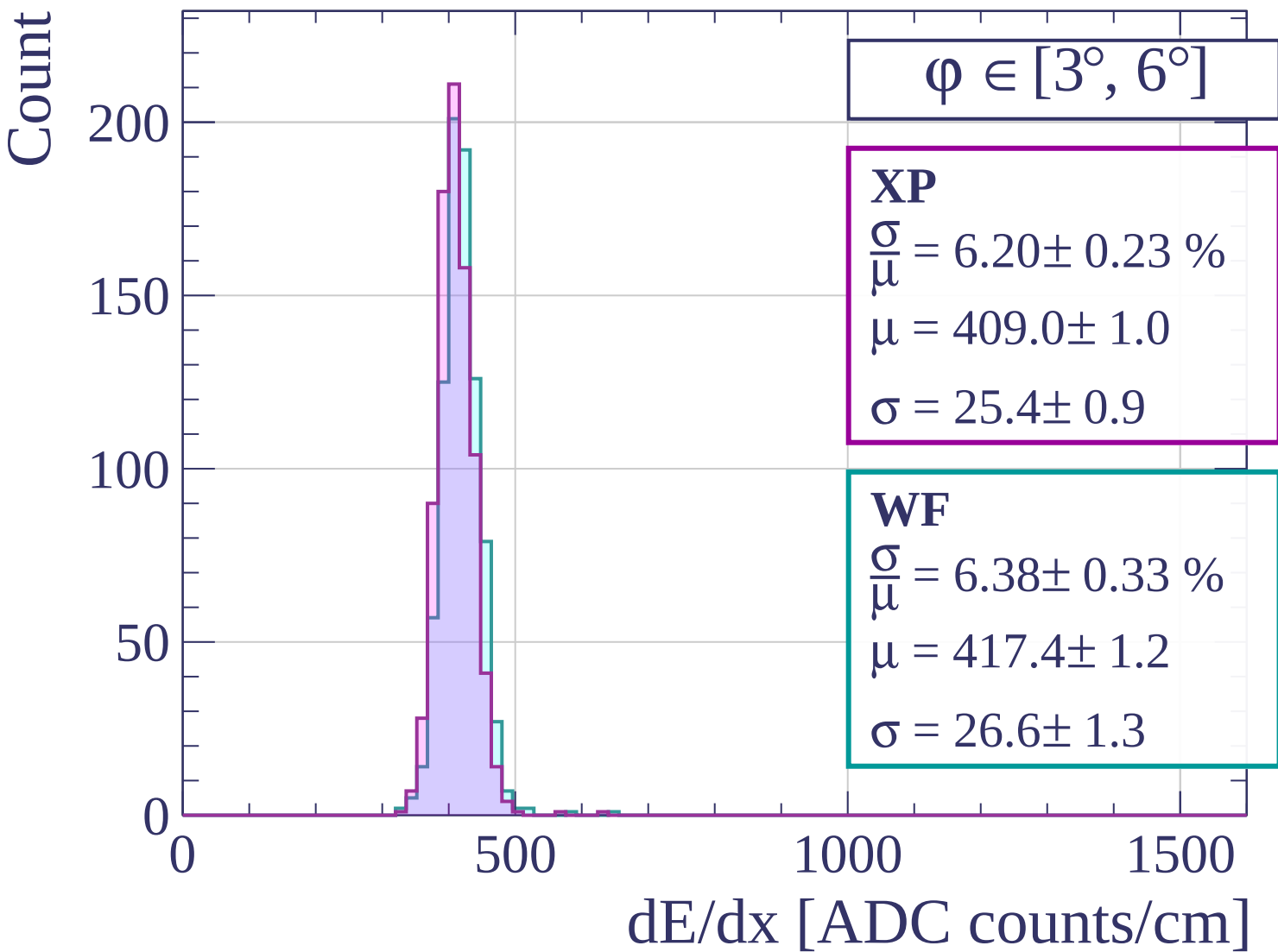
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\varphi \in [6^\circ, 9^\circ]$

XP

$$\frac{\sigma}{\mu} = 6.36 \pm 0.25 \%$$

$$\mu = 409.2 \pm 1.1$$

$$\sigma = 26.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.63 \pm 0.29 \%$$

$$\mu = 417.5 \pm 1.2$$

$$\sigma = 27.7 \pm 1.1$$

150

100

50

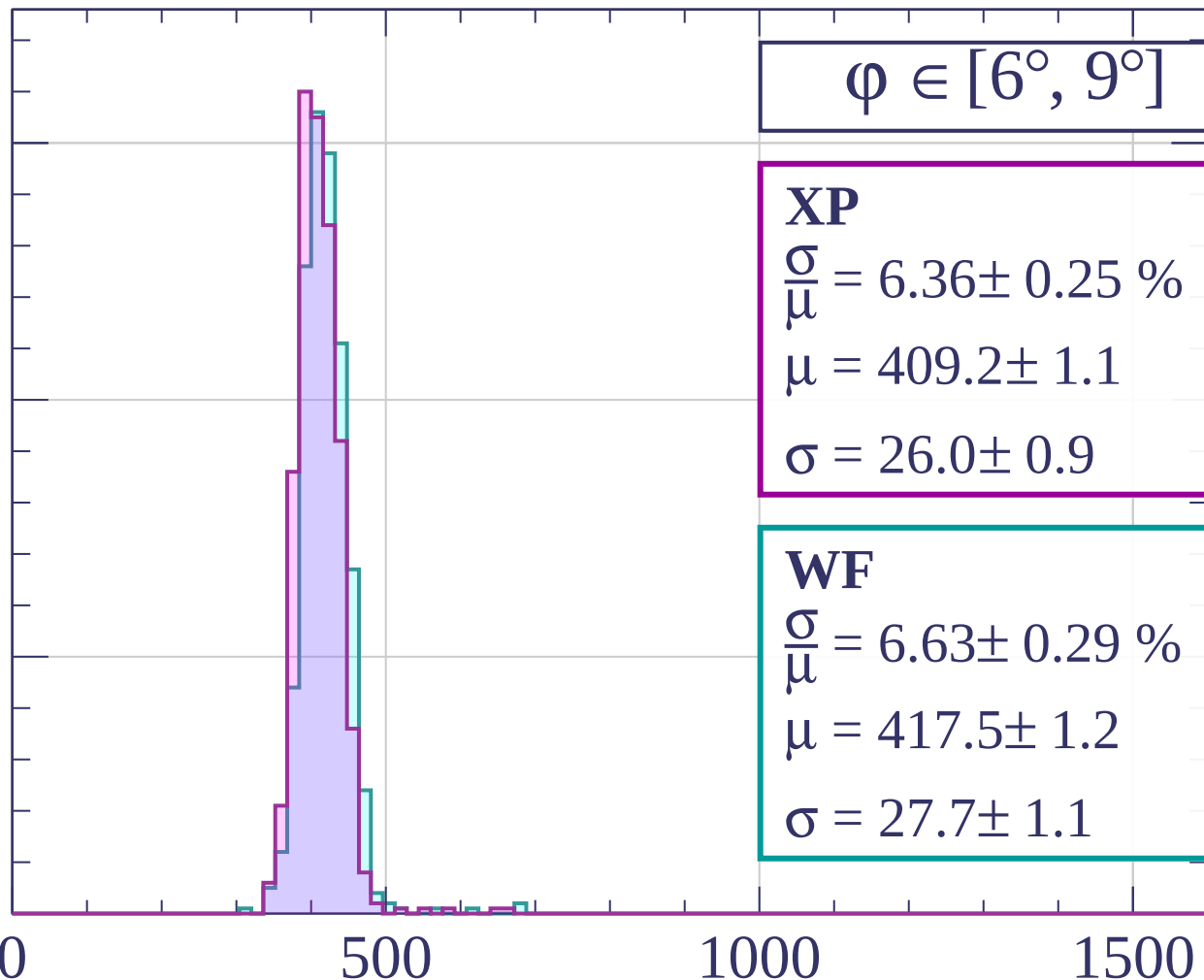
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [9^\circ, 12^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.71 \pm 0.38 \%$$

$$\mu = 408.7 \pm 1.2$$

$$\sigma = 23.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 5.88 \pm 0.30 \%$$

$$\mu = 418.9 \pm 1.2$$

$$\sigma = 24.6 \pm 1.2$$

150

100

50

0

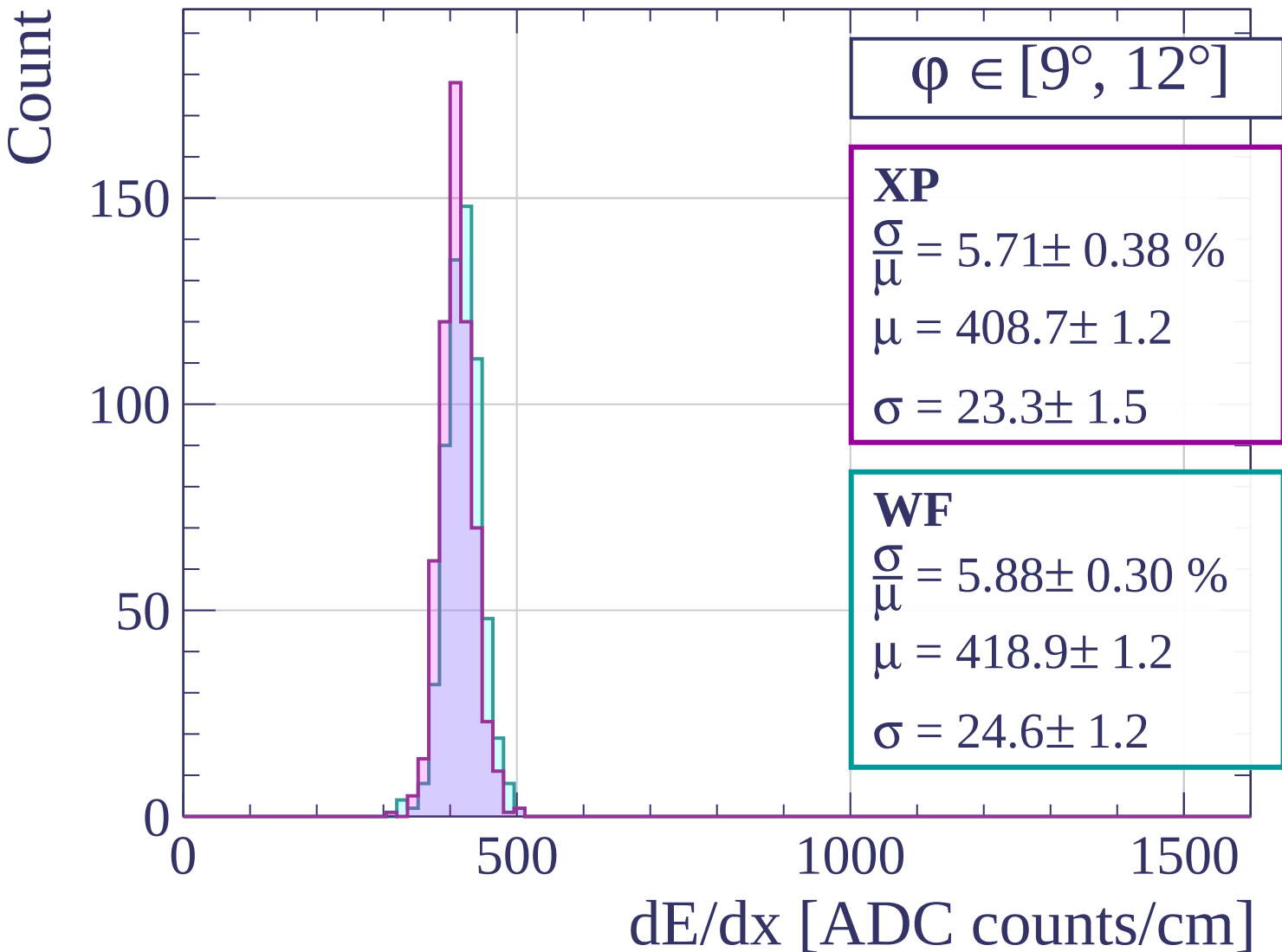
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.66 \pm 0.30 \%$$

$$\mu = 408.0 \pm 1.3$$

$$\sigma = 23.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.97 \pm 0.35 \%$$

$$\mu = 417.6 \pm 1.4$$

$$\sigma = 24.9 \pm 1.4$$

100

50

0

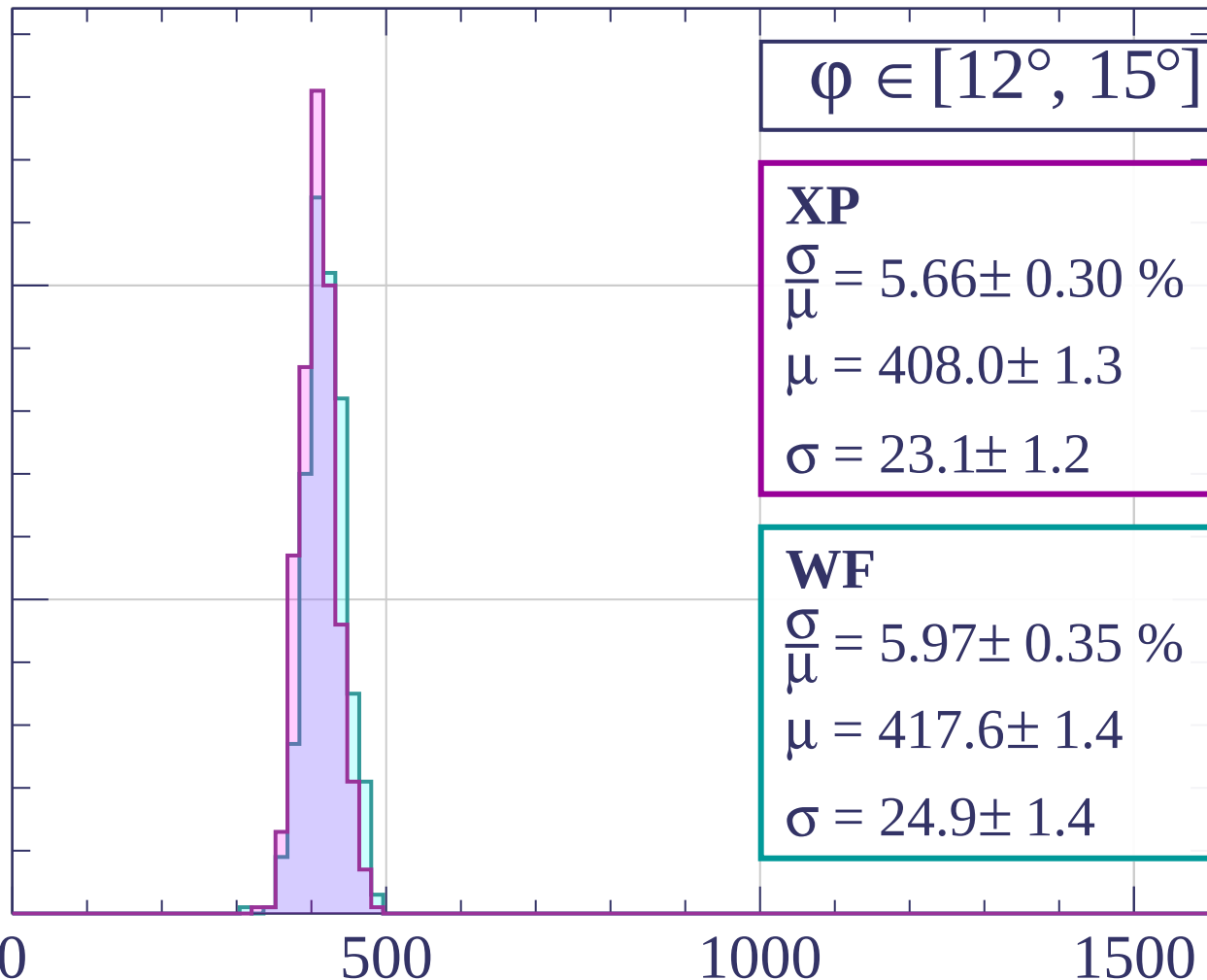
0

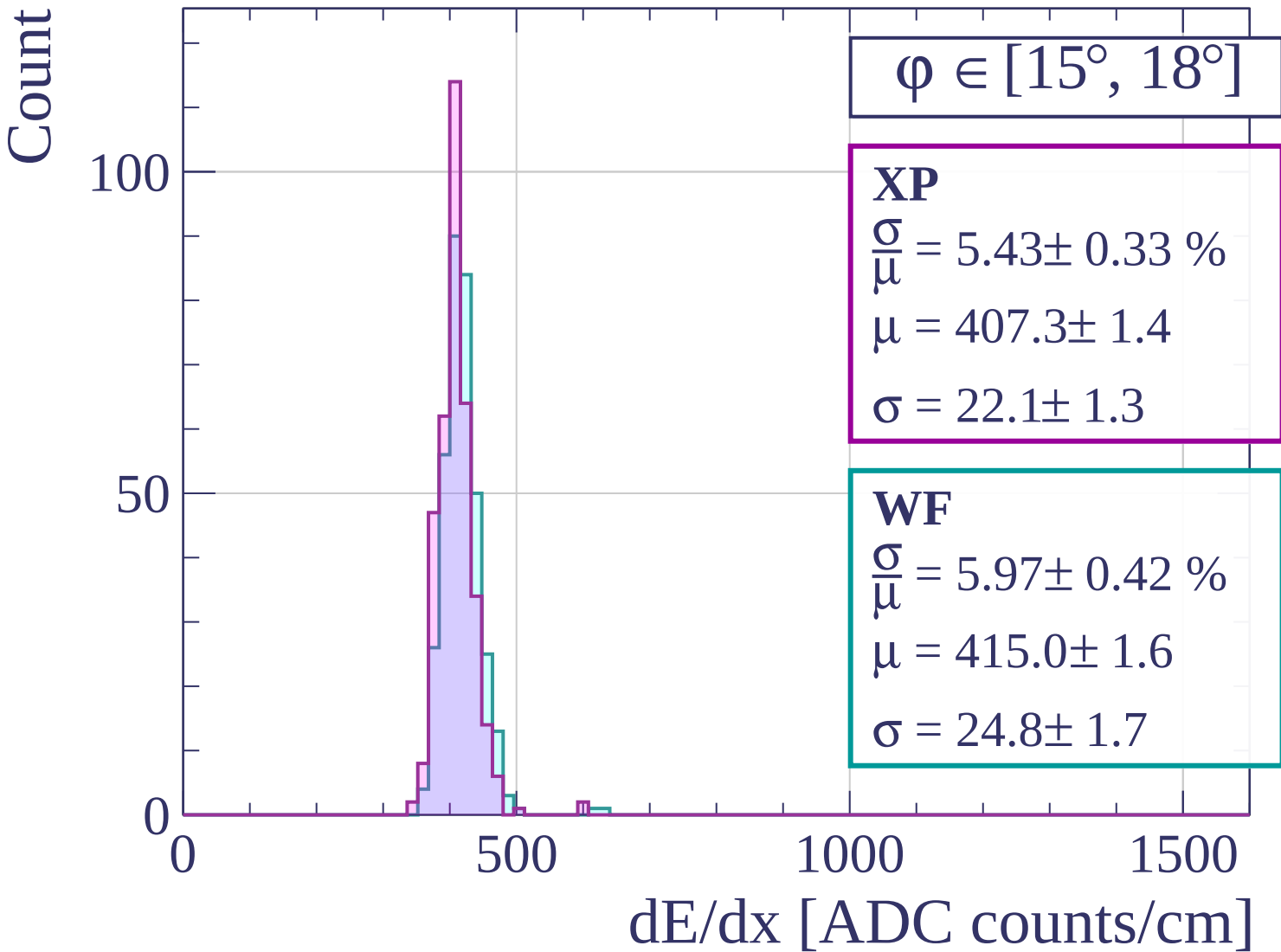
500

1000

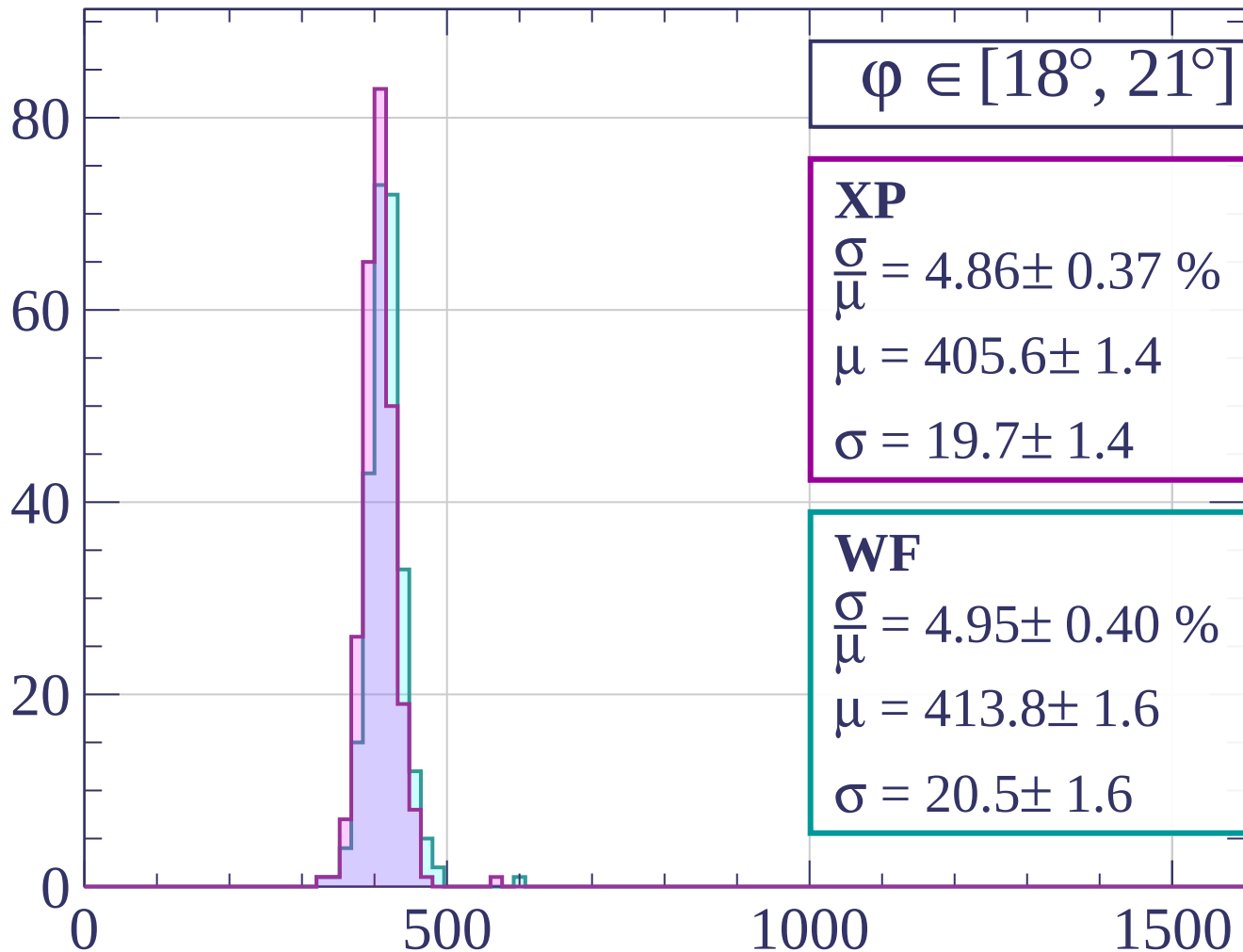
1500

dE/dx [ADC counts/cm]



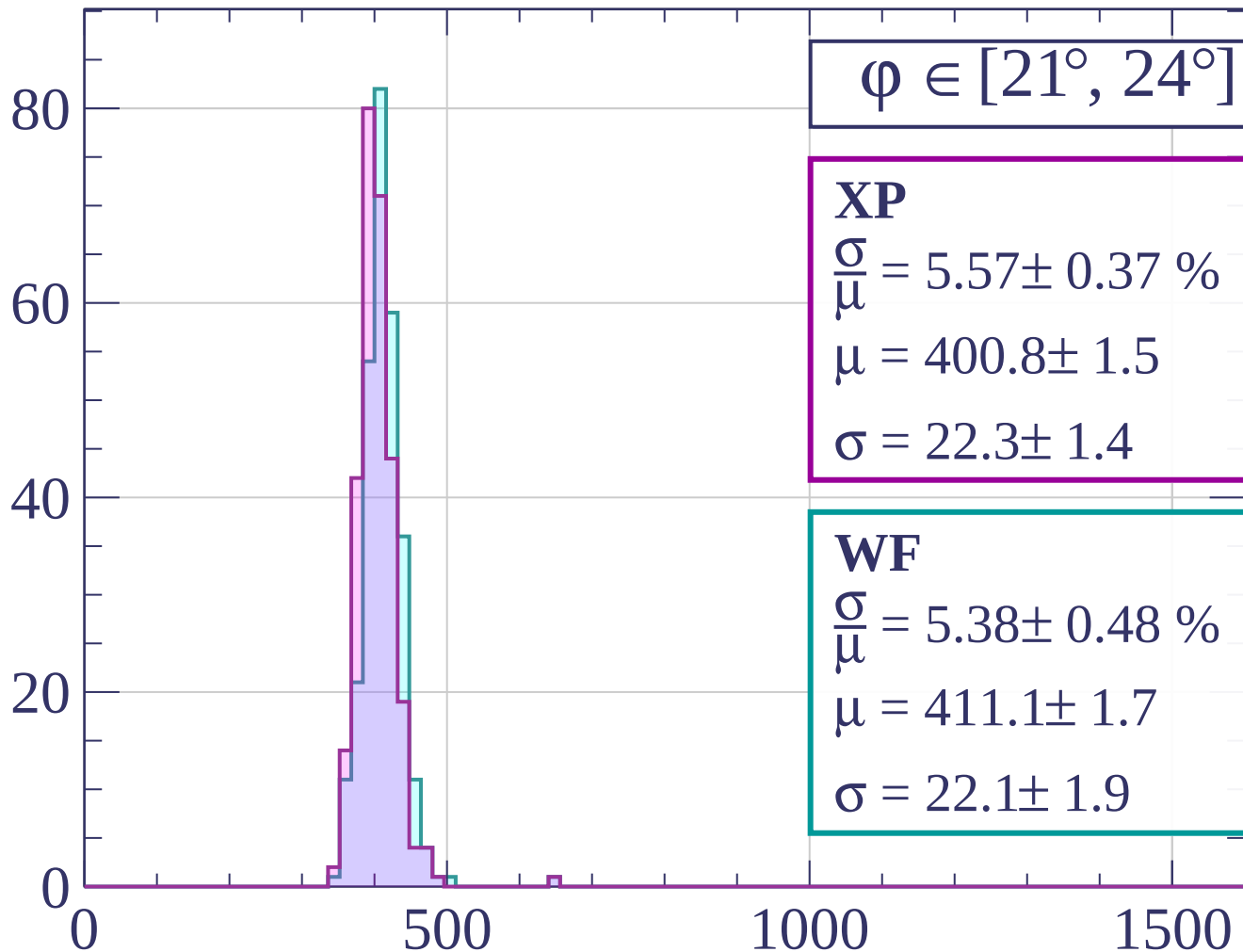


Count



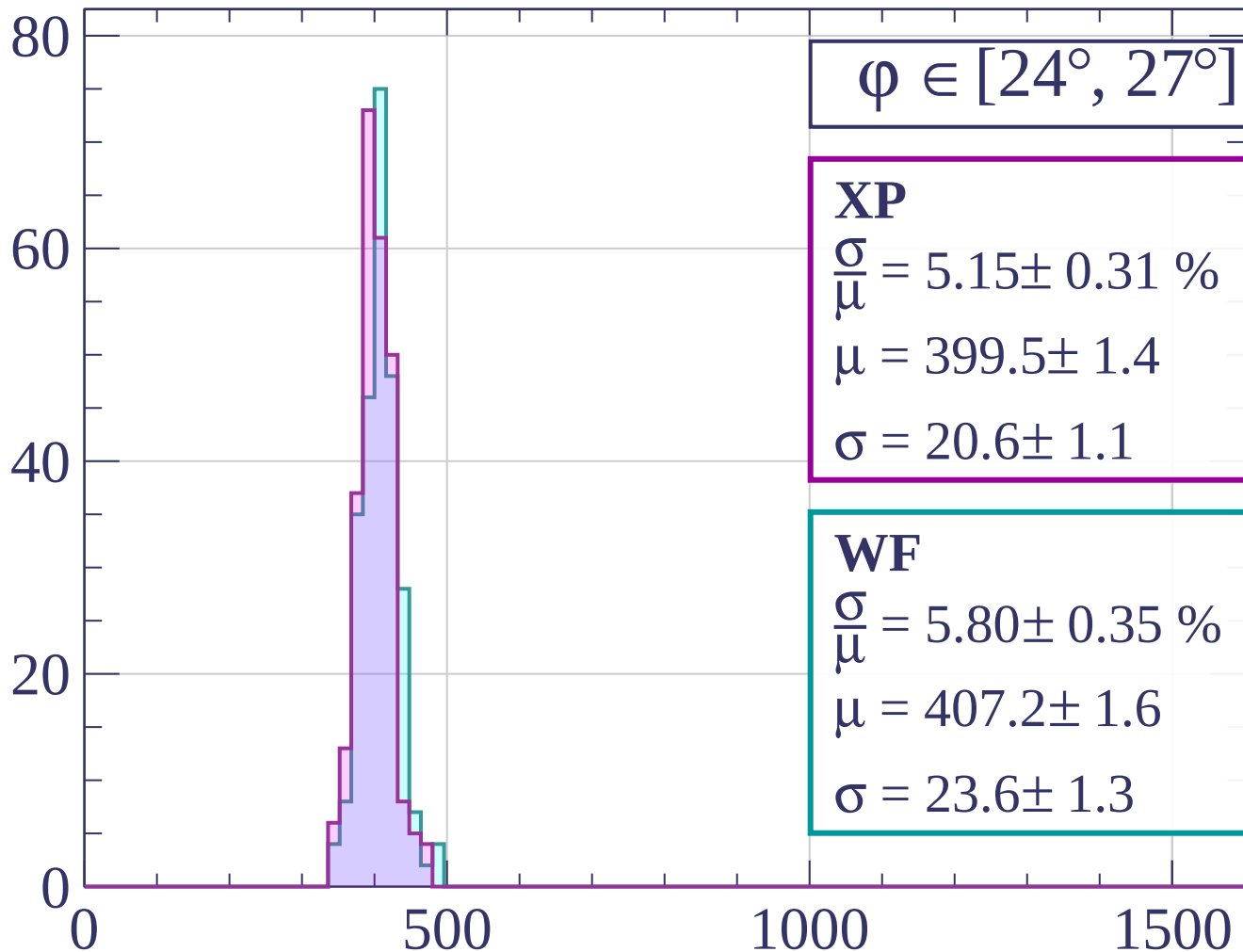
dE/dx [ADC counts/cm]

Count



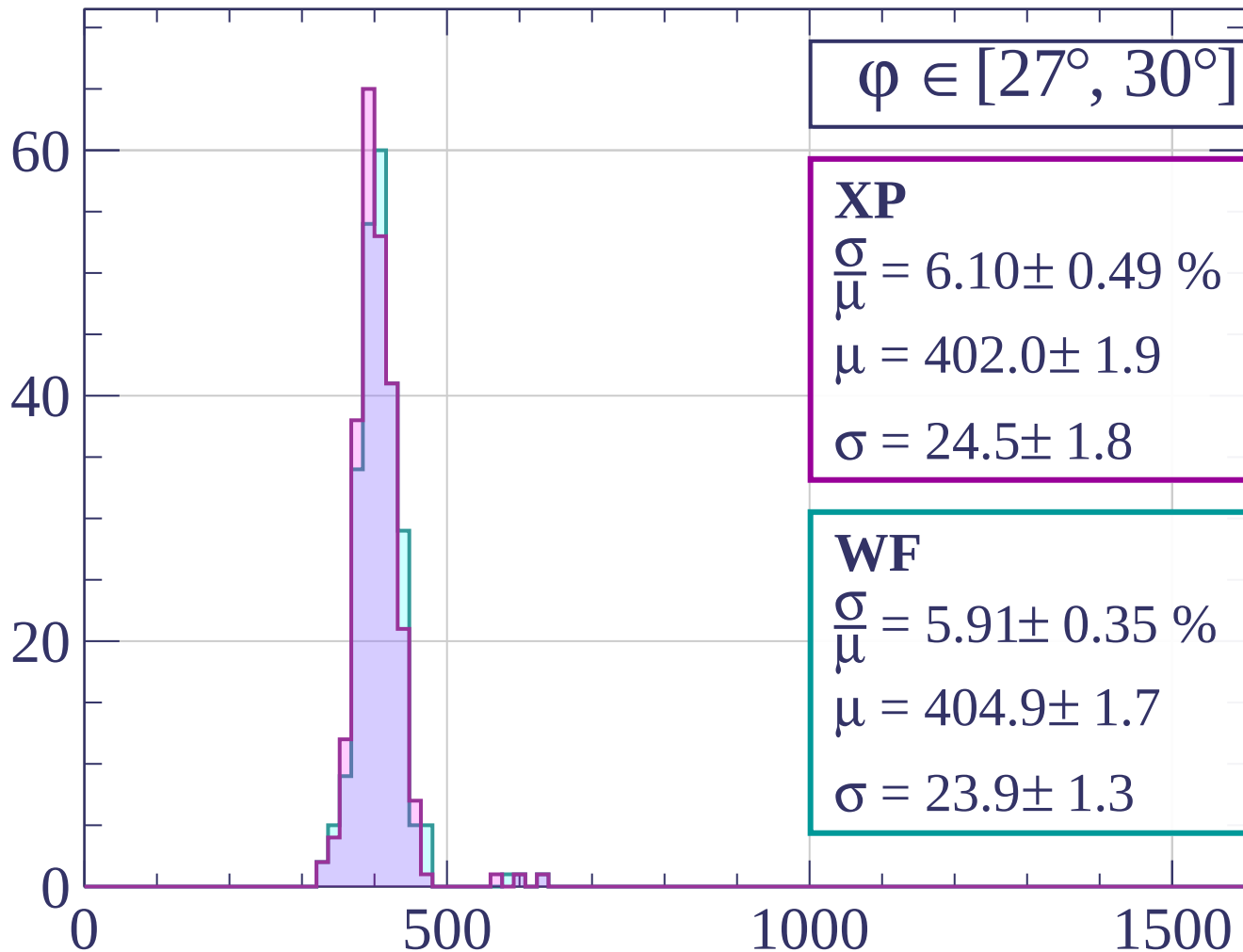
dE/dx [ADC counts/cm]

Count



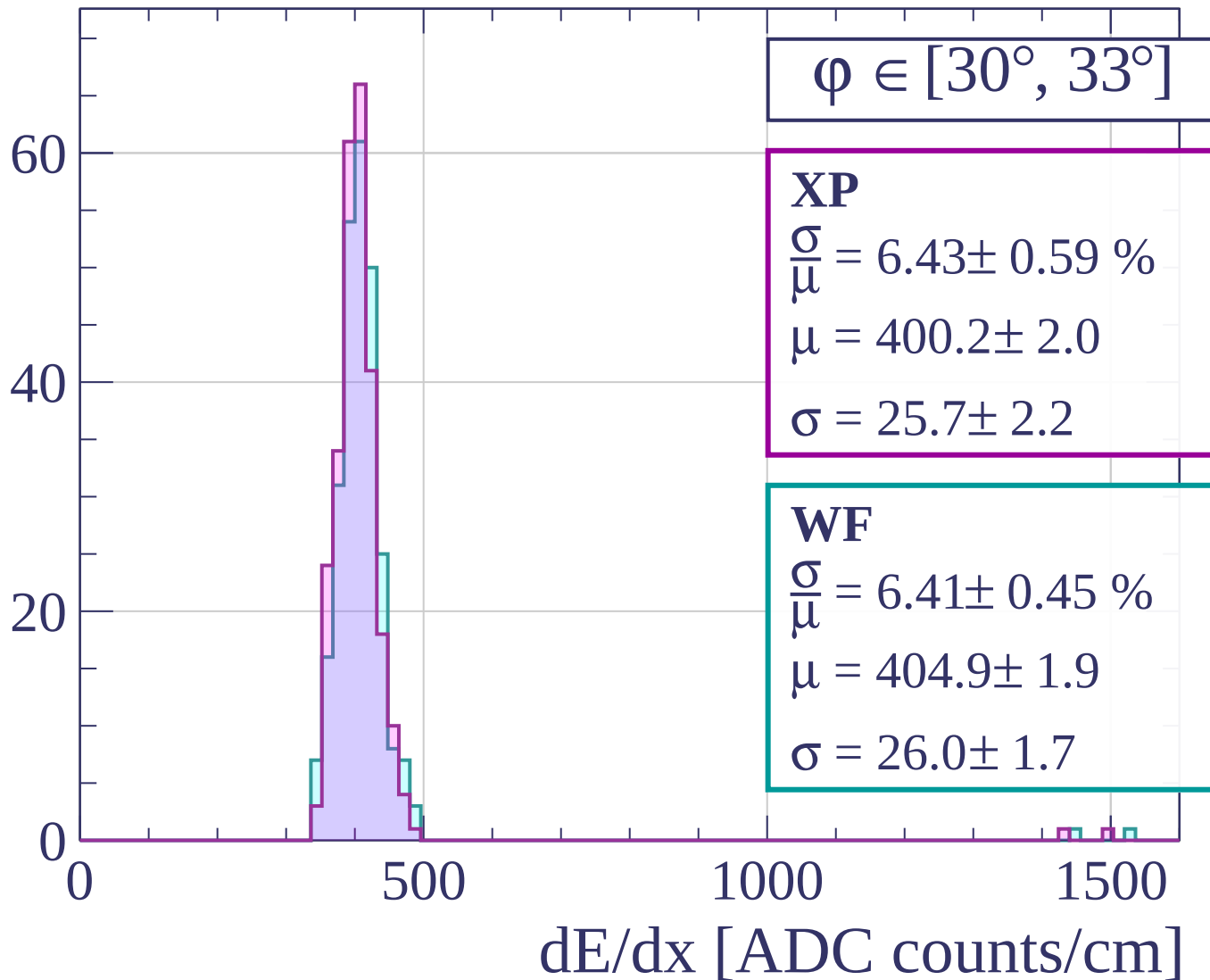
dE/dx [ADC counts/cm]

Count

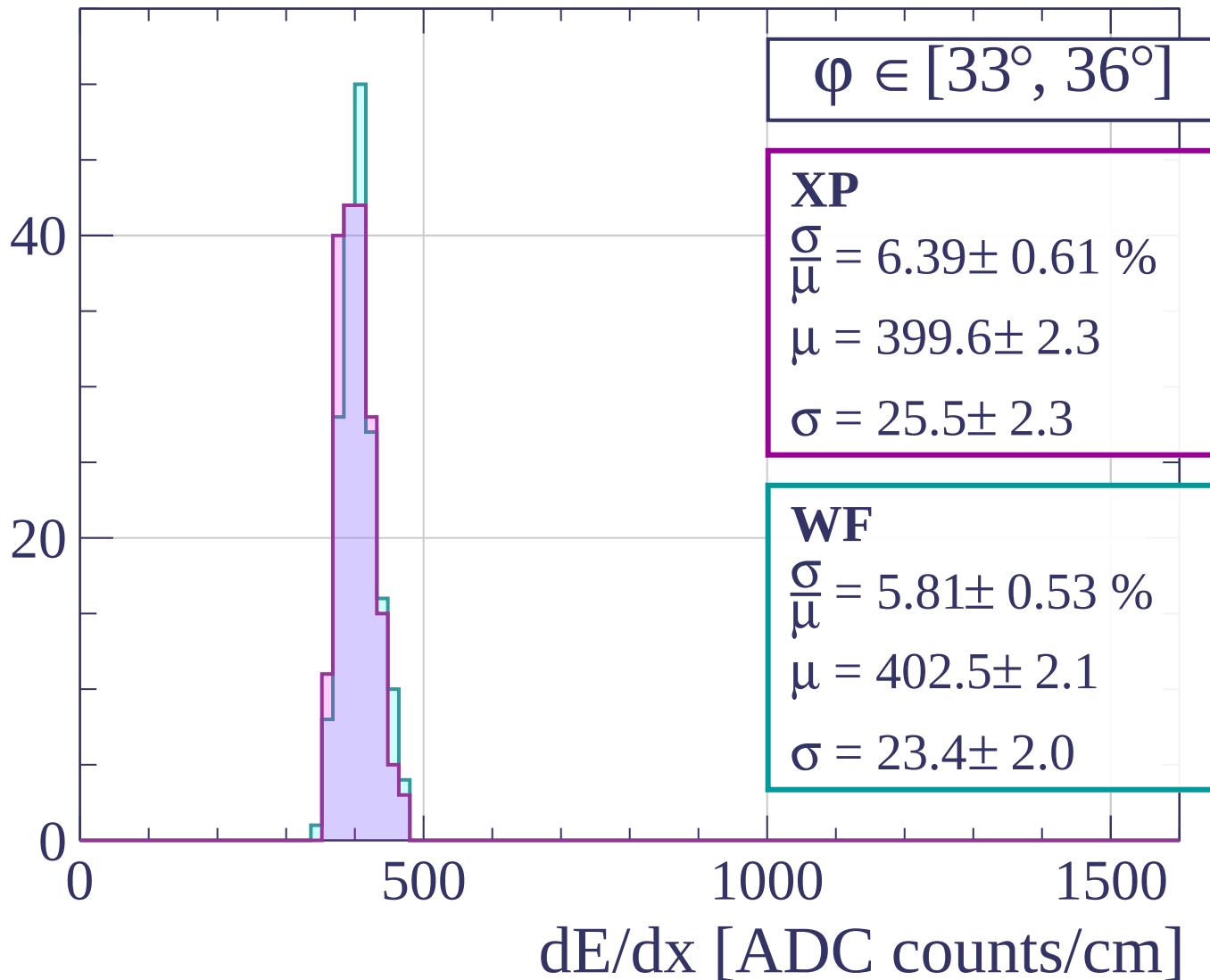


dE/dx [ADC counts/cm]

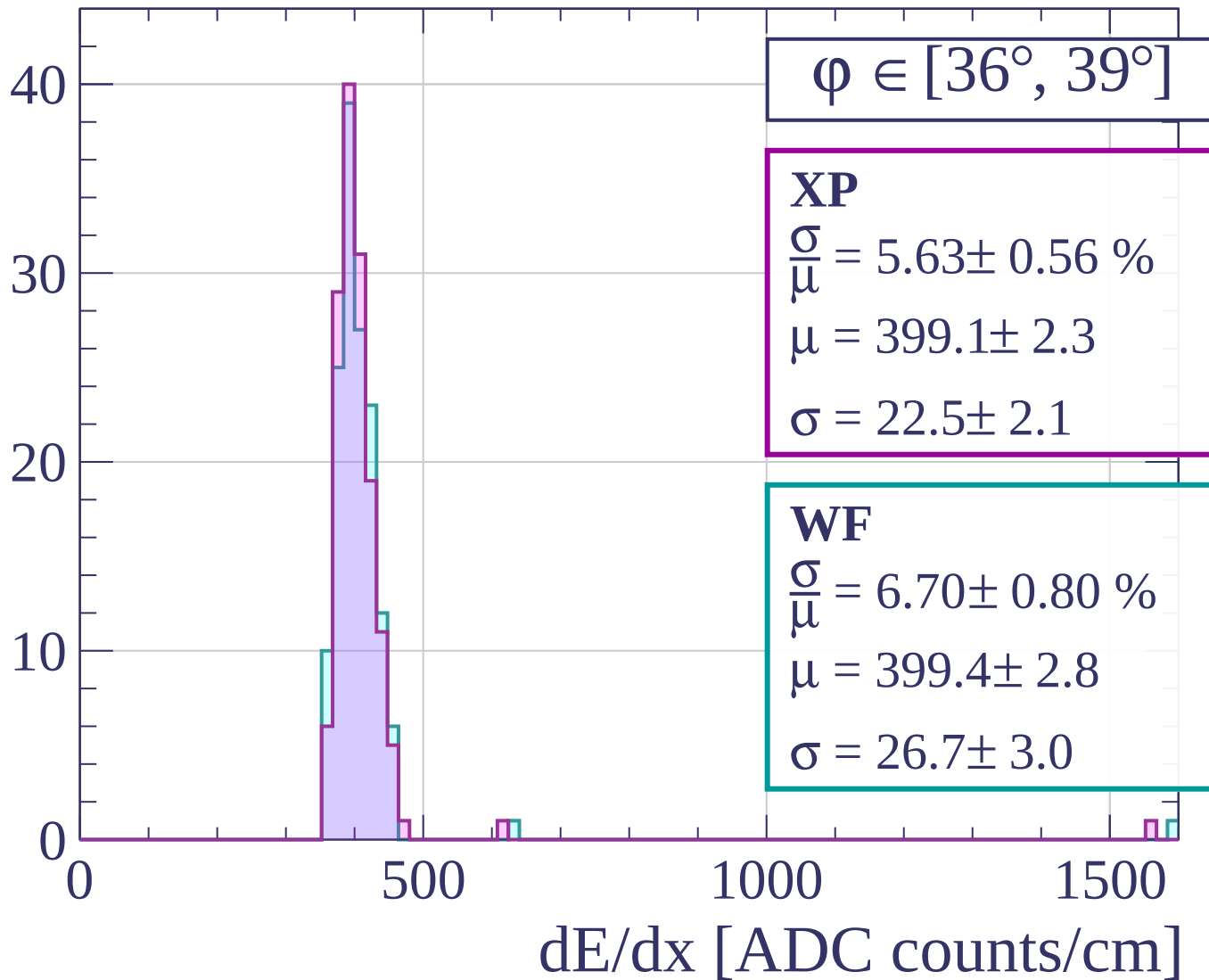
Count



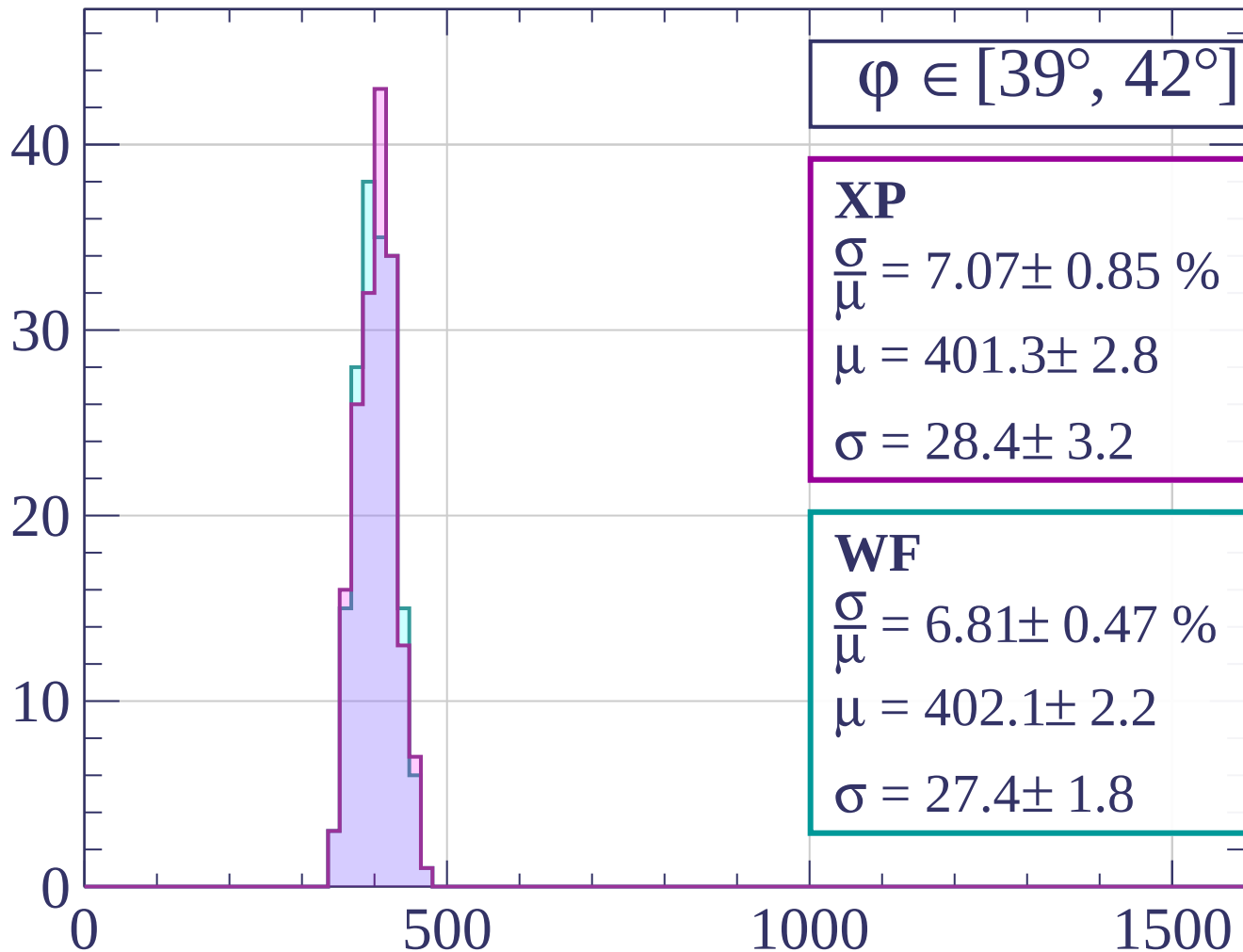
Count



Count

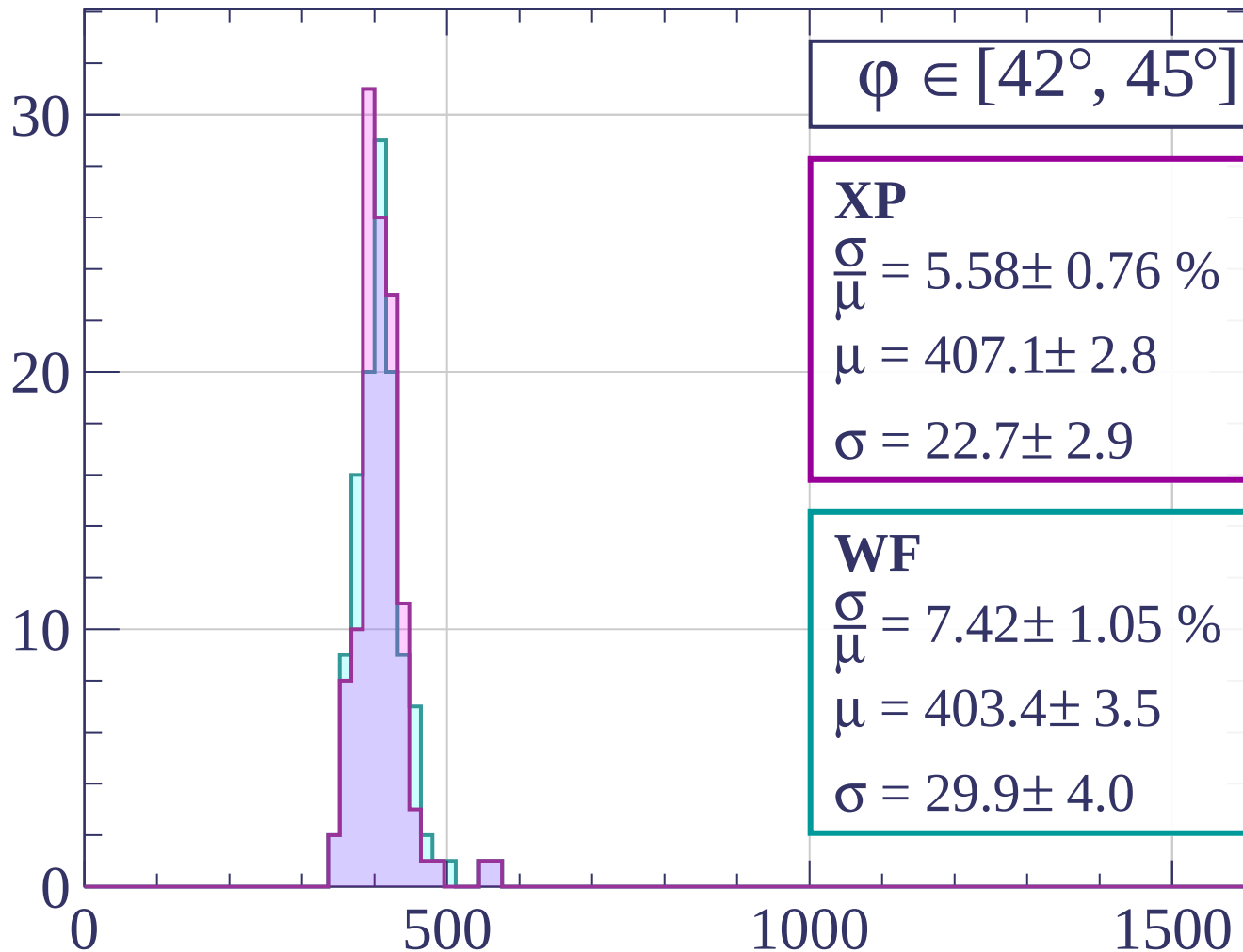


Count



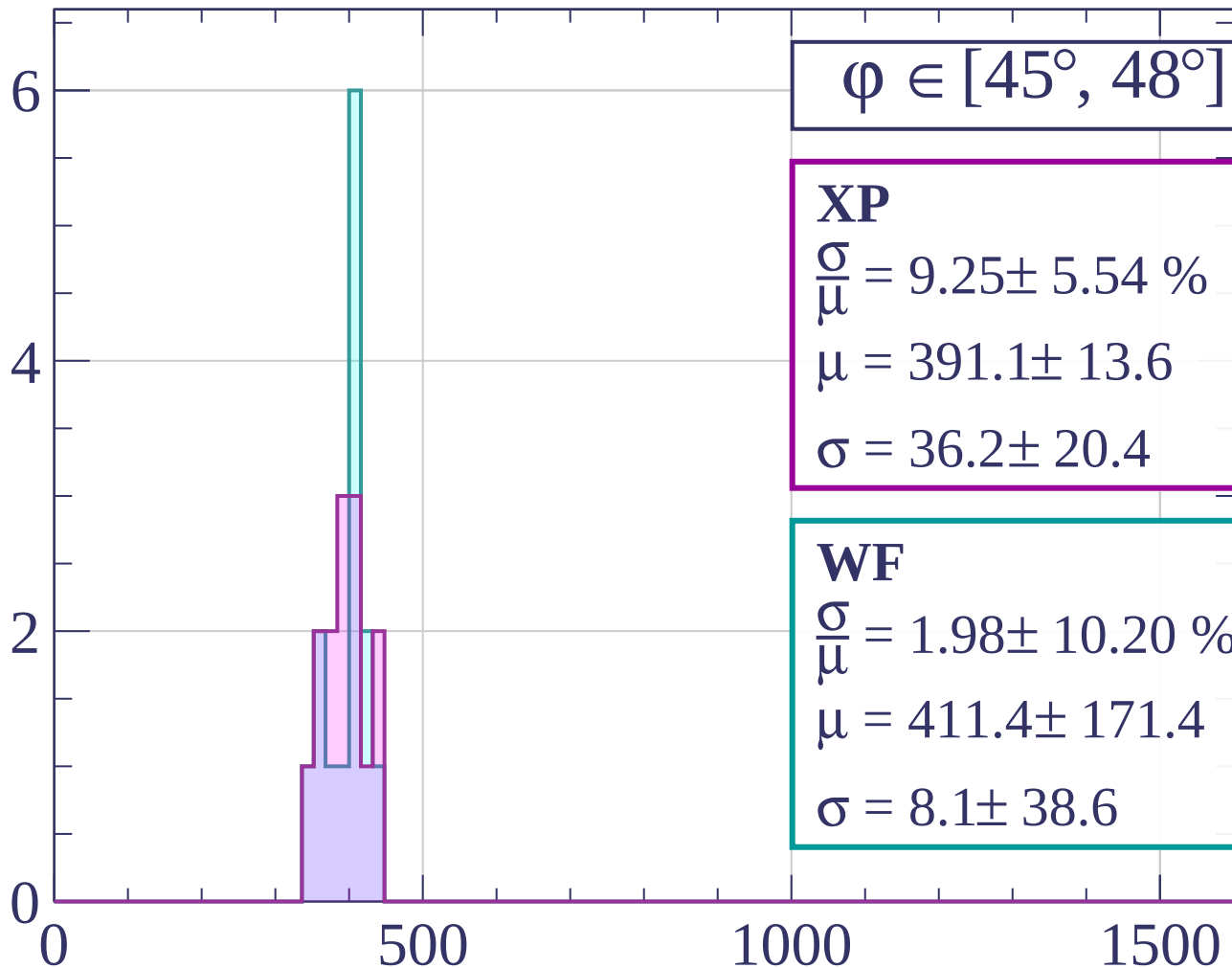
dE/dx [ADC counts/cm]

Count

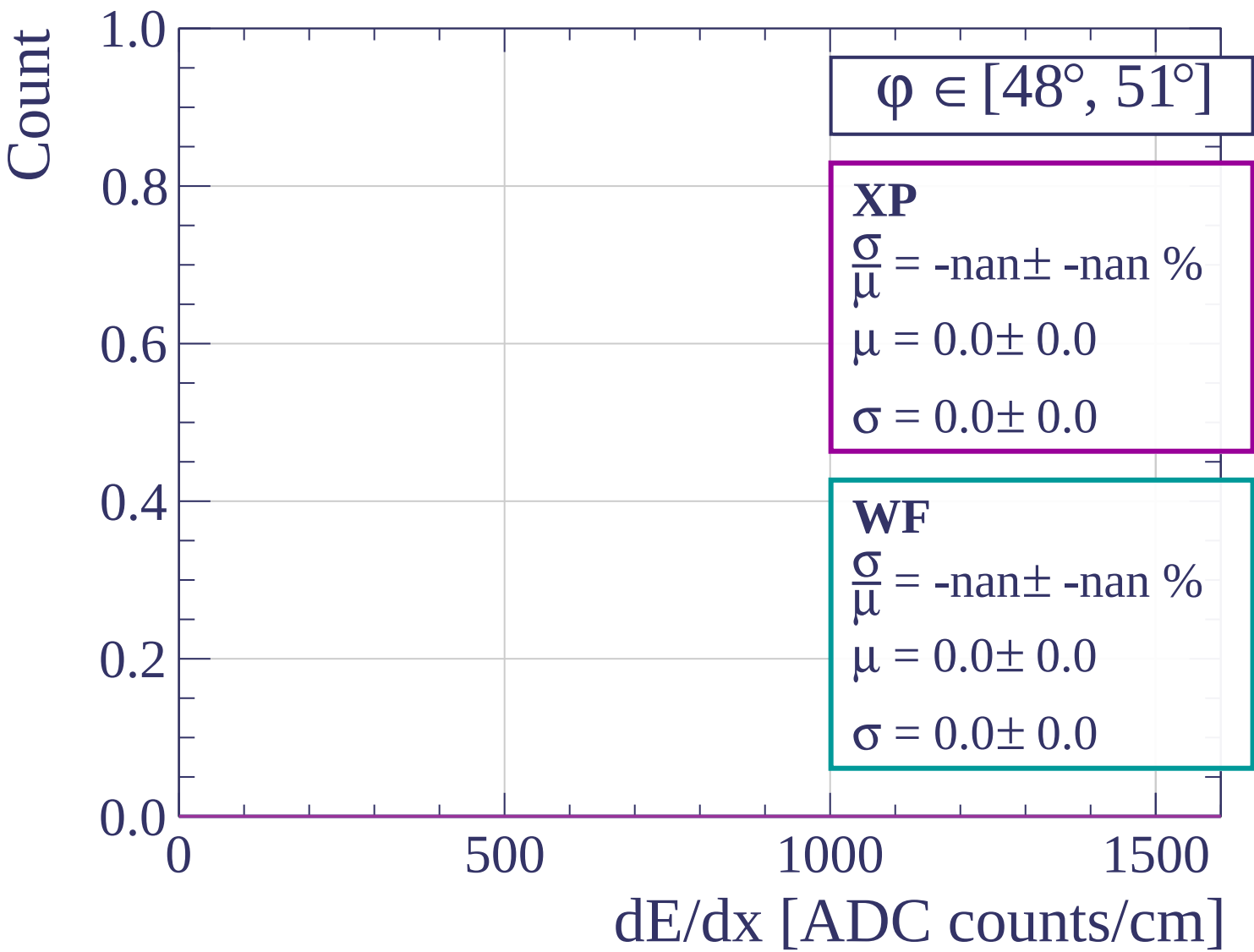


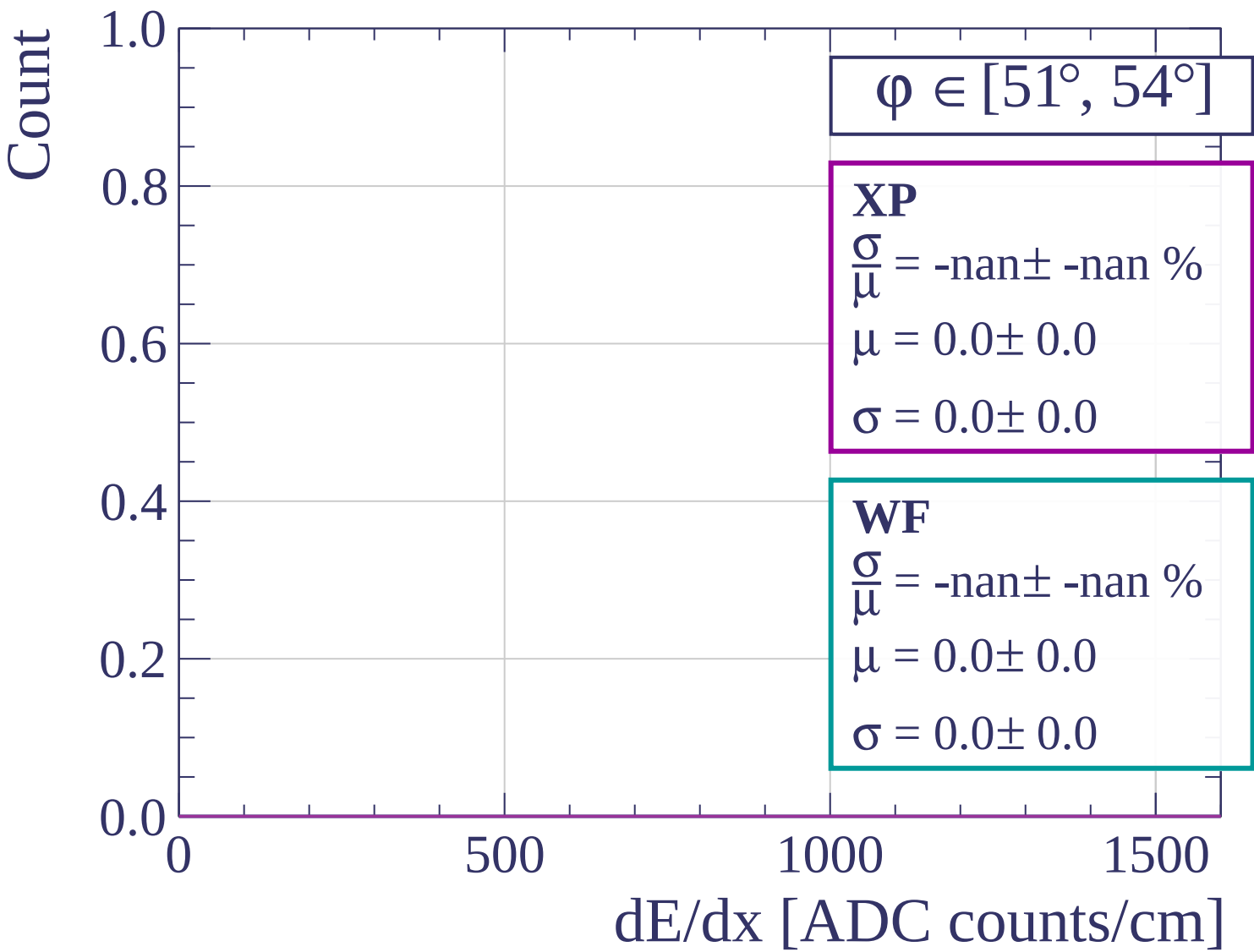
dE/dx [ADC counts/cm]

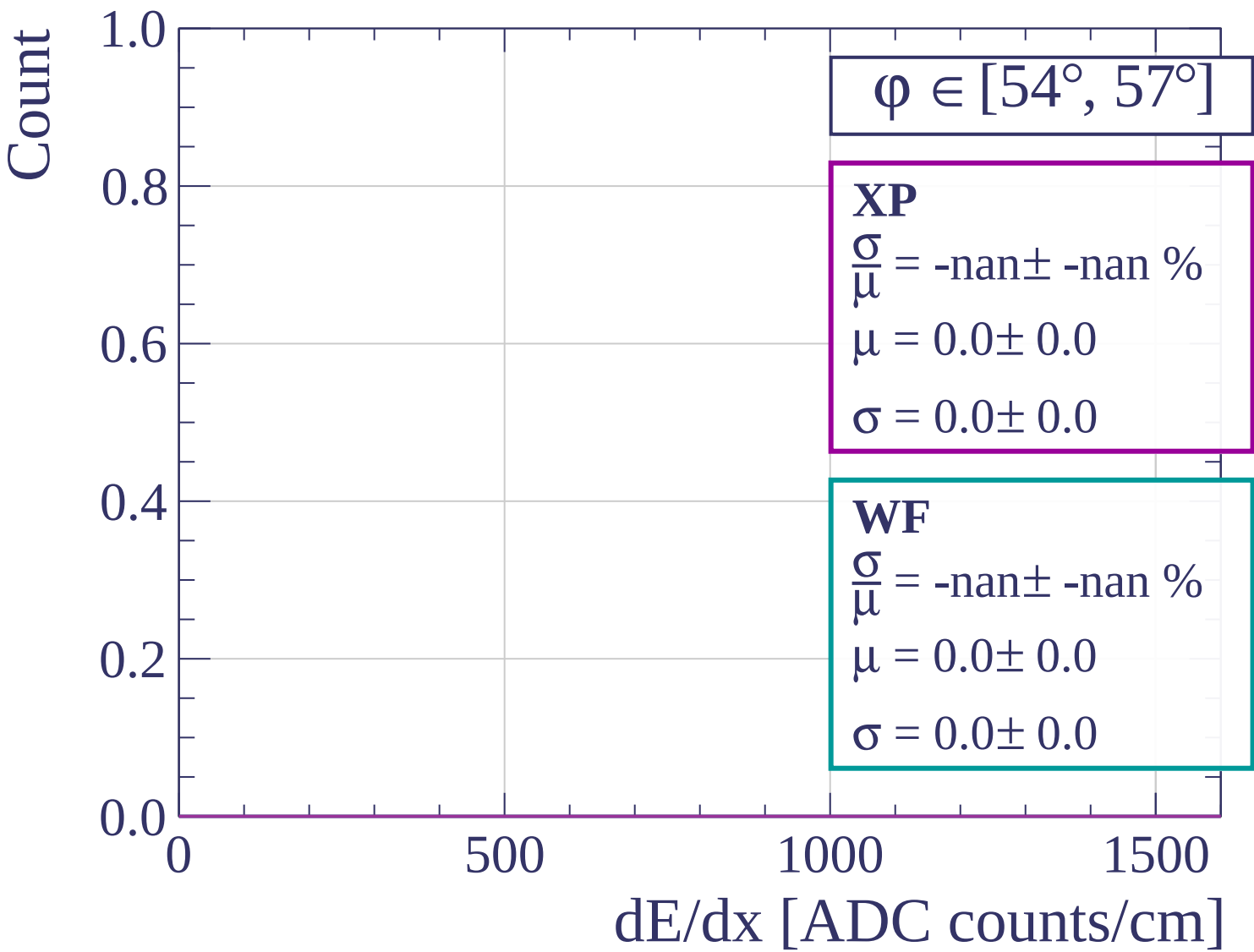
Count

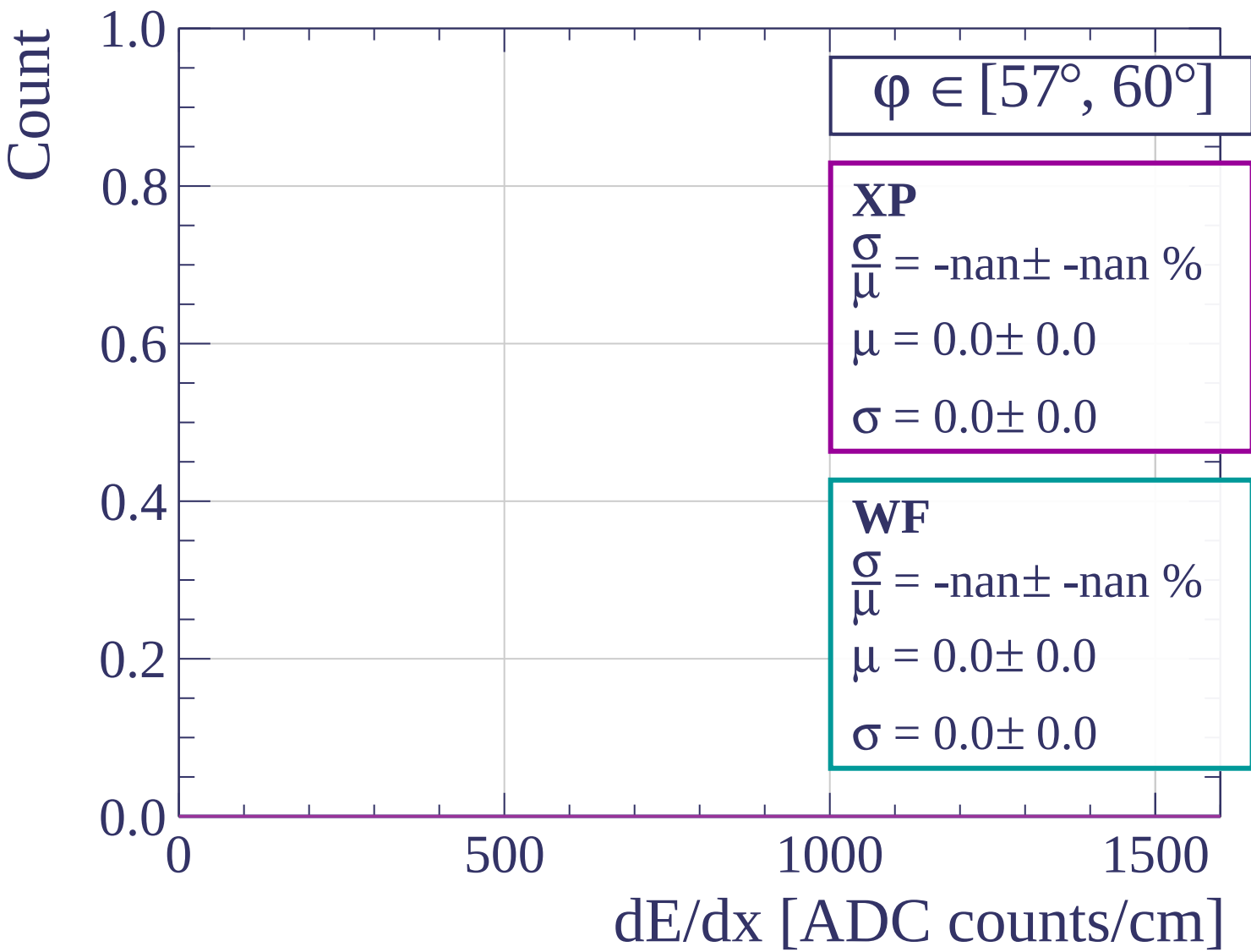


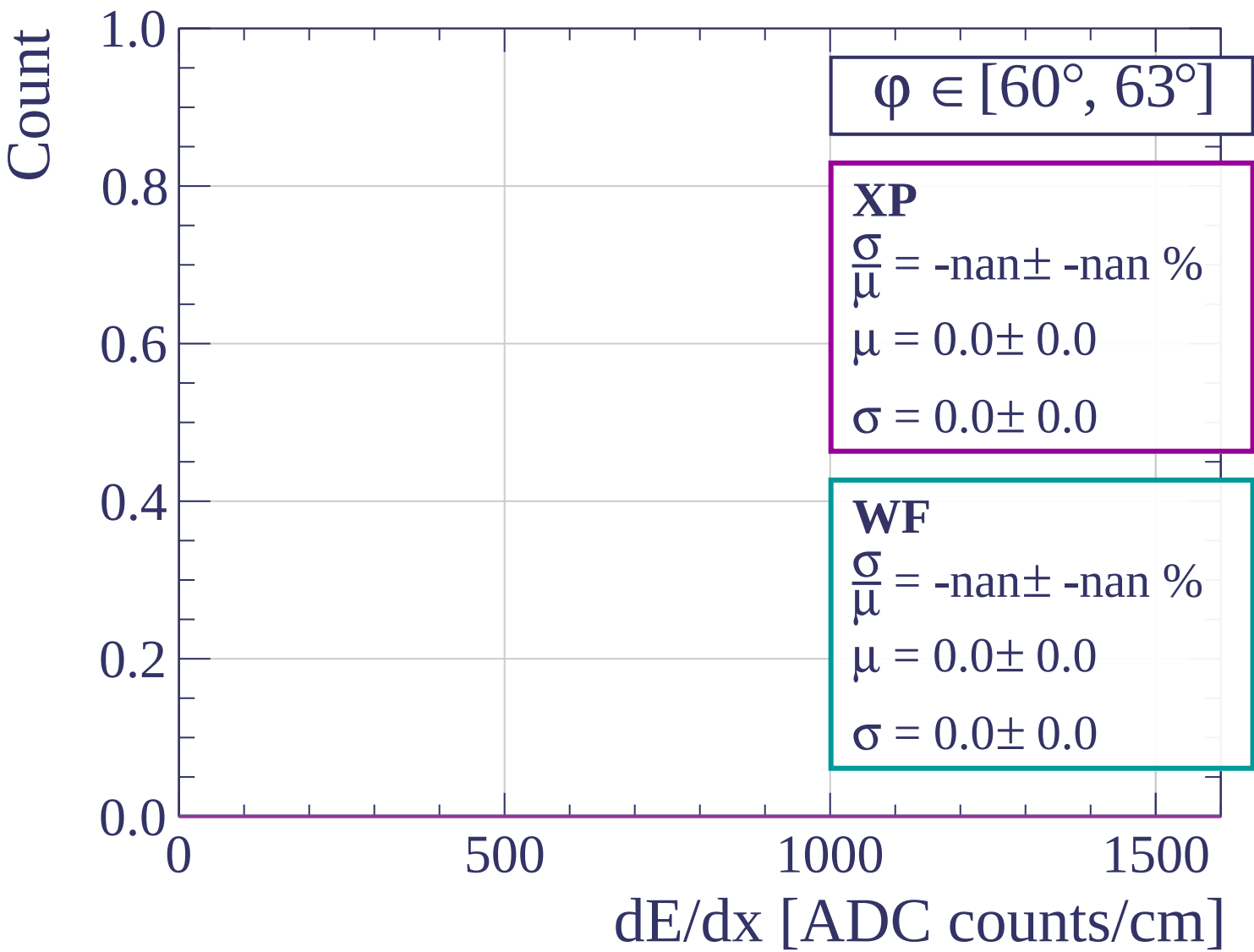
dE/dx [ADC counts/cm]

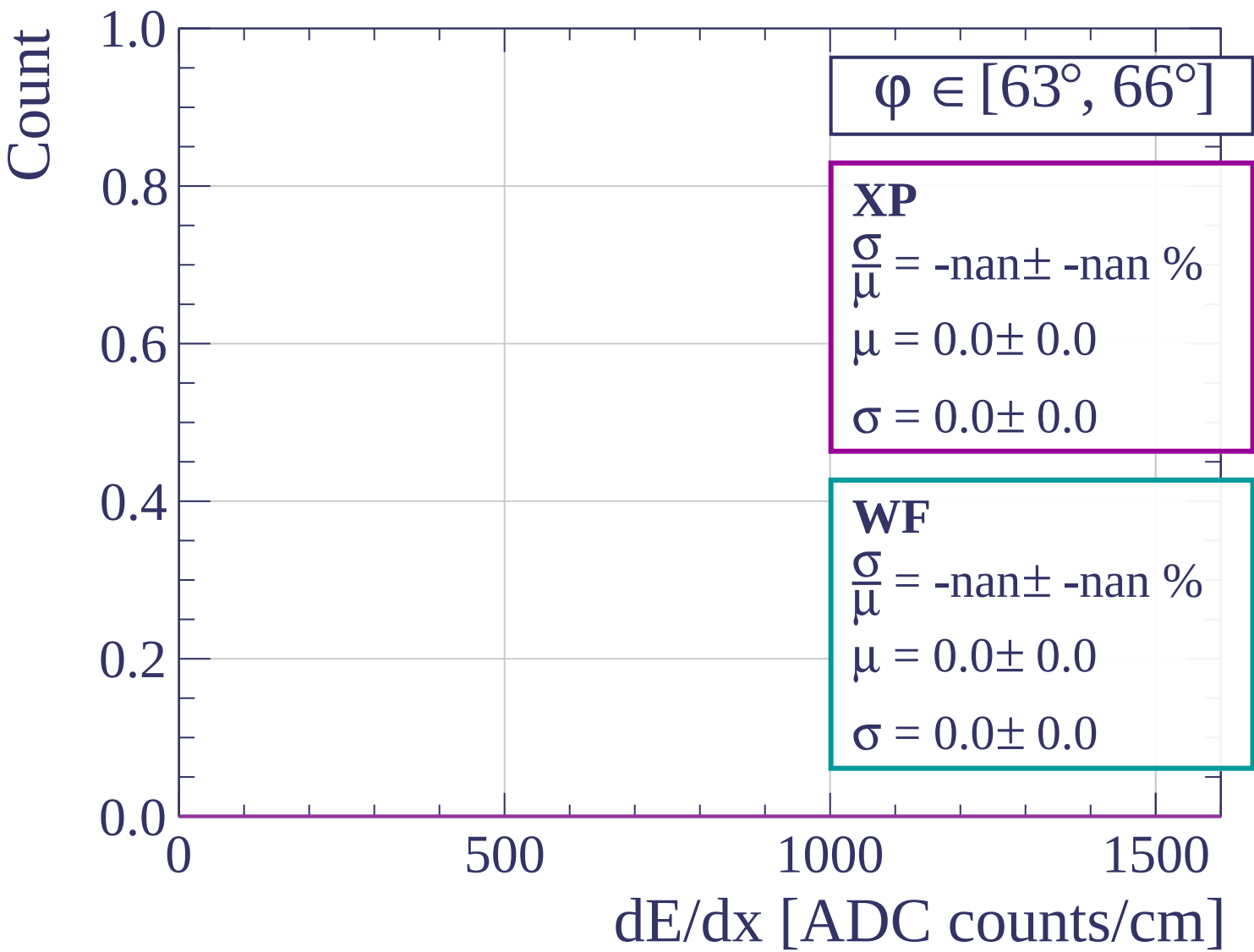


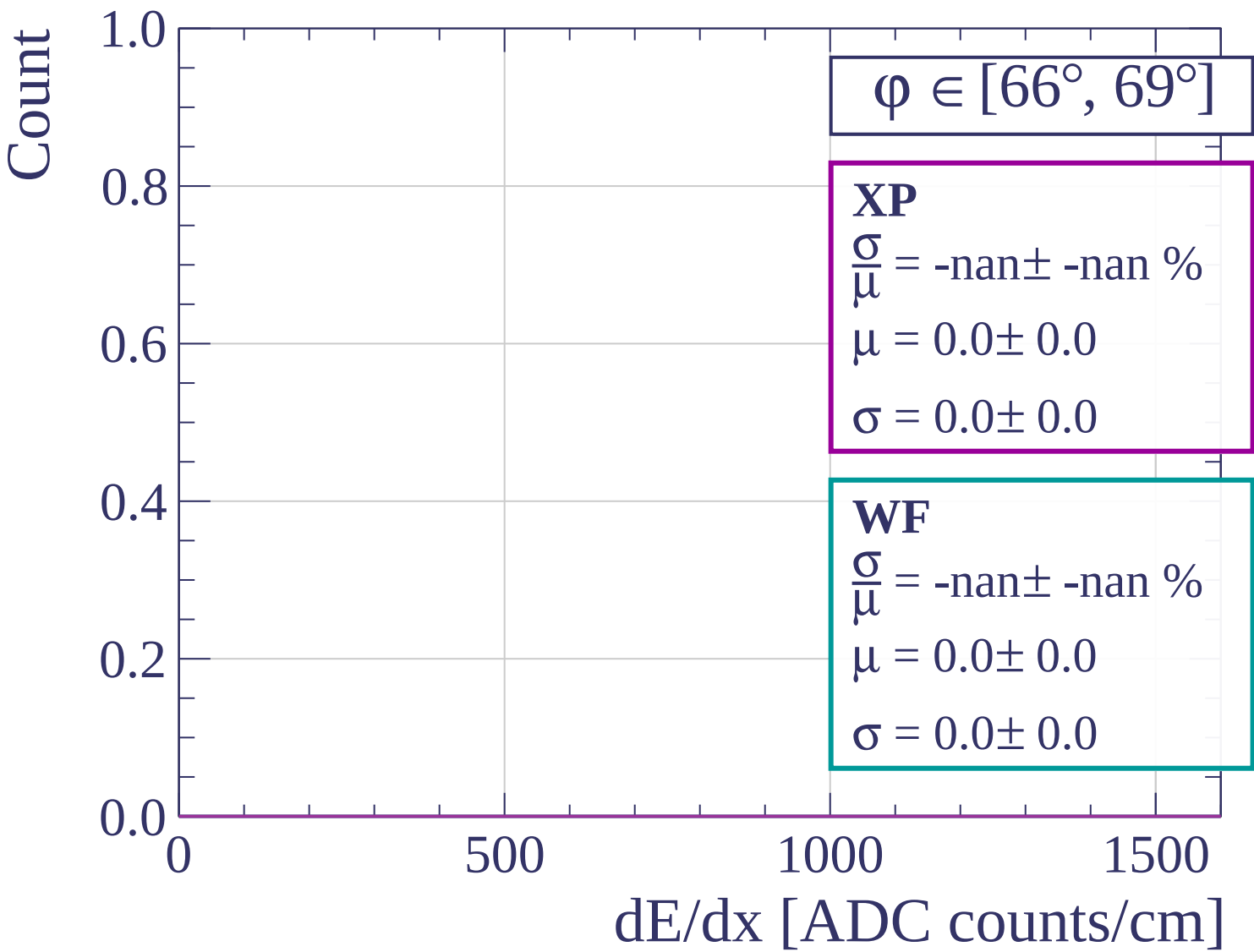


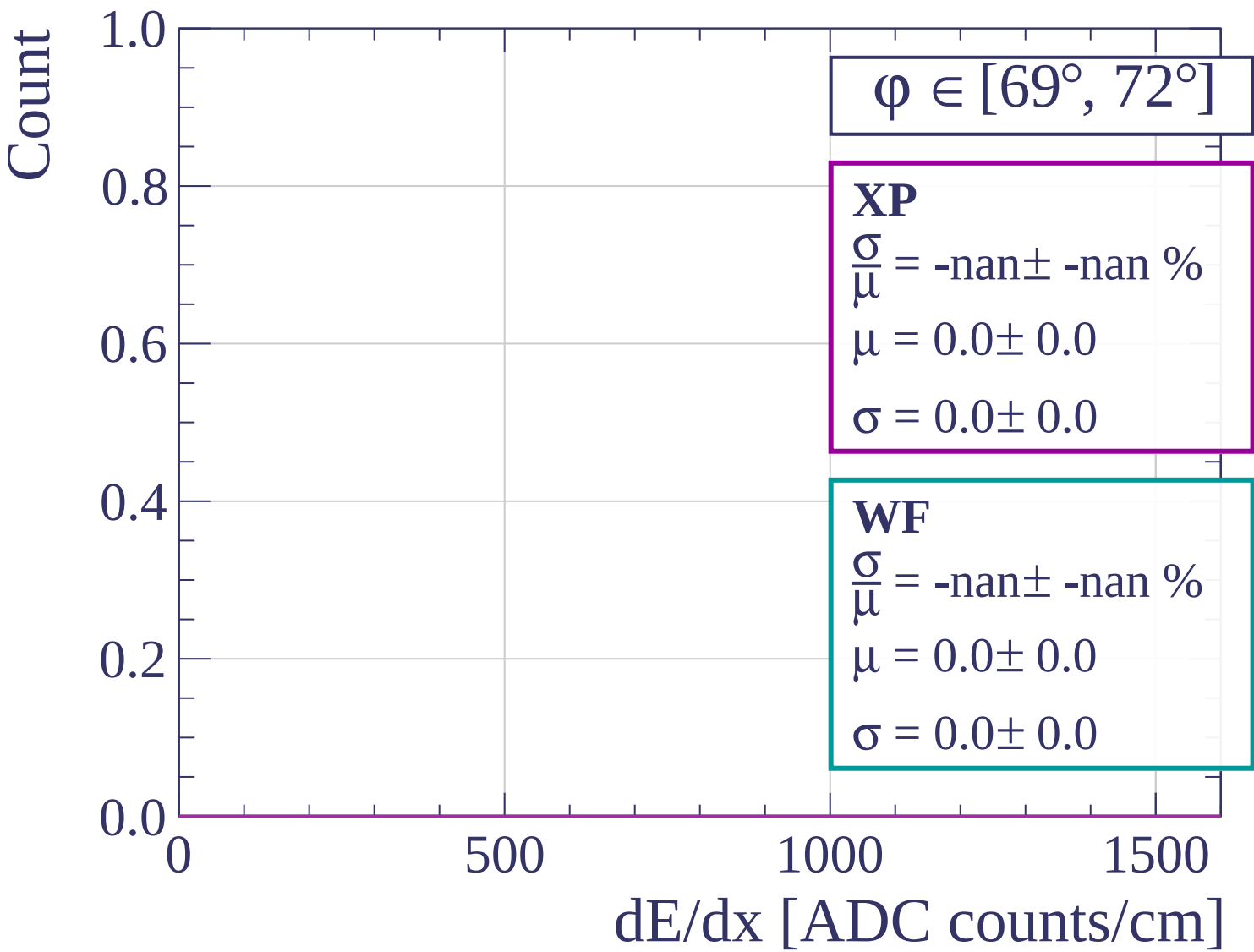


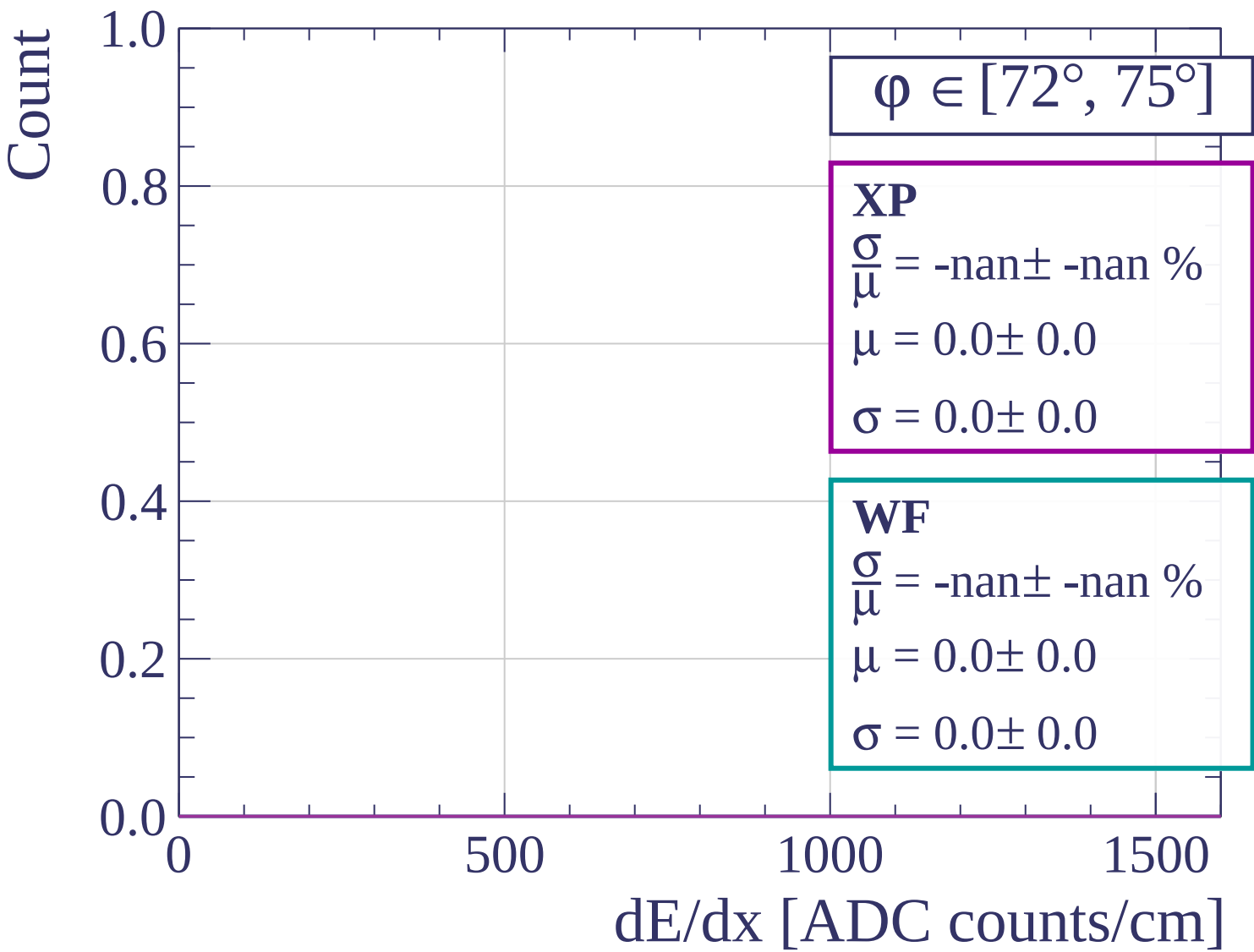


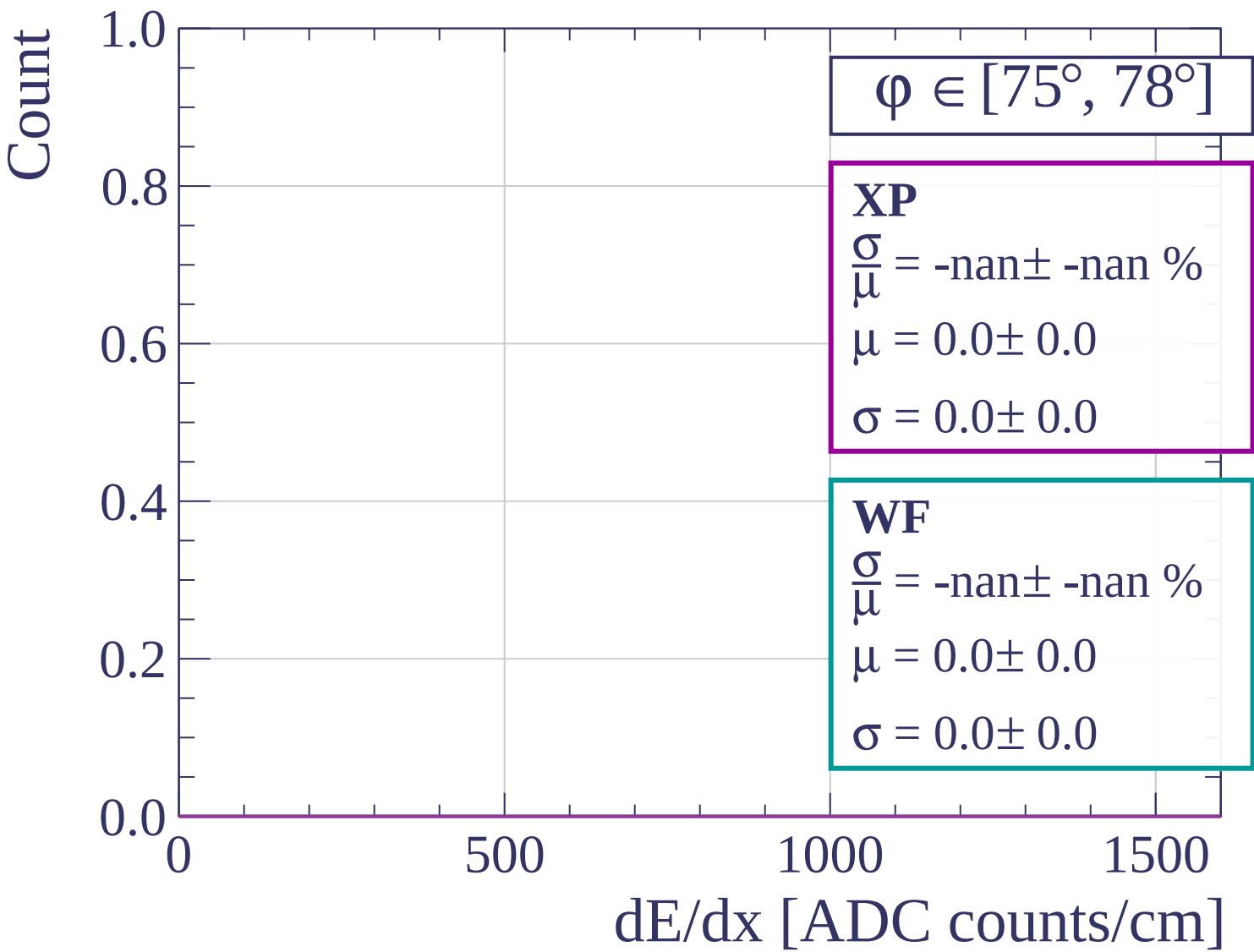


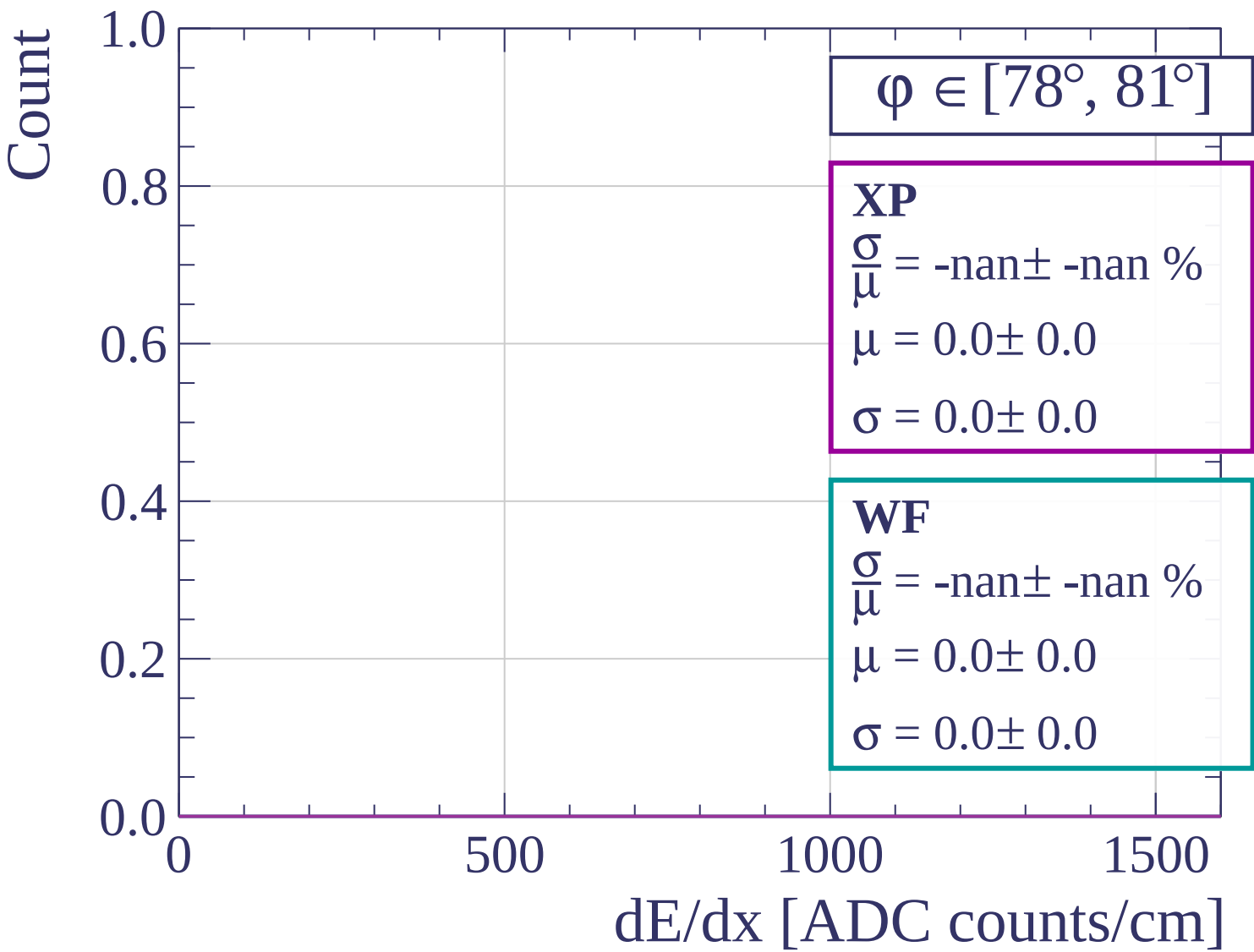


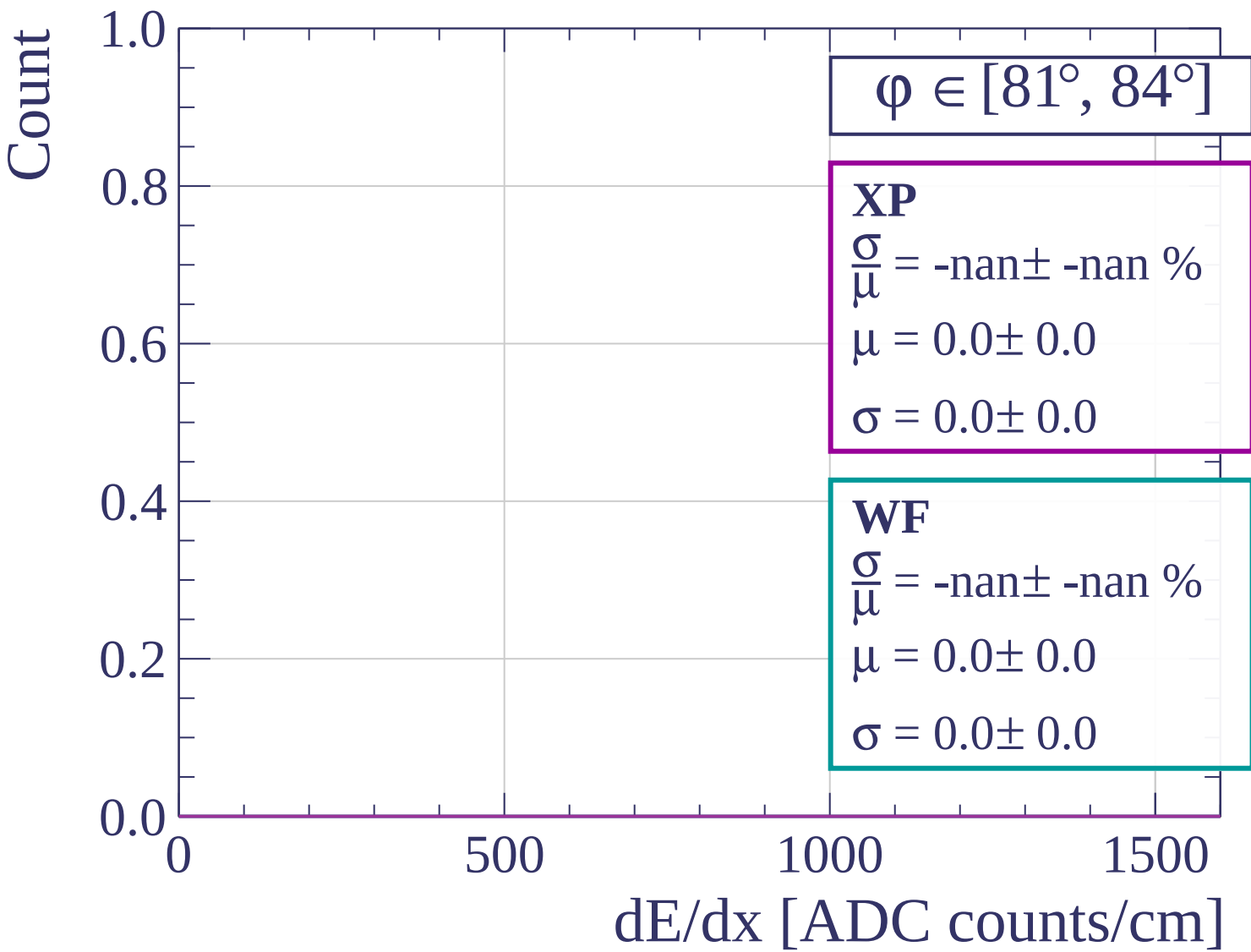


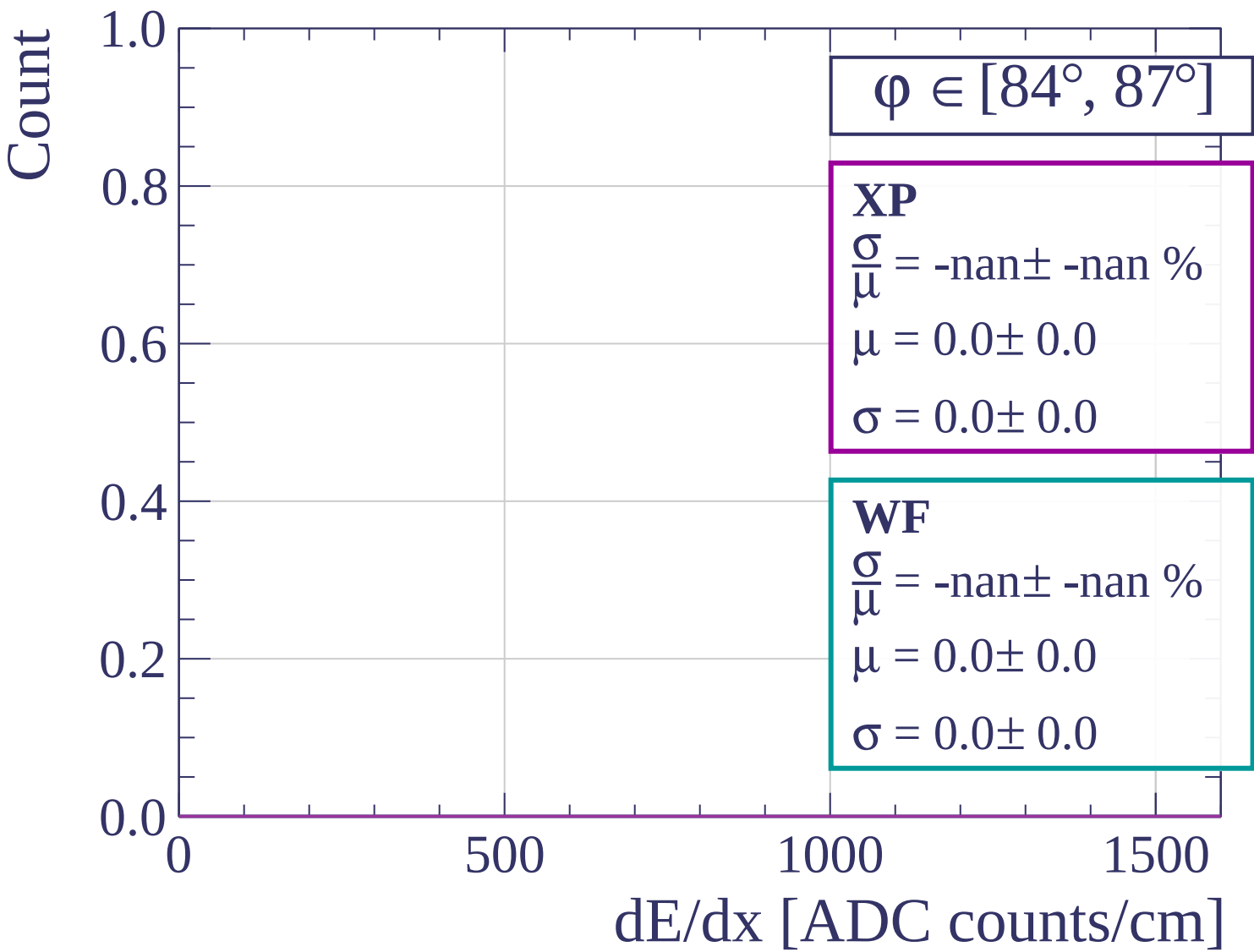


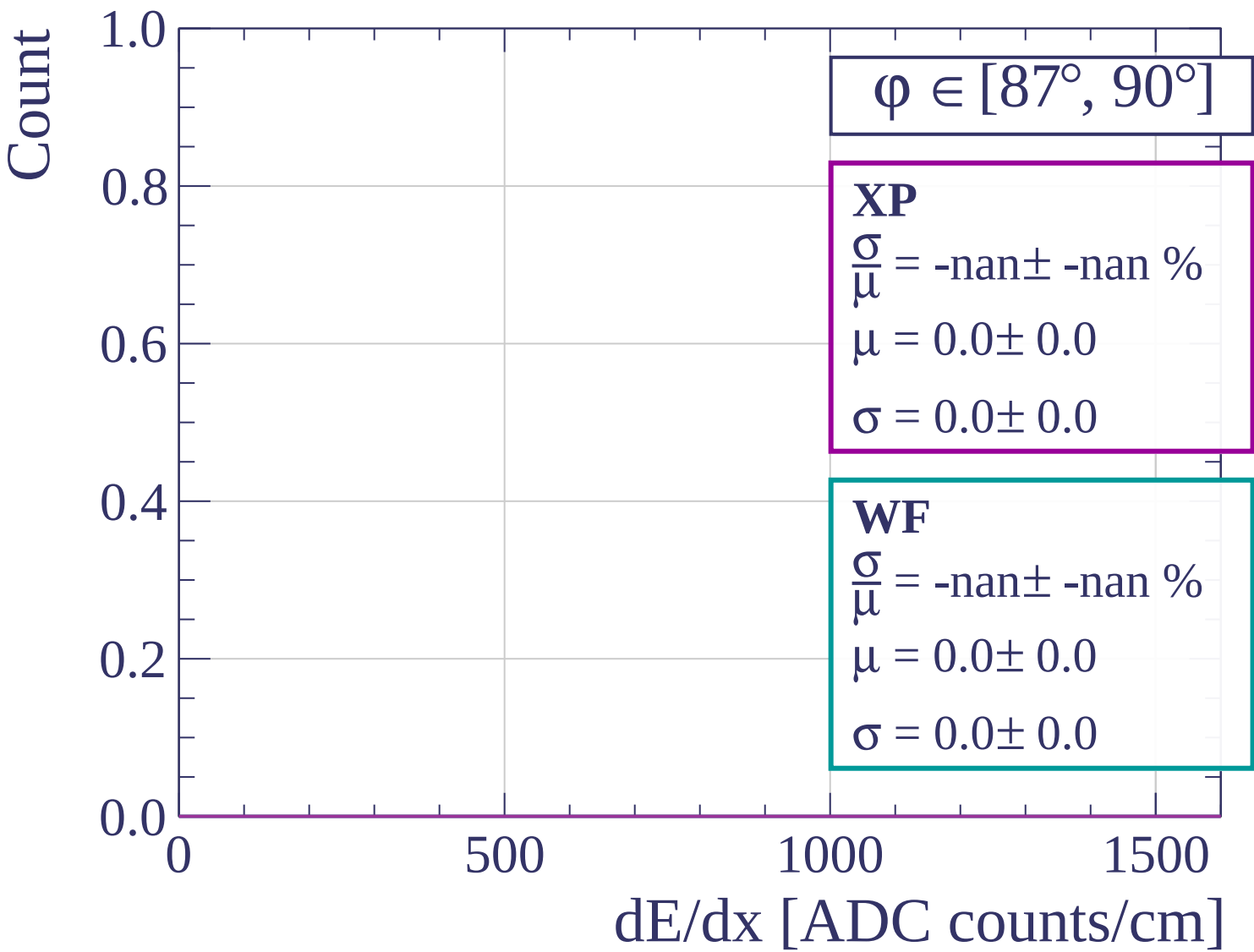


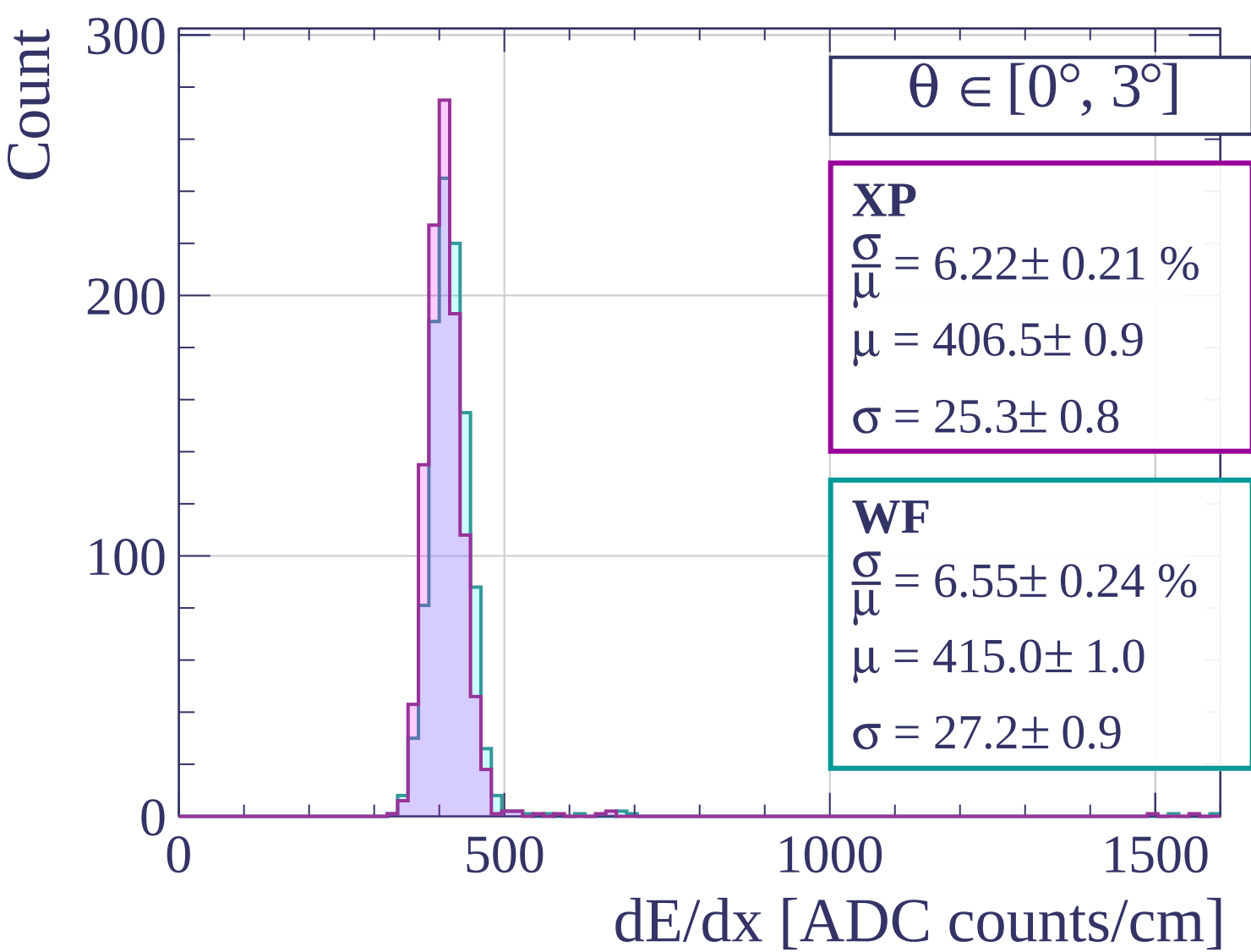




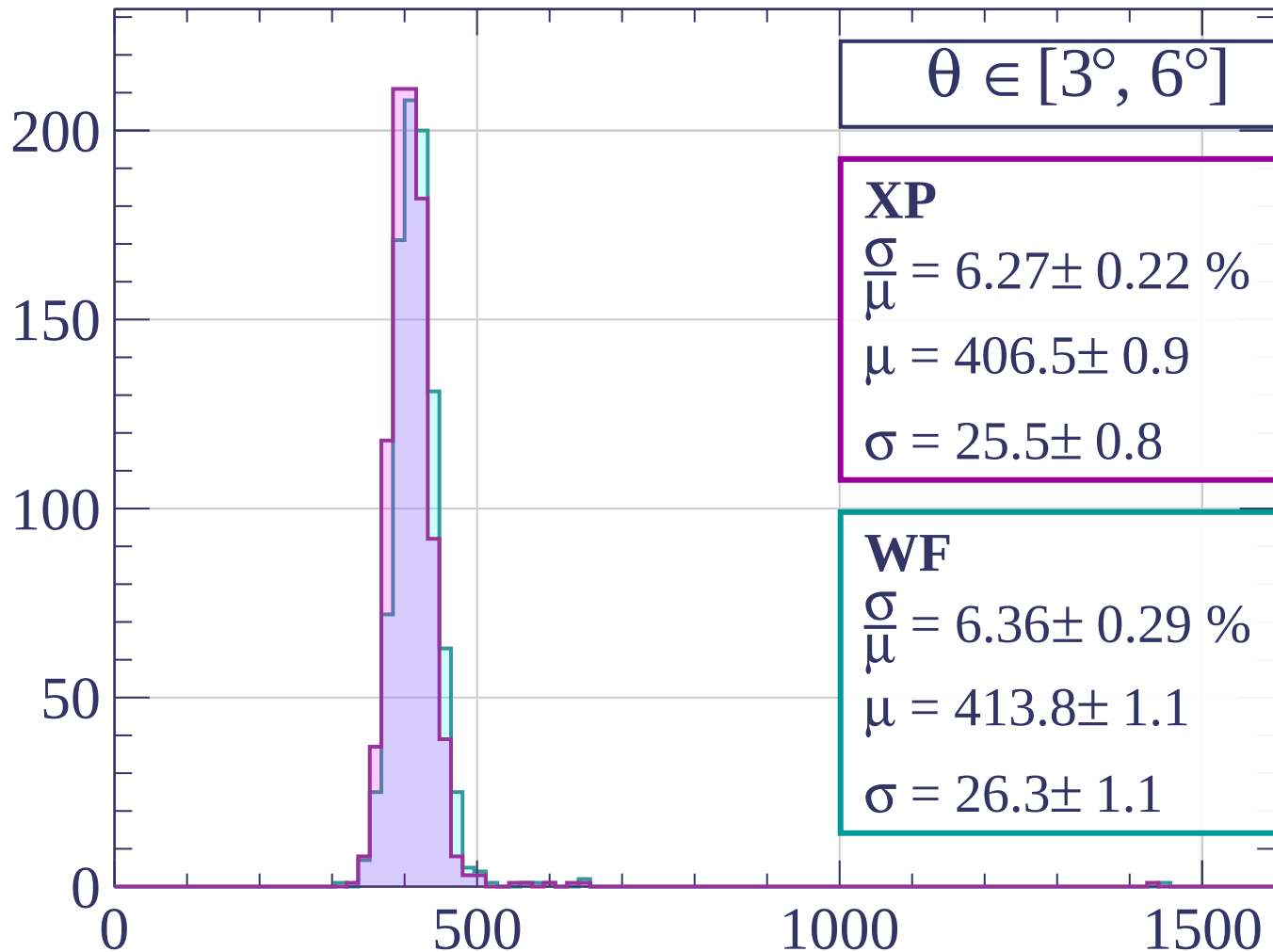






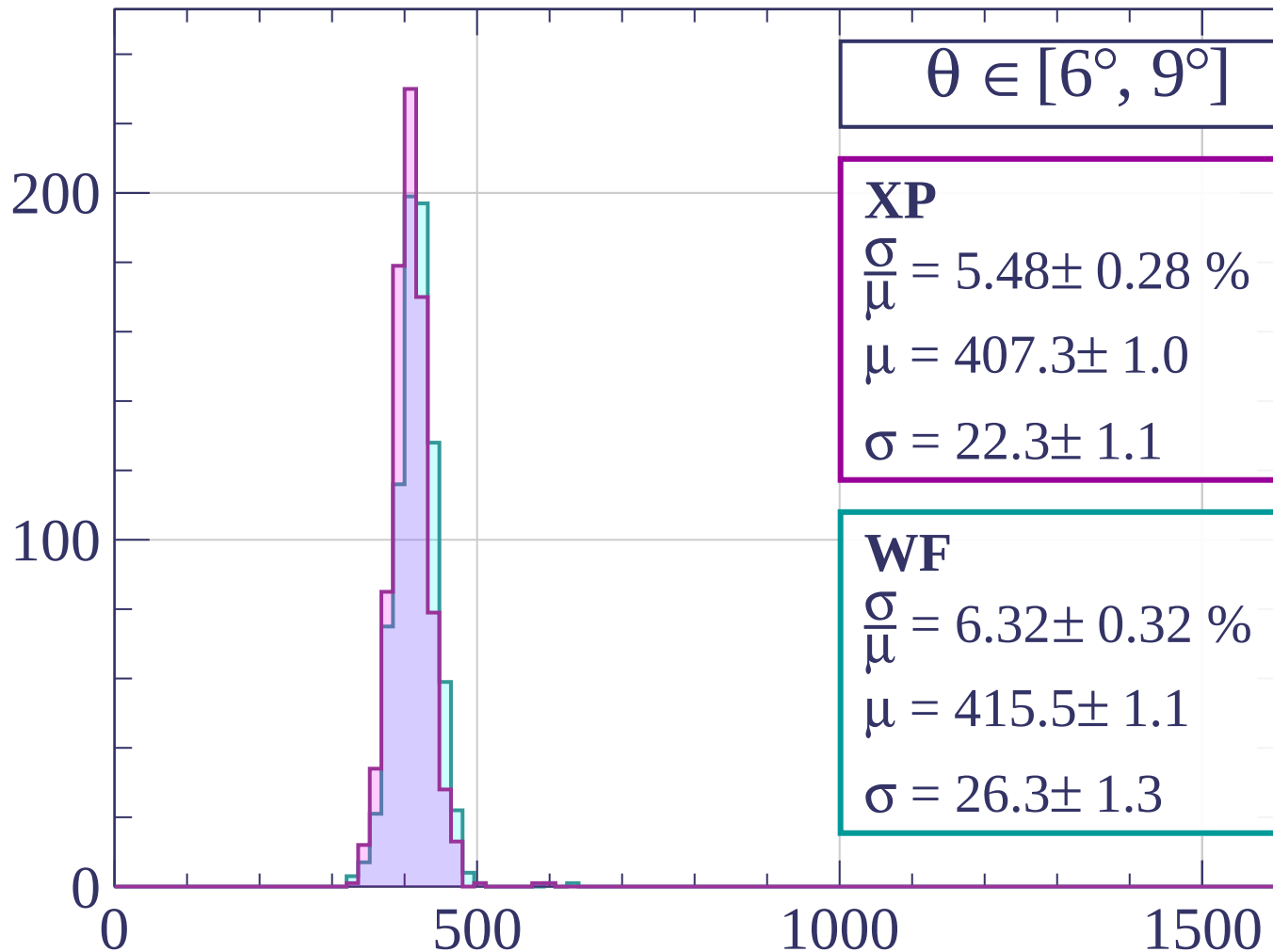


Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

$\theta \in [9^\circ, 12^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.76 \pm 0.30 \%$$

$$\mu = 405.9 \pm 1.2$$

$$\sigma = 23.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.99 \pm 0.33 \%$$

$$\mu = 413.7 \pm 1.2$$

$$\sigma = 24.8 \pm 1.3$$

150

100

50

0

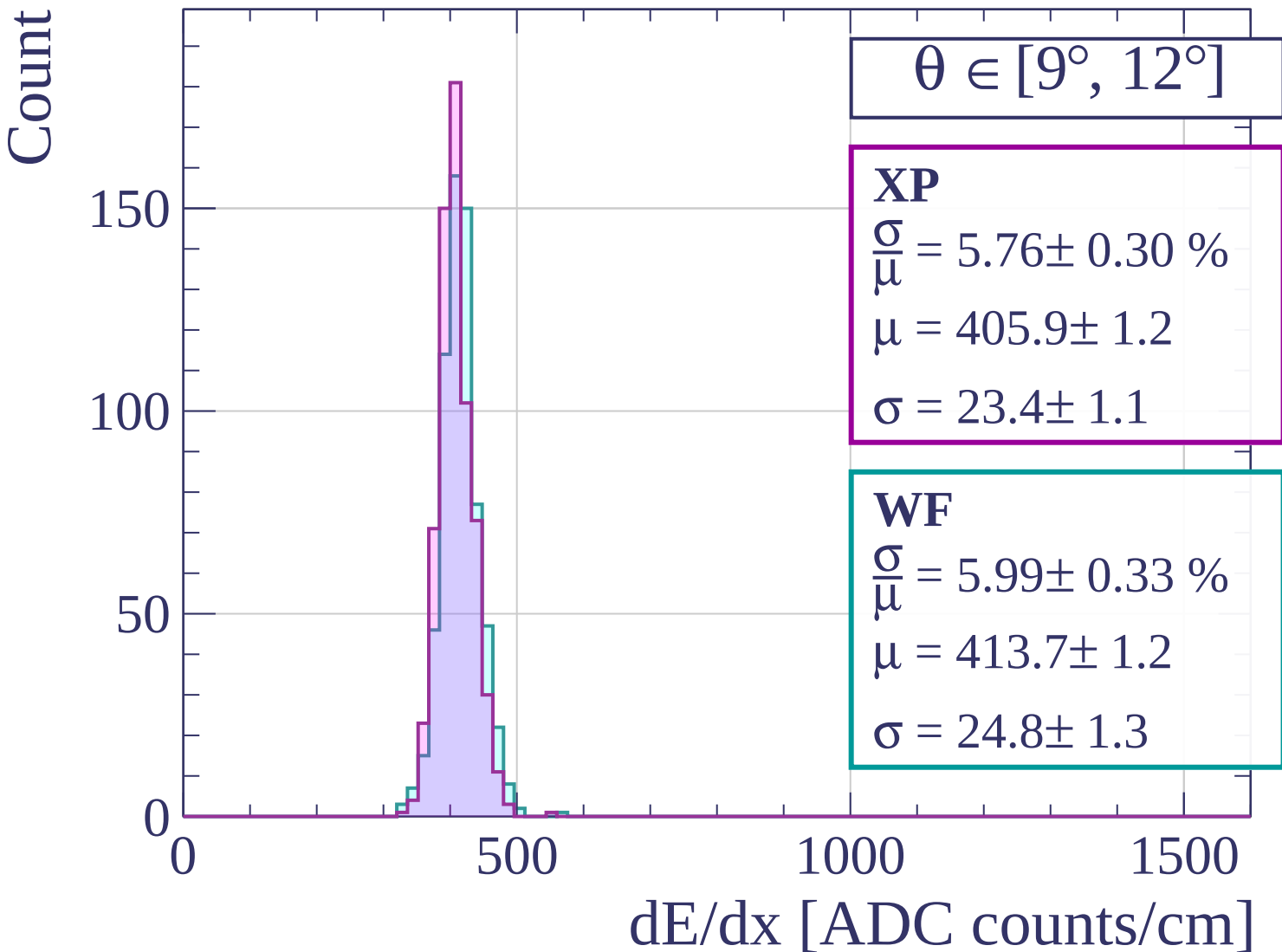
0

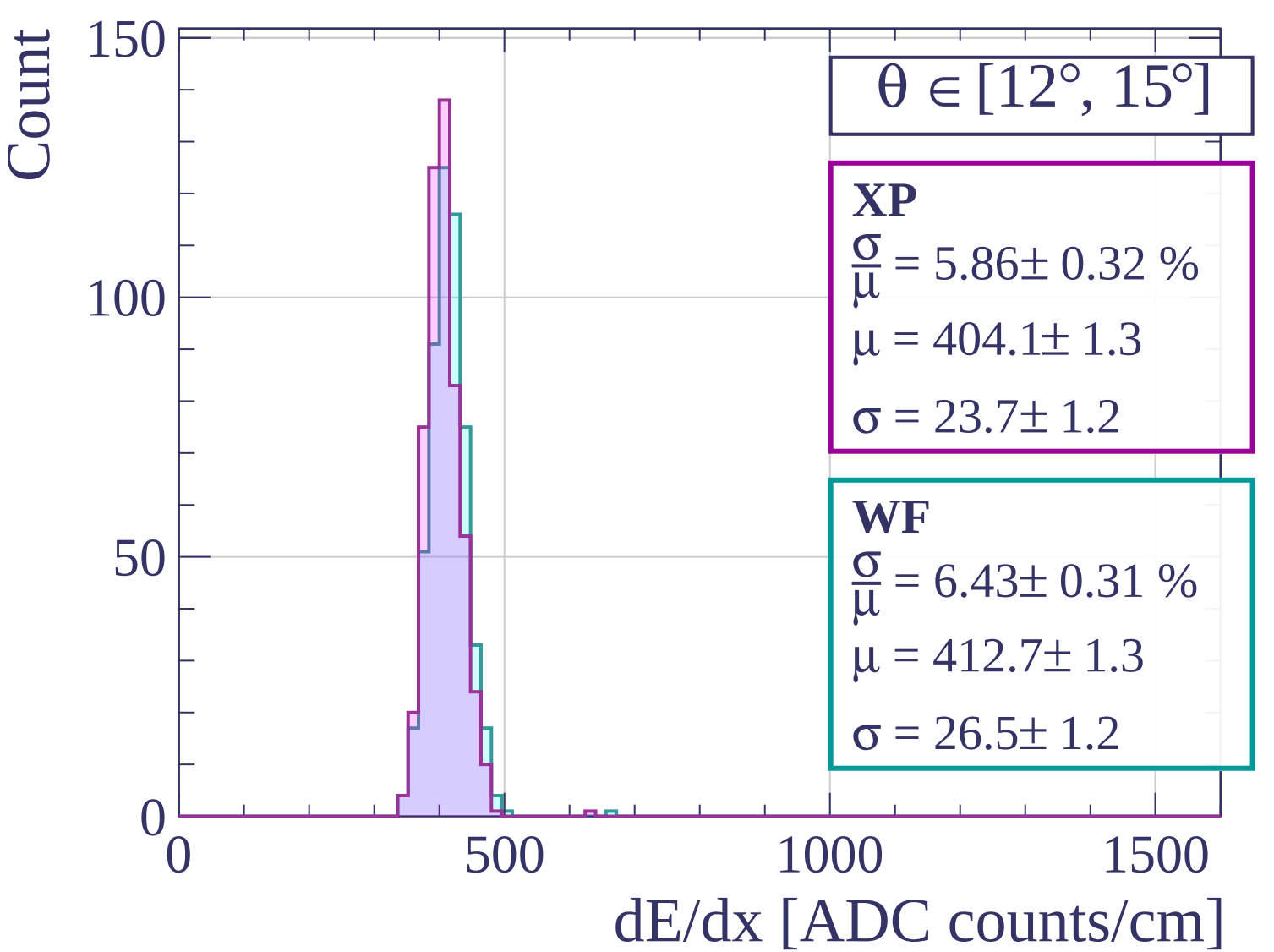
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\theta \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.86 \pm 0.25 \%$$

$$\mu = 407.9 \pm 1.2$$

$$\sigma = 23.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.03 \pm 0.37 \%$$

$$\mu = 416.0 \pm 1.4$$

$$\sigma = 25.1 \pm 1.5$$

100

50

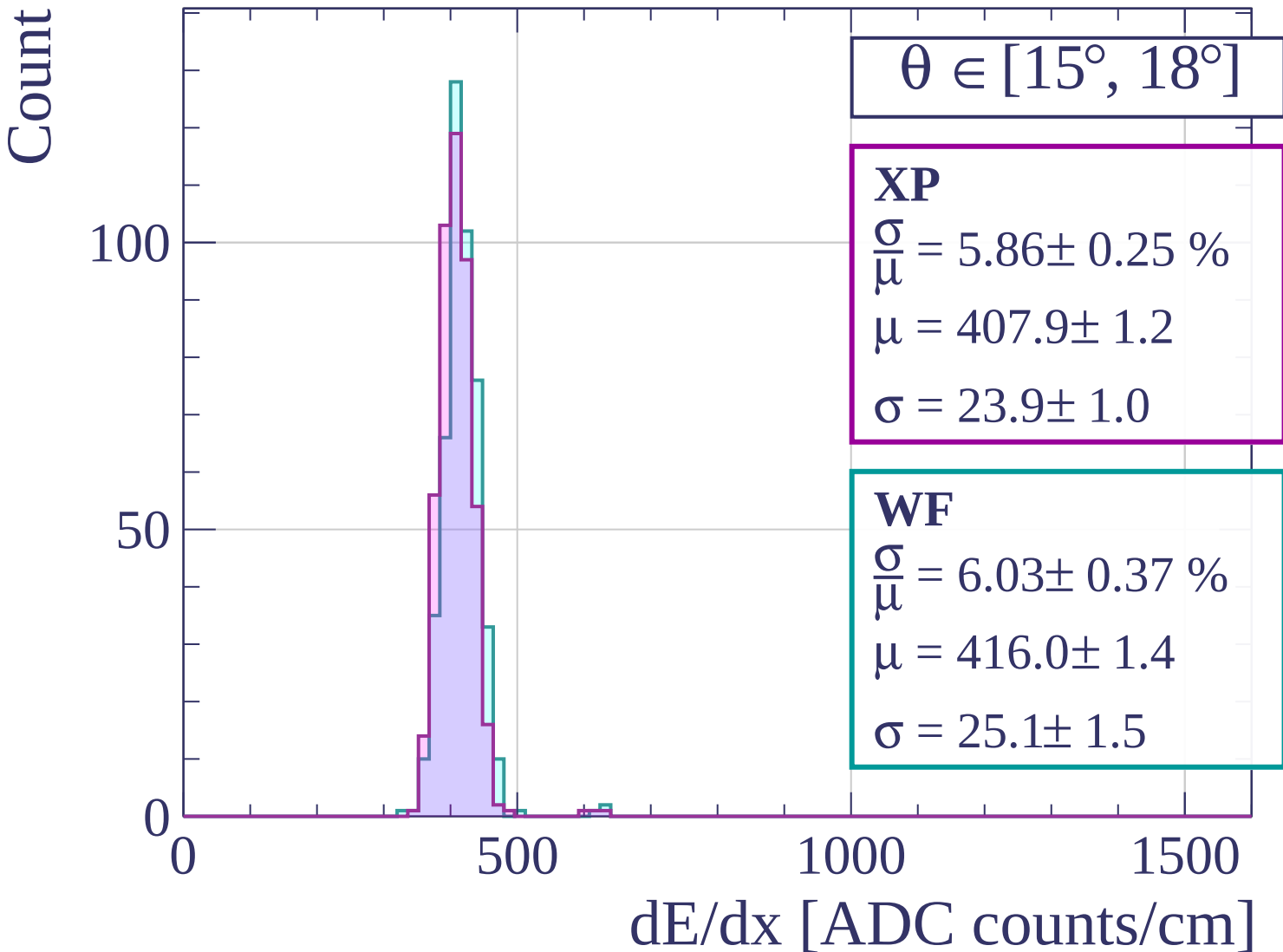
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [18^\circ, 21^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.48 \pm 0.31 \%$$

$$\mu = 408.1 \pm 1.4$$

$$\sigma = 22.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.86 \pm 0.39 \%$$

$$\mu = 414.9 \pm 1.5$$

$$\sigma = 24.3 \pm 1.5$$

0

50

100

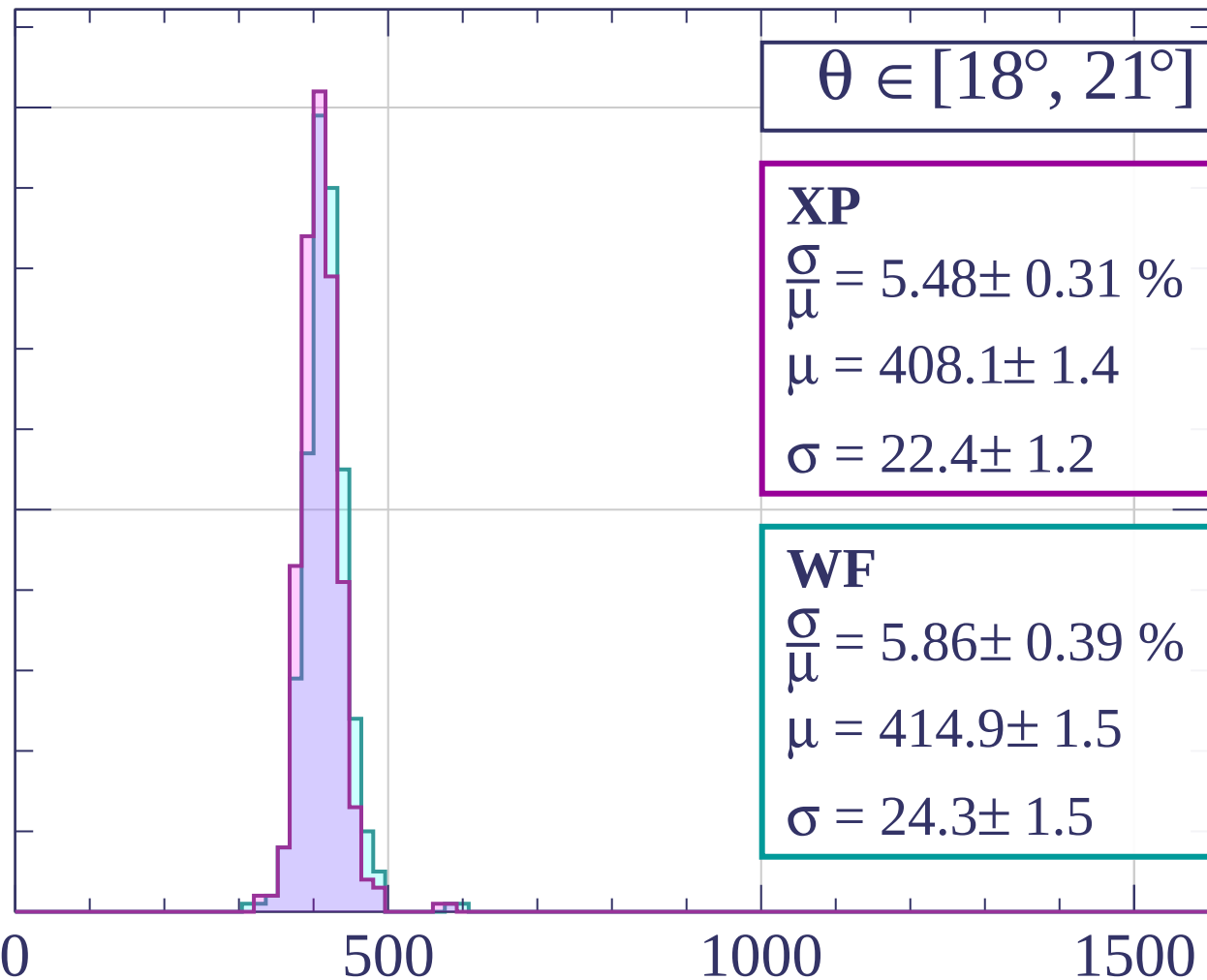
0

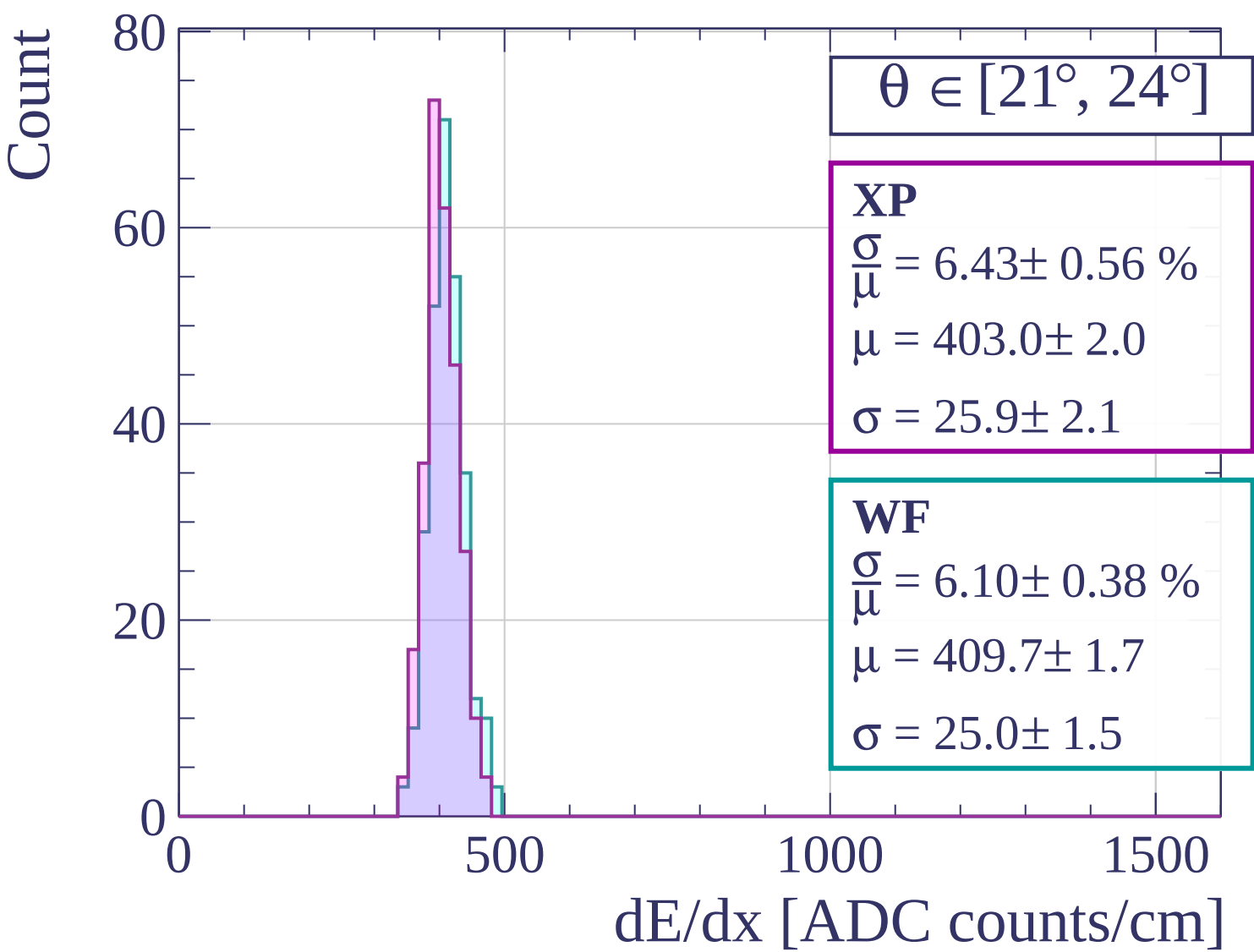
500

1000

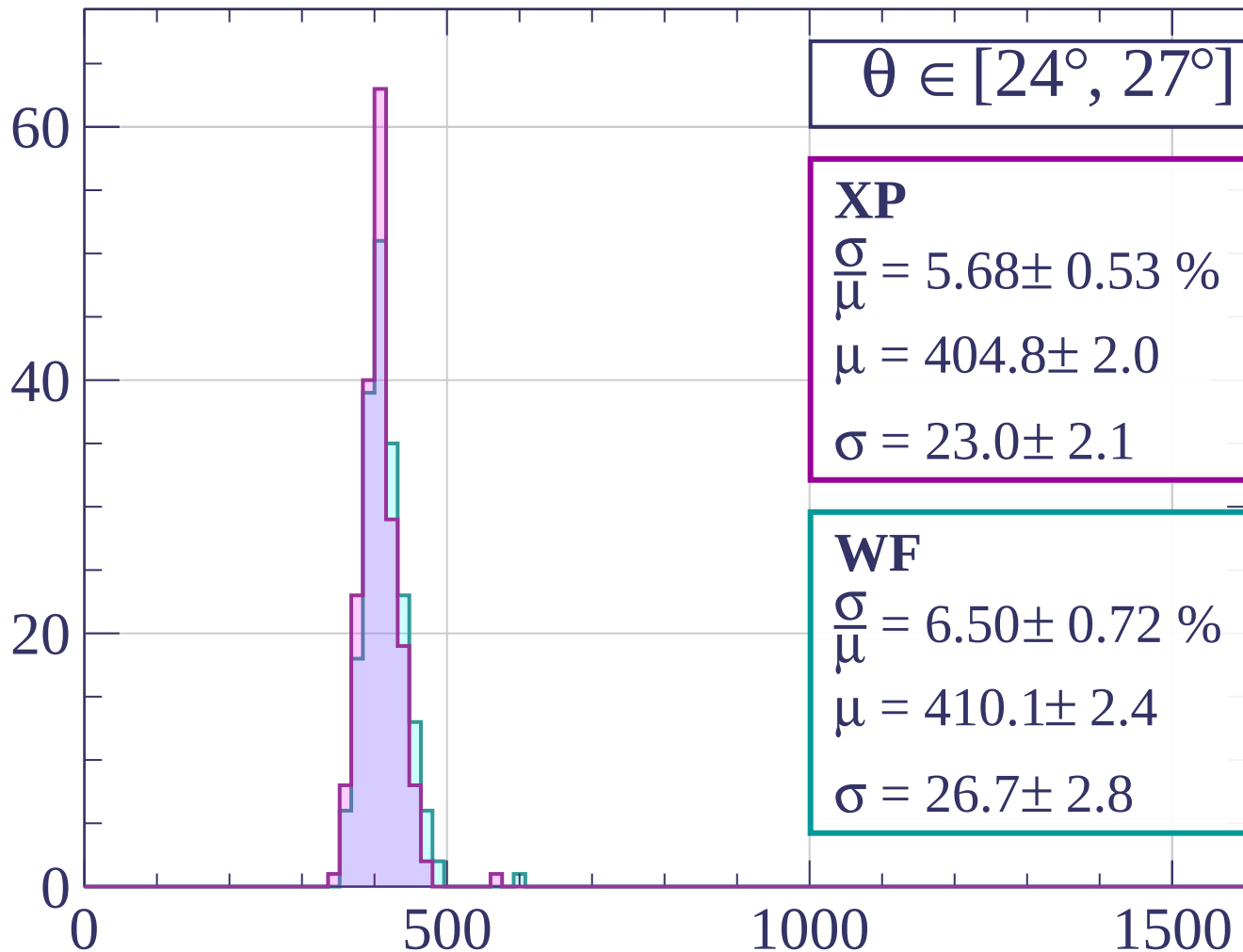
1500

dE/dx [ADC counts/cm]



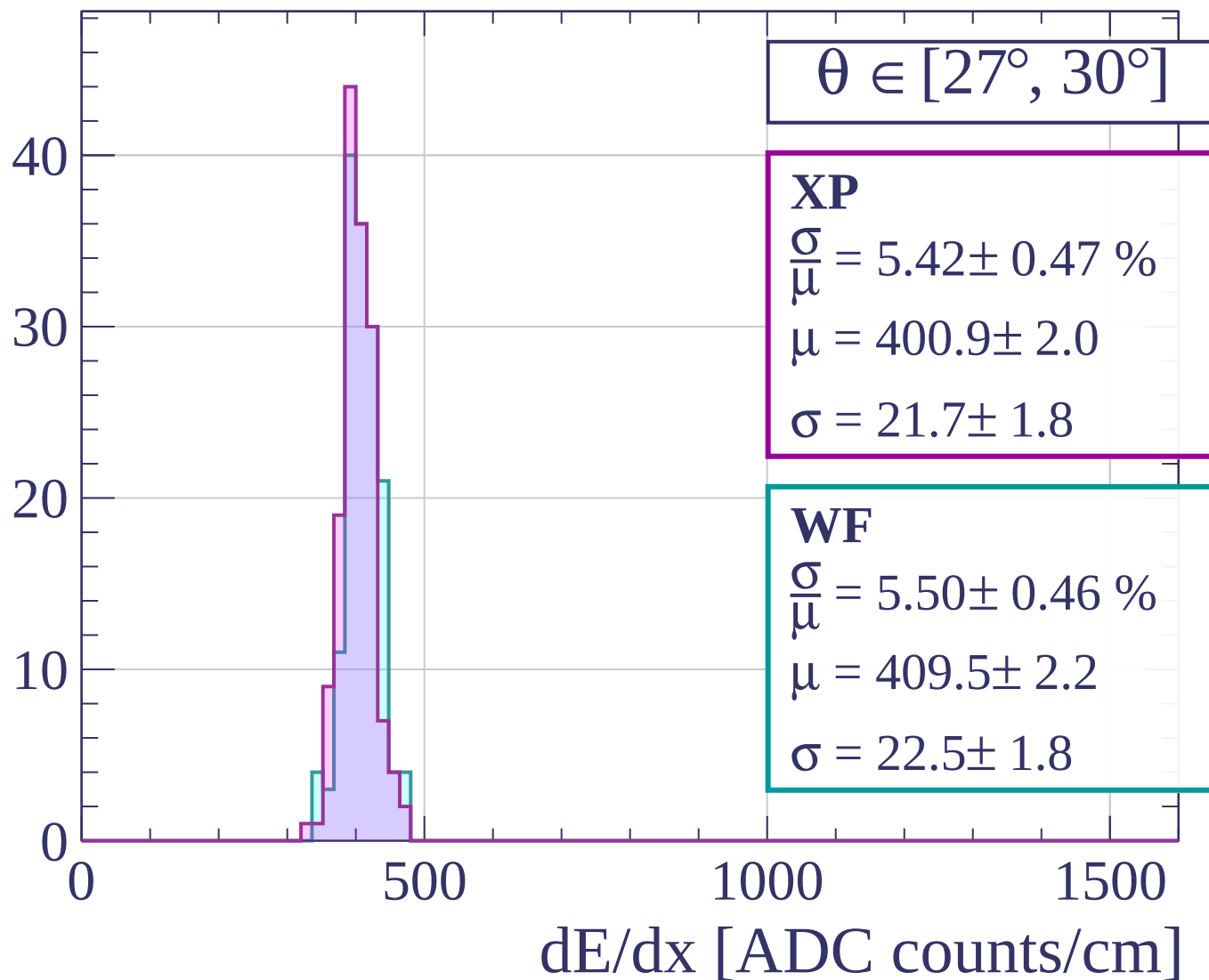


Count

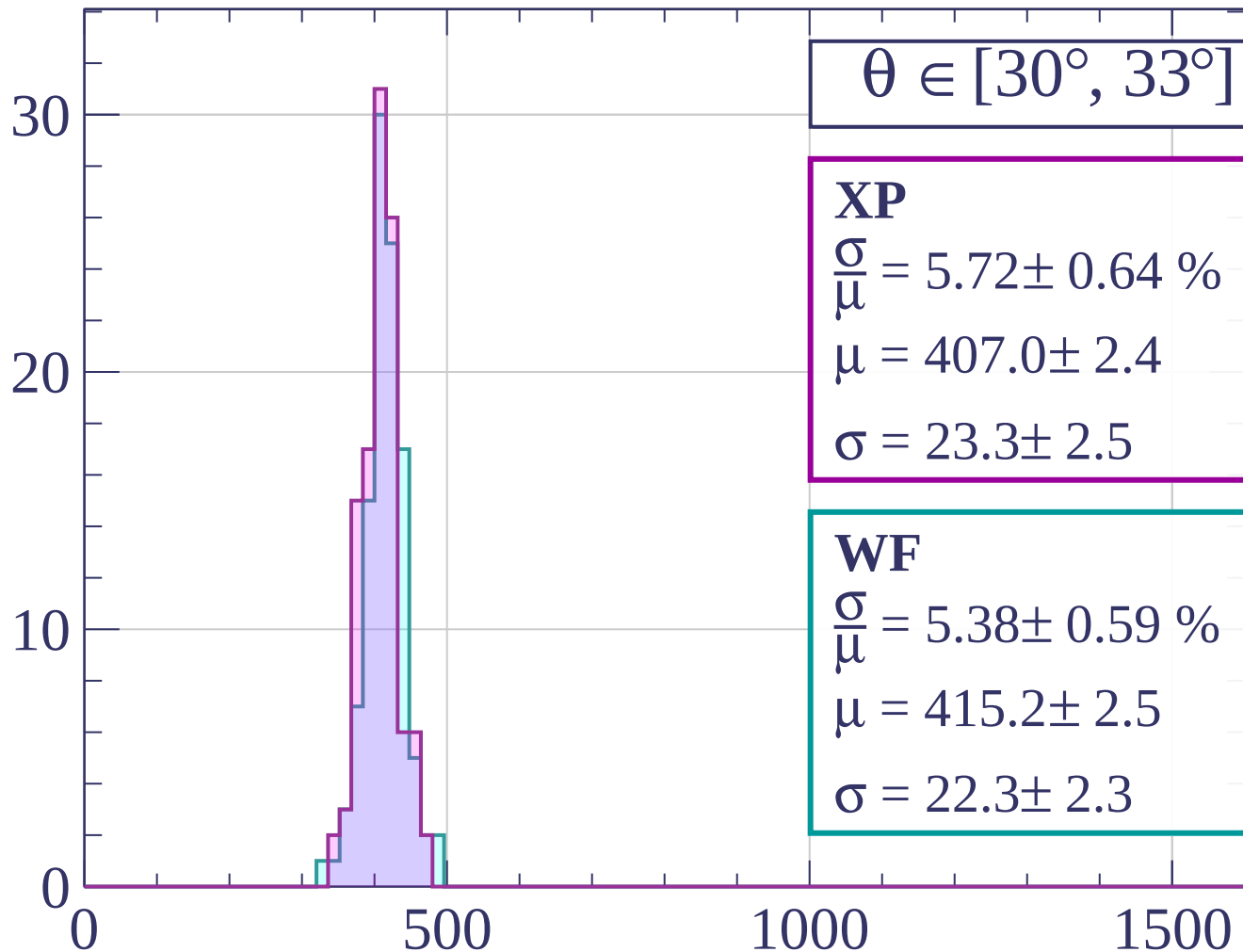


dE/dx [ADC counts/cm]

Count

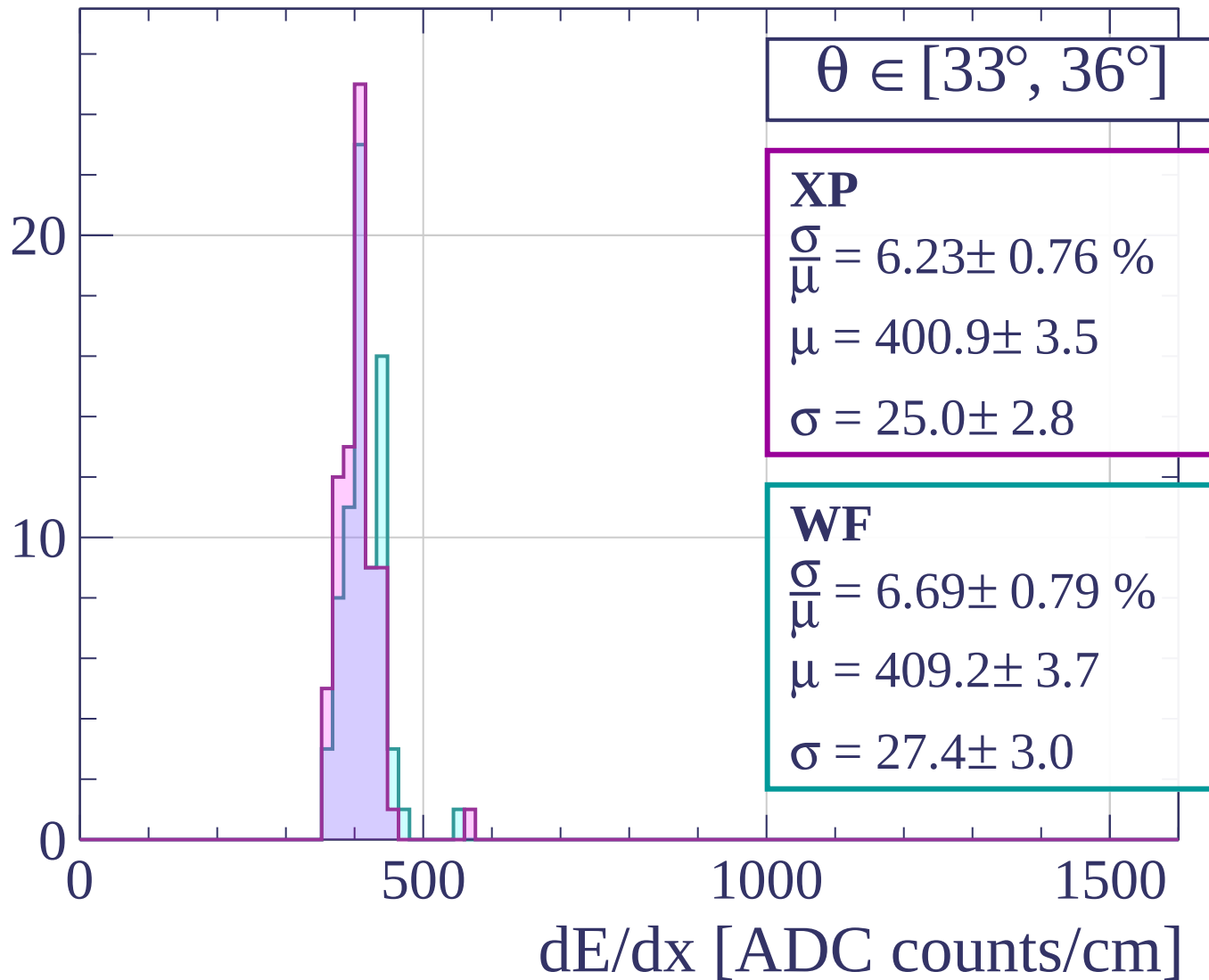


Count

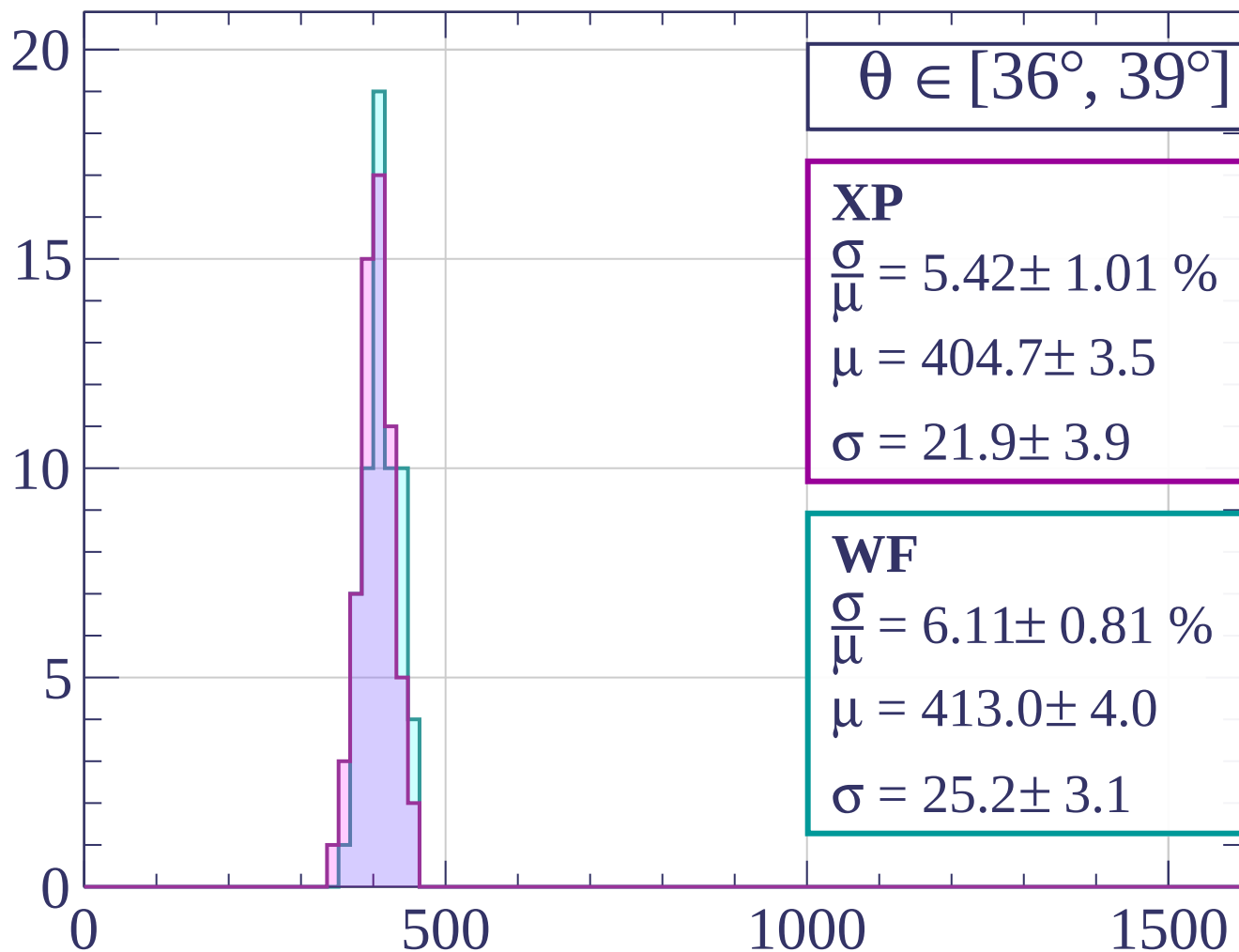


dE/dx [ADC counts/cm]

Count

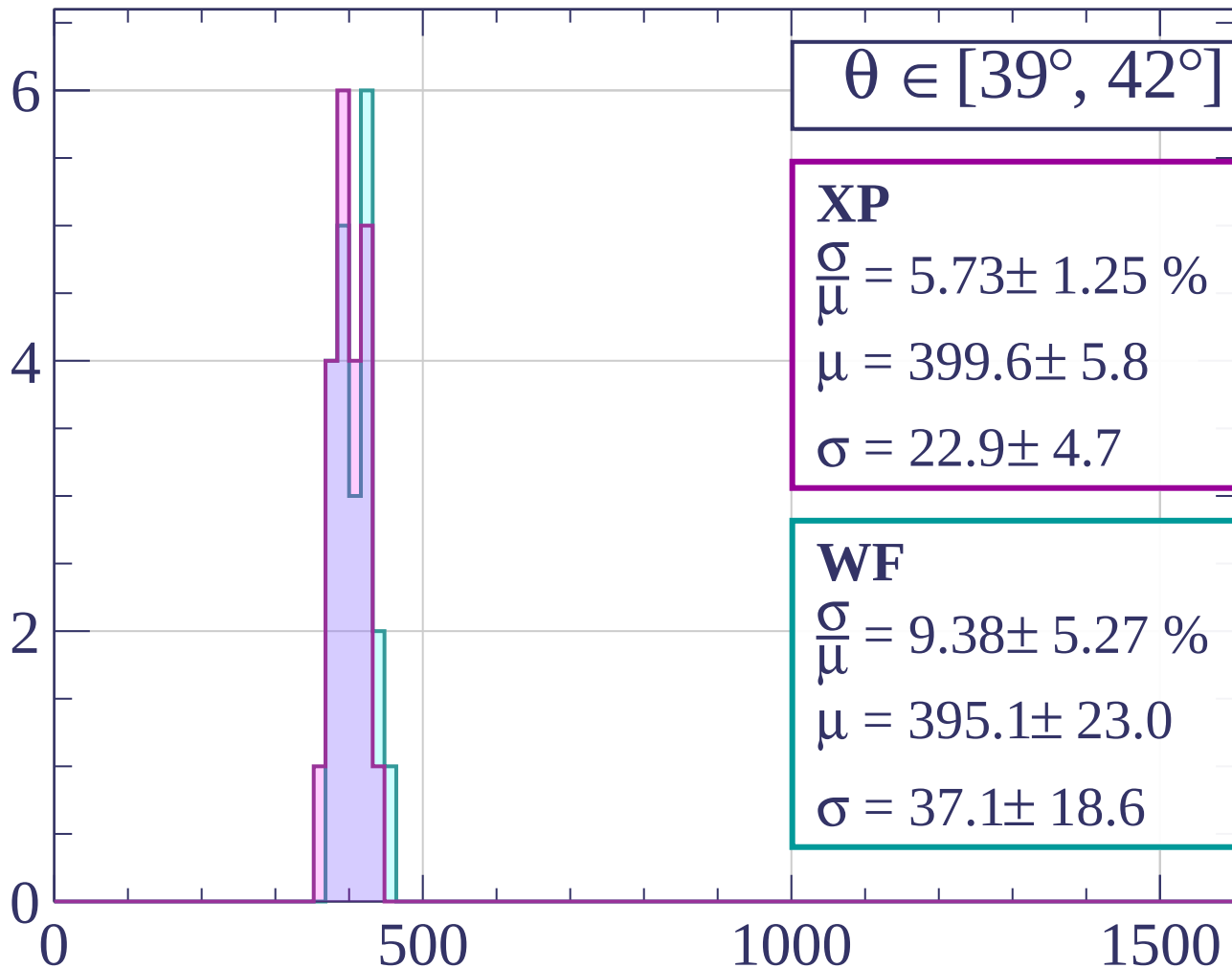


Count



dE/dx [ADC counts/cm]

Count



$\theta \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.73 \pm 1.25 \%$$

$$\mu = 399.6 \pm 5.8$$

$$\sigma = 22.9 \pm 4.7$$

WF

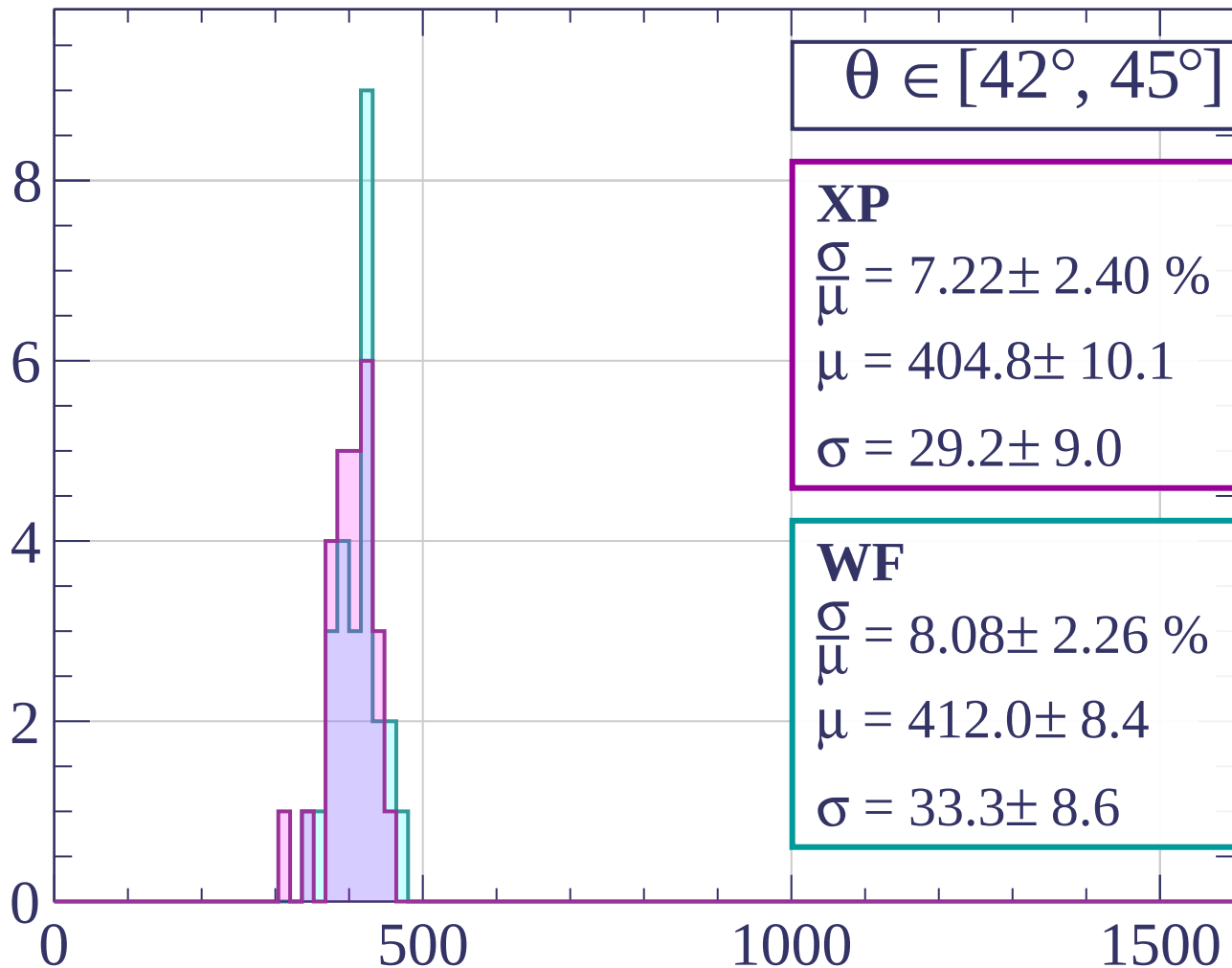
$$\frac{\sigma}{\mu} = 9.38 \pm 5.27 \%$$

$$\mu = 395.1 \pm 23.0$$

$$\sigma = 37.1 \pm 18.6$$

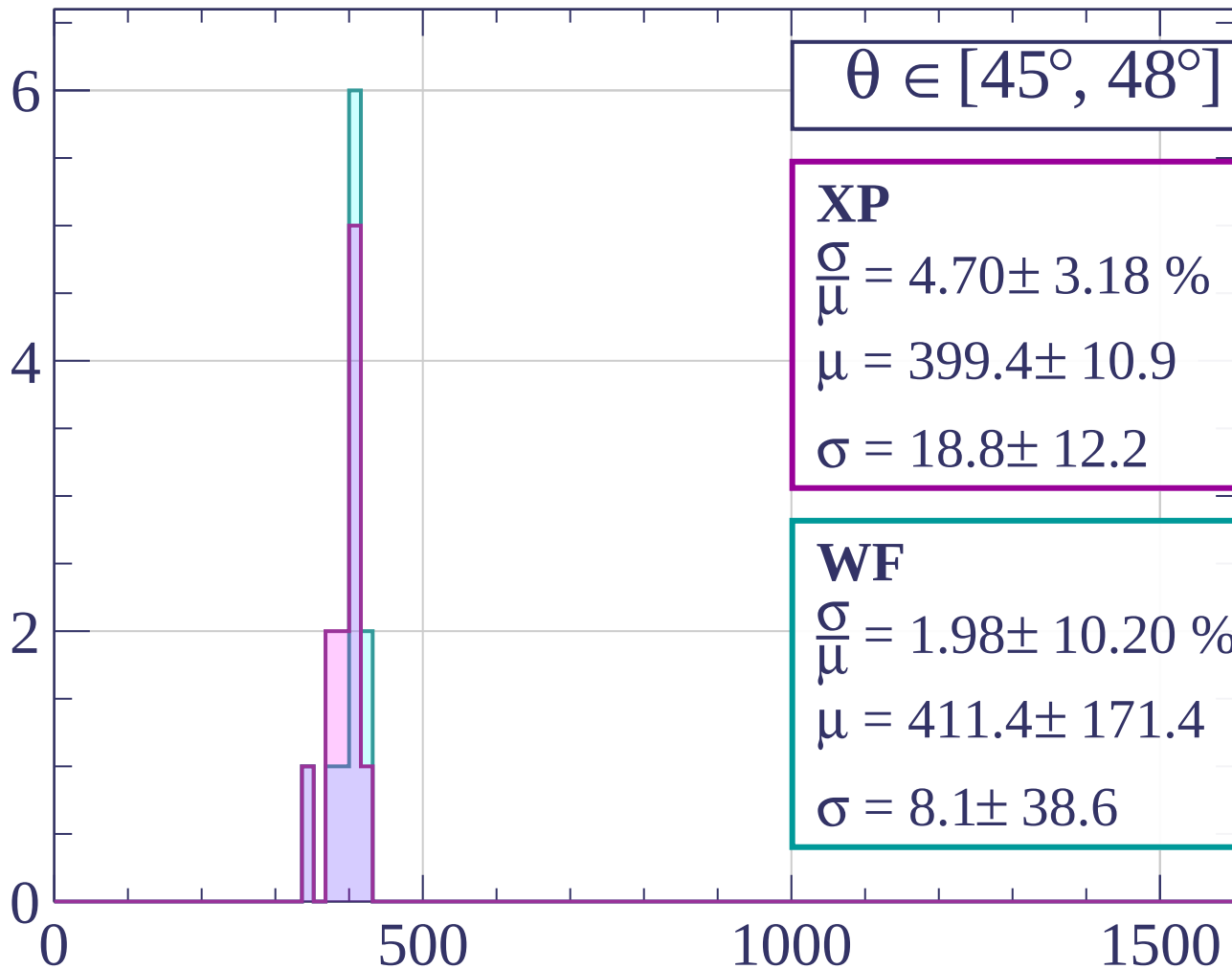
dE/dx [ADC counts/cm]

Count

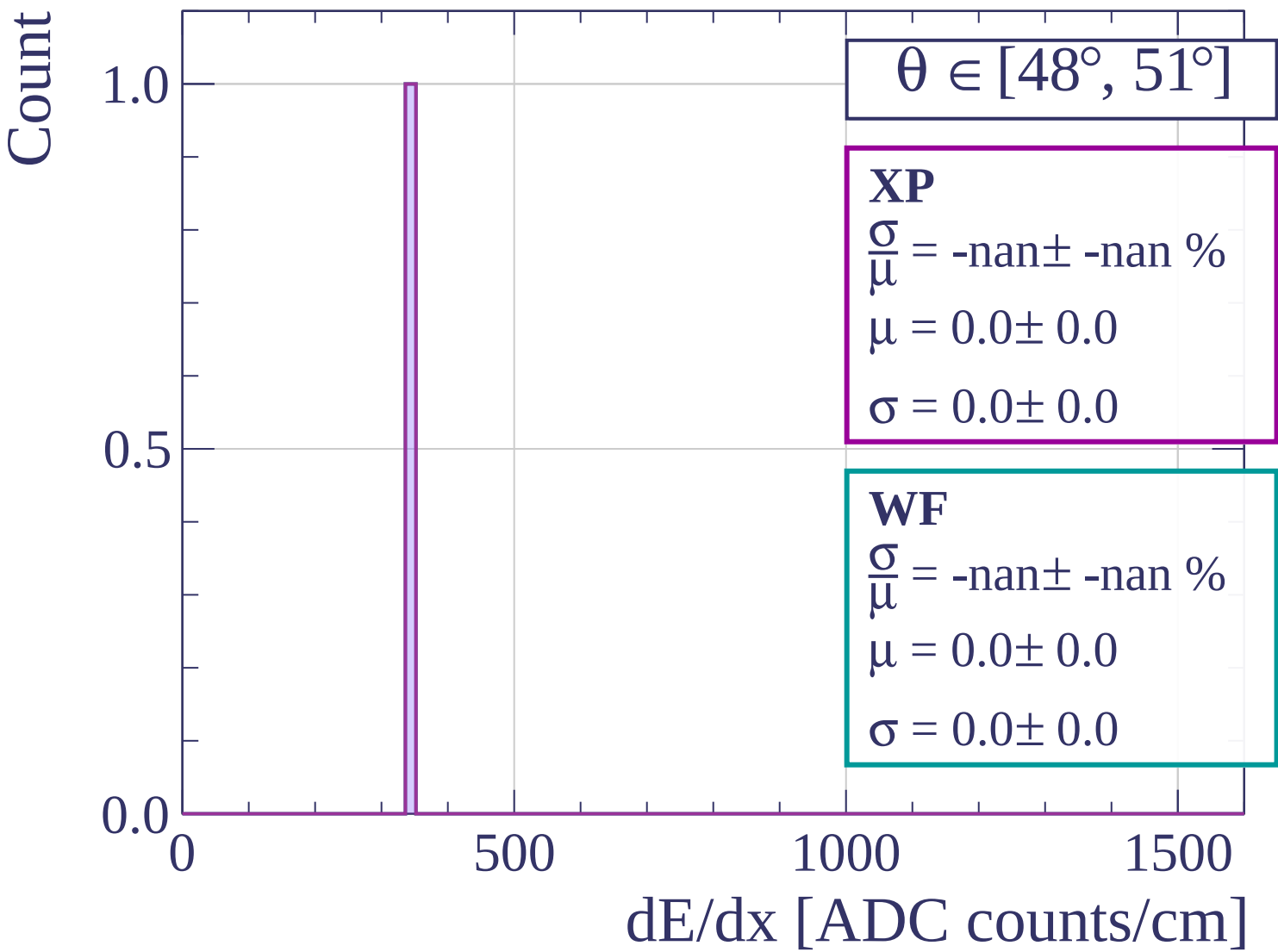


dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]



Count

$\theta \in [51^\circ, 54^\circ]$

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.5

0.0

0

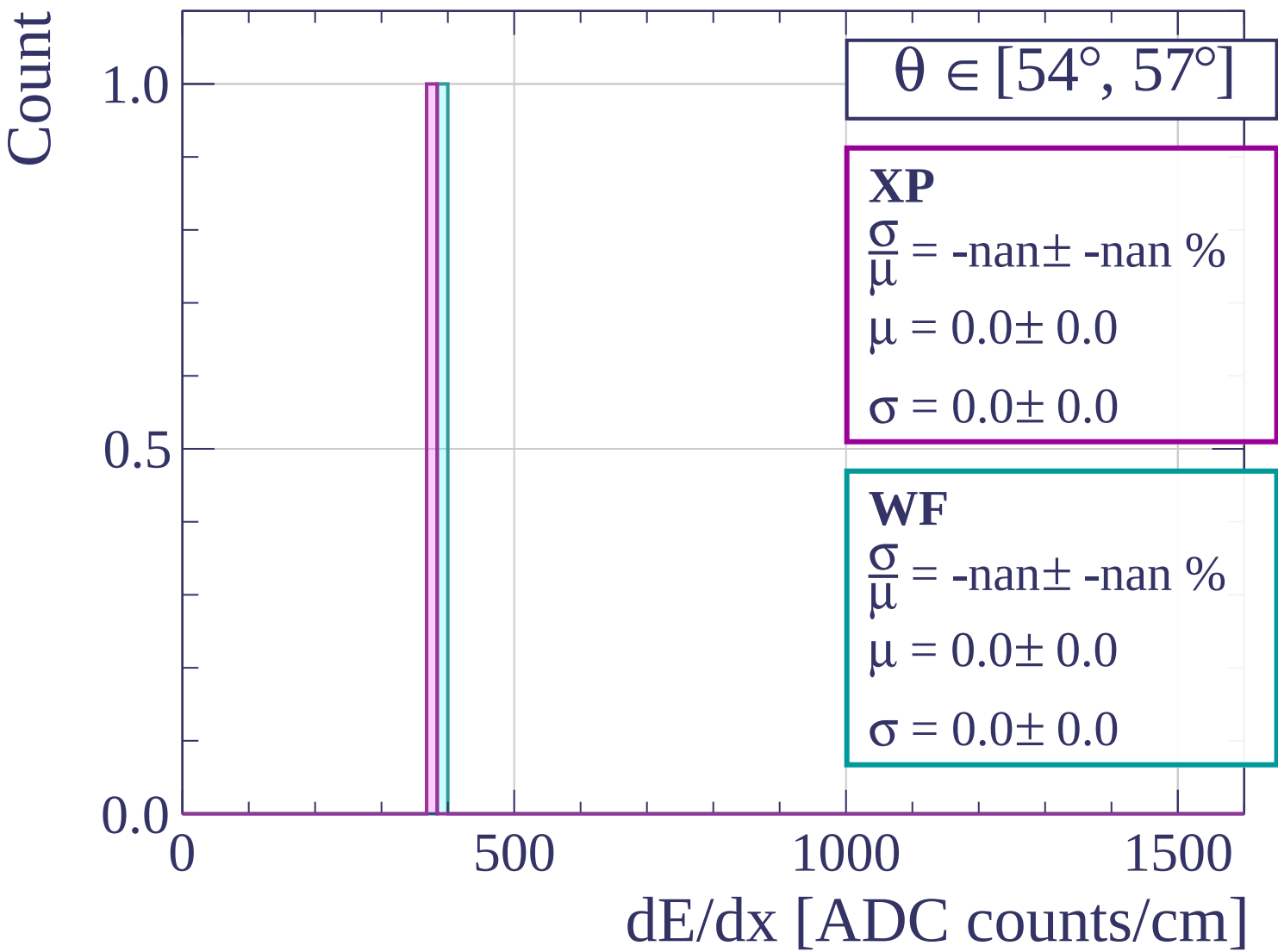
500

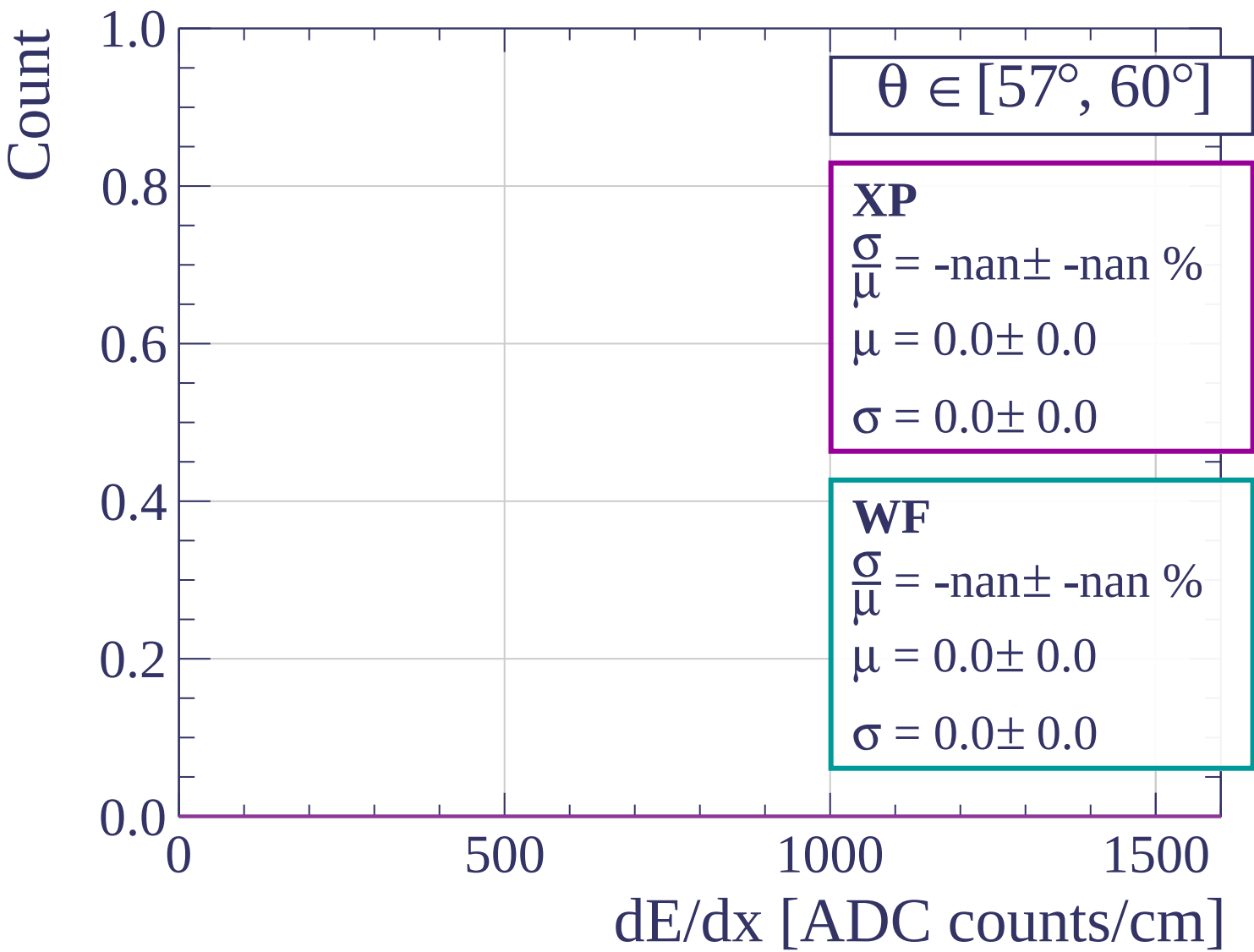
1000

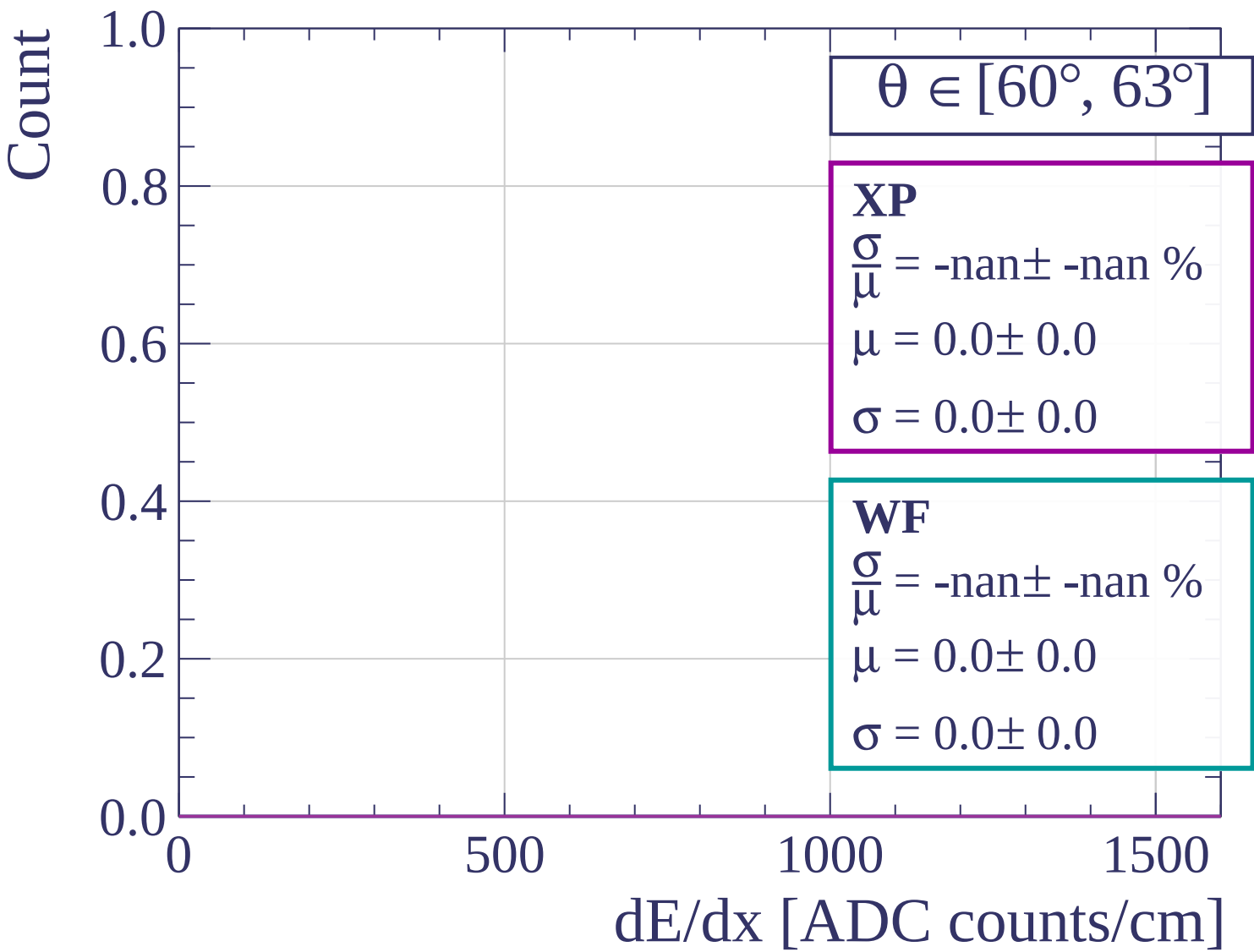
1500

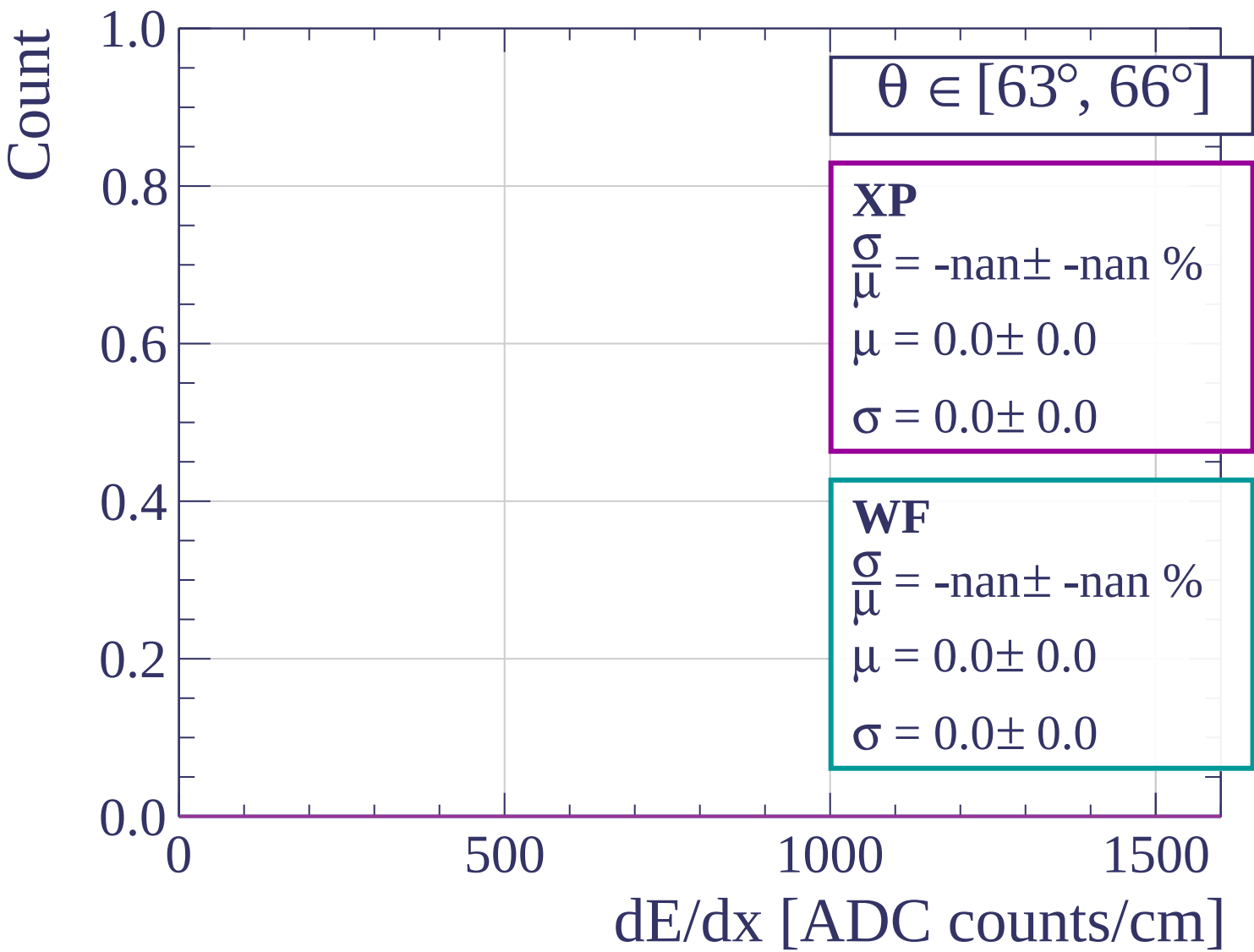
dE/dx [ADC counts/cm]

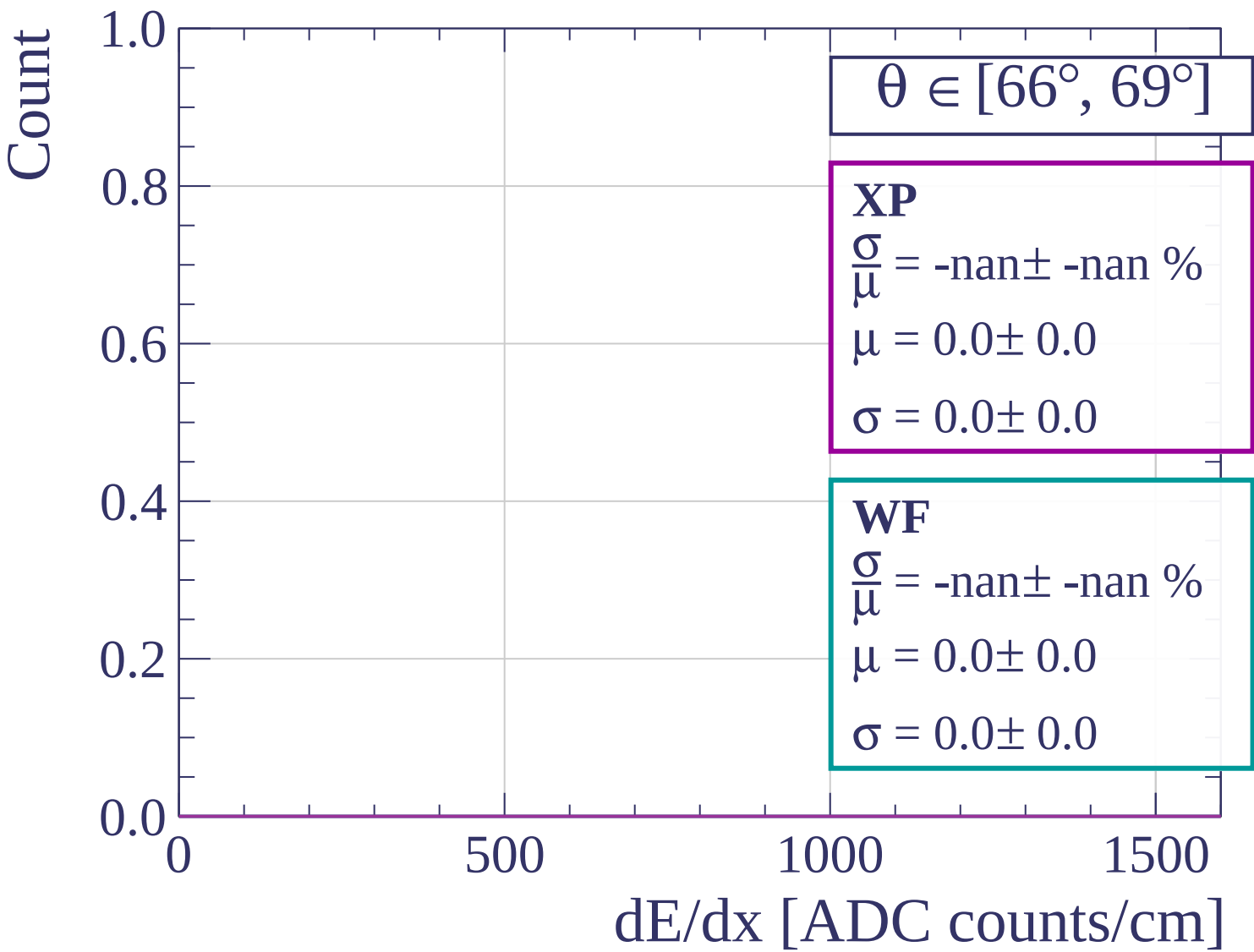


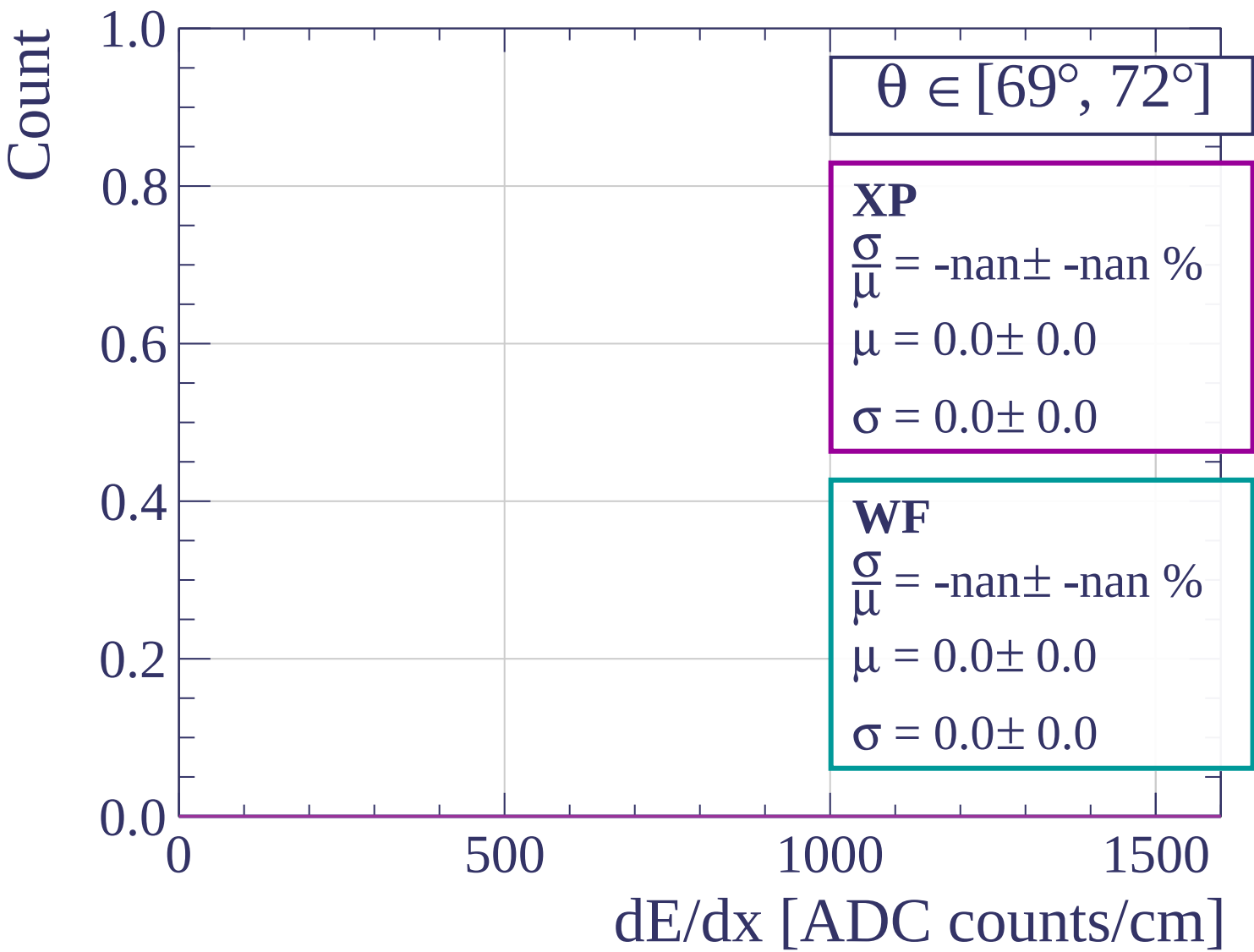


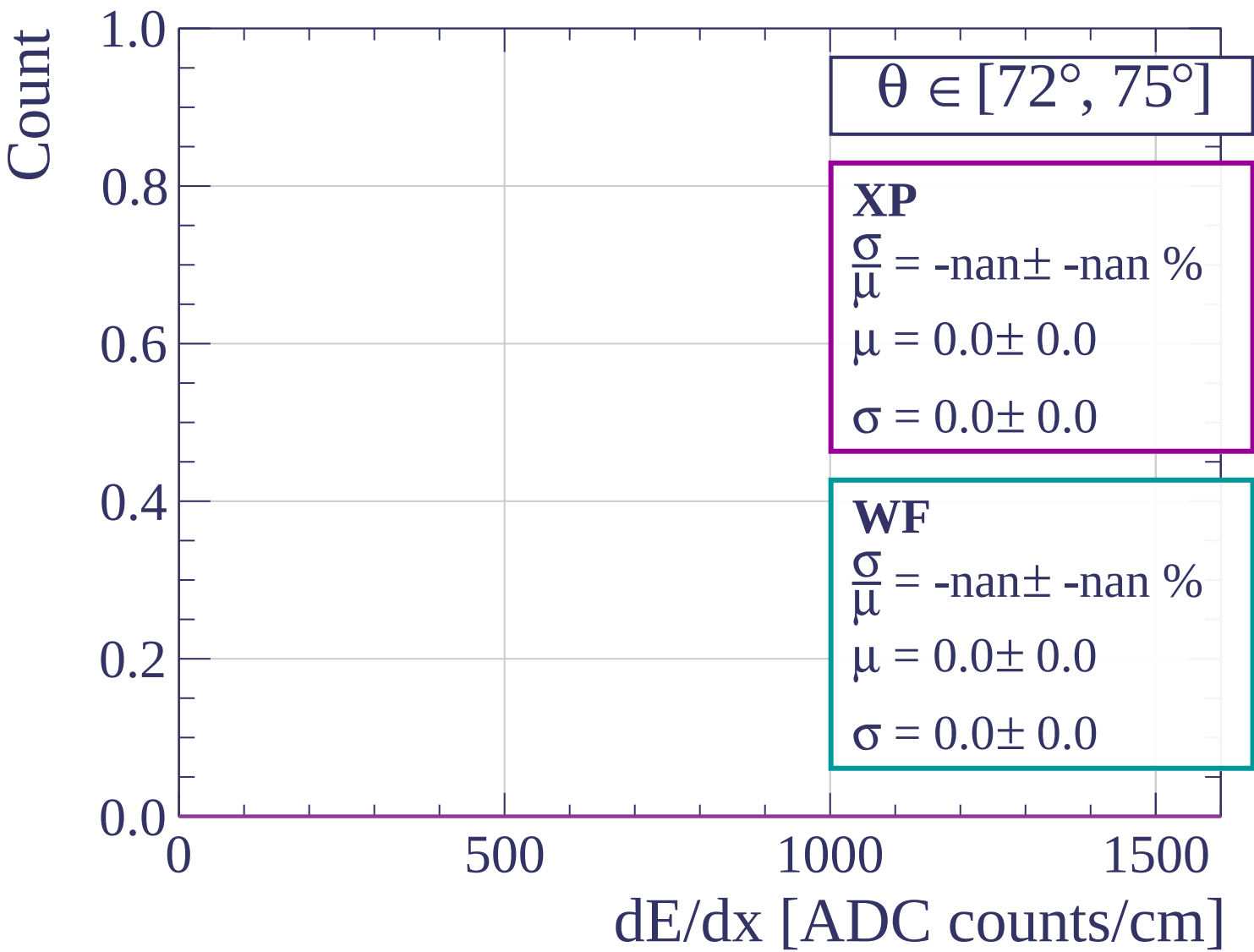


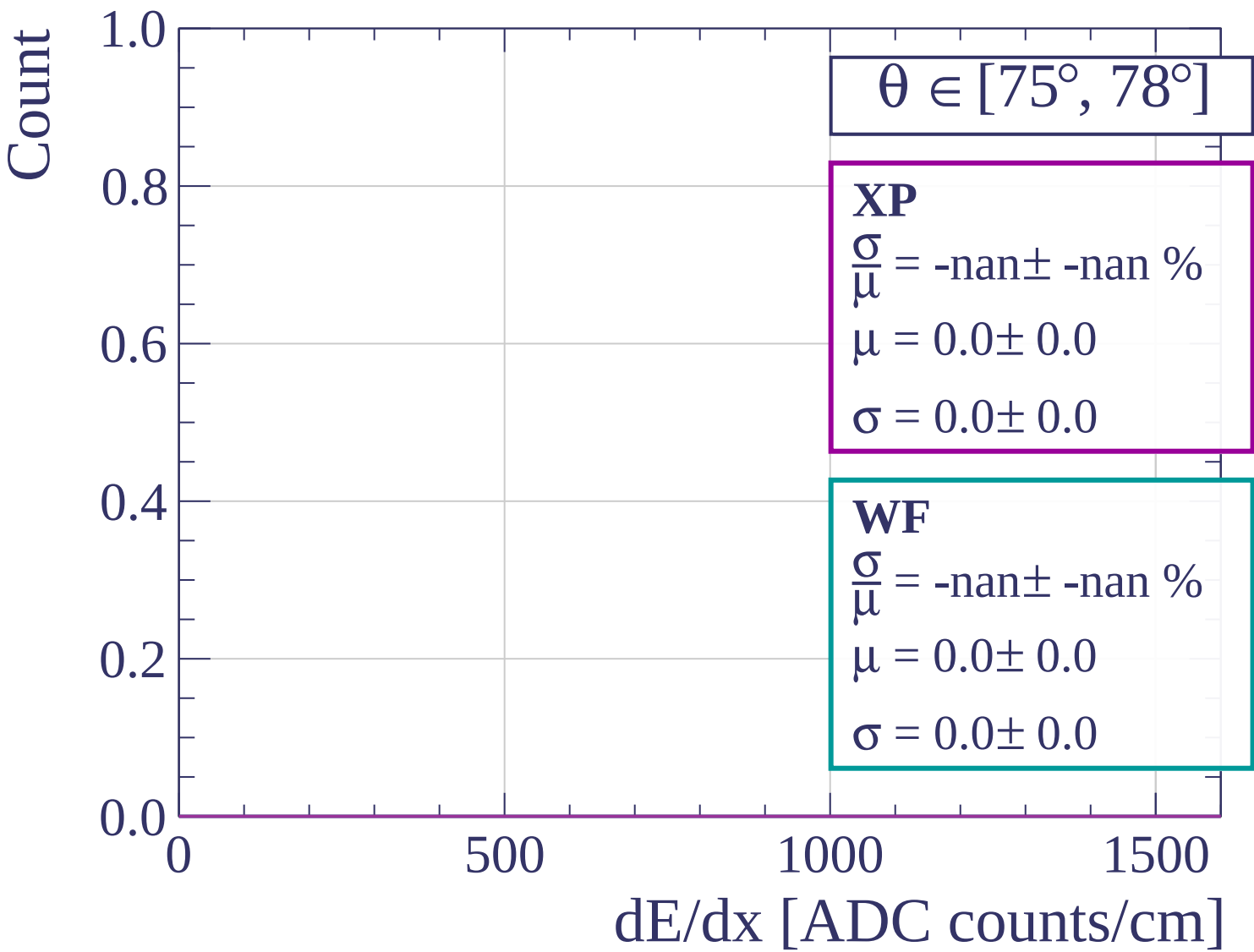


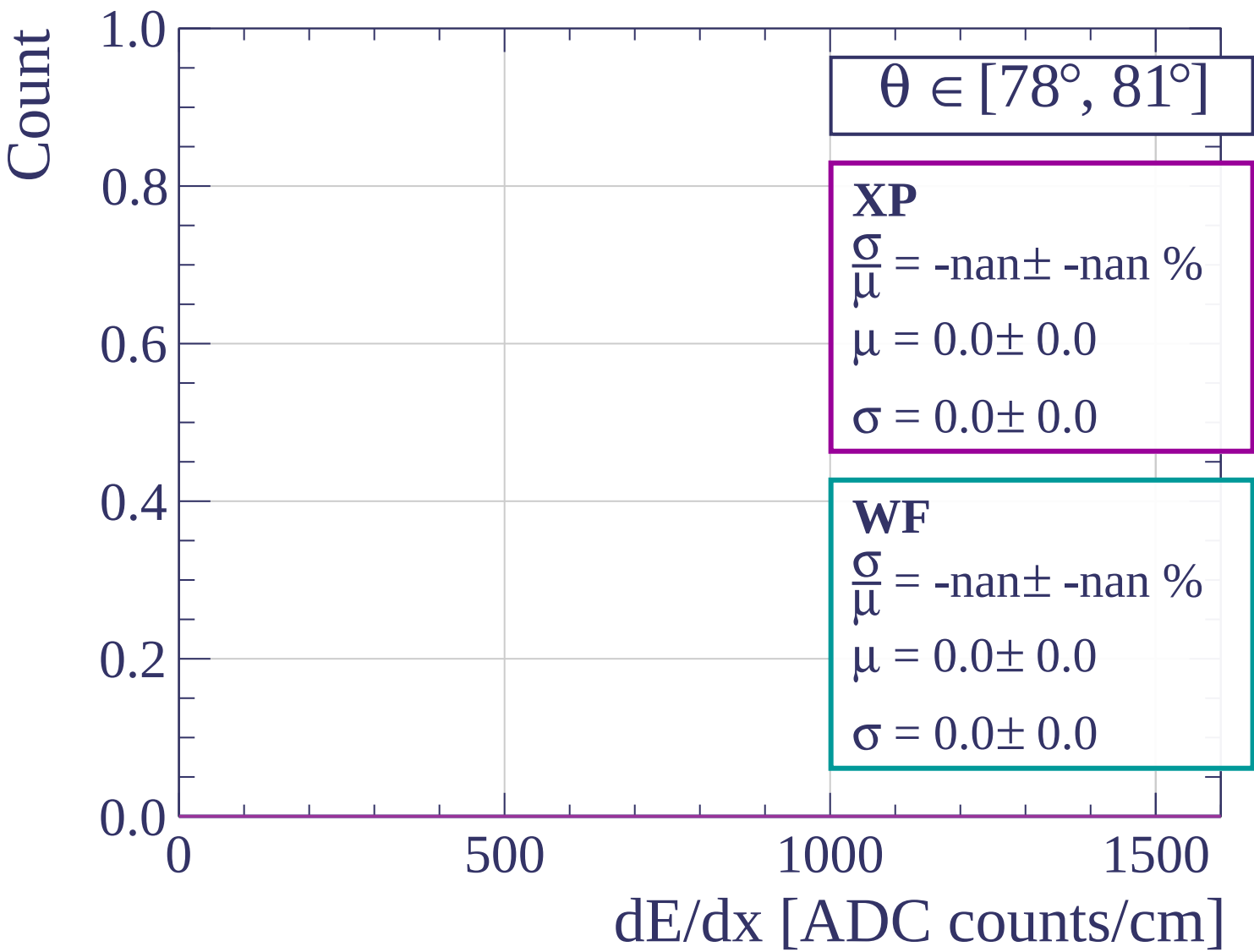


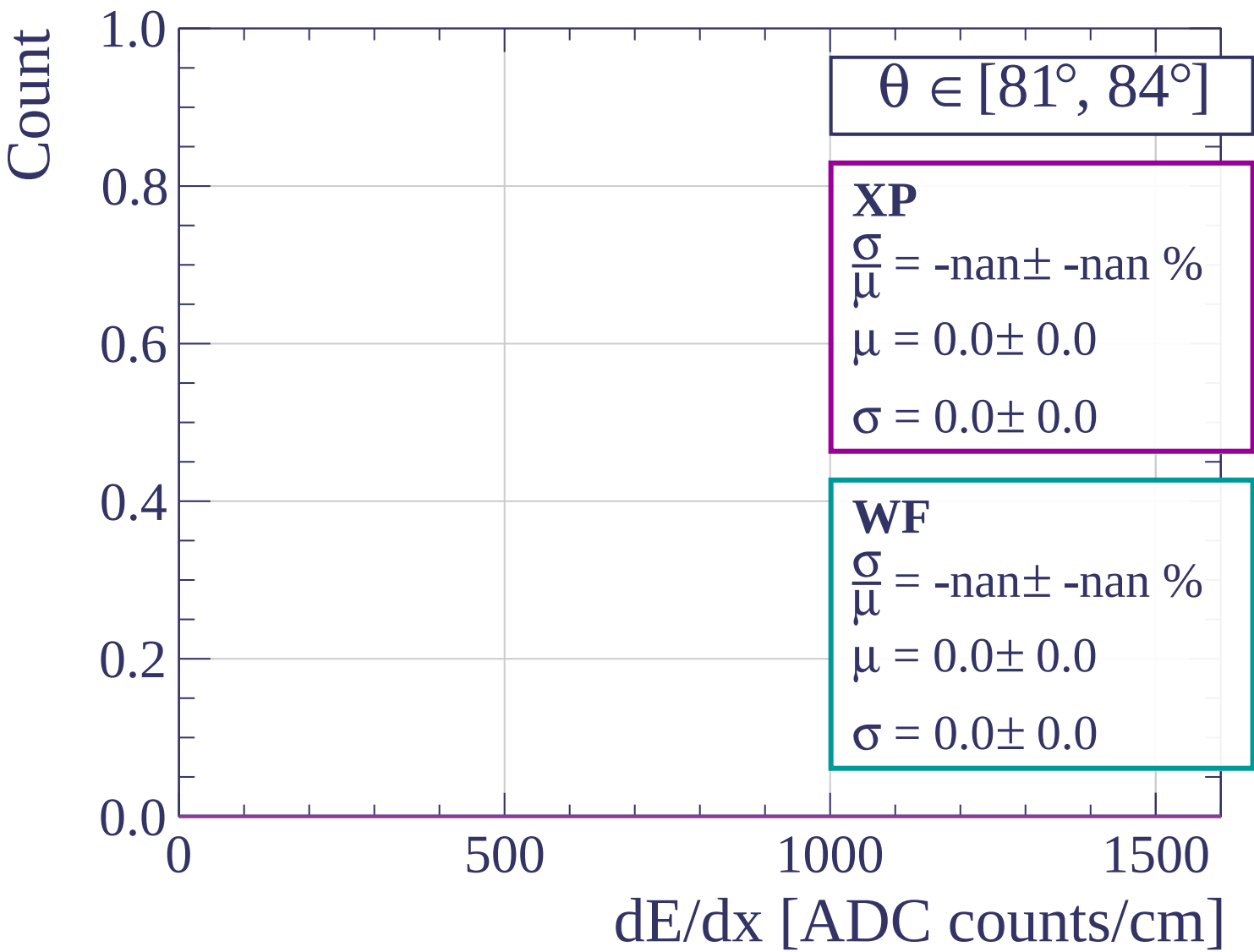


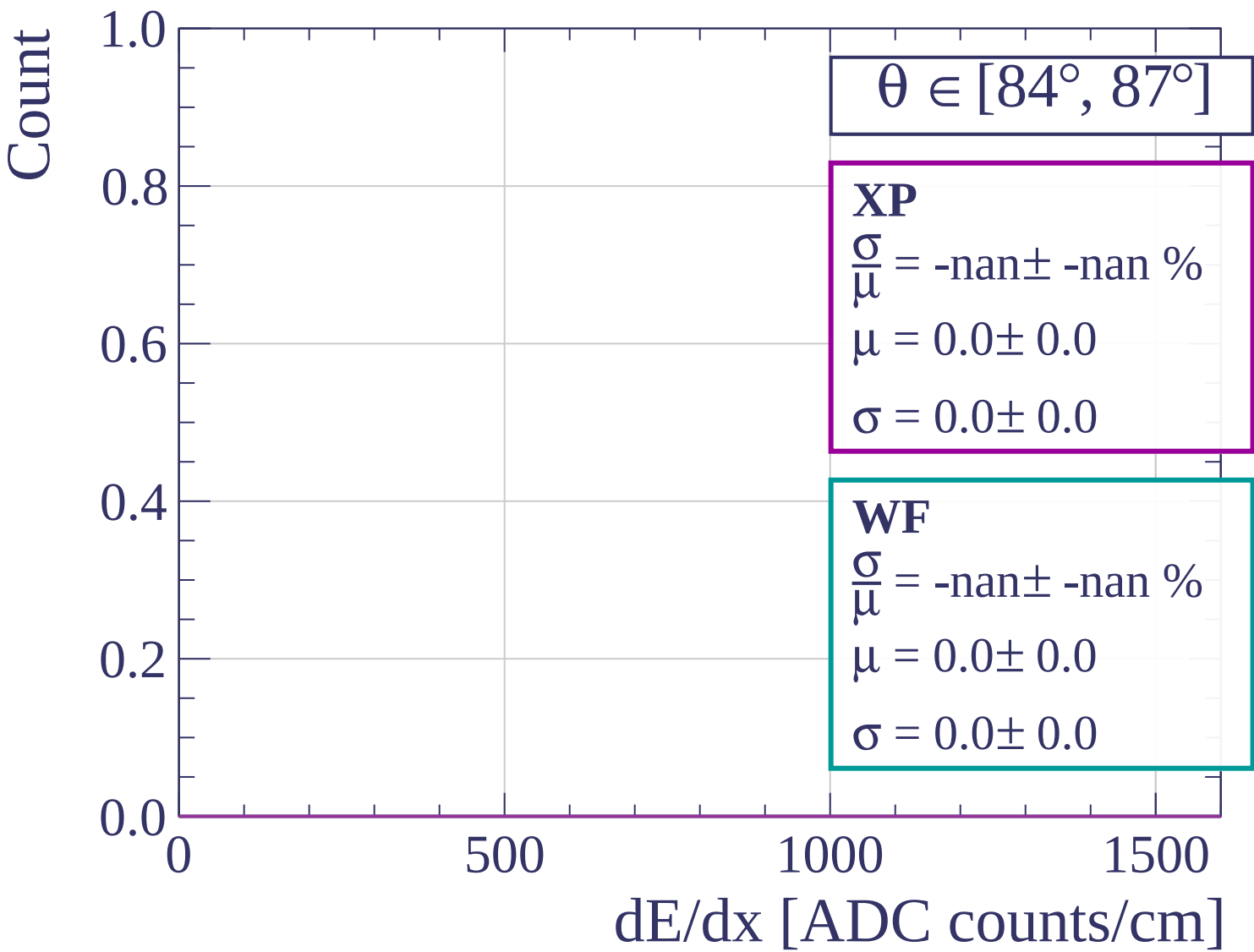


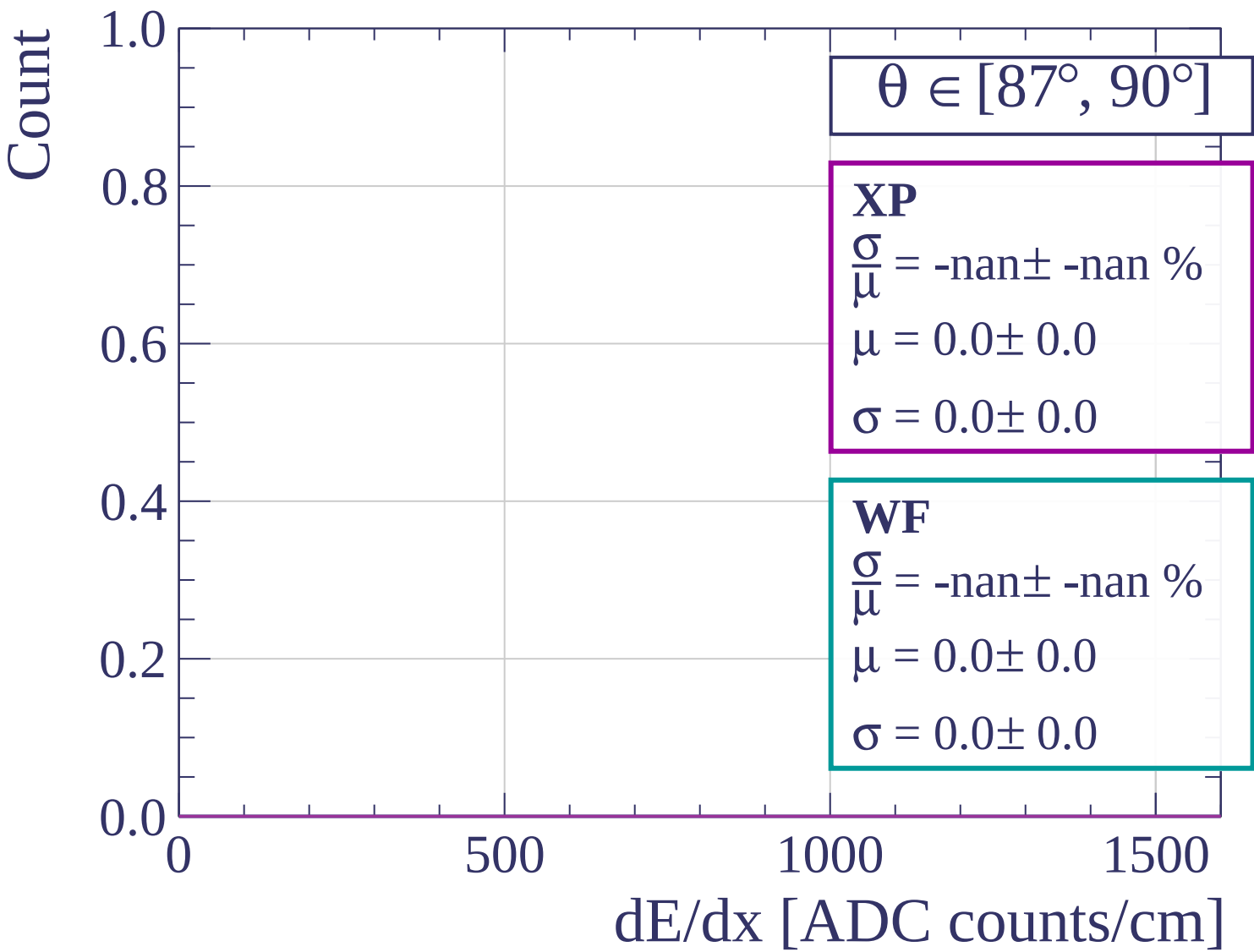




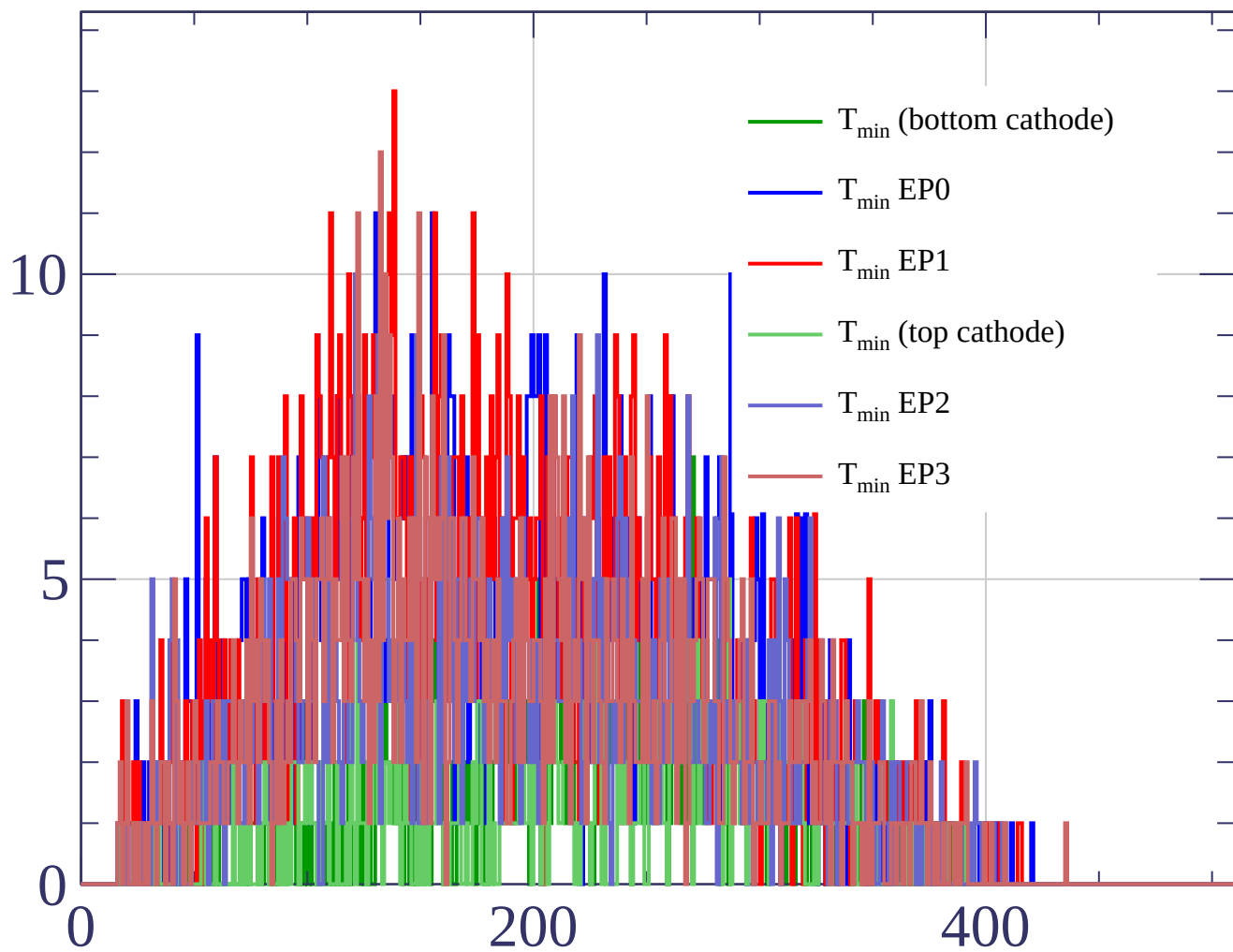








Count



Count

